

SMITHIAN FOLD THEORY - PHYSICS BRANCH PAPER 001

From Fold to Physics

An Exact, Parameter-Free and Machine-Closed Reconstruction of Physical Science from
Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

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Abstract

This paper reports a zero-parameter reconstruction of physical science from the single foundational Smithian One theorem, without imported axioms, fitted coefficients, numerical zero, negative proof magnitudes, irrational or imaginary proof values, floating-point proof equality, or continuum premises. Its lead numerical result is an exact first-principles derivation of the fine-structure constant:

$$\alpha^{-1} = 503846395469/3676744786 = 137.035999177180855\dots$$

$$\alpha = 3676744786/503846395469 = 0.007297352564321794\dots$$

The value was sealed before the registered CODATA 2022 target was released and lies inside the complete interval $137.035999177 \pm 0.000000021$. The same derivational constitution produces exact charged-lepton cubic and Koide relations, electron and muon magnetic anomalies, the on-shell electroweak share, Higgs mass and self-coupling, the Planck/proton hierarchy, proton radius, vacuum and cosmological ratios, nuclear closure numbers, hadron trajectories, inverse-square dilution, relativistic and quantum laws, thermodynamic and extraction boundaries, gravitational-wave ordering, and the remaining Physics inventory documented here.

The publication contains all 349 current engine-admitted Physics claims. Their generated grammars contain 249,072 candidates, 249,072 one-for-one decisions, 349 unique survivors and 1,396 passing mandatory adverse controls. Of those claims, 235 contain sealed post-derivation external-validation records; formal claims are connected to their registered empirical successors or retain their exact structural test boundary. The categorical clean-room audit closes all 488 Physics-owned V1/V2 atoms at the declared current-evidence boundary, with no open Physics atom or gap family. Grand Lock 075 proves the complete root trace and structural perturbation result; Grand Lock 076 reconciles the full empirical vector, including adverse and scope-boundary rows. These locks establish a reproducible release snapshot while leaving Physics permanently open to lawful extension, correction and falsification.

1. Publication and authorship boundary

PUBLISHED SAME-PAPER SUCCESSOR VERSION 1.2. The preceding version 1.1 remains permanently preserved in the Zenodo version chain. This version publishes the Physics full-field roadmap while preserving all 349 admitted claims, both Grand Locks and their exact empirical boundaries.

Maria Smith, independent researcher and founder of Ernos Labs. Contact: Maria.Smith.Sftoe@gmail.com. Submissions and reproducibility reports: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

The paper is prepared for CC BY 4.0 distribution; the engine code remains under the repository's Apache-2.0 license. Copyright preserves authorship while the licenses preserve open inspection, reuse and independent criticism. Use of the Ernos Labs designation requires adherence to the published empirical and community standards.

1.1 Ernos Labs and the authorship of this work

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to ask readers to trust a new institution. Its purpose is to make the work inspectable enough that trust is unnecessary. Every claimed law must expose its premises, generated alternatives, eliminations, survivor, controls, measurement custody, unfavorable evidence and exact receipt. A reviewer needs no credential, subscription, institutional affiliation or permission to reproduce an acceptance or demonstrate an invalidation.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That biography is not offered as evidence for a physical result; the derivations and observations must carry that burden. It is evidence about access. A system that decides who may be heard before their work can be inspected does not merely exclude individuals: it loses questions, methods and discoveries that its credential and capital filters never authorize. This paper therefore refuses both romantic exceptionalism and credentialed dismissal. The same public empirical standard applies to Maria Smith, an established professor, an independent reproducer and an AI system.

1.2 Why this is an open-science publication

The institutional critique is evidence-based. Systematic research has found sponsor-associated differences in reported efficacy results and conclusions, while work on commercial sponsorship shows that funding can influence which questions are pursued

(Lundh et al., 2018; Fabbri et al., 2018). Grant selection is not an infallible oracle: empirical studies report sensitivity to review design and reviewer composition, and a Cochrane review found limited evidence that grant peer review improves the quality of funded research (Gallo et al., 2023; Demicheli and Di Pietrantonj, 2007/2022 archive). Null and negative results remain systematically underrepresented, distorting the visible scientific record (Nature Communications, 2025). UNESCO's Recommendation on Open Science calls for methods, software, source and outputs to be open to rigorous scrutiny and identifies paywalls and high publication charges as sources of inequality (UNESCO, 2021).

Opaque prediction presents a related problem. A black-box system may be a useful instrument and may pass a declared blind benchmark, but predictive accuracy alone does not disclose a law or a derivation. NIST accordingly treats transparency, explainability, interpretability, validity and reliability as distinct properties of trustworthy AI (NIST AI RMF 1.0). Ernos Labs therefore rejects three substitutions: funding success is not evidential closure; professional consensus is not generated uniqueness; and a black-box score is not an explicit scientific law.

The editorial position is direct. Knowledge institutions organized through competitive capital, scarce grants, prestige markets, subscription access and author charges have incentives that are not identical to free human inquiry. Expertise, criticism and measurement remain indispensable, but credentials, funding and consensus cannot select a fundamental law and then stand in for its evidence. The remedy is not weaker science. It is a harder, public standard: derive the law, generate its alternatives, preserve the failures, seal the prediction, expose the observation and permit anyone to reproduce the decision.

1.3 Rights, participation and the Ernos Labs designation

Maria Smith retains copyright and scientific authorship. CC BY 4.0 for papers and documentation and Apache 2.0 for code permit inspection, copying, criticism, modification and redistribution with attribution. The words “Ernos Labs” are a separate standards-conformance designation. A fork may freely test or reuse the open work, but it may represent itself as Ernos Labs only while it follows the published empirical constitution, preserves adverse evidence, submits to critical review and does not weaken the engine or admission route.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Submissions must include the full derivation chain and pass the same engine; reputation cannot open the gate and lack of credentials cannot close it. Contact Maria.Smith.Sftoe@gmail.com, join <https://discord.gg/ucwGryVxGr>, or inspect the public work at <https://github.com/MettaMazza>.

1.4 Published sequence through Physics

The present release stops at Physics. Chemistry, Materials and every later branch remain outside this ordered update.

Order	Paper	Version	Open paper	Archival DOI
00	<i>There Is No Nothing — Methods Paper 00</i>	0.3.0	Markdown	10.5281/zenodo.21627646
01	<i>From Nothing to Fold — Foundation</i>	1.3.0	Markdown	10.5281/zenodo.21627656
02	<i>From Fold to Mathematics</i>	1.4.0	Markdown	10.5281/zenodo.21627708
03	<i>From Distinction to Information</i>	1.3.0	Markdown	10.5281/zenodo.21627717
04	<i>After Turing: The Fold Machine</i>	1.3.0	Markdown	10.5281/zenodo.21627721
05	<i>The Quantum Fold Machine</i>	1.3.0	Markdown	10.5281/zenodo.21627748
06	<i>From Fold to Physics</i>	1.2.0	Markdown	10.5281/zenodo.21627765

2. Lead derivation: the exact fine-structure constant

The paper does not present α as a guessed expression, a regression target or a coincidence selected from nearby fractions. Its construction is a typed chain from earlier admitted Fold structure:

1. The foundational Fold distinction supplies exactly two fibre labels, so $b = 2$.
2. The independently enumerated second nonidentity recurrence closes uniquely at generator $c = 3$.

3. Three-direction support gives the complete generation volume $3^3 = 27$; its successor volume is $3^4 = 81$.
4. The least complete binary covers are forced by exact inequalities: $2^4 < 27 \leq 2^5$ and $2^6 < 81 \leq 2^7$. Therefore the down and up cover depths are exactly 5 and 7.
5. Those typed blocks force the tower $2^7 = 128$, the rank-two boundary block $3^2 = 9$, and the complete directed cover $2 \cdot 5^3 = 250$.
6. Promoting one of the three interchangeable directions from depth five to depth seven at each finite rung produces the complete ladder $(5^3, 5^2 \cdot 7, 5 \cdot 7^2, 7^3) = (125, 175, 245, 343)$.
7. The terminal effective cover is therefore the exact positive rational

$$C = 250 + 1/(175 + 1/(245 + 1/343)) = 3676744786/14706643.$$

8. The tower, boundary, returning One and effective cover then force

$$\alpha^{-1} = 128 + 9(C + 1)/C = 503846395469/3676744786,$$
and hence

$$\alpha = 3676744786/503846395469.$$

The four exact promotion stages converge internally from 34259/250 to 5995462/43751, 1468922449/10719245, and the terminal 503846395469/3676744786. The formal claim generated and decided all 1,024 members of its declared grammar, admitted exactly one survivor, passed minimality and named-shape uniqueness, passed all four mandatory adverse controls, and was independently reconstructed. Its 256-form empirical successor kept the CODATA target inaccessible until the prediction seal, converted the complete published central value and uncertainty into exact rational endpoints [137.035999156, 137.035999198], found the sealed result inside that interval, and rejected the deliberately tampered control. Measurement confirms the result; it does not select any rung, coefficient or survivor.

This derivation is the paper's central numerical result. It is also a dependency rather than an isolated endpoint: the same exact α propagates into the charged-lepton refinement, electroweak share, Higgs relation, proton/Planck hierarchy, magnetic anomalies, proton radius and other terminal precision laws below.

3. Landmark first-principles results

The discoveries are ordered by physical consequence. Evidence identifiers are supplied for reproducibility, but the findings—not filenames or hashes—are the scientific headlines.

Headline finding	Exact first-principles result	External measured comparison	Evidence claim
Fine-structure constant	$\alpha^{-1} = 503846395469/3676744786 = 137.03599177180855\dots$; $\alpha = 3676744786/503846395469 = 0.007297352564321794\dots$	CODATA 2022: $\alpha^{-1} = 137.035999177 \pm 0.000000021$; the sealed exact result lies inside the complete interval	SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
Charged-lepton mass structure	exact cubic invariants (1, 1/6, 1/485, 3/1454), terminal α^3 refinement, and Koide invariant 2/3	both terminal adjacent-mass-ratio consequences and exact 2/3 lie inside the complete CODATA-derived intervals	SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002
Higgs mass and self-coupling	$m_H/v = 1/2 + 6\alpha/5$; with post-seal $v=246.22$ GeV, $m_H = 125.266104978\dots$ GeV; $\lambda = 0.1294167525\dots$	mass prediction lies inside PDG 2025 [125.09, 125.31] GeV; complete ATLAS, CMS and self-coupling rows retained	SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066
On-shell electroweak share	$1930922298157999/8642477221479757 = 0.223422318471252882\dots$	inside PDG [0.22333, 0.22351]; compatible-input W/Z check passes and the all-input tension is retained	SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003
Squared Planck/proton hierarchy	$2^{127}(1 - 2\alpha/3) = 1.693134633261878984\dots \times 10^{38}$	inside the complete propagated CODATA interval [1.693112648161510904\dots, 1.693187333165341130\dots] $\times 10^{38}$	SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003

Headline finding	Exact first-principles result	External measured comparison	Evidence claim
Electron and muon magnetic anomalies	$a_e = 0.00115965218046558296\dots$; $a_\mu = 0.00116592071499941707\dots$, both exact positive fractions	inside the complete CODATA electron and Fermilab world-average muon intervals without fitted coefficients	SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004
Proton rms charge radius	coefficient $4(1 - \alpha/10)$; post-seal scale translation gives approximately 0.840621761 fm	inside the registered 2026 electronic-hydrogen, muonic-hydrogen, CODATA-2022 and conservative PRad intervals; historical CODATA-2014 mismatch retained	SFT-PHYS-PROTON-RADIUS-TERMINAL-029
Vacuum and cosmological scale	local floor $1/2^{20}$; vacuum share $11/16$; normalized $\Lambda(c/H)^2 = 33/16$	$11/16$ lies inside Planck $[0.6833, 0.6945]$; the dimensional Λ interval is transported only after post-seal H and c	SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036
Dark-to-baryon and expansion calibration	dark/baryon $27/5$ with shares $27/32$ and $5/32$, refined $279/52$; late/early expansion $13/12$, refined $3305/3048$	both ratio families lie inside their complete registered Planck and SH0ES observational intervals	SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001
Second nonidentity generator	exact count 3; first-return orbit $1/7 \rightarrow 2/7 \rightarrow 4/7 \rightarrow 1/7$	structural exact result	SFT-PHYS-STRUCT-GENERATOR-THREE-001
Stable spatial dimension	3	independent structural stability census	SFT-PHYS-SPACE-DIMENSION-THREE-001
Boundary rank	2	boundary of the forced three-direction carrier	SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
Source dilution exponent	2; response = source/ r^2	experiment reports $2 + q$ with $q = (2.7 \pm 3.1) \times 10^{-16}$; 2 is the sole positive integer in the interval	SFT-PHYS-VALIDATION-INVERSE-SQUARE-001
Nuclear closure sequence	(2, 8, 20, 28, 50, 82, 126, 184)	exact match to the complete registered IAEA sequence	SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001
Binding-per-nucleon maximum	$A=62$, $Z=28$, $N=34$ (nickel-62)	same unique maximum in the complete 2,548-row positive-composite AME2020 census	SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
Primordial support partition	scalar $31/32 = 0.96875$; tensor $1/32 = 0.03125$	Planck n_s interval $[0.9607, 0.9691]$; tensor bound $r < 0.032$, passed by 0.00075; no equality to conventional e-fold count is claimed	SFT-PHYS-VALIDATION-INFLATION-GROWTH-040
Radiation scaling	fourth-power scale law; exact doubling ratio $2^4 = 16$	post-seal comparison with the measured Stefan fourth-power exponent; no dimensional coefficient is claimed as structure-only	SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042
Physics Grand Lock	349 current claims; pre-lock certificate over 347 claims and 534 dependency nodes; 21/21 declared generator-dependent headline values move under $3 \rightarrow 4$ while half-One, spatial rank 3 and boundary rank 2 hold	234 pre-lock empirical receipts reconcile 147 distinct external source identities; all 14 detected unfavorable-result or scope-boundary claims and all 6 legacy receipt shapes remain explicit	SFT-PHYS-VALIDATION-GRAND-LOCK-076

3.1 Charged leptons: mass structure without irrational roots

Generator-three support forces one three-root carrier with symmetric invariants $\text{sum} = 1$, $\text{pair sum} = 1/6$, leading $\text{product} = 1/485$ and sharpened product $3/1454$. The terminal refinement holds from that product the unique complete

electromagnetic self-coupling $\alpha^3/3^3 \cdot (5 + 7\alpha/3)$. Exact rational root brackets then predict both adjacent squared mass-ratio consequences; both lie inside their complete CODATA 2022 intervals. Independently, the Fold normalization forces the charged-lepton Koide invariant exactly $2/3$, which lies inside the complete CODATA-derived interval. No measured mass enters the cubic, no irrational root is admitted into proof, and no mass-ratio row chooses the invariant.

3.2 Electroweak and Higgs sector

The terminal electroweak construction uses complete binary support sixteen, the three held generator directions, charged support 15^2 , four neutral pair channels of 14^2 , and one terminal $\alpha/17$ self-return. It forces

$$\sin^2\theta_W = 1930922298157999/8642477221479757 = 0.223422318471252882....$$

That sealed value lies inside the complete PDG on-shell interval $[0.22333, 0.22351]$. Its exact One-complement passes the compatible-input W/Z interval; the known all-input tension is preserved rather than deleted or used to change the derivation.

For the scalar sector, half-One is the unique displaced ground and the leading rungs are $1/2$, $1/4$ and $1/8$. Six-cell scalar support, its held return and cover depth five force $m_H/v = 1/2 + 6\alpha/5$. Applying the registered post-seal dimensional reference $v = 246.22 \text{ GeV}$ yields the exact mass $31557437733819647/251923197734500 \text{ GeV} = 125.266104978... \text{ GeV}$, inside the PDG 2025 interval $[125.09, 125.31] \text{ GeV}$. The same relation forces native $\lambda = (m_H/v)^2/2 = 0.1294167525....$ Individual ATLAS and CMS offsets and the present broad direct self-coupling constraint remain explicit in the validation record.

3.3 Precision magnetic anomalies and proton structure

With the admitted exact α and exact terminal turn $355/113$, the finite electron loop forces one exact rational anomaly $a_e = 0.00115965218046558296...$, inside the complete CODATA interval centred on 0.00115965218046 with standard uncertainty 0.00000000000018 . The second-generation correction $\alpha^2(2/17 + \alpha/106 + \alpha^2/853)$ forces $a_\mu = 0.00116592071499941707...$, inside the Fermilab world-average interval centred on 0.001165920715 with standard uncertainty 0.000000000145 . Neither value imports a conventional perturbation series or fitted coefficient.

The proton charge-support law independently forces the dimensionless coefficient $4(1 - \alpha/10)$. Post-seal composition with the CODATA reduced proton Compton scale gives approximately 0.840621761 fm . It lies inside the registered current electronic-hydrogen, muonic-hydrogen, CODATA-2022 and conservative PRad intervals. The disjoint historical CODATA-2014 interval remains in the record as an unfavorable historical row rather than being erased.

3.4 Scale hierarchy, vacuum and cosmological ratios

The terminal squared Planck/proton hierarchy is exactly $2^{127}(1 - 2\alpha/3) = 1.693134633261878984... \times 10^{38}$, inside the complete outward-propagated CODATA mass interval. Vacuum structure separates three quantities that conventional shorthand can conflate: a local boundary-energy floor $1/2^{20}$, a global vacuum share $11/16$, and normalized cosmological magnitude $\Lambda(c/H)^2 = 33/16$. The global share lies inside the complete registered Planck interval; dimensional Λ is obtained only after the sealed coefficient is composed with held observational H and exact C scales.

The three-space generation volume 27 and its least binary cover 32 force dark and baryonic shares $27/32$ and $5/32$, ratio $27/5$, with the period-five refinement $279/52$. A separate Fold calibration forces matter/vacuum shares $1/3$ and $2/3$, a leading late/early expansion ratio $13/12$, and depth-seven refinement $3305/3048$. Both ratio families pass their complete registered Planck and SH0ES interval tests.

3.5 Space, force, nuclei, hadrons and gravitational waves

The independently enumerated generator is three; stability selects three spatial directions; their boundary has rank two; and complete propagation across that boundary forces inverse-square dilation. The exponent is exactly 2 , while the registered experiment reports $2 + q$ with $q = (2.7 \pm 3.1) \times 10^{-16}$, making two the sole positive integer in the complete interval.

The same structure forces the nuclear closure sequence $(2, 8, 20, 28, 50, 82, 126, 184)$ and the unique positive-composite AME2020 binding-per-nucleon maximum at nickel-62. Light-hadron support follows the exact squared trajectory $Q(J) = (6J - 3)/5$; all five registered carriers lie inside their complete resonance-support intervals without fitting a slope or intercept. Gravitational-wave structure forces inspiral with rising chirp, merger, then damped ringdown; the complete GW151226 and GW190521 records validate the three-stage ordering while retaining the short-signal and alternative-source boundaries.

4. What the result changes

The scientific claim is not merely that several fractions approximate familiar numbers. One exact, axiom-free derivational language produces the structural dimension, force exponent, dimensionless couplings, mass ratios, scale hierarchies, thermodynamic boundaries, quantum relations and macroscopic propagation laws under one admission standard. Every numeric result is connected to the same foundational root, generated within an explicit finite grammar, isolated from its measurement target, and tested after sealing. That conjunction—unification, exactness, zero fitted parameters, uniqueness enumeration and external validation—is the work's substantive contribution.

The paper therefore keeps three levels distinct without diminishing any of them: the physical result is the headline; the exact derivation explains why it follows; and the receipt, hash, adverse controls and independent reconstruction establish that the reported chain is the chain actually executed.

5. Complete current Physics status

The current categorical inventory contains 349 Physics claims in 13 ordered subbranches. All 349 have immutable model-admitted receipts. The categorical V1/V2 ownership audit identifies 488 Physics-owned atoms, all 488 closed at the declared same-strength current-evidence boundary, with no open atom or gap family. Formal Grand Lock 075 binds the complete pre-lock ownership surface, verifies its acyclic 534-node dependency dictionary, proves that every Physics route reaches the foundational One and tests generator dependence. Empirical Grand Lock 076 reconciles all 234 pre-lock empirical claims, 147 distinct external source identities, every available measurement receipt, all disclosed legacy receipt shapes and every detected unfavorable or scope-boundary claim.

“Closed” here means complete to the dated corpus, methods and evidence standard. It never means immune to correction or permanently closed to discovery. New findings, stronger evidence and falsifications remain lawful versioned extensions through the unchanged engine; existing receipts remain immutable.

6. Exact constitutional domain

The derivational domain admits the empty One as structural absence but never numerical zero. Proof magnitudes are positive generated counts and exact positive rational parts or ratios. Orientation, opposition and complement are held labels rather than negative quantities. Irrational, imaginary and binary floating values are barred from proof. Completed infinity, an ungenerated continuum, axioms, fitted coefficients and free parameters are also barred. External decimal measurements remain source-bound records. A finite decimal is converted to an exact rational interval only inside the empirical adapter and never gains authority to select the Fold law.

7. The single admission engine

Each claim entered the same `SFTAdmissionEngine`. Registration rejects axioms, free parameters, missing root trace, unadmitted dependencies and source-identity failure before candidate execution. The engine then checks census cardinality and identity, one decision per candidate, exactly one survivor, minimality, named-shape uniqueness, the four mandatory controls, an implementation-distinct external reconstruction and, for empirical claims, prediction isolation, target custody, complete rows and falsification. A failure halts without model admission. Accepted receipts are immutable evidence identities. The engine source is not edited by this publication correction, and the paper cannot confer admission.

8. Why superdeterminism permits uncertainty and quantum weights

No law in this branch installs ontic nondeterminism. The complete Fold state includes preparation, held labels, measurement setting, path and observation record. A probabilistic or quantum weight is an exact census ratio over unresolved support. Each branch execution remains deterministic. Measurement uncertainty records distinctions unavailable to an observation class; it does not assert causeless state selection. Bell correspondence therefore retains setting and preparation records in the complete state, identifies the factorization assumption that fails, and separately preserves the no-signalling marginal through complete remote-fibre enumeration.

9. Empirical constitution

External evidence follows one direction. First the Fold dependencies generate and eliminate candidate laws. Then a data-only Fold program receives only registered inputs and the sealed derivation identity. Its instruction set has no filesystem, network, subprocess, clock, environment, dynamic import or foreign-function capability. A distinct custodian commits target identity before execution and releases content only to the matching prediction seal. The evaluator preserves every registered row and must reject a deliberately altered or displaced control. BIPM, NIST/CODATA, NIST ASD, PDG, IAEA, IAPWS, GWOSC, CERN Open Data and NASA LAMBDA supply the external records.

10. Exact measured-value correspondence

Measured-value forcing means that a zero-parameter Fold relation produces a sealed consequence and a registered exact adapter compares it with withheld observation. The observation is evidence, not a tunable parameter: it cannot enter candidate generation, choose a survivor or add a correction. Finite decimals and uncertainties are parsed as exact positive fractions. Multiplication and quotient propagate interval endpoints in the capability-closed interpreter; target overlap is evaluated only after release. Exact official decimal prefixes with ellipses are bounded by their next decimal place. Any non-overlap halts admission.

The inverse fine-structure result belongs here, in Physics: SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001 forces the terminal exact ratio and SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001 performs its sealed CODATA comparison. The same rule governs every constant, scale and precision claim in the inventory. No redundant handwritten status may override an engine-admitted receipt or a passed empirical certificate.

11. Reading the complete derivation ledger

Each claim section below is a self-contained prose projection of its machine evidence. The complete `candidate_census.json`, `elimination_receipt.json`, `controls.json`, `certificate.json`, external experiment registration and receipt remain authoritative. The paper includes every claim identity and admitted receipt hash so the evidence map can fail closed on omission or substitution.

12. Measurement Metrology

Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain.

1. Capability-closed exact Fold prediction form

Claim identity: SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Within the registered finite prediction grammar, a physical prediction can be target-inaccessible, exact, portable and completely traced in exactly one form: immutable registered Fold inputs pass through a data-only sequence of generated exact operations to one terminal emission, while every ambient host capability and unregistered operation is absent and any violation halts.

Registered exact inputs, a data-only closed opcode surface, no ambient capability, single-assignment state, one final emission, complete trace and no extra rule.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-COMP-FORM-STATE-TRANSITION-001

- SFT-COMP-FORM-OPERATIONAL-PROCESS-001
- SFT-COMP-SEM-EVALUATION-001
- SFT-COMP-SEM-TERMINATION-001
- SFT-COMP-SEM-VERIFICATION-001

Generated grammar and closure boundary. Generate the complete product of input, value, operation, capability, state, termination, trace and extension forms. The declared exact boundary is: All finite data-only prediction interpreters assembled from admitted exact values, registered inputs, generated operations, immutable held states, a declared terminal emission and no host capability. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `registered-input-only__exact-fold-domain__data-only-generated-operation__no-ambient-capability__single-assignment-held-state__one-terminal-emission__complete-step-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
input	registered-input-only	ambient-input	Ambient input can contain the withheld target or an answer-producing channel.
value	exact-fold-domain	host-scalar-domain	Host scalars import forbidden or platform-dependent proof values.
operation	data-only-generated-operations	contributor-executable-source	Arbitrary source can create operations outside the registered grammar.
capability	no-ambient-capability	ambient-capability	An ambient resource can reveal target, comparison or mutable authority state.
state	single-assignment-held-state	mutable-overwrite	Silent overwrite loses the exact predecessor and permits untraced replacement.
termination	one-terminal-emission	implicit-or-open-ended	Implicit or open-ended completion cannot certify a complete prediction artifact.
trace	complete-step-trace	result-only	A result alone hides the operational relation and its resource path.
extension	no-extra-rule	extra-rule	An extra instruction or capability is a free answer-selecting parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d4c5b29b8c8c7fbf677579bb04cb996cdf6878a95ef678bac3998c373374bae6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:101b735f39c0308acff4f1fbb4fc7c0845d09021fb0cada8326ad7a876d77f96`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:06e9ea349c585d18965d1f715617649d780dbd450ab59bce9090d38591920b1b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1814b0879667d0341a6a7cf3b345a60550bcb94d5ad113e00cc9e2db8dc00b6`.
- **boundary:** passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt

sha256:0978722205b9e2a84a6ef2fb7e600cc50bc215a65ae8c025a2e96c53f0e6e3dd.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:2ac5e37d95aea5263c2fcf026b93eb3549e21a04ce7e0e125ad91b62a956c0cd.
Independent certificate:
sha256:060dddafaba441cc92c20cb16ac1c5e960744ea2af940ffac70c6bcce2d0d15. Engine
external-validation hash:
sha256:d6ad95de25a8dd5dd16045baead9da289366c3189e216cff4421a6308e4c7c2e.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- arbitrary Python or native contributor code
- filesystem, network, subprocess, clock, environment, import and foreign-function access
- floating, negative, irrational, imaginary and semantic-zero proof values
- unbounded or untraced execution

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b11e9de76a85a54c63f86af81bcef25db27c06262293fb9236347dcb04fb9d88; engine receipt
sha256:dc597e88dc8d1f725ef158e3e69a1bb6b4b402c4cc138372e40b1df21fe821d8 at receipts/engine/model_admitted/SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001-dc597e88dc8d1f72.json;
empirical-validation hash None; measurement receipt None.

2. Portable target commitment and post-seal custody

Claim identity: SFT-PHYS-MEAS-TARGET-CUSTODY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Within the registered blind-exchange grammar, target content remains unavailable to prediction in exactly one form: a distinct custodian commits the complete named target package and prediction envelope before execution, then releases that unchanged package once to the matching experiment only after a matching prediction seal exists.

Pre-prediction complete commitment, distinct custodian, nonce-bound unchanged content, matching experiment and envelope, one post-seal release and no extra rule.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-COMP-SEC-HASHING-001
- SFT-COMP-SEC-COMMITMENT-001

- SFT-COMP-SEC-INTEGRITY-001

Generated grammar and closure boundary. Generate the complete product of timing, holder, target support, content identity, experiment, envelope, release and extension forms. The declared exact boundary is: All finite portable target exchanges using admitted exact identities, pre-seal commitment, distinct custody and one post-seal release. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `pre-prediction-commitment__distinct-custodian__complete-named-support__nonce-bound-content-identity__matching-experiment-only__matching-envelope-only__one-postseal-release__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
timing	pre-prediction-commitment	post-prediction-choice	A post-prediction choice can select favorable target content.
holder	distinct-custodian	prediction-holder	The predictor holding targets violates structural inaccessibility.
support	complete-named-support	partial-or-open-support	Partial or open support permits selective favorable rows.
content	nonce-bound-content-identity	unbound-content	Unbound content can be replaced after prediction.
experiment	matching-experiment-only	cross-experiment-release	Cross-experiment reuse breaks registration custody.
envelope	matching-envelope-only	unbound-envelope	An unbound seal can certify a different input or relation.
release	one-postseal-release	preseal-or-repeat-release	Pre-seal or repeat release leaks target information.
extension	no-extra-rule	extra-rule	Discretionary release is a free selection parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d531095fb015f927f02c1f40fe9951b973bdf5f6a00934c768a162e60d3eb80c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:9d1ae2f40bb91451031c219678c477468054cc6bfecb398935d7356e3e0e059a`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:18e6f58bb38714edf42a2881d8bf8e19a8c4b4999a5132d7bde03a4a053f20c7`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:53b93b6397aa18513fb73bcd48e2a739a26061f47428ae3790fa8d25f0248087`.
- **boundary:** passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt `sha256:0f1c55cf30f7c60f78e3ff7f089b1c3a641fe34e00e84508ede332bab9d54b03`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:458f3324e7eb0127ec05aab491f9ddd6005718ba642b2d6ed3b67d0bc5adc1d9`. Independent certificate: `sha256:af9b5b3c7fcb2d140f04f3aa902aadd0392c9aec0bc027e7bb517042963abc13`. Engine external-validation hash:

sha256:d4f89d5039a9e646f187e2219dd0b0ee9354ef60010cf9277afb3226e49a0a9d.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- target access by the prediction executor
- release before a matching prediction seal
- selective target-row omission
- host-specific sandbox claims

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:638f04b6b15e15cffa178c9fb8bb8fe84b04a95aec12ff99edee1eae1b07a7ae; engine receipt

sha256:f788edf20c3f0bb019b813cb63534724518765fccaa1afe7610fb736ae6b21a5 at

receipts/engine/model_admitted/SFT-PHYS-MEAS-TARGET-CUSTODY-001-f788edf20c3f0bb0.json;

empirical-validation hash None; measurement receipt None.

3. Hostile empirical-package and authority protection

Claim identity: SFT-PHYS-MEAS-HOSTILE-PACKAGE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Within the registered empirical-package grammar, contributor material can enter official prediction without gaining execution or scientific authority in exactly one form: strict data-only program fields are parsed by the admitted interpreter, the census and model-admitted receipts are identical before and after handling, and every extra field, executable source or protected-tree change halts admission.

Strict data-only Fold document, exact parser, no unknown fields or direct source execution, unchanged authority trees, separate comparison, fail-closed response and no extra route.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-COMP-SEC-INTEGRITY-001
- SFT-COMP-SEC-ADVERSARIAL-001
- SFT-COMP-SEM-VERIFICATION-001

Generated grammar and closure boundary. Generate the complete product of package, parser, field, execution, authority, comparison, response and extension forms. The declared exact boundary is: All finite empirical prediction packages admitted through a strict data-only parser while census and model-admitted receipt authority remain protected. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: data-only-f

old-document__exact-schema-parser__reject-extra-field__interpreter-only-execution__unchanged-protected-tree__postseal-separate-comparison__halt-without-admission__no-extra-route. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
package	data-only-fold-document	mixed-code-and-data	Mixed packages can execute contributor-selected behavior.
parser	exact-schema-parser	permissive-parser	A permissive parser admits undeclared behavior.
field	reject-extra-field	ignore-extra-field	An ignored field creates divergent or hidden meaning.
execution	interpreter-only-execution	execute-contributor-source	Direct execution exposes every ambient host capability.
authority	unchanged-protected-tree	mutable-authority-tree	Changing census or accepted receipts can manufacture admission.
comparison	postseal-separate-comparison	embedded-comparison	Embedded comparison can expose or optimize against targets.
response	halt-without-admission	warn-and-continue	Continuation admits a known authority or capability breach.
extension	no-extra-route	extra-route	An extra route bypasses the protected admission path.

Base and successor. The closure proof is sealed by derivation hash

sha256:3b398ba6ad10467a48f4b6da1b13f21511b7decbb548c176ff7b069acf836de4. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt
sha256:155abd7902abed0cf2e85564b52bafb870f8f8565b09bc3b8c0847134a9399c5.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:9eadbe7d5481a418a910ff9ca6f16ef51c317220a3ac4b47900b0516ec7e419f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a319275abd552274cbf9c75121007066d2ae71142500f2fe884f53cd93276a61.
- **boundary:** passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt
sha256:119b65e69ed287f95df7763be66eb58b7997ef88302cae2a698c8651125a9034.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:b5a6ba9afcebd3f858403bb8a48a8d8623ac2fde50ecb36700c68264453e7103.

Independent certificate:

sha256:2a2bc6e6aadd3b7c2753423f4cd5a8690935d26e4d58a44e1b65a601fef6d8f9. Engine

external-validation hash:

sha256:8ab8a7d3c62932743b19764e0b71fd29d73f24f8b6556fa28d96d47eba360cef.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- contributor Python, native source, shell commands, modules or callables
- unknown package fields
- census or model-admitted receipt mutation
- target comparison within prediction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b6b6f54ba087d71f46fa57a106d83472d89b361c4ccc0bfa6ca6dd4ec914fc1d; engine receipt
sha256:20cd01fae8743456043006ac2f882cbe025b5c0bc5fd1062417ef44a5572b9b0 at
receipts/engine/model_admitted/SFT-PHYS-MEAS-HOSTILE-PACKAGE-001-20cd01fae8743456.json;
empirical-validation hash None; measurement receipt None.

4. Physical observation and retained distinction

Claim identity: SFT-PHYS-MEAS-OBSERVATION-CARRIER-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical observation is forced to retain a distinguishable phenomenon, a held reference condition and the exact comparison record; an answer without those three coordinates is not an admitted physical observation.

The unique observation carrier is a source-bound phenomenon/reference pairing plus its held comparison record, executed without target access and with every row retained.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__distinction-reference-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is

`depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	distinction-reference-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:daf450c3332bf1658e0d2f915c057a3832794c996ea290599a66571ec5979df1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a34141df720d459de4b146698583cf8f8dcde23b929ddcda6e2efd7685bf2a6cd`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99dffccfc8546e03a99d83a3b7a5a24e5a6e352e2a77380cd7904173382c2d60`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f62791c52be506a073e57cb83a2a099e6fa76f0f8e0ecb4df7950d4ceafa44dd`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:eedf3b06cb5ccdc9b7ffad27fa80d7f553e977f01d14994793b7e2530c3cb9a2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4a2fc44538091b1505d1da8b0424f73efbea60a315a3c253ca7307673dbabb08`.

Independent certificate:

`sha256:063d93d38c21349f348acef11572c524fffb601bf2e4b5eba7a05bba5b7e29b29`. Engine external-validation hash: `sha256:b5fdbc401d6760092c30caab2e52b9b20c44144bf03ada1a8fa3721e7ba45861`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- observation-carrier-1: predicted distinction-reference-comparison-record; observed distinction-reference-comparison-record; exact match True
- observation-carrier-2: predicted distinction-reference-comparison-record; observed distinction-reference-comparison-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256 : f9fda2226a2fad7562b3a0dda4144bd637241a274a23a852eee1c9ad9c82e058.

Isolation certificate: sha256 : 09f89a25ba5ccdc6ecee428dff4f3dae996c0d349a76a283f9c359a6141aafa5.

Custody certificate: sha256 : fc49213f497d434380833ef20ec477cd6a96efd7a2c99a5292ea53e2760ffa2b.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf> (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- observation-carrier-1 from BIPM-SI-BROCHURE-9-2026
- observation-carrier-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 99bdb19d7d84e7464f072fc44f536b7167af77971799ea02594035eb5b2632a0; engine receipt

sha256 : 4c4ab1e9623d357d1fcf16813b4c172b508ede8a5216c78c6ad786950b657009 at receipts/engine/model_admitted/SFT-PHYS-MEAS-OBSERVATION-CARRIER-001-4c4ab1e9623d357d.json;

empirical-validation hash

sha256 : 5fc32330a8a436e034c5fb9f31a4cdb46b708bf99cdd29c1edaf118f2979b052; measurement receipt

sha256 : f9fda2226a2fad7562b3a0dda4144bd637241a274a23a852eee1c9ad9c82e058.

5. Physical quantity as reference-equivalence carrier

Claim identity: SFT-PHYS-MEAS-QUANTITY-CARRIER-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical quantity is forced as the equivalence class of observations comparable through one generated kind-preserving reference relation; its measured value is a separate record, not the quantity itself.

The physical quantity carrier is the complete class of observations connected by a kind-preserving reference equivalence, with measured values retained separately.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__kind-preserving-reference-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	kind-preserving-reference-equivalence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f8b6e0a0cb283fec0c60c1d874db053051e2b09c5f425eccd52fee7b7c07ee9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9888ea14f67d6c08a04a1cfc8d37a8522c8222c7855cd64c4a0414328a8486c2.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1ad6611d696ab491614f73e53a56b3a3c1f91931a00215ebdb548ee37e121a47.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:9a060ec9f8710478bdbf4ae6ff1056ad01e38498daa3fa8e712b5b97ec54d29c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:61dc4ec52ba6fb38968ea6c69f29e3b3164d7a336203b9af708cece27e69569a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:8c3c165e5709b23eba3bd85ff8e52ef0582861f3da26491e87b9baf7932be60a. Independent certificate: sha256:e01dbd3531cd39593cd0477f80a7cd7dd4245304c6a3baec8a1a143c92de88c7. Engine external-validation hash: sha256:d46fd4e262c1f7e029cbdfef3f8334fd2495e66eb379acb8f63e8ae7775f6456.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- quantity-carrier-1: predicted property-kind-reference-equivalence; observed property-kind-reference-equivalence; exact match **True**
- quantity-carrier-2: predicted property-kind-reference-equivalence; observed property-kind-reference-equivalence; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256:bea784fbbfa39af43898c8730beb0d3255907a633c0731818d976945654310ae. Isolation certificate: sha256:651961f18efa758979d104f0ba373ce12383ee7e3f99c7e4e411fcb4dc1415cd. Custody certificate: sha256:d1d8736465a0e7dae1ba5cde42947671ee3fa25fdc5153388e07d3a426328e1b.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf> (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- quantity-carrier-1 from BIPM-SI-BROCHURE-9-2026
- quantity-carrier-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 48453cfd4c1b9afda889f62d48c83b3f01b885340ca28def33d9851ac067221d; engine receipt

sha256: 9966065b5a469cd330eca589f63e5d58cc86cb9815f6ff14dc26e620368efa6e at receipts/engine/model_admitted/SFT-PHYS-MEAS-QUANTITY-CARRIER-001-9966065b5a469cd3.json;

empirical-validation hash

sha256: 8ad865ff3aea5f0fec4331d26eff0d499e900f6531868df1a557eb6fc5d60047; measurement receipt

sha256: bea784fbbfa39af43898c8730beb0d3255907a633c0731818d976945654310ae.

6. Dimension composition and independence

Claim identity: SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical dimension is forced as the canonical generated composition path of independent quantity kinds; two descriptions share a dimension exactly when their paths reduce to the same held form.

Dimensions are canonical finite products and exact quotients of independent quantity-kind carriers, compared by form identity rather than numerical value.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__canonical-dimension-path__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	canonical-dimension-path	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:12a26830ac90cd2e04871a7470fd081ea4243cfa12a67c99df9db8dbdfa7c78d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9eed98166f8ac9a14cdf047c4790e53789d749b4d06966057f64d641844c681a`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8f143ffae762666331bd4572a420f8c08eb1eae0abc7edd002800541b3ce5abd`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8ad77de29f4c2d52c5ebe06bb3ac7e97bd096be04e2fd47168eb43346cd9e3cf`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:77c4aa61b2b5b44faae27694a0736b89001989fd4d05a4aa02711ed84ae9f875`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:31540d9c686489dbdc06cde3f9318e569bf13f73ff54208473367ca3ecca16b`. Independent certificate: `sha256:28ead99c9c104d5a6d65eb11f2e9c52ada94502a34c04b76ef4d6aea4a042043`. Engine external-validation hash: `sha256:536b553d70addec336fd9e506e1d4f304920a1b5218c02eb605993bec6c64a6f`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- **dimension-composition-1:** predicted canonical-generated-dimension-composition; observed canonical-generated-dimension-composition; exact match `True`

- dimension-composition-2: predicted canonical-generated-dimension-composition; observed canonical-generated-dimension-composition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256:db439e0a8276e051e917e86afc66345d767cc25a020d3402efc0097722c54b05.

Isolation certificate: sha256:1fa9e925ed9d29be25c4b44d658e81c9f8f3a67e0d2661ad53717869130fd728.

Custody certificate: sha256:920d6b69c7a10197b6f27dcb2b3b5c2aadba67d02327c3f8fbe08c631da762ff.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf> (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- dimension-composition-1 from BIPM-SI-BROCHURE-9-2026
- dimension-composition-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:bf601f63f8539b443eedf791852143e5a38ef3a2054f5c69a766c4a387a4ea7f; engine receipt

sha256:88074780f2b97130169a864d0fe1e45073f0a0bc5544a0f71548c31739c12068 at receipts/engine/model_admitted/SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001-88074780f2b97130.json;

empirical-validation hash

sha256:c5aff7dee978371151afc87e90611e3cbca553be2cf41f07afe32c8ec1fbc29; measurement receipt

sha256:db439e0a8276e051e917e86afc66345d767cc25a020d3402efc0097722c54b05.

7. Unit comparison and exact scale transport

Claim identity: SFT-PHYS-MEAS-UNIT-COMPARISON-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A unit is forced as one held reference quantity used to compare quantities of the same dimension; lawful scale transport is witnessed by an exact positive part ratio and cannot change dimension.

A unit comparison pairs a quantity with a held same-dimension reference and records an exact positive ratio; conversions compose those ratios and preserve dimension.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__same-dimension-reference-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	same-dimension-reference-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d8b385f06f47e3897a6326f3d2cc4bb8da82fed0b97984f823c25d6e8927e71d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fcf38f4b41423e84d4a14af2d289e780fadd2a08dab621a12d3796b71ff65a6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5a03d4dfa95258d4fbc0a242c8209d3aaeb08e107094e45cd864e66d751d880d`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : e96864f5f86da2dd02aab6cbdfb056815ea19b5311500f036264d916ab479c18.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : cf9239eeb6851b1aea6c2989ab384f21ebfebab2c7a62a20ace5d8e4d77f9c52.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 2da4fce7d904328003e628ffb9bc4da6ec55d6f5b30a191ffe73ec2efea7170d.
Independent certificate:
sha256 : cd16c654f4ee555225b946883febceb54e755414770d71342cefdc47d57160ce. Engine
external-validation hash:
sha256 : 35361952b6e72219657f35ade169137ca3ccb3fd3455013b6c9d84b5f62cea74.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- unit-comparison-1: predicted same-dimension-reference-comparison; observed same-dimension-reference-comparison; exact match **True**
- unit-comparison-2: predicted same-dimension-reference-comparison; observed same-dimension-reference-comparison; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256 : 5db572226cdd4c53f814abfd8cb508de3da6cd8c70ed5de26f9e09638d9660af.
Isolation certificate: sha256 : ddd0194e1db1099bfbfb26e5df55d51222dfee70cc7500cc4731b1e417020592.
Custody certificate: sha256 : b400ef792a9d9a4f6a7c1cab9226f8f2ae05e3ef6229cfb7eae37ec0241341f2.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf>
(sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- unit-comparison-1 from BIPM-SI-BROCHURE-9-2026
- unit-comparison-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:e6729d5a5819169aba066bb7477481b2b5949f656944ee10eeafaf344fd0a9ef; engine receipt
sha256:ddea20599918c381ea7bcf8172d46692ae36b76df2bbacbc168d54c1f3a2d5c0 at
receipts/engine/model_admitted/SFT-PHYS-MEAS-UNIT-COMPARISON-001-ddea20599918c381.json;
empirical-validation hash
sha256:36ce553823b5b77257c5425c89595e3353e5a4151d3f29c0019e36b569d595a2; measurement receipt
sha256:5db572226cdd4c53f814abfd8cb508de3da6cd8c70ed5de26f9e09638d9660af.
```

8. Reference realization and traceability

Claim identity: SFT-PHYS-MEAS-REFERENCE-REALIZATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical reference is forced to be operationally realized by a finite reproducible comparison process whose complete trace connects the definition to the observation; a name or assigned scalar alone is not a realization.

Reference realization is a complete finite operational trace from the held definition through each comparison to the observed record.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__definition-to-operation-trace__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	definition-to-operation-trace	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:fd441b4443a34e0d1efc0736e7f7f4981d651f31d27895887510fb09083ca339`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1f2dad9e88737b8f22f19d247b8bfb25c4447d0cac7c1e23c4c1407cf97387bb`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b233c37c78a4e4f34ac70474aceef70cc56604f1f671fc12d620f5ed00716faa`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:209f2b5c736cf23e4b94151005ca3062de1d396d0458739a460a514cd1d3d863`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:023389bb1e2e8f8d47b353faf6147bc91f36156ea0cc9c1f771b60c10e642473`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f78121815a382685c4dcacf3c638f980994cf29225e0e9471841814aad2fc4e12`. Independent certificate: `sha256:fc2c880293991e77dc198e7b11a4e81e14f5c0750c67b42ae86f0b95c6acb3a4`. Engine external-validation hash: `sha256:be2ab45528b95c573dda17a3f3b9eacad8d19a9e86dce508cf9bbef6c28c6b72`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- reference-realization-1: predicted definition-realized-by-reproducible-operation; observed definition-realized-by-reproducible-operation; exact match `True`
- reference-realization-2: predicted definition-realized-by-reproducible-operation; observed definition-realized-by-reproducible-operation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256:53cc527e34d7f176884fc21bd5ab8ae6b52b3d18900539a69c83452cf8af3fa1.

Isolation certificate: sha256:77acdb7059728686161907fa113088985fcd141162a873c11257d910db45534c.

Custody certificate: sha256:47e6159570febd9aa2cd8555a9109d7d654c764bc075a672e9fa151edf62ee9.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf> (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- reference-realization-1 from BIPM-SI-BROCHURE-9-2026
- reference-realization-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:47850de5d0f25c7228c1263b5d41235a42b70bdcea8cf7e57a6f4137fccdb248; engine receipt

sha256:7d7698fb49b9b740e1e29560195980c7dc9d7249a03580bace583c378198588f at receipts/engine/model_admitted/SFT-PHYS-MEAS-REFERENCE-REALIZATION-001-7d7698fb49b9b740.json;

empirical-validation hash

sha256:419dc28cdf09a88c37bc6d4795e65bcda17893ab9e102096558590adad730d5a; measurement receipt

sha256:53cc527e34d7f176884fc21bd5ab8ae6b52b3d18900539a69c83452cf8af3fa1.

9. Measured-value record separated from proof value

Claim identity: SFT-PHYS-MEAS-VALUE-RECORD-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A measured value is forced as a held external record containing the exact reference identity, reported numerical inscription and observation provenance; the inscription never enters the proof as an SFT scalar.

The measured-value record holds the external number inscription, unit/reference identity and observation provenance while the derivation remains exact and separate.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__held-reference-number-provenance-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	held-reference-number-provenance-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ad437e5c37862af878495de4de7ed6dac5c60212dd1485b839c14653282baac0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8cbb8dd6f3a25af27290eb17336e1bb52b737779cc1e8c9fe26d29a3e92bb585`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:51109d21ed7e0a72b1655f6317744d65af2ed906d176ea92cc990b14ff8cbd2b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8dc482be052ba144450746d84f9dfa508643eabd770d8e96fdf38640b7ef8fed`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : bd48eb008e2e79f555cea3790deddd31fd01286d296245b56cdd01ffa97fe127.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 2fd434613bc4467e473b86e1dd9b1720c04bfe2a0deaf8448a1aba855c765f5f.
Independent certificate:
sha256 : 8d58b9c85c1dcf9970b4a62cec01fa6ad66a07b2650d5aff08d6f66d6d7bc803. Engine
external-validation hash:
sha256 : a6eb14b424a10de722279d3a83f80c9841bd8632aeec17339790dc44dfa099a5.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- value-record-1: predicted number-unit-provenance-measurement-record; observed number-unit-provenance-measurement-record; exact match `True`
- value-record-2: predicted number-unit-provenance-measurement-record; observed number-unit-provenance-measurement-record; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256 : 99159d4689f4d920e4a5b194667322505fbd2a174d38608008c73f0e089e6145.
Isolation certificate: sha256 : eeefbb16a836ef1a49a47e5482ad16a3ebab0bc16b97b95da7492ec354520d7d.
Custody certificate: sha256 : ab58f3f8c62e9e5bf2bc4dad61a5c14b9871a832c39f42890eafb89070a0ed42.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf>
(sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- value-record-1 from BIPM-SI-BROCHURE-9-2026
- value-record-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 9fda7c62c521e34a351fb363987d7ee44117239e9ffc558d3b505bb3d4f64b02; engine receipt

sha256:ee71a58e3d4f2f8b3c646c64cb8d3f922a270cb45b751330a85d7b97e69b6d2c at
 receipts/engine/model_admitted/SFT-PHYS-MEAS-VALUE-RECORD-001-ee71a58e3d4f2f8b.json;
 empirical-validation hash
 sha256:c4f2b4cba8722a979be0e2ced471cafd9f1c6bd9fc95f1111c8785467a49f63a; measurement receipt
 sha256:99159d4689f4d920e4a5b194667322505fbd2a174d38608008c73f0e089e6145.

10. Measurement uncertainty as retained admissible alternatives

Claim identity: SFT-PHYS-MEAS-UNCERTAINTY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Measurement uncertainty is forced as the explicitly retained support of observation/reference distinctions not closed by the measurement process; it is not an irrational proof value or permission to choose a favorable result.

Uncertainty is the source-bound set of still-admissible observation/reference relations, reported with the measured record and retained in full.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__retained-unresolved-observation-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	retained-unresolved-observation-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:89948e98f461aa717d8ac035f3f9604dbb9196252996540718f45bf6e989cd36. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:7a0f4a18b8bb0dd172eab8a909284e061cbfbf6ef91fc8722df97f5662cec719.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0020b7b3b47807abfae0edd921d25970bd18f2ef64054c2ecd4e902f8076653.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4bfeb3041e24c6d31b4aac408bd4e3b568816786a88191f65030d9d7b73f0b05.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:46a9d02402b8165db8a231977dd86732974422aa8a0d8e28ba871a18e228aeaa.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:d194e478c4f9b3b77fffb8c7e737f86ddf25d01e64477403d65c8470c9a87413d. Independent certificate: sha256:3ed1ad98c0d3c0ec36d699270786a8ebf315001345b0cf2dd5cf9af2233d0b0c. Engine external-validation hash: sha256:1d61d49b663e4c34c4e77767f65c74ff94ce6a923d04ae6f36423a9158e2bf12.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- **uncertainty-1:** predicted retained-range-and-uncertainty-components; observed retained-range-and-uncertainty-components; exact match **True**
- **uncertainty-2:** predicted retained-range-and-uncertainty-components; observed retained-range-and-uncertainty-components; exact match **True**
- **deliberately tampered unfavorable control rejected**

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256:25fac07fa5ab7c4124704c16ae7cb7b48d99c426206392b5caf3b118629c0ab4. Isolation certificate: sha256:900522af12ba110a676b283546d80ff986104da68eca768cbbd0a29f6c335dfe.

Custody certificate: sha256:d53eb5b926907e899f77d2011528bbd53864fec059e5f5e860c8f767678b016e.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf> (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- uncertainty-1 from BIPM-SI-BROCHURE-9-2026
- uncertainty-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:9dd6c75ade4fb7cf74cc48c436826e9720dfbbe1fd572b4bd8e0bcc73404195c; engine receipt

sha256:b65a1d7557983a7dd30c30fdcde54c55370d7c7ea8c22848abd1ebbc7ced24c at

receipts/engine/model_admitted/SFT-PHYS-MEAS-UNCERTAINTY-001-b65a1d7557983a7d.json;

empirical-validation hash

sha256:fd74fdca51a498a4c37cad7dbd15d36e595ef6a6a9a0af99565dc836f51972ee; measurement receipt

sha256:25fac07fa5ab7c4124704c16ae7cb7b48d99c426206392b5caf3b118629c0ab4.

11. Calibration chain and comparison closure

Claim identity: SFT-PHYS-MEAS-CALIBRATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Calibration is forced as the finite compositional chain of exact reference comparisons connecting an instrument record to the realized unit, with every intermediate identity and uncertainty support retained.

A calibration is the complete ordered composition of source-bound reference comparisons, closing at the realized unit and retaining every intermediate record.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-composed-reference-chain__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	complete-composed-reference-chain	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:582d610ed05178d5192eed4b4fdec549803f63d5301aac65277383bf73bc07fe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:15f6f58bf9fe46ae235575ddc0319dcb4eb91b12dbbfc8866777e205cc5f93`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:382c52e07b6cac7a858f688563fd4f4b9731dc6cabffa6739dfc50f3ac8a60c6`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a1b9be492a77a444a1f989971a243fee651ffd731a8c7290a7dd879b05235af2`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a22c4401b242007fc6be19c41a51e85675303280663309b3fc00510adf2f0cfd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9ab06c6a96ac0cb18688ea26ebb49f23022312c0bf37679c013f208842103c4c`.

Independent certificate:

sha256: eebfa37af6a3e87822fd7a36b7e36c6e8945c251c837ed08a90f5b01e39cef39. Engine

external-validation hash:

sha256: 7cb7a69daec5075021039bbe43cb787566c22df79088c9465d3b6b0142fc6b6d.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- calibration-1: predicted traceable-comparison-chain; observed traceable-comparison-chain; exact match True
- calibration-2: predicted traceable-comparison-chain; observed traceable-comparison-chain; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256: cc8f919e13fbfd2e36a5fc37868c528d5ecbad8e7fc7d236291b7b0a9b0ff5cd.

Isolation certificate: sha256: d3fadf27ff637b82385e7ed2699ecf23603f71a2a739e8cd5ecd5c359ae0c2d0.

Custody certificate: sha256: a60ead04453dca41c3030848ac7bdc7d93e5ddae794edba0d60bbdab72cccaf.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf> (sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- calibration-1 from BIPM-SI-BROCHURE-9-2026
- calibration-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: f3389f0cb33655904ef8ab026b3be2ddb74241cb05f2a92a222c71954fad3f91; engine receipt

sha256: 63c0243e02bdf6d5786ee2ed2ba0fcabfd6ecae5aeeb7d299923a1d2c6728b67 at

receipts/engine/model_admitted/SFT-PHYS-MEAS-CALIBRATION-001-63c0243e02bdf6d5.json;

empirical-validation hash

sha256: 0722575d8b7614615674d68833b5ba5d0a8d957cd8ac71919fac7ff54a2239a0; measurement receipt

sha256: cc8f919e13fbfd2e36a5fc37868c528d5ecbad8e7fc7d236291b7b0a9b0ff5cd.

12. Dimensional consistency and lawful conversion

Claim identity: SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical relation is dimensionally lawful exactly when every compared or joined path has the same canonical dimension form and every conversion is a reversible exact same-dimension reference map.

Dimensional consistency is exact canonical path identity at every comparison, with conversions restricted to reversible same-dimension reference ratios.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-CALIBRATION-001

Generated grammar and closure boundary. Generate the complete product of carrier, relation, provenance, prediction access, measurement record, row retention, finite generality and extra-rule forms. The declared exact boundary is: All measurement relations assembled from admitted Fold distinctions, exact reference pairing, held observation records, target-inaccessible execution and finite generated support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__canonical-path-identity-and-reversible-conversion__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	canonical-path-identity-and-reversible-conversion	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

sha256:9476757ccf62d26eefff26d852ba1adf9048da37c8927ba566c2036c1869529b. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:205006977cc520a84734c06e70ac637ab6c19a87ed851309b146b576a2f9849b.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:fdd99f20271fea3b94c4d86056aa69692355c9e236302ae434dede45f04b99de.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5ebff673a39ef1ba8350015cf0f873f1ec3da528c4748d298b49eecb13fe0a81.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9baeb7fe6d1d1e66374d4e218194d3d58557c0e9dfa1859bfcafcdf895e73fc.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:1d4014ac9c8a6f913458caed124b7f4f6b16f8f2a0d478845ae3bc127e5c8f42.

Independent certificate:

sha256:72a1dbfa77d09de03ce057cf80b490ec72d08154ca662b2b452206c2ada8e128. Engine

external-validation hash:

sha256:3fab1249dba139a25bac4af63b21640b9e7990f01ab1d90ac6c75f2527bf5578.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- BIPM-SI-BROCHURE-9-2026

Observed comparison records:

- **dimensional-consistency-1:** predicted same-dimension-relation-and-conversion; observed same-dimension-relation-and-conversion; exact match `True`
- **dimensional-consistency-2:** predicted same-dimension-relation-and-conversion; observed same-dimension-relation-and-conversion; exact match `True`
- **deliberately tampered unfavorable control rejected**

Falsification condition: The claim is falsified within this registered standards boundary if any committed BIPM target row does not exhibit the predicted structural label, if any row is omitted, or if the tampered control is accepted.

Measurement receipt: sha256:376ba568e7bf5cf0374c42eab5a19efec99fe2a6fd075083d5c32e6d7dd5a770.

Isolation certificate: sha256:0ba1bded32220f294aa19a94d90a7d48af17620e2b6433b46ca39c3c09c1687a.

Custody certificate: sha256:98326d2f719846d08f7ed1689861ad62e03665eb01d595dea56039020dce685d.

Registered source descriptions:

- Bureau International des Poids et Mesures: <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf>
(sha256:5442eea2c680caf77a9d96879205a97f57c7c270b98a0bd0126c18fefe47e02c)

Registered target identities:

- dimensional-consistency-1 from BIPM-SI-BROCHURE-9-2026
- dimensional-consistency-2 from BIPM-SI-BROCHURE-9-2026

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- semantic numerical zero, negative proof magnitude, irrational, imaginary or floating proof value
- measurement target access before prediction seal
- a conventional unit, equation or measured constant selecting the Fold relation
- selected favorable rows, free scales and fitted parameters

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:bc8de8498c6323c70772f812cc5fd8e1398a15eeb11b16c68cb9352fdcf94acd; engine receipt
sha256:5310f3f1c40e05a2c14423b0d0d0d5b1775dc3ed45f585bd0971c05eab333c18 at receipts/engine/model_admitted/SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001-5310f3f1c40e05a2.json;

empirical-validation hash

sha256:cd6b07b2c543de233c54c9eafa40dadcd35a8c41c2797be0be4ae79e57f3e2e; measurement receipt
sha256:376ba568e7bf5cf0374c42eab5a19efec99fe2a6fd075083d5c32e6d7dd5a770.

13. Exact source-boundary growth measurement discriminator

Claim identity: SFT-PHYS-MEAS-BOUNDARY-GROWTH-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After a structural boundary-growth law is independently derived and sealed, a complete source-boundary experiment checks it without a fitted exponent or logarithm by comparing exact positive distance and response ratios, generating every positive exponent by repeated composition, retaining the complete compatible family and closing all later candidates only after a monotonic strict-excess certificate. The observed family cannot supply or alter the structural law.

One source-bound exact ratio experiment, complete response uncertainty, positive successor-power enumeration, all-compatible selection, monotonic closure, post-seal custody and complete trace.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001

Generated grammar and closure boundary. Generate the complete product of source carrier, distance comparison, response comparison, scale, candidate generation, selection, stopping, target access, evidence record and extra-rule forms. The declared

exact boundary is: All finite exact procedures for inferring a positive source-boundary growth exponent from two or more registered distance/response observations under admitted conserved-source dilution, including complete uncertainty support and a depth-independent stopping certificate. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `one-retained-source-boundary__exact-positive-distance-ratio__complete-positive-response-ratio-interval__same-source-ratio-cancellation__positive-successor-power-family__all-interval-compatible-exponents__monotonic-strict-excess-certificate__postseal-custodied-observations__complete-ratio-power-decision-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
source	one-retained-source-boundary	unbound-response-values	Unbound responses may come from different sources or preparations.
distance	exact-positive-distance-ratio	floating-or-signed-distance	Floating or signed distances are outside the exact positive proof domain.
response	complete-positive-response-ratio-interval	point-answer-without-uncertainty	A point answer discards unresolved measurement support.
scale	same-source-ratio-cancellation	fitted-source-amplitude	A fitted amplitude is a free target-derived parameter.
candidates	positive-successor-power-family	borrowed-or-fitted-exponent	A borrowed exponent selects the answer before measurement.
selection	all-interval-compatible-exponents	best-fit-or-first-match	Best fit or first match can erase tied compatible alternatives.
stopping	monotonic-strict-excess-certificate	fixed-search-depth	A fixed depth can hide a later compatible candidate.
target	postseal-custodied-observations	target-readable-method	Target access can choose the exponent rule.
record	complete-ratio-power-decision-trace	reported-exponent-only	An exponent alone cannot reproduce generation, comparison or closure.
extension	no-extra-rule	free-dimension-rule	An added spatial model can force a desired exponent.

Base and successor. The closure proof is sealed by derivation hash `sha256:95eec30df96776e0b54738d1eb8eedabb3ceeccac8b1c7d7c95aff463c93039e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form missing the first preservation; observed reject the generated form missing the first preservation; receipt `sha256:49c0e9ad603063b7a798ce56d84af7fc98dd24c46fca8128895246b531f8598d`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:adebcc98ce249122d2a47909ec040beba1db9e569667015df572c019d289d310`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cc21d02d3416b56c4207d23b16300b4c0e88bcda3089b0151c9f588a8ca1adb8`.
- **boundary:** passed; expected reject excluded capabilities, values or authority paths; observed reject excluded capabilities, values or authority paths; receipt `sha256:47bea616cdeb5da7ba6914b5f7274f3af8f89013addec4dab1c9d32208381f9f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:06b030de0896b998ea78ae969349c43e6a07eac99d60df3c5a93083eb0c83ade.
Independent certificate:
sha256:e6377e2f546a60825cb838c9a3b2ed8969bf99643ecb97fa0690dca5eddc70a3. Engine
external-validation hash:
sha256:4706a2ba009eee6ef2157f511d36fb5722ac65a2105912bb8d28382a03091be5.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Measurement is reconstructed before natural law: observation classes, quantities, dimensions, units, references, uncertainty and calibration remain exact carriers, while decimal inscriptions stay outside the proof domain. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- logarithms, irrational exponents, floating fits and regressions
- semantic numerical zero and negative proof magnitude
- a measured or fitted dimension used to select an inverse-power law
- target access before the complete method seal
- omitted measurements, uncertainties or compatible exponents

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:6b6069bb58cce5f30869ea2ed0cc6228b1809295b42c454f51af592277d61059; engine receipt
sha256:b2d8360be10ba4be5664b59ee4624d91d909962f83927b3bd98d4caf5b02e43b at
receipts/engine/model_admitted/SFT-PHYS-MEAS-BOUNDARY-GROWTH-001-b2d8360be10ba4be.json;
empirical-validation hash None; measurement receipt None.

13. Mechanics Dynamics

Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace.

14. Physical event and generated change

Claim identity: SFT-PHYS-MECH-EVENT-CHANGE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical event is a complete observed Fold state, and physical change is the source-bound transition between two such states; neither an unobserved background coordinate nor an answer-only scalar is required.

The minimal mechanical event/change law is a pair of complete observed states joined by one held, source-bound transition trace.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__state-bound-event-transition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	state-bound-event-transition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:34c6e0bc9be62ec68bc8493c18dcc5f25993cdabfef8ab370dfa43306f402cc3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e5732acdd20a744295dc44bc619a57918fbdc6192e1f90bc424ef81b4205556a`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7602df232de970b2c8237b10af653597ff2791df45f2ce143274cb8982ec8c9b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e6d4253c7ea6e792d78e6130fd2e77d803da3e2aa1130caf377680e0067bcde`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : d3bea2413cac687ffbde7295267a5c87d29d7607d2d9d6d3f0081d7ef424c1f8.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 6211d85755a0d738c265c26c686df476c122fb10f5751a8cfd2fab55c00e3458.
Independent certificate:
sha256 : 8362a480031dc0c7b73082a0266fca0f1d8ca29bf509f8f4c214b3b17c4964d1. Engine
external-validation hash:
sha256 : bd83b1755b683b8add8ed2d67965299e07ec5590dcadc3cebf2744ae07d236d3.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- event-change-1: predicted observed-state-and-transition; observed observed-state-and-transition; exact match `True`
- event-change-2: predicted observed-state-and-transition; observed observed-state-and-transition; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256 : aea2bb497ffa6a123b24025cdb4cc9e236077421681064a77ef96833985d0d38.
Isolation certificate: sha256 : ae46c4cd8c6c401ecb5c048f3de3b56837bea2f5ac84cb47496a5c25b7a0e490.
Custody certificate: sha256 : b077719713ea02b9030b4d92f0991c56dfc9e799128107fa8c6b366b96977893.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- event-change-1 from NIST-CODATA-2022-ALL-CONSTANTS
- event-change-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 8bf0896ea2c1944ef828c95af4fe562623ea5ce9b93e1eab92445e8627965a69; engine receipt

sha256:04874257af857f7573d356ff8ad75f8f85ee0c441752d7f941dbd279bd766613 at
 receipts/engine/model_admitted/SFT-PHYS-MECH-EVENT-CHANGE-001-04874257af857f75.json;
 empirical-validation hash
 sha256:4e8a5951b157520eab6931c979875d2face8326a3089484ecc67a1936225307e; measurement receipt
 sha256:aea2bb497ffa6a123b24025cdb4cc9e236077421681064a77ef96833985d0d38.

15. Duration from ordered recurrence

Claim identity: SFT-PHYS-MECH-DURATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Duration is forced as the exact positive ratio between generated transition count and a held reference recurrence; absence of a transition is structural Empty One, never numerical time zero.

Duration is the exact positive part ratio of ordered event transitions to a reproducible held recurrence.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-EVENT-CHANGE-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__transition-to-reference-recurrence-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	transition-to-reference-recurrence-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3f7e717f257f5dfa9844fd9a8c473b84f806abcd73fc517358ff23478fcac623`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b4f9404fd9b9ea4af729a789bef9b65234be6798a6b44e21e1a5243775bd15d2`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:27378c512ce057e7da835630bcb0de56a2ebde6396583c756fd90916074215bc`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ea92e1601fbcf473a19d93a380e15aa3d0c0a5380af04df94282e814cf88bfb5`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b6acf90f77e4d49ede0f0893e8ece79af4dc9c899fe02e00cff04557a9729da6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a6a15210ce7ca178cfc461603f430df76b1dac99bf6357c9ea0f581f4e1c2ea1`. Independent certificate: `sha256:ea03284dc046e848bd8ea95022c7babbbfefa01cc12ffdf2cdada7ae9e9d18b69`. Engine external-validation hash: `sha256:83cb0ea00cca669c4282da2a7bf29d1976f3c7fc0746218092545591b36f1889`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- duration-1: predicted ordered-recurrence-duration; observed ordered-recurrence-duration; exact match `True`
- duration-2: predicted ordered-recurrence-duration; observed ordered-recurrence-duration; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: `sha256:0ba3539ce8eb8aae69f0b09c3ed790cc19fd75035dcdb15f9d5e5d132b1f2325`. Isolation certificate: `sha256:8b58c16645dc26778a971744c705f888211a21d4e3bb340b4204ca0dde56595b`. Custody certificate: `sha256:b3cf89f143f347b693518369a7fde863595ad083fd6f2a6996c3a1609a7a3a72`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(`sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67`)

Registered target identities:

- duration-1 from NIST-CODATA-2022-ALL-CONSTANTS
- duration-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:6685c1113883053673d628b4aa64960e026993967ee5a5cdf6a15918c75bfc18; engine receipt
sha256:d7c78c834e95a2881236a24ff907323fd5974b1d19eef5b1f5233dcd1fe80bd2 at
receipts/engine/model_admitted/SFT-PHYS-MECH-DURATION-001-d7c78c834e95a288.json;
empirical-validation hash

sha256:eb96aaf54d6ff84876e888c812acde005eec2569729cde7a9ea80088f2007a01; measurement receipt
sha256:0ba3539ce8eb8aae69f0b09c3ed790cc19fd75035dcdb15f9d5e5d132b1f2325.

16. Location, separation and held displacement orientation

Claim identity: SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Mechanical location is a relation to generated reference support; displacement pairs two such relations and holds which endpoint retains the orientation rather than introducing negative length.

Location is reference-relative support identity; displacement is the exact paired separation plus its held endpoint orientation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-DURATION-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__reference-separation-with-held-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__comple

te-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	reference-separation-with-held-orientation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:7f320fe3bfc2aa449ccb0dadfc09b746d5bedb845a698cd2a6c5bb57540283d8. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:546b51f9b2e762bbd20eb2c8f6a9107e39d7be62c1269a1671ba0ada983d4e33.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:29579f288f2f858a4a2927b99128d64207896f55de5c193bc2c0b72278d69f26.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:00b0e58a2e79c26a3ead80fb8930d4fff51228d95a96ddd56183558069228f91.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:7ffe334e6b45268ce30a65f8a96caa4cb7394c356ece583f159260a398aa7448.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:ef7e7c152cab399048000402b884130d8b549982d2d4f4457b8f9d6079cc33b4. Independent certificate: sha256:74f4571e92dce781927dbbf696f483603323a18a90101594093db7831e6317ea. Engine external-validation hash: sha256:24411b1052f4246dd506912be909e6b09589365b094a5ebf0891fa99a31edaa8.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- location-displacement-1: predicted reference-relative-oriented-separation; observed reference-relative-oriented-separation; exact match True
- location-displacement-2: predicted reference-relative-oriented-separation; observed reference-relative-oriented-separation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:14243a9d2dbde5e2ebf8f210c8fd3888ac0e416b5697e32a164017a9846d02e9.

Isolation certificate: sha256:5f0c2dea32097d940570fefbe86f2a3f8182ce1e95a64d622cdc2c933e81d0b4.

Custody certificate: sha256:77790c1790f07e796bda0e5c81ed165ffda99c6166178b953e19716fecf62113.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- location-displacement-1 from NIST-CODATA-2022-ALL-CONSTANTS
- location-displacement-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:164cf71aa30f5139f0b486093b515375dcbc5b1f6354c6f72ab4fd8be2478a90; engine receipt

sha256:651765eae5dc825c8f634ce43690116dbe0a0296306ed19a35ae5a7ff7c4e067 at receipts/engine/model_admitted/SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001-651765eae5dc825c.json;

empirical-validation hash

sha256:896a6f188cb90aacd010827de8d8a0314217cd696bbe2d8610fef7455fe423af; measurement receipt

sha256:14243a9d2dbde5e2ebf8f210c8fd3888ac0e416b5697e32a164017a9846d02e9.

17. Speed and oriented velocity

Claim identity: SFT-PHYS-MECH-SPEED-VELOCITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Speed is forced as exact generated separation per reference recurrence, while velocity additionally retains displacement orientation; no signed magnitude enters the proof.

Speed is the exact positive separation/duration relation; velocity is that relation paired with held orientation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__separation-per-recurrence-with-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	separation-per-recurrence-with-orientation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:282820281c3cfa988386b97dc8e686b3d9265098cabdbb99e77bfe34739c7e42`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:857318859f939a1955cdc6e0ddb44e43ccb563b0898011f7b1fb900a4599267.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:df735e60db928f0a532c8668596c46cb7368b0c6acb8af1cd32c744bbdc6116c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:521586076d2e04c0f020770696dfb7737aba2718e4e5eb467d7d875639871c70.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8a2dcbcb1ee2621c044d84c1c0fd4a9d60bd6265d26eb03a28113d5b0bc332c6b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:5f8459771e9c081691badd81c203fd3d4e2cf19ed97bfffef70892af1881fd68d.

Independent certificate:

sha256:4c406b62a112bbf68a3d9bd28663090c1890c69792279762418ab647e300b45d. Engine

external-validation hash:

sha256:16fd657af776151ecc4cb3d6a5d646936378ac5d2ac23ccb9ad467c79950f3a2.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- speed-velocity-1: predicted separation-duration-oriented-rate; observed separation-duration-oriented-rate; exact match `True`
- speed-velocity-2: predicted separation-duration-oriented-rate; observed separation-duration-oriented-rate; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:110d1d1936053ba41d1250d05ff8169e4f7fac90be528adf73a5d915fc128398.

Isolation certificate: sha256:7514dd49531e595282745b3d7560a52e6fedb867cf332e7a35cbf08eac592066.

Custody certificate: sha256:46fe4826afaf6528b7746829285b44956b7db6d581711cb53812c7f2a3d21b49.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- speed-velocity-1 from NIST-CODATA-2022-ALL-CONSTANTS
- speed-velocity-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation

- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:3ff73715d0314ee18de3427a6de114fb93411e3fb973fdd7bde5239912cde59d; engine receipt

sha256:43e7e8b28583d51949d24a75fd8ca4024ba7c27e2dfa1d31fb39597b4fb2da80 at

receipts/engine/model_admitted/SFT-PHYS-MECH-SPEED-VELOCITY-001-43e7e8b28583d519.json;

empirical-validation hash

sha256:6e45b659df132991bda7ccca0d2cf1d009da3a80435d04306d0afae27e5f24c5; measurement receipt

sha256:110d1d1936053ba41d1250d05ff8169e4f7fac90be528adf73a5d915fc128398.

18. Change of velocity and acceleration

Claim identity: SFT-PHYS-MECH-ACCELERATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Acceleration is forced as the exact oriented change between two velocity carriers per generated recurrence interval.

Acceleration is the held-oriented exact velocity difference per exact positive duration.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-SPEED-VELOCITY-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__velocity-change-per-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	velocity-change-per-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b71ee033260a9717ca14a4d1decdd18a48f40545af022e516d42061323f59df7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:eaa0e0d413902b77efec640ce3898f5916bf70effd6ea8b3ec11d4c59b3afe01`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b6f3db19eb18dbd89342011c6163d5cf7964513f4ed912afd464453f0fa87171`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7ef1899a7ad943756898dbdd1b19194c78cd65367b50c3d006cc0d9073dd043d`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b3658d7236963e88665c6e90bc2f701eff616dd0df6cbc3f2de75eff96476e9b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a780de7d431097d31f0dfdd4e98b4f58f296084b8b5734e3fa3ff9ada31659ca`. Independent certificate: `sha256:6072fa5ba6bdf90653743a0fb99a02994860444b53ea4a2ab965068bdfed3ec6`. Engine external-validation hash: `sha256:aed7ceeb80d25df54062a697c7b5e71690380d422f3a3efb012fa6e75272ba00`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- acceleration-1: predicted oriented-velocity-change-rate; observed oriented-velocity-change-rate; exact match `True`
- acceleration-2: predicted oriented-velocity-change-rate; observed oriented-velocity-change-rate; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:687e98e78211b8fde686f05f2fd64922b33daeb45636e2beb6f3193779025249.
 Isolation certificate: sha256:8129185fb2fc469d365370a1ffb25169a2e57614094556417accbffe1cf37db9.
 Custody certificate: sha256:f9501e2b5c26e272e545fbe9aa34fa9a250d5f4a039dd6e7dcb3ffe03bdc3349.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- acceleration-1 from NIST-CODATA-2022-ALL-CONSTANTS
- acceleration-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:88ffe9c177e9655bc14605137d8db2884c7ffa8df0da7ba13014bf67de29b71f; engine receipt

sha256:cef0caaf43f08f111540c5e96d75a0eac5bbb8c0c0a28d962742677f263e1352 at

receipts/engine/model_admitted/SFT-PHYS-MECH-ACCELERATION-001-cef0caaf43f08f11.json;
 empirical-validation hash

sha256:d9db360cbdc2165cbd9fa54ff6a18a1ba5ce2d2a59ba927768ded5eb0988b730; measurement receipt

sha256:687e98e78211b8fde686f05f2fd64922b33daeb45636e2beb6f3193779025249.

19. Inertial persistence

Claim identity: SFT-PHYS-MECH-INERTIA-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. When no external transfer trace enters a closed state path, the complete motion carrier is forced to recur unchanged; change without a source would erase provenance.

A mechanically closed path preserves its complete velocity carrier from one generated event to its successor.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001

• SFT-PHYS-MECH-ACCELERATION-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-trace-motion-persistence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	closed-trace-motion-persistence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d3415a0f553f18c36c41819694dc800ad3e842d86c555ff5b0a32901e9edb746`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a9b00ca9f70199ff8a9224e7859716a9e3f719eb5a98bf3c81e3babbe0ea5ddb`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c304027d52c9d424f878fbcaed897116ce472297a60b57537bec1fdb21f2932`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:de0614da1cdb04608b176111e85876fe7c84ce21a8cda604ea6681bf5518271e`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2edbb86bf293bad3273f3cb81facff99276ac4a6ba047c3b67f6035a9ea89464`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c75cc345554614e524cddf65683ede97bc16584472df14533e1bd20d8d035aa4`. Independent certificate:

sha256:c5aa0cab1a8a9aee2e9064bcd05a566b64d977b6313cf2ae22f349ab41ad3aa. Engine
external-validation hash:
sha256:a31b2aab0c502670751555546b5e76be40368793ef13ce86a85cb49274f1f210.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- inertia-1: predicted closed-motion-carrier-persistence; observed closed-motion-carrier-persistence; exact match True
- inertia-2: predicted closed-motion-carrier-persistence; observed closed-motion-carrier-persistence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:3453efd2545620b9860a79bdf108ff6a2f03e5f6c30543759843aef72a4495b3.
Isolation certificate: sha256:8ffa8a8ce7156bdc01e028c232c35f485d7e98151c2b0d207cb49c0990e10c91.
Custody certificate: sha256:274df124f0e49177f62ca43481ecf00935c6a0d4090eb243946357823981846a.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- inertia-1 from NIST-CODATA-2022-ALL-CONSTANTS
- inertia-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d5b6704050d0ba9b99388045ce83e92338864f67bd1ab4dfd27ad33dec435cd1; engine receipt
sha256:c3e7e2bccfecec9a476fc32379bcd8b14aaef16ee29fa228ee0a3b4290e5f249 at
receipts/engine/model_admitted/SFT-PHYS-MECH-INERTIA-001-c3e7e2bccfecec9a.json;
empirical-validation hash

sha256:c8da1df56af1f3be2a30f9ecbceddfb70b2bd0de4d7c33bcd804e4157e61fcae; measurement receipt
sha256:3453efd2545620b9860a79bdf108ff6a2f03e5f6c30543759843aef72a4495b3.

20. Momentum carrier and transfer

Claim identity: SFT-PHYS-MECH-MOMENTUM-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Momentum is forced as the complete pair-cell composition of inertia content with oriented velocity, because each inertial part must retain its motion label.

Momentum is the complete exact product of inertial content and oriented velocity, retaining both source coordinates.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-INERTIA-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__inertia-velocity-pair-cell-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	inertia-velocity-pair-cell-composition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5443d18c952f458976a6b5f80fbd10efd11298525ea8cefe172722cd4a0203af`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every

generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e901ba60d3e54a104a96a0940b00151d67825f35111ab394275e3d7374237878.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ab52c54f3cc0dbede483e23d6b4e1eb6c27884892fc23836814945f13ae4daff.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f601ea2f1fc531758f7a20059bf9b4252cf59b4f3d03d16d5b122071afb39aaa.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:428e2d96ef2edbc2a6e12d1f215ef82164580a1670516a026448663b1b4e2cdf.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:d11aff6c886dca731eb6adf5548b7a409bbb7b62d19340b7b6e43f37176d643f.

Independent certificate:

sha256:002be5aa14d8977be1ca48fd14c295a6980caca97a50b3d57aefcac50c3018e9. Engine

external-validation hash:

sha256:7710d8e84a3bb0b4d3b244cce82c56403e3ddba79405da66ab7bbef5ebe3607a.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- momentum-1: predicted inertia-times-oriented-velocity; observed inertia-times-oriented-velocity; exact match **True**
- momentum-2: predicted inertia-times-oriented-velocity; observed inertia-times-oriented-velocity; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:9d61a19063020ef65884d5fdf598a8cbe4b6db21303343f5cb561bafb7d0e1e9.

Isolation certificate: sha256:544649d5b72f19f908dcacbb0aa1667b22998651a99cffa3a389d79fcc6fc83.

Custody certificate: sha256:9b9c407e841b3adfe6741743eeee35fe240b79b4d80e7333c498367826bda7ca.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- momentum-1 from NIST-CODATA-2022-ALL-CONSTANTS
- momentum-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:443c1ce5c004f7f2bbcf8498a3dad47ff763a2df362a1eb20c3448a0dfdbacd5; engine receipt

sha256:5b0049e80287dbca2cb5544338ad288ebc2727bbcb04473a90494926f8693615 at

receipts/engine/model_admitted/SFT-PHYS-MECH-MOMENTUM-001-5b0049e80287dbca.json;

empirical-validation hash

sha256:891b99aa46cbfdf4bdc5233e6f33a4a3434924b797ee894a37f250c8248b9315; measurement receipt

sha256:9d61a19063020ef65884d5fdf598a8cbe4b6db21303343f5cb561bafb7d0e1e9.

21. Force as constrained momentum transfer

Claim identity: SFT-PHYS-MECH-FORCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Force is forced as exact momentum transfer per generated duration; it is an interaction trace, not an independent primitive or fitted coefficient.

Force is the source-bound held momentum change divided by the exact positive duration of transfer.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-MOMENTUM-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__momentum-transfer-per-duration__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	momentum-transfer-per-duration	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:6c262ffceb08893750346dba3ee82ae915141f93ab55ad23bd6f5e557d76b7b2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ca5ca4b9adef57f96c96e099214bb5382deca3bce3fe84c29d618e412193ed69`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:395f9b84abd524ce73fb49e305d8586bce8c48374b6c6700c29f7f7fac00579f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fb2cbd727c7e8484eb6e8c7e29b362cf9fcf2a70db9720fdf12b53c24dcab409`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0f7f4dbc467ffa45c5df7ff588e3db7985e174a71dea3cf8ec581d8604f9bab8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:78246d7975345c0e149c90b0c9f340eab96f6257e25dca8be0149661fcfdb878`.

Independent certificate:

`sha256:a8a6c056543be591933aeea56583421668f6b22d065b6f658424427424e2a125`. Engine

external-validation hash:

`sha256:f2105279f6dfd08f0cda6ee18a127b2c57558c41f5b0bf81272c81bb44bedaf3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- force-1: predicted momentum-transfer-rate; observed momentum-transfer-rate; exact match `True`
- force-2: predicted momentum-transfer-rate; observed momentum-transfer-rate; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:6391b40a7fb5165066b750ff86ec7741197e6b8c32785a6692213beff372df15.

Isolation certificate: sha256:3441927a71b44a759595e95d841563ea460b2057e36216426b48177b2780cacf.

Custody certificate: sha256:e9a80a5fd55021be972f5df3389aba73f7ee7afa14d0e3568e18bfa704c0286a.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- force-1 from NIST-CODATA-2022-ALL-CONSTANTS
- force-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:cb39e08f950e946cd516455c2fc545cd9f22d3a8282e41fd8d8145488f5694db; engine receipt

sha256:bd0576b7dcee5c595698b22128853aeb01eabf681d8e44d8b743c63e8d1f7ae at

receipts/engine/model_admitted/SFT-PHYS-MECH-FORCE-001-bd0576b7dcee5c59.json;

empirical-validation hash

sha256:8f750398faddea1e4f53eddb60dde78b0d80dc69910521cb5a5545632dba0b1b; measurement receipt

sha256:6391b40a7fb5165066b750ff86ec7741197e6b8c32785a6692213beff372df15.

22. Work, kinetic change and energy carrier

Claim identity: SFT-PHYS-MECH-WORK-ENERGY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Mechanical work is forced as the complete composition of transferred force with oriented displacement, and energy is the retained capacity/change record transported through that interaction.

Mechanical work is exact force/displacement composition; the corresponding closed change is retained as mechanical energy transfer.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-FORCE-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__force-displacement-transfer-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	force-displacement-transfer-composition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b4f01c166a34b4becb9a3e43730b6294c1330be023d709f59669ee6a04202f0c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3794d4cdc98beb8e04daa8a8cfd87e57160ee075d437dff47ce14644d7742d68`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cf747a2b9ba20a84e0203edbf6ae22b06374bda74bfff411269e92f9e67d065d0`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e51e82a4d323c5ddbdfb852b651c858a7f99f04750c48968496df4131a56b5c`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:dc17d7767ecf8016267312e2843b86c503602c4f7e3bd803771af774bff6ac7c.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:a6e40d4f25794f17748b16e95cf9cb1d526c6edf9c54c6fb062d5c82ea7d28c9. Independent certificate: sha256:829fed84a305929b33019b69cd3a241bb984b134118246ea556eab8751ca507d. Engine external-validation hash: sha256:f97e670e49ef7954cc946fa4618376c595614659235e849dfee9a75ee842ee2b.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- work-energy-1: predicted force-displacement-energy-transfer; observed force-displacement-energy-transfer; exact match True
- work-energy-2: predicted force-displacement-energy-transfer; observed force-displacement-energy-transfer; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:7c4755e8f9a871aa70157b336b846dfb5f9bb76c858f7989b5a037e6e62953bf. Isolation certificate: sha256:6b1a2aa2aecfbccd197cdd7dff07ba485d23cdb850eda97234d8aaf9cd81dd38. Custody certificate: sha256:d881aeed51852847d890c718bc2dec735fc4a1a8c8a47843bf0ee1e80f85f24.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- work-energy-1 from NIST-CODATA-2022-ALL-CONSTANTS
- work-energy-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:f58fd1918c003907e5f787681c7d21ec3689f7784cafb6b5a53d9f93550f2345; engine receipt
sha256:117d45ba1a18c82dabfffd028c5740f7dfb29924cd0502278ee0b2ec003d681b1 at
receipts/engine/model_admitted/SFT-PHYS-MECH-WORK-ENERGY-001-117d45ba1a18c82d.json;

empirical-validation hash

sha256:905220827ef6a08da36806667e80e88dee60a63e44e4941e99718ccbe311b012; measurement receipt
sha256:7c4755e8f9a871aa70157b336b846dfb5f9bb76c858f7989b5a037e6e62953bf.

23. Power as ordered energy transfer

Claim identity: SFT-PHYS-MECH-POWER-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Power is forced as exact energy transfer per recurrence duration, retaining both the transferred energy record and its ordered interval.

Power is the exact positive energy-transfer/duration relation with held transfer orientation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-WORK-ENERGY-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__energy-transfer-per-duration__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	energy-transfer-per-duration	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256: 39655ad1210f70812df6c2be217edb9e30fed6bfa40e21df534cabd5f336c45a. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: c59e356804c621da0ed401ab1c32733cd8cd8cf1547d3f2ff3a849bc5b34648.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: d38e55fd04c92b8ecef7ee144cde3160c67c590cf0cff9e56407c8cf62dd1699.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 480e4a1e1dd88d9395d3b145f2054cc0d1ba29234dfc29b55d72618b6c99c803.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 8cec23188e4bebe26dbd9eb0d22639e7ceecd2a87295c0d68d526e78ff1ee195.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 0196c246d57db36313a70ba28369b6216fc75ea295113c42d450748df61a0a39. Independent certificate: sha256: 2aad9d67ad4a812c9bfb23eefae164480cc0385d5f466eb3618bbc5bbc161b7f. Engine external-validation hash: sha256: f97953a256ed88a165edcd14c96972d782abc37855c4850ae1d63526909ae096.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- power-1: predicted energy-duration-transfer-rate; observed energy-duration-transfer-rate; exact match `True`
- power-2: predicted energy-duration-transfer-rate; observed energy-duration-transfer-rate; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256: 99998bde324272f38856889bb481160188fcf3ce04c736bd7e4fe5c9b513621f. Isolation certificate: sha256: 3838d74d9ddf45f10bf58b4222bbcbcd2979a63ebc2e52163dddb34e45f7762e. Custody certificate: sha256: d84575e7628b1934b8ec6b93189e0c1cea0d01e4f9fe418ebfb1f29e9bf680e6.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt> (sha256: 77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- power-1 from NIST-CODATA-2022-ALL-CONSTANTS
- power-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:5f0a1752b0cd74fd8e92489b01155f19471eb892c5a17cacb6b3d2049d42f756; engine receipt

sha256:08e14aded1d213c827ff0a0c870fa1cf8d7a2c1058b12237cc90bf34907aa59e at

receipts/engine/model_admitted/SFT-PHYS-MECH-POWER-001-08e14aded1d213c8.json;

empirical-validation hash

sha256:30a30bdf860d1ee3b72357d8bfea69726dc79c96e845f26b83015ed4ebdd5154; measurement receipt

sha256:99998bde324272f38856889bb481160188fcf3ce04c736bd7e4fe5c9b513621f.

24. Angular motion and cyclic orientation

Claim identity: SFT-PHYS-MECH-ANGULAR-MOTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Angular motion is forced by recurrence on a closed ordered path; phase is the exact generated position within that recurrence and orientation is held by traversal order.

Angular position is an exact part of a closed recurrence; angular velocity is its held-oriented phase change per duration.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-POWER-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__closed-path-recurrence-phase__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	closed-path-recurrence-phase	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f9f2e561d109d6e9de81751e7c653921f101f5845c9a78519db2240d4b34ce88`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:91c99573f2c68caa9dab7a90084374ea7f5c76a7cca7604ad7a525068c2fdb20`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8dd93a8a2f3ff99e337aa19195da236de39d77d20d48b13c82e563779cb863ed`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f84f49d5b56f4473cb212dea76ecf6be895a890c59545ba4bb21c54789b4283a`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d8c5f976a54b6985e8ae93a096e8912a7d1284e0d83a52c8b90f3e143682ea3a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:debdc18905c1598597bd9c1c8fc989351f5eba652d479efdadb090aae9d66b0`. Independent certificate:

`sha256:6e1a782602dd60976895fa4f344d672e0edb2fd9ea6a87cdcf3ac298ce39a6bd`. Engine external-validation hash:

`sha256:55e4e9b1082ddb03a7645a8752eff3759c2cc8cc88924962b2f2f8bca26c8297`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- angular-motion-1: predicted cyclic-phase-and-oriented-rate; observed cyclic-phase-and-oriented-rate; exact match True
- angular-motion-2: predicted cyclic-phase-and-oriented-rate; observed cyclic-phase-and-oriented-rate; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:9d02a16ffde6975e36b764ef2f83642c8893a384b75c8058396cbd689f16ef12.

Isolation certificate: sha256:64359ff87112c705b5e648f81d2c9c0bcd9abfac10cc63d66d6823b2e2f4412.

Custody certificate: sha256:f83b8936b522177942eb7779fde1b820da9ca8ac80c5eae67967a88f4a86476.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- angular-motion-1 from NIST-CODATA-2022-ALL-CONSTANTS
- angular-motion-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d8888b5b9776b39c607788489bc8944c59e5d71c42e434e6a6bbdff890f43174; engine receipt

sha256:b07d16797fa48c5cfbc499291558e44e8e02edea85b3730669acb58ce6730323 at

receipts/engine/model_admitted/SFT-PHYS-MECH-ANGULAR-MOTION-001-b07d16797fa48c5c.json; empirical-validation hash

sha256:8438fd76e81ce7158897becd4f186c72d26cae296386f984f7dda9c87f0431; measurement receipt

sha256:9d02a16ffde6975e36b764ef2f83642c8893a384b75c8058396cbd689f16ef12.

25. Angular momentum and torque

Claim identity: SFT-PHYS-MECH-ANGULAR-MOMENTUM-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Angular momentum is forced as oriented separation composed with linear momentum on a closed path; torque is its source-bound change per duration.

Angular momentum is exact oriented separation/momentum composition, and torque is its exact held change per duration.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__separation-momentum-cyclic-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	separation-momentum-cyclic-composition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f8442286386cb2ef239ac8703741fc47e3ec72a40ad5e894a58341bf2fe6e7f5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:232302a821809c4550eb83205445d4fe0816d8317ffa55d47fc0bfe0b8e931a1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3c2cbebd1987ac9cd48919d2c1eaa2540b6431c32946931533f660f03db1551`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6a9e927be1334cbfa1a2a3b9cb661f58799a598a850dd10c16a49c2a643d256b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:897299ac6dc97d35ba690f074a5d8ec1aa9272658f0f6f30e02bad9ea3ce39e5.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:7d49cd2f4c425b1cd3ec28f07c2fb50a5bb3a919834be024fa9dd88a2345cac0.
Independent certificate:
sha256:b3f560d8405276173d31892428ef9dee06ff2c19dc0ee2b925d60ed592cc8913. Engine
external-validation hash:
sha256:24e660a5bbec4afff8b5bab36d2012a84e1910a66d0e0e885b7ed380d506152c.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- angular-momentum-1: predicted oriented-angular-momentum-and-torque; observed oriented-angular-momentum-and-torque; exact match **True**
- angular-momentum-2: predicted oriented-angular-momentum-and-torque; observed oriented-angular-momentum-and-torque; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:01cfac6a131a3f9ac5459dfbba35c567836f69be942c01209c7843d316a28d8b.
Isolation certificate: sha256:3f05090edb4b41ca9175d271216469ec0d7c7b490589b77da9ebe8a8e8520ccc.
Custody certificate: sha256:9c73efffc16a76cadee5e6089cca085746269013460e0796fc1d467e027adc36d.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddb5e71be406f2f26e3f67e67)

Registered target identities:

- angular-momentum-1 from NIST-CODATA-2022-ALL-CONSTANTS
- angular-momentum-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation

- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2d79b541a967090a256c94f8ed25e966d3c64e9ad764283c443b639997b5c91a; engine receipt
sha256:bdb07ed36a0c97019a98b71ae63029df5455948ff7f61ca5f302d7a29d07f5f2 at receipts/engine/model_admitted/SFT-PHYS-MECH-ANGULAR-MOMENTUM-001-bdb07ed36a0c9701.json;
empirical-validation hash
sha256:4c58f6964748badf6194d250a0260df9b904844b08f77eac8cac4394cca65467; measurement receipt
sha256:01cfac6a131a3f9ac5459dfbba35c567836f69be942c01209c7843d316a28d8b.

26. Composite motion and centre relation

Claim identity: SFT-PHYS-MECH-COMPOSITE-CENTRE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The centre relation of a composite is forced by complete inertial-content weighting of every part's separation, divided by complete inertial support; no constituent may be omitted.

Composite centre is the exact complete inertia-weighted location relation and composite momentum is the disjoint junction of all part momenta.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-ANGULAR-MOMENTUM-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-inertia-weighted-part-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	complete-inertia-weighted-part-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:797e257041d29d71bd8fb092cf3630f4b1ee06e17bcd1840555466e54a3a4981`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c511bfc1eab2cad1eea54aa1e729de50494692446e8d8ddc9a2b48351cbdf687`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8d19e87768f3fb35a39532c3a72e89d6cd2201521652f2d23d8f0e38a37f8cb0`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:787aeda12671300fc6430e467f1113186dce377b5dbb79024aed55a65dd28b4c`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1635fa44abf739cc12b3b0b18df0b1a5523b163829a538419f567f9f0264709a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:377e3ae1335d43420ae3d96342bc263578b30766dcf3fc926cb2dcd400536d22`. Independent certificate: `sha256:228be1a5037de6c20a8321e3a332e0ebe67c77d0e395bd9e6233c13878024dcd`. Engine external-validation hash: `sha256:531f607a293e50c7f12d25bade79499eab7c9afb2fc2d3d59b4d591332088fa3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- **composite-centre-1:** predicted complete-weighted-composite-centre; observed complete-weighted-composite-centre; exact match `True`
- **composite-centre-2:** predicted complete-weighted-composite-centre; observed complete-weighted-composite-centre; exact match `True`
- **deliberately tampered unfavorable control rejected**

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256:57931fe0cc00ecf60c476441524aa57d318625ed5a295878643e6dd53b2b4dc1.
 Isolation certificate: sha256:897ae9e5366d65a399997cf6db6287fc023cad0713687d9b5fffd44774d7a73e3.
 Custody certificate: sha256:88392ad05e775f3a15b5611c286af38c935d9ac8500d8cf05bc7ab89943f7526.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- composite-centre-1 from NIST-CODATA-2022-ALL-CONSTANTS
- composite-centre-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d89997a4c2ae4d17dc9639327e560bce38fd65602708dd485856b8f297f9fa61; engine receipt
 sha256:b565dc99daadf7225e7e9a817a731b8a9f7ab22359ea149bbbde3846dc5819c7 at receipts/engine/model_admitted/SFT-PHYS-MECH-COMPOSITE-CENTRE-001-b565dc99daadf722.json;
 empirical-validation hash
 sha256:ab5002712e1f825a0ca0bd12bbc74908248f3b01a15da25beef2157064d3190f; measurement receipt
 sha256:57931fe0cc00ecf60c476441524aa57d318625ed5a295878643e6dd53b2b4dc1.

27. Mechanical conservation under closed transfer

Claim identity: SFT-PHYS-MECH-CONSERVATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For a closed generated interaction network, every lost mechanical carrier at one node is forced to be held at another; complete internal transfer therefore preserves total momentum and energy support.

Closed mechanical transfer is a bijection on complete momentum and energy carriers, establishing exact conservation within the declared boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-COMPOSITE-CENTRE-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-network-transfer-bijection__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	closed-network-transfer-bijection	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7b2d7365612ec55f3bacb18617644577330a71ee92fe7c17c3eafb14d7db3772`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:39dce981a6c1b9efa89ad292f1c4e993835d0c8c80e0a6e962373813517f9756`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f7a65471ba766cb9eaf7aad30613aa0c30042309ce32542d267a35a979d98972`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a3e760a6792fa1b929d65b81fa7a0783ce3d1f4239ee253cb6beeb11a987c5e2`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:87bf710c03122257d4572610f5ad7a93a11d22329112dfd09faebadb29c07d10`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4adb9788db5bc48fa2d40265858e1ffc9ba1e2076bdab833af6e35a6303a0c40`.

Independent certificate:

sha256: b2b1a8e1de1c790784891d0fdf9fb10b142e7b8889317a4860d2d1013e0ea715. Engine

external-validation hash:

sha256: 5703c8d77b0288455bcaafc21fcb52cf0dbb70e98f8dca2497c5264b41d8dd8c.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- conservation-1: predicted closed-transfer-conservation; observed closed-transfer-conservation; exact match True
- conservation-2: predicted closed-transfer-conservation; observed closed-transfer-conservation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256: 815762e2322cb47c3cf1ba62ba6bd075f141894a653579e247717e88a0e0d506.

Isolation certificate: sha256: 79bad415ec7d447d52728a8a38e0a853f64d5c3f5aa149c1cb1b6f805caa7a67.

Custody certificate: sha256: a011ba04ebcca047f26d0eaab21eaa501601fccb6ff7efc0e870329b282ca966.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- conservation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- conservation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 27d63656f9e8b5b72f367ea2348f9d48c42ae7f133f78a3906c7c64fbbc06994; engine receipt

sha256: 01d57e618335bfc6fa3513336d0e1a30aa2fa8cf02bdd18420983b1a29a92093 at

receipts/engine/model_admitted/SFT-PHYS-MECH-CONSERVATION-001-01d57e618335bfc6.json;

empirical-validation hash

sha256: 33093892c053c89f41bc7d34b5957be21dc7d8287e71fcd0f9129c86d21cd20d; measurement receipt

sha256: 815762e2322cb47c3cf1ba62ba6bd075f141894a653579e247717e88a0e0d506.

28. Constrained motion, restoration and oscillation

Claim identity: SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A reversible constraint maps displacement orientation to a source-bound restoring transfer; finite repeated reversal forces a recurrence whose period is an exact transition count relative to the reference recurrence.

Constrained reversible restoration generates a closed oscillation with exact recurrence count, held phase and conserved carrier trace.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001

Generated grammar and closure boundary. Generate every carrier, mechanical relation, provenance, target-access, record, row-retention, generality and extra-rule combination. The declared exact boundary is: All finite relational-mechanics forms assembled from observed events, exact Fold ratios, held orientations, source-bound transfers and no imported equation or fitted constant. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reversible-restoring-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	reversible-restoring-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4dd5fa2a86bb517aed073b8e5bf93a40107b23b1a419c638d67d140154b2e049`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: a5d131fd9eb84ebdd960da36959498a70e1f2399d40f5c72f20e550924de89b3.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f3c229eb365b1fa9e7a68da36f338651cc373ef16a836b6eb2870c663af016f2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 69687c3dd3eb6008dd8e5e5029c70cf19fa4d6f867c34b19d96acb6fa4947ec4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 475a40edbb23e6c9929bd30fd6c8f763a41a1d41b622238548f14a9241f93bea.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: 716edae5dedaef06202415df671c379fdc51dff7d5e60afa73d9879daf953e06.

Independent certificate:

sha256: 733c4ba617c22f1f664f6efb16868a19824da109570788e2d2caf08eda91e44a. Engine

external-validation hash:

sha256: b3c458d7a58a1cc360391e1b2c725a507d31cd3b00c4900519b87411d89157ef.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- constraint-oscillation-1: predicted reversible-restoration-oscillation; observed reversible-restoration-oscillation; exact match `True`
- constraint-oscillation-2: predicted reversible-restoration-oscillation; observed reversible-restoration-oscillation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The registered mechanics correspondence is falsified if any committed CODATA quantity row lacks the predicted carrier/relation structure, any row is omitted, or the changed-row control is accepted.

Measurement receipt: sha256: 3bb6f0375bdece35468b92f96ca24d265ef1d765b0bcfa4d8be15776677dedf0.

Isolation certificate: sha256: 869c675b7fa2e7fdc0a8492e20a6632dec690a4671767db909c1a31b548a0fc.

Custody certificate: sha256: 826366cd43f786fcd9e95b22d77c25aac7e89915e3e9a52fee1ee2e873296b0e.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256: 77fb90e66c40db3e6eb16630bc9c88e4c7c8beddb5e71be406f2f26e3f67e67)

Registered target identities:

- constraint-oscillation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- constraint-oscillation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the

theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- absolute unobserved background space or time
- negative magnitude rather than a held orientation
- floating, irrational, imaginary, semantic-zero or completed-infinite proof values
- a conventional mechanics equation or CODATA value selecting the relation
- target access, fitted coefficients and omitted unfavorable rows

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:f4d4a396cfb3fc87f72258aa02ade12ef76331fd0ddd2e3924cdf0966031edfa; engine receipt
sha256:269bf50548ac623910ae13c882311660ecf1ff58a23a1af3340fc9c6e7c80dab at receipts/engine/model_admitted/SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001-269bf50548ac6239.json;
empirical-validation hash
sha256:766fbe6bfd29d3d36d43619a67424be1ae5a2a33ca9f50346f35fb81854fbb29; measurement receipt
sha256:3bb6f0375bdece35468b92f96ca24d265ef1d765b0bcfa4d8be15776677dedf0.

29. Free-particle Fold phase and exact dispersion

Claim identity: SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A free phase advances by the Fold of its exact momentum carrier, and that advance is identically the composition of two momentum advances on the One-cycle.

For $p=1/4$, Fold dispersion is $1/2$ and phase $1/3$ advances exactly to $5/6$ by either one Fold step or two p steps.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-MOMENTUM-001

Generated grammar and closure boundary. Generate the complete eight-axis product for free-particle fold phase and exact dispersion, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-chain__Fold-momentum-equals-two-cyclic-advances__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.

Axis	Forced form	Eliminated form	Elimination reason
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	Fold-momentum-equals-two-cyclic-advances	borrowed-dispersion-law	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:e218a34d4095d92e7e4bd3609e78cd3a4801b20aa23f06da280a56c29316eaaaf`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8c8abe048b2e0d3c2b40b72cdc0934f6e08fc5372dfb1ddcbd0541dc2a63c3a9`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:bf88bf71af8cd6e1866679adf017e7916401a9e88eb4637e05e30f7407a9597c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6bdefe43465be54912b1b709e50e7d7d8aebab95e2f94b1d23c7144448212750`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:59f5fc8c2264e7dfbc55bd587b4b388cd5eb669df4bdcf418b62391d62681335`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`.

Independent certificate:

`sha256:6cdae2c27ddcf3db943cfa6964247b923e516823c119ec5ff2cd45bfd3e1050f`. Engine

external-validation hash:

`sha256:61029641439e12c80fc08ae3e3e27b5fa2934c47fb3b0c7a850b54ded166caaf`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:6b7380b390388840f329ff9ded65beef257b4c4ecbfabeab412ee16a5915ec2e; engine receipt
sha256:d49c884c63339615c07f1b03712b44675c84df8cf0ebd84299aabc23645cba82 at receipts/engine/model_admitted/SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003-d49c884c63339615.json;
empirical-validation hash None; measurement receipt None.

30. Potential evolution and additive exact phase

Claim identity: SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Sequential kinetic and potential Fold rotations equal one rotation by their exact positive sum, so a path phase retains every energy contribution without an imported evolution equation.

At phase 1/3, kinetic 1/8 followed by potential 1/4 equals one 3/8 advance and ends at 17/24.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Generated grammar and closure boundary. Generate the complete eight-axis product for potential evolution and additive exact phase, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-chain__sequential-rotations-equal-summed-energy-rotation__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	sequential-rotations-equal-summed-energy-rotation	opaque-phase-generator	The alternative loses a required conservation or recurrence record.

Axis	Forced form	Eliminated form	Elimination reason
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:65f3699a00efb37993d3d44d5970ca0ff9daebcddf3b0f4c5f5269499cc6b686`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:b3d149ad76829ca3c63ca9bb36fbb2895c435c84f96096f023ca47d74a787205`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:65156571a85663c80655f18aafef7018aea3a858313aa78c3400061d68e6cd9db`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3d6e70f6e2dfc72a57de3c7bd5cbb299b9958ea4a6499e2183508c7507dd4dae`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:9b6d171f23ae15b03c1c225f0282d8cee0585c8f23a800409bb214abc18475c6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:af7c79d30602d55d60be3b076df0c639d8b4c57fbfee86a611a015a01ed18027`. Engine external-validation hash: `sha256:374155a3e447823649137b289d74d06b0e75fb09bda3d4ed9755bf39b18cd610`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value

- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8e8fed9795cb197cd9489b1041092ff2719682e2956aec072ee53eaa427d69b8; engine receipt sha256:2c0f29888cbfe276f5c1b03d27dc809046841d57b2857fe60a2f2f43c3a341f7 at receipts/engine/model_admitted/SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003-2c0f29888cbfe276.json; empirical-validation hash None; measurement receipt None.

31. Stationary Fold spectrum

Claim identity: SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. At every positive binary depth the unique half-spacing construction yields a finite exact spectrum with ground support half a grid step and every adjacent gap one whole grid step.

At depth k the levels are $(2j-1)/2^{k+1}$; depth two is $1/8, 3/8, 5/8, 7/8$ with gap $1/4$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001

Generated grammar and closure boundary. Generate the complete eight-axis product for stationary fold spectrum, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-v3-dependency-chain__half-step-ground-and-uniform-whole-step-gaps__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-v3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	half-step-ground-and-uniform-whole-step-gaps	selected-or-continuous-spectrum	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.

Axis	Forced form	Eliminated form	Elimination reason
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash
sha256:2fa52725256dfd8f695b8fc1515df473c6d18e1647fc2c9b6d094c0f5dc4bda2. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:2e43e78a4b885264b25e0875c6f9233f7800a358d7aa2942cf7553a0492cd847.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:45ec878430853f47134c7e072166bd943a0c29ba6cd6bb75a7b23142d13cd5a6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b9399927d3ba19ba0e13b5a8a72e6c74986a9bfde3722e6374591b82c4b0231d.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:f5a87e9e424a46e879175b4dca154d2c037f4d8158ddfb0269ef327e85a724c.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f. Independent certificate: sha256:5888a8892bcdfa8895d18ad3977772fe82c4ba3aef3084341d979048fd0b6e32. Engine external-validation hash: sha256:185a3f71d3be406f8edfde089410d1dee66304feb82945edd39b2d60faceb0e5.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:99a553afe97c3c1cd0e31440ea282f901e5e3b57e78eee6fbf3b573b83884dac; engine receipt sha256:b144cf1ec79dc2f2086a89f90d3781f4165344e1170161469f2e800823e584e2 at receipts/engine/model_admitted/SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003-b144cf1ec79dc2f2.json; empirical-validation hash None; measurement receipt None.

32. Odd-lattice all-region occupancy and exact recurrence vector

Claim identity: SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every positive finite odd complete lattice beyond the One, the binary Fold is a permutation of its complete positive residue support. Every positive Fold successor therefore preserves the entire exact occupancy vector under every complete positive region partition having no more regions than members. The vector totals the lattice count and every region count is positive. This closes the all-region law and its recurrence vector at arbitrary positive finite depth, without importing a historical lattice size, chosen Fold count, measured vector or correction.

Every positive finite odd complete lattice is permuted by the binary Fold; every complete positive-region occupancy vector is invariant at every positive Fold depth, totals complete membership and occupies every region when the region count does not exceed the member count. The frozen V1 E3 run independently reproduces (32,32,32,32,32,32,32,31) for 255 members, twelve steps and eight regions.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-ODD-RECURRENCE-003
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of every general, selected or target-assigned lattice domain; every exact, noninvertible or target-assigned Fold action; every complete, prefix or endpoint-omitting support; every complete, selected or target-assigned region partition; every exact-vector, vector-free or target-assigned occupancy law; every successor, historical-only or uncertified depth closure; both target custody states; and both extension states. The declared exact boundary is: Every positive finite odd lattice beyond the One, its complete positive rational support, the binary Fold successor, every positive finite Fold depth, and every complete positive region partition with region count not exceeding member count. Structural absence is an empty support, never a numerical state. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `every-positive-odd-complete-lattice__binary-Fold-permutation-on-positive-residues__complete-positive-lattice-support__complete-positive-region-partition-through-the-One__permutation-preserves-exact-vector-and-all-region-occupancy__every-positive-finite-Fold-depth-by-successor__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
lattice	every-positive-odd-complete-lattice	selected-historical-lattice-only	Rejected by computed Fold predicates: lattice-domain.
Fold	binary-Fold-permutation-on-positive-residues	noninvertible-selected-residue-map	Rejected by computed Fold predicates: Fold-action.
support	complete-positive-lattice-support	selected-orbit-prefix	Rejected by computed Fold predicates: complete-support.
regions	complete-positive-region-partition-through-the-One	selected-occupied-regions-only	Rejected by computed Fold predicates: region-partition.
occupancy	permutation-preserves-exact-vector-and-all-region-occupancy	all-region-claim-without-vector	Rejected by computed Fold predicates: occupancy-law.
depth	every-positive-finite-Fold-depth-by-successor	historical-depth-only	Rejected by computed Fold predicates: depth-closure.
target	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target-custody.
extension	empty-extension	free-occupancy-correction	Rejected by computed Fold predicates: no-free-extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:7428718697ab3ad78dcc7d9b4b4b1ac2e439f159d444ad22c956903f8deb3372`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject a nonpermuting residue rule.; observed The deliberate nonpermutation loses complete support.; receipt `sha256:047005037f2381c91895630630c1f6379614c36210bb6134ad9c99bb45e485f1`.
- **tampered_source:** passed; expected Reject a changed claimant source identity.; observed The changed identity differs from the registered manifest.; receipt `sha256:f9cc802fda9684abe9042aa128f74d5f9a2c098b1846a200fdb61bd404bf4775`.
- **tampered_artifact:** passed; expected Reject duplicate candidate identities.; observed The complete product has unique identities.; receipt `sha256:f8ab5596c700dead84f72de6044f4023b122da6a24c0a8e1bbb6833779ed2a8e`.
- **boundary:** passed; expected Reject pre-seal target access and a free correction.; observed Sealed custody and empty extension are forced.; receipt `sha256:90d112d709b62a1c043a10493f170533fec5e7b22a55c597854a55b009ec6d63`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:46c0f105c76584c7a483e0fb51e60842229e45a7edab165d4dd56d0956544621`. Independent certificate: `sha256:964a744645d5e30ed8fbc09732879d7b17a14e0ab40c432e80d31629cddfd3c`. Engine external-validation hash: `sha256:0dea9b70fb2495b3535ac3ed38aad3ffe5393ba88a942bf71c6158c70a33cb3f`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- SFTOM-V1-E3-EXACT-OBSERVATION

Observed comparison records:

- Both frozen V1 E3 source files are retained byte-for-byte and were opened only after the V3 prediction seal.
- Released historical inputs: complete 255-member lattice, twelve Fold steps and eight regions.
- Complete independently measured recurrence vector: (32, 32, 32, 32, 32, 32, 32, 31).
- Every one of the eight regions is occupied; the exact vector totals all 255 members.
- The same complete vector occurs after the first, eleventh and twelfth positive Fold steps, directly replicating and strengthening the original measured statement.
- The V3 law is depth-independent and applies to every positive finite odd complete lattice and every complete region partition within its member count; the historical run did not select that law.

Falsification condition: Reject if either frozen V1 source identity changes; if the original E3 inputs, status, observation or acceptance rule are omitted or altered; if the independently measured complete recurrence vector changes with positive Fold depth; if any released region is empty despite region count not exceeding the odd complete-lattice count; if total occupancy differs from complete lattice membership; if the general odd-lattice permutation or positive-depth successor certificate fails; or if the historical target or result enters the claimant before its prediction seal.

Measurement receipt: sha256:d594ef374830c0bbaaa7fa872e2bc62d0e69e1559739e76bc3c0618b335f6140.
Isolation certificate: sha256:345afc73832ddb0cc00bd60c847d470bf61a2a413f5b740590726f59cc560cac.
Custody certificate: sha256:2f4f005522a7f137483c4a7872b7754cdbf20f082a870aaa74a422403ec663e3.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, candidate table, answer vector or survivor identity in the formal derivation runtime
- no historical source locator, lattice size, Fold count, region count or measured vector in the claimant
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-null proof state, negative magnitude, irrational, imaginary, transcendental or floating proof magnitude
- no selected orbit prefix, omitted One endpoint, preselected occupied regions or target-assigned boundary
- no historical-only depth claim in place of the positive-successor certificate
- no target access before the prediction seal and no free occupancy correction after closure
- no execution or import of the superseded V1 source during V3 prediction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:bbd11bf580cae4c188a2924b2e35b24ae133985ae39d858462d1f54e9b8e6c8f; engine receipt
sha256:185a5dfb16ba14910625b0e2e94a894e7f422cf8dfe65f824ab4f9f5759b07fe at receipts/engine/model_admitted/SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007-185a5dfb16ba1491.json; empirical-validation hash
sha256:bfd7812d12c91cd25d2c88b7060362652e29a4fd85dfa1f6cfc7ec0a741a9f52; measurement receipt
sha256:d594ef374830c0bbaaa7fa872e2bc62d0e69e1559739e76bc3c0618b335f6140.

33. Half-One pair synchronization and coupled-ensemble terminal law

Claim identity: SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every ordered pair of exact positive Fold parts, moving both members toward one another by the same coupling share of their separation synchronizes them exactly if and only if the coupling is the forced half-One. The two equal moves then reassemble the complete separation and leave an empty structural residual record; every off-half generated coupling leaves a positive residual. An already synchronized finite ensemble remains synchronized under every common Fold successor. Arbitrary-ensemble convergence is not assumed: the complete superseded E4 recurrence is released and reconstructed only after the general boundary is sealed.

Equal paired movement synchronizes every exact positive pair if and only if the coupling is the forced half-One; off-half generated couplings retain positive residual separation, and every synchronized ensemble remains synchronized under common Fold successors. The frozen V1 E4 ensemble independently reproduces final coupling vector (7,7,1,5,5), half-One recurrence (12,10,7,5,4,3,2,1,1,1,1,1,1,1), and terminal common point four-sevenths.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-HALF-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of every complete, selected or target-assigned coupling domain; every paired, one-sided or target-assigned movement; every exact balance, unequal or measurement-selected relation; every empty-record, numerical-null or result-as-premise residual semantics; every conditional terminal, universal collapse or target-recurrence ensemble law; every general, historical-only or uncertified closure; both target custody states; and both extension states. The declared exact boundary is: Every ordered pair of exact positive Fold parts, the complete generated coupling alternatives around the forced half-One, equal coupled movement by a part of exact separation, empty structural equality records, every already synchronized finite positive ensemble and every positive successor depth. Arbitrary ensemble convergence is not assumed; it must be separately measured or proven for its registered recurrence. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `complete-generated-coupling-alternatives__both-members-move-equal-coupled-shares__two-equal-moves-reassemble-complete-separation__half-One-gives-empty-residual-off-half-stays-positive__synchronized-class-remains-synchronized-under-successor__arbitrary-positive-pair-and-successor-closure__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
coupling_domain	complete-generated-coupling-alternatives	selected-half-One-only	Rejected by computed Fold predicates: coupling-domain.
paired_action	both-members-move-equal-coupled-shares	one-member-moves	Rejected by computed Fold predicates: paired-action.

Axis	Forced form	Eliminated form	Elimination reason
balance_relation	two-equal-moves-re-assemble-complete-separation	unequal-free-moves	Rejected by computed Fold predicates: balance.
residual_semantics	half-One-gives-empty-residual-off-half-stays-positive	numerical-null-synchrony	Rejected by computed Fold predicates: residual.
ensemble_terminal	synchronized-class-remains-synchronized-under-successor	universal-collapse-without-condition	Rejected by computed Fold predicates: ensemble-terminal.
closure	arbitrary-positive-pair-and-successor-closure	historical-ensemble-only	Rejected by computed Fold predicates: closure.
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target-custody.
extension	empty-extension	free-coupling-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:b334a0cce278506495097aab5288a05ffd2eba2da739e13ed7dfdfdc2a22396a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject either off-half coupling as exact synchronization.; observed Both leave the same positive residual magnitude.; receipt `sha256:978495b19ea6a78a542087b299beca05d63505e8c7526fdd172d28f492034301`.
- **tampered_source:** passed; expected Reject a changed claimant source identity.; observed The changed identity differs from the registered manifest.; receipt `sha256:f9cc802fda9684abe9042aa128f74d5f9a2c098b1846a200fdb61bd404bf4775`.
- **tampered_artifact:** passed; expected Reject duplicated candidate identities.; observed The complete product has unique identities.; receipt `sha256:c74f2eab8555206eb5f3a49ee2b048122a0dc767178e467e00a155138bf2c2e1`.
- **boundary:** passed; expected Reject universal collapse, pre-seal observation and free correction.; observed Only the conditional synchronized terminal, sealed target and empty extension survive.; receipt `sha256:72ba6bce9aa44e789fee9d4adcd9e58976960c86fe569366903328839152515e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7eb1e3b89d31ce53ad4cf841df1ca561739e92dd7268fa100b3813e10ea17d0`. Independent certificate: `sha256:d5dd037ba8350aaa197d120cbefcb1cdf34c120fb6f2c84655111b2afae7c492`. Engine external-validation hash: `sha256:a818817af6843dd5ed1f9fbdb1c5c1b919968cd06876d35af4ee499b9eefc9e2`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- SFTOM-V1-E4-EXACT-OBSERVATION

Observed comparison records:

- Both frozen V1 E4 files are retained byte-for-byte and opened only after the V3 prediction seal.
- All five registered couplings and all fifteen region-count steps per coupling were independently recomputed exactly.
- Complete final coupling vector: (7, 7, 1, 5, 5).

- Half-One recurrence: (12, 10, 7, 5, 4, 3, 2, 1, 1, 1, 1, 1, 1, 1).
- Half-One uniquely reaches one occupied region, first at step eight, and all twenty members finish exactly at four-sevenths.
- Off-half controls remain visible and finish at seven, seven, five and five occupied regions; they cannot be discarded to manufacture uniqueness.

Falsification condition: Reject if either frozen V1 source identity changes; if any registered ensemble, depth, region, coupling or printed final coordinate is omitted or altered; if the independent exact recurrence fails to reproduce all five final region counts; if half-One does not uniquely reach one occupied region; if its complete stepwise recurrence or terminal common point changes; if any off-half comparison is deleted; if the general pair synchronization or synchronized-terminal proof fails; or if target content enters before prediction sealing.

Measurement receipt: sha256:26c258af8df7b47cb63990a322ad42259c7feae367af3364396472ff127ece79.
Isolation certificate: sha256:b3e64fac2ad2e9c1a88c28ea514125b09d572e5faf258cb8ff120a4de5cd2ac4.
Custody certificate: sha256:ed0a7d1410486928a7bc01d727fabdefc472a961848a815e8551df18a28728bc.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, recurrence, final count, common point or survivor identity in the formal claimant
- no historical ensemble denominator, member count, step count, region count or coupling vector in candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-null equality state, negative magnitude, irrational, imaginary or floating proof magnitude
- no assertion that every arbitrary ensemble converges under half-One coupling
- no deletion of unfavorable off-half coupling rows
- no target access before the prediction seal and no free coupling correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:5316e4ee59f1d5a1abdfb19d98346915438ebd1e23d0c6f72181eb94d3417a1; engine receipt
sha256:c761476abce4bdf4d357022245a0d069f2ca5f30cb34ee917e8a2f7578a5c2d0 at receipts/engine/model_admitted/SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007-c761476abce4bdf4.json; empirical-validation hash
sha256:8630bc2f8c43136cd9e26c429267b05260218a997dfc00c4c1877037ae9e42cf; measurement receipt
sha256:26c258af8df7b47cb63990a322ad42259c7feae367af3364396472ff127ece79.

34. Exact Fold expansion and information-rate carrier correspondence

Claim identity: SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A complete m-label Fold step forces the same exact positive-whole carrier m for local-separation expansion and complete information-support growth; conventional Lyapunov and entropy logarithms remain external correspondence labels under explicitly retained hypotheses.

The exact Fold expansion carrier and complete information-support carrier are the same generated positive whole m at every positive depth; the two-branch external anchor matches carrier two, while analytic logarithms remain external symbolic labels and the entropy-formula scope remains conditional.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-SELF-SIMILAR-CONVERGENCE-002
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete product of every complete, selected or target-assigned label domain; every exact, free or target-assigned separation law; every complete, fixed or selected support law; every common, independent or measurement-equated carrier relation; every exact, untyped or target-assigned information step; every exact-symbolic, imported-analytic or decimal proof boundary; both target-custody states; and both extension states. The declared exact boundary is: Every positive finite complete generated label count m of at least two, every positive Fold depth d , every positive local separation, complete m -label word support and its successor. The exact theorem returns the shared positive-whole carrier m ; analytic logarithmic notation remains solely an external comparison label. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `complete-generated-m-label-domain__uncast-local-separation-multiplied-by-m__complete-depth-support-m-to-d__same-m-carries-separation-and-support-growth__one-m-label-distinction-per-depth__exact-carrier-only-symbolic-external-translation__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
label_domain	complete-generated-m-label-domain	selected-two-label-domain	Rejected by computed Fold predicates: label-domain.
separation_law	uncast-local-separation-multiplied-by-m	free-separation-exponent	Rejected by computed Fold predicates: separation-law.
support_law	complete-depth-support-m-to-d	fixed-binary-support	Rejected by computed Fold predicates: support-law.
common_carrier	same-m-carries-separation-and-support-growth	independent-rate-parameters	Rejected by computed Fold predicates: common-carrier.
information_step	one-m-label-distinction-per-depth	untyped-information-increment	Rejected by computed Fold predicates: information-step.
analytic_boundary	exact-carrier-only-symbolic-external-translation	imported-analytic-proof-value	Rejected by computed Fold predicates: analytic-boundary.
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target-custody.
extension	empty-extension	free-rate-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash sha256: 7749cfe3453324ad469fe98967a5ab6c18491bdce3df2a0c6a3b6165b012fd28. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject a selected binary carrier for a generated three-label Fold.; observed Complete support and separation both return three, not two, as their multiplier.; receipt sha256: 37122691e3a624a1772bee38e44dfa1048c8ed6e88089da0aaa9da8e5173ce0d.
- **tampered_source:** passed; expected Reject a changed claimant source identity.; observed The changed identity differs from the registered manifest.; receipt sha256: f9cc802fda9684abe9042aa128f74d5f9a2c098b1846a200fdb61bd404bf4775.
- **tampered_artifact:** passed; expected Reject duplicated candidate identities.; observed The complete product has unique identities.; receipt sha256: c74f2eab8555206eb5f3a49ee2b048122a0dc767178e467e00a155138bf2c2e1.
- **boundary:** passed; expected Reject imported analytic proof values, pre-seal target access and a free rate correction.; observed Only the exact carrier, sealed target and empty extension survive.; receipt sha256: f84863dd7961828969186e8bb156becee345cb4d435c032810a5de3ac8c5b3df.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 86ac695164dfbe7c792ec43f22a6aa0d730e6234b3efeb74db1e986750e7d607. Independent certificate: sha256: e0b9b6a8b7567be6c502dabf0bc3b6ff21ca9fb455e58fbf92c1f6dd8a436a28. Engine external-validation hash: sha256: a03df497306791d51ed4f5ae3e756129bb5bce84a20220de3a3d6dedf067ee37.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PH2-EXTERNAL-LYAPUNOV-KS-CORRESPONDENCE

Observed comparison records:

- Both primary-source snapshots are retained byte-for-byte and opened only after the V3 prediction seal.
- The direct two-branch map records two pieces, absolute slope two, two preimages and two independent equiprobable output symbols.
- Independent Fold reconstruction gives complete supports two, four, eight, sixteen and thirty-two at depths one through five.
- Every support successor ratio and every tested local-separation multiplier is exactly the same positive-whole carrier two.
- The conventional Lyapunov and one-bit information labels are retained as external labels; no logarithm, irrational approximation or decimal enters the Fold proof value.
- The entropy-formula correspondence retains the source's hypotheses and is not promoted to an unconditional identity.

Falsification condition: Reject if either primary-source snapshot or the source record changes; if any registered map, preimage, distribution, output-symbol, rate-label or hypothesis-scope row is omitted; if the direct map's branch, slope and output counts do not equal the independently generated Fold carrier; if complete support does not grow as m to depth; if the expansion and information carriers differ; if the conditional scope of the entropy formula is universalized; if an analytic logarithm is imported as an SFT proof scalar; if a free correction is introduced; or if target content enters before prediction sealing.

Measurement receipt: sha256: 4acb4635259f330d2a4b434f5439ca3b27a0c4428d5864ce013e7468a4875ae0. Isolation certificate: sha256: f94d6de2c489b95bd239501cb442654b965e53b9a0f7e58cd0db12a85349ab33. Custody certificate: sha256: 598f97ba9038e30b6b6308cf0ae403463b01a10a3825d47b6e1b2e9002c68c99.

Registered source descriptions:

- ARXIV-1211.1234: <https://arxiv.org/abs/1211.1234>
(sha256:716c6b68b4755f47087747f67f4b2c4a7ed0ade30f77391a18129f2ad76de6c9)
- ARXIV-1004.3441: <https://arxiv.org/abs/1004.3441>
(sha256:ea560114c9cc196e18feb4536f63f28c292dadda465aafcb9f41ef12c20ac090)

Registered target identities:

- PH2-WITHHELD-MAP-AND-ENTROPY-CORRESPONDENCE from
PH2-EXTERNAL-LYAPUNOV-KS-CORRESPONDENCE

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- external analytic values as proof scalars
- target-selected carrier
- free rate correction
- unconditional Pesin identity

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d373bc7c4a05f5965eff26e64a699b1c1c82fe6d2a7fb82ab0d3d585ab765abd; engine receipt
sha256:f0b74a8639f9e01be7b9adc31f83b4b24ed82f085e72c17736c38f030b5cae40 at receipts/engine/model_admitted/SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008-f0b74a8639f9e01b.json
; empirical-validation hash
sha256:3c813f65cebb30e294350695b6163319a90d66523742ff9019a0c07abf1681d8; measurement receipt
sha256:4acb4635259f330d2a4b434f5439ca3b27a0c4428d5864ce013e7468a4875ae0.

35. Exact normalized coupled-map synchronization threshold

Claim identity: SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For a complete m-label Fold in the normalized drive-response transverse channel, the exact residual multiplier $m(\text{One}-g)$ uniquely forces $g=(m-1)/m$, expansion below, neutrality at the boundary and contraction above; half-One follows at $m=2$, without universalizing arbitrary topology.

For the normalized drive-response transverse Fold channel, the unique neutral coupling is $(m-1)/m$; smaller coupling expands and larger coupling contracts, with half-One forced at $m=2$ and arbitrary topology factors explicitly retained outside this scope.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008
- SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007
- SFT-COMP-DIST-SYNCHRONIZATION-001
- SFT-FOUNDATION-HALF-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete product of every complete, selected or target-assigned expansion domain; every exact, free or target-assigned residual action; every forced, fixed or measured threshold; every exact, mislabeled or absent regime partition; every normalized, universalized or finite-only scope; every exact-symbolic, imported-analytic or decimal proof boundary; both target custody states; and both extension states. The declared exact boundary is: Every complete generated positive label count m of at least two, every exact proper positive coupling g , the normalized drive-response transverse channel with retained share $One - \text{minus-}g$, and every positive successor depth. Arbitrary coupling topologies retain their separate transverse eigenvalue factors and are not assigned this threshold universally. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `complete-generated-m-expansion__retain-one-minus-coupling-share__unique-m-minus-one-over-m-boundary__below-expands-boundary-neutral-above-contracts__normalized-drive-response-transverse-channel__exact-rational-carrier-symbolic-external-translation__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
expansion_domain	complete-generated-m-expansion	selected-binary-expansion	Rejected by computed Fold predicates: expansion-domain.
coupling_action	retain-one-minus-coupling-share	free-residual-action	Rejected by computed Fold predicates: coupling-action.
threshold_relation	unique-m-minus-one-over-m-boundary	fixed-half-One-for-all-m	Rejected by computed Fold predicates: threshold.
regime_partition	below-expands-boundary-neutral-above-contracts	boundary-declared-contracting	Rejected by computed Fold predicates: regimes.
synchronization_scope	normalized-drive-response-transverse-channel	arbitrary-topology-universal-threshold	Rejected by computed Fold predicates: scope.
analytic_boundary	exact-rational-carrier-symbolic-external-translation	imported-exponential-proof-value	Rejected by computed Fold predicates: analytic-boundary.
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target-custody.
extension	empty-extension	free-threshold-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:d4d7c155c258c90c031af9f9f9233b501169c1d6d3071ca4823ea462f845c32c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject half-One as a universal threshold for every m .; observed The three-label threshold is two-thirds and half-One remains below it.; receipt `sha256:5940cc08cb032c81f0309da13073f53f87d409b435d741e62abf26be88767591`.
- `tampered_source`: passed; expected Reject a changed claimant source identity.; observed The changed identity differs from the registered manifest.; receipt `sha256:f9cc802fda9684abe9042aa128f74d5f9a2c098b1846a200fdb61bd404bf4775`.

- **tampered_artifact:** passed; expected Reject duplicated candidate identities.; observed The complete product has unique identities.; receipt sha256: c74f2eab8555206eb5f3a49ee2b048122a0dc767178e467e00a155138bf2c2e1.
- **boundary:** passed; expected Reject universal topology, pre-seal observation and free correction.; observed Only normalized scope, sealed target and empty extension survive.; receipt sha256: 79e19bb1b8b82b78e5b0a7c3c903c8d1b91154640ddf4c15693314e496b9792b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: e6757c5fd0f980763d0f09c3515997744542ec4afbc5b23dd80c11c27123527e. Independent certificate: sha256: b2d757623e5e3064d4c973f623b7a49707f1a16c41767e78e1f80c1dc5b6d66d. Engine external-validation hash: sha256: 949cc876e33ca0f6b920ec5bc116371cebef67bd9c8d9268c2d56140f4278911.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PH3-EXTERNAL-COUPLED-MAP-CRITICALITY

Observed comparison records:

- Both primary records are hash-locked and opened only after the V3 prediction seal.
- The normalized zero-eigenratio direct case records the half-One lower boundary and the exact Fold reconstruction independently forces half-One for $m=2$.
- Below, at and above controls yield exact transverse multipliers four-thirds, One and two-thirds: expansion, neutrality and contraction.
- The complete topology-dependent lower and upper bounds remain visible; the normalized threshold is not assigned to arbitrary lattices.
- The finite 21-map, coupling 0.8, approximately 0.867 critical-alpha row is retained as an external held record, never a proof scalar.
- Non-Gaussian finite-time fluctuations and intermittent bursting outside the stable window remain adverse critical evidence.
- The NLM record retains the two-quantity linear criterion and its stated high-precision numerical status without converting an approximation into exact proof.

Falsification condition: Reject if either source or source record changes; if any criterion, map, coupling, synchronization, transverse-rate, threshold, topology, finite-example, fluctuation or intermittency row is omitted; if $m(\text{One}-g)$ does not uniquely force $(m-1)/m$; if the half-One direct case fails; if arbitrary topology factors are erased; if an analytic or decimal target becomes an SFT proof value; if a free correction is introduced; or if targets enter before prediction sealing.

Measurement receipt: sha256: db0b7caa12d6b5d1d29c5bdf3c00c359e01c20fcb44993ddea059c76733866cf. Isolation certificate: sha256: 18c48a294f227845b481d1afc35bdf1e062ff767a6e6786ddad9369e7789b78. Custody certificate: sha256: 41b3550dd4fd8c1e438f7c8e72e5831a1320a29b94b5a92264b6a466e6f66604.

Registered source descriptions:

- ARXIV-NLIN-0504012: <https://arxiv.org/abs/nlin/0504012> (sha256: 6ec59cc2f85ed4d90ef3bd457b1f2877d92a4e10059548ab11297e363f588a4d)
- PUBMED-15447575: <https://pubmed.ncbi.nlm.nih.gov/15447575/> (sha256: 08ab034a50e7e6383669d3c6f607634c080cd741018628c4355eeeb8e142d8e1)

Registered target identities:

- PH3-WITHHELD-COUPLED-MAP-CRITICALITY-RECORD from PH3-EXTERNAL-COUPLED-MAP-CRITICALITY

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the

relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- arbitrary-topology universal threshold
- external analytic value as proof scalar
- measurement-selected survivor
- free threshold correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:24af36a052d5a6685bcc0857a378159d968a48280bec0cee65548d64eb99c7c; engine receipt
sha256:845ceed3f993a0523e1f01a5ce9375d469e426be12539941c5e5fd5130b9534f at receipts/engine/model_admitted/SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008-845ceed3f993a052.json;
empirical-validation hash
sha256:35c695aabf5966a39fe78dd6bf7ab75945549270786d974527a84c019ba4d2a4; measurement receipt
sha256:db0b7caa12d6b5d1d29c5bdf3c00c359e01c20fcb44993ddea059c76733866cf.

36. All-dimension orbital-restoration stability law

Claim identity: SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Positive denominator ordering between $q^{(d-1)}$ gravity and q^3 centrifugal response forces restoring circular orbits for $d=2,3$, exact marginality at $d=4$ and nonrestoring response for every d at least five.

Positive denominator ordering between dimension-forced inverse-(d-1) gravity and inverse-cube centrifugal response forces restoring circular orbits only for $d=2,3$, exact marginality at $d=4$, and nonrestoring response for every d at least five.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-VALIDATION-INVERSE-SQUARE-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of every complete, selected or target-assigned dimension domain; every dimension-derived, fixed or target-assigned gravity dilution; every inverse-cube, dimension-matched or target-assigned centrifugal dilution; every positive-magnitude, signed-import or target-assigned balance; every positive-denominator, negative-exponent or measured outward comparison; every complete, selected or universal stability partition; both target custody states; and both extension states. The declared exact boundary is: Every positive whole spatial dimension d of at least two, every exact outward radius ratio q greater than the One, dimension-forced inverse-(d-1) gravity, inverse-cube centrifugal magnitude, and every positive successor displacement. Dimension one is excluded because the registered

circular angular-momentum organization is absent there. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `complete-positive-whole-d-at-least-two__inverse-positive-power-d-minus-one__inverse-cube-positive-magnitude__equal-positive-magnitudes-at-reference-radius__compare-positive-power-denominators__stable-two-three-marginal-four-unstable-above__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
<code>dimension_domain</code>	<code>complete-positive-whole-d-at-least-two</code>	<code>selected-three-space-only</code>	Rejected by computed Fold predicates: dimension.
<code>gravity_dilution</code>	<code>inverse-positive-power-d-minus-one</code>	<code>fixed-inverse-square-all-d</code>	Rejected by computed Fold predicates: gravity.
<code>centrifugal_dilution</code>	<code>inverse-cube-positive-magnitude</code>	<code>dimension-matched-centrifugal-power</code>	Rejected by computed Fold predicates: centrifugal.
<code>circular_balance</code>	<code>equal-positive-magnitudes-at-reference-radius</code>	<code>signed-force-import</code>	Rejected by computed Fold predicates: balance.
<code>outward_comparison</code>	<code>compare-positive-power-denominators</code>	<code>import-negative-exponent-ratio</code>	Rejected by computed Fold predicates: comparison.
<code>stability_partition</code>	<code>stable-two-three-marginal-four-unstable-above</code>	<code>three-space-only-verdict</code>	Rejected by computed Fold predicates: partition.
<code>target_boundary</code>	<code>sealed-before-observation-release</code>	<code>observation-readable-before-seal</code>	Rejected by computed Fold predicates: target.
<code>extension</code>	<code>empty-extension</code>	<code>free-stability-correction</code>	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:167d4ec9cec1fbf7ca5a52db76919bb2d1c3bf11cec0074ae83c5c36f549c15b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject stable five-space orbit under inverse-fourth-power gravity.; observed Gravity falls below inverse-cube centrifugal response.; receipt `sha256:f687597b5fcd7de0c8c037b98df2b6946078182352a98c83f7646ec777826a8`.
- `tampered_source`: passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- `tampered_artifact`: passed; expected Reject duplicate candidates.; observed All identities unique.; receipt `sha256:5f9c0ad545be40ca825517099110bae199281727e06768826c0280cdfae9b73b`.
- `boundary`: passed; expected Reject negative-exponent import, pre-seal observation and correction.; observed Only positive comparison, sealed target and empty extension survive.; receipt `sha256:76d0758795c16363c1a6e8956a090b883ca59eb0503a7800d3814fc4a6728c2c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:94985b6cb2ba2d7ec1a4b3d44c8d8329c6a25dbc957a1237c1299f3f9b3c47ca`. Independent certificate: `sha256:b98bad07fe05e51599604075232fa496c9a7f1d575797d04e6f8e9712ee09ace`. Engine external-validation hash: `sha256:3470c2ebba7aa7c76fe89b5d351eb181d35029d0409f05501b1301bb3b44226`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- D9F-EXTERNAL-ORBITAL-DIMENSION-STABILITY

Observed comparison records:

- Both Cambridge derivations are byte-locked and opened only after the V3 prediction seal.
- The independent census retains dimensions two through twelve at outward ratios three-halves, two and five-halves.
- Dimensions two and three are restoring at every registered ratio; dimension four is exactly marginal; every tested dimension from five through twelve is nonrestoring.
- The successor proof extends the unstable result to every higher positive whole dimension because each dimension successor multiplies only the gravity denominator by q greater than One.
- The external inverse- $(N-1)$ force result, $N < 4$ boundary, potential-power boundary and strict-minimum criterion all match without fitting.
- No signed force, negative exponent, irrational, measured parameter or correction enters the Fold proof.

Falsification condition: Reject if either Cambridge source or the source record changes; if any force-power, balance, perturbation, stability, dimension, marginal, unstable-family, potential or effective-potential row is omitted; if the exact positive-magnitude census differs; if dimension one is silently admitted; if a signed force, negative exponent, measured fit or free correction enters proof; or if targets enter before sealing.

Measurement receipt: sha256:0b1572fc86c9e9d4501efee787ab5388dc4991261d757828f419e19c65ccef09.

Isolation certificate: sha256:cc3c2fd11a02259dbf08fb7cce0e023c9b2386e05bc2e71ff27254e41253c084.

Custody certificate: sha256:15d1d6b2dea21be60367f45f2ec9001b11598445a30182d8edf34f330dfc4fc6.

Registered source descriptions:

- CAMBRIDGE-CENTRAL-FORCES-CH6: <https://www.damtp.cam.ac.uk/user/reh10/lectures/ia-dyn-chapter6.pdf>
(sha256:59de9d143492b1fdf2f1f0b96c9653d861822081f6ee57c7689b943e31210024)
- TONG-CENTRAL-FORCES-FOUR: <https://www.damtp.cam.ac.uk/user/tong/relativity/four.pdf>
(sha256:2d999345dd33fce016498a1ed2dc5085f8e7513530f75dfef94dfbff05aad44e)

Registered target identities:

- D9F-WITHHELD-ORBITAL-STABILITY-RECORD from D9F-EXTERNAL-ORBITAL-DIMENSION-STABILITY

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- signed-force proof
- negative exponent
- dimension-one orbit
- measured fit
- free correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b5f097071692fa729280a05aa9d9c264532f3ff28fda7bb7b689104e130556f1; engine receipt

sha256:9a067d10ece75b8300bec3c77193649313dbc2fcc116f721bd0fef54544a433c at receipts/engine/model_admitted/SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009-9a067d10ece75b83.json; empirical-validation hash
sha256:b545a3186cb862437c92f3d65e46e5570bdf71cbf7b8c2a028280e916f37d03f; measurement receipt
sha256:0b1572fc86c9e9d4501efee787ab5388dc4991261d757828f419e19c65ccef09.

37. Finite Fold symmetry, conservation and least action

Claim identity: SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-TERMINAL-016

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Exact Fold preserves every reduced odd denominator core; complete finite Fold symmetries preserve held invariant fibres; every registered conserved partition generates its preserving symmetry class; the unique dyadic Fold trajectory descends to stationary One; monotone positive descent telescopes to endpoint separation and no finite detour has smaller action.

Fold preserves the reduced odd denominator core of every rational part; complete finite Fold symmetries preserve their held invariant fibres, every registered conserved partition generates its complete preserving symmetry class; every dyadic Fold part follows its unique binary-rank descent to stationary One; every monotone positive descent path telescopes to its endpoint carrier; and no finite detour has smaller action.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-FOUNDATION-PART-001

Generated grammar and closure boundary. Generate the complete product of support, symmetry, incidence, carrier, conserved-fibre, converse, action, path and extension forms. The declared exact boundary is: Every exact positive rational Fold part; the complete dyadic fixed-point basin; every finite generated state support; every bijection of that support; every registered exact transition/carrier relation and held fibre partition; every finite path of distinct adjacent states with exact positive ranked coordinates. The generator produced 512 named candidates and the decision support contains 512 one-for-one decisions. Exactly one candidate survived: complete-finite-generated-support__complete-bijection-enumeration__transition-incidence-preserved__exact-positive-carrier-preserved__odd-core-and-held-invariant-fibre__enumerate-all-fibre-preserving-bijections__sum-of-positive-oriented-step-magnitudes__dyadic-Fold-descent-and-positive-detour-bound__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
support	complete-finite-generated-support	selected-or-continuous-state-sample	A sample or continuum premise is not a complete generated domain.
symmetry	complete-bijection-enumeration	named-transformation-with-out-census	A named transformation does not prove bijection or completeness.

Axis	Forced form	Eliminated form	Elimination reason
incidence	transition-incidence-preserved	state-labels-without-transition-law	Ignoring incidence can turn a non-dynamics map into a symmetry.
carrier	exact-positive-carrier-preserved	free-rescaling-under-symmetry	A free rescaling changes the physical law.
conserved	odd-core-and-held-invariant-fibre	answer-only-scalar	A scalar detached from support has no conservation trace.
converse	enumerate-all-fibre-preserving-bijections	assert-every-conservation-has-unknown-symmetry	An ungenerated symmetry is not evidence.
action	sum-of-positive-oriented-step-magnitudes	signed-or-imported-action-integral	Signed continuum integration is outside the proof domain.
path	dyadic-Fold-descent-and-positive-detour-bound	postulated-stationary-physical-path	Postulating stationarity imports the desired principle.
extension	no-extra-rule	free-Lagrangian-or-Euler-equation	An imported functional or equation is an extra premise.

Base and successor. The closure proof is sealed by derivation hash `sha256:441f216f6c5c0b73c0db50445b2fea60e3796f38f54a4617f5471b65adaf08a5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c993d211ebd09080e4cc3077d8fffee231d05f9ba2a5c0aa80ae292d546f70f5`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:bafd7588c69ca117953fded10f3b6e4f1f381737b86e925f69c249698ec85444`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:250b52554881e0d14d83e9b306f3ce1421a73e9366da9663f93d8509f54115a0`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1839b0fae41ca54f4daa155e56b8d3a75ecc683bd62a3c856453df07a6719066`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4c5efd8efc82310605344c83bfaba79470f3632a90a8dda029a64fcac3cb6b37`. Independent certificate: `sha256:f22863557ec8d4f77008bfd44d42b3885d99c3a8eef7a89bba04d24d8cdae6be`. Engine external-validation hash: `sha256:b16add97905a64f9097085c2ca029a1cfc0c6806bf49c5a65dbee7fff2c4d49`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Mechanics and dynamics are reconstructed as exact state change, recurrence count, relational displacement, closed transfer and coupled evolution. Motion quantities are not imported coordinates; each is the minimal Fold carrier retaining the relevant event and resource trace. In this claim specifically, the forced result identifies the minimal exact relation named in the

theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported Noether theorem, Lagrangian, Euler-Lagrange equation or continuum variation
- no prior V1/V2 proof artifact, answer table or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no assertion that an arbitrary non-Fold physical system has the registered dyadic ranking
- no measured trajectory or target selecting the symmetry, charge or action form

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8e275a82b73779b75647ea049b663e7e733f9318c4d1f8f793f20e7ec758ec72; engine receipt
sha256:3c158a5e6f5ad82f89435d0b00c6542d1230e20706f5cdadad0db202864d4075 at receipts/engine/model_admitted/SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-TERMINAL-016-3c158a5e6f5ad82f.json;
empirical-validation hash None; measurement receipt None.

14. Fields Forces Waves Geometry

Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace.

38. Source, response and field carrier

Claim identity: SFT-PHYS-FIELD-SOURCE-RESPONSE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical field is forced as the complete source-bound map from a localized distinction to every generated accessible response cell; a value detached from source and support is not a field.

A field is the complete finite map of a held source distinction through generated accessible support to source-bound response records.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__source-bound-support-map__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	source-bound-support-map	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ab689d82a4bdef284f4320b91502eab091951f6214e0e2f6f14f46bf220aa998`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:041f78c0bd302fd7b80d81d704c363231a1d9f969d5afa0bb5616744aeeee376`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:744369e625fc3e621706b51f234a09906a2d26b6548cc4fa29f9d5610a8e1b41`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c6481452651e95b15b4bd1d40d946b6b1b6da41e0ce684ca8ccf6b65a4b31b95`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:33b67fad6c33cc5a25ce4f9fd87d6234a1f7c5112d3a4ecdd2f6ac1fb376d3bd`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1eb0754b3023b1f21e490d51ab8ccea125ba60c9247fc197cbd4fa99ede542df`. Independent certificate: `sha256:4557b6f886c13ea1c0718f82f936530fad652700a247c5c61936cdbcec75d8d2`. Engine external-validation hash: `sha256:829fa28c07fdea5f2d37d66e599608a93e54260f49c880d43cf64f2f1fd6003e`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- source-response-1: predicted source-support-response-field; observed source-support-response-field; exact match True
- source-response-2: predicted source-support-response-field; observed source-support-response-field; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:ebc58b140ebc9bb3b1a12b505dff0de9ef6fadb8e9adbdccfe53045a8203daaf.

Isolation certificate: sha256:b422bc0135ec06f820f1fa2b84bd367455372cc06b4ba71b194d8c1ead580528.

Custody certificate: sha256:770b5eec779fcecfc8674e1aef286d851d1821f49bafa4f6a7871be7af944849b.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- source-response-1 from NIST-CODATA-2022-ALL-CONSTANTS
- source-response-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Three-space, rank-two boundaries, inverse-square dilation, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:6a8409307ffe7115cc2e0eedc4f99557d8bb07df360cb88cceb8ec040bb390ff; engine receipt

sha256:203231945e022e79b7ffae5fc692aa84ec9e0cf0acea135fe22ac2bc6838ebc9 at receipts/engine/model_admitted/SFT-PHYS-FIELD-SOURCE-RESPONSE-001-203231945e022e79.json;

empirical-validation hash

sha256:69a79da801e537f1ac5d55c7c785a98140d4a4b89b6f2c30c47a31cd43822e96; measurement receipt

sha256:ebc58b140ebc9bb3b1a12b505dff0de9ef6fadb8e9adbdccfe53045a8203daaf.

39. Generated support and geometric dilation

Claim identity: SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. When one conserved source trace is distributed across a complete generated boundary, response per cell is forced as the exact source support divided by boundary-cell count.

Field dilation is the exact conserved source carrier divided across every generated equivalent boundary cell.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conserved-source-over-complete-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conserved-source-over-complete-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a5ad5c02f73a2f73028454f9df4ce617a86055220c5025e2945f6240c8da7bfb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8d76878d3e63df5947b4e677c368dcbc9f06995db3539663069867031b6026e7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:17870c6c9111f8027c5a04318a92cb02613787d79a5dd91ff6e809bfff8af9604`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:ea33d8af27399c9e8e03ca0d37ed92667744e367f6daf1870754adb4a7a6496e.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:52211b87aa710adc74dd296af6b6cb3f011a2ff8ee05bb75a61c1483607b29c4.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:ff7dc6ce86a1c911c501f6f7dd6fa3d786b3daf1fcd34650b2f0fe9e9b83de57.

Independent certificate:

sha256:31100409c4d7ddd477f1b87fd73fc90377f5755a34b9ecf3c00a30dffdfdf3fa0. Engine

external-validation hash:

sha256:43689921203884f3fa5d4e49909b5f1993a5044ae43aeb130a97af448f5cf1cf.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- geometric-dilution-1: predicted conserved-geometric-support-dilution; observed conserved-geometric-support-dilution; exact match **True**
- geometric-dilution-2: predicted conserved-geometric-support-dilution; observed conserved-geometric-support-dilution; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:7c949ef7a741f0d750e380e39695e428563e895f4056966995f73fa721d4bd3e.

Isolation certificate: sha256:d7b278138765a0a1629cf81e84303985a0b9c8c0d9cdff39b0e5b463f5cf2fb7.

Custody certificate: sha256:92d3436db300e05ebffad7816837749b98ad6b481cbf56d5c8a1cae914c84ef2.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- geometric-dilution-1 from NIST-CODATA-2022-ALL-CONSTANTS
- geometric-dilution-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:a1d9d6acc9d4794e6ca8f81d22c2caf35ff18a9d4876197736779c206b294cfc; engine receipt
sha256:afa50140059d7f2d81931e46b6761c0053f4e77eeee0538c1127843dfa690ceb at receipts/engine/model_admitted/SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001-afa50140059d7f2d.json;
empirical-validation hash
sha256:1984c29055fd7e0858da0bdad73676ee0b11df022da20fef9c51036c1f6cb3c9; measurement receipt
sha256:7c949ef7a741f0d750e380e39695e428563e895f4056966995f73fa721d4bd3e.

40. Gravitational interaction at the weak relational boundary

Claim identity: SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A gravitational response is forced as a universally source-linked relational change of free motion, with geometric support dilution and no held charge-family alternative.

Weak relational gravitation is universal inertial-source response transported over generated support and observed as a free-motion change.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__universal-inertial-source-response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	universal-inertial-source-response	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:98df42a85cb0c69e1cd0d12ceab9721d463fc60730527b17e5eb9bfc6dc677167`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c8f956d03349e201afeea9b8b1d15f3416a3d0728a92431d69d7af3c6cbf8c52`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4c07f5b01b4f9dd95aa95edadd518b8e04541fb9492c9dc9f6d8fb1e42a6a722`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6606681b51c0a9a8c9e7e3af1497013f2f0339e449f349a6ad24519836bebd28`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4bf70ab1a1ab6aa0165875951085c651fda9fdfe6628f26556865be009095ee`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:93e83ead15508eef1a445383097eb16e7061d3dbc4eb5cf3812f52e9070193fb`. Independent certificate: `sha256:6b77d9e624382a50e26aef406e60763d677fd957d4ed398ed5202f15308a76bd`. Engine external-validation hash: `sha256:442209281cc34af2cde089152fb07d5c4d80efcebdd7627edf5ff30777a1ebdf`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravitational-interaction-1: predicted universal-inertial-gravitational-response; observed universal-inertial-gravitational-response; exact match `True`
- gravitational-interaction-2: predicted universal-inertial-gravitational-response; observed universal-inertial-gravitational-response; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:927062d762861272cdd1cbceacf880e129da86595b727c481e7c1c23c1ea5924`. Isolation certificate: `sha256:a230a0bba51fb5940c2035dfd7628114f8649062910724506297f4627960e785`. Custody certificate: `sha256:a3ced0e0bd48469617d4df7b261050ba56cd256bbc3b93a020887649c25f8b20`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: <https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravitational-interaction-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravitational-interaction-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:959b4df999632bda004902de2ef28119033c172f7551c25e097d3e8b949fbd3b; engine receipt
sha256:82395b33bc2287af04a87f7dead671259e3f3cbcb24d0cf507b64379649d7f23 at receipts/engine/model_admitted/SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001-82395b33bc2287af.json;
empirical-validation hash
sha256:f45f6c9954f64dbbdfbf33e554d27c607df83a6c89480ca0ce6442292a4d804; measurement receipt
sha256:927062d762861272cdd1cbceacf880e129da86595b727c481e7c1c23c1ea5924.

41. Electric distinction and held charge orientation

Claim identity: SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Electric charge is forced as a conserved two-fibre source label whose opposite orientations are held distinctions rather than positive and negative proof magnitudes.

Electric distinction is a conserved two-fibre held source label; its orientation changes interaction class without introducing negative quantity.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions.

Exactly one candidate survived: `complete-fold-carrier__conserved-two-fibre-source-label__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conserved-two-fibre-source-label	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2ce0dc22064dab08ad3ffffba1c19bc0ee823b4f9b7eca2e4e6839561390c5233`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:662bd8009ab209cfa1bf46d5a554e8d4b79eb53d9f4f82c88719857fff9fe78b`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:be9c50d45008014f4fdd20aa8bfa18f5e91f9a120a755427781b5ccb608d5d7b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d467869edf735c08251cb166a4868024d2acb9cdeae472dfe01f267faff6511a`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a8d75ee0c59080e854470b31c6db28bb990b07783468a97e7c971f921ebac4b1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9a2814a2eb56fc4b08cd99d8fd678f66d1f3b4ad98b593730e5d5229030fd0da`. Independent certificate:

`sha256:1dc68a09b17229216ebec4249420c082e95bcfc78502b5c053714467bad10b29`. Engine external-validation hash:

`sha256:a7bc39123649080929c475be867911646fee104456cbbba883ae8fc16c46c640b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import

and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electric-distinction-1: predicted conserved-held-electric-orientation; observed conserved-held-electric-orientation; exact match True
- electric-distinction-2: predicted conserved-held-electric-orientation; observed conserved-held-electric-orientation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:22c918d208c1b4bc38aeb319dc9ba5aead21cac8d558392edeb6c341a85e2325.

Isolation certificate: sha256:6d626766c8a56217c2f794556dfd746c7ac879b9ef41e6a5ab2839758785b8d1.

Custody certificate: sha256:90da359acd9cee371bf96463f8116ad13eff990c664b0074301a772c797d1558.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electric-distinction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- electric-distinction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:3b19a83d707e022cee35bb81cd57554e60965da8cd660562699270b1aaa43212; engine receipt

sha256:479be6249f070c662d307aeec59a6ceeeceb4981de428f3200a78575f05e0cf3 at receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001-479be6249f070c66.json;
empirical-validation hash

sha256:fee8a5e70fbcfa86d433625149b2927d948bde0ae51c92ac12783236bdf75863; measurement receipt

sha256:22c918d208c1b4bc38aeb319dc9ba5aead21cac8d558392edeb6c341a85e2325.

42. Electric field, potential and work correspondence

Claim identity: SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Electric field response is force transfer per held charge carrier; potential difference is the exact work transfer per charge between two generated support relations.

Electric field is charge-normalized force response and electric potential difference is charge-normalized work transfer between held endpoints.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__charge-normalized-force-and-work__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	charge-normalized-force-and-work	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ececeb9261cf699b09d96214d6465e648b3ae8179afb60f5c4d5b78771f98817`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:822036fc67f0501f0487767675b89306c3fd296509762afc01915cdf4786acea.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4c3a8711ed4de88b4c8578e65a3f32681900e1c7d93b94a6a46f8b78ecfb38ca.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8c5e7fbf3054bddf4f5a969d20418adfbdd03f419c1f058e8d14e048244a9257.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4da112bfff0e21eb5e42d2bd033cf0da2ce75ca226c5d371ecae8c9e562e422fe.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:29a14819bf54c8423f2fbbdbd136d4462c6698875ac2ae78bb63b70a7aa605f8d.

Independent certificate:

sha256:ba419390f0d39e89fc7bbe511178cb83411759ebdc52d44d4ccd1cdd02c32bee. Engine

external-validation hash:

sha256:d4b8eb4595f2c2c31999a1746c93080d18bfbf03741304e20ef8ee9e12725f07.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electric-potential-1: predicted charge-normalized-field-and-potential; observed charge-normalized-field-and-potential; exact match **True**
- electric-potential-2: predicted charge-normalized-field-and-potential; observed charge-normalized-field-and-potential; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:07aaa4990023f31405ca92090ea2aa9e9ddad31f4cb58413884f1a42cdaead89.

Isolation certificate: sha256:9bc4092f84a6c2a1b24da8eb1ea5466f0c6b7b2f85320d14544ffc2e62351d49.

Custody certificate: sha256:54097d62ec821195f76cd07134ef73de0060c9ad252a1b3370bbb14cc4e699bd.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electric-potential-1 from NIST-CODATA-2022-ALL-CONSTANTS
- electric-potential-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:974168706fd3887f0fd2fb69728d2b38b064d2fa5fcb00130cf5be4220f942d0; engine receipt
sha256:404307768a30ea39fce5e7ab47589d6a5e103841f6c7c532a63328bed4a4feca at receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001-404307768a30ea39.json;
empirical-validation hash
sha256:6af026f60911b1f1a0642a8ac4949daea63e99b0859ba57eaa9876439a5046fe; measurement receipt
sha256:07aaa4990023f31405ca92090ea2aa9e9ddad31f4cb58413884f1a42cdaead89.
```

43. Magnetic response to oriented change

Claim identity: SFT-PHYS-FIELD-MAGNETIC-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Magnetic response is forced when an electric source label is transported with held orientation: the response is transverse to the source-motion path and reverses with either held orientation.

Magnetic field response is the transverse held relation generated by oriented electric-source transport.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__oriented-charge-motion-transverse-response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	oriented-charge-motion-transverse-response	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:46fb7deadb9ab1e5d70004af859171d67cae1ecb18b29c6a4fda07104eb42872`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:30c315164b77365902d50426dc07b08e9aa4c4641eed78a15394675cbcd09921`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:56f4d2c79bb6eeb8f1113d3eca4ef88cf61b28c469e91d55d85c2d6cc55ace57`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ebc25039336454dbcf6bca3ac5bd76621720cc1ad5b7dd049184db31ba6c6239`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e1c50337da9ca2040e691d3aa093e4ae1e50247e0c03fcb77f13512767ba34eb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:92c14267164b0df4326dc155fac09d5a7ebf9b6469774beeaac2999709d34fbc`. Independent certificate: `sha256:63f9580fa4c97bde378def33968315d4581aafcbcfce25c81a9bedc902106a2fa`. Engine external-validation hash: `sha256:c916d84bc22d7175aa68f7d0576f93db52786e5a556780546dbcdf054df39ee1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- magnetic-1: predicted oriented-electric-transport-magnetic-response; observed oriented-electric-transport-magnetic-response; exact match `True`
- magnetic-2: predicted oriented-electric-transport-magnetic-response; observed oriented-electric-transport-magnetic-response; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:7cc2da823d648c756e5c806a61c7dc0b4d5e00d24c847c4b9b6b742962595c52.

Isolation certificate: sha256:c0c7428cdbc6d2a40e2cd5bf2bf774b135424bc54e0951056b7264ba322031ad.

Custody certificate: sha256:abd25cb8c5572b0bf375789e2b66872da1c254ac08ae95f3d8e9209e544a7602.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- magnetic-1 from NIST-CODATA-2022-ALL-CONSTANTS
- magnetic-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:26b1b3c56afb0e115418eb8afb799524ff9b54ca8b2084f02dbedce5e165192e; engine receipt

sha256:4a212db2fc1113d2502f514d023caf210814886a20253226b2b4eeae0328f238 at

receipts/engine/model_admitted/SFT-PHYS-FIELD-MAGNETIC-001-4a212db2fc1113d2.json;

empirical-validation hash

sha256:c064915bfc94f33a0d756ee592e8b77849b75320ea296143a33cdc28380dfda6; measurement receipt

sha256:7cc2da823d648c756e5c806a61c7dc0b4d5e00d24c847c4b9b6b742962595c52.

44. Induction from changing linked support

Claim identity: SFT-PHYS-FIELD-INDUCTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A change in magnetic support linked through a closed generated path forces an oriented electric work-transfer trace around that path, preserving complete source-change provenance.

Induction maps exact change of linked magnetic support to held cyclic electric work transfer on the complete boundary path.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-MAGNETIC-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__linked-support-change-to-cyclic-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	linked-support-change-to-cyclic-transfer	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9a4b3fa00363266d9583a011a13e0da7c501edf7be373eec19dca2ab8cd1792ce`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:effda451435526b806ea9a3e3a39865f35fdd7448e7f0a4af4b255e828947cfe`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:aac1ccdceaf7af651fc29f7cad1a33665c2beabf5356f8529b6dcd4b284becb9`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e69bf6759c052cd10deef515b60131f7cb67a3860c4cb71f8fa7e9f5d045f5b5`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfbbaf6b8db5bfa6776ac91b2499e152d8979e5ee70eed6cc80cd3b22bdb345d1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:d967a5095815ae755cd2805bf7772dbee85d90a62d8f7da10cdd67a2eacc7056.
Independent certificate:
sha256:b16e812eb78b1e7d7dad39616b999b6e02ae2356070dea3fb3e9ce3ae490862b. Engine
external-validation hash:
sha256:007b6be4a433d5f4afd85c2540fbf8eeb3e90f27a4af627b30273915f4de9ee8.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- induction-1: predicted changing-linked-support-induction; observed changing-linked-support-induction; exact match True
- induction-2: predicted changing-linked-support-induction; observed changing-linked-support-induction; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:82ddf178dd9b2f895e85ccfeed9b438ba2c20abbab8d6872acdd92ade0f54328.
Isolation certificate: sha256:bc9122549dfdda1d4dd33778f963a8b6d1eefd1171390a00c9e1311e55eb3438.
Custody certificate: sha256:1cd9be2be5393c86c7b6a4e3f7b11212d3e1bc1971d91703ed7bd8ca5310c4ca.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- induction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- induction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:91048d8bdf43be16b8fe48671f5b7bd714b7ea78acf2903749347e3049beadfe; engine receipt
sha256:65da863a9a1b716d62b7541040863b59ee56c0055a972eaa5ac8c2b5e82959c8 at
receipts/engine/model_admitted/SFT-PHYS-FIELD-INDUCTION-001-65da863a9a1b716d.json;
empirical-validation hash
sha256:a2fc5738662e6375822daa99fca34c25b0efcc44a4cac79f29529db6dd10e3df; measurement receipt

sha256:82ddf178dd9b2f895e85ccfeed9b438ba2c20abbab8d6872acdd92ade0f54328.

45. Electric-magnetic compositional closure

Claim identity: SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Electric and magnetic responses are forced as two observation projections of one conserved oriented source/propagation structure, because each changing projection generates the other on a closed trace.

Electromagnetism is the closed joint carrier of electric and magnetic response projections with common source, orientation and propagation trace.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INDUCTION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__joint-electric-magnetic-propagation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	joint-electric-magnetic-propagation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:0103b751c089c7d16638e42187f95f5e6adff96bff0b0560e0a2996664df99b9. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:bba78f1d8cedfcd5821c132f2d797357cd64b2b029c3826f66f0c65fcf38617a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b0dcc38e2cd08fe6ce5242e906ae9baba21037229d30005d0b0edb1d2d6a282a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:559638b588ba3218eed5a3d470e7e6d6357b9e13f08f851b8ca2495d2385ac2d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:0fdb3b10e60f99983b701490c84f3a76a9ca115ad9ce5af6806024f3401f0fb7.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:82b79da2d7b5c6902bbda52aca1ba49b9602ce9bd1c667d9329cacbc77a7cb05. Independent certificate: sha256:0c9579c39d3c0045693c0c1ed912f75c31f422581621ea0dc0d9c7d25de531c9. Engine external-validation hash: sha256:bcfa813af18126aa970ec5ab55ff4a3272ff0a3ac60232fb485fdbfc0a5a9cf3.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- electromagnetic-composition-1: predicted joint-electromagnetic-field-composition; observed joint-electromagnetic-field-composition; exact match `True`
- electromagnetic-composition-2: predicted joint-electromagnetic-field-composition; observed joint-electromagnetic-field-composition; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:84f2644f6405541a10ec990c1840254adc51d1930d8adeee9ebbbe359f0d20b0. Isolation certificate: sha256:b31f387c1c96ed708d6d43743be58cdcec496efab02aaa77b5b688943cb3bf17. Custody certificate: sha256:bb9967ccb10f141c024eea8f8c0456c3ffaa193c0439001c47d97404de2f60b4.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt> (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- electromagnetic-composition-1 from NIST-CODATA-2022-ALL-CONSTANTS
- electromagnetic-composition-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: dba269304955eec2c12619e9043329c08e61e0ae65151f17c09951511441c2e4; engine receipt
 sha256: 470397ea06dec2c8b33de5b77e4f6d0ba6681999d445b82eec45b93ab9233080 at receipts/engine/model_admitted/SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001-470397ea06dec2c8.json;
 empirical-validation hash
 sha256: 315ce6e7344ac779f48d29f8127718531c10815e571ec6c46a7fa8e4e368742a; measurement receipt
 sha256: 84f2644f6405541a10ec990c1840254adc51d1930d8adeee9ebbbe359f0d20b0.

46. Gauge-equivalent descriptions and retained observables

Claim identity: SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Two field descriptions are physically equivalent when their complete observation and transfer traces are identical; unobserved internal label-path differences remain held descriptive redundancy.

Gauge equivalence is canonical equality of complete observable field traces under changes confined to held unobserved description labels.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__observable-trace-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore

a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	observable-trace-equivalence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:05abd37a18d2cf1e1193aaa56575c12b04dc0245e1d47c30e785a96dd5b95e2c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f145c14ad85295d09e564e20bdeeb87ee7d7536b03cc9e636ec2232c0cf1306e`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3baeec4b82b342a0307585a4f83a80fc186153a8092922a7879bf09be0d9b070`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2b4b3589ac32501ee81c6df1c75b4a840c876e0917f8198373476cef0d4b7d29`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:34fb113f6a06ec0fcbe160ede6077e08f7693735c946d1a0254368c79c94bd63`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:85c93bc7f68ae39b5bae7150435500555cd25db22bb3e4c327ef6a0c5f381644`. Independent certificate: `sha256:fa780de77bbc520df4440a87525dad6b4dee4903aac27d5548bc8087781609eb`. Engine external-validation hash: `sha256:b61301442986bf32d32ff580c85e81da5d8fbfa52bfab2397cab5c70b2e773a7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- gauge-equivalence-1: predicted observable-equivalence-with-held-gauge-redundancy; observed observable-equivalence-with-held-gauge-redundancy; exact match True
- gauge-equivalence-2: predicted observable-equivalence-with-held-gauge-redundancy; observed observable-equivalence-with-held-gauge-redundancy; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256: ee9425b090876521cf61a35a68cf4cf8f8c82707c88083cbd5f841525eb42c27.

Isolation certificate: sha256: cb240904ec157cde290f7f8bcf9ce498ef871efc85231c8487631844e824097a.

Custody certificate: sha256: 7f7d9343777dcde781f49a2337a3624f1929d1d33379f07498d1c8305434115b.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256: ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- gauge-equivalence-1 from PDG-2025-SUMMARY-TABLES
- gauge-equivalence-2 from PDG-2025-SUMMARY-TABLES

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 6ec0cd95c2db07b0f5aed6500a49bbaf23e243af48fa165a86885b944c29678f; engine receipt

sha256: 7c9c2d1366bf84479e96fb53603fa5a94312f5f7606d35b495f6ec03320b5d70 at receipts/engine/model_admitted/SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001-7c9c2d1366bf8447.json;

empirical-validation hash

sha256: 8a968022783129910d6e052c8a995e855f03394964a31d93f7fbd92ae1979651; measurement receipt

sha256: ee9425b090876521cf61a35a68cf4cf8f8c82707c88083cbd5f841525eb42c27.

47. Local propagation and causal accessibility

Claim identity: SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A field response can change only through a generated adjacent transition path from its source; responses outside the completed propagation trace are inaccessible at that event.

Physical locality is source-bound propagation through generated adjacent support; causal accessibility is exactly the completed reachable path set.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__adjacent-source-bound-propagation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	adjacent-source-bound-propagation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:dfa113a0f4fa67a4c725e960962eadb784caa5080a85eafe8123e8becd658a6a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f20e2e9440d80fd6bcae3687d0e494568f05d309366eb8696458443b6d246d9a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:20a8fa5d1d76dfc37bb8cde320c1a4c7d0bfbaed15178472d88f54a43d619f7e`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: faca424ddf42910e2626c8ce34cce75ccb38f8bc320ba7575f90e44bb9fec166.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: e78bbac06cdc159ee09ba8105fb36f3a4d70f60be94c674b07fc110c88834373.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256: e431b510e133b534898a11b27facedb9b7a4bd7f31a177c767a41caa1ad9d93c.
Independent certificate:
sha256: 7dc28b76c3b7fcb33d0f298b40abaa561d8afc3379211f37d9d45b7aad4ded11. Engine
external-validation hash:
sha256: 9bdc983558b526ca79d5ad83c58dab49c7f6d7ed2ab722b68ae214342a389206.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- locality-causality-1: predicted local-causal-propagation-support; observed local-causal-propagation-support; exact match **True**
- locality-causality-2: predicted local-causal-propagation-support; observed local-causal-propagation-support; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256: d7471789b05174ea0fb30ba3ab266534b351b2a406ab1a69b97a4cdc77d0adb6.
Isolation certificate: sha256: 3b05dd2f162d9bad7bce9f9842b6d098c80279fab0309e22eb73103638ce2dd.
Custody certificate: sha256: 6ca63d0bc4075baa504a47497241814a4ef2f89befb82405781a1802540908f1.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center:
<https://gwosc.org/api/v2/catalogs> (sha256: fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- locality-causality-1 from GWOSC-V2-CATALOGS-2026-07-23
- locality-causality-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:377aad9ec09b8b2fc819f404397cc879a95cb81abb380af3a4a88970a6c3c2c2; engine receipt
 sha256:53eddf9f05386d8a26a3de9279a68d1a36451f8dfb290681a1121bfbbd70a6ac at receipts/engine/model_admitted/SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001-53eddf9f05386d8a.json;
 empirical-validation hash
 sha256:4c0da852185c5b19f157ed8b3f7f3a2c2a7abb6a74efba679b195834fab5d18d; measurement receipt
 sha256:d7471789b05174ea0fb30ba3ab266534b351b2a406ab1a69b97a4cdc77d0adb6.

48. Radiative transport and source detachment

Claim identity: SFT-PHYS-FIELD-RADIATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Radiation is forced when a changing field carrier closes into a self-propagating joint electric-magnetic recurrence that transports energy after local source contact ends.

Radiation is detached self-propagating field recurrence carrying source-bound energy, momentum, frequency and polarization records.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__self-propagating-field-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	self-propagating-field-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1de63aec7335d5500c16812437defd664f48d36f81dad6c6f32fc2ab8980adef`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1f6b8207b6701110505f7659295bfe8a2ec2e235b8c808e563443cc5247128cd`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0386c56bd1470024b84aebef849f3212bc179c348936e9d956f0be64faf63d43`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c7bc8f0e89263a161dfc2cd37fadb82e444cbf923916f175b457a998652ed914`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:37e033b8d370e9a37b1fc41f42c70c50360fa36303ae51c724540ffb76e430d0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:653788cbfd634f0f02f407fe78b430ef7621899f2566f0f9f331d61de391f899`. Independent certificate: `sha256:37813a5b21b4330e7fbd979b8c0b4362c6bae623debd74a927f726652b8d0649`. Engine external-validation hash: `sha256:e4a17f72fdaf9e94fd7f9ead60850d3d06914a3ed9c8a0237de30c00c8cc2705`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- radiation-1: predicted self-propagating-radiative-field; observed self-propagating-radiative-field; exact match `True`
- radiation-2: predicted self-propagating-radiative-field; observed self-propagating-radiative-field; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: `sha256:f67a662a2e3f152a9d529c835ad00b1044fd652fcac0664c17aae5a7314e9866`. Isolation certificate: `sha256:b69678067f8b258f8647fe43418f389c40d3a1b601f7f3f0d733bdc072026011`. Custody certificate: `sha256:023f9d4a0ff2081af2fc363af4ceca078b7576fe52ee0ce526dc0212751287b9`.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml> (`sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454`)

Registered target identities:

- radiation-1 from NIST-ASD-5.12
- radiation-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:ab2660f76840eb85ad5503c84c0e2ed5c340caaac7da09d66532d4236964f5c3; engine receipt
sha256:0d61c283fb2b593db4705f03525522b29ed15179ed4b36519d281ca9b34e2fcd at
receipts/engine/model_admitted/SFT-PHYS-FIELD-RADIATION-001-0d61c283fb2b593d.json;
empirical-validation hash

sha256:87d41f9553bcb7449b6c0d58978b6d3c0677da708d44f703e56c56f337813742; measurement receipt
sha256:f67a662a2e3f152a9d529c835ad00b1044fd652fcac0664c17aae5a7314e9866.

49. Empirical closure of physical interaction classes

Claim identity: SFT-PHYS-FIELD-INTERACTION-CLASSES-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Physical interaction classes are distinguished only by non-equivalent conserved source labels, propagation carriers and response traces; classes with identical complete traces merge.

Interaction classes are the canonical equivalence classes of conserved source labels, propagation carriers and observable response traces at the registered PDG boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-RADIATION-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__source-carrier-response-equivalence-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__

`_complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	source-carrier-response-equivalence-classes	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:cea7252af8ef32564d8f0a514734455cdae5a4e6d929166fd65add448517c8a5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e8989ed9853bb54628f49d4906cddab80fbfe215b97e578c08fd1704fb36aebc`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:144c6ff57254d0b52293019574441dcaa66442fc0b6ad1bf6777c8762413ee23`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:78bcc6733218517beeaf151d8b9344b635050b59c70ec5e6198e1720225ce333`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1c60ec0ba523f3de5dc16d6243568dc364cdea5198008b018e0b968fc396a39a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ba2bc950bb6b33ec3c1c0977df1ade1852e9ddab8565b8bc35a812d32017dddc`. Independent certificate: `sha256:4f5d728bf1d7d1d3fb6652019cf0d89a5341c7670ad1b6089960d65e1ada77be`. Engine external-validation hash: `sha256:dc9c45a31eb720f3d0a10515cbfa0749d937fdc4410d223af2f62ca3b63b583d`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- interaction-classes-1: predicted four-observed-interaction-trace-classes; observed four-observed-interaction-trace-classes; exact match True
- interaction-classes-2: predicted four-observed-interaction-trace-classes; observed four-observed-interaction-trace-classes; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:dd3a368b3edf145c3b387532f7c14b2d2607dfd08a62ca953b3c4c338c13e624.

Isolation certificate: sha256:75bc8bc0f83cc04e94ffb22c49e21a3f1e9cb68432fb86fd38e73875926fdae0.

Custody certificate: sha256:c9e3741bcb4889ce945c4524238bbad97b0253a63d1975e91743f2071523a8f6.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- interaction-classes-1 from PDG-2025-SUMMARY-TABLES
- interaction-classes-2 from PDG-2025-SUMMARY-TABLES

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:01c6393d317faab6655b107e5dc2bae30615a8d7c2c0a5046b257e1b14cf5a7d; engine receipt

sha256:ac4ce2d6711e53b55ee78939d1ab2717b841dc92a84a5e0c2af31db12218d322 at receipts/engine/model_admitted/SFT-PHYS-FIELD-INTERACTION-CLASSES-001-ac4ce2d6711e53b5.json;

empirical-validation hash

sha256:8800c5a991882decab96e3717aa028b3a2a12c13bf52b9c3bc88882655e77f1a; measurement receipt

sha256:dd3a368b3edf145c3b387532f7c14b2d2607dfd08a62ca953b3c4c338c13e624.

50. Source continuity and field conservation

Claim identity: SFT-PHYS-FIELD-CONSERVED-SOURCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every closed spacetime support region, a conserved source label can leave only through a named boundary transfer; the interior change and boundary flow are exact paired records.

Source continuity is exact pairing of interior carrier change with oriented boundary flow over complete generated support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__interior-change-boundary-flow-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	interior-change-boundary-flow-pairing	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4bf4acd80b7c78f40c6da6dd796c6ad616bd5587156d37a22c9ef6453015c454`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6879367734b5c7f2286583b5c8739c36b436ccd219dd57c532e16c9774c34680`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e8f40f2e01ff4eb0c12eb93c01a8242f37d8c629af632fb58e5338bf93c54a9b.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:01913905e0c22c6005ce422f4d205530bf82a361f6f38e9892691c85f9b9ae2a.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:0bd5fd4d412afec3911663c85265a74a95472f66fe019bd969830de967cecb13.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:8dc81feb7603dc204bf2e45dbf55e742474325f372bf5cfb173a989d8023133f.

Independent certificate:

sha256:ba915b374fda4dfcddf48182128a3f396ec2f0009d99257fc36f6255c7d37d49. Engine

external-validation hash:

sha256:191b254d9733f0440bf8be1f35fbe5010107246c5155300490675d3e34f834e6.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- conserved-source-1: predicted source-continuity-boundary-flow; observed source-continuity-boundary-flow; exact match **True**
- conserved-source-2: predicted source-continuity-boundary-flow; observed source-continuity-boundary-flow; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:07029ac4d3d3fd288e1146ece1493bbabcf5bb46dd0b829d26dfd6a208db6e75.

Isolation certificate: sha256:1ea5949027540ca2c7b70961e88852bbc50e58e20aa15c8c45e79abc24882caa.

Custody certificate: sha256:9a1dc3a31fa8085db6a80b142d713494366aaf55391203747b7dc22832501c36.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- conserved-source-1 from PDG-2025-SUMMARY-TABLES
- conserved-source-2 from PDG-2025-SUMMARY-TABLES

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:e9267d967e4bbffebe4ad0239802582a994936b5e949d563c55533c0839d54ac; engine receipt
sha256:48f15b7c73add25dba9c828842d81560ccc94d5733518a79f4d96de248334f53 at receipts/engine/model_admitted/SFT-PHYS-FIELD-CONSERVED-SOURCE-001-48f15b7c73add25d.json;
empirical-validation hash
sha256:183057ccc690f97a75b6782893c5236ebd3c122936080aac8c7eff8d4f9120d3; measurement receipt
sha256:07029ac4d3d3fd288e1146ece1493bbabcf5bb46dd0b829d26dfd6a208db6e75.
```

51. Reciprocal interaction and transferred distinction

Claim identity: SFT-PHYS-FIELD-ACTION-REACTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A closed two-body interaction trace forces each transferred momentum carrier to be held once at the source loss and once at the recipient gain with opposite held orientations.

Action and reaction are the two held orientations of one exact transferred carrier on a closed interaction trace.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001

Generated grammar and closure boundary. Generate every field carrier, source-response relation, provenance, target access, measurement record, row policy, successor and extra-rule form. The declared exact boundary is: All finite source/response and propagation relations generated from Fold support, exact reference ratios, held orientation and closed transfer without an imported field equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reciprocal-transfer-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	reciprocal-transfer-pairing	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash
sha256:7351627b625df2d9146aba382796b02791d2d6fd2546a6a81d0a23e76c6bba66. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b874366c3800f4054158f9713a7cde61f1f13dc9a0c6706bf1e49f109b3538ec.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d0ef8c30d8dda7e6d8f0f9488112990148001c81a80e05bda93387c9ab8cf98a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ed8d40efc11b42ee302db2490e861dfb97b376eb07abf21d491efef140de596b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6024740d1e081816ec3b01895ca3341fda0d213857a57c7ce3983ef90c375311.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:297dbef62cbb8d0564d3ab582ef9d773fb2060dc95e4b1253ab052a336e62fbd. Independent certificate:
sha256:40404e78785faacae2f881f4a7d8abe1930ee3a3c3219182d8fa5f8ecd683d63. Engine external-validation hash:
sha256:d5ad4d4b37c295529ccc834f33d1ffeeef1ce619e7581ba3d853dd59a925c8636.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- action-reaction-1: predicted reciprocal-closed-transfer; observed reciprocal-closed-transfer; exact match `True`
- action-reaction-2: predicted reciprocal-closed-transfer; observed reciprocal-closed-transfer; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The field claim is falsified if any committed external row lacks the predicted source/response or propagation structure, any row is omitted, or the tampered unfavorable row is accepted.

Measurement receipt: sha256:dfd339e24fdf144d8825e7ba69c9990478519cc4d372a6c0b2fcaea5f4f99b73.
Isolation certificate: sha256:83b22c70622bef70e54c78af39cfeeff20133a559cbb2ab94244c949f502c4f0.
Custody certificate: sha256:516bfab4f70a7d4cfcac7d25a4aff2c346c61a5fe68856ec2fa484bda82d84c0.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- action-reaction-1 from NIST-CODATA-2022-ALL-CONSTANTS
- action-reaction-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- action at an ungenerated distance without a propagation trace
- negative or floating proof magnitude
- imported field equations, fitted couplings or target-selected constants
- target access and selected favorable measurements

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:6fd0b816fe52b6f475f046aceb099bea7065e9fcbdb8f24befa3eb181c5ac35dd; engine receipt
 sha256:78ed10fe165871136905b7cde51a30df16627b13339643b9b340f69c368fb343 at receipts/engine/model_admitted/SFT-PHYS-FIELD-ACTION-REACTION-001-78ed10fe16587113.json;
 empirical-validation hash
 sha256:be0b4ef2b80815d848e946c18f5c5fe98d11974a1296949fff5f83e5b7dcc7f4; measurement receipt
 sha256:dfd339e24fdf144d8825e7ba69c9990478519cc4d372a6c0b2fcaea5f4f99b73.

52. Period, recurrence and frequency

Claim identity: SFT-PHYS-WAVE-PERIOD-FREQUENCY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A wave period is the exact positive transition count closing one recurrence relative to a held clock recurrence; frequency is the exact count of closed recurrences per duration.

Period and frequency are reciprocal exact positive recurrence ratios with complete phase and reference traces.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__closed-recurrence-count-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is

`depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	closed-recurrence-count-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:601b00b48bd3b97f643a7590795cbcf3f2f51ff535526a7fa531c4651f27f6a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2484c7ec3efe10bd74c654b8c4bbafab282445c5022e2ec5122bcc189c67beda`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9e9a291e31cd136797288998c7d57a7734b99f165857c38894b07da5b609103c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6c93a003263282930c974a19e7419ebc1d785be762ba17f2067b9daa2599c8a6`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d28ae4833fb702e2b45d4b5815b2f139b60e7e315c6ef5dda145286584fa3b9e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:818304b482a4fcf476f693504599edfc466792c241c25186ce113f92a0d6947a`. Independent certificate: `sha256:c281bae88a0df7680ab1638b39dcc30c3badd302974d8dcb05edc9a995ff95f9`. Engine external-validation hash: `sha256:5d4ed5f7998755133cd0b29ef3ad75b9ee0fe331b80168bdb8a8e0d67c243017`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- period-frequency-1: predicted periodic-recurrence-frequency; observed periodic-recurrence-frequency; exact match True
- period-frequency-2: predicted periodic-recurrence-frequency; observed periodic-recurrence-frequency; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:00cf3fbd760d4912e92c38bce1e58c22b3b5abf1a644966e44f430085a9a0cba.

Isolation certificate: sha256:81b4bfa06cf26fb3e492949bd7caba3d3a7ea2951358c131409849d2b92333ee.

Custody certificate: sha256:248bfad8bcbdd6458b8a6e19a43909aabdba5795241d266edf87aaf25c63f4919.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbc0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- period-frequency-1 from NIST-ASD-5.12
- period-frequency-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8c089d92e70f95b50a09fe3ccc510250303d72ce0d7297d31030ac46a0733f87; engine receipt

sha256:55aecf75af90158e67ccf21446f06dc6d604ca0420482afe08637e465da99f93 at receipts/engine/model_admitted/SFT-PHYS-WAVE-PERIOD-FREQUENCY-001-55aecf75af90158e.json;

empirical-validation hash

sha256:9c2404e18eb33aa4056099c5c3dbb9fbc21c0b8fcddbd59cdf2cd4dfc4df951b; measurement receipt

sha256:00cf3fbd760d4912e92c38bce1e58c22b3b5abf1a644966e44f430085a9a0cba.

53. Generated wave propagation

Claim identity: SFT-PHYS-WAVE-PROPAGATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Wave propagation is forced as repeated transfer of one recurrence form through adjacent generated support while retaining source frequency, phase relation and carrier identity.

A wave propagates by source-bound adjacent transport of a recurrent state relation across generated support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__adjacent-recurrence-transport__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	adjacent-recurrence-transport	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d6ac3a6995197c0391122fe470977d800b160fc7a1e54abffe23648d1a188d9f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3af7c76f5029ff2a83b9b23a8d8d1288514912fce6f4735d002320d632f77091`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:026a75a6671c260841c751c47ee49f7c0ffef947edab1117e95a6f68c78cb99d`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:51c5ebc2f063f0349e8df80b91de3ba1e3f140bb831bf587b2831a6cb8a74558`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1f14778ed9b928d9abc8031d0a05a2d11cb2923e5a2586118d92c0a8d8ad4be6.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:0b3cc4a2294b47b1b090f55adc9b3fcc5ff6d6a46be5a52f0559462b58bfb242. Independent certificate: sha256:e9589e5e804c9d00fa36110b5c966679b9c10b80471a1db19aa91ba08120e2d8. Engine external-validation hash: sha256:b302a3e3777ef9c97a1914adb076610527d3e23b43e81bbd3533e63a38051ba1.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- propagation-1: predicted adjacent-wave-propagation; observed adjacent-wave-propagation; exact match True
- propagation-2: predicted adjacent-wave-propagation; observed adjacent-wave-propagation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:ecbdf53b4616a61a417ab3a833b22defe0381ea259b331f5795cf2c77d663f00. Isolation certificate: sha256:66632297b37d6eb268dde1190e813653f9b1b5bd6c23b7901e4dffbedc221fc1. Custody certificate: sha256:01b129f8343c2016bd0a36e1f6b06d2063af396859775dca2451e6e4d3a81368.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- propagation-1 from NIST-ASD-5.12
- propagation-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:583d105d16ec8e339b799ea1bf11abfb70f5a129ed8f38091429bda7f81f7db7; engine receipt
sha256:50a4686ca478198084edb8fed8ddd16cf51d791da3aee98fde34e393b23d0998 at
receipts/engine/model_admitted/SFT-PHYS-WAVE-PROPAGATION-001-50a4686ca4781980.json;

empirical-validation hash

sha256:c69dd8f7d2a4f89cc7b94d6bb085107d89e6fb483a87bbec02aaa683dea47791; measurement receipt
sha256:ecbdf53b4616a61a417ab3a833b22defe0381ea259b331f5795cf2c77d663f00.

54. Propagation speed, wavelength and frequency relation

Claim identity: SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. One recurrence transported through one spatial repeat forces propagation speed to equal exact wavelength support per period, equivalently wavelength composed with frequency.

Wave speed is exact wavelength/period, equivalently the exact product of wavelength and frequency.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-PROPAGATION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__spatial-repeat-times-recurrence-rate__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	spatial-repeat-times-recurrence-rate	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

sha256:84bc6f297507e8ed257a73faea2c6b1251e8c2770fe66ca8154e05f3981593a6. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:721206e1b8d38e686b94457bbb13dba587d49e5f497e5c5c8a57a530960f79ce.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6d6384f19549de2fc0a722fabb0821203feecee12ec799e493473eeabf067477.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1a0ef5a811b9a9e91b38d86e6cf20175aa034745139bc71502307dd236ef5256.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f687c6c853d8362126831975c0bfde0602fe37fae27ca089e39f4c1e5f5da554.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:eebd9cc41ac9e81564b19dc8bac9a4fac725278a96973dd497ce9dcf779bb0cf.

Independent certificate:

sha256:185d968975a383a7b3674347375ff79eb1b104c817038c86f9c0185783888192. Engine

external-validation hash:

sha256:5ec317567459c5d9366b5dfef20a9cca42e9abeb67132e2322a520e25afdd22a.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- speed-length-frequency-1: predicted speed-wavelength-frequency-closure; observed speed-wavelength-frequency-closure; exact match `True`
- speed-length-frequency-2: predicted speed-wavelength-frequency-closure; observed speed-wavelength-frequency-closure; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:e7fa7a2d678af00aa3058d405464161c3862a771103dec03a4e8390ef7d8ffd3.

Isolation certificate: sha256:dede6f8886a6d3ca7742e007a62f8d411f29b215abd7fcc10359c51c7a2939e8.

Custody certificate: sha256:d6b78e7dfb0a31f06bd9e7f65112db6aed1bbec4834edf70fb5c255733519ffb.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- speed-length-frequency-1 from NIST-ASD-5.12
- speed-length-frequency-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:ef50b7f656f86bbb15cd4cb0d3c8172cf7736f419eda1817621fb875f7031a87; engine receipt
sha256:0b1798571aa501d2d2f115ac1863142881c886214b0d3cbada2cc1cc03f47fc5 at receipts/engine/model_admitted/SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001-0b1798571aa501d2.json;
empirical-validation hash

sha256:abd7e043ca2448e874c0bb8b5cf68cb1acef82d8ec018a52d35d065705e51571; measurement receipt
sha256:e7fa7a2d678af00aa3058d405464161c3862a771103dec03a4e8390ef7d8fffd3.

55. Wave superposition

Claim identity: SFT-PHYS-WAVE-SUPERPOSITION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. When distinguishable wave traces share support without interaction, the complete observed carrier is their disjoint labelled junction; no constituent trace may be erased or double counted.

Independent waves superpose as the complete labelled junction of their source-bound traces on shared support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__labelled-disjoint-wave-junction__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	labelled-disjoint-wave-junction	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2287a8073cd6c4f58938d70412d176ec7b4d334e98bd0682e33e49944f99e627`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f1f498bfe822763d05d981ff2cc13c82e3f3097dffef33b20b93d9d8a5822624`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99f36ca76c44e49a26f7202b22981998295b57f995d844b37caebafd3224db03`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6593ae7f0e26039c15d45cbfb55fd71c1b7c22fde364c32e43e5c2725db35932`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f0f1d5f47f0661d51353f05b0af7d28f6e7468e898d8da0d01f07db78038b7fb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:706b5acee19878334d33cfaa3a90accacc00bfe8af9b97731d640cff64a0257`. Independent certificate: `sha256:8759b77eeb48a7cb010732754817c7d3b36337011dfe2c6f1c13fd6e56140010`. Engine external-validation hash: `sha256:aaf37dc4120601e7703e8b255ef4e44249d75be6fce160661265a859e7296413`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- **superposition-1:** predicted complete-wave-superposition; observed complete-wave-superposition; exact match `True`
- **superposition-2:** predicted complete-wave-superposition; observed complete-wave-superposition; exact match `True`

- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:6179415fbcec744b0d304d9e41c8a46334d46232548acf05dcd34bab9dfba677.

Isolation certificate: sha256:400b79d945a54433caac58e8f8f37854c16180a125423c205b6b34f46ac82c73.

Custody certificate: sha256:2fc0372700c99a06f8e135e0113cc733b63888e12f8bbebcbf65da99cab4f9d71.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- superposition-1 from NIST-ASD-5.12
- superposition-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1e084c75cd32f819fb2724505d0d805269b650e12b6cd78784cb2d5aff46c356; engine receipt

sha256:f8421b45a2cf9ca516fca24b3a33fc71e3f9b11e3435c2eaf6a6ccfe471f45e3 at

receipts/engine/model_admitted/SFT-PHYS-WAVE-SUPERPOSITION-001-f8421b45a2cf9ca5.json;

empirical-validation hash

sha256:c8364b6cc5f873f40e52a5c5daf2f57d6deaba3727a12ed8ba4256bda4970cf7; measurement receipt

sha256:6179415fbcec744b0d304d9e41c8a46334d46232548acf05dcd34bab9dfba677.

56. Wave interference and predecessor merging

Claim identity: SFT-PHYS-WAVE-INTERFERENCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Interference is forced where superposed phase-labelled predecessors map to the same observation cell: matching labels retain support and opposed labels close distinctions while the predecessor record remains held.

Interference is exact phase-labelled predecessor merging into common observation support with all source traces retained.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001

- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-SUPERPOSITION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-labelled-predecessor-merging__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	phase-labelled-predecessor-merging	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:46db1b341d3a10315674f6a5e61afa5514f4424ce741e02d9ef56b726d999c95`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:61f255d3b9a55856cf0f48b1e67feba27a0381851562b7f11689609142ee024c`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fd5497bd6c92c00934809764fb5fb06e0b0b89053f4b9178917c650676f37a56`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4a0f57ead6c024248f0960760259083e4d58d81e304758eae4dcfa2963683216`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6488f7f50675f6119b15bfe58e27a1093eebd210f4168384e7fbb19376b1ca40`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d60ddac9afa052b1c537ac69eff482d0be3eba6c7802fac752852e5c6d9a9ace`.

Independent certificate:

sha256:0ccaf4891c616c47996b7881a390fa5561f34a1d1ba36511022dc21a8e4df570. Engine

external-validation hash:

sha256:361bcefd540ebc670a80b6c89e4728a464d21be6916f12ca19c47726b1160cb1.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- interference-1: predicted phase-interference-merging; observed phase-interference-merging; exact match True
- interference-2: predicted phase-interference-merging; observed phase-interference-merging; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:27ee402e11aa853c8e3f3907f43db528e593512879dac3c0bfb19d0e00baa1e0.

Isolation certificate: sha256:78c598145c936624eb2bc701fc6a6f5838e16432aaa1faacee1a75c806745857.

Custody certificate: sha256:ad7a666c859e2f17ccf25cf643e533cf1138ea226ac5cdad93472dd0e51df6eb.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8cf0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- interference-1 from NIST-ASD-5.12
- interference-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:be7b9c926641bd8cabda9e0e20899b92c7e630147cf731a62da44fede8cd2f66; engine receipt

sha256:fecacb47e9a61a33c7df01e0cd569f3306ca1d298ed47232817357fbf247afee at

receipts/engine/model_admitted/SFT-PHYS-WAVE-INTERFERENCE-001-fecacb47e9a61a33.json;

empirical-validation hash

sha256:c77d3e82f8dfd7a43ac79cc5a07ff9d95e0e396dfa0230cc753b4866f51fdb7c; measurement receipt

sha256:27ee402e11aa853c8e3f3907f43db528e593512879dac3c0bfb19d0e00baa1e0.

57. Diffraction at constrained support

Claim identity: SFT-PHYS-WAVE-DIFFRACTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Constraining a wavefront removes propagation paths; complete successor generation from the retained aperture paths forces redistribution across every reachable output cell.

Diffraction is complete reachable-support regeneration from a constrained set of incident propagation paths.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-INTERFERENCE-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__aperture-constrained-path-regeneration__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	aperture-constrained-path-regeneration	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b00f0250f1d752d0d3213d9197a24d07cc28a664ef07afd90f726a7f9c7f14e6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:bba07f5cb14a9f143451e64b7e4ed0521d51a4361bea4867c0cff3c17cc88532.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:004b18f3ae136055c5cc72e2ff8713666ae74bed318be4a7b3e774c9636363ed.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a8c972847fbb73f4a71f95a8b5345ea9606159ce6f9d80b66103bce3da98142b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:12ddb094a78e44398cc42ca75027c12655d74c384edec117c098e8dae4a59684.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:8221e65a8a3ec1dc7622846aec9d52a0a6e961eb4d198c14ba09de5e9faf1b69. Independent certificate: sha256:6f251dc4cda533fb92ec13b262a73815ea0591429ec15e92f200aed0ad8dc027. Engine external-validation hash: sha256:5cfbc19af638d2549d0cc0dff5b3e66c0f487669c5223dee2441ee098c2e9250.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- diffraction-1: predicted constrained-support-diffraction; observed constrained-support-diffraction; exact match **True**
- diffraction-2: predicted constrained-support-diffraction; observed constrained-support-diffraction; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:fddfbfb736f2506f418f6988471945f19870456bd066697ca2e5154e2ed5e891. Isolation certificate: sha256:f6e0757b310fca56631225d0a65abd4a8ec3c21f7b34bab9117caf846b14a7fe. Custody certificate: sha256:30b61e49ddae2bc061146a367ebd3ab6c70c79eb4681ed642a42d5cba370d53e.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- diffraction-1 from NIST-ASD-5.12
- diffraction-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave

- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:7161343d80b1f1dfcca31709a1932fc0c38ac92cc61923966f7f6819137a1c93; engine receipt
sha256:4c9922f000b07cfacacfbac4f820e76af89cc3e83f0e211e5396c5625b61902a at
receipts/engine/model_admitted/SFT-PHYS-WAVE-DIFFRACTION-001-4c9922f000b07cfa.json;
empirical-validation hash
sha256:2e8d10315460c7a7da499c47da3fd3ece4ecf21dd3f19edb6381bd040fcd3b84; measurement receipt
sha256:fddfbfb736f2506f418f6988471945f19870456bd066697ca2e5154e2ed5e891.

58. Polarization as held transverse orientation

Claim identity: SFT-PHYS-WAVE-POLARIZATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Polarization is forced as the held cyclic orientation class of a transverse field recurrence relative to its propagation path.

Polarization is the held orientation of transverse recurrence support relative to wave propagation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-DIFFRACTION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__held-transverse-recurrence-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	held-transverse-recurrence-orientation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a4ad10b48dcea194c75b167d7f7389598c0436845ccb5a7df89a465cbb1eef04`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:41cdd4f74dea4fb6becc4f167cc10573d48f3d6f9d5ade29432ef3ea78cb1c93`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ce6994b049e729471b20447aef4c60376ee5b8704c069dacdb456903a20c7e34`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1f733f085dd0a916578c1b1caae7c363800f0c258ec6aea122c73732dc60d512`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a9e2a076ae00e455be645d7c404747acee5880f3f5bb7674a828ebb80e500453`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:bde18730b7e73c8a93fa972423db6a074505a121a3c8b0a929f8a1d343650232`. Independent certificate:

`sha256:81f3f282e0c3917b2bfbda79bd3bb7ce8644efc67b6c8a920e624b2994b4f702`. Engine external-validation hash: `sha256:5027cfbc5f3d5eef797069fca17a1d5d392c63fd53ea967f47872e6ec56682ae`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- polarization-1: predicted transverse-held-polarization; observed transverse-held-polarization; exact match `True`
- polarization-2: predicted transverse-held-polarization; observed transverse-held-polarization; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:a43d059197b55a8f4b0ec6822538a49fa62203e937ac3c1083dd3cb87af201e9`. Isolation certificate: `sha256:c14780d43a13378b84cf7a70620b262e5890233beefd1c8af26f75452f2b0e35`. Custody certificate: `sha256:390344ae0725188e5f16dba6d73bcd2470369002f84b4da729f66bbd9900f9a8`.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml> (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- polarization-1 from NIST-ASD-5.12
- polarization-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: ff32660b2dd6d39c4658c6320647c05df6c2f984de9ebf2883d67a9628371925; engine receipt

sha256: ed7c412e559c3488049747a95ba94327326d631f7896cdd18ddc33db557ec81e at

receipts/engine/model_admitted/SFT-PHYS-WAVE-POLARIZATION-001-ed7c412e559c3488.json;

empirical-validation hash

sha256: 9191952e3689a1574ad3c1119eabba9143b8df9766c151c1fcfd33e6b816c324; measurement receipt

sha256: a43d059197b55a8f4b0ec6822538a49fa62203e937ac3c1083dd3cb87af201e9.

59. Dispersion and support-dependent propagation

Claim identity: SFT-PHYS-WAVE-DISPERSION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. When support transition rules distinguish recurrence labels, each frequency class is forced to carry its own exact propagation ratio; their phase relations separate with distance.

Dispersion is label-dependent exact propagation induced by the registered support transition structure.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-POLARIZATION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__frequency-labelled-support-

response__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	frequency-labelled-support-response	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:c817a32960319800251cba6ae5313a77bb8e7067abd9c5a1992a92ddb2797006. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:2eba8297442d282c1c34126487ae4d5c2ac8f27b318b97212e55bf0fc9e203cf.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:50935063db3acaf93337341250dd63aef37f4dfb434e90242c843de5ca391af0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:cda9f78a03b2c8254ced238a1568c73e687eb09e119d3ae133d8849d704054cc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:c3a78ce30373f0e5c85cc27b15c81a1804d07b4db02289d2de65f52acf0c2e46.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:c2ce37fc1e3f802f9683fc17d1faf3dbcabcbe30b6db49cd878ca1ccd5a5d9186. Independent certificate: sha256:8ad4fb98343d0ebfa422aefd8e8c1ca0b4bb0b4c729e98daf0abdd1d3f4e9fd3. Engine external-validation hash: sha256:8aa4e151bef9045e9399b16114078d10e1e72479806e4fc3ed9277b09abbdcef.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- dispersion-1: predicted frequency-dependent-dispersion; observed frequency-dependent-dispersion; exact match True
- dispersion-2: predicted frequency-dependent-dispersion; observed frequency-dependent-dispersion; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256: faf14b33a5493c9d3f99a9920b8f33f44d5ba2b2ddb1a04bdf29c3d0f8f7420f.

Isolation certificate: sha256: 62d5c149716fb3b2bcea271c557a24eca7414d34d1f1a1f24b85870a70c5a0ff.

Custody certificate: sha256: 5fd09bde126cebce88449193bb4ab517fffb880d6cab844e1e67f335dcde5093.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- dispersion-1 from NIST-ASD-5.12
- dispersion-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 99dfb753c9092b1a1d9e26c0d60986514bd3b990a3c69ce166e2a40c5802da83; engine receipt

sha256: 72749162a46a877c7d54d3cde6f09848b2d383360ed687b201e6c94475a29fe8 at

receipts/engine/model_admitted/SFT-PHYS-WAVE-DISPERSION-001-72749162a46a877c.json;

empirical-validation hash

sha256: 31b6281252888e47b64ef536c44289125704bed608fdc95a2f077fcbeeb7841f; measurement receipt

sha256: faf14b33a5493c9d3f99a9920b8f33f44d5ba2b2ddb1a04bdf29c3d0f8f7420f.

60. Resonance and recurrence matching

Claim identity: SFT-PHYS-WAVE-RESONANCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A driven recurrence accumulates coherently only when driver and supported recurrence close on a common exact refinement; unmatched phases merge without persistent accumulation.

Resonance is exact common-refinement closure between driving and supported recurrences, producing repeated phase-aligned transfer.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-DISPERSION-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__common-refinement-recurrence-matching__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	common-refinement-recurrence-matching	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bdf5ac36ea7790d3b829fef41996ebe76cc9dfbd4530456db3e643fbf741cb56`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d2df52b8708ce52d02c5f9c8950d5ea9b5b2f9b272fa5f7be3f6a9c9200a7c4e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fc794b574799a1058fffb08f667dc5a42ee0523191d34ae3fcec6bbb143f4e083`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : cc80fe73f2b44d6c50179e804afae5bd6155a2698bee3689f5982707b729a1ec.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 4f8e5411052b53acc7dc47aa78d57b2aa010994dd757abfc6b3ddc9e0bc6440a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 9c5906e5e5607427d0a3ea0a15716c7686f7a6b02ae1e2baa369e857a81c88e8.
Independent certificate:
sha256 : e89d71e94981cf1e8eadbbae1d46929ddc31bb7a5ef70cbfa926c3f173ed58eb. Engine
external-validation hash:
sha256 : 8ccefae22d2afded9378f274eceb0876bdeca43fa4ce14b2a9b0830bc57e08ad.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- resonance-1: predicted recurrence-matched-resonance; observed recurrence-matched-resonance; exact match **True**
- resonance-2: predicted recurrence-matched-resonance; observed recurrence-matched-resonance; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256 : 5587beb44326be1e025b8a6a32d5769ea98ff4821fae2f0a40cf5de33a9ad33a.
Isolation certificate: sha256 : 481d971c9b9bca81cb43c15ef93383756349195ff48f36be010284f866e43a47.
Custody certificate: sha256 : a46b5db8532e537f99537ae6aa82dd634c492b4b76260f535eb232b0fb440bf6.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- resonance-1 from NIST-ASD-5.12
- resonance-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256: 4530e5320e83f41278376ce35c1394e1134d846d07515c70aa7c78fbf5b24e35; engine receipt
sha256: cac57d85a74fcf5b8f323aa8ea98203d242e8e40084465de77b89d33b6db55e3 at
receipts/engine/model_admitted/SFT-PHYS-WAVE-RESONANCE-001-cac57d85a74fcf5b.json;
empirical-validation hash
sha256: 76602285c6259cca45dc32916d234702b3b607e013728be5e2e94f616a0891ee; measurement receipt
sha256: 5587beb44326be1e025b8a6a32d5769ea98ff4821fae2f0a40cf5de33a9ad33a.

61. Wave energy and momentum transport

Claim identity: SFT-PHYS-WAVE-ENERGY-MOMENTUM-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A propagating recurrence that performs work at a remote response cell necessarily transports the corresponding energy carrier and held propagation momentum along its complete causal path.

Wave propagation transports exact energy and oriented momentum carriers, paired at source, path and response.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-INFO-CHANNEL-CAPACITY-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-RESONANCE-001

Generated grammar and closure boundary. Generate the complete carrier, recurrence/propagation relation, provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite wave forms constructed from generated recurrence, adjacent propagation, held phase/orientation, exact ratios and complete observation support without imported continuum equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__causal-wave-transfer-carrier__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	causal-wave-transfer-carrier	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:a2abc6592f4d4db9f1221685ee18b30ec9cd7db7ff109a260d50cb87ea83c9d5. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:703a05f7937d3054ac10d8e2b66119064b82bf4a2891940671546eb6949d7fb4.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6f905acb05ac57cb4fd664d461b08ac448933f7adf404e432f6eaa911be54d65.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:ba67bebcc0dfce85a8cc736d4cfd96f48b98a5617d1fe807d34091e25168e7cd.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:16137d5af11336ef1b1fdacf28e27fe6f63731cb2900c72fbe062bc6d96cea17.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:de27387e232a634fa0a663a09bcaf09a64290408013be504fd5727c5e36537d2. Independent certificate: sha256:e852817a61bb2659ee2d9b24e0d60b77e91ce8f0b8470271e302ab187c1d448e. Engine external-validation hash: sha256:f723a1659fe39ba6ec5f0e2479fd37e1dec3fd195820725fba702a5c04cceaef.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- energy-momentum-1: predicted wave-energy-momentum-transport; observed wave-energy-momentum-transport; exact match `True`
- energy-momentum-2: predicted wave-energy-momentum-transport; observed wave-energy-momentum-transport; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The wave law is falsified if any committed NIST spectral/transition row lacks the predicted recurrence or propagation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:eb45a52316baae6215822fdfb276a428e8883cad50b45dcab87087a8fb9fc7af. Isolation certificate: sha256:0359905dc37a5254d9ee57565fd80b3e660d5a76079decaae9ac6ff8b3f7da41. Custody certificate: sha256:72fee7f977609282e4a6f3102db753b3f19ef7de65429a8a70eebe15ea308622.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml> (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- energy-momentum-1 from NIST-ASD-5.12
- energy-momentum-2 from NIST-ASD-5.12

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite wave
- irrational or floating phase as a proof value
- imported wave equation, fitted medium law or target-selected result
- target access or omitted adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:f58817363c14545510ba7152e7f925e69303e204f13e93e236d43a392980330a; engine receipt
 sha256:1a25f4c7e2b95a50fd6b37e122372952e6563575c716a6cda083ce1d0755e6de at
 receipts/engine/model_admitted/SFT-PHYS-WAVE-ENERGY-MOMENTUM-001-1a25f4c7e2b95a50.json;
 empirical-validation hash
 sha256:337ba7284de709779a228b8d01faeba71e03195569fadbabfe66d18071ba31ab; measurement receipt
 sha256:eb45a52316baae6215822fdfb276a428e8883cad50b45dcab87087a8fb9fc7af.

62. The second structural Fold generator is three

Claim identity: SFT-PHYS-STRUCT-GENERATOR-THREE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact Fold period spectrum forces its second distinct nonidentity generator to be the positive count three: the binary generator is the first-return period of the unit part whose denominator is the positive predecessor of depth-two Fold support, and the least distinct successor period is witnessed by the corresponding depth-three unit part returning in exactly three Fold transitions.

The second distinct nonidentity Fold generator is the exact positive count three, witnessed by the complete first-return orbit $1/7 \rightarrow 2/7 \rightarrow 4/7 \rightarrow 1/7$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete product of Fold carrier, action, recurrence, binary anchor, distinct successor, unit-part construction, orbit record, order closure, measurement direction and extra-rule forms. The declared exact boundary is: All exact positive first-return generator constructions available from finite Fold-word support, positive

predecessors, Fold action, recurrence and positive-count succession. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `exact-positive-period-trace__double-and-cast-complete-One__complete-first-return-orbit__depth-two-unit-part-period__least-distinct-positive-successor__support-predecessor-unit-part__all-Fold-transitions-held__positive-count-discreteness-plus-existence__derivation-before-measurement__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	exact-positive-period-trace	borrowed-integer-name	A borrowed numeral has no Fold recurrence trace.
action	double-and-cast-complete-One	imported-modular-map	An imported modular operation is not the admitted Fold.
recurrence	complete-first-return-orbit	repeated-value-without-first-return	A repeated value can hide an earlier return.
binary	depth-two-unit-part-period	asserted-binary-count	Naming two does not derive its recurrence.
successor	least-distinct-positive-successor	selected-later-period	A later period is not minimal.
unit	support-predecessor-unit-part	target-denominator	A denominator read from an answer is target input.
orbit	all-Fold-transitions-held	period-label-only	A label cannot reproduce the return.
order	positive-count-discreteness-plus-existence	bounded-denominator-search	A denominator bound does not close later candidates.
measurement	derivation-before-measurement	measurement-selected-period	Measurement-to-derivation flow violates the admitted boundary.
extension	no-extra-rule	extra-generator-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:59b8326ad42873e5d8bc8d868f6d8cc795f5757341747de16dd093f92b8f3331`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e990d2e429bd08054cf8b7dabd6f1ef1cd683f13016fd18b84c94b39c72c80ab`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:747052c401ddb2d214104d2c733bf23c0864fbe7b309ea5370b09358385d669e`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:862121326d132cadaef270bd54d159d72e4b223683d7619ed55a95de8cbe7bc9`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:b44f622f2aca2e6f6f17edc34ee0b17e5e7acfb2c16ce2c030843ce1ffbf15ce`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d80e7a81337236193d23dadccc0cbd15fe7c57363ab5e1bbb0c0e021207bb460`. Independent certificate:

sha256:66c249178802488cd20c418f0d1c6f418b549fa541ed22f153d6d5593cbb5b27. Engine
external-validation hash:
sha256:329323f9ce81e7559e72e47ce558f071ad3a619644057a8997a403d700fd87e2.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilation, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- measured dimension, physical target or V2 proof artifact as a premise
- semantic numerical zero, negative, irrational, imaginary or floating proof value
- bounded denominator search standing in for positive-count closure
- selected later period or added generator rule

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c87a7354ead0e1b5834d7da7b269539830b7baab74260db9e0b4205a4152a7de; engine receipt
sha256:3017771dde2eabf649278851cd9e43fb6c5e939b5a052a09f02f6e604d6ed50e at receipts/engine/model_admitted/SFT-PHYS-STRUCT-GENERATOR-THREE-001-3017771dde2eabf6.json;
empirical-validation hash None; measurement receipt None.

63. Three-dimensional Fold stability

Claim identity: SFT-PHYS-SPACE-DIMENSION-THREE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A Fold-stable spatial carrier has exactly three independent generated directions: it must strictly exceed the two fibre roles so their return remains distinct, remain strictly within the four ordered pair-cell roles so a restoring return is retained, and the complete positive-count census of that open window has the sole member three, independently equal to the second Fold generator.

The unique Fold-stable spatial dimension count is three, independently equal to generator three.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate every spatial stability form over carrier, fibre lower bound, pair-cell upper bound, interval status, positive-count census, spatial count, generator cross-lock, measurement direction and extra rule. The declared exact boundary is: All positive finite independent-direction counts classified by the Fold fibre lower boundary, ordered pair-cell closure boundary, return stability and generator-three cross-lock. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `independent-generated-direction-carrier__strictly-beyond-two-Fold-fibres__strictly-within-four-pair-cells__open-Fold-stability-window__complete-positive-count-enumeration__three-directions__equals-generator-three__forced-count-then-observed__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	independent-generated-direction-carrier	coordinate-name-only	A coordinate name has no independence or incidence trace.
lower	strictly-beyond-two-Fold-fibres	one-or-collapsed-fibre-count	At or below the fibre count the common return is identified with a fibre role.
upper	strictly-within-four-pair-cells	unbounded-coordinate-extension	An unbounded extension adds roles not generated by one Fold closure.
interval	open-Fold-stability-window	closed-or-one-sided-window	Including two collapses return; including four exhausts the restoring remainder.
census	complete-positive-count-enumeration	selected-dimension	Selecting a familiar dimension is not forcing.
value	three-directions	two-directions	Two does not remain beyond both Fold fibres.
crosslock	equals-generator-three	no-independent-lock	A single route cannot expose a changed stability bound.
measurement	forced-count-then-observed	space-count-read-from-measurement	Measurement cannot select the spatial law.
extension	no-extra-rule	extra-dimensional-model	An added manifold or conventional dimension rule supplies the answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:30734bd69f1b2f6d01339f06c22dd39fcf6a8c847f1869bdcb5f59752faf81e5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:39008b82c2eb7d8b02749e1f886046630aed4e5affa12b68262488b73ed9eed0`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a86c799b092cf1b396035106b1af9b0b17a95c5d7cc8faeef9bdb4cc80aa6438`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:07466e8d00c6c6eb839d1c6ea54149ccc6771bb4e6d9f62a8112929c702d9255`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:3627147e15586e2083b67ac167af616fad3d399da79173f407f0279cdd30b00d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:3766b1799f4000dca01577afdca82223675061fa7d71a8d597f37ecdfe83de76`.

Independent certificate:

sha256:1c0056d44ff41cd15a9fb973b8f43b7f2979dcf4b31524c5509230c77f42270a. Engine

external-validation hash:

sha256:d8d3ec58597f52d385e4dcc86e394cd7c1375940f4852e69e2b564b8f6690166.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilation, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- observed three-space fact or conventional manifold premise
- V2 proof artifact as a dependency
- semantic numerical zero, signed dimension or nonexact proof value
- target-selected interval endpoint or extra stability rule

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:28ea66b5537d3d9b6adb8241b0ab0863ee82c80179db2ed4b40d2cd020c207ef; engine receipt

sha256:fb06dcdbd49ff61a213ff97420274014c7e2b683cddf0c593f269045b387d589e at receipts/engine/model_admitted/SFT-PHYS-SPACE-DIMENSION-THREE-001-fb06dcdbd49ff61a2.json;

empirical-validation hash None; measurement receipt None.

64. Rank-two boundary of three-space

Claim identity: SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For a complete boundary surrounding a localized source in the forced three-direction spatial carrier, holding one source-normal direction leaves exactly the unique positive predecessor rank two as free boundary organization; scaled equivalent boundary support is therefore the complete pair-cell product.

The complete boundary of the forced three-direction spatial carrier has exact positive rank two and r -by- r pair-cell support at every generated positive scale r .

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001

Generated grammar and closure boundary. Generate the complete product of spatial carrier, source normal, boundary incidence, rank relation, rank value, scale composition, coverage, generality, measurement direction and extra-rule forms. The declared exact boundary is: All complete finite boundaries of a localized cell in the forced three-direction carrier, classified by one held normal direction and the positive predecessor relation on retained independent boundary directions. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: `forced-three-direction-carrier__one-held-source-normal__all-nonnormal-directions-retained__unique-positive-predecessor__rank-two__complete-pair-cell-product__complete-equivalent-boundary-support__pair-product-successor-closure__rank-sealed-before-measurement__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	forced-three-direction-carrier	unbound-coordinate-space	An unbound coordinate collection has no forced dimension identity.
normal	one-held-source-normal	no-held-source-normal	Without a held normal relation the boundary is not source-relative.
incidence	all-nonnormal-directions-retained	selected-tangent-subset	A selected subset can omit a free boundary direction.
rank	unique-positive-predecessor	signed-dimension-minus-one	A signed scalar expression is not the positive proof witness.
value	rank-two	rank-one	One plus the held normal does not exhaust three directions.
scale	complete-pair-cell-product	linear-or-unlabelled-repetition	Unlabelled repetition loses one retained boundary coordinate.
coverage	complete-equivalent-boundary-support	partial-boundary-sample	A sample cannot establish complete geometric dilution.
generality	pair-product-successor-closure	fixed-scale-table	A finite table does not prove the next scale.
measurement	rank-sealed-before-measurement	measured-rank-input	That reverses the derivation-to-measurement boundary.
extension	no-extra-rule	extra-boundary-rule	An added shape or exponent is not supplied by the dependencies.

Base and successor. The closure proof is sealed by derivation hash `sha256:aa3cb73a39bec493be5a0f781d0be770699cf7118d5477952c1e5da90167c185`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:9b50b4f4f1fca1e42ca15c89d7bf1f7c5f533d4484c44c9234deaa5927e87138`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:66896c9dc47e667d9cc368cba3ee78533675b5c097542a9efc91ce2509e114c4`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c0a38d668713c09944f2fce5ca02b238d428333fcb9cd43b4283e6c8f22a3649`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a4c33833ff4802fc48fd89fd2116d367ce1e625e43cffbcf2732fb5fc3fa613a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:6e48fe721547bc6ea1ceb176a81258205bb3c8fa6afd7ada91f787143a911aa7.
Independent certificate:
sha256:503a0420c507563e3b0aadfe67448fb365824653e7436033d4a9e5e49b33d463. Engine
external-validation hash:
sha256:1c2e7a9a392bd2b6eb7d4b6fb3e67322da80b0c102f393fe4d1d6f163140f076.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- measured inverse-power exponent as input
- conventional surface-area formula or continuum premise
- signed subtraction, semantic numerical zero or nonexact proof value
- partial boundary sample or added geometry

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:a888f6a3dab8ef342e57456e170a55a97520a3bb65702e6f7a0ef9e8e613c186; engine receipt
sha256:84ef6397b0fc006ae2ea6525024a9e8920f20afcd42929632db793ae0a1d809b at receipts/engine/
model_admitted/SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001-84ef6397b0fc006a.json;
empirical-validation hash None; measurement receipt None.

65. Inverse-square dilution from rank-two boundary support

Claim identity: SFT-PHYS-FIELD-INVERSE-SQUARE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For one conserved source distributed without preference over every equivalent cell of the forced rank-two boundary, increasing source distance by an exact positive ratio r increases complete boundary support by r paired with r and therefore forces response per cell to be the source divided by r squared. The exponent is derived before and independently of measurement.

*One conserved source over complete equivalent rank-two boundary support forces response(source,r) = source/(r*r) for every generated exact positive distance ratio r.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001

- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of source identity, conservation, boundary rank, boundary coverage, scale transport, response allocation, falloff profile, generality, measurement direction, evidence record and extra-rule forms. The declared exact boundary is: All exact finite geometric-dilution relations for one conserved localized source over complete equivalent boundaries generated from the forced rank-two spatial boundary. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `one-retained-source-carrier__same-complete-source-at-every-boundary__forced-rank-two-boundary__all-equivalent-pair-cells__distance-ratio-paired-with-itself__source-over-complete-boundary__inverse-square__pair-product-successor-induction__sealed-exponent-to-blind-measurement__source-bound-cell-and-measurement-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
source	one-retained-source-carrier	unbound-response-number	A detached number has no conserved source identity.
conservation	same-complete-source-at-every-boundary	distance-created-or-lost-source	Changing total support violates the admitted conserved-source law.
rank	forced-rank-two-boundary	free-or-measured-exponent	A free or measured exponent reverses forcing.
coverage	all-equivalent-pair-cells	selected-boundary-cells	Selection breaks equivalence and conservation accounting.
scale	distance-ratio-paired-with-itself	unlabelled-scale-factor	An unlabelled factor loses one boundary coordinate.
allocation	source-over-complete-boundary	fitted-response-profile	A fit can select the target falloff.
profile	inverse-square	inverse-linear	Linear falloff omits one retained boundary coordinate.
generality	pair-product-successor-induction	finite-distance-table	A table has no successor proof.
measurement	sealed-exponent-to-blind-measurement	measurement-to-exponent	A measured exponent cannot become the law premise.
record	source-bound-cell-and-measurement-trace	reported-exponent-only	A label alone cannot reproduce conservation, cells or comparison.
extension	no-extra-rule	extra-shape-or-scale	An added coefficient or shape is not supplied by complete equivalent support.

Base and successor. The closure proof is sealed by derivation hash `sha256:7ef928a9e2bc5ea3b57e3da3b9728f9e2b50ee36950a261555c7764c5e1c3a27`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : 5126eae290f28749926e1532af4473ba49a075ffe7045e937a67ccb90b2c7afb.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : 73ad9ff5259b5413d9f21ced60ef80c4b9652e711a726e2229f6000c1f2a571c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 6757de96ac570cbdebc3ea1c952bf89e3f2e322455d2086a84a65816abb12181.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : e68c533619348505aa22d14cf741971fab0ad820590878d5234ba19b9603915.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : dd451d0191f043210e3c24969aa8119c07c11f3b67f20c7a71d26f380376e63d.

Independent certificate:

sha256 : 8ecbdb6929c52e1cf21c8229755bab8685a4ccb896c90e4089352870edbf6ccb. Engine

external-validation hash:

sha256 : 9c9b53427e3ac933d30cb465da2fe7e7a4f1cc88e72f1a5738cbed87fd420c6.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- measured exponent, fitted coefficient or conventional inverse-square premise
- partial boundary, omitted response row or preferred cell
- semantic numerical zero, negative, irrational, imaginary or floating proof value
- target access before the forced profile seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 8a3751d86c3485d1aae1196986d3beabc52ad66603ab501e2202fa2d48a00ea8; engine receipt

sha256 : 68fb6f9d11a0c64e08a8d6339dcc5feec4717a76d5d46d600988774ae79ccbf2 at

receipts/engine/model_admitted/SFT-PHYS-FIELD-INVERSE-SQUARE-001-68fb6f9d11a0c64e.json;

empirical-validation hash None; measurement receipt None.

66. Complete prime-sector force and mediator ladder

Claim identity: SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The least binary cover of generator-three successor volume fixes ceiling seven. Complete positive divisibility enumeration then leaves exactly prime sectors two, three, five and seven; each sector p has p charge labels, coupling $(p-1)/p$ and exactly p -squared pair cells less the unique colourless return, hence mediator counts three, eight, twenty-four and forty-eight. The

next prime eleven lies beyond the ceiling.

The complete prime-sector ladder is (2,3,5,7), with couplings (1/2,2/3,4/5,6/7), mediator counts (3,8,24,48), and first excluded prime 11.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sector carrier, dependencies, prime-ceiling relation, enumeration, minimality, measurement direction, evidence and extension forms. The declared exact boundary is: Every positive count through the independently generated upper cover depth, complete proper-divisor tests, complete sector pair cells and the unique colourless return. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__prime-through-cover-ceiling-and-p-squared-less-One__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	prime-through-cover-ceiling-and-p-squared-less-One	composite-or-unbounded-sector-ladder	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:3ec46b64ac79eb7d5890fb933d8591ea0907d57bababe6855736a0fbb4af5301`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:fbc7fcd8393f763160ba4311295df1fbd4295f929c9d213611653f1d6cb54176`.

- **tampered_source**: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:50ef75f995e5860dc10d502e63938d55b5edd6299dbe42475bf436a7aa349fee.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:193ea72fce01861503ea08ada27034731dd469a147bcf39a6a2772f1232697e3.
- **boundary**: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:7402ed46812fea45fa5d719b332b54e32704288ceb51128c7ecfdd0d11a3eb46.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b.

Independent certificate:

sha256:c907917461c9da17b81877c6d36f16e9ef6ccfc8116a9213c80b754c392f2f70. Engine

external-validation hash:

sha256:9b55fe2d38f9aba0b68df47435f6ff39c1d7c7467dbb775855f50378dacbd761.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:a05666ce9ce250c3a40e7f745a59a6e801ca7671082e33aef826d19a102b5c98; engine receipt

sha256:1e22b2eda37dc8afe38d740cf10db286fb2e5f238704a22ccfee7066c8cbaa85 at receipts/engine/model_admitted/SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002-1e22b2eda37dc8af.json;

empirical-validation hash None; measurement receipt None.

67. Complete finite prime-sector force and particle inventory

Claim identity: SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The forced prime ladder two, three, five and seven closes the entire registered force-sector grammar. For each sector p it fixes p charge kinds, p -squared less One mediators, coupling $(p-1)/p$, exact successor running, p -member and antipodal singlets, three-generation mixing shares and fermion mass-part $1/p$. The first excluded prime eleven terminates the inventory.

Sectors (2,3,5,7) have charge counts (2,3,5,7), mediator counts (3,8,24,48), couplings (1/2,2/3,4/5,6/7), gauge-carrier total 83 and 72 penta/hepta carriers. Penta/hepta singlets have 5/7 constituents, mixing diagonals 14/15 and 20/21, electron mass-part ratios 2/5 and 2/7, and the complete named inventory has 98 members including 12 Smithions and

photon/graviton/Higgs. Prime 11 is excluded.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-PHYS-STRONG-RUNNING-DIRECTION-002
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-MATTER-MIXING-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete twelve-axis product of sector, charge, mediator, coupling, running, singlet, mixing, mass, inventory, measurement, prediction-record and extension forms. The declared exact boundary is: Every prime sector through the independently forced ceiling seven; all fibre members, ordered pair cells, finite positive supports, three generated positions, singlet closures and the first excluded prime. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `complete-primes-through-forced-ceiling__complete-p-fibre-members__p-squared-pair-cells-less-One__p-predecessor-over-p__support-successor-in-sector-gap__complete-p-fibre-or-antipodal-pair__three-preimage-offset-over-sector__One-shortfall-equals-one-over-p__complete-count-and-first-excluded-prime__inventory-sealed-before-anchor-check__standing-falsifiable-penta-hepta-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
sector	complete-primes-through-forced-ceiling	selected-known-sectors	Selecting familiar sectors cannot close the ladder.
charge	complete-p-fibre-members	borrowed-group-dimension	A borrowed group label imports its answer.
mediator	p-squared-pair-cells-less-One	listed-particle-names	Names do not generate a complete inventory.
coupling	p-predecessor-over-p	measured-coupling	Measurement cannot select the law.
running	support-successor-in-sector-gap	imported-beta-function	An imported beta function is outside the grammar.
singlet	complete-p-fibre-or-antipodal-pair	imported-hadron-taxonomy	A taxonomy does not prove closure.
mixing	three-preimage-offset-over-sector	free-mixing-matrix	Free entries are parameters.
mass	One-shortfall-equals-one-over-p	invented-sector-scale	A new scale forks the unified carrier.
inventory	complete-count-and-first-excluded-prime	open-ended-particle-list	An open list contradicts the forced sector ceiling.
measurement	inventory-sealed-before-anchor-check	known-counts-visible-before-seal	That would restate observation.
record	standing-falsifiable-penta-hepta-record	omit-fringe-predictions	Omission would erase a forced consequence.
extension	no-extra-rule	free-extra-sector	An extra sector violates the cover ceiling.

Base and successor. The closure proof is sealed by derivation hash
 sha256: 4624b05862724e3285d9f74a45c3ef7142151d86147fc6fd42c25dde3783b8db. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
 sha256: e0da3f1ad608075f24dc7c22e1ba300efc4afff20b1c929fd22f6e086dbf2a91.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
 sha256: cbde686fb626e90fb2ac7838288cdb76d00b198b78063864afb39ae17b18c57e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
 sha256: d9b17a089155cdab4895ea44b5ad732c09127140b3743d6da0f6912bfa2fed2c.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
 sha256: e8e8c91307ad668750ead2ccc08f5eb36976d8c4801fa25ecdd7a5a5436a71f5.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
 Implementation hash: sha256: 319a328268348bb63287e71b45d9eff654c873e1b3f7365f18be09d372169101.
 Independent certificate:
 sha256: 15fd0ea3dd07b0c350fd55c00dc5494379e0a52136632de037958fd9fe6a80c2. Engine
 external-validation hash:
 sha256: a6724da44d789aa5533e609ddc0524c4b1ffc844db35b124d86e07f2f42f4c52.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no omission of predicted penta/hepta sectors because they are presently unobserved
- no imported SU(n), Standard Model, grand-unification or hadron table as a proof premise

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
 sha256: ed90e7c9cd5e1693f0cb76d19db5c57283b70bf94c1841ca79beec9910a10829; engine receipt
 sha256: ee6b271f050a867fbe8d5ddcd10e7dba7f7ccba227b4094efbd5c6e2b48741fb at `receipts/engine/model_admitted/SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003-ee6b271f050a867f.json`;

empirical-validation hash None; measurement receipt None.

68. Charge, Coulomb potential and Gauss closure

Claim identity: SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A conserved held charge distributed over the forced rank-two boundary yields exact inverse-square response; multiplying by complete boundary growth reconstructs the source at every generated radius.

For source 1/8, radii 1/4 and 1/2 retain identical Gauss flux 1/8; fields are 2 and 1/2 and retained potentials are 1/2 and 3/4.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for charge, coulomb potential and gauss closure, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-chain__held-charge-over-rank-two-boundary-with-source-return__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	held-charge-over-rank-two-boundary-with-source-return	imported-Coulomb-profile-or-free-exponent	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:5207acfe629eeb3a1b95422de5b03da37100b21be36667c71a8039d2135b933e. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated

by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: aed03d64d05dd11ac2114c94f12eed64e5fbf8a41ac0785d8c7534a2adbb8a6.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 1db42473facdb53ec56c2ea4e22126862ac3d0e1d7522af4d04004a727102a95.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: a02635acf69c4a02339b1721a416e0629ae94925af07f044f567ebf0e9b41cbe.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: 25ab4a9872d7deb14c1bf89d25eeb44b1ae26eef1151ddb88f65cfd2a1b1bdc.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f.

Independent certificate:

sha256: 7a418f941af171192cbb28438b7d35e08fc2db7203efb9e91f18f193d717e0d6. Engine

external-validation hash:

sha256: 32f4b2183afd1bfba698748c23caea10a2eb5b631239a1f057cc7ccd0cd37c09.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: c881d6a01b046c3069e5b317d7a7da7c0ced55fc2f759078ec68e659befe3dd2; engine receipt

sha256: f30356c4d6f5fc9478100b9626b8303088cead7845844e1dab28c744761b5197 at receipts/engine/model_admitted/SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003-f30356c4d6f5fc94.json;

empirical-validation hash None; measurement receipt None.

69. Magnetic relativistic correction

Claim identity: SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For exact sub-One speed, the motion-held square and positive electric remainder partition the One; the remainder commutes with the Fold and therefore supplies the magnetic correction as a dynamical projection of the electric carrier.

The correction is $C(v) = \text{One} - v^2$ as a positive retained part; at $v = 1/2$ it is $3/4$ and $\text{Fold}(C) = 1/2 = \text{One} - \text{Fold}(v^2)$, likewise at $v = 1/3$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001

Generated grammar and closure boundary. Generate the complete eight-axis product for magnetic relativistic correction, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__One-partition-by-speed-square-and-Fold-covariant-remainder__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	One-partition-by-speed-square-and-Fold-covariant-remainder	independent-magnetic-force-postulate	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9b906c00219993001019af98ebbabb0b464f0ab1677b5ac42feecf983b9c69ae`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: fd58ff0c8faddc6b0127682f22c4ebd1ac5f8c72d78259c3b716b133d804e132.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 27588232bfa00a92166f28965d32ef1fbba4636c5c126546b5c8594c192d178.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: c4cf910271ff874cc8cde959f228e2b7023f1ebb13d6c1c262055bfa49051e52.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: f6341899b006c65deee611e514f5b58ad6c05bf74b11d24504d79e33157946ff.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f.

Independent certificate:

sha256: 8f42c6b9b1037c7964c07564b53d896e18df6bfb2232af192997d2bfc8910a13. Engine

external-validation hash:

sha256: 0b1c27f4c2b5b3cbb82c785e40d18796420620c4d59c743089fde471c056ed5b.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: e07df5d65e828be51bddc3e891f8596ac28e766e55a0ff318f06b6708897176b; engine receipt

sha256: 8039e9c6e59176d5f9a09a7796e95e95fdf2587b1abc870becfdd5d715fe45dc at receipts/engine/model_admitted/SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003-8039e9c6e59176d5. json;

empirical-validation hash None; measurement receipt None.

70. Lorentz transfer partition

Claim identity: SFT-PHYS-FIELD-LORENTZ-TRANSFER-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Motion does not create a second unaccounted force: it holds the exact speed-square share of electric transfer while the positive remainder acts on the observed carrier, and both shares reconstruct the source force.

For electric force 1/4 and speed 1/2, retained force is 3/16 and motion-held share is 1/16; together they return 1/4.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003
- SFT-PHYS-MECH-FORCE-001
- SFT-PHYS-MECH-CONSERVATION-001

Generated grammar and closure boundary. Generate the complete eight-axis product for lorentz transfer partition, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__electric-transfer-partitioned-by-motion-share__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	electric-transfer-partitioned-by-motion-share	unrelated-added-force	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:b05e985dccbb063ceeb2c8d488f431c2d704d76517515de0f2577f48eabebe03a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:6ba16b5ac831f6230b21f169c00a2c6421b2877b8a5a2509054bdc6fdf6876e6`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:57ce2f114d67fb08f7ed01a12d2a3e11e914a41ca46adc9b446baf4680d797c3`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1ab25ba309a9c883998b7016ca264e3f9266c5f540680afde7a5b6214ab10ccb.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:f43103f36eb11d01a281a15a18d16a4554ed54e047faeaa13d6ba187ae38fa.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f.
Independent certificate:
sha256:d94eff429d8b2e205c7d6052f7c03391bedf22267342de6e5ca75dc6cc6dd23e. Engine
external-validation hash:
sha256:02d87952ae8f1554631ea753e3cdbc1cf290092c0e0b1807efa1ab2432a26cdf.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:672db44f64f8909e47215811622cf8a83547dce97fe44a3e301419f1aace10bc; engine receipt
sha256:353351d17f145e0f89d0db4a7250469284dd4bae4732c1961bf519bfcd5f211f at receipts/engine/model_admitted/SFT-PHYS-FIELD-LORENTZ-TRANSFER-003-353351d17f145e0f.json;
empirical-validation hash None; measurement receipt None.

71. Planar Maxwell curvature closure

Claim identity: SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. On the forced rank-two boundary, two spatial axes each carry binary second-difference curvature while one temporal axis carries the same binary curvature; per-axis balance fixes wave speed-square at the One.

Planar spatial curvature is four, temporal curvature is two, total ratio is two and dimension-normalized speed-square is One.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-LORENTZ-TRANSFER-003
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete eight-axis product for planar maxwell curvature closure, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__two-spatial-binary-curvatures-balance-one-temporal-per-axis__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	two-spatial-binary-curvatures-balance-one-temporal-per-axis	imported-planar-wave-equation	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f320aa88ea1c8d20a34b8726f84850e21a6b1ebf418454d0c6f9d711739ec19f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:eafdb99a9d320fec212c4ab36f18c4a4efd682611fef094307d16dc5e1572ebf`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:306d2fa1aeb708f8c68d2b46d8a68a777b72679e9b4af54d77909a4385ff71cd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3ccafd6fdb2f836afd17c5e6fa0162cb1d033ed59894e8163efd758405fc5c94`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt

sha256:74de44412fc44a7fb00345487361b8f0666807a320593e4416a269499189fc7c.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f.

Independent certificate:

sha256:ed9c988d44a697e889d81581c33591eafc4b943c8b7ba4a5ccff6a9fb9304b01. Engine

external-validation hash:

sha256:1f5ba9cfadf5d82ec537e31ce590557050804fae5454fe6ab3c672352a838e14.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1538c42ddea5976e24c0826c734aa61011c750a8b02babef778576e80d51f1f0; engine receipt

sha256:7178eac96fce3c977c64c4e4a9d5e681f1e48a91d8561045c6020979a476460b at receipts/engine/model_admitted/SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003-7178eac96fce3c97.json;

empirical-validation hash None; measurement receipt None.

72. Three-space Maxwell curvature closure

Claim identity: SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. In forced three-space, three spatial axes each carry binary second-difference curvature while one temporal axis carries the same binary curvature; dimension-normalized balance again fixes speed-square at the One.

Three-space spatial curvature is six, temporal curvature is two, total ratio is three and dimension-normalized speed-square is One.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete eight-axis product for three-space maxwell curvature closure, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__three-spatial-binary-curvatures-balance-one-temporal-per-axis__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	three-spatial-binary-curvatures-balance-one-temporal-per-axis	imported-three-space-wave-equation	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f4339ba7d2812e070e981e1a9a6cb9f48a34c34e88cb052d3b16d45fffd9dccc3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ca56f3c99452b5d544cec2cbaef3c385a2cb3ddb8743e8ed6a9689e41f3fac56`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:6886ae9dd1982adaa62e0dae2aaa30a09aa9c47879ae4a814b08d54a8b9a8675`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3bea7400d97d0d0cbc4be33441ab8f200262d450398d7dc457967dd9a94a7a1c`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:d7e4f910ed511c3cf4e320979381cab03750daac9d401bb892c4138b28b34b49`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate:

sha256:58bc2bc515de600a5e83bacd5bef7451a96cb7dd6d3cc19c1c301e29849cc979. Engine
external-validation hash:
sha256:0a4a7e9bcd13ebfe2ad93024e7952f111770f7357c74515e8162ae1f2114caf.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7d91ff057b317ebf0105a8b7789182eb010f224ec331528c6a0f321ab99a1010; engine receipt
sha256:e9314b2725d52aa3136325aee8ba08bca5fbc41e0ee3ee19db391ff98b7c1dac at receipts/engine/model_admitted/SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003-e9314b2725d52aa3.json;
empirical-validation hash None; measurement receipt None.

73. Exact wave, polarization, interference and nonlinear operations

Claim identity: SFT-PHYS-WAVE-EXACT-OPERATIONS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The joint electromagnetic Fold carrier propagates one One per tick, retains two transverse held labels, merges two half-One predecessors into bright One or an empty dark record by orientation, and generates sum, positive-difference and harmonic nonlinear frequencies by exact Fold operations.

Eight One-speed ticks retain phase exactly; polarization has two held transverse labels; equal half carriers give bright One or an empty dark record; 1/3 and 1/4 mix to 7/12, 1/12 and second harmonic 1/2, while Kerr witnesses also return 1/2.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-WAVE-RESONANCE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for exact wave, polarization, interference and nonlinear operations, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and

extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__One-speed-walk-held-polarization-predecessor-merge-and-exact-mixing__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	One-speed-walk-held-polarization-predecessor-merge-and-exact-mixing	imported-amplitude-or-nonlinear-susceptibility	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f1a9fd69f364f55a9d34563887149d0d9e08a914231314207618eadba821b4c5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:da1305278f885cee1927f9a78e3dabf323d17480590bac0f8db653ded1bca5b6`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e85758931f7806bba8989aa4528d1e48278116ffde19d580946b04dd92285d43`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:68a2392e4b4cf80ccaa7c1dcc5c1f691115996e1fc6038100bce876165a5fce0`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:c50fcb605905e42479774ebd71bb662f6ceb2c3915aa07bcb9a91fef5570fa0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:de299aa4bb3048d1dbffc9b0d2ed4302438f6c808ef0f9272afb19f3dd02b2f2`. Engine external-validation hash:

sha256:0f40930d8dd2d4b8acbbdf2f71dbe44827bf4626a6c805cdc4fec8697bbad61b.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:a95c36f5d6be268d83a9555126fbb62c7e647452ee14ba7c72ca938cd9f009f6; engine receipt
sha256:bc9f7fff3079a79ca77a60d0680da1b8474ad13f0effedbf8b5cce9a75513e03 at receipts/engine/model_admitted/SFT-PHYS-WAVE-EXACT-OPERATIONS-003-bc9f7fff3079a79c.json;
empirical-validation hash None; measurement receipt None.

74. Floored finite-loop closure

Claim identity: SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Every generated physical loop support has positive finite Fold depth; summing its complete exact rational terms therefore yields a finite rational receipt, while an ungenerated continuum or unlimited-depth completion is outside the admitted claim.

The depth-n binary loop sum is the exact positive finite sum $1/2 + 1/4 + \dots + 1/2^n$; depths one through six give $1/2$, $3/4$, $7/8$, $15/16$, $31/32$, $63/64$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-PHYS-VACUUM-ODD-RECURRENCE-003
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product for floored finite-loop closure, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-chain__complete-finite-depth-exact-rational-loop-sum__complete-regist

ered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	complete-finite-depth-exact-rational-loop-sum	continuum-divergence-and-fitted-counterterm	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:9cdbf8e9691d002cee7c951d56edef310b246ca461dce7caab05c8113ba368f4. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:643fde1c52ea524dff094a8fb8119cf4e8b1d12be431699739bf7d2213355340.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:160cb7c4a05a174ca6869795f4f8326f525fc3a526deff446b6d42db9af5aecf.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:f40becab2c8048726c80864bb8da877019405784e6fbf5dd8fedef512446ffd4.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:1f71dea4ae509caa698e6abeb41c08f7ce9de7649a8cd002dad8cf54ba06390.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f. Independent certificate: sha256:971a26b748962889407d161cfb2a71d5ad12118f7b6c44aff5b8ff971a766439. Engine external-validation hash: sha256:882dedf3e1de603d4c8c959477d562d7eee7523f66a3aef199521e8662543ed5.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Three-space, rank-two boundaries, inverse-square dilution, force sectors, fields and waves are generated from Fold support and source-bound response. Held labels carry orientation without signed proof magnitude, and locality requires a complete adjacent propagation trace. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed
- no claim about an actually completed infinite support or a measured radiative coefficient

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8e0e61bb5a069e723c60edea5efa42d7f06c7368db6eeb3b7c5f2c103adef023; engine receipt
sha256:654937a341dfef2a5efc3ae1094a90b940e3535923bc1aa465d64e8930549853 at receipts/engine/model_admitted/SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003-654937a341dfef2a.json;
empirical-validation hash None; measurement receipt None.

15. Thermodynamics Vacuum

Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced.

75. Microstate-to-macrostate observation

Claim identity: SFT-PHYS-THERMO-MICRO-MACRO-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A thermodynamic macrostate is forced as one exact observation class of a completely generated finite microstate support; its unresolved multiplicity is retained as the class fibre.

A macrostate is an observation fibre over complete finite microstate support, with exact multiplicity and retained provenance.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-microstate-observation-partition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	finite-microstate-observation-partition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a0bf480b1fb3ed3410aa2c6f693be952019578ecb0c8d769576671e830a5aa74`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c65afd05be1ff96daaf309433e10c3c85d371e4ea512f710f549859f68a4e929`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:214504c9124b489399cc4a308931c0f822a1bbd5a24e492d83ffab44306bccde`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a78d1727daba665726753bb0b20c97d621b23f07d07deabb555620c2535cd011`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:773249981ef18b98d2128f5e7ce4e000e212289760660ea99eeddab18dc06bf1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ff94c98d730d930a00e95a403b5d023e29d8cf70e5e516a7e5f2d03ebe47b6a5`. Independent certificate: `sha256:c245399cfd8062c8a754212fb79e5e6f01f976b0bc62de9ff139118d008eeb03`. Engine

external-validation hash:

sha256: 94302a7d7d0ef8e2194a635506e0940f8e66c3b2121d881c7b3f7e9ea00a2c98.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- micro-macro-1: predicted finite-micro-macro-observation; observed finite-micro-macro-observation; exact match `True`
- micro-macro-2: predicted finite-micro-macro-observation; observed finite-micro-macro-observation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256: 4a564a747c5f3959532436ca15634960455791ad2dfc83a600d2cbebf3a6926.

Isolation certificate: sha256: 1822e678b5f3bd0fd61c6fb633afe806abf85b8e9ae2b40b85d43b4991837226.

Custody certificate: sha256: 5910b505b5bbe5b3906cf20324252a234879ac3133fcdab7971b0081b0b795c5.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html> (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- micro-macro-1 from IAPWS-RELEASES-2026-07-23
- micro-macro-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: caffff80fffa91147655094797dc206b2b2a726fff4aafc70cd55cb66789c48bc; engine receipt

sha256: 11b63417e8a9d91dc1ceb4ecc356fe1083c36cf84cccb0522b42faae15a7e067 at

receipts/engine/model_admitted/SFT-PHYS-THERMO-MICRO-MACRO-001-11b63417e8a9d91d.json;

empirical-validation hash

sha256: adce150365d521a8ea0d96844c6c91803c18a1e6be04b8d7781dc87522e88ce5; measurement receipt

sha256: 4a564a747c5f3959532436ca15634960455791ad2dfc83a600d2cbebf3a6926.

76. Temperature carrier and thermal ordering

Claim identity: SFT-PHYS-THERMO-TEMPERATURE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Temperature is forced as the exact ordering carrier comparing energy-support change with accessible-state multiplicity change relative to one held thermal reference.

Temperature is an exact reference-relative order of energy support per accessible-state change, not an imported continuum coordinate.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__energy-support-to-accessible-state-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	energy-support-to-accessible-state-order	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3740214d264d8442af6b2e386567eaa01e64fad19b15ae07c7984aac0cac2c33`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every

generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 35041c1fc11ce723d4560fe772069cea9216e1ef156e73cddb47a18788d13330.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 2956553dd7226e7cceeceec10db3a4024e4baf3ca57ee01ae13d7df370aadb3a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : d51c89c2466ecb996b33557ce6855bab5fc11ff8b4192e2bdc31e2aa64e9cb78.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 862d8898fc88b7eaa0379763249ab5281a3c34b9a5884ded00a786707a0ba27.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : 47726e82008630482d9781ed4e574ab970284d5e510faaa8965ef1a8d2e85d25.

Independent certificate:

sha256 : ec2560a3be4715f87c1b8831a4f9103a04ee602088163e87d10b20fbb79208f2. Engine

external-validation hash:

sha256 : 26edbb1e3a330662b802e7bae193acf93c95500274f46ce17792801c9ca6be8f.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- temperature-1: predicted exact-thermal-order-carrier; observed exact-thermal-order-carrier; exact match True
- temperature-2: predicted exact-thermal-order-carrier; observed exact-thermal-order-carrier; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256 : 4d1eb98a218d275eaf351ed5538c5981c465ed13a8b3305a46873ef7f85ae315.

Isolation certificate: sha256 : e7f04de4252891257ea2232f95b54ee8a44af79f7d9bca63acf13abf530e1209.

Custody certificate: sha256 : 4de3c72f269ec8861d9837098926011741d57af35c2ed25db40ef5ab0fde6f87.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html>
(sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- temperature-1 from IAPWS-RELEASES-2026-07-23
- temperature-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:fa5a2f29b5b494f06a5addbb0c869a18bff1bf098bdaa2d286a07abe0f05ea23; engine receipt

sha256:d68a2030d9e3e914068a84fb738d109c84f7d1b7be48c88c97d37d26646c09f8 at

receipts/engine/model_admitted/SFT-PHYS-THERMO-TEMPERATURE-001-d68a2030d9e3e914.json;

empirical-validation hash

sha256:ab9efc043970126329e33dc9b07174c9dcdf0277d8e1a347a6ed4a8d0ccc0558; measurement receipt

sha256:4d1eb98a218d275eaf351ed5538c5981c465ed13a8b3305a46873ef7f85ae315.

77. Thermal equilibrium equivalence

Claim identity: SFT-PHYS-THERMO-EQUILIBRIUM-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Two coupled supports are in thermal equilibrium exactly when reciprocal energy exchanges preserve their separate macro-observation classes and common temperature ordering.

Thermal equilibrium is exact equivalence under reciprocal energy exchange at a shared thermal order.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-TEMPERATURE-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reciprocal-exchange-observation-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	reciprocal-exchange-observation-equivalence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:00b0fda7c9c51d799d39cdef0c10cddf1ac3e40d4ef95460cd11551403d23ebd`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1b728c43b61ae39d3405b5c186402610c4eff158d5505ab6fa86eea0692af359`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a51ff7be669db5438ee2d5a09c824f80ad3d9c09bb648c58a7d1fcdefb60223b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:384ad212a4e04dec0e1582702961d3a4b55ddd08593e737c517d8d000edecfbc`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1f8e0ef7d25eff60e11fa3c1a140c4b7ce27cf57558949e98636551681dbdc36`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6a18063c2eca0e9104385448dd15e7c69be3a28b1342459716761e4c4c6808ba`. Independent certificate: `sha256:a1cedfbfbc5f1309184329786c7f8beb97d30fad87fd7c48f7ea1fd7c16180957`. Engine external-validation hash: `sha256:cfb733ced524b6e8b182fcd2d927d1a027f4bc49c4be83605928acf25b292d35`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- equilibrium-1: predicted thermal-equilibrium-equivalence; observed thermal-equilibrium-equivalence; exact match `True`
- equilibrium-2: predicted thermal-equilibrium-equivalence; observed thermal-equilibrium-equivalence; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:25e7ad1d4802cb8ec058e96d9d018212ba832434e8ac2bd424f290cc6960091e.

Isolation certificate: sha256:ae5e7367d98474d0d83c43bfecfd9b3a673c7219bed4fd75484574c93d662e5c.

Custody certificate: sha256:cb1088f4db234ce76c49b76058983fca44445f45939eb61876552df0de22e7c6.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html> (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- equilibrium-1 from IAPWS-RELEASES-2026-07-23
- equilibrium-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:767102333f0f294a68e74238a8acb09acffa668a81f929393389a5462cb2d6be; engine receipt

sha256:6ce39350919988486d96178e51a0bd45264f0c5498519109bdb0e184c29d40e5 at

receipts/engine/model_admitted/SFT-PHYS-THERMO-EQUILIBRIUM-001-6ce3935091998848.json;
empirical-validation hash

sha256:355daffe24741d90270ac9e7b7c41ab85ed0ff2a64c4da56942ea8f9ab0236f6b; measurement receipt

sha256:25e7ad1d4802cb8ec058e96d9d018212ba832434e8ac2bd424f290cc6960091e.

78. Heat and work as distinct energy-transfer traces

Claim identity: SFT-PHYS-THERMO-HEAT-WORK-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Heat is energy transfer whose microscopic carrier label is closed by the receiving macro-observation; work is energy transfer whose organized source-to-response label remains observable.

Heat and work are the closed-label and held-label observation classes of exact energy transfer.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-label-versus-held-label-energy-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	closed-label-versus-held-label-energy-transfer	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:74033a0c324c51cda11dfdce9d2993c05c1a26b239ee6568ff0d7e209098005a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:522c9c0c66247feb3c51b85bb4689e821a7a799531ab9b68ddadb5de1e8f80d6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:501eb3874ae9c5d8b5ed8a38b0de1fada2ca63887a555bc7e0d0e7f68483133c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b43f038351da4176e1465face97b3eac4b4ed9221038d24fef61fb575fe29283`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1d8e132c7d01dd2d917db2c7bbe6b5d32e5957377f81031e2ea0818c34c348f9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
 Implementation hash: sha256:692212d6e9d55652d02fb5791a82569e67f831fa33e471c95251c9fea6ac7afb.
 Independent certificate:
 sha256:d43df8e2076ebc60dc7837d017f9561a29607f73047cab35731c78d92d1ad243. Engine
 external-validation hash:
 sha256:133e197ab18339a13dd805c1401a6ac561901e9c7553c4c7c4df3b1e873dc542.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- heat-work-1: predicted heat-work-transfer-distinction; observed heat-work-transfer-distinction; exact match True
- heat-work-2: predicted heat-work-transfer-distinction; observed heat-work-transfer-distinction; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:52a3381716ae93ecfd8e95729dd33c5d25f7a54d5d21cf059eee11bd2a04cb25.
 Isolation certificate: sha256:ce57ac9fbbe19e765817a3b77dc06793a33bf4a2087598515d47cc9bf6867049.
 Custody certificate: sha256:9326e5f0715eefcea140b21e12fc722576532ec0c53a217088f898ec8bfa2e3d.

Registered source descriptions:

- National Institute of Standards and Technology: [https://webbook.nist.gov/chemistry/\(sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e\)](https://webbook.nist.gov/chemistry/(sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e))

Registered target identities:

- heat-work-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- heat-work-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:433edef20fcdce50fae8e8dc2430c354fdca8dbe3c424b7ab6a24a2ac3f83550; engine receipt
 sha256:3db98f953910f921028c19a7464ad19eb2aa44b2888ec82ab1be56d23576cbef at
 receipts/engine/model_admitted/SFT-PHYS-THERMO-HEAT-WORK-001-3db98f953910f921.json;
 empirical-validation hash
 sha256:7d6de73b7f85e56a2ebbb17473aed9ae2420e614e45b7b9ee48f234ec27e98140; measurement receipt
 sha256:52a3381716ae93ecfd8e95729dd33c5d25f7a54d5d21cf059eee11bd2a04cb25.

79. Closed energy accounting

Claim identity: SFT-PHYS-THERMO-FIRST-LAW-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For a closed support, every change of retained internal energy is forced to pair exactly with boundary-crossing heat and work transfer records.

Internal-energy change is the exact closed composition of all heat and work transfers across the named boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-HEAT-WORK-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__internal-change-boundary-transfer-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	internal-change-boundary-transfer-closure	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

sha256:0b838b25b123c6367b0fca9efe12dd04beac2bd5294d1bd202f587d1264ca8df. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8d3939a5f370964ef42f4dc5c4475f8ec8afcc953289ae724ab7582161ac8dd8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7f43956afb7cc2b419b15ca6cd4271ae8279b8b1b5f55e8ae05a812653290a5c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ff237dd6c3067a027223bba28bc7fc4200700ad5c54bd10a78a92f9e28fc6a28.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:29c891b7564cbca75ba46f533f7644fc9ece22041d405b8c065259d0ac66c14d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:3f7e45454f55f575cf62b0752260ca31edee7d04bd9f7040f96a1d501cc6c0f4.

Independent certificate:

sha256:4f6cdde720d390368b38f62eaa82b47bcc556c7fa987a4fc2ede5941dff9303b. Engine

external-validation hash:

sha256:3f63f7200bab7ab8d23baed640f89c3bdd69366cb901e77416b24c8729b7d130.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- first-law-1: predicted first-law-energy-closure; observed first-law-energy-closure; exact match **True**
- first-law-2: predicted first-law-energy-closure; observed first-law-energy-closure; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:5b600737da737e7769e0b3e0cc4528b75a6cc29c33012af23337de9054fa656a.

Isolation certificate: sha256:4058fbb00286c41adb2382acde521c35dd5555cdd0ea862e2f050985072eb7c4.

Custody certificate: sha256:fa7b84961f5628d8b997a2cb3e53055a7b9b52bd080fa841a40466a6e78027de.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/chemistry/>
(sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- first-law-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- first-law-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness

and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c3e7e60ad32808dc97aaa5544458b40e50de9ca3f17275ec2f17cb2753812fd8; engine receipt

sha256:18e493c59cbd7979db1bcc2b33a97f582329f92707c16bc6455c2ee55ef2b13a at

receipts/engine/model_admitted/SFT-PHYS-THERMO-FIRST-LAW-001-18e493c59cbd7979.json;

empirical-validation hash

sha256:fa52340b68dc82d2ea089378f31bc3aadad7fa494d08bb9794add9ad60c54406; measurement receipt

sha256:5b600737da737e7769e0b3e0cc4528b75a6cc29c33012af23337de9054fa656a.

80. Thermodynamic entropy from unresolved support

Claim identity: SFT-PHYS-THERMO-ENTROPY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Thermodynamic entropy is forced as the exact information required to distinguish the generated microscopic predecessors merged by a macro-observation, conditioned on every retained macro-label.

Thermodynamic entropy is exact conditional unresolved support of a macro-observation over its finite generated microstates.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-FIRST-LAW-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__conditional-unresolved-microstate-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conditional-unresolved-microstate-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d45dec31bfd567235b15d1f0ff5875757474f9ffcdd0cdb26e008daf9cd4c6b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c5269b08d364fbba40bf12531c012b795a1463eba0af221214425a401b2c6b66`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ae449ca3e6d245ae018a6d766684ebbf4a52476c94632b8a3bcfc850e0e1bb54`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a0c251a8959265f3da49548a9f5f190dd0e697a490eb0a57bdbd6dd6be29d33f`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dd028d42a95245598f80ac0fecf0cc08ac43cfe797abcf2f01a5171440f14a24`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0e840231ff85005cf206bde90bc06180ed9a1f798ea988fe4d7e2c5b15f08e98`. Independent certificate: `sha256:8ed2c912ba7e6fac3828cdbc755c8870dbb4f24088884e4f4a4a10f6fa1df3a`. Engine external-validation hash: `sha256:7dc109d00a190a5da182ff19c3fd045a03af04958015ea8c534ec9b50ba13f3c`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- **entropy-1:** predicted finite-fold-thermodynamic-entropy; observed finite-fold-thermodynamic-entropy; exact match `True`

- entropy-2: predicted finite-fold-thermodynamic-entropy; observed finite-fold-thermodynamic-entropy; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:64b76e6801c73b3c9946a0f76f966c56733d9a2e4a2d3f3abb5ee208f98dc6f7.

Isolation certificate: sha256:6de0571242ea40dc7676ce7b5fa60f33665d6c3e4105ec0f4c9894d8ce8bca8f.

Custody certificate: sha256:91264e0a4fba63d40403ce5332242336b735783f3fbe774d3473a88cae9d3e1c.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/chemistry/>
(sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- entropy-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- entropy-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c746cceb4a0536fab9745ff843ca2d6ee21eda56ae9707e3eed6fe98534c0d50; engine receipt

sha256:18dc9c224330e6542798ebde0073e90d982cd04d30fe0e756869f8148983d4f8 at

receipts/engine/model_admitted/SFT-PHYS-THERMO-ENTROPY-001-18dc9c224330e654.json;

empirical-validation hash

sha256:76bb357cdc75b56d5a332823af4523174d6cd06f6ffe87228b6ed832a083ba3a; measurement receipt

sha256:64b76e6801c73b3c9946a0f76f966c56733d9a2e4a2d3f3abb5ee208f98dc6f7.

81. Irreversible observation and entropy orientation

Claim identity: SFT-PHYS-THERMO-SECOND-LAW-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A closed macroscopic process is thermally oriented toward observation classes with no smaller unresolved predecessor support unless the distinctions required for exact reversal are retained externally.

Thermal irreversibility is the orientation induced by closed microscopic distinctions; reversal is lawful only with complete retained records.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-ENTROPY-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__observation-merging-support-orientation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	observation-merging-support-orientation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4d16b7eafead207076240e8118f52debfa4bc8cbeab2ed4c17e3596aa238013f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bd5d44ca5ac75aae6520413625174d04324f813e3ed129902e5e22ffab16e867`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:89b5457505016a0591d72cd0d779469d88001625cf1c604645660e721a61faca`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:705cb309bd4a45d0bd0d43283d8cc3a6b0fd0ca2ae6d049478e3bc0c95927331`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 8a76f82a48d2516bb5d150b93a10eed9104453f1148a75c2790edb45b8119b54.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : ceee18ae030e0b6bda80abd30799585a6fce564b94953c5a27fa18b275e7f2c3.

Independent certificate:

sha256 : 4176b2452cdb078a2fd8544bd43813b1baa59d68052243d8dcd0c60c3badbc8d. Engine

external-validation hash:

sha256 : 033899701ba3b86b2aaf16432b013dfed77cafbfd54d22439c558645423cb15eb.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- second-law-1: predicted second-law-observation-orientation; observed second-law-observation-orientation; exact match `True`
- second-law-2: predicted second-law-observation-orientation; observed second-law-observation-orientation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256 : ceab052d9b5040aef06ba2bef0167154438fa87511c4573a57975a3d29a8d591.

Isolation certificate: sha256 : d95004150364294d059993286d2408d576c4f43b28cd79c7bcb4ebf2fa187602.

Custody certificate: sha256 : 016ddcec986ead9701ba8676bfd5606876df1a981e6d4619d3f662f1a1e51deb.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html>
(sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- second-law-1 from IAPWS-RELEASES-2026-07-23
- second-law-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 02fcf9468b069998405e2ffff909dc5fef14a8c287ad295f5a7c9d454dfc46410; engine receipt

sha256 : b1ec365f5fb8432d4ca9f69ecb6937596ff9e1a144860e58ade0d6a5f6bec17a at

receipts/engine/model_admitted/SFT-PHYS-THERMO-SECOND-LAW-001-b1ec365f5fb8432d.json;

empirical-validation hash
sha256:3c4186c69899746c010d7918efdf89c44bdf75c4cfadeb977486c08f111d6b05; measurement receipt
sha256:ceab052d9b5040aef06ba2bef0167154438fa87511c4573a57975a3d29a8d591.

82. Finite unattainability boundary

Claim identity: SFT-PHYS-THERMO-THIRD-LAW-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. No positive finite transition sequence can erase the final generated recurrence distinction of a material support; complete recurrence closure is therefore a limiting boundary rather than a reachable destructive state.

The minimum thermal recurrence boundary cannot be reached by erasing all distinction through a finite lawful process.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-SECOND-LAW-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__positive-transition-unattainability__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	positive-transition-unattainability	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:19f775c6ec56ce1a2e15c29316400e65b1259c388873f0a478027c79b66253c6. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:14b4815387ff86bbd60b37b5ce2e8305f1297484efb231a9ab76b087ee47487b.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:14a67126b7e513be27af787df4da5ee968000098cfcbe746cce45b8ef944f2ec.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:ac5e1bef5db2e4e07e3e2a331595bb74aa93a46b431b846471762b2ed6bd1fa.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:4b5790435184a81b32eaab8dc32a905f9c80c104ab0201d738c3cab6266f62ad.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:a4ba57ed42270c8a7ae69418ff067aab619e429c483b71d05d0af94c1c6e73cb. Independent certificate: sha256:c7c50729009b9c1eaf672794a191934ca83f87bfd1bffb88d1d982f12e5f3016. Engine external-validation hash: sha256:08bfa798c6ae950f3e0ec03b335948a9a18e1bfbf69103a80c7c546b50668437.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- third-law-1: predicted finite-third-law-boundary; observed finite-third-law-boundary; exact match True
- third-law-2: predicted finite-third-law-boundary; observed finite-third-law-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:855ece6421af57b129259ade0b6f65c572b248e809fcb3fcb0fddab1bdc8042d. Isolation certificate: sha256:f7d709472482e63022e731f1232abbd3653bde50d8d3769f3100078149a4d3ff. Custody certificate: sha256:b97280c0c1752f272b35f17c1ec6f9ba57ce7ef4383f5500e3cfbe4f1b31de66.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html> (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- third-law-1 from IAPWS-RELEASES-2026-07-23

- third-law-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:41b7d4b59030442a33336e31cb3ccfc9db10f430919f43cf47bf513ab0612b18; engine receipt

sha256:a66152169839178e093243202041879920e7d82b50d08fae42cf6c14f464e15c at

receipts/engine/model_admitted/SFT-PHYS-THERMO-THIRD-LAW-001-a66152169839178e.json;

empirical-validation hash

sha256:411250ec2b850cb5347f9bfb0852a26191fbb8d018f34cf93a43e9bc9645d153; measurement receipt

sha256:855ece6421af57b129259ade0b6f65c572b248e809fcb3fcb0fddab1bdc8042d.

83. Exact finite statistical weight

Claim identity: SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The statistical weight of a macro-observation is the exact positive count of its generated microstate fibre relative to the complete accessible support count.

Statistical weight is an exact positive rational census ratio over complete generated support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-THIRD-LAW-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__exact-fibre-count-over-support-count__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-fibre-count-over-support-count	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d4ac1efa48dd0e7bada3923e37cd64c6633edfa64a89544daf2a0adad56efebc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:20300f111d9dca3814ee908c92f5c0f65a94e71b60770634d02dda75ad148e34`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ae3eb0351361990cc53e46cf3b098f2d373d6267b064dd204dd937ac88f323e5`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ddaf3b086f0fbba7a731397924b840c450b7228f3903a413bd2d8a5c251bbcd5`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b69676863ee2ff8e9d958e57780df3bff7ef82ba16637994944469df9d05ec0a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:81a65c0c1d85723d19ca7c0b2e9103bc8be2e1a3834fd17f20a283a3deb4c33a`. Independent certificate: `sha256:a56d8972ed6beb1ff28103ab146793a9dd7dfa9bf2dde2f83b4cc487035176c6`. Engine external-validation hash: `sha256:4555be56f36f5930731595107d40d12a984c37b19a242640f5797e98052bf823`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- statistical-weight-1: predicted exact-finite-statistical-weight; observed exact-finite-statistical-weight; exact match True
- statistical-weight-2: predicted exact-finite-statistical-weight; observed exact-finite-statistical-weight; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:1d77769023931c2dbdd4012815b6bc4d9b3f7763b52c1c0ef9ede5da8aaec85a.

Isolation certificate: sha256:833d7b71bb32d1f464d62c543bc9e0b2cd2baf36a33f206913169e194544c6f9.

Custody certificate: sha256:5f497c555a4ce6696dfc7da414e29eca11ab8a8ca8ac463c5e8897bcfebf6c8b.

Registered source descriptions:

- National Institute of Standards and Technology: [https://webbook.nist.gov/chemistry/\(sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e\)](https://webbook.nist.gov/chemistry/(sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e))

Registered target identities:

- statistical-weight-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- statistical-weight-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:e5c8a8c575ea73a768b25a408c73950578a9dded6d59d781b870fb94d2ae5691; engine receipt

sha256:fc2fdc5352d172755221a6fb15a61dc121038e01f7390aae19cdbc4c9334c9d0 at receipts/engine/model_admitted/SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001-fc2fdc5352d17275.json;

empirical-validation hash

sha256:b473c92dbc371b98ccfa9e4b7f5577449e1de07f76492138709dda2596d14a4e; measurement receipt

sha256:1d77769023931c2dbdd4012815b6bc4d9b3f7763b52c1c0ef9ede5da8aaec85a.

84. Thermodynamic state relation

Claim identity: SFT-PHYS-THERMO-STATE-RELATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A thermodynamic state relation is the minimal closed dependency among independently observed support content, boundary response, thermal order and energy carriers after redundant coordinates are removed.

A thermodynamic state law is the unique minimal exact relation closing all registered state carriers on its declared finite domain.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__minimal-closed-thermodynamic-coordinate-dependency__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	minimal-closed-thermodynamic-coordinate-dependency	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d656e11b809197c78f6cd210f98b11a2baeef1bcb0e8d3742d89d2678b5adc40`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2d4ff98f4e05c186b576d6a5c15c2c6d56d46197243299fbaaf7a2b79a4cb926`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:aac21fc9d70da35afbc04c2f4f02367f34bbc59cf10440b79644448a989ccb05`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 7eada2f344e23049e34b2caf66a6170bfff87d85a1303e1f3b8f5cb05bd36cdf.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 6e2ab2f551a6a20796a7d3d6e7a0cb38cf926f3c1b05915952674190323de8b5.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 46e58c0847d9c306c641b73ccbfad62f69e06ebb0b62a15605a20decfbcf32d.
Independent certificate:
sha256 : 32e7ab8e33d2cdaaa90269d5a5e530a00c55c2cbb74767f7e5f31813ff3e1e07. Engine
external-validation hash:
sha256 : 7fa13052c0e6557792214b988f1077ff30292dd6c436f65c7fa857aa81ba07b7.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- state-relation-1: predicted minimal-thermodynamic-state-closure; observed minimal-thermodynamic-state-closure; exact match
True
- state-relation-2: predicted minimal-thermodynamic-state-closure; observed minimal-thermodynamic-state-closure; exact match
True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256 : ab0aa384e9b13ed17f21031e5eba406bc1c9b93bfd5c5f5de5fd1c6a868c454e.
Isolation certificate: sha256 : 65dc12ea74c2f7cc9e26067c0fecb61984579c9f0afae668f2a0b4f9ee7c2589.
Custody certificate: sha256 : f10be60f8dba1c824a1b30e71edd30cd47b66a9bcbd77abdaca784c51cccb958.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html>
(sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- state-relation-1 from IAPWS-RELEASES-2026-07-23
- state-relation-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:83596a70b3b622fc58c4f2694fb31f764aea5895333f19dddec9148a9d267987; engine receipt
sha256:e74262c55559b31cb22638ee24e9203e34a582937ebb8053d54e55268cb707a8 at receipts/engine/
model_admitted/SFT-PHYS-THERMO-STATE-RELATION-001-e74262c55559b31c.json;
empirical-validation hash
sha256:2a5bd1c195abe5b66e19ebb70d273a115ed3ee1aaf309aa3dbd37240053b73a2; measurement receipt
sha256:ab0aa384e9b13ed17f21031e5eba406bc1c9b93bfd5c5f5de5fd1c6a868c454e.
```

85. Phase coexistence and transition

Claim identity: SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Distinct macroscopic phase classes coexist when their exchange boundary preserves a common thermal order and reciprocal transfer balance; a phase transition changes the selected observation fibre while retaining total provenance.

Phase coexistence is balanced exchange between distinct structural observation fibres; transition is provenance-preserving fibre change.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-STATE-RELATION-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__coexisting-observation-fibres-under-balanced-exchange__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	coexisting-observation-fibres-under-balanced-exchange	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2ea8645395888cb98611995be975d320e16cfbe1355bdc5cdd20b9f8e342c64f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f42cfff2d91851d91717f421aeae259d02959ed2a9a736586d1d333e968d9723`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e8b2e3d4a7c16422e8b1867ca2d6dcbc36f3f90ecbf6962b6ccec29e53d2d3ed`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:312eb8573d58e786edfd4a14765ad2fcb8bad4ec82de9488659ff013dfa0e92b`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b7592e2613534291ef6f2b6dd187cc4a2a88ce205c213f5c8265ce95707b5998`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c5ad6f16c8ccb1c0195b7dcd450794abf0a6582d0801d755e1d1cee852687b5b`. Independent certificate: `sha256:46defb37258a5ca944a8d3954d8d25998ccb82efeea199a0c98458f785896355`. Engine external-validation hash: `sha256:5cac7be29a9861ed564fa20e12133e3b1e74e3c52fa323c9bd088c55c20ec720`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- phase-equilibrium-1: predicted phase-coexistence-transition; observed phase-coexistence-transition; exact match `True`
- phase-equilibrium-2: predicted phase-coexistence-transition; observed phase-coexistence-transition; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:8d7e1705c61cb89e775e56f2bd970529f4c51505e8cdda5a4966c3972736e18c.
 Isolation certificate: sha256:e65dc19612a5d5e476aa80c965b602422ee83334fd6f5e73163134f47edff48.
 Custody certificate: sha256:63365f3340988042c81718784307802ae1e5fd1d5c10b7d3a792710962be5582.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html>
 (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- phase-equilibrium-1 from IAPWS-RELEASES-2026-07-23
- phase-equilibrium-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:500813f6ac91be300d730b71689ca9330667887f4ebd6d97ac3b77914245ad2f; engine receipt
 sha256:804bbbce151a9807129d17226d039f46d4b5ad9fb4cd7d90135bbb01fc50da79 at receipts/engine/model_admitted/SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001-804bbbce151a9807.json;
 empirical-validation hash
 sha256:3507c316bca528f72c7e066c7e9b8da9bcf31bccfdd292be2258f80106b712e0; measurement receipt
 sha256:8d7e1705c61cb89e775e56f2bd970529f4c51505e8cdda5a4966c3972736e18c.

86. Kinetic transport and collision mixing

Claim identity: SFT-PHYS-THERMO-KINETIC-TRANSPORT-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Kinetic transport is the exact propagation of held motion and energy labels between generated cells; collision mixing is their closed local permutation subject to conserved transfer records.

Thermal transport is adjacent carrier propagation with locally conserved collision permutations over complete finite support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001

• SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__cellwise-carrier-propagation-and-local-permutation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	cellwise-carrier-propagation-and-local-permutation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:970188e713877d4ca242224e6ee4bdbc89d64d5cdcfd9a024fa03e9b76183a55`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d30fbb38fb12fda20dddfb163e4dfd0434415120f7118b225279220c17463cf7`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cd84dd2bc9be81e40abe6345768b97a1c0f4d9fdcd7a310121d3935a08eb4d9c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b510d9c52a16063ce072ec1ae96bbc6a002abe947f74d692ed979787fecfb4f6`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d94f1208accd846d0e7148bf3577939ac944619e4279e87006453d03f5b2f579`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:83e6a6d6a4a84aa01fc03b09986a11fdce29c319f2018c0f199a7f2fa0cade8c`.

Independent certificate:

sha256:3d10eba440c2b0fc41c95cdf04539a180db711dde6c55e71a42c60bc9cbb9d6a. Engine

external-validation hash:

sha256:8819d0ab3dc2bd3bec32854c33559c8a448d22a05cbaa76d386092e589127985.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- kinetic-transport-1: predicted finite-kinetic-transport; observed finite-kinetic-transport; exact match True
- kinetic-transport-2: predicted finite-kinetic-transport; observed finite-kinetic-transport; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:8e81fb917a5769ce9d1477e1fb113e43add31d4cdd6a267e5f8a7e8b0114e98.

Isolation certificate: sha256:bd4b00e6c3dc34dee8920efb54c94fe0707ca3f051c04097f64f4eaad9dde8bc.

Custody certificate: sha256:f9ebde4d1266eb476bf8be09200374a0ab79672ae4e591700006df918d88e4fc.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html> (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- kinetic-transport-1 from IAPWS-RELEASES-2026-07-23
- kinetic-transport-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:27dc7b5558723466a12162ff23a4c83eb32bb10ce9053af043919dae22f6d8d4; engine receipt

sha256:0001a2d7176cae05e6b1ad4ef9942e474d14916872262d377350c7a465823c18 at receipts/engine/model_admitted/SFT-PHYS-THERMO-KINETIC-TRANSPORT-001-0001a2d7176cae05.json;

empirical-validation hash

sha256:aa0023c1ca2b4b546c7f8680153d6c14d572396674e067622c2a5ee39399302e; measurement receipt

sha256:8e81fb917a5769ce9d1477e1fb113e43add31d4cdd6a267e5f8a7e8b0114e98.

87. Finite fluctuation and recurrence

Claim identity: SFT-PHYS-THERMO-FLUCTUATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A thermal fluctuation is a finite departure of an observed fibre count or transfer count from its recurrence class; complete deterministic support fixes every departure and return path.

Thermal fluctuations are exact finite count changes across deterministic Fold paths, weighted only by complete support census.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-count-departure-and-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	finite-count-departure-and-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1018628409f455d8d475562031fbc08345c74b74652c2d2532dfa4f754b2164f`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:00138e10f0a178a63731e18278eccfda8318dbc158bd82da5420e550a30a5c76.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b0e40ef20b9a4436701c2356257b61ba295912e51754d039135fdf2fad587680.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c1f74423658eddae370554c1aeb672cd2b0738d8683ca9eba6e8d6e78ef88e87.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:087c5bd21117695a5b7de997a677d07134aaa675ae69aed071669d6feba944eb.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:03fa0b411cd53d9dae61566fb1f06b5e587d565c9384c0f8799eb196854cb1fe.

Independent certificate:

sha256:9022eaebaff4f04b2b66185a6fa3642d43f39cb420d9e29868b084491ee72d67. Engine

external-validation hash:

sha256:21cd8ec2459dfd3b1b895655ea8ef38fef71ae97b210d79505e9e3e3bb2bec34.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CHEMISTRY-WEBBOOK-2026-07-23

Observed comparison records:

- fluctuation-1: predicted deterministic-finite-thermal-fluctuation; observed deterministic-finite-thermal-fluctuation; exact match
`True`
- fluctuation-2: predicted deterministic-finite-thermal-fluctuation; observed deterministic-finite-thermal-fluctuation; exact match
`True`
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:c24fce3c578e4d987bd8cd1890a1888daa511269141b9f1dfac7e3d153619c23.

Isolation certificate: sha256:15a12ba1249c8fdb3325c1b9cff562958f45b7c8eb979ca716e0569508249c02.

Custody certificate: sha256:2a54b66f71698bd7cb2234ceee3b35417c2eb375d1a8722952bc82f17346b888.

Registered source descriptions:

- National Institute of Standards and Technology: <https://webbook.nist.gov/chemistry/>
(sha256:21a9b51ff6b5b0f4de59982d177359207d1d00b519dcfd00f6a5c044bf47ca9e)

Registered target identities:

- fluctuation-1 from NIST-CHEMISTRY-WEBBOOK-2026-07-23
- fluctuation-2 from NIST-CHEMISTRY-WEBBOOK-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness

and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:9e7385a5320d843028b9c3cfff918c0cbcbc4b5ef29a98d10c68f48b2183297a; engine receipt

sha256:c39f22eaa3f319ad2c29d582ffc035e542f8df1e7665956e9e3d50506ce26d07 at

receipts/engine/model_admitted/SFT-PHYS-THERMO-FLUCTUATION-001-c39f22eaa3f319ad.json;

empirical-validation hash

sha256:0193016a0f6bdb0b5d0349f7a7e4562b81d57c23bc26cc8a9d05f83923ab49ef; measurement receipt

sha256:c24fce3c578e4d987bd8cd1890a1888daa511269141b9f1dfac7e3d153619c23.

88. Macroscopic irreversibility with retained microtrace

Claim identity: SFT-PHYS-THERMO-IRREVERSIBILITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Macroscopic irreversibility occurs when observation merges distinct microscopic histories; the complete Fold evolution remains reversible exactly when their fibre labels are retained.

Macroscopic irreversibility and microscopic reversibility are corresponding projections distinguished by retained predecessor information.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-FLUCTUATION-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__macro-history-merging-with-microtrace-retention__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	macro-history-merging-with-microtrace-retention	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:06cf6ba1530f9c8dfac16b38c8aec8731e075a8f595e8680295fc0906b8c57ad`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a951e838910cc42db069eb2b5f345c610e61552ae58bee73b9c3555a03f303b3`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e4a5fa3374d35a5a5b6c7b05e6ec9a06ad4ce856e8f25bbf68fe2c43704dff22`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c58c5081d1760e12d16d32be04067b042d8b4489f64006681cc3ae974d5fc0dd`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4c4cbcf15512184cf5cc9dc5122d793badd4085ca55356feda189d5f580cfa8e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:208cabe0f6efc2a2901bb5510b1559bd30ab859ffdc6a6f1d9e5bada19fac7c0`. Independent certificate: `sha256:2b6ba3c9dbbc8057e7d929f2e3bc6c310019c4451f1f1e0000d3de1e6209457d`. Engine external-validation hash: `sha256:848fef281db085300df71fe607ec80a438182f91d4d6037e05c84183bb7fd321`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- **irreversibility-1:** predicted macro-irreversibility-micro-reversibility; observed macro-irreversibility-micro-reversibility; exact match `True`

- irreversibility-2: predicted macro-irreversibility-micro-reversibility; observed macro-irreversibility-micro-reversibility; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256:ac90cb1396e2257b13162cdd1249089686047e24f83b2226b1357f983a0710ba.

Isolation certificate: sha256:efdba3d89f4b065bdeedeafb72ce9fccf4b3d2f662fb7d108cb6e25012a09ea6.

Custody certificate: sha256:ece97206b53986eb944a1e4d39735d49efaae16fbf27920fb982a958d5a837e5.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html> (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- irreversibility-1 from IAPWS-RELEASES-2026-07-23
- irreversibility-2 from IAPWS-RELEASES-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:47d423e50a3b67c092517b42de9ef930ab20226a2be315c511c8d4e1ce7f4078; engine receipt

sha256:10ae5d449758f74df3bba03643a4b2f3f48f58096812e5052dffe56e1b0ade69 at receipts/engine/model_admitted/SFT-PHYS-THERMO-IRREVERSIBILITY-001-10ae5d449758f74d.json;

empirical-validation hash

sha256:62adf31b0648d8e60afcc5d5009c749c3f5e9b5f409f3c878323efd055f95f03; measurement receipt

sha256:ac90cb1396e2257b13162cdd1249089686047e24f83b2226b1357f983a0710ba.

89. Susceptibility, response and transport correspondence

Claim identity: SFT-PHYS-THERMO-RESPONSE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Thermal response is the exact change of one registered observation carrier per exact change of its source carrier, conditioned on the same held support and recurrence class.

Susceptibility and transport are exact source-conditioned response ratios on a declared finite observation support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

Generated grammar and closure boundary. Generate the complete microstate support, observation partition, energy-transfer provenance, target-access, measurement-record, row, successor and extra-rule product. The declared exact boundary is: All finite thermal forms generated from exact state counts, held distinctions, observation merging, recurrence and closed energy transfer without importing a statistical distribution or continuum equation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-source-response-change-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-source-response-change-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d65cb90346dfc94d9cd5a3654979e1098dd16aaba415bdb46f3024797e32d35c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:eccd392239638a2f850d68b176fe90134f718e514c3d996a6575f82bfd601890`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:415c4fc3248360a54fff8b3877d91378f6901ddb21ccb24a44977cd058c71cd3`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1f5e7ec3a67a74630e6a6955565c03bb4cda052fb88b4976c9c91db0458b06af`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 554006a67b3012c6996aefa6658c90e5eb362ee1467800525bbdac3741313e81.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : aba9c526a54975949c1a2800c2ea4f06406cea4ef6a10c7e60679e922e20e072.

Independent certificate:

sha256 : 554a7ad79538b78eb9bc6ff390b0bc27dc162c20a834a55108f16d138ef8eec9. Engine

external-validation hash:

sha256 : eec549428de2dc92dac63d5312e065c53c037e7ac0f9d8dcccfd83926826b9.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- response-1: predicted exact-thermal-response-correspondence; observed exact-thermal-response-correspondence; exact match **True**
- response-2: predicted exact-thermal-response-correspondence; observed exact-thermal-response-correspondence; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The thermal law is falsified if any committed external row lacks the sealed finite-state, transfer or ordering structure, if a row is omitted, or if the tampered row is accepted.

Measurement receipt: sha256 : e5ca7d358c3abb7496c905cf1f57e75196354b55b78e843ac022e7f01592c1ae.

Isolation certificate: sha256 : b1134e2384835c5a56f4b0c2eed231e084de016745ff87843ef63a3483c2122d.

Custody certificate: sha256 : eb1eb989aafa52313661a345fe973c6a70b2893a9374fca124d5a221fadeec90.

Registered source descriptions:

- National Institute of Standards and Technology: <https://www.nist.gov/srd>
(sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcb34f6ffcb8364d6)

Registered target identities:

- response-1 from NIST-SRD-INDEX-2026-07-23
- response-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or equilibrium ensemble
- floating, irrational or signed proof magnitude
- imported thermodynamic equation, fitted coefficient or target-selected value
- target access, omitted adverse row or unrecorded microscopic predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 6e79304fa667cd6f93ef31a4f8ff35ec930d6cafa75c59f2bbbf91dfe3ab074d; engine receipt

sha256:b5a02f181350f74161c9c1041572b66e4a180034f99401bdad07bb9ae177c262 at
receipts/engine/model_admitted/SFT-PHYS-THERMO-RESPONSE-001-b5a02f181350f741.json;
empirical-validation hash
sha256:742f32e96d6a7c55270447d00d5410b54ddb51f4daf50b60f1522d80e87efd1c; measurement receipt
sha256:e5ca7d358c3abb7496c905cf1f57e75196354b55b78e843ac022e7f01592c1ae.

90. Half-One vacuum floor and oscillator spectrum

Claim identity: SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Complete binary-support enumeration leaves exactly the half-One as the positive part whose self-pair reassembles the One. It is self-antipodal and Folds to the One. At every positive depth k , complete binary support then forces 2^k uniformly separated oscillator parts $(2j-1)/2^{(k+1)}$, beginning one half-spacing above structural absence.

The exact vacuum floor is 1/2. At depth k the complete oscillator spectrum is $((2j-1)/2^{(k+1)})$ for $j=1..2^k$; depth two is $(1/8,3/8,5/8,7/8)$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-FOLD-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of vacuum carrier, dependencies, self-pair relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact positive parts of complete binary support, their self-pair return, and every finite positive-depth half-spacing spectrum generated by complete binary words. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__unique-half-One-self-pair-and-half-spacing-spectrum__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	unique-half-One-self-pair-and-half-spacing-spectrum	empty-ground-or-selected-offset	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.

Axis	Forced form	Eliminated form	Elimination reason
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:47e38dd7eaa368b1e979aebef76e91579daff609fdf61b9ef32a33c2875a5cf1. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:bea5aa10ba513963a5dab594a7cf5fedcc49e0fb4293997f2e449476bc5c068e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:62f320a5284328b3166bebfc4539d8be9e4b0e61f68993ce26527f6faa8c66ef.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e0829af2a4f920e6232904f8e8050ccaa95d0d2462796fe996d19efddd69f013.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:247b00c3e6325d3cd0d0d728ea395597d854ec60c03ae9a22cc542adabbea6d4.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f. Independent certificate: sha256:f18ad879d0ebc3445c8ffc4cdaed9a18d4a0687f72a511b80358bd42958e0c71. Engine external-validation hash: sha256:71b8cdc47818b3759bb6a994559a41edcdcbf0e319c73073cd42f110fe560127.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d81fef2090a7156a7624c032cbb33b0fc51f93ad76a4599b456de0f0166cb089; engine receipt
sha256:288674203497f376e9e6dbd0cb8aba169f1f2343b3b7717dbac789c886759548 at receipts/engine/model_admitted/SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003-288674203497f376.json;
empirical-validation hash None; measurement receipt None.

91. Perpetually live odd-denominator vacuum recurrence

Claim identity: SFT-PHYS-VACUUM-ODD-RECURRENCE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every generated odd denominator beyond the One, pairing a positive residue with the binary Fold cannot acquire the denominator's factor and therefore cannot reach structural absence. The finite positive residue support forces a first return, so each such vacuum mode remains on an exact live cycle.

Every positive odd-denominator unit part beyond the One remains strictly positive and returns exactly; examples $d=3,5,7,9$ have periods 2,4,3,6.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of recurrence carrier, dependencies, odd-denominator relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every positive finite odd denominator, its complete positive residue support and the exact first-return Fold orbit of its unit part. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__odd-denominator-first-return-without-empty-state__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	odd-denominator-first-return-without-empty-state	dead-ground-or-finite-pre-fix-only	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.

Axis	Forced form	Eliminated form	Elimination reason
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:beac2f0dcbbbea64fa9ef4de654304b6d50c8785ea331c4c2ea36e86d48e22ea. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:d48480aab6d50f9557755c7cd1705cb0c395baafa0443759b305697e7b4ce0fb.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:db203330f3989affb79b99144ea157d9253410c257c1baf9074eda9a4f70ed07.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:293da57c4fdb0fb1d60b3a401553b7b2f9e7f607dc420f506897370c1d7bdb5e.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:2258042d5ee3201b476efeea0a71b013701220c514831e59652022498af9c254.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f. Independent certificate: sha256:c974d7ea59b08cd92bd148e7aa9873df3a813262c50bb773f0f489257c258bf0. Engine external-validation hash: sha256:4d8faab743747422b7d527e47b7024c0d49f9213e200d6acfb419b87ed95d1cc.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:23af938da6534560c91d70cb33221166d64ef31b77fdc08f249b3891f426a21b; engine receipt
sha256:dad4e023733464a86c1fcad9913bbe661064c10e67d270e47cc1430d3846a821 at receipts/engine/model_admitted/SFT-PHYS-VACUUM-ODD-RECURRENCE-003-dad4e023733464a8.json;
empirical-validation hash None; measurement receipt None.

92. Vacuum polarization and running direction

Claim identity: SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The live vacuum retains the half-One screened support at the far carrier. One exact Fold exposure returns that support to the bare One, forcing electromagnetic support to increase as screening layers are removed. This fixes the running direction and endpoints without importing a renormalization model.

Vacuum polarization forces the structural running direction screened 1/2 -> exposed One; closer probing removes screening and increases effective electromagnetic support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-PHYS-VACUUM-ODD-RECURRENCE-003
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001

Generated grammar and closure boundary. Generate the complete eight-axis product of polarization carrier, dependencies, screened-to-exposed relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: The complete exact Fold fibre from the half-One screened carrier to the exposed One, with every intermediate finite screening layer retained if subsequently generated. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__half-One-screened-to-One-exposed-running__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	half-One-screened-to-One-exposed-running	fitted-running-curve-or-reversed-direction	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.

Axis	Forced form	Eliminated form	Elimination reason
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:1a88e248c5def845615f245bd8e27f9e3903fd5e69e18dff5870a0cb7dcb063e. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:3a155046e477a1c21adb3ffb90fd772408c346db409d0e172ccf39cfd5ab5327.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:b663484e15866b5b05d67a3e40919d253ee0e0ca5163c05a4d15b4738a15ca84.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:955c2f7f51a571fe90a3bf935ab10fe474d19eb89bbdfd22931a16be42c18f83.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:a21e93bb9aa177cdefd411a060c7ff1d0387205d46dc5c4172327f39666a4397.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f. Independent certificate: sha256:9e397be6732edd641d37f0536a3dc3da18ef2f55d4eb82958f0c67b98e06a98b. Engine external-validation hash: sha256:ba1411492596f29e4dec72c7348bc3526469b3f48945e996fdc8a4ffb9e4e728.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger
- no claim that the two structural endpoints alone fix a measured beta coefficient

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:4db98f792906b7ce138c295954059ec4063c32f4a3e987f8ec0f2dc0fbfb9b20; engine receipt
sha256:40f95e89489d596b961aeab5d234dca5938b41b0dd6de02ffae7da90e2d66d76 at receipts/engine/model_admitted/SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003-40f95e89489d596b.json;
empirical-validation hash None; measurement receipt None.

93. Exact vacuum-to-inertia unity relation

Claim identity: SFT-PHYS-VACUUM-INERTIA-UNITY-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The independently forced vacuum displacement and native Fold coupling are both the half-One. Their exact exchange ratio is therefore the One and both complete in one Fold. Any lawful change of one carrier must be held identically by the other; this is a structural proportionality and a standing experimental discriminator, not an observation of engineered inertia reduction.

The exact vacuum-to-inertia exchange ratio is the One: $(1/2)/(1/2)=1$, with both carriers completing in one Fold.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-PHYS-MECH-INERTIA-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of vacuum/inertia carrier, dependencies, exact exchange relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact ratios between the independently admitted vacuum displacement and native Fold coupling, including changed-carrier controls and the one-Fold completion trace. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__half-One-over-half-One-exchange-at-One__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	half-One-over-half-One-exchange-at-One	free-inertia-coupling-or-uncoupled-carriers	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.

Axis	Forced form	Eliminated form	Elimination reason
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:96417536c5fcbe4bce4ee91ee82f4662cf0c3a400b059b5fad4990d61a9216f2. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:542db57def3e4bfff7ea9bef689d05ad43bf9028730be6baecd66ae677629997e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:6ac3b4036cc44e3d7031304280c0b51b3cfee0c50d06f2f38512f0f110045c9b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e4cda41b2cfa5455673da0c1b740f5b85cd92d5d6a0111d8b896dd6e4829c3ad.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:288ea6ec8c1414873c1d31997f3a967b584036fb81a538d8b291e865ea54acad.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f. Independent certificate: sha256:918411a91aea85ce591208e168de2386c4e4001f9bd7960a9351b8600db0da97. Engine external-validation hash: sha256:2654b9a181407cf63138b72d2cd2dda030cd62a48492557c618915266a90b528.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger
- no UAP report, propulsion observation or engineering performance used as validation

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:00072adfe4dcd71e07d644bfd46420ec9f6b4b8588902be3bab08bb5abeb554a; engine receipt
sha256:3e1b000691688a8bec6ab6301c21ae05136256e08f10d0e0e36babb0dc602d71 at
receipts/engine/model_admitted/SFT-PHYS-VACUUM-INERTIA-UNITY-003-3e1b000691688a8b.json;
empirical-validation hash None; measurement receipt None.

94. Positive asymmetric vacuum-beat extraction

Claim identity: SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The half-One vacuum carrier and generator-three unit carrier have the unique exact positive separation one-sixth. Transferring that beat to a held work carrier leaves one-third in the local vacuum carrier, and one-third plus one-sixth reconstructs the original half-One exactly. Thus a positive outward vacuum-to-work transfer is structurally admitted without creating support.

The forced asymmetric vacuum beat transfers positive work 1/6: vacuum 1/2 -> retained vacuum 1/3 plus held work 1/6, conserving the original carrier exactly.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-PHYS-VACUUM-ODD-RECURRENCE-003
- SFT-PHYS-VACUUM-INERTIA-UNITY-003
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-MECH-CONSERVATION-001

Generated grammar and closure boundary. Generate the complete eight-axis product of vacuum-beat carrier, dependencies, exact transfer relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact beat transfers between the independently forced half-One vacuum carrier and unit generator-three carrier, with source, retained reservoir and work support held separately. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__half-One-versus-one-third-positive-beat-transfer__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	half-One-versus-one-third-positive-beat-transfer	asserted-free-energy-or-erased-outward-transfer	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.

Axis	Forced form	Eliminated form	Elimination reason
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:2f47106944b3e20b732cc0595935b9a514f9dcd24a2d4f15de3c6b4e589229e9. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:4bf356fe9c24fe022293f5a5ed5c1500682d3b164da201fa5959b89b18d8112b.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:b6de76683aea01da6edd411b7e57d7d6a6eb1488190604c7ba3887150ea85ce3.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:91bd04981098412ed0e07bda628244e221c20bc8043ffec95e8a4aacaec2e393.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:d13dd0b618e4449e4f46631bd8b8e922aed44be2978f532e283bdfa191ffae7b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f. Independent certificate: sha256:793c10b8b4aed4ee54bfc17343955c4ee50a8d51a0730d743ea662aad26e0d42. Engine external-validation hash: sha256:4dd6db68e462a6363b9a827baff37f5910702a00dcea9f6a3fdb06f74fed5058.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger

- no claim of repeated net work without restoring the changed vacuum carrier

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:f6f2c4c55430e81592b9cf311250acda2491ce4cfd32da5f042e6cd92b055985; engine receipt sha256:de7c651c4eccf51ac10efdc04853bcd36fcb3ef0dcbe3ef58bba0cf52efc0e1d at receipts/engine/model_admitted/SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003-de7c651c4eccf51a.json; empirical-validation hash None; measurement receipt None.

95. Complete returned-cycle vacuum energy ledger

Claim identity: SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The admitted outward beat transfers one-sixth to work and leaves one-third in the vacuum carrier. Returning the apparatus and vacuum carrier to their exact initial state requires transferring the same one-sixth back. The vacuum is restored to one-half and the work carrier becomes the empty structural record. A complete returned cycle therefore retains the outward extraction event while forbidding an unrecorded net-support gain.

A complete returned cycle is: outward vacuum $1/2 \rightarrow 1/3 + \text{work } 1/6$; restoration $1/3 + 1/6 \rightarrow \text{vacuum } 1/2$; residual work is the empty structural form, not numerical zero.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-ENTROPY-001
- SFT-INFO-CONSERVATION-LOSS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of returned-cycle carrier, dependencies, full ledger relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: The complete two-leg process returning vacuum, apparatus state and retained information to their initial records, with every positive transfer held and structural absence represented only by the empty form. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-trace__outward-one-sixth-and-equal-restoration-cost__complete-registered-product__all-omitted-carrier-forms-rejected__exact-result-sealed-before-comparison__complete-source-transfer-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	borrowed-vacuum-number	A borrowed vacuum number has no Fold trace.
dependency	admitted-V3-dependency-trace	asserted-prior-answer	An asserted prior answer is an unregistered premise.
relation	outward-one-sixth-and-equal-restoration-cost	partial-cycle-net-gain-or-erased-extraction	The alternative loses a required carrier or imports an extra law.
enumeration	complete-registered-product	selected-candidate-neighbourhood	A selected neighbourhood cannot establish uniqueness.

Axis	Forced form	Eliminated form	Elimination reason
minimality	all-omitted-carrier-forms-rejected	survivor-without-lower-controls	A bare survivor does not prove the required carriers are minimal.
measurement	exact-result-sealed-before-comparison	target-visible-before-seal	Target access before sealing reverses the scientific direction.
record	complete-source-transfer-control-record	answer-only-record	An answer alone cannot reproduce the source and transfer ledger.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:e1fc93d8c434b200be0d721922f3f8ad2852f1af09dff8227be458caaaefb916. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:6c9a7d833b3b0110709bda928a42da2f53e18eedb4a33baa3e9381c747fc78c0.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:2bde9e9f906aaad134ebdb7a39a62da7fb987ba0ce794556b05363a2043a64ce.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:071b8351889521815841af93def5f161d15d6488e583a10c04adaa6d41bf3687.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:791700b22bce99bfb57c65e063896b88220dee949e16479c67f75d784e7d7d7a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:46b4fc1bc5063437791f86dec18f6804cb3c98174c289b1500bc51ab6c880a7f. Independent certificate: sha256:c49bcf20bf0c88eb8ebb0994347a7692fb3046cce229aefc8e8a252f9c5739b7. Engine external-validation hash: sha256:8bd9231535784a0b227c3b86d1731fbf499c432eb53cd3d3eead47c5143faded.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, candidate table or answer artifact as a premise
- no external measurement, conventional vacuum model or apparatus result selecting a survivor
- no fitted coefficient, tunable phase, chosen reservoir value or answer-selected harmonic

- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no deletion of the positive outward extraction result or of the complete restoration ledger
- no suppression of reservoir depletion, apparatus reset, information record or restoration cost

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:98996bf1b2d39caa17f257315a7068e396ef95e789608d452d163589cac20c45; engine receipt
sha256:a0ae88fcb49661c576a54741af33c09c88fb11d125f48e41edf2658b08ede835 at receipts/engine/model_admitted/SFT-PHYS-VACUUM-COMplete-CYCLE-LEDGER-003-a0ae88fcb49661c5.json;
empirical-validation hash None; measurement receipt None.

96. Exact one-distinction erasure cost and Maxwell-demon ledger

Claim identity: SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete quarter-One/three-quarter-One reset fibre merges at half-One, closes one distinction with native throw half-One, requires one reverse label, and forces a Maxwell-demon cycle to export exactly one environment record.

Resetting the complete quarter-One/three-quarter-One Fold fibre to half-One closes exactly one binary distinction with native throw half-One; reversal requires one predecessor label; and a closed Maxwell-demon cycle exports exactly that one label to its environment.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-FOUNDATION-HALF-ONE-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of reset support, image, distinction, native throw, reverse record, sorting, memory, environment, thermal translation and extension forms. The declared exact boundary is: Every exact reset of the complete two-preimage Fold fibre to one held ready image; every finite closed sorting-memory-reset-environment cycle; and every retained predecessor-label assignment. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `complete-two-preimage-fibre__common-half-One-image__exactly-one-binary-distinction__half-One-preimage-separation__retain-one-predecessor-label__one-held-gas-distinction__two-label-Fold-memory__one-environment-reverse-label__postseal-dimensional-correspondence__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
support	complete-two-preimage-fibre	sampled-memory-state	A sampled state cannot establish a two-to-one reset.
image	common-half-One-image	selected-reset-state	A selected state has no Fold trace.
distinction	exactly-one-binary-distinction	unquantified-information-loss	An unquantified loss cannot balance a cycle.
throw	half-One-preimage-separation	free-or-vanishing-cost	A free or vanishing cost violates the exact fibre separation.
reverse	retain-one-predecessor-label	infer-erased-predecessor	The common image cannot identify its source.
sorting	one-held-gas-distinction	unrecorded-sorting-gain	An unrecorded distinction is outside the closed system.
memory	two-label-Fold-memory	memory-without-two-states	A one-state memory cannot encode the decision.
environment	one-environment-reverse-label	erased-without-external-record	That would destroy the predecessor distinction from the closed ledger.
thermal	postseal-dimensional-correspondence	import-kT-log-two-as-premise	A conventional thermal formula cannot select the Fold law.
extension	no-extra-rule	free-cost-or-demon-exception	A free exception defeats closure.

Base and successor. The closure proof is sealed by derivation hash sha256:86744c170537a1f26f665bd8afa02395f86f81ee0fc6d48028eb7c6e2a61b6ad. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:556549826b012b3cd71e8b80542015d3224b773d5fa6f5869057a4b2d549c6ac.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:f085eeb647249724d91ed525c9cccd368fe1fbbaa6b20ac7accc21ce81840518.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:9edfe8f63e326a077581b89b692a4cd4105839ff44467908a825fdb9b2dcc58e.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:372c69b1e6718117746a72b16bdb6732ac79dab6e19b75d13f20b81b065a1275.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:1829d0eb2c8ca53323d38289398a7be8cf250eee451bab6ef02f7c374cfe7758. Independent certificate: sha256:7acc7a70718cf879102de06754364697ed9fc271868177acb62f393306874ac8. Engine external-validation hash: sha256:07a875d566587e561700d4f1eccc50c80b2fea02cfbb581955e3bc3a0adb54d3.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no conventional Landauer formula, logarithm or experimental target as a derivation premise
- no claim that native half-One is numerically identical to a dimensional energy ratio
- no numerical-zero, negative, irrational, imaginary or floating proof value
- no stochastic demon decision, free reset or omitted environment record
- no V1/V2 executable, answer table or stored survivor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:19be5baaaefa68250f055175ebd66cf2fee7fdadf3332a82c1ba8c77e1ae0575; engine receipt
sha256:b130990c39c87ac65a1f5728a4c471bc05f8f5720d0413feebfb5b03760f88b at receipts/engine/
model_admitted/SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018-b130990c39c87ac6.json;
empirical-validation hash None; measurement receipt None.

97. Post-seal Landauer erasure and Maxwell-ledger comparison

Claim identity: SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The admitted one-distinction Fold reset predicts nonfree environmental transfer; primary theory gives the conditional $k_B T \ln 2$ average bound and primary experiment reports long-cycle mean-heat saturation, without equating native half-One to a dimensional energy number.

The exact Fold one-distinction reset predicts a nonfree environmental transfer for physical erasure. Primary theory identifies the symmetric thermal-reservoir ensemble bound as $k_B T \ln 2$ and primary experiment finds long-cycle mean heat saturating it; all distribution, reservoir, average and long-cycle scope rows are retained, while native half-One is not falsely identified with a dimensional energy number.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete eight-axis product of physical carrier, erasure relation, provenance, prediction isolation, measurement separation, row retention, successor closure and extension forms. The declared exact boundary is: The admitted exact two-preimage Fold reset and every registered primary source row concerning a symmetric one-bit memory, thermal-reservoir erasure, environmental heat, ensemble averaging and long-cycle approach. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__one-distinction-reset-with-environment-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and

named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	one-distinction-reset-with-environment-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:16eae314150f4ef5415826275a0bd5bffffe62cd95ee60d16ecb10273a76d196a. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:2047f7bb211fa66dcc8744e437d3c453a04625798a8920fd39b5b1c28d88c8c8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:befd819dcb8d77edbc8ef2a6234a79acd3f70a14eacffbc2c769f917bbea35f2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4050b7eadd8cccadd44897f07254dae6869044a51fd68cf3c1728b0892fb4fa.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e55ed114da671c1b05fc0e9a9e9b6abb33f388712ed08769dbff27ddd4f4edc0.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:84839a2508413f560a01deddae908cb9c312f00d19c256c0e60601972e2ddff5. Independent certificate: sha256:66b4c5a652c1c49075027abfbc6a3ded3c0d080cdc3462e286624478182376b8. Engine external-validation hash: sha256:8970e6e8ed2f188332f346b0959e9ad84999cd7ca38f0c33053b7ec85b7306e9.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PIECHOCINSKA-PRA-2000
- BERUT-NATURE-2012

Observed comparison records:

- LANDAUER-TWO-TO-ONE-RESET: predicted two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; observed two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; exact match True
- LANDAUER-THERMAL-BOUND: predicted two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; observed two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; exact match True
- LANDAUER-EXPERIMENTAL-SATURATION: predicted two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; observed two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; exact match True
- LANDAUER-SCOPE-CONTROL: predicted two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; observed two-state-bit-reset-to-one__finite-positive-environmental-heat__long-cycle-mean-saturates-Landauer-bound__thermal-reservoir-and-ensemble-scope-retained__no-half-One-to-joule-identity; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: Reject if the source snapshot changes; if a registered reset, heat, bound, experiment or scope row is omitted; if physical reliable erasure has no positive environment transfer; if long-cycle mean heat does not approach the registered bound; if the thermal coefficient enters the Fold proof; or if native half-One is misreported as a dimensional numerical equality.

Measurement receipt: sha256: 22ca198ca9538ec8ce25f9a63769c520fa423768c6e6d2500c3df4c2e45a3476.

Isolation certificate: sha256: d02b3f8cfaafaa6e038f114f6b54b919ec8f2aca69aaaf284193952e0d3c778d.

Custody certificate: sha256: 4df01ed99678ac1bfe03b87d41d775043d9767c29e7136890eb4d0d697521820.

Registered source descriptions:

- Nature: <https://www.nature.com/articles/nature10872>
(sha256: 76a746eb8760816598724a2db17fb7cd1f3a07325e69193525d313fecea4d27f)
- American Physical Society: <https://harvest.aps.org/v2/journals/articles/10.1103/PhysRevA.61.062314/fulltext>
(sha256: 76a746eb8760816598724a2db17fb7cd1f3a07325e69193525d313fecea4d27f)

Registered target identities:

- LANDAUER-TWO-TO-ONE-RESET from PIECHOCINSKA-PRA-2000
- LANDAUER-THERMAL-BOUND from PIECHOCINSKA-PRA-2000
- LANDAUER-EXPERIMENTAL-SATURATION from BERUT-NATURE-2012
- LANDAUER-SCOPE-CONTROL from BERUT-NATURE-2012

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no source text, $k_B T \ln 2$ coefficient or experimental result in the formal predecessor law
- no numerical identification of native half-One with joules or with $\ln 2$
- no omission of ensemble-average, reservoir, distribution or long-cycle limitations

- no numerical-zero, negative, irrational, imaginary or floating SFT proof value
- no target access before the prediction seal and no fitted correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d624b93544e7d34754e530739caa4c2e54f6d204bb2b8e12fa027b643db37cc3; engine receipt
 sha256:eb264a10c1fe3666cf25a5e18beb9bab271534333bfdee25fadc0ffafcc3a93e at receipts/engine/model_admitted/SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019-eb264a10c1fe3666.json;
 empirical-validation hash
 sha256:774f63c5cce8ca0eee88a9f9cda0d0bf6095d9d105d90bc43c45f8b2e6b0c0c8; measurement receipt
 sha256:22ca198ca9538ec8ce25f9a63769c520fa423768c6e6d2500c3df4c2e45a3476.

98. Terminal vacuum floor, cosmic density and scale-transport law

Claim identity: SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The unique complete local vacuum boundary-energy floor is $\text{One}/2^{20} = (\text{One}/2^{10})^2$. At every finite radiative depth k , the complete 2^k half-spacing modes sum to 2^{k-1} and have common-scale mean $1/2$. The terminal cosmic vacuum fraction is separately $11/16$. The normalized cosmological magnitude is $\text{Lambda}(c/H)^2 = 33/16$, and dimensional transport is $\text{Lambda} = (33/16)H^2/c^2$ after held external scale references are supplied. Equating the local floor, a raw mode total, the global fraction or a dimensional cosmological density is rejected as a type error.

The unique complete local vacuum boundary-energy floor is $\text{One}/2^{20} = (\text{One}/2^{10})^2$. At every finite radiative depth k , the complete 2^k half-spacing modes sum to 2^{k-1} and have common-scale mean $1/2$. The terminal cosmic vacuum fraction is separately $11/16$. The normalized cosmological magnitude is $\text{Lambda}(c/H)^2 = 33/16$, and dimensional transport is $\text{Lambda} = (33/16)H^2/c^2$ after held external scale references are supplied. Equating the local floor, a raw mode total, the global fraction or a dimensional cosmological density is rejected as a type error.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-EXACT-OPERATIONS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-PHYS-COSMO-COMPLETE-BUDGET-001
- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032
- SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete product of generator volume, binary cover, boundary-label scope, observable composition, local floor, finite radiative ledger, cosmic share, spatial rate multiplicity, typed prior-claim correction, scale transport, measurement custody and extension form. The declared exact boundary is: The generated volume 3^3 ; every binary support until the unique least cover; one-label versus complete-two-label boundary support; amplitude versus two-leg

energy composition; every finite positive-depth complete half-spacing mode ledger; exact 11/16 terminal vacuum share; three-space rate transport; every exact positive rational held rate/speed reference; and no untyped equality between a local floor, density fraction and dimensional cosmological constant. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `generator-three-cubed-volume__least-cover-depth-five__both-held-labels-per-depth__two-leg-energy-self-composition__complete-boundary-energy-floor__complete-finite-ledger-and-half-One-mean__terminal-eleven-sixteenths-share__three-space-squared-rate-carrier__distinct-typed-quantities-and-prior-correction__postseal-rate-squared-over-speed-squared__postseal-only-comparison__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
volume	generator-three-cubed-volume	borrowed-cosmic-volume	A borrowed volume imports a model.
cover	least-cover-depth-five	selected-depth-ten-or-twenty	Selecting the desired exponent is circular.
boundary	both-held-labels-per-depth	one-label-per-depth	One label omits half the Fold fibre.
observable	two-leg-energy-self-composition	amplitude-relabeled-as-energy	A one-leg support is not a complete energy observation.
floor	complete-boundary-energy-floor	named-one-over-two-to-twenty	Naming the prior value does not force it.
radiative	complete-finite-ledger-and-half-One-mean	unbounded-mode-sum-as-local-density	An unbounded sum erases finite support, common scale and quantity type.
cosmic	terminal-eleven-sixteenths-share	local-floor-equals-cosmic-fraction	The quantities have different supports and types.
geometry	three-space-squared-rate-carrier	borrowed-continuum-coefficient	A conventional field equation cannot be a premise.
typing	distinct-typed-quantities-and-prior-correction	untyped-number-identification-or-rubber-stamp	Equal-looking numbers without scale carriers are not one quantity, and preserving an old label cannot excuse a type error.
transport	postseal-rate-squared-over-speed-squared	fitted-dimensional-value	A fitted unit magnitude is a parameter.
measurement	postseal-only-comparison	measurement-readable-before-seal	That would fit the magnitude.
extension	no-extra-rule	free-vacuum-scale	A second vacuum ruler violates the one-axis law.

Base and successor. The closure proof is sealed by derivation hash `sha256:c01238ae67c8185018e5b6be9ad1cf39d8c2dfc241b7e6e0845381f81fa87586`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4aeb43be5f9e2b45452280cbd30496eddbbc0607aba3ac2a1b9de4e54fcdc820c`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:81167a736ebfd0209293af293c2d9751cb0eff2a31cf7a415da2653ef0f2ad19`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:74a9f3aaf140a89bddd338da261541701151f82dd1ed6490a59310d95e21db5c.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:77bde99e13a432ec1df6a953152db7ae57c1fc3aaf061198814f68ce4cfd4cad.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:ea8f27dc1760e402a959b6da81c14c9bbc8ecb3c20de37ede2abfc9a0f41fc10.
Independent certificate:
sha256:b4c62c3d58a90ce9af0c1db9053aed8b9962d5cab16e2adf5502500d15bb31ad. Engine
external-validation hash:
sha256:0f107bf8bfbd842816b120cd750e391cd712a567ecd649cad20db2163b818f52.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, stored exponent survivor, dark-energy measurement or conventional field equation in forcing
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar
- no fitted cosmological density, Hubble value, gravitational value or second vacuum scale
- no direct identification of the local $One/2^{20}$ cell floor with Ω_v , Λ or a dimensionful density
- no reuse of the withdrawn measurement-influenced $2\alpha^4/H^3$ proposal
- no claim that a continuum zero-point mode sum is an SFT derivational object

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b6466fba1a682b36391e848e061aa30773de6cb511b44b1767e23cb677346772; engine receipt
sha256:c7b4777b12fc70628b0fd9a2f7d957274b61ede7f3624a665781364c8c9f7723 at receipts/engine/model_admitted/SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035-c7b4777b12fc7062.json;
empirical-validation hash None; measurement receipt None.

99. Terminal thermal-history, freeze-out and recombination law

Claim identity: SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every exact positive scale growth g , $T_{\text{later}}=T_{\text{earlier}}/g$ and $T_{\text{later}}*g=T_{\text{earlier}}$. Distinct binding thresholds are crossed in unique descending order. The depth-seven binary orbit $(1,2,4)/7$ forces freeze-out neutron/proton shares $1/7$ and $6/7$ and ratio $1/6$. The least complete binary-cover successor forces capture-entry shares $1/8$ and $7/8$ and ratio $1/7$. Complete retained-neutron pairing forces helium-family mass share $1/4$ and hydrogen-family share $3/4$. Recombination has an exact half-One classification midpoint and a finite positive visibility ledger at every generated radius. Acoustic standing modes are positive whole labels with exact odd/even compression-rarefaction parity; observed angular positions retain their projection and

are not proof identical to the internal whole labels.

*For every exact positive scale growth g , $T_{\text{later}} = T_{\text{earlier}}/g$ and $T_{\text{later}} * g = T_{\text{earlier}}$. Distinct binding thresholds are crossed in unique descending order. The depth-seven binary orbit (1,2,4)/7 forces freeze-out neutron/proton shares 1/7 and 6/7 and ratio 1/6. The least complete binary-cover successor forces capture-entry shares 1/8 and 7/8 and ratio 1/7. Complete retained-neutron pairing forces helium-family mass share 1/4 and hydrogen-family share 3/4. Recombination has an exact half-One classification midpoint and a finite positive visibility ledger at every generated radius. Acoustic standing modes are positive whole labels with exact odd/even compression-rarefaction parity; observed angular positions retain their projection and are not proof identical to the internal whole labels.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-EXACT-OPERATIONS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-PHYS-COSMO-REDSHIFT-001
- SFT-PHYS-COSMO-BACKGROUND-001
- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032
- SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004
- SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005
- SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006
- SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006
- SFT-PHYS-PLASMA-COLLECTIVE-001
- SFT-PHYS-PLASMA-OSCILLATION-001

Generated grammar and closure boundary. Generate the complete product of provenance, temperature transport, threshold order, freeze-out carrier, complete-cover transport, nuclear capture, recombination support and acoustic observation form. The declared exact boundary is: Every exact positive rational temperature and scale carrier; every finite set of distinct positive binding thresholds; the complete binary orbit of the least part on depth seven; the least complete depth-three binary cover; every finite positive visibility radius; every finite positive internal acoustic mode count; exact generated-finite sound-horizon segments; and strict separation of internal mode labels from observed projection. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `admitted-threshold-and-transport-dependencies__exact-inverse-scale-transport__descending-distinct-binding-thresholds__least-live-neutron-share__least-complete-binary-cover-successor__paired-neutron-helium-and-hydrogen-families__half-One-midpoint-with-finite-visibility__internal-whole-modes-parity-and-projection-record`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
provenance	admitted-threshold-and-transport-dependencies	named-cosmological-chronology	A familiar chronology cannot select a Fold law.
temperature	exact-inverse-scalar-transport	same-direction-or-fitted-cooling	A named cooling curve or fitted exponent adds a rule.
threshold	descending-distinct-binding-thresholds	named-epoch-list	Names do not prove order.
freezeout	least-live-neutron-share	one-seventh-ratio-rubber-stamp	The old label conflates a constituent share with a pair ratio.
decay	least-complete-binary-cover-successor	selected-decay-correction	A fitted decay fraction is inadmissible.
capture	paired-neutron-helium-and-hydrogen-families	named-quarter-abundance	Naming one quarter does not force it.
recombination	half-One-midpoint-with-finite-visibility	instantaneous-zero-width-collapse	A zero-width event erases physical visibility support.
acoustic	internal-whole-modes-parity-and-projection-record	observed-multipoles-assumed-exact-integers	Projection and driving cannot be erased to preserve an old slogan.

Base and successor. The closure proof is sealed by derivation hash

sha256:e2b318586427283e1e61c49e481e1879341de04ac670c7fafd5eb773e6d1e4b5. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:ca700bf092d66ef62469796c6034f11ce2c3fbf15077e333c745944e70df2965.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:5344c8a8ce62d060bb86c94ede6ce9e478829115c9b53de55441b0605db23a08.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:69a86c8830048d9b18e28fe284a06eb763f1c1810f4670bc959c5dc6be95bc50.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:1835665e3693287862638f0ef309fb7a625762c9048a4d8df2297913b141d8ca.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:22a9e8259b3e561ef7cc79cdf28846e652a16bd4fbdcbcea06323462bd46acfe9.

Independent certificate:

sha256:11a443134f16d7f8946c96ad3f94487aa48fddef4ad560f0526a378441a582ed. Engine

external-validation hash:

sha256:bce1224d0bcc023c31e2ca981ca1963b0f59601c8016ec9dfa07f848b3450972.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no conventional hot-big-bang equation, continuum fluid integral, stochastic freeze-out or fitted thermal exponent as a premise
- no measurement value, helium abundance, recombination redshift or CMB peak location available to candidate selection
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar
- no untyped identification of neutron share with neutron/proton ratio
- no instantaneous zero-width recombination event
- no claim that observed CMB angular multipoles are exact integer multiples
- no free reaction rate, decay fraction, sound speed, baryon loading or projection coefficient

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:9e3ef14860b7a7675b9c189a6512168ec5815fdd4289c6652283ba0aad5c186c; engine receipt
sha256:25dda78644ebc04ad43cd3e416950ad76dae6b124ed023f41364f0661eae9c00 at receipts/engine/model_admitted/SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037-25dda78644ebc04a.json; empirical-validation hash None; measurement receipt None.

100. Terminal exact temperature, finite canonical equilibrium and fluctuation-response law

Claim identity: SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every finite positive population, $T=(\text{sum throws})/N$ and $\text{total}=N*T$. Every paired binary population uniquely maximizes multiplicity at half-One. For L levels, exact canonical counts are $(2^{L-1}, \dots, 1)$, weights divide by 2^{L-1} , sum to One and are the unique fixed-count/fixed-throw multinomial maximum. The least fluctuation/response pair is $3/4$ and $1/4$ around equilibrium $1/2$, with equal departure $1/4$; the complete deterministic cycle $(1/4, 1/2, 3/4, 1/2)$ has mean $1/2$.

*For every finite positive population, $T=(\text{sum throws})/N$ and $\text{total}=N*T$. Every paired binary population uniquely maximizes multiplicity at half-One. For L levels, exact canonical counts are $(2^{L-1}, \dots, 1)$, weights divide by 2^{L-1} , sum to One and are the unique fixed-count/fixed-throw multinomial maximum. The least fluctuation/response pair is $3/4$ and $1/4$ around equilibrium $1/2$, with equal departure $1/4$; the complete deterministic cycle $(1/4, 1/2, 3/4, 1/2)$ has mean $1/2$.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-HALF-ONE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001

- SFT-PHYS-THERMO-FLUCTUATION-001
- SFT-PHYS-THERMO-RESPONSE-001

Generated grammar and closure boundary. Generate the complete product of temperature, total throw, equilibrium, canonical weighting, maximum-count, fluctuation, response and noise-orbit forms. The declared exact boundary is: Every finite nonempty tuple of exact positive throws; every positive paired binary population; every finite generated level count at least two; every exact fixed-count/fixed-throw population vector; and every positive repetition of the complete deterministic response cycle. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `exact-positive-mean-throw__population-count-times-mean__unique-binomial-multiplicity-maximum__normalized-dyadic-successor-ladder__complete-fixed-count-throw-multinomial-maximum__three-quarter-complementary-departure__quarter-One-antipodal-response__finite-periodic-orbit-readout`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
temperature	exact-positive-mean-throw	primitive-continuum-coordinate	A primitive continuum coordinate is not Fold-generated.
total	population-count-times-mean	fitted-Boltzmann-coefficient	A fitted coefficient adds a parameter.
equilibrium	unique-binomial-multiplicity-maximum	selected-half-label	A named midpoint alone is not a census.
canonical	normalized-dyadic-successor-ladder	continuum-exponential	An exponential is outside exact Fold arithmetic.
maximum	complete-fixed-count-throw-multinomial-maximum	asserted-geometric-shape	A shape assertion does not eliminate alternatives.
fluctuation	three-quarter-complementary-departure	ontic-random-kick	Ontic randomness is not forced.
response	quarter-One-antipodal-response	free-dissipation-rate	A free response adds a parameter.
noise	finite-periodic-orbit-readout	untracked-stochastic-source	An untracked source violates determinism.

Base and successor. The closure proof is sealed by derivation hash `sha256:5c35b4405e5642b59fd905d0bd2234f993c9044d79992afda6fe01fb378c4272`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c3b1b82ed7cd92c6c8ae26c3af6308311cccecdab0dbfeafae63c7f3bf346418`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:0ab83da6161f67c5bc0714e98e112e553c81dac404123f06a41c4ffe1b564811`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:32db5e2c2486690c946989ea76680a858768efbe3539a292f37f2cf8b5904fc4`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt

sha256:eb6624a1ed05a67939cf033f893fbff90f4c850fc29c8cd206d14db219e44a4e.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:89170070cce40d01eecb4b5d96ff33dba1dca2d04353cab3ee8f7a00348197d5. Independent certificate: sha256:5f0562864209036b569059443c080ce156913ae27a8fd0bfff4482b4a9ee8c4. Engine external-validation hash: sha256:5d8e46a19f68ad5d44097d3f450594420c4cbd4f87e765f4c2371a568378f891.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no Boltzmann exponential, continuum bath, Lagrange multiplier or fitted distribution
- no thermometry, noise voltage or measured k_B available to candidate selection
- no ontic randomness
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof magnitude
- empty combinatorial counts are structural absence records only

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:85f556f93bf5dfcf96da823aaa9cafe4cdecf73a781e4bdfc6d03f5f5f95dd22; engine receipt sha256:cf536f7516eb1c85bb075500d9c1a0ebd64bddd544878f26a27424d08de25bf at receipts/engine/model_admitted/SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043-cf536f7516eb1c85.json; empirical-validation hash None; measurement receipt None.

101. Terminal physical helium-isotope closure

Claim identity: SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete four-position by fifteen-nonempty-subword capture ledger has sixty cells and exactly one collective composite-identity record. The unique physical isotope conversion is therefore 59/60. Composed with the admitted analytic helium-family share One/four, this forces primordial physical helium-isotope share 59/240 and complementary hydrogen-family share 181/240, closing exactly to the One.

The complete four-position by fifteen-nonempty-subword capture ledger has sixty cells and exactly one collective composite-identity record. The unique physical isotope conversion is therefore 59/60. Composed with the admitted analytic helium-family share One/four, this forces primordial physical helium-isotope share 59/240 and complementary hydrogen-family share 181/240, closing exactly to the One.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037

- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

Generated grammar and closure boundary. Generate the complete product of analytic-family, constituent, subword, capture-product, composite-identity, isotope-conversion, physical-partition and target-boundary forms. The declared exact boundary is: The admitted analytic helium family; exactly four helium-four constituent positions; every nonempty subword of those positions; their complete incidence product; the single global composite identity; exact positive rational partitions; and no measurement in candidate selection. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `admitted-quarter-baryon-family__helium-four-complete-arity__all-fifteen-nonempty-subwords__complete-four-by-fifteen-product__one-global-composite-identity__fifty-nine-of-sixty-isotope-cells__fifty-nine-over-two-forty-and-one-eighty-one-over-two-forty__target-closed-until-formal-seal`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
family	<code>admitted-quarter-baryon-family</code>	<code>measured-decimal-called-family-law</code>	A measured decimal cannot select the analytic predecessor.
constituents	<code>helium-four-complete-arity</code>	<code>selected-arity</code>	A selected isotope arity is an extra premise.
subwords	<code>all-fifteen-nonempty-subwords</code>	<code>selected-proper-subset</code>	A selected subset erases a physical composition channel.
product	<code>complete-four-by-fifteen-product</code>	<code>partial-incidence-ledger</code>	A partial ledger breaks permutation completeness.
identity	<code>one-global-composite-identity</code>	<code>empty-or-multiple-composite-identities</code>	No identity leaves the labels unbound; multiple identities add an ungenerated split.
conversion	<code>fifty-nine-of-sixty-isotope-cells</code>	<code>fitted-isotope-correction</code>	A fitted correction reads the measurement.
partition	<code>fifty-nine-over-two-forty-and-one-eighty-one-over-two-forty</code>	<code>quarter-rubber-stamp-or-decimal-copy</code>	Neither the analytic quarter nor a copied decimal is the completed isotope partition.
target	<code>target-closed-until-formal-seal</code>	<code>target-readable-candidate-selection</code>	Target access could manufacture a correction.

Base and successor. The closure proof is sealed by derivation hash `sha256:ef909127179759e20899c1a50e206b4eba113be6e79c7f1cf06c05c14dfc7327`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ecce4667ef692326a59c782ce85a23682f9c1d0a82679d13ce636264cd914663`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:8d4c010fe7c9e3c154a34ec6d86d4704f8b5340c827720ad6f08ba8a152e2e`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6ed685241657e4ba96441f453f3f0318495b23ec540abd4cdfa60e2235d11e08.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:ee2859ae1b79e5ae1f35d866602c3857fb690467ae4dc1972b81c11cbb2f9d48.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:4546969d129794116becb23ba047631a9235a6af3675cdfa5639e9bdc26ece39. Independent certificate: sha256:6005685a689e8c82d6ec3c8079e8949bc5a2d84150cc802332ad387a95b52c6d. Engine external-validation hash: sha256:1e59614b5080af79330ce3d00e48e3da143d488a925dc3e0f94eb9385e72d31e.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Thermodynamics and vacuum structure are reconstructed over complete finite support. Temperature, entropy, equilibrium, irreversibility, zero-point structure and extraction limits are exact observation and transfer relations; ontic randomness and unaccounted energy are not introduced. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured helium abundance, uncertainty or central value in formal survivor selection
- no fitted decay rate, reaction rate, binding coefficient, abundance correction or selected denominator
- no omission of a constituent position or nonempty subword
- no multiple ungenerated composite identities
- no numerical-nothing, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:324b4b482bf10a8ae53f9b27a1814344b7405e5638423d292672166b74a3c1; engine receipt sha256:a3b3a44a0d3add7032680427c3b2b504147c0eb824d05fa5d044cef454c5ebc4 at receipts/engine/model_admitted/SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057-a3b3a44a0d3add70.json; empirical-validation hash None; measurement receipt None.

16. Physical Quantum Relativistic

Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic.

102. Physical quantum state correspondence

Claim identity: SFT-PHYS-QUANTUM-PHYSICAL-STATE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A physical quantum state is the complete generated support of distinguishable Fold words together with held phase/action and preparation provenance; it is not an answer-only amplitude list.

The physical quantum state is complete finite Fold support with held phase/action and source-bound preparation trace.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-word-support-with-held-phase__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	complete-word-support-with-held-phase	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a66a1c035a02aac6b4092403632107eeacca1d696e5f6dda52c4848a19601c81`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f11e96e98ea962641d29ecde80b00416bd1674538139bbe194dd49859d02988f`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 672a07dff9f246bfddb32a064c4aa643ffdc3d7a7f1b0603c5631ea289fcf7.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 815054d543d23cb8762114e29be753c6caafd2edb3526a773cb4c792281373f8.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 8cd47b523aba7d4c02f62e90f0dbf213234e179910d3affff1ab8752f8ddcf89.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: fa75461c4e288e36ede8428abbda45cf621831355500de7054dc6b6f4e0868ac. Independent certificate: sha256: 5efc8bd57bf4ead54b60a7a9f2b23c13a85041c03edc946693ded1dc62e03b6e. Engine external-validation hash: sha256: 05266c971285714a284201a1b440e9c615418c345e0c843036039631cffa8c5b.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- physical-state-1: predicted physical-quantum-support; observed physical-quantum-support; exact match `True`
- physical-state-2: predicted physical-quantum-support; observed physical-quantum-support; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256: 557cb140756b84937cfff842ca98bc46a5ba17f33b66ecd390541eea96ab3e8c8. Isolation certificate: sha256: bc978552bd11060de1127ea1f2c8b39fe01b33a7496f7fadda7066d388989b25. Custody certificate: sha256: 6aa6b08641be66190a4907283b531f844e8d10f1bbd5387be9dbc70a336ba149.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml> (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- physical-state-1 from NIST-ASD-5.12
- physical-state-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:646d05ecdd35f77c6440a8725ed08c85e6a8b59ed17c7a999e30c649417ad9e0; engine receipt
sha256:66bb63008bf5c34e9b3d260cf59890ac2d05f4474830c998296449725445145b at receipts/engine/
model_admitted/SFT-PHYS-QUANTUM-PHYSICAL-STATE-001-66bb63008bf5c34e.json;
empirical-validation hash
sha256:0b3e2b8b03aa236e15c72bb453856c5055d6b5996c4f8c5cf9b987ec2ca2f54f; measurement receipt
sha256:557cb140756b84937cff842ca98bc46a5ba17f33b66ecd390541eea96ab3e8c8.
```

103. Reversible physical quantum evolution

Claim identity: SFT-PHYS-QUANTUM-EVOLUTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Closed physical quantum evolution is a one-to-one Fold transition on complete support; its held predecessor labels reconstruct every prior state exactly.

Closed quantum evolution is exact reversible transition of the complete physical Fold support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__bijective-complete-support-transition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	bijective-complete-support-transition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:73d98d6a2c61130e167ae21e7c0dc1228a43f3734d500dc67b73b587ee99e228`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5aa3839cb9d2fa990c5eef22d1b1bb2a10642f726af0ffdbf0a071f8fc7ba3ed`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7577729d7390f62b2fdb9ae943374aba6baaead28cb6bb4a7caec07bd7d79837`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:646833aea82f4db8f7b9a7b40a8b4d8d66ccd6a60744c865a76651b68d27500d`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:03664f9fbe104e20bee6d232f787bdfddaecf7eb2fcbb41a903dd035e1bc078f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b88b6661761e1575417e3ced3cbb436daefa0efb420a72ffd6e9d29cf0d87d6`. Independent certificate: `sha256:c5f7cee6cb315e49e0e49a0d5bd83897c9fe20a141537f38d1d96ada4f489557`. Engine external-validation hash: `sha256:3b7ede270308eaa7871b044a15f9f4cc8dc31d8439b88625aeb358ff71b8af67`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of time: exact predicted interval [81120909/335364978631600000000000, 527285909/21798723611006000000000000]; external interval [12094421632919/5000000000000000000000000000, 2418884326589/10000000000000000000000000000]; overlap `True`
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact action/energy interval does not overlap the released CODATA atomic-time interval or the displaced control is accepted.

Measurement receipt: `sha256:41780708e62d7216e0211cf8c8c7e026ec72ec34530c9d843f5e0051cc445edc`. Isolation certificate: `sha256:74a7416c4ed12235fb4c2277ebc1a12fd5d38ffb17665047605bc0a9c643486b`. Custody certificate: `sha256:f5660469cc66b1b704b11ab57d4d9f7a704e73ce923e5b194332bac8c74f0530`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- evolution-1 from NIST-CODATA-2022-ALL-CONSTANTS
- evolution-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c5f29458763791f86a617642bcdcf6b2e8e97f01f8baf5d689e702d136424a983; engine receipt

sha256:0744b73d5436d955436b9dc7c39a4a63f229d0ab019c41409dccc37663148fdc at

receipts/engine/model_admitted/SFT-PHYS-QUANTUM-EVOLUTION-001-0744b73d5436d955.json;

empirical-validation hash

sha256:7665bd0ab9d73de01f2b4ef090839dc749773336cf8a7d28585bb22bcfe5b1f4; measurement receipt

sha256:41780708e62d7216e0211cf8c8c7e026ec72ec34530c9d843f5e0051cc445edc.

104. Observable and measurement record

Claim identity: SFT-PHYS-QUANTUM-OBSERVABLE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A quantum observable is a generated partition of physical support into distinguishable record classes; measurement creates one retained source-bound record for the observed class.

A physical observable is an exact support partition and measurement is its provenance-retaining record transition.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-EVOLUTION-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__support-partition-to-retained-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	support-partition-to-retained-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:12d0fb73d6f5134644ce0e8a9714cb56e7d21e113e4502e9e1d3a8a92238249f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:31e681c5a7afc9a3487c9cf195df85aa683e030ef0fb4c5d807f5b18271593f4`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:958782c82e68ed494612530e55434209d317595a5d8bf1f8ddfb1058552a7d5e`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b9d57ba62cdefebfb656007ac936b9d0af7c745999bfa855115292ca30a28787`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f82689bab0a8c4eb4d3bf59e677a2f10f20e0007a87075032177ebb2324c442`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5b205bb4f52f6c844d0dd0b60e252d9c5fe684eba16878c5d62c82453b782472`. Independent certificate: `sha256:d856dda98523631331ca9e6d52d2825b95a3d832172d3fff08ed4158e6978057`. Engine external-validation hash:

sha256:eafb213ceb10b58f0c1608cb8302ad1e1f376def86e2786fbbf9bb0460904d58.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- observable-1: predicted quantum-observable-record; observed quantum-observable-record; exact match `True`
- observable-2: predicted quantum-observable-record; observed quantum-observable-record; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:2871381c6705550221b73749a4698fea5746c13528f010e96af8cc00cdb27470.

Isolation certificate: sha256:40eba5543b5dfbb2f077246045519bfea6f4025844baf9529c4430cf00919662.

Custody certificate: sha256:5059ae0c8ccc527ccee63e88fc8f8bf516b812e8ed5c90d7ca724f77f02081dd.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml> (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- observable-1 from NIST-ASD-5.12
- observable-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:a15964f14ab058b547e8eb3b2e3238bfa7a09a9ed447f178e78f39196dee4579; engine receipt

sha256:a7ffffafe809c0f14f4eeaf1b32dbb4f54057f22c2ec06450c9f6d42450df1be1 at

receipts/engine/model_admitted/SFT-PHYS-QUANTUM-OBSERVABLE-001-a7ffffafe809c0f14.json; empirical-validation hash

sha256:93cbfc103be9c659299bdfbe057e2d895f8d7026be8e8dee44eb7b363d2f1087; measurement receipt

sha256:2871381c6705550221b73749a4698fea5746c13528f010e96af8cc00cdb27470.

105. Measurement weights from exact branch support

Claim identity: SFT-PHYS-QUANTUM-WEIGHT-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The weight of a measurement record is the exact positive count of its phase-compatible predecessor fibre relative to complete generated predecessor support.

Quantum measurement weights are exact rational branch-support ratios over a complete deterministic census.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-OBSERVABLE-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__compatible-predecessor-count-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	compatible-predecessor-count-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9f0e7a4dd0856f15844b0912c6abfcc6289787e87abbb1de36cd1c994a45c7b3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:bf51dde0d20e4b318bb6a0714ec5fc0cdce3ab2b91178e3c43fdeb812f2d19cc.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d9fc7d487a086cbfa2b9d9e69dddafb926af1837fc1d891ddae7054521e63f3b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7e5b72118aedf409fbf527aa4d17e4b4fceab7d8803cf0bcfd802da9548754db.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9869064001c5cfb9d3cd21d88947abeeacaebb42c5b91b62494c58a07c5f5b69.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:f9af1074be87591526a6071e6a4804494c8b408d6d5e87a104a67f0d57271c1f.

Independent certificate:

sha256:daf893f865417ee3cdfc25605e989b9074346c5345fe92ab2cfc268bb782aa78. Engine

external-validation hash:

sha256:5f5a4400f4a09026d6210fb06e32594bedfb54bed54a299568b95d1d54bba89c.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- weight-1: predicted exact-quantum-branch-weight; observed exact-quantum-branch-weight; exact match **True**
- weight-2: predicted exact-quantum-branch-weight; observed exact-quantum-branch-weight; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:535fc025ce1999bed3c30aef0f2ae77cfba6395ee37f9ffe816aa59b2c736dae.

Isolation certificate: sha256:6cc773d44b9bbde7ca65d80e0e7ff2f9a18a859e52f9d8cc5a1d184c566646b3.

Custody certificate: sha256:d0a75a812468b077f05fbf9695e654ee8be64cee230eaec818d6b21f110591c6.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- weight-1 from NIST-ASD-5.12
- weight-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude

- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:3bde7c0d35d4ce9a0fd5c01b7fb760d6fbf328b60b0152b2e97f962527dc810a; engine receipt

sha256:ac65de8a653d0efe53bca999f07d44dcac07e42d121a74974e8ea0ecf49ad101 at

receipts/engine/model_admitted/SFT-PHYS-QUANTUM-WEIGHT-001-ac65de8a653d0efe.json;

empirical-validation hash

sha256:e92f51b52deb77493b1cdc52cd55bc05c358afa0c818375b3a9db0d54132d265; measurement receipt

sha256:535fc025ce1999bed3c30aef0f2ae77cfba6395ee37f9ffe816aa59b2c736dae.

106. Incompatible observations and uncertainty

Claim identity: SFT-PHYS-QUANTUM-INCOMPATIBILITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Two observations are incompatible when their support partitions do not share a common refinement preserving both held phase/action records; resolving one necessarily closes distinctions needed by the other.

Quantum incompatibility is absence of a joint record-preserving refinement of two generated observation partitions.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-WEIGHT-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__noncommuting-observation-partition-refinement__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	noncommuting-observation-partition-refinement	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:35d09d2429fe2535d01ae8f4c3af7575b3a3a80ae8da112567c935fc4752dc2f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:158811357c5a538c1fca4b7fdaf9bc1fb2c2753248c18b4a1b9f19a16d714de2`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:30262f754cbdae3c420a4809a86515b4fc9ec2e6b0f393d644af5f266f1b82b2`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0de745408f2d5a993eb485c54819897e0b5ff04972894f1a2a0d6b42961c55`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a2c93fe9bb2f155be22a81fa63369500fcd12366413eeba1836870243c1f7595`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c35b641f13953583e4e48f397470b0a9d440416f2c0bd73e993ac6d7033b8464`. Independent certificate: `sha256:bd6f7030405d0751463d75d2a47c86bb88207d4f8e6cff53d7048eb753b2927d`. Engine external-validation hash: `sha256:fdbd79494dee7ffb5e8061edd0a2c011155d467053a97cc2f97ba7e3d86d4fd1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- incompatibility-1: predicted quantum-observation-incompatibility; observed quantum-observation-incompatibility; exact match `True`
- incompatibility-2: predicted quantum-observation-incompatibility; observed quantum-observation-incompatibility; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:3c0bc44b43868fe92dff7a8b258cdf6895f9adfd65f43532e528e7e1c3fbd205.

Isolation certificate: sha256:8249451c3e3afa9a18b70a43af14ead80b8c1da21fc7b8bf6975d4722f8d8501.

Custody certificate: sha256:f7dea63a65493f7ca9a1e266646684725b37edd5536f07d3e931c8d2616ca79e.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml> (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- incompatibility-1 from NIST-ASD-5.12
- incompatibility-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:a35454ea1d2cc3324b041bfa1e5a343337b71ce6fc3ef953d24d47cdd192b269; engine receipt

sha256:fd15b51c35d435c048c1f8d3d05bb80e67fe5ab8f706df40e6a2af13e04fd430 at receipts/engine/model_admitted/SFT-PHYS-QUANTUM-INCOMPATIBILITY-001-fd15b51c35d435c0.json;

empirical-validation hash

sha256:f925f8612be1d3bb5ebac1b74f17e66b9c7a3c4d00859d8425e64e5c8cc8d04d; measurement receipt

sha256:3c0bc44b43868fe92dff7a8b258cdf6895f9adfd65f43532e528e7e1c3fbd205.

107. Spin and finite cyclic label action

Claim identity: SFT-PHYS-QUANTUM-SPIN-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Physical spin is the finite cyclic held-label action of a constituent under generated orientation recurrence; opposite readings are fibre labels, never signed proof magnitudes.

Spin is finite cyclic orientation action on a physical Fold constituent with exact held measurement labels.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001

- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-INCOMPATIBILITY-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-cyclic-orientation-action__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	finite-cyclic-orientation-action	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:86b0dd83b47ba632c10f4e9c87ce9c6bf978a392949102cb2243f619da59f0cc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9abc35415e4749e0e224f44a961bf7e30269d12a9fb7256953f3c14186f3364b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:db916532b2cb153c2f330fa206cf12fb490d61af66dc238e59275ecb4bc77611`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d17d43683fff65b3f9dfdc5236679043af5f399108fa0cc78c365dc5861b49c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:963828635e8ecbdeb873622872f9b3cc1a16fff70c650a55b7339c0e0c695a3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ecd65100aa6616719ee338cf1e710ec95cded237bb9586bdb45c8f532803a5ce`.

Independent certificate:

sha256: 371f7e7b225aa68088362c332d7f831a371291b1bf96280fd57333b2cb9ea08b. Engine

external-validation hash:

sha256: f24c73aefbbefb32a9b57b99522822d4dc424d4bdb1d6078ae1b96e8b518d240.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- spin-1: predicted finite-cyclic-spin; observed finite-cyclic-spin; exact match True
- spin-2: predicted finite-cyclic-spin; observed finite-cyclic-spin; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256: 7d20ee9ac2ed5843b1a09f6ffa54fce08fb7bb59c53e19903c735c772dbd3ffa.

Isolation certificate: sha256: b820f1e1246c234f6aea5595ded40d7f1e2fb323ad4a542047f8da18f584269d.

Custody certificate: sha256: d8d97712140f7e6c0713dc45d85143e04903db389200f5cf3f127d18daea222e.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256: ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- spin-1 from PDG-2025-SUMMARY-TABLES
- spin-2 from PDG-2025-SUMMARY-TABLES

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: b9ec52945330138564cb3a577cb2440ccc4d25c872102ed9e1362f569832a8e7; engine receipt

sha256: f233aa5e16a279c3803209c902993252d6631531d91fc5959f1e3123c13f5351 at

receipts/engine/model_admitted/SFT-PHYS-QUANTUM-SPIN-001-f233aa5e16a279c3.json;

empirical-validation hash

sha256: 0f6db2e5307e2cfc3bfb7007770f189e38455b9b82594f11e4ce3c2d2f95e5da; measurement receipt

sha256: 7d20ee9ac2ed5843b1a09f6ffa54fce08fb7bb59c53e19903c735c772dbd3ffa.

108. Identical physical constituents

Claim identity: SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Constituents are physically identical exactly when exchanging their unobserved identity labels preserves every complete observable transition and composition trace.

Identical quantum constituents are the exchange-equivalence class of complete physical Fold traces.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-SPIN-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__constituent-exchange-observable-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	constituent-exchange-observable-equivalence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:94f7c002b3ea517a33a036afd2a422103c08e0dbbd2c5874b4c564fc97ebe8ce`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every

generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : aab2c92e7b762d31cb25bc68bcb80ed971da4eb2051a50fa55e461fafd318bde.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 0efa8b7889317c183594f5168fd1f282182dab407ced9e46aee80da270edb6d0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : d87f052ae46189313214dd64b3e37c5561f0f0ed21d4e72b9fb960bd430e254c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : c1e08af79027067b143e65b5f88a280c20655a915fba624dabe47979298896f4.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : 85d4bb3454e1a15386cb5030a57a6e6279dc9115b659e2f00377e143c77df2c4.

Independent certificate:

sha256 : 7384ac457d91ab3248d9d5bdf1921a272b92984ed1cb3ea17355a246af5627b9. Engine

external-validation hash:

sha256 : 391793959adb5df11e16b2f1581e0e5ff20a62094a3163eb54ea6175044ba4ae.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- indistinguishability-1: predicted quantum-constituent-indistinguishability; observed quantum-constituent-indistinguishability; exact match **True**
- indistinguishability-2: predicted quantum-constituent-indistinguishability; observed quantum-constituent-indistinguishability; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256 : dad17c56682caf0c7113d28d618515d4903b9790953166b4479fdcb8208cc46b.

Isolation certificate: sha256 : bf9f554f65c9b889862ea7a292c05ef0e64d626223009e593fd887dc644bd453.

Custody certificate: sha256 : 48c4eef98bbd600d413d9dd9786e4b9f37d859086c1cc930c77a1848c03013b2.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html
(sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- indistinguishability-1 from PDG-2025-SUMMARY-TABLES
- indistinguishability-2 from PDG-2025-SUMMARY-TABLES

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its

grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4f56f583db68635a253fb89c468c68303a45f12623b0f532f3b24b756bc27dab; engine receipt
sha256:7e67d989399d97c898d0a1aad60aa27d5f62224f323adaec9159f8f38dc378d8 at receipts/engine/model_admitted/SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001-7e67d989399d97c8.json;
empirical-validation hash
sha256:89f7c99135492c64d116908f9bab5ebf019a4b4d4e8a89c7b034ca1613bdb830; measurement receipt
sha256:dad17c56682caf0c7113d28d618515d4903b9790953166b4479fdcb8208cc46b.

109. Exclusion from antisymmetric composition

Claim identity: SFT-PHYS-QUANTUM-EXCLUSION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For the alternating held exchange class, two identical constituent labels cannot occupy the same one-constituent observation cell because exchange would erase the only distinguishing orientation trace.

Fermionic exclusion is absence of a distinct same-cell joint word under alternating held exchange composition.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__alternating-exchange-composition-exclusion__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	alternating-exchange-composition-exclusion	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0cd2200a9acc1547dbc642a6ea043db99f30ec0df5cdf1fa1972a8138af8947a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8bfe4d1671a6378a9659881a08b2ec43b7e8ea2de48f7d6256b5b2e2f13fa1e7`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5a37d4decf9f8cfb19f4158b0ec17548bb010b57567a48cb269d18f322ccf018`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fda1464760f59392dbe0dd19fc04d4affad8b926f222732fff7b158093adc211`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0c3883c36033a3fc149220284e3b199a88357eee0385de508566454e6fe4b5da`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d3c390a960261b1252d41e7ce98d06d23696d6ca0d172141b31fd3a59f3c354f`. Independent certificate: `sha256:26695bc4bce93fcb1cb6ff194422727859a22fa8a4589dca768f75e6abc9536`. Engine external-validation hash: `sha256:1746eec6ce34fc98f992695265b38dd23d513a4ada8ed3ca065b48b256cfe8e7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- exclusion-1: predicted fold-exclusion-law; observed fold-exclusion-law; exact match `True`

- exclusion-2: predicted fold-exclusion-law; observed fold-exclusion-law; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:9d4c868e9c114a7b9fcd325b6ca8cea8a1497590cdad5e6ef70e03aa66c382a1.

Isolation certificate: sha256:b624470e88af677e4189745c8206e482595e421d8bbd783c47349f9cb05ad724.

Custody certificate: sha256:bfac40a5753c85004ab5cae4e30e48e561ab4fc458180e1dc5e232910baf6251.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- exclusion-1 from PDG-2025-SUMMARY-TABLES
- exclusion-2 from PDG-2025-SUMMARY-TABLES

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4ab498dbd8b0d9aa4e62db6bca8e8e87eb80c4ed58fea206c06eb538c21f1973; engine receipt

sha256:27239f90141d1287616ee28f368bcd94cf1f94b549e9b65a0a0982afab765eaf at

receipts/engine/model_admitted/SFT-PHYS-QUANTUM-EXCLUSION-001-27239f90141d1287.json;

empirical-validation hash

sha256:3802d875f3aef69b91e7bbac3eb7f17bee7c6b599c0fabaa61b92e5bdca5f129; measurement receipt

sha256:9d4c868e9c114a7b9fcd325b6ca8cea8a1497590cdad5e6ef70e03aa66c382a1.

110. Barrier traversal on complete support

Claim identity: SFT-PHYS-QUANTUM-TUNNELLING-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Barrier traversal occurs when complete reversible support contains phase-compatible paths through generated barrier cells even though the corresponding coarse classical observation closes those paths.

Quantum tunnelling is observation of phase-compatible generated barrier paths present in complete support but absent from the coarse classical projection.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001

- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-EXCLUSION-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-compatible-barrier-path-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	phase-compatible-barrier-path-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8354dd1efda50ff11c80bace128dec0d52c753e8e592b3ceff7237c5a4d3c975`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:054d47cba32d67859017762ff6bf14c2f0b544c41418c091bd0983deca5446c7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:10a111e522af07c23409f5512cbfae2bf9b26e5c9b79a06caa1a9cfc1889f632`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a4364fd900c8a3a7d2940706b9158a806370b044965df03083717e810be4dd1e`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9a065d74a8c2e4777619141c741c06b2d5fc12f379b8ec06fdc2aa9170236900.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:f68922fe77299126dd1219fda584ddd255588262561a49164d3d2d6baa4febb6.

Independent certificate:

sha256:6c13114affef6115891454d71a84bb6273b1ddd16c4d33f8929c0847e9f3de7e. Engine

external-validation hash:

sha256:3912d0e1650e14bb5a40a2160052126d40f35d9b5b89e9a121c52a01a24883ee.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- tunnelling-1: predicted quantum-barrier-traversal; observed quantum-barrier-traversal; exact match `True`
- tunnelling-2: predicted quantum-barrier-traversal; observed quantum-barrier-traversal; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:74ad6683ed699277b5bec118e07af8ca5d095679bce121b28a5991cf24d5b1b2.

Isolation certificate: sha256:1e114351c80ee50e3895aca59986d5ba43b4c10ad0cf6572780c8c419dfb08ad.

Custody certificate: sha256:c7debcaa62135bfbbffff29279e828139b24ead3be533bd7bbf1d0044e5892784.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- tunnelling-1 from NIST-ASD-5.12
- tunnelling-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d4dcc2f2b5cbbe69ab4db94e0813a7745106099cda2bad310876b7d29526d5b1; engine receipt

sha256:a07368612ae686daeb2371484fa843364d9da57a9c8a8ade54dc8957c5841fe3 at

receipts/engine/model_admitted/SFT-PHYS-QUANTUM-TUNNELLING-001-a07368612ae686da.json;

empirical-validation hash
sha256:86d898a1e550e1d68a3c624d7d6e1994a7960621a6129f49e0112a1b7f2f6701; measurement receipt
sha256:74ad6683ed699277b5bec118e07af8ca5d095679bce121b28a5991cf24d5b1b2.

111. Discrete bound-state spectra

Claim identity: SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?
Theorem. A bound physical support admits only recurrence classes whose phase/action closes exactly on the generated boundary; transition records are exact distinctions between such closed classes.

Bound-state spectra are the generated exact phase-recurrence classes closed on finite physical support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-TUNNELLING-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__boundary-closed-phase-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	boundary-closed-phase-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:895e71892702f582fb925e974066e9a49464f785fe2b0ebd94c167b255deead2. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:422b87a303403ab306d1efc983d71bf64e249005e7626ec235958c406b1d08c3.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6919d017b7fdd311bd79696d8ebe35689cd2ae098a3c871b3d9d95876d9084ab.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e3d63ad84f86ab21de692f6bb037b798dfe1b99d2fc9e29964caa3a00c5b7680.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:a4286e85fa7e5a0c26c0671a240ca4971459ae6c3f05323ecdcbf5fd916a0fd7.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:c92ea98d77b307cc0c0a2b4fad119b5a338d78ab94afcb130153cc1ae7ff6330. Independent certificate: sha256:a1ff6a07fb1629118374bb4fd5751dedff5fd2b41f04b407bade1fdaff80f379. Engine external-validation hash: sha256:2f31eb94c1b9bd95852527400d37c08126ecf50866aaaeb219ab5bb8b02f6a43.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- discrete-spectra-1: predicted discrete-quantum-spectra; observed discrete-quantum-spectra; exact match True
- discrete-spectra-2: predicted discrete-quantum-spectra; observed discrete-quantum-spectra; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:b674c815c986f459beac866858a0cc50e67908b3044aa2d8a5e68e72e13cd720. Isolation certificate: sha256:4923a9f29b96d66b7f725d7d78b01ef5979d4e2816a594db85234d4c8cb3bb5b. Custody certificate: sha256:6c8738676619e3e7adf23fb5a3f544cf6630b181f48edf89bf2afe219c936119.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml> (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- discrete-spectra-1 from NIST-ASD-5.12

- discrete-spectra-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:a8581a5d686b58f9a0bfee85d11be0a91a13e3b6d1ebc5fbfb40da70d49c68f8; engine receipt
sha256:40bcc8d05aea67f6b2812e426772bdf7b973d0410a1a24129afb79ab1650a57 at receipts/engine/model_admitted/SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001-40bcc8d05aea67f.json;
empirical-validation hash
sha256:bbc82779ca094ae3dec89dd18b17ec8c653228a0e5072e95cf2dcd1169e8ff1c; measurement receipt
sha256:b674c815c986f459beac866858a0cc50e67908b3044aa2d8a5e68e72e13cd720.
```

112. Physical entanglement correlations

Claim identity: SFT-PHYS-QUANTUM-ENTANGLEMENT-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A joint physical state is entangled when its complete word support cannot factor into independently prepared constituent supports while its joint preparation trace remains held.

Physical entanglement is nonfactorable complete joint Fold support with common preparation provenance.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__nonfactorable-joint-word-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	nonfactorable-joint-word-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:89c06b1637317cb2d824dd49c5780e50123ba82a8291b611eeb302a094b10e01`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a885931cb76ddb45fb63b53ba90f23207ca655e03daa2d7193e42a798446a8c9`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b3951cec8c44b8bd6c767ca92ee5c91df5cec9d2f5b115a19ebc6c778779c0cf`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fdd02410e8f55f94f279bb3939bf2828f51933a4b9cd9ff275b0e6d36c1b8f96`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2221140dc52e9b9d8091d2b19d0e34c3f6027d8d989fc7d1c13426540cf8ba54`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:69341d35615e16f99eb1c1728d5507acfe896271668491a4a0e883f961d85069`. Independent certificate: `sha256:cb2b51584c68f7af8a20930aa1af9382548a8b30fa84bc2f501af83b1f3f3054`. Engine external-validation hash: `sha256:49486b7ca96bdf4e55459e39a8480e81e8fb55cc6dad29964eb0a29e6b40cec9`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- entanglement-1: predicted physical-quantum-entanglement; observed physical-quantum-entanglement; exact match True
- entanglement-2: predicted physical-quantum-entanglement; observed physical-quantum-entanglement; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:738d451159d76be70e460472ad139ae20cad76767c7de8959369e12ffbd2f299.

Isolation certificate: sha256:5cca2cfbf4c5742d79331a3151501296e75890450fab93eed2fc8b52f98c2c2e.

Custody certificate: sha256:484506d3fad808fbf7ab825c09c56342ef01fc3b192a50bec882d71909afcac1.

Registered source descriptions:

- European Organization for Nuclear Research: <https://opendata.cern.ch/api/records/?type=Dataset&size=1>
(sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- entanglement-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- entanglement-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:9c33166d4650e2b66a1daf713804f5c025a21c157730a80bee1645131ebca6f8; engine receipt

sha256:4fe300f9b7c68fa7877536f5a6b517ad868f51f653621adf0ca0a16db177cebb at

receipts/engine/model_admitted/SFT-PHYS-QUANTUM-ENTANGLEMENT-001-4fe300f9b7c68fa7.json; empirical-validation hash

sha256:2a22e56f177817dff8555448a08e6bdb6f9c740557a06fff6e4873f380843f6b; measurement receipt

sha256:738d451159d76be70e460472ad139ae20cad76767c7de8959369e12ffbd2f299.

113. Bell correlation and local-hidden-record boundary

Claim identity: SFT-PHYS-QUANTUM-BELL-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Bell correlations arise from joint generated support in which preparation and measurement-setting records belong to the complete deterministic state; a factorized local record omitting that dependence cannot reproduce the full joint census.

Bell correlation is exact setting-inclusive joint support; the excluded local-hidden-record model is the incomplete factorization that omits a generated dependence.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__setting-inclusive-joint-support-correlation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	setting-inclusive-joint-support-correlation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:758ea149915b6ed73f409d98a8fb8d1f4f618e42e9c5ecfecda05a31e027cf98`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b09299bc18140e5c855348082952106725ba8320a83c8cafd1ab997351ddf377`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:538825205848b5bdac60332d5d6e1737736e12e6a7bd2068badb9f674bbd2c93`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 939c55bbbf66a6d3fd049812da4f9361ad4b43066d21432229ec2ae79449d78.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 9fde24c371247082d6f4f220726cbb0fb1a12324e0645156b6a6bf934926108b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 7367a60e4c819fe426e6407c fb2dd2c0a0ac39d3ce82aec4540244a4f6914860.
Independent certificate:
sha256 : 03bf661944d56afc47ced421769684b15f988ec4f3b90cd89073e1ceca635ede. Engine
external-validation hash:
sha256 : 9fadca435dfdb4e3059108d93f1fc5358c80359ec812367e7c2bf90c0945e344.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- bell-1: predicted bell-joint-support-boundary; observed bell-joint-support-boundary; exact match **True**
- bell-2: predicted bell-joint-support-boundary; observed bell-joint-support-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256 : 36ee32c42f1452ca3d8556227dc36fe48b04c0e2f7875de74807377bbfe4636d.
Isolation certificate: sha256 : fc7b631cebb31e370d7d74e5ee1046cfc152743fb555cbb7df6952ce1b22fad5.
Custody certificate: sha256 : 294fc3d6fa9b521818de1299efb46f279048a6964320b9e90aa7a24aa81cdf98.

Registered source descriptions:

- European Organization for Nuclear Research: <https://opendata.cern.ch/api/records/?type=Dataset&size=1>
(sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- bell-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- bell-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8480c03a3eb947112387ce6883c18aa824b617284a2bb64c67170422b924097f; engine receipt
 sha256:2c368d4a200d2566c3c78014d6927a9a64017b2fde91972501b7cf865fb47aa7 at
 receipts/engine/model_admitted/SFT-PHYS-QUANTUM-BELL-001-2c368d4a200d2566.json;
 empirical-validation hash
 sha256:130c9d0eef669e97bdb02886a33ad59beebfb42924b6b27463c70fc42c9ea016; measurement receipt
 sha256:36ee32c42f1452ca3d8556227dc36fe48b04c0e2f7875de74807377bbfe4636d.

114. Context-dependent joint observation

Claim identity: SFT-PHYS-QUANTUM-CONTEXTUALITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. An observed value is contextual when its record class depends on the complete compatible partition measured alongside it; no context-erasing assignment preserves every joint trace.

Quantum contextuality is dependence of an observation record on its complete compatible partition, with no universal context-erasing assignment.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-BELL-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__context-bound-observation-partition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	context-bound-observation-partition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5f7527ff4f10d355ab298a5f09703479616714faa258ee1703d3a5948921ae41`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f66538e9706174e08adb910288533ae32057f5077a78e705b803aad5888f1a1b`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0feebd50f773f44a1e5e8c31a4ad6fee0aaae9cb2b1af74b004e7899bcb4495c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:24a148617bb3fb6972e20eeaeaaa4f7130699c648ebefb76023bdb68b76a0823`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:098b5f19652af1197d4c456245a7ce41033ce299cdec9b34a3a6a35fe53d4d28`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e5437e0655b540bc5cbbb7e119227c0519503d0644c3a3e649ca6b3cca31c80`. Independent certificate:

`sha256:b00a502d9248194ee83d3dd6f506e0298891aef1c17ecfbff08c39e13e7f113e`. Engine external-validation hash:

`sha256:b6251d620ac893b10e3651fa00d8758ba9716ac0347d768d4fbca62e6ace2e56`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- contextuality-1: predicted quantum-contextual-observation; observed quantum-contextual-observation; exact match `True`
- contextuality-2: predicted quantum-contextual-observation; observed quantum-contextual-observation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:009ae275af7ed879270ba6b817d4e6db440ee62d4b6123a81bb8a026c8101bb3`.

Isolation certificate: `sha256:f12380f13a0918d84a2862bdeefaf7418abe87793b793c6bdc601c6c48a83303`.

Custody certificate: `sha256:ce623c10e91a407067a43df273267540c6917b73d734454445a41e42ff10d8b3`.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml> (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- contextuality-1 from NIST-ASD-5.12
- contextuality-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:dba339b5c3b9f5f131240bc554ec7d013eec62b16b1163a31a7c94c5291437e3; engine receipt
sha256:8b6083f43638b7416c9639697971b3e176c77f6db2858f00b927bcf0d8942946 at receipts/engine/model_admitted/SFT-PHYS-QUANTUM-CONTEXTUALITY-001-8b6083f43638b741.json;
empirical-validation hash
sha256:51a645d0762908514a2c86f29988424b8858318ccc8ff71aa2ccfc6907325d10; measurement receipt
sha256:009ae275af7ed879270ba6b817d4e6db440ee62d4b6123a81bb8a026c8101bb3.

115. Decoherence through environmental record distribution

Claim identity: SFT-PHYS-QUANTUM-DECOHERENCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Decoherence occurs when phase-distinguishing labels are reversibly distributed into inaccessible environmental words, closing interference in the reduced observation while retaining it globally.

Decoherence is reversible distribution of phase records into environmental support and their closure under reduced observation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-CONTEXTUALITY-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records

without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__environmental-phase-record-distribution__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	environmental-phase-record-distribution	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:16e18e9d72bb357c0e45f9521e474ff93dd8648a2356404332daca261aaf80b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a3cf5ecafd15bc2a406db28ad3834d285d09fc7a8c70a3b1bdde10fab6ea314b`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b0d671dbe216be2ae2b9dbfb167004711fd0b12487cf958d5d4f49741f6ac73b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b7aac10ce1b3e9e4e646b126c82c729b0878336745423529f2acdcf6ea8621c3`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8ea6ddfbc48eb65b1fdd13fe12f252413f61491c7067bcd5cad06b43e64ad4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:92fe5a2bbcc2e642707032bfaed0f2db0f7f0705c2f481da74f3282d0b1ec3e9`. Independent certificate:

`sha256:c00a9d191b43079e1bcc5c4a74d8d902662444b646199961a5ba04ea60a50252`. Engine external-validation hash:

`sha256:e3587715eab421a9b3582588febd463ff2cf8ee725e0ceac3032e1fbb5ccc550`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- decoherence-1: predicted fold-physical-decoherence; observed fold-physical-decoherence; exact match `True`
- decoherence-2: predicted fold-physical-decoherence; observed fold-physical-decoherence; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:0a29356b051ea36807814c65d4ced93db31a45d9c91278e0c43a2ad417c67a10.

Isolation certificate: sha256:4ea72fb7a3ae7fb8881d4e9b23307e635f0a74181c0b113729ea53511f24d3ba.

Custody certificate: sha256:01ab93b3d8fd6ba6807073233ea08cd3f7fb72800659856c586708ab9d732aa5.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- decoherence-1 from NIST-ASD-5.12
- decoherence-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:40b2ad33193d36d486b8c6fb4edae22fbc2632b36e355b165a2296ea25070939; engine receipt

sha256:d55a9454999245b5fa0177305095d9fcd1e8c0bc6f0d75c9541ea1cfa58d8325 at

receipts/engine/model_admitted/SFT-PHYS-QUANTUM-DECOHERENCE-001-d55a9454999245b5.json; empirical-validation hash

sha256:8b313ff949f29e4e2e1c59211c96794d2a79f01ed2dfb32897411483ee92830e; measurement receipt

sha256:0a29356b051ea36807814c65d4ced93db31a45d9c91278e0c43a2ad417c67a10.

116. Operational quantum-to-classical physical limit

Claim identity: SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The classical physical limit is the observation quotient in which environmental records distinguish all interfering predecessor classes relevant to the retained macroscopic variables.

Classical physics is the record-stabilized observation quotient of complete quantum Fold support on the declared scale.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-DECOHERENCE-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__environment-distinguished-classical-quotient__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	environment-distinguished-classical-quotient	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:14f71f5fff82287d9285688725670bdfd11b58231e86c98d6eaea79271206cb8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 548a2a84e76b692c08a361e060e76119136c52b19bc29e977c10e85ba345eca1.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : e34d504d275c746a0589c9eae6f8c894578a1f6691b08044db822366b58b22a2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : bb4d0b3743489900ecc2943b7e7a02be2a24b9c4135e523ce740f2e15ed6a769.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : b194e483b537990a34a2cfec1a0b01fad9bb869aff1f526858c510ad6b1a1d7.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : 8c3401dc997205e7cd4e8a8ea930ddc09f7f56f8800443b67cfb2483a1333a3c.

Independent certificate:

sha256 : 4585eff2f4b1d7856d596067f9882bea67f448f44cfb2848221722c3c214803f. Engine

external-validation hash:

sha256 : 2fa0cee18eae4479641c0766b9a98f829c23e45475af16c20e12415ec1d6fade.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- classical-limit-1: predicted operational-quantum-classical-limit; observed operational-quantum-classical-limit; exact match **True**
- classical-limit-2: predicted operational-quantum-classical-limit; observed operational-quantum-classical-limit; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256 : 51666f38ecfba3745a6e04f824e24e0e2cf0c95104722ab8b5f1dccd6d2e59e5.

Isolation certificate: sha256 : cc6dcabadc29290b38300f49dc4b819220dfca95cd7ba16be977e05b592f2c28.

Custody certificate: sha256 : 742d020801434c924e3d391c6f88bf836adee9e7e59bb48612aeef2e3732de4b.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- classical-limit-1 from NIST-ASD-5.12
- classical-limit-2 from NIST-ASD-5.12

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude

- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:316655c872665410113abbe9d5e347660917473f2df1466f53c8ad0fd014f64e; engine receipt
 sha256:b87f06d0e0edbee677f693e0480d07cbad931e2a07a64841f47a204726cc9020 at receipts/engine/model_admitted/SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001-b87f06d0e0edbee6.json;
 empirical-validation hash
 sha256:4813ad6d27220db24bec1ea9c440157ef9a5ded7fec9cce1f48408ea7d69d5b1; measurement receipt
 sha256:51666f38ecfba3745a6e04f824e24e0e2cf0c95104722ab8b5f1dccd6d2e59e5.

117. Entanglement and no-signalling boundary

Claim identity: SFT-PHYS-QUANTUM-NO-SIGNALLING-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A local observation of joint entangled support sums the complete remote fibre; changing a remote partition only relabels that fibre and cannot change the local exact support count without a causal transfer trace.

Entanglement correlations preserve exact local observation support under remote setting changes and therefore cannot signal without a generated causal path.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001

Generated grammar and closure boundary. Generate the complete physical-support, composition, phase/action, observation, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite physical quantum forms generated from exact Fold word support, held cyclic labels, reversible transitions, predecessor merging and observation records without imported amplitudes or stochastic collapse. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__remote-fibre-relabeling-invariant-local-count__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	remote-fibre-relabeling-invariant-local-count	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:74316a1ebb4a384bf4ea3fd4122b4f47673c690dc609d9de48750c88c8c25bbc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:50ef85a22d59552c7f977b851d3dfe881ee9bf2e8353f75918709b57389cbe51`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3aaedc89dcc240da20f0214d2aaf19790e070bdde550e9c704f7d7f159c0c460`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7fafa2cec4d1912ec287601653ce89a461e18341a346ad5c38fe17d0bda33c0b`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9070a95e7db91df8731327226a744baa43d0621b5df358e3b261a617508160c4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:80893f62bbf80046828c9dc9c41c2c93c22d00d2c6270a5dbfa60785047a03c4`. Independent certificate: `sha256:d1f318fb1e4a7bb9503e6359296cfd7b940932aa9767cb17a5eb8642efbbe15a`. Engine external-validation hash: `sha256:e7f1bb01eb155da13830bf10b3d0d0494a9aac0e343141440f3ca19d5bd4fc79`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- no-signalling-1: predicted quantum-no-signalling; observed quantum-no-signalling; exact match `True`
- no-signalling-2: predicted quantum-no-signalling; observed quantum-no-signalling; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The physical quantum law is falsified if a committed external row lacks its sealed support/composition/observation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:450eab3defda05b7ad82cccc90bbe07e93564b75c6117d69cf907932c4348215.
 Isolation certificate: sha256:b206d1d5ed8ea02f5e9078fa88e2f325493da9c8572c6c08030d9d8c3c1856ef.
 Custody certificate: sha256:90256980f5bb8d51a621c5d719aa16cce1f907225e2012a6791de649fef8a3f0.

Registered source descriptions:

- European Organization for Nuclear Research: <https://opendata.cern.ch/api/records/?type=Dataset&size=1>
 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- no-signalling-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- no-signalling-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, completed infinity or continuum wavefunction
- floating, irrational, imaginary or signed proof magnitude
- imported quantum postulate, fitted amplitude or stochastic oracle
- target access, omitted adverse outcome or unrecorded measurement setting

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:38c8e09824b751f13b2c7841ce8b49995fc78f0cb862bf570a4909621599fbe1; engine receipt
 sha256:984ce2195db3fdf46ca0370f06565f79c7f1469bdf581f7c2ba99020827cd2fc at receipts/engine/model_admitted/SFT-PHYS-QUANTUM-NO-SIGNALLING-001-984ce2195db3fdf4.json;
 empirical-validation hash
 sha256:173268462d4ff03fc8b60527fcf3a464e24cda0ac2a7dc90e9c654901931c602; measurement receipt
 sha256:450eab3defda05b7ad82cccc90bbe07e93564b75c6117d69cf907932c4348215.

118. Two-hand exact Dirac square

Claim identity: SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The first generated spatial, binary-square and covering counts force the exact 3/5 motion and 4/5 substance partition whose squares close on the One.

The exact two-component square is $(3/5)^2 + (4/5)^2 = 9/25 + 16/25 = \text{One}$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product for two-hand exact dirac square, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision

support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-dependency-chain__generated-three-four-five-square-closure__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	generated-three-four-five-square-closure	chosen-Pythagorean-or-imported-relativity	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:14cd7fd94b9bb6afd346c888b62ae33ba1f90035e904dee2e3a757b047361db5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:0e161f33d1805a12c969a8cc038baecdaa508c8bc47e56e42eaa615d3e4e8cc7`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:5d41e32b7af5ceec2942ff0aa452a89a1255d38be5635a607dc739ee18c0d350`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9994533327142e264468046737dcfa5405b4e6f13f58f89023e341b80db7767d`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:29601296f6d95d18124fa4cf18fc2c1bb4f5e2f56b924666b04044d43e1c0632`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f`. Independent certificate: `sha256:f55b035b32807c70f3a92f777c92a1c40f96e03798a3e3e0e79a93fef8d8d5cc3`. Engine external-validation hash: `sha256:0c034b9e432d4f47abb3ba8de039f84408330d03eebf8ee98cb6c0886d16448`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:40ef972c0547c26a2117edca910060284b765d0fe367cab1d5eba616e3b88eea; engine receipt
sha256:8781e3f2ddd1b3980b2ab928c942ccc4db1bccac6527d0723c0a797855991279 at receipts/engine/model_admitted/SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003-8781e3f2ddd1b398.json;
empirical-validation hash None; measurement receipt None.

119. Full three-plus-one Dirac square

Claim identity: SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Three momentum generators and one mass generator exhaust the binary-square count four; placing each at the unique half-One balance partitions total square support into four quarters and closes on the One by direct and polarized routes.

Four half-One generators give direct square $4(1/2)^2=One$; the complete polarized pairing gives the same One, with equal differences retained as empty structural records.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003

Generated grammar and closure boundary. Generate the complete eight-axis product for full three-plus-one dirac square, including carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All finite exact Fold carriers named by this claim and every product of the eight registered binary form axes; conventional correspondence and physical calibration remain downstream. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-dependency-chain__four-half-One-generators-close-by-two-routes__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-trace-and-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore

a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-continuum-object	The object lacks a generated Fold support trace.
dependency	admitted-V3-dependency-chain	asserted-prior-result	A prior answer cannot act as a V3 premise.
relation	four-half-One-generators-close-by-two-routes	matrix-postulate-or-omitted-generator	The alternative loses a required conservation or recurrence record.
enumeration	complete-registered-product	selected-neighbourhood	A selected candidate neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	A missing carrier could silently select the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access would reverse the empirical direction.
record	complete-trace-and-controls	answer-only	An answer without trace cannot be independently reproduced.
extension	no-extra-rule	free-extra-rule	An added selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256: fb9666924defb309efaaa50a045ec539122821b5f4d2eb50158c0d54dd012bb3. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256: a60105939901d4902a76dc299b1c441c0300e5ff6c8cb260c034d2a4bf5bba7f.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256: 317013de33c572c047fbff80898c6778e0840f3085ebe3bb8fe9f6b0ac46ec70.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: ed415a81d14c83964e841006bee07f884fca7d49a8cd2cb46ed2ace966ee939d.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256: eaa2f57d6422d763fdaaf2ae8dd4d01fdc74c335477a82917c4ebee3770ed36d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: e0b85e4b3274842d4dfd26b1342bf57200f692d8fa9a7363194d38c4bd31e95f. Independent certificate: sha256: 686f1d09dd22e4d7e1c05f264785bf904d03475349cab3b62b8158abfa273efb. Engine external-validation hash: sha256: 17c4bee45b0cb9de04ae29d85aa89721aa5e93a5ec5be40715047e7fef9b4750.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no conventional field equation, continuum, matrix algebra or measured constant selecting a survivor
- no fitted coefficient, adjustable phase, imported normalization or target-selected value
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no target access before the complete exact result and candidate census are sealed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c3987de00e4203965aaa1d5be22f2e3271663bee9a5b794a0e3dff741afc3b5f; engine receipt
sha256:8b58f344d672ea8b89164ee8c5e46e4faacb1eda64aa4d0d682fa4931cd30a4f at receipts/engine/model_admitted/SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003-8b58f344d672ea8b.json;
empirical-validation hash None; measurement receipt None.

120. Electron/muon magnetic-anomaly structure

Claim identity: SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The bare gyromagnetic value is the binary count. The sole terminal electromagnetic self-return forces a strictly positive leading phase-normalized anomaly $\alpha/2$, and any additional mass-scale channel has muon/electron sensitivity equal to the squared sealed mass ratio.

Bare g is 2; the exact phase-normalized leading anomaly is terminal $\alpha/2$; muon/electron mass-scale sensitivity is the square of their sealed mass ratio. Physical full-turn normalization is a post-seal correspondence operation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product for electron/muon magnetic-anomaly structure from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__terminal-self-return-plus-squared-mass-sensitivity__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	terminal-self-return-plus-squared-mass-sensitivity	imported-QED-series-or-measured-anomaly	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:13fb57b80197a484c12cbeca599ef193619cb0a4caa78515ed3f5979124f01c3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:da61bd0221b876d5634545923539e31de5a18725e77c483e5fc7f18af6b775e6`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:5f9e94716ea096e3a19b9ff1a5e6df3c3ff4ceca1a0efe8ce6765892643f257f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:79e36b113aa82cea01532d5533b9c68787a4dd962316654c81683f2ac4fc4f48`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:0bbbfcad33b1878ef498271a836fdd97767fb5829ec55956e07f3c0a66c31a3d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:2407c8dc639d27ece8eaa73b14d817e086be9ab667c44e286515c6a27b36d973`. Engine external-validation hash: `sha256:5ec140d88f7693f93cdd1531cbc15358344fb9d9ebca83b685580015252e1ba9`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its

grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no imported Schwinger coefficient, QED perturbation series or irrational pi inside the formal derivation

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:46732443f32f146935ec1afede91a6c9bacad389cea20ef8150b7f224d41cec6; engine receipt
sha256:e0a6844ac70a6e929c5e23ed576df2ea5c13eb4a05d33b148b2b3ad75a4c4f6f at receipts/engine/model_admitted/SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003-e0a6844ac70a6e92.json;
empirical-validation hash None; measurement receipt None.

121. Terminal exact finite turn projection

Claim identity: SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A diameter-normalized finite turn has three complete generator sectors. Its sole remaining return passes through the forced up-depth seven and the complete binary support sixteen of the four-rung terminal promotion object, forcing the exact continued ratio $3 + 1/(7 + 1/16) = 355/113$. This is an exact rational Fold projection, not an imported irrational proof value.

The unique exact finite turn projection is $355/113 = 3 + 1/(7 + 1/16)$, with every count generated and no irrational or floating proof value.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-EXACT-OPERATIONS-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete nine-axis product of turn carrier, complete sectors, return, up-depth, terminal support, join, exact domain, provenance and extension forms. The declared exact boundary is: All diameter-normalized finite turn projections using the complete generator sectors and the sole terminal return through already admitted up-depth and finite promotion support. The generator produced 512 named candidates and the decision support contains 512 one-for-one decisions. Exactly one candidate survived: exact-generated-turn-carrier__generator-three-complete-sectors__one-positive-return__forced-up-cover-depth-seven__four-rung-binary-support-sixteen__nested-positive-return-ratio__exact-rational-projection__registered-observational-prediction-protocol__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	exact-generated-turn-carrier	imported-continuum-circle	A continuum circle imports an ungenerated object.
sectors	generator-three-complete-sectors	selected-sector-count	A selected count is a parameter.
return	one-positive-return	free-decimal-remainder	A decimal remainder is fitted.
depth	forced-up-cover-depth-seven	selected-return-depth	A selected depth can tune the projection.
support	four-rung-binary-support-sixteen	unbounded-or-selected-support	An unbounded or selected support has no closure.
join	nested-positive-return-ratio	untyped-sum-or-product	Untyped composition loses return order.
domain	exact-rational-projection	irrational-or-floating-turn	That violates the exact Fold domain.
provenance	registered-observational-prediction-protocol	target-readable-shortcut	A shortcut cannot seal a generated prediction.
extension	no-extra-rule	extra-turn-term	The four-rung object is terminal.

Base and successor. The closure proof is sealed by derivation hash sha256:f2657ac1f51936321cda999e843ac9e3ce9f1ed73660a101ff8327fb033764e4. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:fd56fc882a3efad690547e28802033e3b23b3f9adf5805eb9ed0f520c4fefc04.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:059bb4f1c25cfbb09faa9e08abbd8553ab8d5a3da61fc4d9aefb047624b9a9bd.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:8005e9b53d57a84a4674f3b907b538684147a103fb7a7570fc298a603f75fd89.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:2cbbc5c74c328822ca54699f1081bed118e454368ae3b97c4e8a1cb23112db91.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b7540aebff9282350d9117b62faf26ec592c8c187c0c870d64985bea3fab0e0b. Independent certificate:

sha256:64ea64c3bf1b107130d7a657b4228556fe9887c34d70489d4fbb8ea077a251ee. Engine external-validation hash:

sha256:e58eb683402b1b1280ad28d4d7d1200655e0c425ef68a23356a3f957d8c21a2b.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported pi or continuum circumference
- no measured magnetic anomaly
- no floating or irrational proof value
- no selected remainder or extra term

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4fa0407163c8c6f8f346b78cf12d51e9a74c28c6fb123488b4abd84b5528d1a2; engine receipt
sha256:b85f5162d1cf55490a4bc8666fa02b354029ff4a297d9013c61dddd93491629d at receipts/engine/model_admitted/SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004-b85f5162d1cf5549.json;
empirical-validation hash None; measurement receipt None.

122. Terminal exact electron magnetic anomaly

Claim identity: SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The admitted leading alpha-over-binary phase anomaly is projected through the terminal exact turn. Its complete finite radiative ledger alternates two positive held/returned pairs: down-depth five over the open colour bulk 27 held by 3, against the binary alpha return over 5 times 3; and the single alpha-squared return over 5 times 3 against the up-depth-seven alpha-cubed terminal return over the closed bulk-plus-boundary support 30. Holding alpha times that complete loop carrier once from the leading projection forces one exact electron anomaly.

With alpha the admitted exact coupling and $T=355/113$, the electron anomaly is exactly $[\alpha/(2T)]$ times $\text{positive_take}(\text{One}, \alpha L)$, where L is the sum of $\text{positive_take}(5/24, 2 \alpha/15)$ and $\text{positive_take}(\alpha^2/15, 7 \alpha^3/30)$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003
- SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis product of leading carrier, turn, open bulk, first pair, second pair, orientation, retention, target custody, provenance and extension forms. The declared exact boundary is: All first terminal finite-loop completions of the admitted leading electron anomaly using the down/up covers, complete colour bulk/boundary supports, binary labels and terminal alpha exactly once in their typed loop roles. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `retain-admitted-leading-alpha-carrier__terminal-exact-turn-projection__complete-volume-held-by-boundary-channels__down-over-open-bulk-held-by-binary-alpha-over-down-colour__One-alpha-s`

quare-over-down-colour-held-by-up-alpha-cube-over-closed-support__two-positive-held-pairs__hold-alpha-loop-once-from-leading__target-inaccessible-until-prediction-seal__registered-observational-prediction-protocol__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
leading	retain-admitted-leading-alpha-carrier	replace-bare-or-leading-law	Replacement erases the admitted Dirac and leading receipts.
turn	terminal-exact-turn-projection	imported-irrational-turn	An irrational carrier violates the domain.
bulk	complete-volume-held-by-boundary-channels	selected-bulk-subset	A subset can tune the result.
first	down-over-open-bulk-held-by-binary-alpha-over-down-colour	free-first-loop-coefficient	A free coefficient is fitted.
second	One-alpha-square-over-down-colour-held-by-up-alpha-cube-over-closed-support	free-terminal-loop-coefficient	A free coefficient is fitted.
orientation	two-positive-held-pairs	signed-or-complex-coefficients	Signed or complex proof values violate the Fold domain.
retention	hold-alpha-loop-once-from-leading	append-unbound-anomaly	Appending creates an unnormalized extra carrier.
target	target-inaccessible-until-prediction-seal	measurement-readable-execution	That cannot issue a target-inaccessible seal.
provenance	registered-observational-prediction-protocol	unregistered-fit	An unregistered fit lacks custody and prediction receipts.
extension	no-extra-rule	extra-radiative-term	Every down/up, bulk/boundary and terminal carrier is already consumed.

Base and successor. The closure proof is sealed by derivation hash sha256:c5e16e178fab26ccfa5e068d9de209fb44b5d6ea6f649b5eae6ffd2b9170d0bd. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:4c45ca44eff5ba01eb8cdfb6e718ad8db95256754310a489f1e8e3e4b742d8b5.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:203aea523fb5f22f9bc4cc8826785f1a833bd83b4ff31b2f4e29c2d1c28b88f4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:0058b40f58796261ba5911a35e9595e46d4334501a4941f0dd88bca00797cde8.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt

sha256:37cab5cef2b623790f23f0f8196876dce1cca03189b966df24c70d7d34182887.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:b7540aebff9282350d9117b62faf26ec592c8c187c0c870d64985bea3fab0e0b.

Independent certificate:

sha256:023c041f450da88f8ad1d9c7b1645f4bf2b6b4fb0dba728fff3dc84974ad0335. Engine

external-validation hash:

sha256:2b2b7b40257196a6a79a9bd4b28efe2695b79499ea1c4388bdd914ec6c5231ca.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026

Observed comparison records:

- ELECTRON-MAGNETIC-ANOMALY-COMPLETE: predicted terminal-target-inaccessible-electron-anomaly-prediction-inside-complete-CODATA-interval__observational-prediction-protocol-passed; observed terminal-target-inaccessible-electron-anomaly-prediction-inside-complete-CODATA-interval__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The exact sealed electron prediction leaves the complete registered interval, the target becomes readable before sealing, a measured value selects the survivor, an uncertainty changes, or observational provenance is hidden.

Measurement receipt: sha256:0333ea4cb3caa128658e08eed7d087f6d3cfcb8cadaa10587a0bd6636ee84495.

Isolation certificate: sha256:6a6ad1ed45d2103fae47829f41f089c7ab7fabe5419ddae873ee8b4ebbc1c09e.

Custody certificate: sha256:e1f3782858855e1c2f8ed1701d1d1609e24e36b38e465668e2429c714fdf5f12.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants and Fermilab Muon g-2 Collaboration: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>; <https://muon-g-2.fnal.gov/result2025.pdf> (sha256:4b506052fc5037576cd6ba4de8ee5afce74460f7a5cb8da4bc58cbb823f6732f)

Registered target identities:

- ELECTRON-MAGNETIC-ANOMALY-COMPLETE from MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no electron measurement in the executable relation
- no imported QED series or coefficient
- no signed, irrational, imaginary or floating proof value
- no selected loop truncation or extra term

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:fc418f64a16c0e62b0f737b92a281a91f405cd5415bfd40060a8db8db965b663; engine receipt

sha256:1901b85dc30105470cbf50fd13560402dd359f21aab9be6f324db678318087e4 at receipts/engine/model_admitted/SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004-1901b85dc3010547.json;

empirical-validation hash

sha256:27d7c447bf883365a58cdc46832155aa9fac6b0c928cf8df52f113ccf8bf41c7; measurement receipt
sha256:0333ea4cb3caa128658e08eed7d087f6d3cfcb8cadaa10587a0bd6636ee84495.

123. Terminal exact muon magnetic anomaly

Claim identity: SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete electron loop is universal. The second charged-lepton generation adds the positive mass-scale correction at alpha squared: both Fold labels over the depth-two complement 17, then one alpha return over the binary-distributed depth-three complement 2 times 53, then one alpha-squared terminal cell over four-rung support 16 times 53 plus down-depth five. This forces the exact added carrier alpha squared times $[2/17 + \alpha/106 + \alpha^2/853]$.

*The exact muon anomaly is the terminal electron anomaly plus $\alpha^2 [2/17 + \alpha/106 + \alpha^2/853]$, with $17=23^2-1$, $53=23^3-1$ and $853=16*53+5$.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003
- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis product of universal base, muon generation, coupling order, depth-two complement, depth-three return, terminal support, composition, target custody, provenance and extension forms. The declared exact boundary is: All first terminal positive mass-scale corrections to the complete electron loop for the second charged-lepton generation, using depth-two/three complements, Fold labels, four-rung support and down-depth exactly once. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: complete-terminal-electron-loop__second-charged-lepton-depth__two-charged-end-alpha-square__both-labels-over-depth-two-complement__one-alpha-over-binary-depth-three-complement__four-rung-times-depth-three-plus-down-depth__complete-positive-sum-appended-once__target-inaccessible-until-prediction-seal__registered-observational-prediction-protocol__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
base	complete-terminal-electron-loop	independent-fitted-muon-base	A separate base loses electron/muon correspondence.
generation	second-charged-lepton-depth	selected-generation-label	A label alone cannot force a carrier.
order	two-charged-end-alpha-square	selected-alpha-order	A selected order can tune the anomaly.
two	both-labels-over-depth-two-complement	free-leading-muon-weight	A free weight is fitted.
three	one-alpha-over-binary-depth-three-complement	free-successor-weight	A free return is fitted.

Axis	Forced form	Eliminated form	Elimination reason
terminal	four-rung-times-depth-three-plus-down-depth	selected-terminal-denominator	A selected denominator is fitted.
composition	complete-positive-sum-appended-once	signed-or-cancelled-series	Cancellation imports signed proof values.
target	target-inaccessible-until-prediction-seal	measurement-readable-execution	That cannot issue a target-inaccessible prediction.
provenance	registered-observational-prediction-protocol	unregistered-fit	An unregistered fit lacks custody and prediction receipts.
extension	no-extra-rule	extra-muon-term	Depth two, its successor and terminal cell exhaust the registered muon grammar.

Base and successor. The closure proof is sealed by derivation hash `sha256:e548339e60cec55e97e251b8a6d87bf45438161236d2975cce969b1b699cd5b1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:10a2e4a82eb07ef38e6087da7c06c66650945ce4b639df23ef71a2dde2502d86`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:de33e0625fde4d4b5ce4bef9b77098f827c6633db8de0b93119ba13a62a1773a`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:63b3a7c81f925eb38705876db1b21d3437e0d29d9a9330b7b23d6a452c5502a6`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:dd6d713d8248a0c754b67ca8dd4bf5fc7fb70b7bd6bb5de63933f47489cbd8a6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7540aebff9282350d9117b62faf26ec592c8c187c0c870d64985bea3fab0e0b`. Independent certificate: `sha256:7e90529df25555089225ae8f5d7c3dbfb5e5b0ecaaaf2b5af9fcdd65a6b6e470`. Engine external-validation hash: `sha256:61dd5befefb3ff2afee04e2182439a727a391b88cf5b9f8cf4233dac0fc83090`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026

Observed comparison records:

- MUON-MAGNETIC-ANOMALY-COMPLETE: predicted terminal-target-inaccessible-muon-anomaly-prediction-inside-complete-Fermilab-world-average-interval__observational-prediction-protocol-passed; observed terminal-target-inaccessible-muon-anomaly-prediction-inside-complete-Fermilab-world-average-interval__observational-prediction-protocol-passed; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The exact sealed muon prediction leaves the complete registered interval, the target becomes readable before sealing, a measured value selects the survivor, an uncertainty changes, or observational provenance is hidden.

Measurement receipt: sha256:c7a96120cd4417a4c2ed2c0017302fae283a054f57360a89a5b43ab7edcbb3ad.

Isolation certificate: sha256:549d8ef04e06e174dbe1f88f250bb09073f24515c7bc8c918259055d243495af.

Custody certificate: sha256:95dde804edf72a44a0f9b7ca077455ee3778f40b5797e8bfaa1d1ea60db0e3a6.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants and Fermilab Muon g-2 Collaboration: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>; <https://muon-g-2.fnal.gov/result2025.pdf> (sha256:4b506052fc5037576cd6ba4de8ee5afce74460f7a5cb8da4bc58cbb823f6732f)

Registered target identities:

- MUON-MAGNETIC-ANOMALY-COMPLETE from MATTER-FLAVOUR-TERMINAL-ANOMALY-AUTHORITATIVE-2022-2026

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no muon measurement in the executable relation
- no imported QED or consensus anomaly
- no fitted mass scale or coefficient
- no negative, irrational, imaginary or floating proof value
- no extra terminal term

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7cf1d27c83bb525f718b3e39b081d7055af231b5098a3160a117479fc1b476e9; engine receipt

sha256:8f5637b4d0189746e09a6e7d05937cefeef8d493c095e90162c3b3c8e458dd5a at receipts/engine/model_admitted/SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004-8f5637b4d0189746.json;

empirical-validation hash

sha256:fb81d1a4113a2a96a692624b1b55b6bb57d00d09fe913b2e927d46e4f71922a4; measurement receipt

sha256:c7a96120cd4417a4c2ed2c0017302fae283a054f57360a89a5b43ab7edcbb3ad.

124. Terminal finite spin-statistics, occupation and Bose-condensation law

Claim identity: SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For positive finite N and L , preserving exchange has exactly $C(N+L-1, N)$ occupation words and admits N occupants in one cell for every N ; alternating exchange has exactly $C(L, N)$ words for $N \leq L$, an empty support for $N > L$, and never exceeds one occupant per cell. Every finite canonical occupation weight is exact rational and sums to One. The two-label spin census is $3/4$ preserving and $1/4$ alternating; alternating spin first returns after two turns, while a pair is preserving and returns after one. For every N, L and $m \geq 2$, the ground-orbit share crosses $(m-1)/m$ at a uniquely first finite cold depth, fixing an exact rational critical mean throw; the unique minimum-throw preserving word places all N occupants in the shared ground orbit.

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preserving and returns after one. For every N, L and $m \geq 2$, the ground-orbit share crosses $(m-1)/m$ at a uniquely first finite cold depth, fixing an exact rational critical mean throw; the unique minimum-throw preserving word places all N occupants in the shared ground orbit.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-HALF-ONE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006
- SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008
- SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043
- SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041

Generated grammar and closure boundary. Generate the complete product of exchange, exclusion, accumulation, occupation-weight, spin-return, pairing, lock-threshold and ground-orbit forms. The declared exact boundary is: Every positive finite particle count N , positive finite level count L , preserving or alternating exchange class, positive finite cold depth, and Fold factor m at least two; empty cells are typed structural-absence records. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `preserving-versus-alternating-held-trace__one-cell-distinction-exclusion__every-positive-finite-multiplicity-admitted__complete-exact-dyadic-occupation-census__typed-one-turn-two-turn-held-orbits__alternating-pair-preserves-exchange__first-exact-crossing-of-forced-m-minus-one-over-m-share__unique-minimum-throw-shared-ground-word`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
exchange	preserving-versus-alternating-held-trace	named-particle-species	A species name does not force its exchange law.
fermion	one-cell-distinction-exclusion	selected-occupancy-cap	A selected cap is a free rule.
boson	every-positive-finite-multiplicity-admitted	finite-selected-ceiling	No generated ceiling follows from preserving exchange.
weights	complete-exact-dyadic-occupation-census	continuum-statistical-distribution	A continuum distribution is outside Fold arithmetic.
spin	typed-one-turn-two-turn-held-orbits	imaginary-signed-phase	Imaginary or signed proof scalars are inadmissible.
pairing	alternating-pair-p-reserves-exchange	independent-boson-postulate	An independent postulate breaks the single derivation chain.

Axis	Forced form	Eliminated form	Elimination reason
lock	first-exact-crossing-of-forced-m-minus-one-over-m-share	measured-or-fitted-temperature	A measured or fitted onset would select the law.
ground	unique-minimum-thrown-shared-ground-word	asserted-macroscopic-ground-mode	Naming a condensate is not an enumeration.

Base and successor. The closure proof is sealed by derivation hash `sha256:5ee1579388bfe3edb1a8772aee7d8c0c09a7eb05308730458a52e33091e85f95`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:605f4063b74d23be764144a1028dc3b1319b22982a21a8c6b3ed31a9d1ed22d3`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:890c659e1760dc96d83ab9ee8051afa91a93668f46e03415fb11ba1e9ecbd6d4`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0a855eb9c0225038937d4f3f46327ecc7dc1005cb7c237489b76ee28b7435148`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:dedd5d8e13991355ca1ba3ec94cd7c41b87841659d51696b3cf828e1f85d3fdf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:274d954c4cd12e6b469d51421e40869b90a3aef3fcc2f7eea6a22b54d18442c1`. Independent certificate:

`sha256:18f0ad787e6f6a48fb7eed4308f6dce85e09c74d6b815b4c60db0b81df9de0ed`. Engine external-validation hash:

`sha256:cd4226bd92f2cc114d70530b517280e799828647c49c75afc8d5af64ff4e9bab`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no Fermi-Dirac or Bose-Einstein continuum function imported as a premise
- no measured transition temperature, particle species or 360/720-degree observation available to candidate selection
- no selected occupancy cap, chemical potential, thermodynamic limit or infinite population
- no ontic randomness

- no conventional numerical-nothingness, negative, irrational, imaginary, floating, NaN or continuum proof magnitude; visible 0 in an occupation vector denotes an empty One cell

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:231ff1d3b8535154ad55b040e0317d77b3f2f2059a8ec02cc46eb1dad41782d3; engine receipt sha256:f71da5b86f99d6569a1f33dc6fc37024cc5d458b12e625c5dbc4faf3c33ccda7 at receipts/engine/model_admitted/SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045-f71da5b86f99d656.json; empirical-validation hash None; measurement receipt None.

125. Terminal preparation-depth, support-uncertainty and joint-correlation law

Claim identity: SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every complete dyadic preparation $N=2^k$ and every nonempty position support, exact Fold-parity observation forces $s_{ts_f} \geq N$, unit-free support-spread product $(s_{ts_f})/N \geq \text{One}$ and squared normalized support-spread product at least $1/N^2$; depth two attains $1/16$ and an eight-branch preparation uniquely has depth three with branch unit $1/8$. The complete 2-by-3 census has 63 nonempty supports: 21 factorable and 42 nonfactorable. The complete 3-by-5 census has 32767: 217 factorable and 32550 nonfactorable. Both full products are factorable, correcting their prior relabelling as entanglement; entanglement is the nonfactorable class. Complete projection preserves local counts under remote relabelling. All sixteen local deterministic Bell-response pairs obey the three-of-four bound, while setting-inclusive joint support can exceed it with invariant exact local marginals and no signalling.

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Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ALGEBRA-001
- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-QUANTUM-MEASUREMENT-001
- SFT-PHYS-QUANTUM-INCOMPATIBILITY-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-QUANTUM-BELL-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001

Generated grammar and closure boundary. Generate the complete product of preparation depth, Fold-parity observation, support bound, spacing-weighted spread, joint factorability, projection, Bell-factorization and extra-rule forms. The declared exact boundary is: Every nonempty subset of complete binary support through exhaustive depth four with depth-independent

orthogonality induction; every nonempty 2-by-3 and 3-by-5 joint subset; all sixteen deterministic local two-setting response pairs; and the complete setting-inclusive two-outcome support. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `preparation-derived-complete-binary-depth__held-returned-parity-count-table__orthogonality-Parseval-support-count__exact-unit-free-support-spread__complete-factorability-subset-census__complete-marginal-count-invariance__incomplete-local-factorization-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
depth	<code>preparation-derived-complete-binary-depth</code>	<code>universally-selected-depth-three</code>	A universal depth is an unforced parameter.
observation	<code>held-returned-parity-count-table</code>	<code>imported-complex-Walsh-amplitudes</code>	Complex amplitudes are outside the proof grammar.
uncertainty	<code>orthogonality-Parseval-support-count</code>	<code>named-or-selected-uncertainty-bound</code>	Naming a prior theorem does not force it.
spread	<code>exact-unit-free-support-spread</code>	<code>support-width-called-statistical-variance</code>	Support extent is not automatically a statistical central moment.
joint	<code>complete-factorability-subset-census</code>	<code>product-size-relabeled-entanglement</code>	A full Cartesian product factorizes exactly.
projection	<code>complete-marginal-count-invariance</code>	<code>remote-label-change-relabeled-signal</code>	A bijective relabelling changes no local count.
bell	<code>incomplete-local-factorization-record</code>	<code>ontic-randomness-or-superluminal-message</code>	Neither is forced by the finite correlation census.
extension	<code>no-extra-rule</code>	<code>measurement-selected-correction</code>	A fitted correction would select the law.

Base and successor. The closure proof is sealed by derivation hash `sha256:b87466e8c836feaea657b3d35f530a9de80e70f92fb663dd7a449b119c7fb469`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:f62a278d0ac5036bccf72d4a00e15ad07756b8bb607da8a238cb7c4886b34b63`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a3c93700d600582544c0cb2f801144a1d23eee3c2ca588338760d5aab12bd18f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:090198dc748d1481a485cefd5c34157d91fd5f1e9346d266361e10f6f1085502`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:b6008e573ace6486049207789bfcabb1459212302db171f5fa384ad7a819c2ad`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e78b5c1aaf806e5969295fa2d7cea0f8bea225528621eca464e3c54cba1a5afc`. Independent certificate: `sha256:25877a95c51e203cd290ecafaffaa47587c6082300cd3454b337b60470a1344d`. Engine

external-validation hash:

sha256:b3339e6b86d61ef06190b2f4cc4c1dd778e3b039e34c0d6b35cf2bd7701176bf.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Physical quantum and relativistic theory provide empirical correspondence for exact Fold support, phase, composition, observation and causal propagation. Complete-state evolution remains deterministic. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported Fourier, Donoho-Stark, Heisenberg, tensor-product or Bell theorem as a premise
- no universal depth-three measurement rule independent of preparation
- no relabelling of support extent as statistical moment variance
- no relabelling of a complete Cartesian product or product-greater-than-sum arithmetic as entanglement
- no claim that fifteen is irreducible to its three-by-five product factors
- no ontic nondeterminism, stochastic setting oracle, superluminal signal or omitted preparation/setting record
- no measured Bell or uncertainty value accessible to candidate selection
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:246abe6ea59ec47581adf7eb2cc9552ad5cf703fe88d5e65b404a8d1ad83a672; engine receipt
sha256:1560f2e0de3870abac2bdc6575aa9811c4dc013d5a6d705e729a832dc451b79f at receipts/engine/model_admitted/SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049-1560f2e0de3870ab.json; empirical-validation hash None; measurement receipt None.

17. Constants Scales Precision

Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt.

126. Terminal exact inverse fine-structure Fold ratio

Claim identity: SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The binary predecessor and generator-three recurrence force cover depths five and seven. Their typed tower, boundary, complete three-direction cover, returning One and finite one-direction-at-a-time promotion ladder force the exact terminal ratio 503846395469/3676744786 before measurement.

The terminal inverse fine-structure Fold ratio is exactly 503846395469/3676744786; its leading rung is exactly 34259/250.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete typed product of cover, depth, tower, boundary, return, join, promotion, termination, target custody and extension forms. The declared exact boundary is: All type-correct assemblies that use each forced binary/generator cover block in its generated role, admit only the One as recurrence correction, and promote each of the three interchangeable cover directions at most once. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `least-complete-binary-covers__cover-generator-cross-lock__binary-at-up-depth__generator-at-boundary-rank__one-return-over-complete-cover__tower-plus-dilated-boundary__one-remaining-direction-per-rung__all-three-directions-promoted__measurement-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
cover	least-complete-binary-covers	asserted-depths	Asserted depths have no support census.
depth	cover-generator-cross-lock	single-route-depths	One route cannot expose a changed carrier.
tower	binary-at-up-depth	free-leading-whole	A free leading whole is a parameter.
boundary	generator-at-boundary-rank	free-boundary-factor	A free coefficient is not structural.
return	one-return-over-complete-cover	fitted-small-correction	A fitted correction reads the target.
join	tower-plus-dilated-boundary	untyped-arithmetic-permutation	Permuting typed carriers erases their roles.
promotion	one-remaining-direction-per-rung	selected-refinement-series	A selected series adds unfixed terms.
termination	all-three-directions-promoted	infinite-or-truncated-by-fit	A fit-selected truncation is not closure.
target	measurement-inaccessible-until-seal	measured-alpha-visible	That makes measurement an answer-bearing premise.
extension	no-extra-rule	extra-scale-rule	An extra scale is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9dfbee307fc2a9b96180abd41ec62bdf2a131a6e7aa702951e7acbf0ad0459b1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: ca5b4e49ab3bedece92f0ec11667e4305baafe8624283410439868e1b79e151c.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 0d516338a2c561300eae04d43e460a74cae268ab52d41ce1ace635214b048ab5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 937d252d67fbb9ad1f3a8bcbdb5000226d4d7bde3b511429cbe0ee917e4986843.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: debfd802d48d0d9c9b5e1df0a6bdbf950f19bd012c0bdf52cfeec2505494619b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 97d0cdb7077804e63fb1bab56244287f15fbc1ee6fdeece167b59a309604117f. Independent certificate: sha256: ec1a7c48b4269d74564b029d286350d600f1feca69383bf655ad7415714f71f4. Engine external-validation hash: sha256: d2a29ed3e7a828cd1adc21abfff803d43cce478e12935dc83818e0a79af26a67.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 624239a4a1a47a06acec655b1a55db66675f5d32fe9f67c6a4305d52eb56b99a; engine receipt
sha256: 4999a85b5862afebcbf24604c2191c0ce8b96aa57f988b3fab83b469c3506a7c9 at receipts/engine/model_admitted/SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001-4999a85b5862afeb.json;
empirical-validation hash None; measurement receipt None.

127. Exact charged-lepton cubic invariants

Claim identity: SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The forced generator-three partition supplies one exact three-root mass carrier whose symmetric invariants are root sum One, pair sum one over the two Fold fibres paired with generator three, leading product one over the positive predecessor of twice the colour fifth-power support, and sharpened product obtained by the unique generator-three channel part. The resulting exact

invariants are 1, 1/6, 1/485 and 3/1454; no irrational root is formed.

The exact charged-lepton cubic invariant tuple is (sum=1, pair-sum=1/6, leading-product=1/485, sharpened-product=3/1454), with the sharpened colour channel uniquely retained over channels two, three and four.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ALGEBRA-001

Generated grammar and closure boundary. Generate the complete product of cubic carrier, generator, partition, pair invariant, leading support, predecessor, sharpening channel, sharpening form, root boundary and extra-rule forms; enumerate channels two, three and four without measurement access. The declared exact boundary is: All exact positive-rational symmetric three-root invariant constructions made from the admitted Fold fibre count, generator three, complete colour fifth-power support, positive predecessor and one neighbouring generated channel part. The generator produced 2304 named candidates and the decision support contains 2304 one-for-one decisions. Exactly one candidate survived: `one-complete-symmetric-three-root-carrier__admitted-generator-three__root-sum-is-the-One__one-over-two-colour__twice-colour-fifth-power__positive-predecessor-of-support__colour-channel__positive-take-one-channel-part__retain-exact-symmetric-invariants__no-extra-rule`. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	one-complete-symmetric-three-root-carrier	three-names-without-joint-invariants	Names do not retain one cubic relation.
generator	admitted-generator-three	selected-three	A familiar generation count cannot be assumed.
partition	root-sum-is-the-One	unnormalized-root-sum	An unnormalized sum adds a free scale.
pair	one-over-two-colour	one-over-colour	This omits the two Fold fibres.
support	twice-colour-fifth-power	colour-fourth-power	This stops before the complete fifth colour succession.
predecessor	positive-predecessor-of-support	support-count-itself	The complete support retains the terminal One rather than the massive interior.
channel	colour-channel	binary-channel	The binary channel is not the carrier's generator identity.
sharpening	positive-take-one-channel-part	add-channel-whole	A whole channel count changes scale rather than resolves one channel part.
roots	retain-exact-symmetric-invariants	solve-and-round-roots	Root approximation introduces brackets and stopping choices.
extension	no-extra-rule	extra-coefficient-or-measured-selector	An added number or target-selected channel is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f7852dece1ada6b22875ee694a35d8ffd9684c8c78d7184f762f86de35f99420`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 4d3a47f711ee8d5a5a8d8fdf4560ffaa1c3bf66dec3d732c3f8e4a487346f1c0.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: d3ed7f51b7c2112714cc8982209527b165276b283cc66442bfee554236d5254e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 6b6025c4aab652df50047f8a0beb5ebfdf807ba155258b19cd8c59f1ab05daa9.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: c20ff35677c6755b47078ff9779fca59f0c3e5e1b7413ae6ecbab57bb49ce3c0.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: 5903885dd237b470dd8f94410d2195eacca778ce56fc3e923f356a4322e94432.

Independent certificate:

sha256: 1b1ffd5bd2d7ab0367b2cc600933def92af65edd7479ed8ab5d33bf6eb4e0f37. Engine

external-validation hash:

sha256: 1092d58c7305f5ba8787b449888ceaaaf7294355ec8a88de3e52558e39577f2e2.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate or answer table in the derivation runtime
- no measured lepton mass or ratio may select a coefficient or channel
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted root, bracket, tolerance, iteration count or extra coefficient

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: c24fd6454b7ad9b50590f83fff13e4e005094537cd495720210047085a28056f; engine receipt

sha256: d45737a631dbf57c19fb433817cdb6e443f9a6e598f7d0cbb469d47819ee3ae2 at receipts/engine/model_admitted/SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001-d45737a631dbf57c.json;

empirical-validation hash None; measurement receipt None.

128. Terminal self-coupling refinement of the charged-lepton cubic

Claim identity: SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The admitted charged-lepton cubic is refined by the complete terminal electromagnetic self-coupling of its three-root product: alpha to the generator-three order is distributed over the complete three-cubed carrier and weighted once by the forced cover-depth pair five and seven, with the up-depth transported through one alpha and one colour channel. The exact correction is held from 3/1454.

The terminal product invariant is exactly 3/1454 held by $\alpha^{3/3^3}$ times $(5 + (7/3)\alpha)$, with alpha the reciprocal of the admitted terminal inverse fine-structure ratio.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ALGEBRA-001

Generated grammar and closure boundary. Generate the complete product of base invariant, interaction order, support, down-depth role, up-depth transport, channel distribution, correction orientation, target custody and extension forms. The declared exact boundary is: All exact first terminal self-coupling refinements of the admitted three-root product constructed from its generator order, complete generator volume, terminal alpha and forced down/up cover depths. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: retain-admitted-sharpened-product__one-alpha-per-three-root__complete-generator-volume__forced-down-cover-depth__one-alpha-transported-up-depth__one-complete-colour-share__hold-correction-from-product__measurement-inaccessible-until-seal__observational-derivation-disclosed__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
base	retain-admitted-sharpened-product	replace-cubic-product	Replacement erases the admitted cubic.
order	one-alpha-per-three-root	selected-linear-or-square-order	A lower selected order does not act once on every root.
support	complete-generator-volume	single-channel-support	One channel omits the complete three-root carrier.
down	forced-down-cover-depth	free-depth-coefficient	A free coefficient is a parameter.
up	one-alpha-transported-up-depth	bare-up-depth	A bare successor depth ignores transport between orders.
channel	one-complete-colour-share	undistributed-up-depth	An undistributed successor counts every colour channel as one.
orientation	hold-correction-from-product	append-unheld-product	Appending creates an unaccounted product carrier.
target	measurement-inaccessible-until-seal	measurement-readable-relation	That would fit the failed target.
provenance	observational-derivation-disclosed	mislabel-as-blind-forward-discovery	The failed comparison was already observed.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	extra-fit-term	An additional term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash

sha256:9dbba8bd548a404e48c29b19742b586a071c3883e6950d4496ca6f16bf8ef54c. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:33b65a0122091153ecdbdc9e848361db0f7e008ac0b95d3b1a90ae77b8693346.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:f9e2d38d95ab449a15dcdee72eec078045e21abea84bbae18516bc9ef7166e2a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a46afb9f969a7e23e67abe378cd7479d441a69d41227779f0b7d65dbda921e56.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:18a7245b2f1e01acfae3356fbd8e7aab39fb5d715893dd98f85030edbfef8f7f4.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:99ce35598da99fedf661a7d92e1bfc40adeb85861204eb5196a13aa30f7a0746.

Independent certificate:

sha256:05d0e8eebbcb2b6dcff81df8eef6f8f3aa7f9fa838ed210a6b035720858fc4c3. Engine external-validation hash:

sha256:65703bcf3b046bf3d385c0e9177a6d8bdabb3afcc63a226e0a86d8ee615fe2f.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-MUON-ELECTRON-MASS-RATIO: predicted terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; observed terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; exact match `True`
- NIST-CODATA-2022-MUON-TAU-MASS-RATIO: predicted terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; observed terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: Either terminal prediction interval fails to overlap its complete source interval, source custody changes, or any failed row is omitted.

Measurement receipt: sha256:0c73be05888de167157643d1b1773ade8b7cb32fa0c4eb0fcc0e8aeef4c4b731.

Isolation certificate: sha256:ce6192af110a16a013d2a9733358725fe8f18cabcb61d32634529ee0cdb5bb049.

Custody certificate: sha256:0d6a80603ccf392a0caa760b048f143bc620c60b50241eb632b8e1e1e3b76014.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured lepton mass or ratio in the executable relation
- no floating, irrational, imaginary, semantic-zero or negative proof value
- no fitted coefficient, tolerance, selected root or extra correction term
- no claim that this post-failure construction was a blind forward discovery

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:15f02146523652fee8ca84a6d2d023e1a9ba5e8764956c7cf6c221ded12db04d; engine receipt
 sha256:c74f9c45eab7c232ebf85fe2fd5aea24f07d167df3857dad50ffcc5c34732294 at receipts/engine/model_admitted/SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001-c74f9c45eab7c232.json;
 empirical-validation hash
 sha256:3df224f1ca64f8ccc26c3fd5d785d4ee2abdea51c783550acce714a96eb52d5; measurement receipt
 None.

129. Squared proton-to-Planck hierarchy

Claim identity: SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Generator-three successor volume has least binary cover depth seven. Complete binary support at that depth contains 128 states and its unique positive predecessor contains 127 massive positions. The half-One gravity carrier therefore makes the unsquared exponent $127/2$; no irrational root is formed, and the exact forced comparison object is the squared hierarchy 2^{127} .

The exact squared Planck/proton hierarchy is $2^{127} = 170141183460469231731687303715884105728$; the unsquared exponent is retained as $127/2$ and no square root is formed.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of scale carrier, cover dependencies, full-support predecessor relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: The complete binary support at the least cover of generator-three successor volume, its unique positive predecessor and the half-One coupling, with squared comparison only. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-dependency-trace__depth-seven-support-predecessor-at-half-One__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore

a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	depth-seven-support-predecessor-at-half-One	formed-irrational-root-or-fitted-hierarchy	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:5d2f08b5bd30059644269313d7cf73d173e43997a6468844a174ed239ed9f278`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:b549ad7c5df8d84d32a49d45dc61b0945f14d569357542f2c56c9429c07fbcf5`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:578efe7a44cf4d163751a387457c2a7da657fb15273bc86a041f4d076216195c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a780f43acd149fd24c88aef7fb706fc433b85dae8864d13b618ed88c488b339f`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f7d411c65f442f9ef1a9dad47831ca171d1d44522bc8b142b2c69d3db4cab9b8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate:

`sha256:ae1fdcb1dde94d52082aae299929e6590d7e392cfd5650004fa19d075e7dcddd`. Engine external-validation hash:

`sha256:d3bb3974eb182a550c2e34f963a9c47fa105747b4186d187c5d86fac25e86e59`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no irrational square-root formation; empirical comparison must square the measured positive interval instead

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d847dbf538c92b07c9f710c9f33887685b40dca9eef56f3943e855ac2f49b5bb; engine receipt
sha256:d1cf13e86e2d9aeaa5f6993456b5f594a303d75e446ea1d9c80d41906b984d9f at receipts/engine/model_admitted/SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002-d1cf13e86e2d9aea.json;
empirical-validation hash None; measurement receipt None.

130. Terminal charged-colour proton-to-Planck hierarchy

Claim identity: SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The admitted lower hierarchy counts complete depth-seven massive support as 2^{127} in the squared comparison domain. A proton is the hierarchy carrier that retains both colour labels over generator three and one terminal electromagnetic self-coupling. Transporting that exact $2/3$ alpha share once and retaining its positive One-complement forces the terminal squared hierarchy before measurement.

*The terminal squared Planck/proton hierarchy is
25592393460381748800840516069019941843257249470880/1511539186407, equal to 2^{127} times $(1 - (2/3)$
 $\alpha_{\text{terminal}})$.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002
- SFT-PHYS-NUCLEAR-COLOUR-COUPPLING-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis product of lower hierarchy carrier, terminal stage, complete support, charged-colour relation, alpha transport, retained orientation, enumeration, minimality, measurement direction and extension. The declared exact boundary is: All exact squared hierarchy successors assembled from the admitted 2^{127} support, the admitted two-over-three colour carrier and exactly one terminal electromagnetic self-coupling, with the bare hierarchy retained as a lower-order control. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: admitted-lower-law-carrier__complete-promotion-termination__complete-binary-terminal-support__bare-hierarchy-times-One-complement-of-two-thirds-alpha__single-typed-terminal-alpha-transport__orientation-fixed-by-share-or-retention__complete-registered-product__bare-and-intermediate-forms-preserved__exact-result-sealed-before-target-release__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	admitted-lower-law-carrier	detached-measured-number	A measured number has no forward Fold dependency trace.
terminal	complete-promotion-termination	selected-intermediate-stage	An intermediate stage chosen for agreement is a fit.
support	complete-binary-terminal-support	free-scale-support	A free scale is a parameter.
relation	bare-hierarchy-times-One-complement-of-two-thirds-alpha	free-hierarchy-exponent-or-untyped-correction	The alternative loses a required carrier or adds an unforced operation.
correction	single-typed-terminal-alpha-transport	fitted-additive-offset	A fitted offset reads the target.
sign	orientation-fixed-by-share-or-retention	target-selected-orientation	A target-selected sign is not structural.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	bare-and-intermediate-forms-preserved	successor-without-lower-controls	A terminal answer alone hides whether the lower form was necessary.
measurement	exact-result-sealed-before-target-release	target-visible-before-seal	That reverses empirical direction.
extension	no-extra-rule	free-extra-correction	An extra term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash
sha256:13aa1c5380e62ab000cab2f898e5dd6f7cfc1498d20da4fad9f6c8da470da133. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:f1523a85cb053c03b2ec81e22f6b82a870fcbdbc30f320e8a456043f04dae4af.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:404dfca4694089557d07640290478cded4b0e0e92f193fb5c090f99c9015f913.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b303be1b9682c63e30540ac74f82d51581cef9b2ea130460d417efb1ce732c4f.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:1ff75c1743c0e2f721404dc2311de33c6d45dcf0234d98a0f9149ecf42879a56.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:61bd427364f94266efdc7681e8bb7fb1211cf8a628b54ac1d8dab2b2841c504. Independent certificate:
sha256:b9425f0a9d11c8870d321a8252c8d951729310b3cf6f088ecad7fd1ae499ea5a. Engine external-validation hash:
sha256:c7af4edd3accefe40f6324979f2b366eed9106156270a7b98419aed2f80eaf71.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no external measured value, uncertainty or best-fit table as a derivation input
- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no fitted level, coefficient, correction sign, truncation or target-selected candidate neighbourhood
- no irrational square root; authoritative mass intervals are squared only after the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:5eca6d00fe54b7afbabb728a9dab3716786ae87790266d141e993221dddce0d6; engine receipt
sha256:affd1dc6754854b15d7f286291611e481f0572f50a11589dd88cb019289f88d7 at receipts/engine/
model_admitted/SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003-affd1dc6754854b1.json;
empirical-validation hash None; measurement receipt None.

131. Terminal colour-channel alpha-squared proton-energy fraction

Claim identity: SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The terminal structural proton-energy fraction is exactly $108147617771025486368/253861190227103943729961 = 8 \alpha_{\text{terminal}}^2$. The superseded leading-rung prediction $500000/1173679081$ is retained as an exact historical control. External proton mass and Parker observations remain inaccessible until the derivation seal.

The terminal structural proton-energy fraction is exactly $108147617771025486368/253861190227103943729961 = 8 \alpha_{\text{terminal}}^2$. The superseded leading-rung prediction $500000/1173679081$ is retained as an exact historical control. External proton mass and Parker observations remain inaccessible until the derivation seal.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of proton identity, alpha stage, all four forced mediator-sector counts, amplitude/energy/cubic powers, typed composition, rest-scale carrier, prediction custody, provenance, external comparison and extension forms. The declared exact boundary is: The admitted proton, all forced sectors two/three/five/seven and their mediator counts, the leading and terminal exact alpha rungs, powers one through three, and a post-seal proton-rest-energy/Parker comparison. The generator produced 3072 named candidates and the decision support contains 3072 one-for-one decisions. Exactly one candidate survived: `admitted-three-colour-proton__complete-terminal-alpha-rung__colour-eight-channels__alpha-squared-energy__complete-channel-count-times-energy-share__proton-own-rest-energy-postseal__capability-closed-before-Parker-release__observational-derivation-explicit__complete-reported-range-and-limitations__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
particle	<code>admitted-three-colour-proton</code>	<code>named-proton-without-composition</code>	A name does not generate its interaction support.
alpha	<code>complete-terminal-alpha-rung</code>	<code>selected-leading-alpha-rung</code>	The leading rung is retained as a historical control but is not the completed promotion object.
channels	<code>colour-eight-channels</code>	<code>weak-three-channels</code>	The binary-sector mediator count does not match proton colour.
power	<code>alpha-squared-energy</code>	<code>linear-alpha-amplitude</code>	A single amplitude is not an energy self-composition.
composition	<code>complete-channel-count-times-energy-share</code>	<code>add-or-average-channel-shares</code>	Addition or averaging changes the complete channel multiplicity.
scale	<code>proton-own-rest-energy-postseal</code>	<code>local-plasma-fit-or-free-energy</code>	A local field or density would be a fitted environmental parameter.
target	<code>capability-closed-before-Parker-release</code>	<code>Parker-range-readable-before-seal</code>	That would permit formula selection by the observation.
provenance	<code>observational-derivation-explicit</code>	<code>claim-new-blind-discovery</code>	V2 and the Parker report were already known during V3 reconstruction.
comparison	<code>complete-reported-range-and-limitations</code>	<code>treat-approximately-400-as-exact-cutoff</code>	The paper reports an approximate energy reach and no uncertainty on 400 keV.
extension	<code>no-extra-rule</code>	<code>free-local-factor-or-correction</code>	A free local factor would fit the event.

Base and successor. The closure proof is sealed by derivation hash `sha256:ba3f74d46f0fa4868ea25ce160affd879952128cb5dc07f451750f83598b38eb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:8a698572330616bbe1d93d09c5416ec8ec01e583ff1effdff412eb40647ff207`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:41ecf3347f1f4aa20f85a7e123fd66850f15f7934d3a2f4cc54fdab2dd949a29`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a6fe0f61a16b5f1e132529709d14156c2714bc883d3d10553f14cb207b45484b`.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : a6d0935252f735483db99db340353132969c9bb58a1ee253c53965bb1e111847.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256 : 9e8af8c582652edc7816af89b7d3254a39c2a7cabcd838f7371d5cdc8d36029a. Independent certificate: sha256 : 65903e28af5143824e401feddc2b1261fe3bdd81a8f13aa4e7ab9a3c8b50bc11. Engine external-validation hash: sha256 : 4adb9cda9fd0ffa1ba246a731e1009ea3a9d8ef1815222bacdac90ca92279438.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- DESAI-ET-AL-2025-PARKER-HCS-PROTONS
- NIST-CODATA-2022

Observed comparison records:

- The primary Parker manuscript, complete CODATA source and curated row record are hash-locked and released only after the exact Fold prediction seal.
- The terminal prediction from the complete current alpha rung is approximately 399.714077 keV after post-seal CODATA scale translation; the retained V2 leading-rung control is approximately 399.714072 keV.
- Both exact propagated prediction intervals lie inside the paper's complete 67-527 keV analyzed proton spectrum.
- The primary paper reports protons up to, above and upwards of approximately 400 keV; it does not register 400 keV as an exact cutoff and gives no standard uncertainty for that approximate label.
- Therefore the earlier 'within 0.1 percent' wording is not admitted as a precision measurement claim; only range-level correspondence is supported.
- The complete forced-sector/power control census is retained. Both three-alpha-squared and eight-alpha-squared land inside the broad observed range, so the Parker event alone does not select the eight-channel formula.
- Eight channels remain structurally selected by the independently admitted proton colour sector, never by the Parker target.
- Changing the external upper range to 300 keV rejects the prediction as an unfavorable control.

Falsification condition: Reject if any source identity or registered row changes; if the terminal fraction or NIST-propagated prediction leaves the complete 67-527 keV Parker analysis range; if approximately 400 keV is presented as an exact cutoff or given an unreported uncertainty; if the broad observation is claimed to empirically select eight over every adverse formula; if a local plasma value enters the derivation; or if targets are accessible before prediction sealing.

Measurement receipt: sha256 : f934b85069bf77672ff16ee8acf662add04f6e652d0fe2a3807076f7bc9a5f36. Isolation certificate: sha256 : 260875f77819464ff90d563f292a61190470d0d2f647199ccdb69270c40af7c9. Custody certificate: sha256 : d64484e3eef9294ffde93e19b562a48f1eb47bbd49e7e9990e0c78da943a2d6c.

Registered source descriptions:

- Astrophysical Journal Letters accepted manuscript via arXiv: <https://arxiv.org/abs/2410.16539>
(sha256:6c950a6e142edf57df600d06e1ce40445fc18c84d42c3d2b916d63679a100fb5)
- National Institute of Standards and Technology: <https://physics.nist.gov/cuu/Constants/index.html>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- WITHHELD-PARKER-PROTON-RANGE-AND-CODATA-SCALE from None

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim

specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no local plasma density, magnetic field, Alfven energy or guide-field parameter
- no proton mass-energy or Parker spectrum in the candidate generator or survivor decision
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude
- no claim that approximately 400 keV is an exact measured cutoff or has 0.1-percent precision
- no omission of alternative forced-sector/power formulas at the post-seal comparison boundary
- no claim of historical blindness; V2 and the external result were known before reconstruction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:003e85d7036062243cbc4a81865ab60247779d486516d64533b64f681eedb21; engine receipt
sha256:41f17af1f94905b0e93d67d7dfc7a399fd9063a53e6db4d21d486e1a8c5f1673 at receipts/engine/model_admitted/SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028-41f17af1f94905b0.json;
empirical-validation hash
sha256:ba3f659435ad35064dfb8c2e5fc085b84dc24edf6109f8d4090878e770a67c6b; measurement receipt
sha256:f934b85069bf77672ff16ee8acf662add04f6e652d0fe2a3807076f7bc9a5f36.

132. Terminal probe-independent proton rms charge-radius relation

Claim identity: SFT-PHYS-PROTON-RADIUS-TERMINAL-029

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The unique terminal dimensionless proton rms charge-radius coefficient is exactly $10069574419808/2519231977345 = 4(\text{One-alpha_terminal}/10)$. A dimensionful radius is obtained only after the seal by composing this coefficient with the registered reduced proton Compton wavelength. The coefficient contains no electron, muon or scattering-probe input.

The unique terminal dimensionless proton rms charge-radius coefficient is exactly $10069574419808/2519231977345 = 4(\text{One-alpha_terminal}/10)$. A dimensionful radius is obtained only after the seal by composing this coefficient with the registered reduced proton Compton wavelength. The coefficient contains no electron, muon or scattering-probe input.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-NUCLEAR-COLOUR-COUPPLING-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-EVOLUTION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of proton carrier, tripling orbit, colour/Fold edge support, internal/external charge support, electromagnetic transport order, retained orientation, one-action scale boundary, target

custody, provenance and extension forms. The declared exact boundary is: The admitted colour-three charged proton, its one-third/two-thirds tripling partition, both Fold labels, every ordered colour pair cell, the unique external unit charge, terminal alpha and the post-seal reduced proton Compton carrier. The generator produced 2304 named candidates and the decision support contains 2304 one-for-one decisions. Exactly one candidate survived: `admitted-uud-colour-three-proton__outer-two-thirds-complement__all-three-colours-times-both-Fold-labels__ten-internal-pairs-plus-unit-charge__one-alpha-spatial-traversal__hold-inward-share-from-edge__reduced-proton-Compton-postseal__capability-closed-before-target-release__observational-derivation-explicit__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
particle	<code>admitted-uud-colour-three-proton</code>	<code>named-proton-without-composition</code>	A particle name does not generate a radius.
orbit	<code>outer-two-thirds-complement</code>	<code>inner-one-third-as-outer-edge</code>	The inner constituent site is not the composite edge.
edge	<code>all-three-colours-times-both-Fold-labels</code>	<code>partial-colour-or-single-fibre-support</code>	Omitting a colour or Fold label loses a charged path.
charge	<code>ten-internal-pairs-plus-unit-charge</code>	<code>eight-nonreturn-mediators</code>	The mediator count omits the return and external charge roles.
transport	<code>one-alpha-spatial-traversal</code>	<code>no-terminal-transport</code>	This retains only the earlier leading structural radius.
orientation	<code>hold-inward-share-from-edge</code>	<code>append-outward-free-share</code>	Appending a free share breaks the held bound-composite ledger.
scale	<code>reduced-proton-Compton-postseal</code>	<code>full-cycle-or-fitted-length</code>	A fitted length or full-cycle circumference double-counts the one-action radial carrier.
target	<code>capability-closed-before-target-release</code>	<code>probe-results-readable-before-seal</code>	That would fit the known radius puzzle.
provenance	<code>observational-derivation-explicit</code>	<code>claim-historical-blindness</code>	V1/V2 and the measurements were already known.
extension	<code>no-extra-rule</code>	<code>free-form-factor-or-probe-correction</code>	An added term would be a fitted parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:1f5b55c75de20a987a553421413cf4ad54e259b5da55649402bce332cf7d926f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:a7a6a49aa74db2977d54690d7a0ce9f676adb4bcc085662d318f3380d6d3a760`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:b2b88486d92413ea48792aecfd581c0f6708d599a9e42be2cc007085c3604811`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0990ec3b4ad7b9cd1ce8c9d4bd44fdb8747b9ef8e7ef9846327c49c945ccfa50`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ca75083edfef55fc18c3d5e70d7091287a3eac7767e4bba931ae56b02e6002fc`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
 Implementation hash: sha256:b7ae5cd32b920bc549c3cb64b718a2ea64732691be02ccf8cb0118764207f7de.
 Independent certificate:
 sha256:1bdcc6ce6b78572a95ca6fc1b89b728840d2624720824904ffb9b1f1179ace31. Engine
 external-validation hash:
 sha256:c30497188e21b2cf74496f02ce05fd120478ff49a586f643a660ac364c916451.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NATURE-2026-ELECTRONIC-HYDROGEN-PROTON-RADIUS
- NATURE-2019-PRAD-PROTON-RADIUS
- NIST-CODATA-2022
- NIST-CODATA-2014-HISTORICAL

Observed comparison records:

- The exact coefficient and all structural alternatives seal before the four primary/authoritative source identities and radius rows are released.
- The exact terminal coefficient is $10069574419808/2519231977345 = 4(\text{One-alpha_terminal}/10)$, with no probe or radius measurement in its derivation.
- Post-seal composition with the complete CODATA-2022 reduced-proton-Compton interval predicts approximately 0.840621761 fm with exact outward uncertainty propagation.
- The prediction interval lies inside the registered 2026 electronic-hydrogen 0.8406(15) fm interval and the current muonic-hydrogen 0.84060(39) fm interval.
- The prediction interval lies inside the CODATA-2022 0.84075(64) fm interval and the conservative PRad 0.831 +/-0.007(stat) +/-0.012(syst) outward interval.
- The historical CODATA-2014 0.8751(61) fm interval is disjoint and remains an explicit unfavorable historical row.
- The primary 2026 paper's partly discrepant earlier electronic results and scattering-analysis dependence remain explicit; probe independence is not relabelled as universal agreement among all extractions.
- All three linear-support alternatives land inside the present muonic interval, so measurement does not select ten; the complete proton charge-support derivation does.
- Changing the current electronic-hydrogen row to a disjoint interval rejects the correspondence.

Falsification condition: Reject if any source identity or registered row changes; if the exact terminal prediction leaves any registered current electronic, muonic, CODATA or outward PRad interval; if the historical CODATA-2014 conflict or current extraction limitations are omitted; if a probe, radius or Compton value enters before sealing; if observation is claimed to select ten charge cells; or if target access precedes the seal.

Measurement receipt: sha256:6a828b91cce631a0caa5a33a44284ade02f779931199d0e9fc96e42ecfd3607f.
 Isolation certificate: sha256:f9112b893f35f1a69d98ea1e06da8339cd9df96de0c830502c76fe6896853a6b.
 Custody certificate: sha256:7bf579b9c3ac12c02353d9692b0a29ccaede0dcc3e7ff0599a263d402c23ca06.

Registered source descriptions:

- Nature: <https://doi.org/10.1038/s41586-026-10124-3>
 (sha256:834ddf04fdb3064d07307cc743544990e086601948903fb73390144d2a3ae253)
- Nature manuscript via arXiv: <https://doi.org/10.1038/s41586-019-1721-2>
 (sha256:66925b870f8391be1a940b741ff60322877df7551cfa1db58b59add002e15ad5)
- National Institute of Standards and Technology: <https://physics.nist.gov/cuu/Constants/index.html>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

- National Institute of Standards and Technology: <https://physics.nist.gov/cuu/Constants/archive2014.html> (sha256:48c31408c79d3a9aa7c16efdbf5ef729d8ad162f2b9dbb8029d4d8cdcbe3568)

Registered target identities:

- WITHHELD-PROTON-RADIUS-COMPLETE-VECTOR from None

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no proton radius, reduced Compton wavelength, electron, muon or scattering result in the candidate generator or survivor decision
- no fitted coefficient, probe-dependent correction, selected uncertainty or hidden form factor
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude
- no claim that rms charge radius is a hard material edge
- no claim that every historical extraction is mutually consistent or systematics-free
- no historical-blindness claim

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d39039deed7c3890466a520e0c899d59b6dc8e7b70600f46697fb153e0142e37; engine receipt
sha256:77a31b069cc1cdd4bdf84f3ca08fb4849ae33fddb02945727f1809d9e4c2373d at receipts/engine/model_admitted/SFT-PHYS-PROTON-RADIUS-TERMINAL-029-77a31b069cc1cdd4.json;
empirical-validation hash
sha256:39c3b0612ff4b5ddc4ec5c86b7194a9d3f814b3f3af85dc4dad38d03093d8465; measurement receipt
sha256:6a828b91cce631a0caa5a33a44284ade02f779931199d0e9fc96e42ecfd3607f.

133. Common Fold scale axis and terminal electroweak transport

Claim identity: SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. One exact common scale axis is forced: supports 1,2,4,8,...; spacings One, half-One, quarter-One, eighth-One,...; and every forced prime sector uses the same support. The electroweak leading curve is $w(R)=(R+2)^2/[4(R+1)^2+(R+2)^2]$. Terminal support sixteen holds three directions, forcing active level thirteen, $w(13)=225/1009$ and the separately typed $\alpha/17$ return, exactly reproducing the admitted terminal on-shell share. The unique internal power anchor is support two. Common exact unit rescaling leaves every dimensionless result unchanged; physical magnitudes are transported only through sealed exact ratios and a post-seal held reference.

One exact common scale axis is forced: supports 1,2,4,8,...; spacings One, half-One, quarter-One, eighth-One,...; and every forced prime sector uses the same support. The electroweak leading curve is $w(R)=(R+2)^2/[4(R+1)^2+(R+2)^2]$. Terminal support sixteen holds three directions, forcing active level thirteen, $w(13)=225/1009$ and the separately typed $\alpha/17$ return, exactly reproducing the admitted terminal on-shell share. The unique internal power anchor is support two. Common exact unit rescaling leaves every dimensionless result unchanged; physical magnitudes are transported only through sealed exact ratios and a post-seal held reference.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-COUPPLING-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003
- SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003
- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

Generated grammar and closure boundary. Generate the complete product of foundation origin, scale successor, order/spacing relation, local propagation transport, sector domain, electroweak squared-support form, terminal held-support completion, internal anchor, unit transformation, absolute-ratio placement, target custody, provenance disclosure and extension form. The declared exact boundary is: The One, every positive finite binary support successor, reciprocal exact spacing, one-cell causal transport, all four forced prime sectors, binary charged and neutral Fold fibres, terminal support sixteen with three held directions and one alpha return, every common positive rational unit rescaling, the exact proton-Planck scale ratio and the complete registered post-seal coupling vector. The generator produced 12288 named candidates and the decision support contains 12288 one-for-one decisions. Exactly one candidate survived: One-origin__binary-complete-support-successor__reciprocal-One-over-support-spacing__one-local-act-per-support-cell__complete-prime-sector-ladder__neutral-square-over-charged-plus-neutral-squares__support-sixteen-hold-three-return-alpha-over-seventeen__unique-support-two-source-four__common-positive-rational-rescaling-cancels__forced-ratio-then-postseal-held-reference__capability-closed-before-target-release__observational-development-explicit__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
origin	One-origin	numerical-zero-origin	Semantic numerical zero is not an SFT scale carrier.
successor	binary-complete-support-successor	linear-or-continuous-free-step	A free or continuous step is not generated by Fold.
spacing	reciprocal-One-over-support-spacing	untyped-arbitrary-spacing	Arbitrary spacing loses the common exact ratio.
transport	one-local-act-per-support-cell	independent-free-running-axis	A second arbitrary axis breaks one-cell propagation.
sectors	complete-prime-sector-ladder	selected-known-sectors	Selecting familiar sectors omits forced penta/hepta predictions.
weak_curve	neutral-square-over-charged-plus-neutral-squares	linear-channel-ratio	Channel amplitudes require squared support.
terminal	support-sixteen-hold-three-return-alpha-over-seventeen	identify-active-level-with-full-support	Three held generator directions would be omitted.

Axis	Forced form	Eliminated form	Elimination reason
anchor	unique-support-two-source-four	target-selected-depth	A measurement cannot assign the internal landmark.
units	common-positive-rational-rescaling-cancels	unit-name-changes-dimensionless-law	A held reference cannot change a like-dimension ratio.
placement	forced-ratio-then-postseal-held-reference	measured-rung-or-free-unit-selects-law	A target-selected rung or calibration would be a free parameter.
target	capability-closed-before-target-release	target-readable-before-seal	That would fit the known running vector.
provenance	observational-development-explicit	claim-historical-blindness	The V1/V2 observations and physical measurements were known.
extension	no-extra-rule	free-scale-or-running-correction	An unforced addition would violate zero parameters.

Base and successor. The closure proof is sealed by derivation hash
sha256:688facc545ef6d4e7cbbf86ea70d5400432086ec1b86e421fb5118c1661dfb2c. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:8f4540da7dcef139c22150d7490b7a29ff18f3a4c5ff0127fa9412ad3e212d47.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:efeba3f23ea1e5900da498c95bedf19655677ebe0f090f0aff8873ced57feab8.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:39f4f41bb376072deaa4b774f3f7255ab3410bd25ef449aa6c0155a38e496d80.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:4c5aabb4e2c5926719fd737be86a2211bb0f6cbde8e77e2ed0e0f59f889e8bdf.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:c39e4a115db04ee65c879d06d98342923668e286b615eac1805203ab76def489. Independent certificate:
sha256:a4215f5464a8f8e2dca60674b41b77841a71c70ab155afddd9dc379be9110b. Engine external-validation hash:
sha256:0bbcd4db30a68a6a99843f5a139129d19cda5330d4a32f62b6b9c1c4e0ef93b3.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2026-ELECTROWEAK-SCALE-VECTOR
- PDG-2025-2026-STRONG-EM-COMPLETE-RUNNING-VECTOR

Observed comparison records:

- The exact common axis, terminal held-support chain and scale-ratio law sealed before release of any coupling or dimensional target.

- The terminal on-shell weak share lies inside the complete PDG 2026 on-shell interval without a fitted rung or correction.
- The exact leading support-eight weak share 25/106 lies inside the complete registered cesium APV interval; this is a post-seal correspondence, not the selector of support eight.
- The full SLAC E158 and cesium APV intervals lie above the MS Z interval, matching the sealed sub-W descending direction.
- The eDIS interval overlaps the Z interval and remains non-discriminating; the NuTeV on-shell interval lies above the terminal interval and remains an explicit adverse row with its interpretation concerns.
- The complete PDG weak-running figure is retained by immutable source hash; no unprinted coordinates are reverse-digitized.
- The PDG W-threshold minimum and sign change are retained, so the finite monotone Fold branch is not overclaimed as one universal all-energy curve.
- The already admitted complete strong and electromagnetic multi-scale vectors, terminal electroweak comparison and proton-Planck hierarchy comparison remain hash-bound dependencies of this one-axis result.
- Changing the on-shell target to a disjoint interval rejects the correspondence.

Falsification condition: Reject if any source or inherited receipt identity changes; if the terminal on-shell prediction leaves the registered on-shell interval; if the registered sub-W low-transfer direction is reversed; if the NuTeV adverse row, scheme distinctions, W-threshold sign change, strong/EM complete vectors or proton-Planck scale receipt are omitted; if a measured value selects a rung or law; or if target access precedes the seal.

Measurement receipt: sha256:14ba878aa37865afc13d58c3f89581aed5136172f4acfc4141078da959c50a20.

Isolation certificate: sha256:05a792a4b6f8bdcc6b6aec0fb266de053d1072eeaa3f4b8d4fba8cdc49bb8e4b.

Custody certificate: sha256:882a236b6d1f3e176438b58b8addefc46a30e4e6348e2a5aa90222f9ddbfb8e30.

Registered source descriptions:

- Particle Data Group: <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-standard-model.pdf> (sha256:a102f6252b7190dc423200271dffa7c805cd15a50391b1c578853d2f777611cb)
- Particle Data Group: [experiments/external_sources/physics/snapshots/coupling-running-convergence-source-record.json](https://pdg.lbl.gov/2026/reviews/rpp2026-rev-standard-model.pdf) (sha256:b83331089d96c073fbd5101753ba5c4716ae1a8b1b891e068684b6f7246d9953)

Registered target identities:

- WITHHELD-COMMON-SCALE-AND-COUPLING-COMPLETE-VECTOR from None

Meaning. Dimensionless constants, mass scales and precision values are kept in Physics. Each reported value is tied to its zero-parameter Fold relation, sealed prediction, exact rational measurement adapter and immutable receipt. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no numerical-zero scale origin, negative exponent, irrational, imaginary, floating or completed-continuum proof value
- no imported renormalization-group equation, beta function, logarithmic energy coordinate or continuous scale
- no measured energy, weak angle, coupling, Planck value or selected rung in the formal survivor decision
- no identification of the earlier full-support running level with the terminal held-support level
- no claim that one monotone weak curve applies above the registered W-threshold boundary
- no fitted offset, calibration constant, scheme correction or hidden target-selected depth
- no historical-blindness claim

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:41008d57dd39bcbccd3e2bb5c023c4ab96e0cf7e8e1666a2ce689507cca8fc7f; engine receipt

sha256:44d542f87d12844c02c321ac651008ae7ddd738a3b16e7d2c3d8700ec89eca55 at receipts/engin

e/model_admitted/SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030-44d542f87d12844c.json;
empirical-validation hash
sha256:626bb4f32ea61fd62a404aaafc02a12c1fa95c2b64e116eafdfb1ef9c30aaf5ba; measurement receipt
sha256:14ba878aa37865afc13d58c3f89581aed5136172f4acfc4141078da959c50a20.

18. Matter Interactions Flavour

Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed.

134. Mass-energy correspondence

Claim identity: SFT-PHYS-MATTER-MASS-ENERGY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A retained inertial recurrence composed with two limiting-speed propagation transfers is the equivalent transferable energy carrier; conversion preserves the complete source trace.

Rest mass-energy is exact composition of the inertial carrier with two limiting propagation-speed carriers.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__inertial-carrier-times-limit-speed-square__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	inertial-carrier-times-limit-speed-square	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f1dcc64ee584cc011237e0a5238afcdb7d5581b36b3f37cc5360b07f38daaeca`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ed0bec154485f63255b26601114771f2175e5c9c127aa857b0e159ebb53d3819.
- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0d3b8bf23c1d3242a5b5801dad3467f0698bae611dfca54dd64fc741e69eb04e.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:aa11d50ff9abd65f109a711ae145bcaf5ce16355a23aabea67ce51f63cc71a1b.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5fe31bb7a21737a2dbf16c212f31f9e68ee0f67bb7ecd1330de2ae306d6c9945.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:40cd4a67fb937a98d196b2c5a836586e9cfb46c3a74c27fe4d1692c39c27cb7e.
Independent certificate:

sha256: e977626f21f6e22faac2b47e0d2f32e48bd60458129716ffff62d1771b3cd8e2e. Engine
external-validation hash:

```
sha256:cee15ea8bac655817aa738d2fec855214375d63b62026c25f19645a42ba8cd20.
```

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- [illegible]

Falsification condition: The sealed exact mass/speed prediction does not overlap the released CODATA mass-energy interval or the displaced control is accepted.

Measurement receipt: sha256:8bdb0451dcbb1ef48b8ff41f8847fd33e3892e527e0b6a7eaa3c2b62513f8a6d.

Isolation certificate: sha256:342a9584b5be2165873a97779bbe074cb9dd04bdb169d8d9ef0335d39b9660ba.

Custody certificate: sha256:6f8b83b11a77161dab0cbc0eed7e4edaffafd241915c40423b159f015a1d602c.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- matter-mass-energy-1 from NIST-CODATA-2022-ALL-CONSTANTS
- matter-mass-energy-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:205e2ecd215cf16dae0cb8afaecb503ddde7c5ec132379c728020e8f87c80147; engine receipt

sha256:f812b0091bd3e7f6fac435253a1e3dcf81af7a36de5b57f25c4b641cd3fa13cd at

receipts/engine/model_admitted/SFT-PHYS-MATTER-MASS-ENERGY-001-f812b0091bd3e7f6.json;
empirical-validation hash

sha256:c92bb3e6d747fab7f9feccef2a19ea88774b68e10431891b9b22ca0af330677b; measurement receipt

sha256:8bdb0451dcbb1ef48b8ff41f8847fd33e3892e527e0b6a7eaa3c2b62513f8a6d.

135. Particle-antiparticle held orientation

Claim identity: SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Particle and antiparticle are opposite held orientations of the same constituent word under conserved-label reversal, with identical positive inertial support.

Antiparticles are conserved-label orientation reversals of particle Fold words.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-MASS-ENERGY-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision

support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__conserved-label-orientation-reversal__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	conserved-label-orientation-reversal	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d0c020e5bbbb2a863a14652173767dc88149cef47ddfbf32699d56cc043b5601`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:991779226ade641f899769964c870af0f0c4de28a230876d1cd633ba35b19e5f`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4eaff1ede6a4c124348e1a40c781d70d96d04dd6ba35566819589270cf3ca56d`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bf21d14f456d343ce6a02b72bccb0b561869c44afa8d3dfd6b77c4f0e0e7956`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1cd2a34c6959a22862cd410adb85dd16cf0885b2d555bcfa80ed0b66811afb0f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:66fbd5e270b64c302f3c7c195598eb54e596e438cf610d36a1a5fcde0b818096`. Independent certificate: `sha256:746cfe41e7346ca3f8fe778d89bd096037143f9bef8199d0fd4871711ef88768`. Engine external-validation hash: `sha256:861e1634065625f513cfc6fb505d7e7d77493a092f049c219057f75b4be87da7`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import

and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-particle-antiparticle-1: predicted particle-antiparticle-orientation; observed particle-antiparticle-orientation; exact match True
- matter-particle-antiparticle-2: predicted particle-antiparticle-orientation; observed particle-antiparticle-orientation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:114ac7cfe3c8afe08e4ded55d36915cd0f9c6b736def7d0c28cc69522b3df520.

Isolation certificate: sha256:fb1def96c1f0e769619cfa260e5cd5438abc0821762523f235e569d98494c9a1.

Custody certificate: sha256:2a2a0cc1a5d8001f326b7c04495dacd65be095c29a4b7dfee0320cceb8828c9c.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-particle-antiparticle-1 from PDG-2025-SUMMARY-TABLES
- matter-particle-antiparticle-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:893742fb4e72ad76d3698e0dc95990485d8a2282321c2fc79ee6fc6528535dac; engine receipt

sha256:4840b7076ad83c2c76882af2e81a8309d6ee80028e61796c2f91a896fea2ebfd at receipts/engine/model_admitted/SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001-4840b7076ad83c2c.json;

empirical-validation hash

sha256:21fadf4dd9de421fc4ef535a9d2f551fb9dfd5d59a0365c85c79cff5d3f8dd1b; measurement receipt

sha256:114ac7cfe3c8afe08e4ded55d36915cd0f9c6b736def7d0c28cc69522b3df520.

136. Fermion and boson composition classes

Claim identity: SFT-PHYS-MATTER-FERMION-BOSON-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Constituents divide into alternating and preserving exchange-composition classes according to their finite cyclic spin action; the former exclude duplicate one-cell words and the latter admit them.

Fermion and boson are the alternating and preserving exchange classes of physical Fold composition.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__spin-exchange-composition-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	spin-exchange-composition-classes	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ebb1423777328bdf7e4eace13e90f246ffbc726ffff5156fd72f7b219994860c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:156fcb775b2aff5eca6d29287f58eeab4ab91b456fb3d64093f8169f35b9f222.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:39c0be052afd07273ef0ad9a013e145adfab5e66c08807f1c72adc015286643f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6213645d9d40a9519d4cc65ec8a058d98c3f73b8e70cabf0db75f458b8c8b1b1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:9f77efc02091736508973db62a1945e1adc476f5486ec713a061d46cb177c231.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:347ed8537673fef317ddf0f5ab470dfb67c15efb12d37358fc4db2d8fc5e4773.

Independent certificate:

sha256:9de3b2995bd56763940473afbf734a3bf257024a3d7614121e2bbdfc890fa227. Engine

external-validation hash:

sha256:1206b2c448c117ea0104acef16c31c9400275abc30c3beddc7a1334de5825a91.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-fermion-boson-1: predicted fermion-boson-exchange-classes; observed fermion-boson-exchange-classes; exact match **True**
- matter-fermion-boson-2: predicted fermion-boson-exchange-classes; observed fermion-boson-exchange-classes; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:b62d84281d62a3bd43e014f64e0fc119c7173a712656a2e1c54a5305b7a089e4.

Isolation certificate: sha256:bdcf6d556e0634792d59829d64ea70f32749a0eebe036193b94775b8e9cb26ee.

Custody certificate: sha256:47ff6011a01dfff2f7d7726254ec7a0d7dd02c9013e874699e4fe314b292c58c.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6feff8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-fermion-boson-1 from PDG-2025-SUMMARY-TABLES
- matter-fermion-boson-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude

- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:84e502c03f5b1e7f8ffdd767e97ebba9c71a066725b2500625ed3fa8817213c9; engine receipt

sha256:8e8738ba7238b2b8216d04aec21eb1d15ab6ce2e919b3b265f5a65d50c3951 at

receipts/engine/model_admitted/SFT-PHYS-MATTER-FERMION-BOSON-001-8e8738ba7238b2b8.json;
empirical-validation hash

sha256:a27494b99485e534973865ca0648ff11ae6cce9e9cc3f0ec7bedabf42782cba3; measurement receipt

sha256:b62d84281d62a3bd43e014f64e0fc119c7173a712656a2e1c54a5305b7a089e4.

137. Conserved physical labels

Claim identity: SFT-PHYS-MATTER-CONSERVED-LABELS-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A material label is conserved when every complete generated transition pairs its disappearance on an input word with its appearance on an output word or named boundary carrier.

Conserved particle labels are exact held-label flow invariants of the complete transition grammar.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-FERMION-BOSON-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__transitionwise-label-flow-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	transitionwise-label-flow-closure	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3250981cf6d63ab0da7638f97c224fc22a0c3276f9655b03f6ffdecae9314435`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:baf66a25d81ccf81aaa130e0732edf6d70758e68cac88ca87f1b06e3beaa1602`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6f149ce11d6661b1bd2c56ef88117bd66516923c70eba8e0377be09be835918d`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b3c1dddcc5408913462dfdbe70490ee0d8e0144e9f670c69d992958154620a30`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6f479d740b225e7286d71a8a4d28f81e285bb80433f72f011ad1cb78006a0e72`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7ebe6010d8a85c0fb10f770a73d7ca4fbd8db55ac00f6cb454d8fca33966cce2`. Independent certificate: `sha256:09fbb8d4b6458db759a9b634a65db73ce94ab658a72b96bddd9923c5621465`. Engine external-validation hash: `sha256:fd5e7a453ae51737b1b9750c5a3062a9700c0e108971f0cbe639a64c78c17863`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- **matter-conserved-labels-1:** predicted particle-label-conservation; observed particle-label-conservation; exact match `True`
- **matter-conserved-labels-2:** predicted particle-label-conservation; observed particle-label-conservation; exact match `True`
- **deliberately tampered unfavorable control rejected**

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:a37fbf21c2c1532ceb7477f56a5acd1e4ce9cb6c5588899a05175aec3ac4e588.
 Isolation certificate: sha256:fd1debefc10a612117b4a30ddb4ba04b1a4ad115b85018c2a0f4f84bc46dd08c.
 Custody certificate: sha256:ebdba44b843f9833a0db8b9ee44a89464fc6aa1cbcb27a787ebab8368fb0ec1.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-conserved-labels-1 from PDG-2025-SUMMARY-TABLES
- matter-conserved-labels-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:73ef40384dac7f3c76495576bf9f808e57c5fdc63c81a989c914d82c090bc2d1; engine receipt
 sha256:83772ecab34a176c0e87265993183d8c445be8f49ec0d86e615c9d0c1740e945 at receipts/engine/model_admitted/SFT-PHYS-MATTER-CONSERVED-LABELS-001-83772ecab34a176c.json;
 empirical-validation hash
 sha256:b44d6261da8de14792040b7131fda6cf2219d91d5632dfc2b4b3ca60fd0ba8c0; measurement receipt
 sha256:a37fbf21c2c1532ceb7477f56a5acd1e4ce9cb6c5588899a05175aec3ac4e588.

138. Observed elementary-particle spectrum

Claim identity: SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Elementary particle kinds are the non-equivalent minimal recurrent constituent words under complete observable transition traces at the registered measurement boundary.

The observed elementary-particle spectrum is the finite census of minimal non-equivalent recurrent Fold trace classes at the PDG boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__minimal-recurrent-trace-equivalence-classes__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	minimal-recurrent-trace-equivalence-classes	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:30471ffa0c2b9f94dc70357fa22a67e4d1ff45623cb94929b781ac28f146f923`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1111b793deef38ccd05162c2a85883eef2417a3e7762c5c86e6d8f3d38c03cf4`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ddcc358eb534bdd1f240bc3694532fb8db0b67e6281b95bc4c3c6eae0c6e5db`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:512ad96f068b575ad0544dd3c8797dd7f8893144162266f9d2f3d8abb440f5c1`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c110db659ea14ee96663e2ed3e8c5dbdcf57d20d7a4d67e624d15c7cd6be233`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5b513a50d02c9708abf487a28febbe002b36f0f797c14df1948fb0be98f34af`. Independent certificate: `sha256:f5851afd8e7790d039c120b8a89d75a1b5f34eb15ec0422ad89d16aa202a775d`. Engine

external-validation hash:

sha256:d10ff651bcccdbc2894bfb2cbcf8e09a7eebaba0885a3ccb0d008c36cac0d491.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-particle-spectrum-1: predicted observed-particle-trace-spectrum; observed observed-particle-trace-spectrum; exact match `True`
- matter-particle-spectrum-2: predicted observed-particle-trace-spectrum; observed observed-particle-trace-spectrum; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:c6c667a66b77d96b117f8e97fafdf3cb9c40c0423908f5f9c6b8ea5f27fa763f.

Isolation certificate: sha256:8487cf41ea71e8a870132223c040819a2b4e46c48e13136b029d6f2272a0728f.

Custody certificate: sha256:3a1e08e0415632f358fe4ba66ee3fce7d967e81d128d2a5c7029a51f299707c1.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-particle-spectrum-1 from PDG-2025-SUMMARY-TABLES
- matter-particle-spectrum-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:17408d0e7ea308694f773c1adfbdb532ab0a6f5304c8e9088f7eff63cdddecfca; engine receipt

sha256:a5d2f4b06e01f8d5acabb36577883d2ad220e3dd5aa6aa2e170b52ff5273fff0 at receipts/engine/model_admitted/SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001-a5d2f4b06e01f8d5.json;

empirical-validation hash

sha256:72e141aef98afe3d189fe1fd563e8f46123cbe94f14447d62931d82925f53772; measurement receipt

sha256:c6c667a66b77d96b117f8e97fafdf3cb9c40c0423908f5f9c6b8ea5f27fa763f.

139. Composite hadron organization

Claim identity: SFT-PHYS-MATTER-COMPOSITE-HADRONS-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A hadron is a closed joint constituent word whose internal interaction labels are not separately observable outside its boundary while total conserved labels and inertial support remain observable.

Hadrons are boundary-closed composite Fold words with confined internal labels and observable total carriers.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__confined-joint-word-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	confined-joint-word-composition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

`sha256:7d15922136da75a509275a76b6cef216c7ac8a155b8833094706a0558283d24b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 794134875ca65cc8d22045a544dc4a94357c2b1cc7965734eaedfae649186334.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8fc2f5f15a7a26710f8f241bc15eaba1833240845edaf6b7a334130a1450ad09.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: d5ce26b1e74bebc5ff35a429fd4f17b9b392d39ed94a7a0bfc31b39b17b64f85.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: fc3f7576bde48ef11fbe9449ae4006d0b914605fa8f399aa5036a7ab3829ed40.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 0bfc340f875b4d4ee94d7602b2670fe3e8c1a1b8dfd0814b98bdb78bbc2b9876. Independent certificate:

sha256: 22d961672ae7e26b6566dada76865b9771a018c67e77c2c2299f96c1a96be5fb. Engine external-validation hash:

sha256: bfe345229b0d776a8998d0458b0f1c52795ecdb8d76ec08edd612f2ee907c1f7.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-composite-hadrons-1: predicted composite-hadron-closure; observed composite-hadron-closure; exact match **True**
- matter-composite-hadrons-2: predicted composite-hadron-closure; observed composite-hadron-closure; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256: 84b6999676d9e3786fb32a7bdca5efdc93ca0d9596b75fbfeae52b80a0bb3bb6.

Isolation certificate: sha256: e487257b9dfa92658e4d73467970da58fc7f9d08e27ce42049a23cb593ef567a.

Custody certificate: sha256: 60521bbad490f550ac641aba82b204003ff0f812c605a61197607ca7070d9581.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256: ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-composite-hadrons-1 from PDG-2025-SUMMARY-TABLES
- matter-composite-hadrons-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set

- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:98f3c1986b69b272fabf8a95f88c72cb8ecfb48edd659776bd5c16fa73fe7d74; engine receipt
sha256:f99b0600faea8923a9b335972b511b488ad0a252566da40911ad5fd2d9136697 at receipts/engine/model_admitted/SFT-PHYS-MATTER-COMPOSITE-HADRONS-001-f99b0600faea8923.json;

empirical-validation hash

sha256:02b815af214cb37ac677f126abf0c880ed63cebd6a6397dc7704230c5fb42b682; measurement receipt
sha256:84b6999676d9e3786fb32a7bdca5efdc93ca0d9596b75fbfeae52b80a0bb3bb6.

140. Scattering and cross-section measure

Claim identity: SFT-PHYS-MATTER-SCATTERING-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Scattering is the complete map from prepared incoming joint words to distinguishable outgoing channel words; cross section is the exact positive successful-channel count relative to registered incident support and geometry.

Scattering is a complete transition-channel map and cross section is its exact source-normalized observed support ratio.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__incoming-to-outgoing-channel-census__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	incoming-to-outgoing-channel-census	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bde4ec9cbc784e2c29a227b2901c09e7461898de7cecaf6aeba557b173d80f10`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5c420945a230b02e98d4dfa2a708774bb8984f0abebc9d5c98f193adaf32def3`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:999d57cb70e017713c5af6e41f6f7e0459044c050f9ddd56eddaba7cdb1220b6`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:096cbfc59e330a05b88b0554bfa19788f10aa302e7d08849e159857d7ac7e063`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:61ca7df4431fdfa7b0090df0baab5bf4c030853e341a00f3fcb1d36be9f00886`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:80d091c6fc94b6f1aced9db1f5b7b7d92be9b0fffb9cea55c5347a8d0dd789c84`. Independent certificate: `sha256:3f803d9f31e1300c8f38daf92f03201cfbb1a1ef21d0f0565a17db3cc385108d`. Engine external-validation hash: `sha256:1aa9fb8df8af6ae11e239bfd8544ce8b8cb82d5a6e3539c4c6cb51acae494b2a`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- CERN-OPEN-DATA-DATASET-API-2026-07-23

Observed comparison records:

- matter-scattering-1: predicted scattering-channel-measure; observed scattering-channel-measure; exact match `True`
- matter-scattering-2: predicted scattering-channel-measure; observed scattering-channel-measure; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:9fce4306e8bd1dc729a61f34e3166eb388153ff8bd3fda93842ffd1258dd5f6e.
 Isolation certificate: sha256:9fb6b4e31148bb7309035fce9e6ad14a1c21ddd52d7e086bcce6b31d4d220ef1.
 Custody certificate: sha256:25e982819e4e752fe6991214a1c164a26c8d34058ad2adc25dba38093cabae8b.

Registered source descriptions:

- European Organization for Nuclear Research: <https://opendata.cern.ch/api/records/?type=Dataset&size=1>
 (sha256:4e232ef39985c02005828f97c43d903381411e7c0a762af768060343d17bd57d)

Registered target identities:

- matter-scattering-1 from CERN-OPEN-DATA-DATASET-API-2026-07-23
- matter-scattering-2 from CERN-OPEN-DATA-DATASET-API-2026-07-23

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:cf6d74c28ced7dfb9302c3ef44c582c57bf8fc35d77a6525e9f426c250072bb1; engine receipt
 sha256:01fb73df0849a188dfd668d8f8549cd1b64ed201e5fe706bcaa391967ab61f0f at
 receipts/engine/model_admitted/SFT-PHYS-MATTER-SCATTERING-001-01fb73df0849a188.json;
 empirical-validation hash
 sha256:910bbc6b8cb2daca6e00dc9b9792851b70d7126484b3fa78a522d2928594fd60; measurement receipt
 sha256:9fce4306e8bd1dc729a61f34e3166eb388153ff8bd3fda93842ffd1258dd5f6e.

141. Particle transition and decay

Claim identity: SFT-PHYS-MATTER-DECAY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Decay is a recurrent constituent word transitioning into a lower-recurrence composite of output words while every conserved held label and total energy-momentum carrier is paired.

Particle decay is exact conserved transition from one recurrent word to a generated output composition.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-SCATTERING-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__closed-constituent-transition-channel__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	closed-constituent-transition-channel	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:7c3d62e4e9f23f4c61a96a2f4a3c7fc7179dd71964b6da4dc7ba44e1b86d4f7f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:794db5bad533dc878ce7f5091dd8534c2f9b9037ae26c75f69e0107f804a4582`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c7a9c9cbc99ddf70f9a58b2de21043f733beb858f4329a69b6178a07b623d150`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a924a3d1c129b90a5c746a6d723c15627d3abe0de4c5e58cbb5c0e489706ef33`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d6c967315e23cdb063fdc12735b33154fa9cf0db3bcc1c548c6e25c3b7f57343`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:809d2705df4cf4ab7132c2df98d03b54a449771aa0de6420c9061b0900e27e3a`. Independent certificate: `sha256:7cc8dd2f74fdd1f9b4bce38797fef768190f35f07dc57353e11c917977d4847d`. Engine

external-validation hash:

sha256:14bb288aac3e051b88c97a368e17c106739741086958601638bdf6689f262df.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-decay-1: predicted particle-decay-transition; observed particle-decay-transition; exact match `True`
- matter-decay-2: predicted particle-decay-transition; observed particle-decay-transition; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:e38dcce46ece40beeadc86694d24f4621e7b92a5886b414d250b085528428231.

Isolation certificate: sha256:92af8eeef129f8f13e711bb0fd2216e05afaa7ed77ad3938c508bff22e997ee0.

Custody certificate: sha256:9b426857927415273d745cd432c067b61e5dfbcdffe6cbb352fc1a9ede77c1d7.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-decay-1 from PDG-2025-SUMMARY-TABLES
- matter-decay-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c01f0e0c1c08c62ff3c0f887f1b65d8a94002e2e887cd018a30b2700ffd4c9ac; engine receipt

sha256:67296b745325db1768a99db563e6a6966b0201b9fb107133746aaac222ae8291 at

receipts/engine/model_admitted/SFT-PHYS-MATTER-DECAY-001-67296b745325db17.json;

empirical-validation hash

sha256:7081680de252cc8971336c5ab0e2296c5092c49e1dc9d4bba6cd0eaf4663fe3e; measurement receipt

sha256:e38dcce46ece40beeadc86694d24f4621e7b92a5886b414d250b085528428231.

142. State mixing and oscillation

Claim identity: SFT-PHYS-MATTER-MIXING-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Mixing occurs when preparation and interaction observation partitions select different refinements of the same finite recurrent support; cyclic phase action then changes their exact overlap with recurrence count.

Particle mixing is cyclic recurrence of one finite support viewed through nonaligned preparation and interaction partitions.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-DECAY-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__nonaligned-partitions-with-cyclic-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	nonaligned-partitions-with-cyclic-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:146a2aa9a27a200c26d676c30ca782b2f8f302898f0512e0fca2ec46a910d9b5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8aad8060dbde458070e15cd15fa55313d0a0fb1efb7f79e6dab95208c93c6834.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9af76d3b28e78224c81f28b597e1a3b05c573059bf8de8f128315deb08295b87.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:763ea04baba4dfffed25ed238304c89f82e5f9d97a3ebf05d1e6842ed4f40dd8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:65352a8468399e495fccb9ef4943f1e3c475b298ca8954d56f4a9c9d1567f479.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e2076b010706a7bcf0e70455f5f7701bc2b71fb07a36cabab6a08dab7e7d88f8. Independent certificate:

sha256:c44e862daedd1331de9f2b39221975753d321a594023e4c0872c2e30a9205e8b. Engine external-validation hash:

sha256:26022d6f191095471ef38e32bd069ab5a8b3a1ec98310673fc3841d0f5bb6ce1.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-SUMMARY-TABLES

Observed comparison records:

- matter-mixing-1: predicted particle-state-mixing; observed particle-state-mixing; exact match **True**
- matter-mixing-2: predicted particle-state-mixing; observed particle-state-mixing; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:29c2514184ac55bea194bf4dd3a54c68dc30cb837dfc48626be37458aff7e7c4.

Isolation certificate: sha256:6d7e34542666520a3ef0f24a90082fc7b5389d7e61f0eba3a7ade7abdc20afa.

Custody certificate: sha256:f8c15bb32625847abee7f91d00e1d52865182049cc4eda5ccecf8defb4525de6d.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: https://pdg.lbl.gov/2025/tables/contents_tables.html (sha256:ae10b6a7c6fefb8a6d90d35dc341d3ae8284a1e3cf2c94c656ae2e8957df4d56)

Registered target identities:

- matter-mixing-1 from PDG-2025-SUMMARY-TABLES
- matter-mixing-2 from PDG-2025-SUMMARY-TABLES

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set

- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1d735f19a4e97bfe8341691b055e6b7274fc5380833ef335cd4e314677047ee4; engine receipt

sha256:6e42de870a32d7dd47c1d24d1410009498854662fc9b7b85ff2aae24642a3e2b at

receipts/engine/model_admitted/SFT-PHYS-MATTER-MIXING-001-6e42de870a32d7dd.json;

empirical-validation hash

sha256:462921308b12d9ad2785f45aac4e115da26e8e33f0cbad99fd658cc2809adb4e; measurement receipt

sha256:29c2514184ac55bea194bf4dd3a54c68dc30cb837dfc48626be37458aff7e7c4.

143. Electroweak Fold-fibre mixing law

Claim identity: SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The binary Fold balance carrier is the half-One. Complete depth-two preimage enumeration uniquely returns one-quarter and three-quarters; both fold to the half-One and together partition the One, forcing the bare squared channel split one-quarter and three-quarters.

The bare electroweak squared channel split is sin-squared = 1/4 and cos-squared = 3/4 over unified half-One support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-PART-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of Fold carrier, dependencies, complete preimage relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact depth-two Fold parts tested as preimages of the independently generated half-One image. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: **generated-exact-carrier__admitted-dependency-trace__complete-half-One-preimage-fibre__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule**. Closure is **depth_independent** with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	complete-half-One-preimage-fibre	selected-angle-or-continuous-phase	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.

Axis	Forced form	Eliminated form	Elimination reason
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:c5ec8856eab3b04f44906f0218eab9d35a9703bea4a849a669249f0888072907. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:a9147e40197c34a7e912d27f0c0586c4843d24382c80f5ea5d2bb4a0346d38fb.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:9a58f1c2fb85881b5b8088644bf9f0d15c13e7b2dd3268d4923c5dfdd5adf345.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:3ce203371d382384d04ff1f196998476e98f70f9c124c314690cf7a7c90b29ee.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:7d7db1c9ee3e4272ddd76896fc6914357e518fbe5820395a6d0875ce3b639785.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b. Independent certificate: sha256:d087df08ebbe7c5b0a27827e56677841cd5a130cb8b3fe1a41814468ffca14c0. Engine external-validation hash: sha256:6aa4b5dddadccdf0bcc0f23aa8f91a599c1d3b68028193db720d03811da50cb.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:43210de9ea1472d62b32f5dc7639fb5c7bc7dced272aa3925d66563505d057f8; engine receipt
sha256:c15ae82a6cb030fc2b13b36ab6099bdbaf8faaffb0092a7537e44aaa0d160db0c at receipts/engine/model_admitted/SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002-c15ae82a6cb030fc.json;
empirical-validation hash None; measurement receipt None.

144. Exact Fold PMNS squared-support triple

Claim identity: SFT-PHYS-NEUTRINO-PMNS-ANGLES-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The binary half-One separation, generator-three unit separation and complete binary support at generator depth force atmospheric, solar and reactor squared-support parts $1/2$, $1/3$ and $(1/2 \text{ times } 1/3)/2^3 = 1/48$.

The exact Fold PMNS squared-support triple is atmospheric $1/2$, solar $1/3$ and reactor $1/48$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of neutrino support carrier, dependencies, binary/generator/tower composition, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All exact products of the admitted binary and generator-three separation carriers over complete binary support at generator depth. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-dependency-trace__binary-generator-separations-over-depth-three-support__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	binary-generator-separations-over-depth-three-support	selected-mixing-triple	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash
sha256:9cc8b42876a4a25a440f99c8a3a4070e722f065e3db4445c0f07c976c3af550b. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:d340934bee8714c7a750e6e93d98dc73a496991650cd02aa0d6c4877d1aff5c3.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:1177452085ab8e621fe3c07979fd647400be7695804ef0e945889c5af6257c38.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b47a2bfe918a8458685fc8ed0439bb5f20ce2279ba54416ddbea536368569cfb.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:be4fc2aee75404b3370cc9878eb7e0da27e33d584bc0b1b7f6900f40858a2229.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b.

Independent certificate:

sha256:27da38f682958d47ff250304ddbf78cc55909cc68659ed8274e58b01d7b1401a. Engine
external-validation hash:

sha256:8563a746be3cb01682bdfd656d3d56e5b952476f88e420f5417cbdf28685b27e.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c9b825a0bb567ea8b3160782c0b05d414b543a41b685136c5ef9dc2cd0f23f05; engine receipt
sha256:f15bedd63f3cddc56f4ca7397633b6081ad47086cf77496cf834bb82f5a5287 at

receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-PMNS-ANGLES-002-f15bedd63f3cddc5.json; empirical-validation hash None; measurement receipt None.

145. Exact squared W-to-Z Fold relation

Claim identity: SFT-PHYS-ELECTROWEAK-WZ-RATIO-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The massive-to-complete weak-channel squared relation is the upper member of the already forced half-One preimage fibre. It is therefore exactly three-quarters. The irrational unsquared root is outside the SFT proof domain and is never formed.

The exact squared W-to-Z Fold relation is $(M_W/M_Z)^2 = 3/4$; no irrational unsquared ratio is formed.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of mass-ratio carrier, admitted mixing dependency, upper-channel squared relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: All type-correct squared weak-channel ratios assembled from the complete admitted electroweak Fold fibre. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-dependency-trace__upper-channel-squared-mass-relation__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	upper-channel-squared-mass-relation	formed-root-or-untyped-channel-choice	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:653437257d12094f6b927b80227ef4e366a6ad704664391321d66004cb9b52eb. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:8d6af386d5b72c4c47d3dd7a1848b4a0f3584f60af3b4ba7687e661525b1b8d3.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:e9e31cfdeb7d97be8f51a2b49f90bf2e6a70416d3a0bb5cb2faf88a8963c40a9.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:de0a2b689c020172a062678a5cc32c5150549f0c4cbace73a3a04fd533e488c3.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:bbcfefeb66ed945c758a4f68eb1995ebefcc6a0f312fe32c5312d2ef8f86c48bc.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b.

Independent certificate:

sha256:0ebf286ad43f2ffab15f028c0d28b1bdfdf2cb0a463769bee01d12320fee204de. Engine

external-validation hash:

sha256:7c72af9f1edd25f8f3ab11a0d85f745794f54d9028d0af63d9a4e9aa80fe0dad.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no square root of three and no unsquared irrational proof value

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4469cf54ba2db54b9670721e12c705b796c564846d172ffed40c2bf5d7dc0abb; engine receipt

sha256:a69443396f1fdf1f9beb11e65d2cc318b1c01dc5864e52d52a22eaebee1a4e89 at

receipts/engine/model_admitted/SFT-PHYS-ELECTROWEAK-WZ-RATIO-002-a69443396f1fdf1f.json;

empirical-validation hash None; measurement receipt None.

146. Self-sourced colour running direction

Claim identity: SFT-PHYS-STRONG-RUNNING-DIRECTION-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A neutral carrier retains the One source count at every positive level. A colour-carrying mediator appends both Fold labels at every successor level, forcing exact source-count prefix 1,3,5,... with successor difference two. The claim fixes the structural direction and binary slope, not a measured renormalized beta coefficient.

The colour self-source count is $1+2k$ after k successors (prefix 1,3,5,7,...) and the neutral-carrier count remains the One; the exact structural successor slope is two.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of running carrier, dependencies, charged-versus-neutral successor relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every positive finite running level generated by retaining the bare source and appending either the complete binary colour feedback pair or no charged feedback. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__binary-self-source-successor-versus-neutral-hold__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	binary-self-source-successor-versus-neutral-hold	imported-beta-function-or-fitted-slope	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:3fc1da0842f8e087db7c17007cc1bddb63e58c41666b27f7be09d4658af8e4b51`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

sha256:e4c3f3abf5b5e710909a9adefd750f1c01f4cc47f41940c7ef05792408bf9781.

- **tampered_source**: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:9fdc2a6e7b63ad431ae9d6257d442d5d9e295e2e4cf961a3c24e13ac5fbf56aa.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:a0be9ed3ea3361185ee50b9e73e9dd1ca1dc4282cea5044a5649275a28b4e6e0.
- **boundary**: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:28fd98942d7a6e881ac50f9963c51a8ebfc5b500e12eadcc60d0b8fa34f14c6f.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b.

Independent certificate:

sha256:c055cc182bf27e19cd2acd94c18a3c516c3ee5f1e4e563f096daa85289a95076. Engine

external-validation hash:

sha256:9f9008c81386f465f6ad81f8c1b05fa69b119aedd92340afcfc46348dde6d92.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no claim that the exact host count two is a measured QCD beta-function coefficient

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b8e20f3f9f64ffa2fbc29048ea7c06fef38df706fb17093d6961a68da8d7a6ab; engine receipt

sha256:c9628d8af62fc85936b91328ab2d539f4a5def412e3a61c4cf36c0b6d5a8d412 at receipts/engine/model_admitted/SFT-PHYS-STRONG-RUNNING-DIRECTION-002-c9628d8af62fc859.json;

empirical-validation hash None; measurement receipt None.

147. Self-antipodal maximal Fold phase carrier

Claim identity: SFT-PHYS-NEUTRINO-CP-PHASE-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Complete binary Fold separation has one self-antipodal part: the half-One. It folds to the One and its complement is itself, forcing the exact maximal-separation phase position one-half before any neutrino phase measurement.

The exact maximal Fold phase carrier is the half-One, 1/2 of a complete turn.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-OBSERVABLE-001

Generated grammar and closure boundary. Generate the complete eight-axis product of phase carrier, dependencies, self-antipodal relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact part of complete binary Fold support tested for return to the One and equality with its held complement. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__unique-self-antipodal-half-One__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	unique-self-antipodal-half-One	continuous-or-measurement-selected-phase	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:ed0153a7e2ba490f785c56a0e04ee548c41ff9c83400e46081d5d83ed4e591f6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:6c1ddde82e7659ae5b118b6580d6dadef2d1d3c96dbf9df974ee84f1a034f6ad`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:56bd3cd53e0e6910301ef3ebc2650de127be2662a86aa64d903b64f0c7d47656`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1c2b025a43a1375fc277813882f8b7f2a175ef8199cc59ab70d082e61639d99e`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt

sha256:13fc8fc10826ccdd187cc7aad171da44978f483fdefb5d59c901e9ea215fe4c3.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b.

Independent certificate:

sha256:373a3307dc34dbb7e93cca107defbba08df51b4b1801237fc0c473c6ba9d012d. Engine

external-validation hash:

sha256:38e01fd64b49253c63cabbdcc293d64a62f6a27e9720baf2b368d340c2bd82f5a.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:cbd6a5fd6b8bd47abce44109d527dde714c4d0290c70d4f7c666c3360ceb6578; engine receipt

sha256:66a61437fe9e5f7101002e716423413dbeb111125e7c9c4d540f179a8c1eb12f at

receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-CP-PHASE-002-66a61437fe9e5f71.json;

empirical-validation hash None; measurement receipt None.

148. Bare Dirac gyromagnetic Fold value

Claim identity: SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The binary electromagnetic balance carrier is the half-One. Its exact reciprocal comparison to the One is the positive whole count two, forcing the bare Dirac gyromagnetic value two. Radiative anomaly corrections are separate empirical Physics and are not claimed by this law.

The bare Dirac gyromagnetic Fold value is exactly 2; this claim excludes the measured radiative anomaly.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-MATH-EXACT-RELATIONS-002

Generated grammar and closure boundary. Generate the complete eight-axis product of gyromagnetic carrier, dependencies, reciprocal balance relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary

is: All exact positive reciprocal relations between the admitted electromagnetic half-One balance carrier and the complete One. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-dependency-trace__One-to-half-One-reciprocal__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	One-to-half-One-reciprocal	measured-anomaly-or-fitted-g-value	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9af2478ed0735a7411cae7c10aa6e8eb2cd1334d25bb633fbf36afdf2d0e6e97`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c74885975149e72dfdd85c229f5f15e3e4eb5fcc064cdfd2f60cf02f6665a6f4`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:889413928203edd2333b8faf1d7a0f2f4090f0e33e531a0c29597ba07f19b86`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:26db1a75752722428065c5f51f5f9aa1ca599c763fd30441907777416bb2ac84`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:59ece166f57582dfae45fd8c546c2003f4121b783ca55c8248b827d4ff10921b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate:

`sha256:3615d9d1afc305b42da5a1f86140e1c9742f96920315c82e00cc74b7c7a51ba4`. Engine external-validation hash:

`sha256:e54dbe2f2a807d76ab57647f6bc5e7d8e5e2857e4ca9477cb044bbb636599d65`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule
- no claim that the bare exact value equals the anomaly-corrected measured electron g value

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2cfa8f4103b3f5c5d8096bac8874a6aa3e3f953c75d4bcf9578ea6333ac3ed79; engine receipt
sha256:2d4d2cf74bd6c8b589961f0ad65b4ebe3f614317493daa65c4024de4ef2ad66e at receipts/engine/model_admitted/SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002-2d4d2cf74bd6c8b5.json;
empirical-validation hash NONE; measurement receipt NONE.

149. Weak parity from the held Fold fibre

Claim identity: SFT-PHYS-WEAK-PARITY-FIBRE-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Complete half-One preimage enumeration forces one lower part one-quarter and one upper part three-quarters. Both map to the same half-One image while remaining distinguished by held side. A channel restricted to exactly one held side is therefore parity asymmetric; a channel retaining both is parity paired.

The weak handed Fold fibre is left-held 1/4 and right-held 3/4 over common image 1/2; a one-side channel violates parity by construction.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001

Generated grammar and closure boundary. Generate the complete eight-axis product of handed carrier, dependencies, held-side fibre relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: The complete exact half-One Fold fibre with lower and upper held-side labels and all one-side or two-side channel restrictions. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-dependency-trace__held-lower-and-upper-common-image-fibre__complete-registered-product__all-predecessor-forms-rejected__derivation-sealed-before-comparison__complete-trace-controls-and-census__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	borrowed-physical-label	A borrowed label has no Fold trace.
dependency	admitted-dependency-trace	asserted-input-values	Asserted values are parameters.
relation	held-lower-and-upper-common-image-fibre	unheld-or-imported-chirality	The alternative loses a required distinction or adds an unforced rule.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	all-predecessor-forms-rejected	survivor-without-predecessor-controls	A survivor alone does not exclude a shorter form.
measurement	derivation-sealed-before-comparison	target-visible-before-seal	Target access reverses empirical direction.
record	complete-trace-controls-and-census	answer-only-record	An answer alone cannot reproduce the derivation.
extension	no-extra-rule	free-extra-rule	An extra selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:6b246cba470a9223d2bab7c00bf774485a142d4d1b8d51bb22f489b29bdd5f60`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:542ff56ad805d9ccb73fd88585057f41fa166d015d115020abc575757988e7f4`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:84020e1fd0a86146b16c09234542f86ccdba61dca53fb9cac0ed7f7253baaac`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d0b9d1f721ff2993580bb827eb317c677485a274083ae3949b75c092d32c2e09`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:94af8f923be880482a518d812a7665411c8af59256168a285fedecb1814dc24`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b0cbc777be51372a0484fd4cc4db16a013a5d1ccee251ff5b3ac1b731af8af4b`. Independent certificate: `sha256:cb069b78dc7b2f79c029832c7b232caccbd903c23f02f85f283896e14972ea26`. Engine external-validation hash: `sha256:d69283078c43fab728f08fc6f7bdd22a5d4908c30de30711abd7968c97cf7eb9`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, proof certificate, candidate table or answer artifact as a premise
- no external measurement, conventional equation or fitted parameter selecting a survivor
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no unregistered force, particle, phase, scale, channel or correction rule

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:258fdfa55042a4bc6030320a2913118ad436dba74602e50835a0a87c3cdbeb96; engine receipt
sha256:5e33439d3af142c055b5455921783e65a846e8d3d9bbf588409022480f263454 at
receipts/engine/model_admitted/SFT-PHYS-WEAK-PARITY-FIBRE-002-5e33439d3af142c0.json;
empirical-validation hash None; measurement receipt None.

150. Terminal exact on-shell electroweak Fold share

Claim identity: SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The four-rung terminal alpha object generates complete binary support sixteen. Holding the three generator directions leaves running level thirteen. At that level the charged squared support is fifteen squared and the four neutral pair channels carry fourteen squared each, forcing 225/1009. The sole terminal self-return crosses the complete support successor seventeen once, adding exact alpha/17 and forcing the terminal on-shell share before measurement.

The terminal on-shell electroweak share is sin-squared = 1930922298157999/8642477221479757 and its exact One-complement is cos-squared = 6711554923321758/8642477221479757.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete ten-axis product of lower carrier, terminal stage, support, running relation, self-coupling transport, orientation, enumeration, minimality, measurement direction and extension. The declared exact boundary is: All type-correct exact channel shares formed from the admitted bare weak fibre, the complete terminal alpha promotion object, its complete binary support, held generator directions and one returning terminal self-coupling. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: admitted-lower-law-carrier__complete-promotion-termination__complete-binary-terminal-support__level-thirteen-charged-over-four-neutral-pairs-plus-alpha-over-seventeen__single-typed-terminal-alpha-transport__orientation-fixed-by-share-or-retention__complete-registered-product__bare-and-intermediate-forms-preserved__exact-result-sealed-before-target-release__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	admitted-lower-law-carrier	detached-measured-number	A measured number has no forward Fold dependency trace.
terminal	complete-promotion-termination	selected-intermediate-stage	An intermediate stage chosen for agreement is a fit.
support	complete-binary-terminal-support	free-scale-support	A free scale is a parameter.
relation	level-thirteen-charged-over-four-neutral-pairs-plus-alpha-over-seventeen	free-running-level-or-imposed-renormalization	The alternative loses a required carrier or adds an unforced operation.
correction	single-typed-terminal-alpha-transport	fitted-additive-offset	A fitted offset reads the target.
sign	orientation-fixed-by-share-or-retention	target-selected-orientation	A target-selected sign is not structural.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	bare-and-intermediate-forms-preserved	successor-without-lower-controls	A terminal answer alone hides whether the lower form was necessary.
measurement	exact-result-sealed-before-target-release	target-visible-before-seal	That reverses empirical direction.
extension	no-extra-rule	free-extra-correction	An extra term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:fd697d15e5dc304d1dd7895cfa4522eaa3fc93a1b9fd6995ba07ce08eeae201b. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:7bd7c74304217234fef6ff08648317974648a26071821cbf9cbcb2e2eb295fd5.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:69655e626f057180c9028f67846d4bd9e9b47e0898272bd1b4251aba71ad4462.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:68d67b360ce6aa76d71bad675c8c5ddb071bb7a8f955c58a2230b85102f7c3d6.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:8ee95740ea9bfb696e2ff420311da10309620d1e8c88218bbb19a138c0f556fe.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:61bd427364f94266efdc7681e8bb7fb1211cf8a628b54ac1d8dab2b2841c504. Independent certificate: sha256:22c924c06b1b2156027f3ae232e3fa3381b6b2054efff2e3f8a467a6cbafca16. Engine external-validation hash: sha256:861287114903a56a4014cf411543ed557c0568200a560ab63d833113e7b41402.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no external measured value, uncertainty or best-fit table as a derivation input
- no V1/V2 executable, candidate table, certificate or answer artifact as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no fitted level, coefficient, correction sign, truncation or target-selected candidate neighbourhood

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2e86543fe719e208cbcea24ccb6205531713d8b4fac915803978c313f77b048e; engine receipt
sha256:263cf79e79ed1d7ba0246f84abe869084aa4db7b195dd0c07300d1dabdef72f4 at receipts/engine/model_admitted/SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003-263cf79e79ed1d7b.json;
empirical-validation hash NONE; measurement receipt NONE.

151. Quark channel invariants and cover depths

Claim identity: SFT-PHYS-MATTER-QUARK-INVARIANTS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Complete colour, hand and electroweak channel counts independently cross-lock with exact Fold products, forcing up pair sum 1/12, down pair sum 1/8 and least binary cover depths seven and five.

Up pair sum is 1/12, down pair sum 1/8, up cover depth seven and down cover depth five.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for quark channel invariants and cover depths from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__channel-count-cross-locked-with-Fold-product__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore

a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	channel-count-cross-locked-with-Fold-product	asserted-quark-coefficients	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:8367541eba8e54604067cba4a24815100b7d8072711ed5b29587e97d75776c2a. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:701e492976ce746f77d3d3a66e774e734480b45b9f5e0e584766a2cf3f0b35c1.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:5aea5f9105680ae4d40b236454cf2d039c658066468b3f344b9509043ea2114a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:58a26c819a69b30efc6ddc9625bac241135323a3d711a154728da7ea2071c3ef.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:8912df316fc73ceefe8f14f5001820bf1b8b08c75e789fd2e7b88903d677baed.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c. Independent certificate: sha256:79d399f00e09cdd5ae132fc3c8c49400d96a5149cde57432821ed9416d0ab8fb. Engine external-validation hash: sha256:1cafc16eba0153762c7ab9508d3c29d4c3b2df7d08e047070c87fdee5a34ab76.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:45d72cc60c27211a20d2f8cb0b327b506990007b9e7714637893e4303450c507; engine receipt
sha256:8dad6c72f49f916be9fd5d4aba1c3837e86795580ef246403098aed2a2b34b63 at receipts/engine/model_admitted/SFT-PHYS-MATTER-QUARK-INVARIANTS-003-8dad6c72f49f916b.json;
empirical-validation hash None; measurement receipt None.

152. Dual exact quark mass cubics

Claim identity: SFT-PHYS-MATTER-QUARK-CUBICS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Exchanging the generator and Fold-fibre roles in the already generated cubic carrier forces one down-type and one up-type three-root polynomial; roots remain exact algebraic objects through their invariants and first complete rational isolating census.

Down invariants are $(1, 1/8, 1/383)$; up invariants are $(1, 1/12, 1/3071)$; each polynomial has three disjoint positive rational isolating intervals.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001
- SFT-MATH-ALGEBRA-001

Generated grammar and closure boundary. Generate the complete eight-axis product for dual exact quark mass cubics from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__dual-colour-binary-cubic-invariants__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	dual-colour-binary-cubic-invariants	selected-root-table-or-fitted-polynomial	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:547c264a9ae3c10878f9521af81a09dea7616e7d04461585a7e136d284ce6043`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:61410ccc1145885dfc11eca3770442ad2bef5e540aa5dbb9fb8cd32a7ee1ae69`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:b32992ae11355f4c1222ac08e26ec9285ac6c9f5f1a3cb45281bdac6feb5bd24`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:85127dfdb5b409fc2dae381443a60cbe2f809ea31e7d5355d5dbb886018d8687`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:d70b91bb4102715e1376abb8ad5828a0e39ba029cdacba5fbd54608c46667cdb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`.

Independent certificate:

`sha256:6172e60c7ad6c11777a9e1ee9f69cf506516e6fdded501ef3aab125e24d5c5360`. Engine

external-validation hash:

`sha256:2c90259c314ae4ef5a15450aec766bcfb4c699a7665b9684a4a51502606b7728`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed

consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:af7d8959524fca444e74bf7fb69c4ea963a99a5275a7dd3f6bdddffc18a68033d; engine receipt
sha256:e6fccd6c428a5ee8c9126ae25ef145d8c5780d69465d1bf878d3ff39b34969e3 at
receipts/engine/model_admitted/SFT-PHYS-MATTER-QUARK-CUBICS-003-e6fccd6c428a5ee8.json;
empirical-validation hash None; measurement receipt None.

153. Terminal quark self-coupling dressing

Claim identity: SFT-PHYS-MATTER-QUARK-DRESSING-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The sealed terminal electromagnetic ratio transports once through the exact cover-depth roles: the depth gap two lifts only the central down-type carrier, while depth seven is retained over the upper up-type pair.

The exact terminal dressing factors are $(\text{inverse-alpha}+2)/\text{inverse-alpha}$ for the central down carrier and $\text{inverse-alpha}/(\text{inverse-alpha}+7)$ for the upper up pair.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-CUBICS-003
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for terminal quark self-coupling dressing from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__central-down-lift-and-upper-up-retention__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.

Axis	Forced form	Eliminated form	Elimination reason
relation	central-down-lift-and-upper-up-retention	measurement-selected-mass-correction	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:095d27329f4dc0c6e6af69bfb9e6a17083f65569b33024d30306d6edcd6b235c. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:8e45d733f0daf6f32844e1b081dea2d4b75ab3ef5e8279902d26d8a35d0920cd.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:c75971596e5fc57077c4e999c0278481059070b7fa51bd9d8aa728d33cb73879.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:7476de235ba285c3b9a7028eeb12744f1a2860fa297e15b8475989a3bf9293b5.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:74c6d4ac53311668b08b2c242aafeb88eaf351e2ce8352ab9f29f755632b7af2.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c. Independent certificate: sha256:154aff97fbb624726151edecbd57b074b85836b80cc01859a1da22b51ea7e511. Engine external-validation hash: sha256:6156bee434308622c9f3c8246c2844886da5403bbcf401d7304e00227d87b475.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise

- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:5a1c9cd707d96b02bec6217a0dfb4c49cca00839f5f431ed0c747ce89cdea3e4; engine receipt
sha256:4156d17fa99d73374151d8abebba074fb43f2d9852ae3c3e15564f8a49dd7f6e at receipts/engine/model_admitted/SFT-PHYS-MATTER-QUARK-DRESSING-003-4156d17fa99d7337.json;
empirical-validation hash None; measurement receipt None.

154. Exact CKM fibre alignment

Claim identity: SFT-PHYS-MATTER-CKM-FIBRE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The tripling fibre of the holding point and the tripling fibre of the One force a complete asymmetric three-by-three overlap object with uniform diagonal, unequal adjacent bands and one unique far corner.

The exact fibre matrix is ((8/9,5/9,2/9),(7/9,8/9,5/9),(4/9,7/9,8/9)).

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete eight-axis product for exact ckm fibre alignment from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__complete-two-fibre-overlap__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	complete-two-fibre-overlap	nine-independent-mixing-dials	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.

Axis	Forced form	Eliminated form	Elimination reason
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:a2b0b741ae35367bbdc0ed310a52292efde749b5093b19407a1c1f72ee3487e4. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:fb783664d4c25bc1c9750e58db0b6a44bccf89429900d588b5bfa11c9332232e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:241bb19700236e9f6d925fcd6c75e492842a29aa65bdc70f88ff7b5df0ae3763.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:9ced90ecd8ba83d54f14e70ccc9a5fb74243661882b990c4bf4031419ec70901.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:9298b0f05f61cb7339d06fd6a1c1ab0fa267a2538f5fe2cbd4d46502c14c3944.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c. Independent certificate: sha256:e6a1e4fe62c67dcdb0ffb60c2c53f3c15fb23a653d6e661542950dade1fb1d9a. Engine external-validation hash: sha256:32c89b09e1387092b89d67b6a906c0eeac5aebc65c9e612b8ce0c060ecd11da9.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:8dd6650b21e8c5fcdec9f7539ab0861eab3981ab189243d3826d62f4a5afb674; engine receipt
sha256:2cf6948eb88b328bc6efb24b394a37cb8c6d2c32044196fa39ae189488976ea3 at
receipts/engine/model_admitted/SFT-PHYS-MATTER-CKM-FIBRE-003-2cf6948eb88b328b.json;
empirical-validation hash None; measurement receipt None.

155. Mass-root CKM physical mixing law

Claim identity: SFT-PHYS-MATTER-CKM-PHYSICAL-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The sealed quark polynomials determine the three physical mixing relations: light-to-central down mass, positive difference of down/up ladder slopes, and their joined six-channel dilution; the unitarity apex square is identically 1/6.

The physical relation graph forces s12 squared, s23, s13 squared and exact apex square 1/6 without forming an irrational root or reading a CKM target.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-DRESSING-003
- SFT-PHYS-MATTER-CKM-FIBRE-003
- SFT-PHYS-NEUTRINO-CP-PHASE-002

Generated grammar and closure boundary. Generate the complete eight-axis product for mass-root ckm physical mixing law from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__polynomial-root-mixing-graph-with-six-channel-apex__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	polynomial-root-mixing-graph-with-six-channel-apex	independent-fitted-CKM-angles	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:9647ee9a8a9168aae99a4ef0a91bf849fc7cddb4984e41d88eadf0a0caa067b9. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:842d818dd1dbaf0401793167348adb23e6abea95833937abc2bf48eb3a93fe57.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:3a019d7e9e72f37435f1ff61184c85a4ee92346f81820b84cb230784c00f5fe1.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:eb3d77e0b92ea05333d15ca76558fc74f3841a4bbbeb90f5feebe039a0d7f2684.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:65d542f27268e01e6bcb82974ac06fe70d55b8ff75c81786e9001fd2c563fb1d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c. Independent certificate: sha256:d8e02283338b80713372ae755c0e7014d7b1214c701e119d83fbb5a618092de3. Engine external-validation hash: sha256:bcea55c9efa76d66b08ec396a8b55e8dc5b4c6e2d4953da69cd1a07e398f9aa1.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:90a16983817869e8773eadc5d7820bb5216766026cb0a4649adbb1cfd0b489fd; engine receipt sha256:beaa25910b1a8fb944c90bca43b8cf07fec997756bb3cf3b96724133eee18d4a at receipts/engine/model_admitted/SFT-PHYS-MATTER-CKM-PHYSICAL-003-beaa25910b1a8fb9.json; empirical-validation hash None; measurement receipt None.

156. Proton-to-electron hierarchy reconstruction

Claim identity: SFT-PHYS-MATTER-PROTON-ELECTRON-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The leading exact charged-lepton cubic and the colour-bound one-third baryon carrier force a dimensionless proton/electron ratio graph; its algebraic roots remain sealed by exact rational intervals and are compared only afterward.

The ratio is (1/3) times (muon mass minus electron mass) over their product, where both masses are squared roots of $x^3-x^2+(1/6)x-1/485$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-PHYS-MATTER-QUARK-CUBICS-003

Generated grammar and closure boundary. Generate the complete eight-axis product for proton-to-electron hierarchy reconstruction from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__one-third-baryon-over-lepton-polynomial-mass-graph__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	one-third-baryon-over-lepton-polynomial-mass-graph	measured-proton-electron-ratio-as-input	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:917fbf15ac520dc18357a4db061cdaa98bf1dc415617977c922ba28cf407f8f3. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : 850ea0d650a082e616ebcee9b5f92d0a3aae5663cad409e0d31d849411efcbb3.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : 4b173236c3f303a862a783db2b06a8e11251661e35916f8e02934273bfbfd62c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 71d55a7bb4d79fb1bc40b9039b64c7c49ea33f495a91a82bb6ef5b1f462eeee.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : ddfe5ccdacb15c5b2a6f890302d3fd4ba72453a56e27965b63f159f4675eef41.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c.

Independent certificate:

sha256 : e8384dd3306ead33578b769b21fcc66aece0efee93133f2913509ea805908b9e. Engine

external-validation hash:

sha256 : 3a41fb146747eeda946530c9234e0136982bad4435796ad327506b499ec6629d.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : d041effc72083dd57b57e76ec98be80792624f76010a05243e8ecd06fa3aae0d; engine receipt

sha256 : 456305f867a1afc75a47afa355364f48132b901c4c1f0f7a7ace9a40be64b988 at receipts/engine/model_admitted/SFT-PHYS-MATTER-PROTON-ELECTRON-003-456305f867a1afc7.json;

empirical-validation hash None; measurement receipt None.

157. Exact neutrino splitting ladder

Claim identity: SFT-PHYS-NEUTRINO-SPLITTING-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The non-unison counts at down depth five and doubled depth ten force exact rung ratio 33; the independent complete depth-five cell and covering-prime-square predecessor translation forces solar-to-atmospheric ratio 3/100.

The Mersenne rung ratio is $1023/31=33$ and the physical translation ratio is exactly 3/100.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-NEUTRINO-PMNS-ANGLES-002
- SFT-MATH-ORBIT-NUMBER-THEORY-002

Generated grammar and closure boundary. Generate the complete eight-axis product for exact neutrino splitting ladder from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__Mersenne-rung-and-cover-translation-cross-lock__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	Mersenne-rung-and-cover-translation-cross-lock	measured-splitting-ratio	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:45c5d6311f12a8fa4be64f48939e8206b745daeb23c69a4ce35f769af7ac11a0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:bffb647896f78e999ca001940f829ac69bbf25f858dc9617d81c14aa5e9bc883`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7f28bd60d0a82607fc553c23fb62ef1e6155ac819f6d5d01c6b66c0adb0e6580`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8dc26c8a1ca7c3362efda86a14f0caf220a47611534baf669342b4c26812cfa.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:6ef1d94bf5cc0d6a631a199c3dc817a203e04bd4d546d6075b4a50dd476d3d91.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c.
Independent certificate:
sha256:9e429034da7aa71d771683095c72f5ae308a9e29fbbd2d2dd564019420987cdc. Engine
external-validation hash:
sha256:2abac81fa7e8033a78190bcdbac185f9728f4132246ffa779e85e5b282b5530c.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:122cbab6e170a47c4af1e4f210294c097929f323ea0ab21baf046a9d36516032; engine receipt
sha256:037d94c3ae94a015aa5177ee829fcd77e1137f5cdb10cfb2a9476f7a4bf4f5b7 at
receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-SPLITTING-003-037d94c3ae94a015.json;
empirical-validation hash None; measurement receipt None.

158. Positive absolute-neutrino mass structure

Claim identity: SFT-PHYS-NEUTRINO-POSITIVE-MASS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. No-Zero requires the lightest mass-square carrier to occupy a positive cell rather than a numerical zero. The first complete depth-five cell forces 1/32 of the solar anchor, normal ordering and exact mass-square coefficients 1/32, 33/32 and 3203/96.

In solar-splitting units, mass squares are lightest 1/32, middle 33/32 and heavy 3203/96, forcing positive normal ordering; eV translation requires one post-seal solar anchor.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NEUTRINO-SPLITTING-003
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-PHYS-NEUTRINO-PMNS-ANGLES-002

Generated grammar and closure boundary. Generate the complete eight-axis product for positive absolute-neutrino mass structure from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__first-positive-depth-five-cell-plus-forced-splittings__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	first-positive-depth-five-cell-plus-forced-splittings	massless-lightest-or-fitted-absolute-mass	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256: fe8cf8148443fd9c8620c22d46163b89e349a08ad5211c4232de07f73d978a98`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256: 8510ba04a8e69e1f4652a0e63a376de644e03eef374d598228c13a6b460294be`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256: 0493c74e4a1ecaf84fd3b1964d6a115318efe74a7e17ffef3d229040f86cdb41`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256: 96922b9c6430d09b84f4d43e58901bfbc4a5ed4a58dd031e9e423703f858c41f`.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : 9fb3ab6932fccb343d2bd9671f11e9bfc5f26dceee9e13463328ccff2c521dfd.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c.
Independent certificate:
sha256 : 99751f2439c5e78d2176bba578d7e9aa3cf00c9eb6fa73b164ec27295d277777. Engine
external-validation hash:
sha256 : 5273caa93f658cdb1c01da093ffb06d9252bf6c102b21ff0ca7b95c2ba4813e4.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no claim that an SI/eV scale is produced without one declared post-seal dimensional anchor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 3dace26ddd3e2dedad0068cc4aa8d87eb183e2a14c7d4830f0f7c4d823a36604; engine receipt
sha256 : 049c0ce9ac0b05591b8636141be26c976a31f39aabb5941ad6aca5ffb60dd7fd at receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-POSITIVE-MASS-003-049c0ce9ac0b0559.json;
empirical-validation hash None; measurement receipt None.

159. Physical PMNS and CP support

Claim identity: SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Binary, generator-three and complete depth-three support force the squared PMNS triple and its normalized electron-flavour weights, while the unique self-antipodal Fold position forces the maximal phase carrier.

The squared triple is (1/2,1/3,1/48), CP phase carrier 1/2, and electron weights (47/72,47/144,1/48), which sum exactly to the One.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NEUTRINO-POSITIVE-MASS-003

- SFT-PHYS-NEUTRINO-PMNS-ANGLES-002
- SFT-PHYS-NEUTRINO-CP-PHASE-002

Generated grammar and closure boundary. Generate the complete eight-axis product for physical pmns and cp support from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__complete-PMNS-support-with-self-antipodal-phase__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	complete-PMNS-support-with-self-antipodal-phase	measured-PMNS-fit	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:69117b5094f9aa163f4de46be2856a742cc23239cf79d4480b99ab5a9fbd0c92`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:22174211eab4cf8c4c4e0a5cbe2c5142b73bc60e9f24c9f9e94f611df751695a`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:ad207c162a66d6615c0f9aa00e03d44fe60d0c90115e59527b864b203ec11aa8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:97d462bfe73083fb33887ac148411ef5b9984290fc705805d438f11c77118546`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7bcda760e7ab87c86ff371ad2e7db7461326aecedcaa6399a00436813df78ab81`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`.

Independent certificate:

sha256: 4de43a0c939ef6bc4d56fa2fc3351199f3a3251355970dcb19c91651f5ceab09. Engine

external-validation hash:

sha256: 9dbee323d92016a6ad0800a8b5ae424bd07e558bffa0112f71016d90883fcd06.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 3de5e79626b2ffb9a4f60b807960030d601dd222fbf903ece4cb28cccaadaf25; engine receipt

sha256: 6cc0b2828e8a304fd80c274f6e9a2e722cd1bebd68cff071f4c394bf25c872bd at receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003-6cc0b2828e8a304f.json;

empirical-validation hash None; measurement receipt None.

160. Single-hand Majorana discriminator

Claim identity: SFT-PHYS-NEUTRINO-MAJORANA-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete half-One fibre has two held hands for a Dirac pair, but the neutrino census retains one hand. The unique self-antipodal half-One is therefore the only generated massive one-hand coupling, forcing the Majorana class within the declared grammar.

The charged two-hand separation is 1/2; one hand cannot instantiate that pair; the half-One alone is self-antipodal, so the massive one-hand carrier is Majorana.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001
- SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002

Generated grammar and closure boundary. Generate the complete eight-axis product for single-hand majorana discriminator from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The

generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__single-hand-unique-self-antipodal-coupling__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	single-hand-unique-self-antipodal-coupling	asserted-Dirac-or-Majorana-label	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:c1b7f7d253a9e819ee91010eb08d33ef1aadd06fcd56d502b29fac54dfcb4daa`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d3c3f79e6df43e5e3b7d36e9c42f47854bd3154b913c378663dfe7419f541299`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:9685fc6d1764e908cc4e6bf7996f595af4a419e60774341668d3ced6e23efac1`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a9fcf706f3f16f7ff18c246a0781d23242be16ef3812a0d71ff8ad3cf7aa7bc4`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:d237227957ff1354bd3173bfbb78ca96b4f85f559c40b3544188ad8145c2b4d6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c`. Independent certificate: `sha256:4b5bd24e4fd47ab15c9c86fa8e760fe7117978ec97365a248c9059b7e711e23a`. Engine external-validation hash: `sha256:84790de07d96b9f6f86311bc80a14ebb6db112d9b4c37bb1262b6bd75cb30056`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:820ff5018ea73ac86f2793fd7cea4a77c11ed4f9cb13203250761f724496f355; engine receipt
sha256:da6688fb7442a5d3f4e3aa39f745c04aebc87102af60186b6e1bcd4e7078345e at
receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-MAJORANA-003-da6688fb7442a5d3.json;
empirical-validation hash None; measurement receipt None.

161. Neutrinoless-double-beta noncancellation prediction

Claim identity: SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Positive Majorana masses and nonempty electron-flavour weights force a nonempty neutrinoless-double-beta amplitude. Exact rational root bounds certify that the middle contribution exceeds the other two combined, so arbitrary held phase opposition cannot close it.

The effective Majorana amplitude has a strict positive coefficient floor and finite ceiling relative to the square root of the post-seal solar anchor; therefore zero-nu-beta-beta must occur within this grammar.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NEUTRINO-MAJORANA-003
- SFT-PHYS-NEUTRINO-POSITIVE-MASS-003
- SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003

Generated grammar and closure boundary. Generate the complete eight-axis product for neutrinoless-double-beta noncancellation prediction from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__positive-Majorana-weighted-amplitude-with-noncancellation__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	positive-Majorana-weighted-amplitude-with-noncancellation	asserted-decay-or-zero-mass-limit	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:a4c5d66b5cdb644eeb61f969e08b3444becb787c7e611d980e6d1e0331c9503c. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:43091e29d7f6004930a664a339cbb34a1409fd515c90a39dd0874d7dd589f83e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:30bb52c6abc8abe8c4c29636b11083992c43dfdc43748017ba89930f62f43eea.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:a39dfb3fb09593934949b7d2f859ba4ec217eea11831e11f93ff39e56870701e.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:302276b0bcc22dcdbbb6230b3729a98dafae1bcf77ac17f90218922a36bd56ac.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e5c0c734f062ac7768bce6d36d04ce2ce2e0969c80ef1386d074f0e6a09b357c. Independent certificate: sha256:92eff69bcd9d1e2eb7f9fea469f76d2ccec2663776bbf19a5df9316e2ac57aae. Engine external-validation hash: sha256:0ab95cf6291c1264dc5baef83e7b41aace1e91e2426833a8a18204d38ce7edd5.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no claim that neutrinoless double-beta decay has already been observed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:20d78c28b28c0d7c6a524a6b86ca039ec35f6d448a4dc7d9d3542aa70d8c15f4; engine receipt
sha256:3b7bc18e1160fe2a3633ce7cba15b9c3001f55ed73516ef9a6a6db474292f6fb at receipts/engine/model_admitted/SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003-3b7bc18e1160fe2a.json;
empirical-validation hash None; measurement receipt None.

162. Baryon-to-photon Fold transport law

Claim identity: SFT-PHYS-MATTER-BARYON-PHOTON-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The sealed quark-root mixing graph supplies the squared CP transport measure, and the unique Fold imbalance share transports exactly half of that measure into the surviving baryon residue; the dimensional-free prediction is $\eta = J^2$ times half-One.

The exact algebraic prediction is $\eta = J^2/2$, with J^2 generated from the sealed quark-root CKM graph and no cosmological target in the relation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-CKM-PHYSICAL-003
- SFT-PHYS-NEUTRINO-CP-PHASE-002
- SFT-PHYS-COSMO-COMPLETE-BUDGET-001

Generated grammar and closure boundary. Generate the complete eight-axis product for baryon-to-photon fold transport law from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__sealed-Jarlskog-square-through-half-One-imbalance__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	sealed-Jarlskog-square-through-half-One-imbalance	measured-baryon-abundance-or-free-efficiency	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d78a4c8e523d4eec8dc217efc67feb5db8b491739cf7ce8e107ca1b82e16ead7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:61754f4c80f9c2d1d531657d2ff64cc81659f0bb7dad4c8a0199e552c6019deb`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:fc4b67af7d485e826654614c7398cfa4945a3f0aeef3a3ce2468ee239fe3682f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5a40398d19173de4ad04e171f084e2c2a6c7a258786ff1a22f7a2e328517badf`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:4a08608f58003ada72897fa2a1ec1e3ea7542ab307baaf2cdb9311fa451e049e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4`. Independent certificate: `sha256:56b560437ea8d0b326b2cc9c184b18800ba227bd9345c6513b647eb4e14a2675`. Engine external-validation hash: `sha256:02e2867f1b41f8ecacc85ada35978b7a13d22be159345e88ebca62fba0c7be41`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed

consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:46e32e2d1115d1e27342a153fe03037e3fc0bb0108893096576f3a65d1f7d763; engine receipt
sha256:2455218c2d7918d015bcacceffca6308da79a73beedf33f9ecdbd5a715413ceb at
receipts/engine/model_admitted/SFT-PHYS-MATTER-BARYON-PHOTON-003-2455218c2d7918d0.json;
empirical-validation hash None; measurement receipt None.

163. Common quark/lepton mixing construction

Claim identity: SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Quark and lepton mixing are the same complete alignment construction between one generated mass fibre and the channel fibre; only the already forced generating lock differs, producing orbit-class quark alignment and self-antipodal lepton alignment.

The quark lock is 2/3 with its exact ninth matrix; the lepton lock is 1/2 with first row (5/6,1/2,1/6), and all three lepton mass sites return to half-One under tripling.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-CKM-FIBRE-003
- SFT-PHYS-NEUTRINO-PMNS-ANGLES-002
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002

Generated grammar and closure boundary. Generate the complete eight-axis product for common quark/lepton mixing construction from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__one-alignment-law-two-forced-locks__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.

Axis	Forced form	Eliminated form	Elimination reason
relation	one-alignment-law-two-forced-locks	two-independent-fitted-matrices	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:9f8798415792c18a220f22546a3a66e5aa8970fe657200da378e560ada41d60a. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:544ba03b8295520e583bd75510bb66fe137d4cf2adaf4aaa3907356b8c6ca15b.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:884c08243e089403b23da33978a75ed4bdf166628c02a3f5376472440b3fdc37.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e787729f2a04937fb31c7ee5c25ef545f24e2159b10fe1ce1e94816c6074e97d.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:1a51052cc2b9218db55cdd85e71a5facaee6179886ed860b66147e578c755c44.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4. Independent certificate: sha256:b1fe3faa319dacf95e0e4c92a4fcb9b16e261e415f2a8319b921a2f4601dc63a. Engine external-validation hash: sha256:397c4b8561c45d1467a62a387f3bd59f2c1d7f0ec85d459d8918b6b6439bf0af.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise

- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:78b6a1f0ad3701b6fb0e0afc8cc150d00c7627cd5f86967fbb952c458810baac; engine receipt
sha256:8a8a498160b16bb65ebfc1071d404a5db6de08b6c9f3499765889ca17677b501 at receipts/engine/model_admitted/SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003-8a8a498160b16bb6.json;
empirical-validation hash None; measurement receipt None.

164. Depth-independent Fold mass-ratio family

Claim identity: SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. At every positive combined-ladder depth d , complete binary-by-generator support forces light part $1/N$, heavy complement $(N-1)/N$ and heavy-to-light ratio $N-1$ for $N=2$ times 3^d .

For every positive d , heavy/light = $2 \cdot 3^d - 1$; the first three values are 5, 17 and 53.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-COUNT-001
- SFT-MATH-DISCRETE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for depth-independent fold mass-ratio family from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__complete-support-complement-ratio-family__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	complete-support-complement-ratio-family	finite-ratio-list-or-measured-masses	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.

Axis	Forced form	Eliminated form	Elimination reason
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash
sha256: 4935a33d1117e20d1f7406bc3248124286d104819de2be5ded94e0f8d34f5ebe. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 0fd23506fa091043c2ffbb00c877129a878ed7bfc9039e22232b0e150c911fe4.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 53f6dcf22d65eb551c53d783151f4fa1423644b716bd727a77b3be791b629c2f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: a74885a7ccf6c29ceaf9b181da5e8b7e865ecc387392964b9f7e7633c19b99c.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: 1f5516a3e74e9f6694e1a6c3fb1cae91f2b9d4621c52e914ce52e888ffb7f691.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4. Independent certificate:

sha256: eb8a922de1e53cc992e2efaa71385e066a860bdcd47f714c07090d2a87ee5dfc. Engine external-validation hash:

sha256: e03703957018b7727249aaed6bca1b756fbc240afd891ad14a02753da02774ee.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:0cd36623ca774b0159e22354a79d4ccaa6de4cf617236e3d7a788443bb772a65; engine receipt
sha256:2bd73d71dcee6d09c2d427da1e1ac79c1866ef75949c5571509fd37f8fd21168 at receipts/engine/model_admitted/SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003-2bd73d71dcee6d09.json;
empirical-validation hash None; measurement receipt None.

165. Mirror-closed lepton mass carrier

Claim identity: SFT-PHYS-MATTER-MIRROR-MASS-CLOSURE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The generated lepton positions and their exact One-shortfalls are the same complete multiset in opposite order, so every symmetric invariant is route-independent before any cubic-root comparison.

The position triple is (1/6,1/2,5/6), its shortfall triple is (5/6,1/2,1/6), and both sums equal 3/2 exactly.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product for mirror-closed lepton mass carrier from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__position-shortfall-multiset-identity__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	position-shortfall-multiset-identity	independent-position-and-mass-tables	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:9ec46ff5c95b2fd48a31bc690e9bf84766a3b1ffdf1341ee4dee16431b5b479a. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:c1dfc1914fc49f2342ab4c3ef9c0275934af2f03335bbd7983d791f51a3aed09.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:ae8180a6ae41198e1486c567c366381774c279ac9099ec8943d4dc1978381f32.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:abf06429178c30c9d1499e9968d6db8547a9181d561da41d39885f03fa8cf4b4.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:003e0a517d302c9b660ea8a3dcc7083bc8eae53d3b9480c6eaf1bcb61e975b0.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4. Independent certificate: sha256:b4ec2c038542041df9d42be1edf37a296c1686a090288661014740f84bee158c. Engine external-validation hash: sha256:731e0a163bfec04a79482e3ccdc3c93914404c41b83735890fc18e3ba6ea4824.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:cc112ad075219f223756ab606e34c4979a6fff9bbd4733570352cb652916ae03; engine receipt sha256:bd01069f06d421e37672d6a58657824c529f0bf33ce2286dc723a40d1d7855e4 at `receipts/engine/model_admitted/SFT-PHYS-MATTER-MIRROR-MASS-CLOSURE-003-bd01069f06d421e3.json`; empirical-validation hash None; measurement receipt None.

166. Mixing-row coupling return law

Claim identity: SFT-PHYS-MATTER-INTER-ENTRY-COUPLING-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The first row of each exact alignment matrix returns its own generating lock after one complete One is held out: the quark row returns two-thirds and the lepton row returns half-One.

The quark row sums to 5/3 and returns 2/3; the lepton row sums to 3/2 and returns 1/2.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002

Generated grammar and closure boundary. Generate the complete eight-axis product for mixing-row coupling return law from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `g-generated-exact-carrier__admitted-root-trace__first-row-residue-returns-generating-lock__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	first-row-residue-returns-generating-lock	independent-row-normalizations	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:534a23bfab502320e3aec41886fcc96ea1acbb81bce1303e015beb7551aecbf4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:53ee82af1f684449f3abdbea5fb6b136bd55d010c973619074fd546b78f7cc25.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:ac116620d6b74cfb550484b4131b1a9fa0fba58e35916d6a8d81e9f4275dd119.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:83f7e568165c611deb04ab0918666b221ed83002a3ad2c8353d25272e14e97e3.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:1193336759f76dc52770c3acb8a4a80fc07c783c87fb25d19049083f4b6fdc08.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4.

Independent certificate:

sha256:7159d522ec42e11a1789670be8544263e7a917ab561037dff7da0076b2980c08. Engine

external-validation hash:

sha256:6cf97db02aa7a8085d86430429ecf1b6b628fce538a784427eb354983117a76b.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:f5d787dfeaa767ad67d5d3b4389c64742ffe4dfcf50619513936a8e372f9ef6d; engine receipt

sha256:c1ef281840dac57df5ac309768874c793c7880fcbbfe5299d1b4deff3f2531ea at receipts/engine/model_admitted/SFT-PHYS-MATTER-INTER-ENTRY-COUPLING-003-c1ef281840dac57d.json;

empirical-validation hash None; measurement receipt None.

167. Common generation return depth

Claim identity: SFT-PHYS-MATTER-GENERATION-DEPTH-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. All three generated lepton sites traverse one generator-three action to the shared half-One waypoint and one Fold action to the One, forcing equal two-operation depth despite distinct positions.

Sites 1/6, 1/2 and 5/6 all reach half-One after tripling and the One after one Fold; their common generator-operation depth is two.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001

Generated grammar and closure boundary. Generate the complete eight-axis product for common generation return depth from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__one-tripling-plus-one-Fold-common-return__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	one-tripling-plus-one-Fold-common-return	generation-specific-coupling-depths	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:c34f9858c4d40073f21cc0b619c683df8e417496b6d2a5958de06080ab2de1b5. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:d070d1a38a9f793bdd976358859df7f00dfa35d239432ba46d07788fe33f8b89.
- tampered_source: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:70b95a2d2cd3e50f23d8a7154f674325af2057d6c6438c0a699983301409c5a4.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8f30e25a78d4faca49dcfce018fe0ea6ea0624d82dacefcb37083e78adf35754.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:5dadf6ad9b24c7e8e3dacea28aca0a939fb72cfc51dcb835109a18d1c2d52661.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4.
Independent certificate:
sha256:1e31a785a752e048f5a2c7d57f7b952bd5a1004a5e5761dae18c924c06eefc5a. Engine
external-validation hash:
sha256:a0e248347df405deb85786191034fda3b58477c2d13e646489df5f6b05006bf9.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:173a98a17650553e593e97fa9882cc6f5ba74b783a0ede98df20e631e81cf5d8; engine receipt
sha256:a120ad261a4308fc3008ccaab35d38939f6152f82c31f4f00c773582e3f2d726 at receipts/engine/model_admitted/SFT-PHYS-MATTER-GENERATION-DEPTH-003-a120ad261a4308fc.json;
empirical-validation hash None; measurement receipt None.

168. Lightest-quark confinement lift

Claim identity: SFT-PHYS-MATTER-CONFINEMENT-LIFT-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The generator-three sharpening acts on the complete colour-tower predecessor at the independently forced down and up depths, while the sole lightest unclosed generation receives the complete two-fibre confinement action once.

The sharpened down/up products are 3/1454 and 3/13118, and the lightest-carrier confinement lift is exactly the Fold fibre count two.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-CUBICS-003
- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001

Generated grammar and closure boundary. Generate the complete eight-axis product for lightest-quark confinement lift from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `g-generated-exact-carrier__admitted-root-trace__colour-channel-sharpening-plus-one-two-fibre-light-lift__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	colour-channel-sharpening-plus-one-two-fibre-light-lift	measurement-selected-quark-rescaling	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:8239d56c52092676eb989075291d6a8299930949e94df3fb3eed399b5c1f1d1c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:1a448fe27f4aecb62ef3cb8dec4b0eb060555ce3b287baa87957c8f5684268fe`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e1a7d6501000853476073c8b6663ed8593169d41136066d9580091ff11a6dd83`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:83a2544c740cb324dd0ad054cec80e316c0837608cadb041659e6d114a3a3912`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:00730b4fbf6ba7db829ad6ceac234db8610661ec657d2905518fe8270d76b617`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:e297ecaa54e2a0b727ffaecfa9f83c02bed5e49311010dc13417e083ff294dc4.
Independent certificate:
sha256:5f6bd05f4e59b83fcf89d425a3c85270175298598347f164027a77a8b588d3a0. Engine
external-validation hash:
sha256:525e79a933241afd45233f11cbe657ff63e801318f98d9b28975b38974ba9491.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:022b73b16a6366287fb7a90cbd6c836e9eea19e3255568c74500f32e5ed79102; engine receipt
sha256:53f2509cb96b422753fb8211e2d7fdb93b351d64fb85eb1a90a7dacc717e646a at receipts/engine/model_admitted/SFT-PHYS-MATTER-CONFINEMENT-LIFT-003-53f2509cb96b4227.json;
empirical-validation hash None; measurement receipt None.

169. Terminal physical CKM transport

Claim identity: SFT-PHYS-MATTER-CKM-TERMINAL-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The admitted physical CKM slope named terminal quark dressing as a dependency but its leading function transported only the two bare root slopes. Completing that declared carrier forces one terminal electromagnetic share across three colour channels, retained through the already forced depth-seven upper-quark factor and appended to the positive slope separation.

The terminal CKM contribution is exactly α times $[A/(A+7)]$ divided by three, equivalently $1/[3(A+7)]$, where A is the admitted terminal inverse- α ; it appends to the leading positive s_{23} slope gap, while the existing s_{12} and six-channel s_{13} relations are preserved.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-CKM-PHYSICAL-003
- SFT-PHYS-MATTER-QUARK-DRESSING-003
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for terminal physical ckm transport from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-carrier__admitted-root-trace__leading-slope-gap-plus-colour-shared-terminal-alpha-under-up-retention__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	leading-slope-gap-plus-colour-shared-terminal-alpha-under-up-retention	leave-declared-terminal-dependency-unused-or-target-select-a-mixing-shift	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:a79816273fff55c7bcd2390028da1b04e651200a164badcf80b09db7a91a696d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:fe3eec0be13f9f525a03306c6a999497d0c9d2ff2f7d19173b14744317dd4f25`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c4352e2e942341ffdda9795e3022e1b5890d2380334047acd6ff14740b1342e9`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0417c11fe4f2247781f74107ca13f86aa39fb1d49dfb7f52dc430422fc153837`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:e511a716293d36318f800889ed8ff7ff3f38074d0680c24c953cfee5ad1a9501`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ce3435925d6dd0b23b93b50097de951cb6dae8b0dc62b00589ede5117dcc9ae`. Independent certificate: `sha256:fbd90e63b1844637cba91aa60a1bad8c85899c974a05f279d805b6140f598077`. Engine

external-validation hash:

sha256 : 42365683488a296c6cd9e7339f86eedcbfccb52ac3cb8ba3bb5bea0a2cf49d21.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026

Observed comparison records:

- PDG-CKM-S12-COMPLETE: predicted
terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed;
observed
terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; exact
match `True`
- PDG-CKM-S23-COMPLETE: predicted
terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed;
observed
terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; exact
match `True`
- PDG-CKM-S13-COMPLETE: predicted
terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed;
observed
terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; exact
match `True`
- PDG-CKM-J-COMPLETE: predicted
terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed;
observed
terminal-CKM-s12-s23-s13-and-J-all-overlap-complete-PDG-three-sigma-intervals__observational-derivation-disclosed; exact
match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: Any sealed CKM interval misses its complete registered source interval, a row is omitted, the target enters the relation, or observational provenance is hidden.

Measurement receipt: sha256 : 604170b5df7b720c1094ccc24ab02cb9afd8f1901f3f6a3548e1bfdec23b8977.

Isolation certificate: sha256 : 9ac0111812fe52190eda155f69b04eb3a97bb61b3341d4c70c2cbee5566acd23.

Custody certificate: sha256 : 0f1998e5f3ae4dcdbd9e86256180b9c09db9f313352e4a06ba279fff35862ca79.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-cosmology.pdf>
(sha256:652b186e29c835c9cf8dfd3393fe19c6a81917666989331c5ca060dc789ae9e)

Registered target identities:

- PDG-CKM-S12-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- PDG-CKM-S23-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- PDG-CKM-S13-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- PDG-CKM-J-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed

consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no claim that this post-adverse completion was a blind forward discovery

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7d72dcc15e07636e51dfe544a4e23ae594982850e38eed19a8104cb33c43d8; engine receipt
sha256:745bc9840a7f5396866ff2e581fe700d9f254bff2538ef3c31eb74fc8683513e at
receipts/engine/model_admitted/SFT-PHYS-MATTER-CKM-TERMINAL-004-745bc9840a7f5396.json;
empirical-validation hash
sha256:0559e001acd3bc3927566d610d059744d52960b7fa419ae2a8518fc8c87896f5; measurement receipt
sha256:604170b5df7b720c1094ccc24ab02cb9afd8f1901f3f6a3548e1bfdec23b8977.

170. Terminal baryon-to-photon transport

Claim identity: SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The already admitted baryon-residue law consumes the complete terminal CKM graph rather than the leading graph; its unique Fold imbalance share remains half-One, so no abundance scale or efficiency is introduced.

The terminal baryon-to-photon prediction is $\eta = J_{\text{terminal}}^{1/2}/2$, with $J_{\text{terminal}}^{1/2}$ formed from the terminal CKM squared-support graph and no cosmological value in the relation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-CKM-TERMINAL-004
- SFT-PHYS-MATTER-BARYON-PHOTON-003
- SFT-PHYS-COSMO-COMPLETE-BUDGET-001

Generated grammar and closure boundary. Generate the complete eight-axis product for terminal baryon-to-photon transport from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every exact finite carrier named by the law and the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-carrier__admitted-root-trace__terminal-Jarlskog-square-through-half-One-imbalance__complete-product__every-omission-rejected__seal-before-comparison__full-polynomial-trace-census-controls__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-carrier	imported-parameter-table	A table has no Fold provenance.

Axis	Forced form	Eliminated form	Elimination reason
dependency	admitted-root-trace	prior-answer-premise	A prior result cannot select V3.
relation	terminal-Jarlskog-square-through-half-One-imbalance	retain-superseded-leading-mixing-or-import-baryon-abundance	The alternative loses a forced count, type or retained distinction.
enumeration	complete-product	selected-subset	A subset cannot establish uniqueness.
minimality	every-omission-rejected	uncontrolled-shortcut	A shortcut may hide a free choice.
measurement	seal-before-comparison	target-visible-before-seal	That reverses empirical direction.
record	full-polynomial-trace-census-controls	answer-only	An answer cannot reproduce the result.
extension	no-extra-rule	free-extra-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:67668cac8068a8186be11f4c617ae4bb504f4c08f0b9b4b0f265863f7e6d4c13`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4fc7e50c9e4299e64e95d3adae7f384d9e89d29279995e653d71944758b95642`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c0f168f6e37b6e74d121a47297bbd2f979d86045cfc9f58c17b3e289b15e4939`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ceaf7ee28f7412d9d5c4c5aee94c1a103ab8559bf141aebb801bcaaf8aaf87ee`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f27cc9c43a309ae202de93ac8ae420824339ecf0ed864a0b717dea8bac3b79a0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ce3435925d6dd0b23b93b50097de951cb6dae8b0dc62b00589ede5117dcc9ae`.

Independent certificate:

`sha256:a0ac7c3918ab6a5f35f5cd67b7017b1efe2ddf3c4f0cc65e8c31d6c1906d558d`. Engine

external-validation hash:

`sha256:fb52c027008664ad6b0684437f449f3c5daf6d70397d4302eeff36e55a4d1b32`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- `MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026`

Observed comparison records:

- `PDG-BBN-ETA-COMPLETE:` predicted terminal-baryon-to-photon-prediction-lies-inside-independent-PDG-BBN-and-Planck-derived-intervals__observational-derivation-disclosed; observed terminal-baryon-to-photon-prediction-lies-inside-independent-PDG-BBN-and-Planck-derived-intervals__observational-derivation-disclosed; exact match `True`

- PDG-PLANCK-ETA-COMPLETE: predicted terminal-baryon-to-photon-prediction-lies-inside-independent-PDG-BBN-and-Planck-derived-intervals__observational-derivation-disclosed; observed terminal-baryon-to-photon-prediction-lies-inside-independent-PDG-BBN-and-Planck-derived-intervals__observational-derivation-disclosed; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The sealed eta prediction leaves either complete registered interval, a route is omitted, or a measured abundance enters the law.

Measurement receipt: sha256:8eb2337c426dc25c6c9cb35803b78804dc2ab707f729dba7139fa5f8c37bd26c.

Isolation certificate: sha256:badde313c11651ad1456bc3a14f5f5f1ca4306dc1a1b3e1e65d46d8784a8a81b.

Custody certificate: sha256:e184126cf70ac4284a964d2ebdc73995aafb2c57dc72aae335f309e73eb113d5.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-cosmology.pdf> (sha256:652b186e29c835c9cf8fd3393fe19c6a81917666989331c5ca060dc789ae9e)

Registered target identities:

- PDG-BBN-ETA-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026
- PDG-PLANCK-ETA-COMPLETE from MATTER-FLAVOUR-TERMINAL-AUTHORITATIVE-2025-2026

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, result table, certificate or measurement as a derivation premise
- no target-selected coefficient, bracket, iteration count, correction or candidate neighbourhood
- no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof value
- no solved irrational algebraic root; exact polynomials and rational enclosures only
- no external value accessible before the formal seal
- no claim that this post-adverse completion was a blind forward discovery

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:e0b821d56719441b9abf5e31bc3681ea338d0b05e502f9d6978560d72b6c81b1; engine receipt

sha256:21ba6f25f064602267b8e415ceb3ed6f985e318a3b29e7c8626b1cc71ea4c7cc at receipts/engine/model_admitted/SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004-21ba6f25f0646022.json;

empirical-validation hash

sha256:63ebdda8d0667627c3a7ece5e018c0def2e0bab906e49bf8ff4d9d0f247af74c; measurement receipt

sha256:8eb2337c426dc25c6c9cb35803b78804dc2ab707f729dba7139fa5f8c37bd26c.

171. Terminal composite proton/electron precision law

Claim identity: SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The admitted proton/electron algebraic graph receives the complete electromagnetic self-coupling of its three-depth colour composite. Binary-by-colour-cubed support has heavy complement 53; the complete colour bulk and its three carried boundary channels supply support $27+3=30$; both charged Fold ends force alpha squared; and the sole terminal return transports the two Fold labels through down-depth five and colour volume 27 once. The retained ratio is therefore the admitted algebraic ratio

times One held by $53 \alpha^{2/30}$ times (One plus $2 \alpha/135$).

*Let $R3$ be the admitted exact algebraic proton/electron ratio and α the reciprocal of the admitted terminal inverse- α . The unique retained successor is $R4 = R3 * [1 - \delta]$ in conventional notation, represented only as $\text{positive_take}(\text{One}, \delta)$, where $\delta = (53/30) \alpha^{2/30} [1 + (2/135) \alpha]$.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-PROTON-ELECTRON-003
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003
- SFT-PHYS-MATTER-CONFINEMENT-LIFT-003
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPPLING-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete eleven-axis product of base graph, composite depth, complement, coupling order, bulk/boundary support, terminal return, transport, orientation, custody, provenance and extension forms. The declared exact boundary is: All first terminal electromagnetic dressings of the admitted proton/electron algebraic graph that use each already admitted composite-depth, colour-volume, boundary-channel, Fold-label, cover-depth and terminal-alpha carrier exactly once in its typed role. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: `retain-admitted-proton-graph__generator-three-composite-depth__complete-depth-three-heavy-complement__one-alpha-per-charged-Fold-end__complete-colour-bulk-plus-carried-boundary__one-terminal-alpha-successor__both-labels-through-down-depth-and-volume__hold-dressing-from-existing-ratio__measurement-inaccessible-to-executable-relation__registered-observational-prediction-protocol__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
base	<code>retain-admitted-proton-graph</code>	<code>replace-or-refit-base-ratio</code>	Replacing the admitted algebraic graph erases its receipt.
depth	<code>generator-three-composite-depth</code>	<code>selected-composite-depth</code>	A selected depth can tune the correction.
complement	<code>complete-depth-three-heavy-complement</code>	<code>free-multiplicity</code>	A free multiplicity is a parameter.
order	<code>one-alpha-per-charged-Fold-end</code>	<code>selected-alpha-order</code>	A selected order can read the mismatch.
support	<code>complete-colour-bulk-plus-carried-boundary</code>	<code>partial-bulk-or-boundary-support</code>	Omitting either carrier loses part of the composite.
terminal	<code>one-terminal-alpha-successor</code>	<code>leading-order-only-or-extra-series</code>	Leading-only omits the declared terminal carrier; an open series has no closure.
transport	<code>both-labels-through-down-depth-and-volume</code>	<code>free-return-weight</code>	A free return weight is a fitted coefficient.

Axis	Forced form	Eliminated form	Elimination reason
orientation	hold-dressing-from-existing-ratio	append-unbound-mass	Appending a new mass carrier breaks the normalized composite ledger.
target	measurement-inaccessible-to-executable-relation	measurement-readable-relation	That would fit the known mismatch.
provenance	registered-observational-prediction-protocol	unregistered-target-readable-fit	Target-readable execution cannot issue an independent prediction seal.
extension	no-extra-rule	additional-fit-term	A further term is not supplied by the exhausted first-terminal grammar.

Base and successor. The closure proof is sealed by derivation hash sha256:e558533a1d64a278c2bc3ed093f1c27e07d06e7ba8fa92ea29a73e69a180c1d8. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:6da8830d7839592209d46483244091a89ada278cce758eff2c7649e1befccb46.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:318343b5344c2afffea78d447779ff1015f5a3f9e225721fa927d0eccfc730e2.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e44a69a0e2643fe6c150a20a38555ecfe2a7dc20ecb80d43f25f58d492107161.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:5ad270ae68f84ba1ed0883d3935585922e73076242ad3e3205aed8fe99461965.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:0fa8728b5e9260f04e84a1d640d54dadffcdca7181dec03acfb9a8b1abc2d55. Independent certificate: sha256:9502cb05b282e7b7f7c7a40405577cf03d6363dbe77e5bd6a43e203d0eab275d. Engine external-validation hash: sha256:bd9b4f287e3ad3f3e49e3a70b1bc228e2ac9189504037c906171dcc8e54c0042.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-TERMINAL-PROTON-AUTHORITATIVE-2022-2026

Observed comparison records:

- NIST-CODATA-PROTON-ELECTRON-COMPLETE: predicted terminal-target-free-proton-electron-prediction-contained-inside-complete-CODATA-standard-uncertainty__observational-prediction-protocol-passed; observed terminal-target-free-proton-electron-prediction-contained-inside-complete-CODATA-standard-uncertainty__observational-prediction-protocol-passed; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The sealed prediction enclosure leaves the complete registered CODATA interval, any uncertainty or adverse receipt is omitted, the target enters the relation, or observational provenance is hidden.

Measurement receipt: sha256:7c86525e64500a2255c15a28bc21011e883d2dd54243756c91b504a6e2d43b2f.
 Isolation certificate: sha256:994f4feabdc9bbdd7caa16df5a3801fe4a157eb7d890d6615e63a3d9ffdc52ab.
 Custody certificate: sha256:8c07d0c6d1e8f2d2839f45d28ab551399eebfadb94ff73050656caf872879ec.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:a2fd002521902242e3db10eac896389d2190f88169b827c3575640e9a31f2790)

Registered target identities:

- NIST-CODATA-PROTON-ELECTRON-COMPLETE from
 MATTER-FLAVOUR-TERMINAL-PROTON-AUTHORITATIVE-2022-2026

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured proton/electron value or uncertainty in the executable relation
- no fitted coefficient, selected correction neighborhood or added scale
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no conflation of observationally informed derivation with target-readable execution

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:caec57525d68d96feb9a78bc73f427a1ab961c3b131c950b5579008ee730ef79; engine receipt
 sha256:c52208ec2212ffa3fd16e34b90abfdd3adc083411d7a19914d3591b4f8ea9d23 at receipts/engine/model_admitted/SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004-c52208ec2212ffa3.json;
 empirical-validation hash
 sha256:aa1a599db84034f15fa144add92d2af133935ab2fd3848c8fde56da5a96a79d1; measurement receipt
 sha256:7c86525e64500a2255c15a28bc21011e883d2dd54243756c91b504a6e2d43b2f.

172. Terminal Fold scattering, Rutherford angular law and Compton transfer

Claim identity: SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A scattering amplitude carrier is the exact phase-compatible overlap support of one transition leg; an observed differential channel pairs the preparation-to-transfer and transfer-to-record legs, so its weight is their exact self-composition without an imported complex magnitude. For a conserved Coulomb source the rank-two transfer boundary forces differential support proportional to the inverse square of the exact direction-transfer part, with charge-product-squared and incident-energy-inverse-squared scale. At the comparison boundary this is the Rutherford inverse fourth-power half-angle law. For closed photon-electron transfer, wavelength increases by two direction-transfer parts of the action-over-inertia-speed carrier; outgoing energy is incident energy divided by One plus twice the incident/rest-energy ratio times that transfer part. Right-angle and backscatter high-energy ceilings are respectively one and half-One electron rest-energy support. Forward nondeflection remains the structural empty transfer class.

Paired phase-compatible Fold overlap legs force exact Coulomb differential support proportional to the inverse square of the positive direction-transfer part, with charge-product-squared and incident-energy-inverse-squared scale. Closed photon-electron transfer forces wavelength increase by two transfer parts of the action-over-inertia-speed carrier, exact outgoing-energy loss, and right-angle/backscatter high-energy ceilings of One and half-One electron rest support; forward nondeflection is structural absence.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-SCATTERING-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-PHYS-QUANTUM-WEIGHT-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001
- SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of every exact overlap-leg account, every cross-section measure, every Coulomb angular power, every Coulomb charge/energy scale, every photon wavelength-transfer law, every photon energy-transfer law, both target-custody states and both extension states. The declared exact boundary is: Every exact positive rational nonforward direction-transfer part not exceeding the One; the structural empty forward class; phase-compatible incoming/transfer/outgoing legs; source-normalized differential boundary support; conserved charged-source and incident-energy carriers; one photon, one initially held inertial word and closed energy-direction transfer; sealed or exposed targets; and empty or free extension. The induction certificates are independent of a finite Fold depth. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `paired-phase-compatible-overlap-legs__paired-weight-per-incident-boundary-support__inverse-transfer-part-squared__charge-product-squared-energy-inverse-squared__two-transfer-parts-times-action-over-inertia-speed__rest-over-rest-plus-two-energy-transfer-parts__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
amplitude	paired-phase-compatible-overlap-legs	single-unpaired-channel-count	Rejected by computed Fold predicates: <code>amplitude_carrier</code> .
cross_section	paired-weight-per-incident-boundary-support	unnormalized-outgoing-count	Rejected by computed Fold predicates: <code>cross_section_measure</code> .
coulomb_angle	inverse-transfer-part-squared	inverse-transfer-part-first-power	Rejected by computed Fold predicates: <code>coulomb_angular_law</code> .
coulomb_scale	charge-product-squared-energy-inverse-squared	charge-product-linear-energy-inverse	Rejected by computed Fold predicates: <code>coulomb_scale_law</code> .

Axis	Forced form	Eliminated form	Elimination reason
photon_shift	two-transfer-parts-times-action-over-inertia-speed	one-transfer-part-times-action-over-inertia-speed	Rejected by computed Fold predicates: photon_shift_law.
photon_energy	rest-over-rest-plus-two-energy-transfer-parts	incident-energy-preserved	Rejected by computed Fold predicates: photon_energy_law.
target	sealed-before-release	readable-before-seal	Rejected by computed Fold predicates: target_boundary.
extension	empty-extension	free-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:dfe407aa03c635803c44c57974d102983d3873a525baa241ca3cc4c31e913180`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject an angle-independent Coulomb response and elastic photon energy.; observed Exact transfer-part and conservation predicates reject the false angular law.; receipt `sha256:611a3831f0a7d21d03f350bb88a985e400da613244a25f9c57546a545c5b7962`.
- **tampered_source:** passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt `sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae`.
- **tampered_artifact:** passed; expected Reject a duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete candidate identity.; receipt `sha256:85f7f2fccaa8a82e12423de0043e0cde43f60a6c59bbf25b4c3079af293bc64a`.
- **boundary:** passed; expected Reject a numerical forward null, pre-seal target access and free correction.; observed Forward transfer is structural absence; sealed custody and empty extension are required.; receipt `sha256:7f83757007f5533bab0a7d8d12ed5c9237eeba52b2c9c1626a8cbf5ebd8d3ff0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0d35b75c4957b1b77081f92b1cfcf6cfda6596302a2ae3b324a7242b1a03accb`. Independent certificate: `sha256:633aaa0ff4b45f924cfe3a7ef076b2ddac8083b464eada0999874de472b80e99`. Engine external-validation hash: `sha256:2dfad50881002854fd881204b6420b5c76d8e524fa066ab8b1c028b92b14eda5`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-STI-PUB-1196-RADIATION-PHYSICS
- NIST-CODATA-2022

Observed comparison records:

- IAEA complete Rutherford record retained: inverse-square Coulomb source, charge/energy scale and inverse-fourth-power half-angle differential law
- IAEA historical Geiger-Marsden record retained: approximately one in 10^4 alpha particles above 90 degrees versus the obsolete diffuse-model order 10^4 -3500
- IAEA complete Compton record retained: free stationary electron boundary, closed energy-momentum transfer, delta-lambda relation and $h/(m_e c)$ carrier

- NIST CODATA complete seven-row scattering vector retained with every reported uncertainty
- exact $h/(m_e c)$ interval m: (Fraction(6310543, 2600880509804446332), Fraction(18931629, 7802641524616659668))
- NIST Compton wavelength interval m: (Fraction(121315511731, 5000000000000000000000), Fraction(121315511807, 5000000000000000000000))
- NIST right-angle high-energy ceiling interval MeV: (Fraction(51099895053, 100000000000), Fraction(10219979017, 200000000000))
- NIST backscatter high-energy ceiling interval MeV: (Fraction(51099895053, 200000000000), Fraction(10219979017, 400000000000))
- first-power, angle-independent, elastic-energy, numerical-forward, incomplete-row and changed-source controls rejected

Falsification condition: Reject if the complete IAEA record does not retain inverse-square Coulomb force, the inverse-fourth-power half-angle Rutherford relation, nonempty large-angle Geiger-Marsden support, closed Compton energy-momentum transfer or the wavelength-shift relation; if exact $h/(m_e c)$ does not overlap the NIST Compton-wavelength interval; if the right-angle and backscatter energy ceilings disagree with the NIST electron rest-energy vector and IAEA limits; if any registered row, uncertainty, scope condition or adverse control is omitted; or if target content enters before seal.

Measurement receipt: sha256:d6c7057f63d6f29411d7fbdddef78e3a8cd96aeeadcf8f8ef1348b0c395c17cd0.

Isolation certificate: sha256:e90f382c158c571f9386300054242fa1cdf5aaf259ee1b5183c1fadee9e181f.

Custody certificate: sha256:6345f82fc72048f621af014d5e3ce39930ffe9c02d03cbf94d60ccfb50e0f53a.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no Rutherford formula, Compton formula, measured angle, cross section, wavelength, energy or source locator in formal candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-zero proof value, negative, irrational, imaginary, floating or imported complex proof magnitude
- no path integral, fitted coupling, target-selected angular power, selected correction or stochastic oracle
- no evaluation of the forward empty class by a divergent numerical expression
- no target access before prediction sealing and no free extension after closure

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:e5d2d0eef6a42f2e3af50b8fa33d8312b17c518a603b44ee9c1bf05046f9edc5; engine receipt
```

sha256:2c2e6212e5bb27a2b7647e8eca1583bc75775c886f2185885a7925d83e195b91 at receipts/engine/model_admitted/SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006-2c2e6212e5bb27a2.json; empirical-validation hash

```
sha256:4bd3b98fb21c2f4758f64c4b2bba3df272ab97ea863c67477988b192679cf56e: measurement receipt
```

```
sha256:1b3d3b9c9f32112f1f3c9131b2b3a3d1272ab07ca3c3333f1f733331c12b733f3333;
sha256:d6c7057f63d6f29411d7fbbddf78e3a8cd96aeaadcf8f8ef1348b0c395c17cd0.
```

173. Terminal Fold decay widths, exact branching partitions and lifetimes

Claim identity: SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Every distinguishable open decay channel has exact positive partial width equal to its paired transition weight times available generated output support per action carrier. Closed channels are structural empty forms. The total width is the complete ordered positive sum of the open partial widths; each branching part is its partial width over that total, and the mutually exclusive complete branching family sums exactly to the One. Duration is the action carrier over total width, so increasing transition support increases width and strictly shortens lifetime. The base and channel-successor laws hold for every nonempty finite generated family.

Every distinguishable open decay channel has exact positive partial width equal to paired transition support times available generated output support per action carrier; closed channels are structural emptiness; total width is the complete positive sum; branch parts are partial width over total and partition the One; lifetime is action over total width and strictly decreases with width. The complete PDG 2026 W-width and decay-mode vectors and the NIST reduced-action transport pass the registered post-seal comparison without subset double counting or a fitted correction.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-DECAY-001
- SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-PHYS-QUANTUM-WEIGHT-001
- SFT-PHYS-MECH-DURATION-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of every paired or unpaired transition carrier, every open/closed channel domain, every total-width aggregator, every branching normalization, every completeness partition, every width-lifetime relation, every induced ordering, both target-custody states and both extension states. The declared exact boundary is: Every nonempty finite generated family of mutually distinguishable open decay channels; exact positive rational overlap, available-output, action, partial-width and duration carriers; structurally empty closed channels; complete channel composition; sealed or exposed targets; and empty or free extension. The base and successor certificates are independent of a finite Fold depth and introduce no numerical null state. The generator produced 8748 named candidates and the decision support contains 8748 one-for-one decisions. Exactly one candidate survived: `paired-transition-weight-times-output-support-per-action__positive-open-channels-closed-as-empty__ordered-positive-sum-of-partial-widths__partial-width-over-total-width__complete-exclusive-partition-of-one__action-over-total-width__greater-width-shorter-lifetime__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
partial	paired-transition-weight-times-output-support-per-action	unpaired-transition-leg-times-output-support	Rejected by computed Fold predicates: partial_width_carrier.
domain	positive-open-channels-closed-as-empty	numeric-null-closed-channels	Rejected by computed Fold predicates: channel_domain.
total	ordered-positive-sum-of-partial-widths	largest-partial-width-only	Rejected by computed Fold predicates: total_width_law.
branch	partial-width-over-total-width	total-width-over-partial-width	Rejected by computed Fold predicates: branching_law.
partition	complete-exclusive-partition-of-one	incomplete-open-channel-subset	Rejected by computed Fold predicates: partition_law.
lifetime	action-over-total-width	total-width-over-action	Rejected by computed Fold predicates: lifetime_law.
ordering	greater-width-shorter-lifetime	greater-width-longer-lifetime	Rejected by computed Fold predicates: ordering_law.
target	sealed-before-release	readable-before-seal	Rejected by computed Fold predicates: target_boundary.
extension	empty-extension	free-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash sha256:42d6264dd3d0731a3920f4c904f688ccf53e028f356a0be3ff910c6041dc0f6d. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject incomplete branching and a width-proportional lifetime.; observed Complete normalization and exact inverse duration reject both false laws.; receipt sha256:1f11a1f6c70596215f0cd6e40f1fc35fac6a7542f354c765400f4b1a3d562c69.
- **tampered_source:** passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae.
- **tampered_artifact:** passed; expected Reject a duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete candidate identity.; receipt sha256:85f7f2fccaba82e12423de0043e0cde43f60a6c59bbf25b4c3079af293bc64a.
- **boundary:** passed; expected Reject a numerical null channel, pre-seal target access and free correction.; observed Closed channels are structural absence; sealed custody and empty extension are required.; receipt sha256:89137a6f44d4b9f7604a1d09999b21276005254103306653d01e9d7fc79e424a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b0291ebf59a77a2abc9bb3ba2774893f5738d22b048abf4b280be93fd0788e16. Independent certificate: sha256:29c5bc317aac67c022d87e417828f668fd1d0748855e6d0eab968f7047c00910. Engine external-validation hash: sha256:72baee80b3adaeac88de833ce60ce9ab2f0c8885919a6d53aad763fd5a44068.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2026-W-BOSON-LISTING
- NIST-CODATA-2022

Observed comparison records:

- [illegible]

Falsification condition: Reject if the complete registered PDG W-width table, thirteen-row decay-mode table or either exclusive branching organization is incomplete; if PDG does not identify each branch as Gamma_i/Gamma or does not close the hadronic branch as One take three leptonic branches; if the complete measured exclusive intervals do not contain the One, if the exact complements forced from the printed leptonic values miss the printed hadronic interval, or if inclusive subsets are double counted; if the NIST action-prefix enclosure and PDG width interval do not yield a finite positive inverse-width lifetime interval and ordering; if any registered row, uncertainty, source identity or unfavorable control is omitted; or if target content enters before seal.

Measurement receipt: sha256:d48b51e08f77bfd7850e90dcc4f0bea4b17a68f8588730baa1d0627dd9376f36.

Isolation certificate: sha256:6cefc1a860ab53a4ffd00d2c2dace7b786342d3fb7cd0d33829d4b63766e7083.

Custody certificate: sha256:95a979a4ed2600d2d93d1a25e76b28f6e9b5fa9a9afdd8a2e19e5a14d980ef65.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no particle identity, measured width, branching fraction, lifetime, source locator or PDF text in formal candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-zero proof value, negative, irrational, imaginary, transcendental or floating proof magnitude
- no continuum exponential, Breit-Wigner line shape, imported perturbative amplitude, fitted coupling, selected correction or measurement-selected candidate
- no double counting of aggregate, inclusive-subset, exclusive-subset or limit rows as independent primary branches
- no promotion of the NIST ellipsis to an exact rational proof value
- no target access before the prediction seal and no free extension after closure

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d2750e9b099ed37535180507e4ef1eca7b678d6c4ce14a16b21c9cd8ec5ee688; engine receipt
 sha256:a4e14f20a7c1f6920115f0e5ad0b911b5462b690aad35ee79aac39b691d12d3f at receipts/engine/model_admitted/SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006-a4e14f20a7c1f692.
 json; empirical-validation hash
 sha256:5f55eaf606e42e1746e073343e52e184ebe55bd62288aa5c0339e4834511eef7; measurement receipt
 sha256:d48b51e08f77bfd7850e90dcc4f0bea4b17a68f8588730baa1d0627dd9376f36.

174. Terminal Fold running functions and common-support convergence

Claim identity: SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete forced prime sectors p in two, three, five and seven share one generated scale carrier with One base and binary successor $R_{(n+1)}=2R_n$. Each sector's exact finite running share is $g_p(R)=(p+R-1)/(p+R)$, with positive shortfall $1/(p+R)$. Every binary successor raises g_p by $R/[(p+R)(p+2R)]$. For $p < q$ the exact gap is $(q-p)/[(p+R)(q+R)]$, it strictly shrinks at every successor, and for every finite positive tolerance a generated binary support gives a smaller gap. Thus all four sector functions converge constructively on the common support axis without a completed infinity, imported beta function, fitted parameter or measurement-selected correction. The sealed self-source-range and screening-exposure laws determine the opposite physical energy orientations used in post-seal comparison.

The complete prime sectors two, three, five and seven share One-base binary support R . Each exact finite running share is $(p+R-1)/(p+R)$, its positive shortfall is $1/(p+R)$, its binary successor take is $R/[(p+R)(p+2R)]$, and every ordered pair gap is $(q-p)/[(p+R)(q+R)]$. Every share rises, every gap strictly shrinks, and every finite positive tolerance has a generated binary-support witness. Current authoritative PDG comparison retains the complete QCD and electromagnetic vectors: α_s decreases with increasing transfer scale while electromagnetic α increases as screening is removed, exactly as the separately sealed carrier-specific coordinate orientations require. The common-support four-sector convergence is an exact standing prediction, not a fitted value or a falsely claimed direct measurement.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-STRONG-RUNNING-DIRECTION-002
- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003
- SFT-PHYS-MEAS-BOUNDARY-GROWTH-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-MATH-LIMIT-CONTINUUM-002
- SFT-MATH-SELF-SIMILAR-CONVERGENCE-002
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of every closed or selected sector domain, every binary, linear or target-assigned scale carrier, every holding-share, fixed or target-assigned coupling form, every successor direction, every exact, constant or target-assigned pair-gap law, every constructive, completed-infinity or visual convergence form,

every carrier-specific, universal-sign or measurement-selected physical translation, both target-custody states and both extension states. The declared exact boundary is: The complete independently forced prime-sector ladder two, three, five and seven; every generated finite positive depth level with One base and binary successor; exact positive rational holding shares, shortfalls, successor takes and ordered pair gaps; every finite positive tolerance denominator; the already forced self-source-range and screening-exposure translation classes; sealed or exposed targets; and empty or free extension. No completed infinity, continuum scale or numerical null state is admitted. The generator produced 8748 named candidates and the decision support contains 8748 one-for-one decisions. Exactly one candidate survived: `complete-prime-sectors-through-seven__One-base-binary-depth-successor__holding-share-of-sector-plus-support__binary-support-successor-raises-holding-share__exact-generator-gap-over-paired-sources__finite-positive-epsilon-witness-for-every-pair__carrier-specific-self-source-range-and-screening-exposure__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
sectors	<code>complete-prime-sectors-through-seven</code>	<code>selected-familiar-sectors-only</code>	Rejected by computed Fold predicates: <code>sector_domain</code> .
scale	<code>One-base-binary-depth-successor</code>	<code>imported-linear-scale-grid</code>	Rejected by computed Fold predicates: <code>scale_generation</code> .
coupling	<code>holding-share-of-sector-plus-support</code>	<code>fixed-bare-sector-share</code>	Rejected by computed Fold predicates: <code>coupling_form</code> .
successor	<code>binary-support-successor-raises-holding-share</code>	<code>binary-support-successor-lowers-holding-share</code>	Rejected by computed Fold predicates: <code>successor_law</code> .
gap	<code>exact-generator-gap-over-paired-sources</code>	<code>constant-sector-gap</code>	Rejected by computed Fold predicates: <code>pair_gap_law</code> .
convergence	<code>finite-positive-epsilon-witness-for-every-pair</code>	<code>completed-infinity-as-proof-of-value</code>	Rejected by computed Fold predicates: <code>convergence_law</code> .
translation	<code>carrier-specific-self-source-range-and-screening-exposure</code>	<code>one-imported-energy-sign-for-every-carrier</code>	Rejected by computed Fold predicates: <code>physical_translation</code> .
target	<code>sealed-before-release</code>	<code>readable-before-seal</code>	Rejected by computed Fold predicates: <code>target_boundary</code> .
extension	<code>empty-extension</code>	<code>free-running-correction</code>	Rejected by computed Fold predicates: <code>extension</code> .

Base and successor. The closure proof is sealed by derivation hash `sha256:ba74e810bf7fdb5eaba6092188c295e809742ad65a33f8e4bc192f3a01326964`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject reversed running and a completed infinity as a proof value.; observed Exact positive successor takes and finite epsilon witnesses reject both false laws.; receipt `sha256:615b161de849d45bd7c2991db2cae052a01ca04b8213f4229e19d60b8dae8c55`.
- **tampered_source:** passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt `sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae`.
- **tampered_artifact:** passed; expected Reject a duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete candidate identity.; receipt

sha256:85f7f2fccaaba82e12423de0043e0cde43f60a6c59bbf25b4c3079af293bc64a.

- **boundary:** passed; expected Reject pre-seal target access, free correction and a numerical null base.; observed The scale begins at the One; sealed custody and empty extension are required.; receipt
sha256:1282d4e9761ba33212fc7dbf3768f3ad708c64cb4bcc97ef5eae8e4168396988.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:d6afafe211ebca3297f1e684cae091e39ca0cc80931b70c3cf11924360fdb9a1.

Independent certificate:

sha256:dfa5fe2d5d8b63a7964d01f099c74abfb48ba490275147c4286b07f1b4645280. Engine

external-validation hash:

sha256:305e1659813dc637a2e779b3578008bc1144c3318872ad7f8d9a6c0ee6efe283.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2026-QCD-REVIEW
- PDG-2026-ELECTROWEAK-REVIEW
- PDG-2025-ELECTROMAGNETIC-RUNNING-ENDPOINT

Observed comparison records:

- PDG 2026 QCD complete Table 9.1 retained: seven rows, every reported averaging column and uncertainty
- PDG 2026 QCD Figure 9.4 retained by immutable source hash and visual audit: nine measurement classes across the displayed multi-scale range
- PDG strong low-scale interval at $Q=1.37$ GeV: (Fraction(331, 1000), Fraction(559, 1000))
- PDG strong m_Z -scale interval: (Fraction(1171, 10000), Fraction(1189, 10000))
- The separated exact intervals confirm that α_s decreases as transfer energy rises; the full figure and page-43 conclusion retain the wider multi-scale evidence
- PDG 2026 electromagnetic scope retained: five $\alpha(0)$ determinations, eleven hadronic Delta-alpha evaluations, the published running relation and direct L3/OPAL observations
- PDG explicit prior Thomson inverse-alpha interval: (Fraction(137035999063, 1000000000), Fraction(27407199821, 200000000))
- PDG explicit prior Z-scale inverse-alpha interval: (Fraction(63961, 500), Fraction(63969, 500))
- The inverse-alpha intervals decrease, hence electromagnetic alpha increases toward the Z scale
- Opposite physical energy directions pass only under the sealed carrier-specific self-source-range and screening-exposure translations; a universal imported sign is rejected
- The exact common-support convergence of the four Fold sector functions remains a standing falsifiable prediction and is not mislabelled as a direct measurement

Falsification condition: Reject if any of the seven PDG QCD Table 9.1 rows, nine Figure 9.4 measurement classes, exact low- and m_Z -scale strong-coupling rows, five current $\alpha(0)$ rows, eleven current hadronic Delta-alpha rows, published electromagnetic running relation, direct LEP observation or explicit prior Z-scale endpoint is omitted or altered; if the separated strong-coupling intervals do not decrease from 1.37 GeV to m_Z ; if the explicit inverse electromagnetic endpoint intervals do not decrease from the Thomson limit to the Z scale; if a universal physical energy sign is imposed despite the sealed carrier-specific coordinates; if the exact Fold support successor, running-share increment, pair-gap shrinkage or finite tolerance witness fails; if the formal convergence prediction is falsely labelled a direct measurement; or if any target content enters before the prediction seal.

Measurement receipt: sha256:f7b8eb2dd10c0c689ee5dedb5b1add4a6b69ff8466d70e2727964d94289993bc.

Isolation certificate: sha256:11efa3cdf27c3e412e3a48235b1c6bfb258890a6b4778cad35c3bbc6e6eb794.

Custody certificate: sha256:4252ba538115e1ee971b038a463ef6f1d24b97cebda59e8a96094076c93aabcb.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no measured coupling, energy, beta function, renormalization-group curve, unification group, source locator or PDF text in formal candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-zero proof value, negative magnitude, irrational, imaginary, transcendental or floating proof magnitude
- no completed infinity or continuous scale used as a proof object
- no fitted coefficient, selected scale, running correction or measurement-selected orientation
- no universal physical energy sign imposed on physically distinct self-source and screening-exposure coordinates
- no plot digitization inventing coordinates that the authoritative source does not print
- no promotion of exact common-support convergence to a direct experimental observation
- no target access before the prediction seal and no free extension after closure

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8e5b0d59dd3626f39dcb398cba57819bac31eca897583d65bb5b99084d67f3f9; engine receipt
 sha256:b9263f585f01a2401261f465ad250f405a9914da5c53e57ba113f076aeb4f309 at receipts/engine/model_admitted/SFT-PHYS-COUPPLING-RUNNING-CONVERGENCE-TERMINAL-006-b9263f585f01a240.json; empirical-validation hash
 sha256:2f1893b335cbcd3be69e125cc935f6b69ab16c56d65aedcc028cfe9c3e85d3cb; measurement receipt
 sha256:f7b8eb2dd10c0c689ee5dedb5b1add4a6b69ff8466d70e2727964d94289993bc.

175. Simultaneous massless, One-speed and confined strong-carrier law

Claim identity: SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The generated colour-three sector has eight non-singlet carrier states and coupling two-thirds. Its carrier has an empty mass and rest-capture record, so its local phase advances exactly one support cell per tick. The same carrier bears the colour it mediates and therefore self-sources a fixed half-One transverse tube: every positive separation successor adds the same two-thirds work carrier, exceeding every positive exact finite bound at a generated finite witness. Local massless One-speed propagation and asymptotic confinement are thus simultaneous non-equivalent records, not contradictory alternatives.

Three colour labels force eight non-singlet strong carriers. The carrier has empty mass and rest-capture records and advances one support cell per tick with retained phase. Because it carries the colour it mediates, it self-sources a fixed half-One tube whose work rises by exact two-thirds at every separation successor and exceeds every positive exact bound at a finite witness. Local massless One-speed propagation and asymptotic confinement therefore coexist without a fitted parameter or free correction.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-STRONG-RUNNING-DIRECTION-002
- SFT-PHYS-SPACETIME-EXACT-INTERVAL-003
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of every generated, free or target-assigned colour sector; every empty, numeric/fitted or target-assigned mass law; every One-speed, free-speed or target-assigned causal law; every colour self-source, chargeless or target-assigned carrier law; every fixed-tube, spreading/bounded or target-assigned confinement law; every simultaneous, exclusive or target-assigned composition; both target custody states; and both extension states. The declared exact boundary is: The generated colour-three carrier at every positive finite causal tick and separation depth. Masslessness is the empty mass/rest-capture record and One-cell-per-tick local phase propagation; confinement is fixed-width self-sourced colour transport whose positive separation work exceeds every nominated positive exact bound at a finite witness. No completed infinity or claim about an observable isolated colour carrier enters. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `generated-colour-three-sector-with-eight-nonsinglet-carriers__empty-mass-label-with-no-rest-capture__One-support-cell-per-tick-with-retained-phase__colour-carrying-mediator-resources-its-own-channel__fixed-half-One-tube-and-two-thirds-work-successor__local-massless-One-speed-and-asymptotic-confined-records-coexist__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
sector_law	generated-colour-three-sector-with-eight-nonsinglet-carriers	free-sector-or-mediator-count	Rejected by computed Fold predicates: sector.
mass_law	empty-mass-label-with-no-rest-capture	numerical-zero-or-positive-fitted-mass	Rejected by computed Fold predicates: mass.
causal_law	One-support-cell-per-tick-with-retained-phase	free-sub-or-super-One-speed	Rejected by computed Fold predicates: causal.
self_source_law	colour-carrying-mediator-resources-its-own-channel	chargeless-linear-carrier	Rejected by computed Fold predicates: self_source.
confinement_law	fixed-half-One-tube-and-two-thirds-work-successor	spreading-flux-or-bounded-separation-work	Rejected by computed Fold predicates: confinement.
composition_law	local-massless-One-speed-and-asymptotic-confined-records-coexist	masslessness-and-confinement-treated-as-exclusive	Rejected by computed Fold predicates: composition.

Axis	Forced form	Eliminated form	Elimination reason
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target.
extension	empty-extension	free-carrier-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash sha256: 40144181c27ae46335d82275c66df9cba4743451c49ed8a8700e0fb4e3eceebe. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject treating empty mass and empty isolated-colour records as the same claim or as numerical zero.; observed Local propagation survives while isolated asymptotic colour remains empty.; receipt sha256: de98ae67df337f751884db61a7bffa6821a5b74354eeaae2f79cdd62fa9a451c7.
- **tampered_source:** passed; expected Reject changed source identity.; observed Identity differs.; receipt sha256: 7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83.
- **tampered_artifact:** passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt sha256: 0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049.
- **boundary:** passed; expected Reject numerical/fitted mass, false exclusivity, pre-seal target access and free correction.; observed Only empty mass, simultaneous records, sealed custody and empty extension survive.; receipt sha256: e74908f15eb59a896903336653189ab9920bb7e4dd97ca7a41b9cbf680c116cf.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 9fb24ffba02632292d1b725b216db5ebcd2511a9b53aecefd27b54ea9647efdc. Independent certificate: sha256: a9b8d1ca2b0cf318f9c1107d3e76f790f8c924d4751876c7d8cb4d638a829a90. Engine external-validation hash: sha256: 5cc5486ed41238fd730f87a266c7456cc8c07e4d6c6063c694b49ea4fb690dd4.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- D10F-STRONG-CARRIER-MASSLESS-CONFINED-COMPARISON

Observed comparison records:

- The complete V1 D10f row, ten admitted prerequisite certificates and three complete current PDG sources are hash-locked and released only after the V3 formal commit and in-run prediction seal.
- V3 independently forces three colour labels, eight non-singlet carriers, empty mass and rest-capture records, retained phase advancing one support cell per tick, and no fitted propagation value.
- The same colour-bearing carrier self-sources a fixed half-One tube whose exact separation work increases by two-thirds at every successor and exceeds every positive exact bound at a finite witness.
- The PDG gluon listing records an SU(3) colour octet and $m=0$, matching the frozen structural classification after release.
- The PDG explicitly classifies $m=0$ as a theoretical value and says a mass as large as a few MeV may not be precluded; this qualification is retained and no direct free-gluon time-of-flight measurement is claimed.
- The PDG QCD review records $N_c=3$, $N_c\text{-squared-minus-One}=8$, the adjoint gluon representation, nonlinear field term, three-gluon vertex and four-gluon vertex.
- The same review states that neither quarks nor gluons are observed as free particles and that hadrons are colour-singlet combinations, matching the frozen asymptotic observation boundary.

- Collider colour-factor measurements $CA=2.89\pm0.03(\text{stat})\pm0.21(\text{syst})$ and $CF=1.30\pm0.01(\text{stat})\pm0.09(\text{syst})$ contain the published $SU(3)$ expectations 3 and $4/3$ within exact conservative summed uncertainty boundaries.
- The PDG also records clear experimental evidence that gluon jets are softer and broader than light-quark jets; the quark-model review describes physical states as colour singlets and colour as believed permanently confined.
- External measurements and classifications test only the already sealed result: they do not select its survivor, enter its proof arithmetic, fit a parameter or add a carrier correction.

Falsification condition: Reject if the V1 D10f row, any admitted prerequisite or any complete PDG source changes or is omitted; if three colour labels, eight non-singlet carriers, empty mass/rest-capture, One-cell-per-tick propagation, self-source, fixed half-One tube, positive two-thirds work successor or empty isolated-colour record fails; if the PDG theoretical $m=0$ listing is misreported as a direct measurement or its few-MeV caveat is omitted; if measured colour-factor intervals cease to contain their published $SU(3)$ expectations; if a free coloured state is observed; or if any external value selects the survivor, becomes a formal proof scalar, is fitted, or supplies a free correction.

Measurement receipt: sha256:798dc0308ac250c03b0c1a8685e9192e95bb1f950344edc37b3c1fa1defd4447.

Isolation certificate: sha256:1d8ad63be3229d00c3c6fcd4137e4271b221fd4045c96c7aed217c209e3865dd.

Custody certificate: sha256:be921ad0f8b3b28c013100c797ddb677217e66cda7ae90ec4f7de764a0ac8c8.

Registered source descriptions:

- PDG-2026-GLUON-LISTING: https://pdg.lbl.gov/encoder_listings/g021.pdf
(sha256:fcc152780d2efb6cb66d2f9351abd034d30cab0e914fd87a0daee46a0344353d)
- PDG-2026-QCD-REVIEW: <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-qcd.pdf>
(sha256:c04c628d76b18610c5fa2a919c6081918a25b55fb971b6af5829f4ca2baa386f)
- PDG-2026-QUARK-MODEL-REVIEW: <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-quark-model.pdf>
(sha256:6aa98fa53857122f27b638c59081af1a2857d787c4205370d8fa38fcb6b70ff0)

Registered target identities:

- D10F-WITHHELD-HISTORICAL-PREREQUISITE-AND-PDG-RECORD from
D10F-STRONG-CARRIER-MASSLESS-CONFINED-COMPARISON

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- imported QCD Lagrangian
- imported gauge-group axiom
- numerical zero as the empty mass record
- free carrier mass
- free causal speed
- free confinement potential
- completed infinity
- direct observation of an isolated colour carrier
- irrational or imaginary proof value
- measurement-selected survivor
- measured fit
- free carrier correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:c523e40c2f6b2f01c6c825148b83d9dfe2d0c20fffe768152a77196764da1742; engine receipt
sha256:bae91c8dc0b88c6130260c7e10ea22e9433ca322bd5ad3a8f9077653315f57e1 at receipts/engine/model_admitted/SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013-bae91c8dc0b88c61.json; empirical-validation hash
sha256:76113e290c7e36ae64b9325312733981bc88d5dd06a3620245b74140a7a16b05; measurement receipt
sha256:798dc0308ac250c03b0c1a8685e9192e95bb1f950344edc37b3c1fa1defd4447.
```

176. Strong self-source iteration and finite fixed-point boundary

Claim identity: SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A chargeless carrier has an empty self-source correction and holds its matter source. The admitted gravitational scalar self-source has one admissible quarter-One fixed point with shrinking corrections. A colour-bearing carrier instead remains inside its own source update: every successor retains the prior source and appends the complete binary carrier, so the correction is exact two and never shrinks. For every positive exact finite F , F plus two is strictly above F , leaving the strong finite fixed-point record empty; the same successor law exceeds every positive exact finite bound at a generated finite witness. This is the depth-independent confinement boundary.

The exact three-way field-update census admits only the chargeless held-source, gravitational quarter-One contraction and colour self-source F -plus-two persistence discriminator. The strong correction is exactly two at every positive finite successor, so no positive exact finite fixed point exists and every positive exact finite bound has a generated finite witness above it. Post-seal PDG and lattice-QCD comparison supports non-abelian self-interaction and the linear unbroken string branch while explicitly retaining full-QCD string breaking, ground-state saturation, uncertainties and single-ensemble scope.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRONG-RUNNING-DIRECTION-002
- SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of every colour-self-sourced, external-only or target-assigned source; every binary-successor, free or target-assigned update; every persistent, shrinking/free or target-assigned correction; every empty, selected or target-assigned finite fixed-point law; every exact three-way, conflated or target-assigned comparator; every arbitrary-bound, finite-prefix/completed-infinity or target-assigned confinement law; both target custody states; and both extension states. The declared exact boundary is: Every positive finite iteration depth of the admitted colour self-source successor, every positive exact finite fixed-point candidate and work/source bound, the admitted gravitational scalar self-source channel, and the chargeless held-source control. The result distinguishes structural source recursion from polynomial degree and does not claim a conventional renormalized Yang-Mills solution. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: colour-carrier-retained-inside-its-own-source-update__each-successor-retains-prior-source-and-appends-complete-bina

ry-carrier__persistent-exact-binary-correction-at-every-successor__empty-positive-finite-fixed-point-record-by-F-plus-two-order__neutral-hold-gravity-contraction-strong-persistence-three-way-discriminator__finite-witness-above-every-positive-exact-bound__sealed-before-observation-release__empty-extension. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
source_composition	colour-carrier-retained-inside-its-own-source-update	external-matter-source-only	Rejected by computed Fold predicates: source.
update_law	each-successor-retains-prior-source-and-appends-complete-binary-carrier	free-linear-contracted-or-oscillatory-update	Rejected by computed Fold predicates: update.
correction_law	persistent-exact-binary-correction-at-every-successor	shrinking-or-free-correction	Rejected by computed Fold predicates: correction.
fixed_point_law	empty-positive-finite-fixed-point-record-by-F-plus-two-order	selected-finite-strong-fixed-point	Rejected by computed Fold predicates: fixed.
comparator_law	neutral-hold-gravity-contraction-strong-persistence-three-way-discriminator	conflate-all-self-source-iterations	Rejected by computed Fold predicates: comparator.
confinement_law	finite-witness-above-every-positive-exact-bound	bounded-prefix-or-completed-infinity	Rejected by computed Fold predicates: confinement.
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target.
extension	empty-extension	free-strong-field-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash sha256:9764f4e0b2309d8a96191eff92927bb5cd80838266b70108e8f94ea5bbf7713a. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject conflating empty, shrinking and persistent correction records.; observed Neutral, gravity and strong updates retain distinct exact correction classes.; receipt sha256:e4a7af5577b7ed98d709925f9db75a53724b7348abaf318dffdb4857ab64d3b4.
- **tampered_source:** passed; expected Reject changed source identity.; observed Identity differs.; receipt sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83.
- **tampered_artifact:** passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049.
- **boundary:** passed; expected Reject selected fixed point, finite-prefix/completed-infinity inference, target access and free correction.; observed Only universal positive-order exclusion, finite witnesses, sealed custody and empty extension survive.; receipt sha256:01fbecf349784fbb255dfecf26b57add49eece9e3bc3fee506da4ac196466b6e.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:a27960b8343ae01d0fafd4fd7327d118319bd93b780b87b43142e8a304d20759.

Independent certificate:

sha256: 4009d2e7479a082ceda554a357c72b83f4c0b9056676c4fa7539c21ed758b4a1. Engine

external-validation hash:

sha256: 35920b4ede6319913138c6ef2d31e39e51545b7e4412f6a86af0fbb90dab9d07.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- D10E-STRONG-FIELD-NONLINEAR-FIXED-POINT-COMPARISON

Observed comparison records:

- The complete V1 D10e row, nine admitted prerequisites, the complete current PDG QCD review and complete primary lattice-QCD paper are hash-locked and released only after the frozen V3 commit and in-run prediction seal.
- V3 independently forces an empty chargeless correction with held One source, the admitted gravity quarter-One fixed point with shrinking corrections, and a colour self-source successor F plus two with the exact binary correction retained at every successor.
- For every positive exact finite F, F plus two is strictly above F, so the strong positive finite fixed-point record is empty; for every positive exact finite bound, the same law constructs a finite source witness above it without a completed infinity.
- The PDG QCD review independently retains non-abelian colour, colour-bearing gluons, three- and four-gluon self-interactions and the absence of observed free quarks.
- The lattice study independently reports that the gluons-only static energy grows linearly at asymptotically large separation with string tension σ and models the unbroken string state as $V\text{-hat}(r)=V\text{-hat-0}+\sigma*r$ in the breaking region.
- The same study observes the required adverse boundary: dynamical light-quark pair creation mixes the string with broken-string states, and the full-QCD ground state tends to two noninteracting static-light mesons rather than growing without bound.
- The reported light and strange breaking distances, 1.224(15) fm and 1.293(16) fm, are retained with lattice spacing 0.06426(76) fm and the 280 MeV pion and 460 MeV kaon masses.
- Those distances come from one ensemble with nonphysical quark masses and are not promoted to universal physical constants; they neither select the Fold survivor nor enter its formal arithmetic.
- The exact correspondence is therefore branchwise: persistent Fold colour self-source matches the unbroken string branch, while pair-created state mixing supplies a distinct full-QCD saturation channel that the validation retains explicitly.
- No external value fits a parameter, supplies a free correction or changes the frozen generated grammar.

Falsification condition: Reject if the complete V1 D10e row, an admitted prerequisite, the current PDG review or the complete primary lattice study changes or is omitted; if the exact chargeless hold, quarter-One gravitational contraction, persistent binary strong correction, empty positive finite strong fixed-point record or finite witness above every positive exact bound fails; if QCD ceases to carry non-abelian gluon self-interaction; if the unbroken string branch ceases its linear growth; if pair-created broken-string mixing and full-QCD ground-state saturation are omitted; if the single-ensemble, nonphysical-mass or non-universal-threshold limits are hidden; or if an external target selects the survivor, enters formal arithmetic, fits a parameter or adds a correction.

Measurement receipt: sha256: ef22864dfb7236904932ef92516fddedfd7b22bd5d2b471a9a1c3ad1d6e0fbf95.

Isolation certificate: sha256: fb7f7d48bf0d7cc2239a427718dbca7e59534210dc2face57fa133e5c8bb36793.

Custody certificate: sha256: c0eadbbe6154dbe534b7602036251380ac7f2caaf9b121e1725283ecd6ceb5ea.

Registered source descriptions:

- PDG-2026-QCD-REVIEW: <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-qcd.pdf>
(sha256:c04c628d76b18610c5fa2a919c6081918a25b55fb971b6af5829f4ca2baa386f)
- BULAVA-ET-AL-2019-STRING-BREAKING: <https://arxiv.org/pdf/1902.04006>
(sha256:6c6de4ebdfc86d7976fb343a35f490ff108ab0d56dd1450a1ee27539e2641bb5)

Registered target identities:

- D10E-WITHHELD-HISTORICAL-PREREQUISITE-PDG-AND-LATTICE-RECORD from D10E-STRONG-FIELD-NONLINEAR-FIXED-POINT-COMPARISON

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- imported Yang-Mills field equation
- imported gauge-group axiom
- numerical zero as the empty fixed-point record
- negative, irrational or imaginary proof value
- completed infinity
- finite-prefix inference presented as an unbounded law
- measurement-selected survivor
- measured fit
- free strong-field correction
- unbounded full-QCD ground-state potential

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4af939c69d947be443ff772e7030b49ca3104b4b090febee3ec396240c01ff61; engine receipt
sha256:641cdf7a00a56d3889b1a31731dff04c70dbe068465924f005d000d3fc100cdc at receipts/engine/model_admitted/SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014-641cdf7a00a56d38.json; empirical-validation hash
sha256:6a20eb16725a453c3f37f22c5b0b63324e1a9cc9cc3912738b6584eb73650d4f; measurement receipt
sha256:ef22864dfb7236904932ef92516fde9d7b22bd5d2b471a9a1c3ad1d6e0fbf95.

177. Finite accumulated Fold-coupling separation

Claim identity: SFT-PHYS-COUPPLING-ACCUMULATED-SEPARATION-TERMINAL-015

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For the admitted binary and generator-three running sectors, every positive finite partial accumulation of their exact common-support gaps lies in $[1/12, 11/60]$; every successor from the second gap onward is below half its predecessor, and every positive rational tolerance has a generated finite tail witness.

Every positive finite partial sum of the binary/generator-three Fold coupling gaps lies in the exact bracket $[1/12, 11/60]$; after the second level every successor gap is below half its predecessor, and every positive rational tolerance has a generated finite tail witness.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-COUPPLING-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-MATH-SELF-SIMILAR-CONVERGENCE-002
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-COUNT-001

Generated grammar and closure boundary. Generate the complete product of predecessor, gap term, first-step, contraction, envelope, finite-boundary, tolerance and extension forms. The declared exact boundary is: Every positive finite common-support level generated from the One by binary succession; the exact binary/generator-three pair gap; every positive finite partial accumulation and every positive rational tolerance represented by a positive denominator. The generator produced 512 named candidates and the decision support contains 512 one-for-one decisions. Exactly one candidate survived: `admitted-common-support-gap-law__exact-adjacent-sector-gap__retain-one-twelfth-and-one-twentieth__exact-strict-half-contraction-after-level-two__eleven-sixtieths__every-positive-finite-partial-support__generated-finite-tail-witness__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	<code>admitted-common-support-gap-law</code>	<code>imported-or-selected-coupling-table</code>	A conventional or selected table is not an admitted predecessor.
term	<code>exact-adjacent-sector-gap</code>	<code>unsigned-or-fitted-difference</code>	A fitted difference can encode a target.
first	<code>retain-one-twelfth-and-one-twentieth</code>	<code>assert-half-contraction-from-first-gap</code>	The transition 1/12 to 1/20 shrinks but is not below half.
contraction	<code>exact-strict-half-contraction-after-level-two</code>	<code>visual-or-imported-convergence</code>	Visual approach and an imported series theorem are not Fold proof objects.
envelope	<code>eleven-sixtieths</code>	<code>one-sixth</code>	Later exact partial sums exceed this candidate.
boundary	<code>every-positive-finite-partial-support</code>	<code>completed-infinite-sum</code>	Completed infinity is outside the SFT domain.
tolerance	<code>generated-finite-tail-witness</code>	<code>limit-symbol-without-witness</code>	A limit label alone supplies no finite certificate.
extension	<code>no-extra-rule</code>	<code>free-tail-correction</code>	A free correction is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:5ab1dace9fee93e8153b8079d416bcb0d8a7b6f77af74ad8c27931ddf9ce1beb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e6b9d33fb638063e5ba0234bcd82d4893ce84bd7de2e3cd9bdc1a4800c024ad4`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:59fc2a7bc6963288e74ddf7ecce05a7d87ab2912550cda0b8100f98a348f930b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9895bbbd82df65ced4fbb2385eced8ba8abee2c31bdf0ae5725ff30b43ea464`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a563d412cbe3b2652716fb85b51c254c26c5a2dec563675a60639c7eaa467bdc`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2cc5ffedbd8c84347cf51184f9d36cfc05a3e969dacad6592956690b77d3f34b`. Independent certificate: `sha256:180c8807b7d767624480aae2662b1589e08ec8c7bbffff5d132d609f47a84ce47`. Engine

external-validation hash:

sha256: 525894b57a38373dfc48ecad834d5b588803d3b69779fc2101a093beac584626.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, stored partial-sum table or prior survivor in execution
- no imported coupling equation, infinite-series formula, logarithm or continuum limit
- no numerical-zero, negative, irrational, imaginary or floating proof value
- no measured coupling, target value, fitted coefficient or selected stopping depth
- no completed infinity and no claim of an attained terminal sum

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: b12b82925d6cfe4cc7d55b920957f5a3afceb5362164da3818782126d0d3e85b; engine receipt
sha256: 723ff785cf91a40c287d2f2b38394a63fd8bfaa7a0be7a34be2e9e973202f4bf at receipts/engine/
model_admitted/SFT-PHYS-COUPLING-ACCUMULATED-SEPARATION-TERMINAL-015-723ff785cf91a40c
.json; empirical-validation hash None; measurement receipt None.

178. Exact two-fibre scattering partition and reciprocal mean free path

Claim identity: SFT-PHYS-SCATTERING-PARTITION-PATH-TERMINAL-017

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The registered one-target Fold support partitions into scatter half-One and pass half-One; cross section is successful/incident support; and mean free path is the reciprocal of positive density times cross section, with larger section forcing shorter path.

The registered one-target Fold support partitions exactly into scatter half-One and pass half-One; cross section is successful/incident support; and mean free path is the exact reciprocal of positive density times cross section, with larger section forcing shorter path.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CLASSICAL-PROBABILISTIC-001
- SFT-PHYS-MATTER-SCATTERING-001
- SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006

• SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of target geometry, outcome support, exclusivity, outcome measure, cross-section measure, density, path, causal interpretation and extension forms. The declared exact boundary is: The one-target geometry with exactly two equipotent Fold outcome cells; all positive finite incident/successful support counts; and all exact positive target densities and cross sections. The generator produced 512 named candidates and the decision support contains 512 one-for-one decisions. Exactly one candidate survived: `registered-one-target-Fold-geometry__complete-scatter-pass-pair__mutually-exclusive-held-labels__equipotent-two-cell-support__successful-over-incident-support__positive-density-times-section__reciprocal-encounter-support__deterministic-held-outcomes__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
geometry	registered-one-target-Fold-geometry	arbitrary-scatterer	Arbitrary geometry does not force equal outcome cells.
outcomes	complete-scatter-pass-pair	sampled-or-omitted-channel	An omitted channel cannot establish certainty.
exclusivity	mutually-exclusive-held-labels	overlapping-outcome-cells	Overlap double-counts one event.
measure	equipotent-two-cell-support	assumed-equal-probability	An assumed probability imports the answer.
section	successful-over-incident-support	fitted-dimensional-number	A fitted number is not the structural measure.
density	positive-density-times-section	density-omitted	Omitting density makes reciprocal path dimensionally incomplete.
path	reciprocal-encounter-support	selected-path-length	A selected length has no encounter trace.
cause	deterministic-held-outcomes	stochastic-transition-oracle	A random cause is not supplied by deterministic Fold structure.
extension	no-extra-rule	free-geometry-or-correction	A free correction defeats zero-parameter closure.

Base and successor. The closure proof is sealed by derivation hash `sha256:e429e239bc740c04992863e402d178f69f6a39f6e0b4a4f98c0df4cce80e7b4a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:f12cced3188c8fd3c53734121199b399596f46210cd1717cc718adc4c6ee6b4a`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4caec307ab95c9decfce61920cbc4a6d0a92d5ad49da1793b3dc81f27f72b6c4`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f3e060037ddaf61b1b2fe110639e9d5b88d3f9f254bb77ec77a195940bf4f9b8`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ba7b87b2808814ec99f967b0dfdd6c22be77466992cc5052594752fb451071f8`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:965070318a614d6cb29833c963c07a4aefa2d05f692222ac1a1f74855deeb597.
Independent certificate:
sha256:3bfa663478d1cbc7a15fe9223a42e5d8b023a36cc1ab063d2f9ed05fc39927aa. Engine
external-validation hash:
sha256:6534bb6de6dd67d5b84d7d490b5a06d74692e33d1585e2820dd60c143238dfefb.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no assertion that arbitrary scattering has equal scatter/pass measures
- no stochastic causal selector or nondeterministic transition premise
- no numerical-zero, negative, irrational, imaginary or floating proof value
- no fitted cross section, measured path or target-selected correction
- no omission of target density from the general reciprocal law

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b741febb3bdb061e559fa6ba2ff10235d96a2d0ddf4952acfb5954ef0cb4a316; engine receipt
sha256:b4b42d9e9e0a52a703a059c265cc94325c4f56907da3422c4cb83b0facd9efdf at receipts/engine/model_admitted/SFT-PHYS-SCATTERING-PARTITION-PATH-TERMINAL-017-b4b42d9e9e0a52a7.json;
empirical-validation hash None; measurement receipt None.

179. Preserved/broken Fold channels and inverse mediator range

Claim identity: SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Complete-One channels have empty mass record; proper channels have positive shortfall, exact conserved forward/rest transfer, finite native trace and reciprocal mass range; larger mass means shorter range while massless inverse-square response remains positive at every finite radius.

Complete-One channels have an empty mass record and no finite transport endpoint; proper channels have positive shortfall, exact conserved forward/rest transfer, a finite native trace and reciprocal mass range; larger mass means shorter range, while massless inverse-square response remains positive at every finite radius.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002

- SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003
- SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of carrier, channel completion, mass record, forward/rest transfer, conservation, range scale, mass ordering, massless transport, post-seal comparison and extension forms. The declared exact boundary is: Every exact positive proper Fold channel and every complete channel sum; every positive finite mass share, its complete positive subtraction trace and reciprocal range; every exact positive finite radius; and the already sealed W/Z, electroweak-share and mediator-range comparison records. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `held-forward-and-rest-carriers__combination-reassembling-the-One__positive-shortfall-or-empty-record__exact-positive-forward-to-rest-transfer__forward-plus-rest-reassembles-One__positive-trace-and-reciprocal-mass-scale__larger-mass-shorter-range__held-forward-and-positive-finite-radius__inherit-sealed-electroweak-and-range-records__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	<code>held-forward-and-rest-carriers</code>	<code>untracked-field-value</code>	An untracked value cannot conserve transferred support.
channel	<code>combination-reassembling-the-One</code>	<code>named-unbroken-channel</code>	A name does not prove completion.
mass	<code>positive-shortfall-or-empty-record</code>	<code>assigned-mass-label</code>	An assigned label imports the classification.
transfer	<code>exact-positive-forward-to-rest-transfer</code>	<code>sink-or-imported-exponential</code>	A sink loses support and an exponential is not generated.
conservation	<code>forward-plus-rest-reassembles-One</code>	<code>forward-loss-without-ledger</code>	That violates closed transfer.
range	<code>positive-trace-and-reciprocal-mass-scale</code>	<code>selected-cutoff-distance</code>	A selected cutoff is a free parameter.
ordering	<code>larger-mass-shorter-range</code>	<code>independent-range-order</code>	Independent ordering can contradict the carrier.
massless	<code>held-forward-and-positive-finite-radius</code>	<code>finite-hard-stop</code>	No generated positive mass exists to supply a stop.
comparison	<code>inherit-sealed-electroweak-and-range-records</code>	<code>W-Z-photon-selected-before-law</code>	Named targets cannot select the discriminator.
extension	<code>no-extra-rule</code>	<code>free-Yukawa-or-cutoff-rule</code>	A free profile or cutoff adds a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:99da829e577bcb3cc137f224acb0f9d8e17b1bfb14a3ae8896dc950fb8cb0a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : 69e754afee5837b3eda670ba0f811559a1aa35d07851f1adbac32f6519d9a306.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : c03d0aeca1d46b9f7df23ddee15a2125eadf000da02f0240e71dd48d2404c1fd.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 6db4abf41d52e7f0e129e09fd4099afe11c4a198267f17d86fb05bc06a405de1.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : f49676a98e78127dfcfb2b2f1a2ecc870035f253200fd40bbcb708e7ac3b451a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : 8ab6340e93e19ee8a863181f592c6834e42deb555ca7472fa81c525ff2ea7a32.

Independent certificate:

sha256 : ecbb5b36bc6221e4a891fbca2720ed86be8747bddd32314ce1cf3836ea5a0ee3. Engine

external-validation hash:

sha256 : 573b48a888abd5f135aa27b44db4f4d9b8be0eb3521a2443ff71710c185cd9e9.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, answer table, selected particle name or stored survivor
- no imported exponential, fitted range, hard physical numerical-zero cutoff or sink
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no claim that the finite native subtraction trace is a universal physical field profile
- no target access before the predecessor electroweak and mediator-range seals

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : e2c3444e331c2693253b4e161eed9d6566d0edd036f774a26749476b033dd86a; engine receipt

sha256 : 3ad2016a90d7adc2c1c5c5afdfdaf5510f73ade146e2ea2e0a097d9158621f54 at receipts/engine/model_admitted/SFT-PHYS-MEDIATOR-RANGE-CHANNEL-020-3ad2016a90d7adc2.json;

empirical-validation hash None; measurement receipt None.

180. Fold baryogenesis dependency and retained-residue law

Claim identity: SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. From exact paired particle/antiparticle support, a positive retained baryon residue exists exactly when baryon tally change, CP-distinguished conjugate paths and a nonequilibrium reverse-completion hold are all present; the other seven complete combinations return the empty-One residue record.

From exact paired particle/antiparticle support, a positive retained baryon residue exists exactly when baryon tally change, CP-distinguished conjugate paths and a nonequilibrium reverse-completion hold are all present; the other seven complete combinations return the empty-One residue record.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-CKM-TERMINAL-004
- SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of initial carrier, baryon-tally action, conjugate-path relation, reverse-completion relation, residue ledger, dependency composition, enumeration, physical correspondence, provenance disclosure and extension forms. The declared exact boundary is: Every finite minimal process beginning from one positive paired particle/antiparticle carrier, with each of baryon-tally change, conjugate-path distinction and reverse-completion hold either present or absent, plus the already admitted terminal CP and baryon-abundance comparison records. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `exact-paired-particle-antiparticle-carrier__one-explicit-tally-changing-transition__CP-carrier-distinguishes-conjugate-paths__nonequilibrium-reverse-completion-held__positive-oriented-residue-or-empty-One__single-three-condition-process-composition__all-eight-presence-absence-combinations__inherit-sealed-CKM-and-baryon-photon-records__observational-reconstruction-explicit__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	exact-paired-particle-antiparticle-carrier	preloaded-unpaired-residue	A preloaded residue assumes the result.
tally	one-explicit-tally-changing-transition	every-transition-preserve-s-baryon-tally	A preserving transition cannot create a tally difference from paired support.
conjugacy	CP-carrier-distinguishes-conjugate-paths	exactly-paired-conjugate-paths	Equal conjugate appends preserve the paired tally.
reverse	nonequilibrium-reverse-completion-held	complete-reverse-recurrence	Complete reversal closes every newly appended carrier.
ledger	positive-oriented-residue-or-empty-One	signed-or-unrecorded-net-number	A signed or missing ledger violates positive exact accounting.
composition	single-three-condition-process-composition	named-independent-conditions	Names alone do not prove their joint necessity.

Axis	Forced form	Eliminated form	Elimination reason
enumeration	all-eight-presence-absence-combinations	selected-successful-combination	Selecting the familiar triple is not uniqueness.
comparison	inherit-sealed-CKM-and-baryon-photon-records	abundance-selects-dependency-law	A measured abundance cannot choose the formal conditions.
provenance	observational-reconstruction-explicit	mislabel-as-blind-discovery	The dependency triple was known before this V3 reconstruction.
extension	no-extra-rule	free-efficiency-or-extra-condition	An inserted efficiency or condition is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:97be399ff1fc4749dff9e890d336b1888f8ce4735e8a5fd642f065ef5acbffffa. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:e3bae526e83cc28f339b9d5a9119cad60a18dc63bdb69e86fceeafa06ba8ceb5c.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:0e3b154bfe87d76d7f8129c5874028dc59692ca2a8aa7d3b2635f6b3cdb7a4c0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:29ad32c527daca782aa19ff642ad10b565934de369d47aac2730a7ff659fbf1b.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:0f15328abb659835ff35fdb4efba0235867f24814e58f0230bd90c896b51d2e8.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:c8f6b747b46a795c765a24c8f311aa2d4d7f8e4e4082ed396e7290bf6a168091. Independent certificate: sha256:098fcbe06ef474f48600931558aa9ca3ba817711e07989ed650609ec262d57df. Engine external-validation hash: sha256:9acd0a3e91324547d90bd78059493ffe57c7a37963659598ad08e2afd459738f.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported Sakharov theorem or conventional kinetic equation as a premise
- no V1/V2 executable, result table, measurement or stored survivor

- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no measured baryon abundance, fitted efficiency or selected mechanism in the derivation
- no claim that the V3 reconstruction was a blind historical discovery
- no cosmic-history claim beyond the universal physical dependency law

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b51006c613c918a6983ff9d857f3e515f4c955917ce1aeca6552c7a1316f25bb; engine receipt
sha256:738dd11360b3cb43280ce6b595023c6f6408166ab6ef96ad4f8e9a6d4592c7b4 at receipts/engine/model_admitted/SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021-738dd11360b3cb43.json;
empirical-validation hash None; measurement receipt None.

181. Prime-sector standing-mode, interaction and period unification

Claim identity: SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The forced sectors (2,3,5,7) share one exact m-indexed interaction table; half-One uniquely unifies the odd-sector standing modes; all finite common-support sector gaps are positive and shrink without finite coincidence; the gravity/EM/strong Fold periods are 1/2/3 with joint recurrence 6; and the inconsistent flat-EM slope bundle is rejected in favour of the sealed terminal carrier-specific running laws.

The forced sectors (2,3,5,7) share one exact m-indexed interaction table; half-One uniquely unifies the odd-sector standing modes; all finite common-support sector gaps are positive and shrink without finite coincidence; the gravity/EM/strong Fold periods are 1/2/3 with joint recurrence 6; and the inconsistent flat-EM slope bundle is rejected in favour of the sealed terminal carrier-specific running laws.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-STRONG-RUNNING-DIRECTION-002
- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003
- SFT-PHYS-COUPPLING-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003
- SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of sector domain, standing-mode carrier, coupling table, mass relation, common support, sector order, coincidence boundary, period dictionary, empirical succession and extension forms. The declared exact boundary is: Every forced prime sector through seven, every interior standing mode, every positive binary common-support successor, every ordered sector pair, the exact Fold periods of One, one-third and one-seventh, and the already sealed complete force/running comparison records. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: complete-prime-ladder-through-seven__unique-self-antipodal-half-One-mode__single-m-indexed-exact-table__positive-shortfall-from-unison__common-positive-binary-support__exact-positive-m-difference-gap__finite-

triple-coincidence-forbidden__complete-one-two-three-period-dictionary__terminal-carrier-specific-running-and-anchors__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
sector	complete-prime-ladder-through-seven	selected-familiar-sectors	Selection omits generated prime sectors or the exclusion boundary.
standing	unique-self-antipodal-half-One-mode	named-unification-point	A name does not prove membership or uniqueness.
table	single-m-indexed-exact-table	independent-fitted-couplings	Independent values violate the one-generator law.
mass	positive-shortfall-from-unison	inserted-mass-parameter	An inserted mass is not forced by the holding relation.
support	common-positive-binary-support	independent-scale-axes	Independent axes cannot define exact separation or convergence.
order	exact-positive-m-difference-gap	target-named-physical-order	A physical label cannot choose an inequality.
coincidence	finite-triple-coincidence-forbidden	asserted-unification-crossing	A crossing contradicts the positive pair-gap identity.
period	complete-one-two-three-period-dictionary	selected-force-periods	Selected labels provide no Fold traces.
comparison	terminal-carrier-specific-running-and-anchors	flat-EM-and-inconsistent-slope-bundle	It conflicts internally and with admitted electromagnetic running.
extension	no-extra-rule	free-group-or-crossing-scale	An added group or fitted crossing scale is a free premise.

Base and successor. The closure proof is sealed by derivation hash sha256:286814bb194316d6ca1b4ba21051191f7ae874f2b884b018593d5a189349f367. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:11291e339eeb79bc8fd240fa7f224091ae8d90181454b0e5f7dc54036db11e0f.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:3f3679f7090946e7e35c8c24b3404ea6d728db8441fbcee8d15c9340a77c896c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4bc46d67f83fe4f2fd275512e8d0fe0b5fbf7b4d79bd1c8c8a2af05dff6ad616.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:f15b3d34a2b7c0ee7df313404be18a2a1be08f98b252e38425e7f7c638322e0c.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:912c5c89d9d541e061c7403105de6f544a261de230b53815c44e381fc3a03606. Independent certificate: sha256:761d3dec7f1740758885b49c4b939885091bc84bd9b04ba311257a8cecd52ee. Engine

external-validation hash:

sha256:b3654b21b34c6f939fa95016665043430242b00e09da2b02541fd3988504c9a8.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported gauge-unification group, beta function, coupling crossing or mass model
- no V1/V2 executable, answer table, measured target or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no physical-sector ordering substituted for the exact common-support m ordering
- no retention of the internally inconsistent slope= $m-1$ and slope(2)=0 bundle
- no target access before the inherited force-sector and running seals

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:33d4c53c23e99c7b5b4b44662cc05b6531060e42157a5153505898281139c14a; engine receipt
sha256:464f20ad41d716a52836a9d5952a62991e76b488a2ea70539ea6f4c0ee8c8f0e at receipts/engine/model_admitted/SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025-464f20ad41d716a5.json;
empirical-validation hash None; measurement receipt None.

182. Finite Fold colour-singlet positive excitation gap

Claim identity: SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Within the complete finite Fold colour grammar, the least observable colour singlet has normalized gap one-third above the empty excitation record at every positive finite depth; confinement work advances by two-thirds; the two shares form a period-two partition of the One; and the locally massless confined carrier is retained without being mistaken for a free gapless physical singlet.

Within the complete finite Fold colour grammar, the least observable colour singlet has normalized gap one-third above the empty excitation record at every positive finite depth; confinement work advances by two-thirds; the two shares form a period-two partition of the One; and the locally massless confined carrier is retained without being mistaken for a free gapless physical singlet.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013
- SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014

- SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of colour sector, observable support, vacuum boundary, gap carrier, confinement work, local-carrier distinction, Fold orbit, depth closure, prior-claim correction and extension forms. The declared exact boundary is: The generated colour-three Fold sector, its complete antipodal and three-member singlet supports, every positive finite separation/depth successor, the locally massless confined carrier and the normalized One partition. No continuum field action or completed infinite-volume limit is asserted. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `generated-colour-three-sector__complete-colour-singlet-support__empty-excitation-record__exact-positive-one-third-complement__fixed-two-thirds-positive-work-successor__retain-local-massless-physical-gapped-distinction__complete-period-two-Fold-partition__positive-finite-successor-induction__correct-free-singlet-spectrum-boundary__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
sector	<code>generated-colour-three-sector</code>	<code>named-Yang-Mills-sector</code>	A conventional name does not generate colour structure.
support	<code>complete-colour-singlet-support</code>	<code>isolated-colour-carrier</code>	The admitted confinement law leaves the isolated-colour observation record empty.
vacuum	<code>empty-excitation-record</code>	<code>numerical-zero-vacuum</code>	Numerical zero is outside the Fold proof domain.
gap	<code>exact-positive-one-third-complement</code>	<code>massless-or-fitted-gap</code>	A fitted value or an empty mass label conflates local carrier and physical singlet.
confinement	<code>fixed-two-thirds-positive-work-successor</code>	<code>bounded-or-free-separation-work</code>	Bounded or free work permits an unrecorded free colour state.
carrier	<code>retain-local-massless-physical-gapped-distinction</code>	<code>conflate-local-carrier-with-physical-singlet</code>	The local carrier is not an observable isolated colour state.
orbit	<code>complete-period-two-Fold-partition</code>	<code>selected-gap-coupling-pair</code>	A selected pair does not prove closure or recurrence.
depth	<code>positive-finite-successor-induction</code>	<code>bounded-depth-census-or-completed-infinity</code>	A bounded sample or completed infinity cannot certify every finite successor.
prior	<code>correct-free-singlet-spectrum-boundary</code>	<code>repeat-no-massless-strong-excitation</code>	That wording contradicts the admitted locally massless strong carrier.
extension	<code>no-extra-rule</code>	<code>free-scale-action-or-continuum-limit</code>	An added scale or action is a new premise.

Base and successor. The closure proof is sealed by derivation hash `sha256:17d2c3576cd428a73269198c334acb0f1cf802b25f6e497ed6bacfca9b0190a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:c99ca2dfdaa551521dbaf64ffa4fe813fb7ebf7aefd9fe92a43cc9acc3490815`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7dfce3e24867adf20b37bd68a2974b3187df30d0a3fd70d622a21cb0ad1425ef`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : f0a61704c36c3b494f221da8100787eed72ce7f23f01f0bf4ccc2768b06fa77c.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : 7cb2b037e883a22a7ffc40dc3dfc3ce55a2407fe78599ed0360d879f2713a4e6.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 9036da93a060fe7853f080b732a716bc5660a07b3dff95439a56848c43e86f33.
Independent certificate:
sha256 : 7e8adb8391a99dcbd0a57c2801f6a9929eedd49ef4ebe7aeb5419dda306c6226. Engine
external-validation hash:
sha256 : fcf8862fd656cdf3479ca59a90b686dd0ad9ed1d8d1b5844e42682bfd549305.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported Yang-Mills action, gauge-group axiom, Hamiltonian or continuum mass-gap proof
- no V1/V2 executable, answer table, lattice mass or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no fitted dimensionful glueball scale or claim that one-third equals a mass in MeV
- no completed infinity or inference from a finite simulation to a conventional continuum theorem
- no conflation of the empty local carrier mass label with the least physical singlet gap

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256 : 98fe4e9d3c65ed14b2416232187dd76ef9816f3774b984860d87138ad775fb82; engine receipt
sha256 : 899e904600c1018041286e947de6acefe6c1afb50f1f8da202cfda587fa9568a at receipts/engine/model_admitted/SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026-899e904600c10180.json;
empirical-validation hash None; measurement receipt None.

183. Post-seal lattice-spectrum boundary for the colour-singlet gap

Claim identity: SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The sealed Fold prediction of a positive physical colour-singlet gap matches every retained PDG pure-gauge lattice interval: 0^{++} at 1653 ± 26 MeV, 2^{++} at 2376 ± 32 MeV and 0^{-+} at 2561 ± 40 MeV are each strictly positive and their intervals are ordered and disjoint. These are external lattice records, not fitted Fold values. Quenching, state mixing, full-QCD requirements and the lack of an unambiguous experimental glueball remain explicit boundaries.

The sealed Fold prediction of a positive physical colour-singlet gap matches every retained PDG pure-gauge lattice interval: 0^{++} at 1653 ± 26 MeV, 2^{++} at 2376 ± 32 MeV and 0^{-+} at 2561 ± 40 MeV are each strictly positive and their intervals are ordered and disjoint. These are external lattice records, not fitted Fold values. Quenching, state mixing, full-QCD requirements and the lack of an unambiguous experimental glueball remain explicit boundaries.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete eight-axis product of physical carrier, positive-gap relation, provenance, prediction isolation, measurement separation, complete spectrum-row retention, finite successor closure and extension. The declared exact boundary is: The admitted finite Fold colour-singlet gap and every registered PDG 2026 pure-gauge glueball mass/uncertainty row, ground/excitation ordering row, quenched, mixing, full-QCD and experimental-identification limitation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__positive-colour-singlet-gap-with-complete-spectrum-scope__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	positive-colour-singlet-gap-with-complete-spectrum-scope	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b91011e3dc72496d920bcc5b4aed304377a9c66b2f4de7e181c4ecb7a3175197`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 67966cb00d6a0c9fd1864193d149e7e29bc3e4a4678bb14be657f7527f7419db.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f303362e4cc90b1b2c95b5098b87e69832a55e960e23b88ca6df495dd0df0dae.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 8feb5e24377ab231eb8515c33743564d296e3e046311f170e02121a0cef32202.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f2b80835649f5c2ba9eb104f5c7dbe64338493657e15e5b076ce067128d5a8d8.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 4f4480d8fbbdb8825c6729c2869fd8f455314664312e2be313925fafec7b9ad. Independent certificate: sha256: b6b4b4ec429281b86ec1e5485d020d152dcb0bb939f415377433ec4648bbde77. Engine external-validation hash: sha256: 241e721b4221701046a50bdd437bb7c219336e8e53cf0d6ab0f93767d5b1f364.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2026-QUARK-MODEL-GLUEBALL-BOUNDARY

Observed comparison records:

- The PDG 2026 source record and complete PDF are hash-locked and released only after the Fold prediction seal.
- All three registered pure-gauge lattice mass and uncertainty rows are retained and evaluated with exact rational inscriptions.
- The 0++ interval 1627-1679 MeV, 2++ interval 2344-2408 MeV and 0-+ interval 2521-2601 MeV are each strictly positive and mutually ordered without overlap.
- The source identifies 0++ as the lattice ground state and 2++ as the first excitation; these quantum-number labels are retained external records, not separately forced by this Fold claim.
- The quenched approximation, neglected quark loops, glue/quark-state mixing, full-QCD requirements and high-statistics limitation are retained.
- The PDG nonidentification boundary is retained: no scalar below 2 GeV is established as predominantly glueball, and no unambiguous experimental glueball is claimed.
- No dimensionful lattice value selects, fits or numerically identifies the exact normalized Fold gap one-third.
- A deliberately altered lowest-mass row breaks interval ordering and is rejected.

Falsification condition: Reject if either source snapshot changes; if any registered spectrum or limitation row is omitted; if the least retained pure-gauge mass interval reaches the empty boundary; if registered intervals lose their stated ordering; if an external mass is imported into the Fold proof; if a lattice prediction is relabelled as direct detection; or if the target is available to the capability-closed predictor before sealing.

Measurement receipt: sha256: 3835a4b4efc277cbd7a7a69ec3afb546254ca82d841b405b6f5abb3bb9100b0f. Isolation certificate: sha256: f0903f9280b06201a1a69b2e3297c8741e7ad4ef85af1ae70c9956984e919fa8. Custody certificate: sha256: 6e92af1f8314d310047e62c9a970652cb5ce748cdf6a635c001b9934a40c35b0.

Registered source descriptions:

- Particle Data Group: <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-quark-model.pdf>
(sha256: 6aa98fa53857122f27b638c59081af1a2857d787c4205370d8fa38fcb6b70ff0)

Registered target identities:

- PDG-2026-WITHHELD-COMPLETE-GLUEBALL-BOUNDARY from
PDG-2026-QUARK-MODEL-GLUEBALL-BOUNDARY

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no PDG, lattice mass, uncertainty or quantum-number ordering in the formal predecessor derivation
- no claim that normalized one-third equals 1653 MeV or any dimensionful mass
- no omission of quenched, mixing, full-QCD or unresolved-identification limitations
- no claim of direct experimental glueball detection or conventional continuum Yang-Mills proof
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude
- no target access before the prediction seal and no fitted correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:24e898b09f1ff360265f53be7b7fb3c7ef2555cdc64056bc0118c2c0fa00f61b; engine receipt
sha256:d85984ba94789e15d9fdb97588e63d318b014b3e84315ace56cb50c16cbeb2fe at receipts/engine/model_admitted/SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027-d85984ba94789e15.json;
empirical-validation hash
sha256:b96e76915046971454a96f7d502c2a1f81c6b84d810d1c21cab293c1e99a1e6c; measurement receipt
sha256:3835a4b4efc277cbd7a7a69ec3afb546254ca82d841b405b6f5abb3bb9100b0f.

184. Depth-independent composite confining-sector orbit law

Claim identity: SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete composite-sector census is: sector 8, denominator 7, orbits $(1,2,4)/(3,6,5)$, three antipodal pairs and coupling $7/8$; sector 12, denominator 11, one ten-mode orbit, five pairs and $11/12$; sector 18, denominator 17, two eight-mode orbits, eight pairs and $17/18$; sector 24, denominator 23, two eleven-mode orbits, eleven pairs and $23/24$; sector 30, denominator 29, one twenty-eight-mode orbit, fourteen pairs and $29/30$. The general odd-support inverse proves denominator confinement and complete recurrence at every positive finite Fold depth.

The complete composite-sector census is: sector 8, denominator 7, orbits $(1,2,4)/(3,6,5)$, three antipodal pairs and coupling $7/8$; sector 12, denominator 11, one ten-mode orbit, five pairs and $11/12$; sector 18, denominator 17, two eight-mode orbits, eight pairs and $17/18$; sector 24, denominator 23, two eleven-mode orbits, eleven pairs and $23/24$; sector 30, denominator 29, one twenty-eight-mode orbit, fourteen pairs and $29/30$. The general odd-support inverse proves denominator confinement and complete recurrence at every positive finite Fold depth.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003
- SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025

Generated grammar and closure boundary. Generate the complete product of sector scope, held support, standing-mode census, Fold action, confinement, orbit partition, antipodal closure, pair-count law, holding coupling, external correspondence boundary and extension form. The declared exact boundary is: Every positive finite even sector beyond two; its odd positive predecessor denominator; every nonempty numerator label; every exact Fold successor and first return; every complement pair; the five registered composite-sector instances; and the already sealed force-sector anchor boundary. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: `every-generated-even-sector__positive-predecessor-denominator__all-nonempty-predecessor-labels__double-and-cast-denominator-wholes__odd-support-exact-inverse__complete-disjoint-first-return-partition__unique-positive-One-complements__predecessor-of-denominator-over-two__all-but-one-predecessor-over-sector__sealed-eight-anchor-and-explicit-standing-cases__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
scope	every-generated-even-sector	five-selected-case-table	A selected table does not prove the next even sector.
support	positive-predecessor-denominator	borrowed-or-measured-denominator	A named or measured denominator can select the orbit.
modes	all-nonempty-predecessor-labels	selected-mode-subset	A subset cannot establish exhaustive confinement.
action	double-and-cast-denominator-wholes	selected-orbit-permutation	A listed permutation imports its answer.
confinement	odd-support-exact-inverse	finite-table-assertion	A table alone does not close arbitrary successor depth.
orbits	complete-disjoint-first-return-partition	named-or-overlapping-orbits	Named or overlapping rows omit a complete partition proof.
antipodes	unique-positive-One-complements	borrowed-pairing-rule	A borrowed pair rule does not prove exhaustiveness.
pair_count	predecessor-of-denominator-over-two	ambiguous-sector-minus-one-over-two	That notation is not a whole count for even s.
coupling	all-but-one-predecessor-over-sector	measured-or-fitted-coupling	Measurement cannot select a holding law.
comparison	sealed-eight-anchor-and-explicit-standing-cases	unobserved-cases-counted-as-confirmed	A standing prediction is not a measurement.
extension	no-extra-rule	free-orbit-or-coupling-correction	An inserted row or correction is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:f2a635c4e730cc95000af2f718ae9c0ea451904511d39264b7aa4669be821395`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:d64de949d0073feb311cb6b076043542586f4bb0f6d164c1c5ae451b8344c2f8.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:de67bce63f980db5ef3b82782a6980729c913a1abb9e175ed3ce896e696d48ce.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8720f86993e5ca432ed6f38fd4bc5f64f288b42723ba14aa0f57729cc611249a.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:11ce4408c46af0dca4b297e5207ff15539732f5323461ae6ca66338a157e79e4.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:d0a35a4a0cf61ded5a2a3438c5811e48274991518e20380c4132a18b441f638c.
Independent certificate:

sha256:d3cb445ee39882793a1669e1a0519589f5c1d7353d1f4f8f2c1d1822de9d0cac. Engine
external-validation hash:

sha256:a4b6ff145b49bdf9e7752f6b380faa2559e021103010f6efd6932681e2fe9e7e.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, stored orbit table or prior survivor in the derivation runtime
- no imported gauge group, particle taxonomy, measured coupling or target-selected sector
- no semantic numerical zero, negative, irrational, imaginary, floating or completed-infinity proof value
- no claim that every arithmetic composite sector is already an observed independent physical force
- no relabelling of the standing sectors twelve, eighteen, twenty-four or thirty as measured confirmations
- no ambiguous half-whole pair count: the exact whole formula is $(s-2)/2$

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:fdfab054033e97c405ab0a5d9237425c3b5ee4acaf1e3ccc154e47160c5454ab; engine receipt
sha256:eb8b53e1662db23eff80d371b5b6db094e3c7d4665b5ea0a6882ba3072e29ec8 at receipts/engine/model_admitted/SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031-eb8b53e1662db23e.json
; empirical-validation hash None; measurement receipt None.

185. Terminal general Fold-fibre, particle-mode and generation-transport law

Claim identity: SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every positive finite $m \geq 2$, the m -Fold has exactly m positive preimages and $m-2$ interior fixed modes. The four generation coordinate records are complete ordered three-label systems and are not masses. The exact 27-cell state volume forces binary cover depth five and colour cover depth three. Every finite recurrent particle class injects into internal Fold modes without adding a spatial dimension. Generation order transports to the admitted terminal lepton, quark and neutrino polynomial carriers; superseded site-as-mass readings are rejected. The exact dual seeds are $1/95$, $1/383$ and $1/485$; at every positive depth the heavy/light ratio equals the subtraction reach $N-1$; and the quarter-mode 1:1:4 record is a structural squared-separation multiplicity, not a universal measured mixing-rate ratio.

For every positive finite $m \geq 2$, the m -Fold has exactly m positive preimages and $m-2$ interior fixed modes. The four generation coordinate records are complete ordered three-label systems and are not masses. The exact 27-cell state volume forces binary cover depth five and colour cover depth three. Every finite recurrent particle class injects into internal Fold modes without adding a spatial dimension. Generation order transports to the admitted terminal lepton, quark and neutrino polynomial carriers; superseded site-as-mass readings are rejected. The exact dual seeds are $1/95$, $1/383$ and $1/485$; at every positive depth the heavy/light ratio equals the subtraction reach $N-1$; and the quarter-mode 1:1:4 record is a structural squared-separation multiplicity, not a universal measured mixing-rate ratio.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003
- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-PHYS-NEUTRINO-POSITIVE-MASS-003
- SFT-PHYS-MATTER-GENERATION-DEPTH-003
- SFT-PHYS-MATTER-CKM-TERMINAL-004

Generated grammar and closure boundary. Generate the complete product of fibre, fixed-mode, generation-coordinate, cover-depth, particle-placement, mass-transport, dual, reach, transition and extra-rule forms. The declared exact boundary is: Every positive finite m -Fold by depth-independent index induction; all four complete three-label coordinate systems; the exact 27-cell cover; every finite recurrent particle-trace census; all admitted terminal charged-lepton, quark, neutrino and CKM carriers; and every superseded site-as-mass and separation-as-observed-rate alternative. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `complete-m-offset-fibre__complete-fixed-index-range__three-order-isomorphic-coordinate-systems__least-cover-of-generator-volume__internal-recurrent-trace-mode__order-transport-to-terminal-polynomials__complete-colour-binary-dual__exact-structural-reach-identity__structural-separation-multiplicity__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
fibre	complete-m-offset-fibre	selected-binary-or-colour-count	A selected special case is not a general law.
mode	complete-fixed-index-range	named-three-mode-pattern	Naming three modes does not force the general count.
generation	three-order-isomorphic-coordinate-systems	one-coordinate-system-selected-as-mass	A coordinate value is not automatically a physical mass.
cover	least-cover-of-generator-volume	chosen-depth-five-or-seven	A selected depth is a free choice.
placement	internal-recurrent-trace-mode	extra-spatial-dimension	An extra spatial coordinate is not generated.
mass	order-transport-to-terminal-polynomials	site-fraction-is-mass	This superseded identification conflicts with the terminal spectra.
dual	complete-colour-binary-dual	lookup-depth-fractions	Looked-up fractions are not forced.
reach	exact-structural-reach-identity	lifetime-equals-mass-ratio-postulate	A dimensional lifetime equality is not forced by the count.
transition	structural-separation-multiplicity	universal-observed-rate-ratio	The old 1:1:4 physical-rate claim omits interaction carriers.
extension	no-extra-rule	extra-dimension-or-fit	Either addition is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash
sha256:c475546316355bcfdc77aac50982f522bb2e5d2837eec8b3175f6691668e0508. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:89d95f25da0998ae0e2c12ef972877ed435ba65c15ba347f4ad8cc2464ef27.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:49d418baf34f44b4527e9317cc194779bc21f294b0dfd0d94461154a0bb2fd63.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:847552ca2ada1c1c2f5999d26c3c4e4278c5ba52eae77a5d4ff309df95948836.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:4c316148e986b887cbd73d1b0bbce7b8c7dd31124b99dca5d77a7040247ff17f.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:d7e58bd3b815e0c502770da64b136f2235d69191a2118750f48392f9020b1738.
Independent certificate:
sha256:e3a787096b77a1d8c83910dd1d3de2908658797d5c75947f31aa4abc1c01b97e. Engine
external-validation hash:
sha256:c7a4c26b64bed04fe7a5cb92b586a03223af71bd5f707908a514762f11d6b4e8.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 result artifact, string-theory dimension rule, particle table or measured mass as a formal premise
- no identification of $1/6$, $1/2$, $5/6$, $1/3$, $2/3$, One, $1/4$ or $3/4$ with terminal measured masses
- no claim that 1:1:4 is a universal observed mixing or decay-rate spectrum
- no extra spatial dimension, compactification choice, landscape count or target-selected mode
- no fitted mass coefficient, lifetime coefficient, mixing shift or omitted terminal sector
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4e6de203282d2e77866724630a7ca291d01785551605bda4768e703600efb464; engine receipt
sha256:18e4aac0792a01c613d31f4d980951b0e2da69107155c4e50967ed80c9eaa700 at receipts/engine/model_admitted/SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051-18e4aac0792a01c6.json;
empirical-validation hash None; measurement receipt None.

186. Terminal dark relic, Smithion spectrum and flavour-transition law

Claim identity: SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Four exact penta/hepta cubics at depths (7,10,9,12) each have three disjoint positive roots, yielding twelve exact rationally enclosed Smithion mass-ratio predictions. Their coloured product family cross-locks the two quark values $3/1454$ and $3/13118$. Complete neutral p -member singlets and unique lightest roots force the stable relic class within the closed grammar. Dark/baryon is $27/5$ and matter/baryon $32/5$. Flavour weights are exactly $1/32$, $5/96$ and $5/24$, reducing to 3:5:20 with tau-to-e over tau-to-mu equal to 4. Sector beta slopes are $p-1$.

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Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-MATTER-CONFINEMENT-LIFT-003
- SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051
- SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001

- SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

Generated grammar and closure boundary. Generate the complete product of sector, kind, depth, invariant, root-census, mass-ratio, neutral-singlet, stability, abundance, flavour-weight, measurement-boundary and extension forms. The declared exact boundary is: Prime sectors five and seven; down/up cover powers three/four; the complete exact cubic grid; all three root intervals; complete neutral sector fibres; all three lepton transitions; exact positive whole/fraction carriers; and no physical target before seal. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: `both-forced-prime-sectors__complete-down-up-dual__least-binary-cover-of-sector-power__same-coloured-product-family__complete-grid-and-rational-halving__squared-rational-enclosures-with-light-lift__least-complete-neutral-singlet__generation-volume-over-cover-depth__squared-separation-times-parent-part__all-targets-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
sector	both-forced-prime-sectors	selected-particle-sector	A selected sector is not a complete census.
kind	complete-down-up-dual	one-selected-kind	One kind omits half the generated spectrum.
depth	least-binary-cover-of-sector-power	chosen-depth	A chosen exponent fits a hierarchy.
invariant	same-coloured-product-family	independent-mass-parameter	A mass parameter forks the law.
roots	complete-grid-and-rational-halving	chosen-decimal-roots	Chosen decimals import an irrational model.
ratios	squared-rational-enclosures-with-light-lift	floating-central-estimates	A central decimal loses exact root custody.
relic	least-complete-neutral-singlet	named-dark-particle	A name does not force stability or neutrality.
abundance	generation-volume-over-cover-depth	freezeout-fit-or-density-input	A cross-section or density would be a parameter.
flavour	squared-separation-times-parent-part	fitted-branching-fractions	Fitted rates cannot force a ratio.
measurement	all-targets-inaccessible-until-seal	search-result-readable-before-seal	That would tune the theory to current limits.
extension	no-extra-rule	extra-scale-sector-or-correction	Any added choice violates zero parameters.

Base and successor. The closure proof is sealed by derivation hash `sha256:3351ece9cf5b191e15416884e44dce897be9cb7cfaa2359d088604779ae3602e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
`sha256:3084e9fee79537fb19204951d2bc8e82d13335ddf0ad5d68a2c9054f0c7d738d`.

- **tampered_source**: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:3187efb26cdc55782a7d746a3e39775a4088b3aa65b92c31e188588d54803f10.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1a08b1cc107e3eb8504a130861084b840018b107d069e3991364b95b3eda5f90.
- **boundary**: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:5d51b9f8d04a4587584e88486b2a63c0fc0677fb27663430bf131394d110606a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:4eb56b948d11e8219bfddf29dac76fae3eb2a7d8e6cc994888e2fec4432e29c4.

Independent certificate:

sha256:f53233046528aef9158e3a637b87e7a874856c29aa86d1808a8568bc89eb5699. Engine

external-validation hash:

sha256:0d0180e4437cf2371bdd914f9b6b5519bbeaf33ba5d192dc4761bc66689728b7.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 code, answer table, survivor identifier or measurement in formal execution
- no fitted mass, coupling, cross-section, freeze-out efficiency, density, branching fraction or search limit
- no negative, irrational, imaginary, floating, NaN, infinite or continuum proof scalar
- no claim that an unobserved Smithion mass has already been experimentally measured
- no claim that non-detection alone identifies the relic

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:dfbaa171ffbee41651dcf56b1a0e0097d2cd088376d003686d56f094f7c8113a; engine receipt

sha256:71a131119ffd152690ec772019177fb06599474cd5e89b60a62c1b9b69ec5762 at receipts/engine/model_admitted/SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061-71a131119ffd1526.json;

empirical-validation hash None; measurement receipt None.

187. Terminal strong-CP alignment and proton/baryon stability law

Claim identity: SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Strong-sector CP is the aligned One; the strong-violation, electric-dipole and added-compensator records are empty-One. The weak one-hand phase remains the self-antipodal half-One. All 83 generated mediators are fibre-preserving; all 202 ordered cross-fibre label pairs and all six colour-to-binary pairs lack a generated carrier. The proton carries baryon One = $3 \cdot (1/3)$, all 27 generated colour-three images retain it, and every finite composition remains in the colour fibre. Proton decay is absent within the complete declared grammar.

Strong-sector CP is the aligned One; the strong-violation, electric-dipole and added-compensator records are empty-One. The weak one-hand phase remains the self-antipodal half-One. All 83 generated mediators are fibre-preserving; all 202 ordered cross-fibre label pairs and all six colour-to-binary pairs lack a generated carrier. The proton carries baryon One = $3 \cdot (1/3)$, all 27 generated colour-three images retain it, and every finite composition remains in the colour fibre. Proton decay is absent within the complete declared grammar.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-WEAK-PARITY-FIBRE-002
- SFT-PHYS-NEUTRINO-CP-PHASE-002
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

Generated grammar and closure boundary. Generate the complete product of immutable predecessor, CP carrier, handed support, strong-sector product, compensator, sector-action typing, baryon tally, finite transition, baryogenesis boundary, measurement custody and extension forms. The declared exact boundary is: All nonempty subsets of the exact two-hand fibre; complete colour-three times hand support; every mediator cell in prime sectors two, three, five and seven; every ordered cross-fibre label pair; every colour-three word; and every finite composition of the generated typed actions. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: `compose-immutable-predecessors__aligned-One-or-self-antipodal-half-One__complete-two-hand-support__paired-parity-returns-aligned-One__empty-One-no-extra-compensator__all-generated-actions-are-fibre-preserving__three-one-third-parts-close-to-One__finite-composition-preserves-the-fibre__retain-explicit-cosmological-boundary__targets-inaccessible-until-seal__no-extra-rule`. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	compose-immutable-predecessors	rewrite-or-bypass-predecessors	A successor cannot change an admitted receipt.
cp	aligned-One-or-self-antipodal-half-One	continuum-angle-or-fitted-small-value	A continuum angle or small fitted value is inadmissible.
hand	complete-two-hand-support	selected-single-hand	Selecting one hand omits half the colour-times-hand product.
strong	paired-parity-returns-aligned-One	weak-antipode-reused-for-strong	A one-hand weak carrier contradicts complete strong support.
compensator	empty-One-no-extra-compensator	add-axion-or-free-compensator	An extra carrier has no discrepancy to act on and forks the closed inventory.
sector	all-generated-actions-are-fibre-preserving	invent-cross-fibre-mediator	A two-sector signature is absent from every complete sector action table.

Axis	Forced form	Eliminated form	Elimination reason
baryon	three-one-third-parts-close-to-One	untyped-particle-name	A name alone does not establish conservation.
transition	finite-composition-preserves-the-fibre	assume-an-unregistered-decay-step	An unregistered fibre crossing inserts the desired decay.
baryogenesis	retain-explicit-cosmological-boundary	conflate-distinct-process-grammars	A cosmological tally-changing premise is not a low-energy mediator.
measurement	targets-inaccessible-until-seal	target-readable-before-seal	That would tune alignment or stability to observation.
extension	no-extra-rule	free-angle-carrier-or-operation	Any such addition is an unforced parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:d9f8979ae65fb3c20f78a9ab79262c68e94e265ee6c4ca439f4dfe72eae49a84`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:290a02b4e4f473a3ce4642758d4802afc041ce55fa1bdd67b6bfa67362291492`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:52f4575af714aac932f9d9b34d2ffa0e5c9f931c5e2344837a94e396c9dad408`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:552ae613a5e02f35cf12b163fba2b2bab42f8d26c5f5fa288400676c03132929`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:35261f06a81f2466962ec820e1b6b4ec55dd52c60bdf13960ff56c23e2f0db76`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:158f6d0db3af5495b5377fbaa556c17f4010b3ef24b7274d6c60f7e0c4608e67`. Independent certificate: `sha256:3fe766ec11ea0dba5bf00a0d2acd889ca3523c97ef518c6f35d6379d0c32fcab`. Engine external-validation hash: `sha256:fc2a863920358c2c6138acdbecd364cce3472795ef6a85c45b2f9695b2e4e9fe`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table, survivor identifier or measurement as a premise

- no conventional theta continuum, fitted small angle, negative, irrational, imaginary, floating, NaN or infinite proof scalar
- no neutron electric-dipole bound or proton-lifetime limit in formal execution
- no imported Peccei-Quinn, axion, grand-unification or Standard Model mechanism as a proof premise
- no contradiction of the separately admitted cosmological baryogenesis dependency law

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:751d04511116f683c0771afbc6752af128a9360cedbb8b98ad9ba08a2cbc6818; engine receipt
sha256:6bd69d2eb78150c11fefc82f512bdfd775eaa2f3a2667127407d7a34bd695657 at receipts/engine/model_admitted/SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063-6bd69d2eb78150c1.json
; empirical-validation hash None; measurement receipt None.

188. Terminal displaced-ground and Higgs mass/self-coupling law

Claim identity: SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The displaced ground is uniquely half-One; the retained leading Higgs rungs are $1/2$, $1/4$ and $1/8$. Complete six-cell scalar support, its unique held return and the independently equal cover depth five force terminal $m_H/v = 1/2 + (6/5) \alpha = 2563352914777/5038463954690$. The exact native self-coupling is $\lambda = (m_H/v)^{2/2} = 6570778165695741824959729/50772238045420788745992200$. Both routes cross-lock in the squared domain; no irrational root is formed.

The displaced ground is uniquely half-One; the retained leading Higgs rungs are $1/2$, $1/4$ and $1/8$. Complete six-cell scalar support, its unique held return and the independently equal cover depth five force terminal $m_H/v = 1/2 + (6/5) \alpha = 2563352914777/5038463954690$. The exact native self-coupling is $\lambda = (m_H/v)^{2/2} = 6570778165695741824959729/50772238045420788745992200$. Both routes cross-lock in the squared domain; no irrational root is formed.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-ONE-001
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

Generated grammar and closure boundary. Generate the complete product of ground carrier, displacement, leading rungs, scalar directional support, held return, cover cross-lock, terminal transport, coupling composition, route equality, measurement custody and extension forms. The declared exact boundary is: Structural empty-One, half-One and One ground forms; every binary successor of the displaced ground through the leading coupling rung; the complete binary-hand times generator-direction product; its unique held return; the least cover of generator volume; exactly one terminal alpha transport; exact positive rational operations; and no physical target before seal. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: **positive-proper-ground-carrier__unique-self-antipo**

dal-half-One__retain-half-quarter-eighth-controls__complete-two-by-three-product__hold-unique-return-leaving-five__least-cover-of-generation-volume__one-six-over-five-alpha-r
 eturn__squared-excitation-shared-over-two-fibres__exact-two-route-cross-lock__all-targets-inaccessible-until-seal__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
ground	positive-proper-ground-carrier	absence-as-numerical-ground	Structural absence is not a field value.
displacement	unique-self-antipodal-half-One	unison-or-selected-part	Unison is not displaced and a selected part is a parameter.
rungs	retain-half-quarter-eighth-controls	discard-or-rewrite-leading-rungs	A terminal successor cannot erase its lower-order receipt.
directions	complete-two-by-three-product	selected-direction-subset	A subset omits a generated hand or generator direction.
return	hold-unique-return-leaving-five	retain-all-six-as-active	The invariant return cannot also be an active displaced direction.
cover	least-cover-of-generation-volume	free-denominator	A denominator chosen for agreement is a parameter.
transport	one-six-over-five-alpha-return	fitted-offset-or-repeated-series	An offset or truncation is a fit.
coupling	squared-excitation-shared-over-two-fibres	import-potential-or-free-lambda	An imported potential or free lambda would add an axiom.
route	exact-two-route-cross-lock	independent-mass-and-coupling-routes	Independent routes introduce a dial.
measurement	all-targets-inaccessible-until-seal	target-readable-before-seal	That would fit the coefficient to the measured mass.
extension	no-extra-rule	extra-mass-potential-or-correction	Any added term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:b032f36759d5585227839385874ee6a097037af0b1142ebf0738a290b480bff5. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:92eb8365d3cc30c6263094989f3acef1ac062d7e661401ab949da0b36db56255.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:c1821578d2ccf8237255f446bbde7e3117de5edcd59b00b37071f3e485c6a946.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:7f5b272fccfbef1f1ce7eabace8825d5f3c8073c754e1e4b9ab230a2b07b530d.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:ea6d7a5ec8dbbde9369629b7a0919e4233a1aec722a5894371da443dd9818951.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:89f6565f52c17b5f9eb1f217a3dabd38f0142336cd4e23900204d370aa4e78bc.
Independent certificate:
sha256:348c2daebd8443b15a0943ea97c6336d0767529492929a71a15fa788a4a30d53. Engine
external-validation hash:
sha256:3482618408366fc7c558674ede8b2c5ee86d7b0d6e80b42a2373df5bc0ca70db.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Matter, interaction and flavour sectors are classified by minimal recurrent physical words, exchange classes, conserved labels and complete transition channels. External tables test membership and value only after the classification law is sealed. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, target value, candidate table or stored survivor as a premise
- no measured Higgs mass, VEV, Fermi constant or self-coupling in formal execution
- no imported Higgs potential, Standard Model mass relation, fitted coefficient, uncertainty or correction series
- no semantic numerical zero, negative, irrational, imaginary, floating, NaN, continuum or infinite proof scalar
- no claim that the leading 1/2 mass ratio or 1/8 coupling is the terminal result

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2c07deac3b179ef81471403a48877e0da72f003cae44166043302a1f0f0a973b; engine receipt
sha256:3f84e4ede48a489bc78a8cc47eac935bbff3566d1e561c26ef0059d7bbcbfb1e at receipts/engine/model_admitted/SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065-3f84e4ede48a489b.json;
empirical-validation hash None; measurement receipt None.

19. Atomic Molecular

Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry.

189. Atomic orbit-cell capacity from three-space boundary orientation

Claim identity: SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. At positive orbit rank r , three-space supplies one central orientation and the rank-two boundary supplies one held pair for every predecessor step. Two exclusion-distinct Fold labels occupy each orientation, forcing capacity $2(1+2(r-1))$ and the sequence 2, 6, 10, 14, 18 without observation.

For every positive orbit rank r , capacity is exactly $2(1+2(r-1))$; ranks one through five have capacities 2, 6, 10, 14 and 18.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of carrier, central class, successor extension, boundary multiplicity, Fold label, exclusion, capacity, generality, target custody and extension forms. The declared exact boundary is: All orbit-cell capacity laws generated from one central orientation, positive orbit succession, the complete rank-two boundary, two held Fold labels and exclusion. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: held-orientation-cell__one-central-orientation__on e-boundary-orientation-pair__forced-rank-two-pair__both-Fold-labels__one-of-each-label-per-cell__labels-times-complete-orientations__constant-boundary-pair-successor__widths-inaccessible-until-seal__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	held-orientation-cell	unlabelled-cell-count	An unlabelled count loses orientation identity.
central	one-central-orientation	empty-numerical-origin	Numerical zero is not an SFT carrier.
successor	one-boundary-orientation-pair	free-rank-increment	A free increment is a parameter.
boundary	forced-rank-two-pair	single-or-three-direction-addition	One omits a boundary side and three re-adds the held normal.
label	both-Fold-labels	label-erased-or-selected	Erasure or selection violates complete Fold support.
exclusion	one-of-each-label-per-cell	duplicate-cell-label	Duplication destroys distinguishability.
capacity	labels-times-complete-orientations	linear-or-doubling-without-boundary	Those former survivors ignore the newly admitted boundary discriminator.
generality	constant-boundary-pair-successor	finite-width-list	A list has no next-rank proof.
target	widths-inaccessible-until-seal	width-list-visible	That reverses derivation and test.
extension	no-extra-rule	extra-degeneracy-rule	An added degeneracy is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:6cd7fec86b24f0f4624a70894b507f836d4358eb2229515b61498407d958f993. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- false_premise: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:1ed87436e21f59d043967ca432026eaebd71e5e1f8be1907692eab24d946c6d5.

- **tampered_source**: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256: 625315de8974967651ba198f65bcbdd903e36bfac7a0cc349d449aed6298445b7.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 4bf7aa1fcc6d37bda661b6b898af0d9b3a48ba75d0f12afa04f8db6b363f4d61.
- **boundary**: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256: 1c45ec73b2de2f2c722af6fcddecc09f844cab496f9bb77bdbdda1f0c85f80ab.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 86c0b909d7fb32a35993571e1c3646d129604a21264f1da55aabad2b205c428e. Independent certificate: sha256: dd12a5fc73eacfd47d2eeaaaf16cc2925d756b0b88926041d958ecc2867474ec0. Engine external-validation hash: sha256: 4ac2066f98eb5a6803d8f83eb315180dfaccda40a4fdf6575c2208668792f67a.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 7b624a62e7d39d55c7a235163b8d0f93f6ffeebf172a7b94bf8fd4790be796f8; engine receipt sha256: 2b8928b0ddf5b87a7bf56813a5f430ba4a99f715b1cfc1dd1db98106216e3d70 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001-2b8928b0ddf5b87a.json; empirical-validation hash None; measurement receipt None.

190. Exact atomic existence boundary

Claim identity: SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A neutral atomic coordinate remains structurally admissible only while its positive whole charge does not exceed the exact inverse fine-structure Fold ratio. The unique greatest whole below that sealed ratio is 137, with 137 below and its successor 138 above the boundary.

The unique greatest positive whole atomic coordinate under the exact Fold binding ceiling is 137; $137 \leq 503846395469/3676744786 < 138$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete product of charge carrier, coupling, ceiling, endpoint construction, lower witness, upper witness, generality, target custody and extension forms. The declared exact boundary is: All greatest-positive-whole atomic-coordinate conclusions under the One binding ceiling and the independently sealed exact inverse fine-structure ratio. The generator produced 512 named candidates and the decision support contains 512 one-for-one decisions. Exactly one candidate survived: `positive-held-charge-count__charge-over-sealed-inverse-ratio__complete-One-ceiling__greatest-whole-division-certificate__exact-lower-inequality__successor-above-plus-order__depth-independent-greatest-whole-law__observations-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
charge	positive-held-charge-count	unbound-number	An unbound number has no element identity.
coupling	charge-over-sealed-inverse-ratio	fitted-binding-equation	A fitted equation imports the endpoint.
ceiling	complete-One-ceiling	external-critical-charge	An external critical charge is target input.
endpoint	greatest-whole-division-certificate	bounded-neighborhood-walk	A chosen search interval does not close all successors.
lower	exact-lower-inequality	endpoint-label-only	A label has no inequality witness.
upper	successor-above-plus-order	single-next-value-check-only	One check without order closure does not exclude later values.
generality	depth-independent-greatest-whole-law	finite-coordinate-census	A finite census leaves later coordinates open.
target	observations-inaccessible-until-seal	observed-elements-visible	That would turn an observation boundary into a law.
extension	no-extra-rule	extra-size-parameter	A size correction is a different, parameterized model.

Base and successor. The closure proof is sealed by derivation hash `sha256:6baa991e8391efbd50b0d7235617136cbf6b14f2df03464d9e05d075d2bd3476`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : 4160fd712b17226fbf3a8cc047e5446f5375e7bd3fa14fd1c64641c517a5e38e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : eba05c35e0e176c852c05a3f09315b28f7bd0cbe56b2511e8b5f44780bd2472a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : fe957e7c888b14fab4f760796244349be17a05c08bb8a440bb09417b28d8a928.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : a37c7574414b2ddd6b071742183623ac6758ef060b905ba5bbf38a6617730b62.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 55d5c09b7d7696f1d73843966a64fb278a666537289fd9472b7c3e557a83e55c.
Independent certificate:
sha256 : d4b358d1dc307a24df8800cec04b9db1f575e9e52f920356fabfbf7739b7e2d2. Engine
external-validation hash:
sha256 : eb370cb694ac8bb024e4020fbcd03b0cfbec1a28e9e0af2d2abf7141d17c999c.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256 : 97c4ce7770ec307c488fa3e908f713ef94eb5057771ebba5848675592d99d879; engine receipt
sha256 : 5cedeb1cb99ec83a668db0c80839b248c93e83bdeefbd067fa379fb8e45eb17 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001-5cedeb1cb99ec83a.json;
empirical-validation hash None; measurement receipt None.

191. Cubic atomic support and exact neighbour balance

Claim identity: SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Three-space supplies three axes and the binary Fold supplies the two held directions on each axis, forcing six nearest neighbours. Dividing the half-One balance equally over those complete neighbours forces weight one-twelfth per neighbour and their exact sum is the half-One.

*The unique cubic support has $2*3=6$ nearest neighbours, exact weight $1/12$ per neighbour and complete balance $6/12=1/2$.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-HALF-ONE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis cubic-support product. The declared exact boundary is: All nearest-neighbour organizations using every held direction on every forced spatial axis and an exact equal partition of the half-One balance. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash `sha256:98ba9e4afaabf76679c1d6f1011d96fca9d2bad2cf9728b0bf3bc3cdda30eef8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:3962c8446b0f2c5cc4da2bb6f72bca60722b395df7888a482d679d9dcfe83efe`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:19b698c707c9a455606734e8059b4e78d4f8095991c62ad7d34ff516beb53611`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:b0cf6ad66e665745c70a806342def3ec57c78117c33486c7415675a8a3fce98f.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:47d6285a90253f07f50b5b152957fb5d8f652f63123263ad7b805d1e763c2bab.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabedb13aa09b1e6.

Independent certificate:

sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98. Engine

external-validation hash:

sha256:ca1b84299c80439ee2c7550a2094b18db30925cc5f6674754054af51357a960d.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported cubic lattice
- no measured coordination number
- no floating or signed proof value
- no selected neighbour subset

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:afb1739770387458474abfbc467386a6746ceb7ab03ecf7a3ff9630a08b0da6; engine receipt

sha256:8fa8b8bb81ea31d715fae3c5f3ceb869b5e4d8afad6c959ff86f12ab2ea7471a at

receipts/engine/model_admitted/SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004-8fa8b8bb81ea31d7.json;

empirical-validation hash None; measurement receipt None.

192. Depth-independent exact hydrogen spectral ladder

Claim identity: SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every supplied positive finite principal count n , the binary Fold forces the exact bound support one over n squared. An emitted line is the guarded positive difference between the lower and upper supports, forcing Lyman-alpha three-quarters and Balmer-alpha five-thirty-sixths without a continuum solver.

For every positive finite n , $H(n)=1/n^2$; $H(1)-H(2)=3/4$ and $H(2)-H(3)=5/36$, with every emitted line an exact guarded positive difference.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001

- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis hydrogen-support product and the positive finite principal-count successor. The declared exact boundary is: All exact positive finite Coulomb-bound principal supports generated by the binary exponent and all ordered emitted transition gaps between them. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash `sha256:2324379b0dded9987f1c5298650eca06fe03487674562cd942e8c3aa1c7a6170`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:9ad1dcf544500f7bfee7f57d704b81e8ba36b249fe9ec348be5bcd7e91042fa8`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a503e335734dce29a3e66cd48009361f4756281238383caef59dc7d2e9b1ff2c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:10c26242805a62fe6560b570a7d524ba4713f416ee3fcfa13d43c478064220a1`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:23d2dd786a0239ae2cd67d3cf3a0805e5115eca5a39de1aaa1d0d6c96baa857c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabcd13aa09b1e6`.

Independent certificate:

sha256: 2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98. Engine

external-validation hash:

sha256: 10f3e86c883efea5fe0bb93c9b0353434dac24e8becaa30091736bfd1450a9ae.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured wavelength or Rydberg value
- no continuum Schroedinger solution
- no completed infinity
- no numerical zero or negative energy

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: a0d511dc4a594dc2724d449a85150580a244d8b41e08614e5f5009855fa5c5bb; engine receipt

sha256: ea9b3b9f34311e1f1f41962a5405ffad6250fcd74d75cf107a3b92d35dd2d464 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004-ea9b3b9f34311e1f.json;

empirical-validation hash None; measurement receipt None.

193. Exact atomic gross, fine and first bound-return hierarchy

Claim identity: SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The forced inverse fine-structure carrier supplies alpha exactly. Two held Fold directions force the relative fine scale alpha squared below gross; one complete bound-state self-return applies the same fine carrier once more, placing the first Lamb-family support at alpha to the fourth below gross.

Gross is the One scale, fine/gross is α^2 , and the first complete bound-return support is α^4 of gross; the gross Rydberg carrier in electron-rest units is $\alpha^2/2$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003
- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003

Generated grammar and closure boundary. Generate the complete eight-axis atomic-correction hierarchy product. The declared exact boundary is: The first two complete binary-direction correction layers on the admitted gross atomic support, with alpha transported once per direction and the first closed bound-state self-return applied once. The generator produced 256 named

candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash `sha256:4164bd671351dd706a0ec7b5d8cb71bd90ecd2dec356d7127759caa3a9e6246a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ef9efe1ac0aa2d75e506bdefba5a10692f86c3d7e5077a18a3c075d0c2f44264`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:ecdb546f6df8f215021e8573c3a3a547043042fb47bc7dc4855956e3bb3b495a`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1f9d9869ec2d0fd20479b04fb2d82a6bb36cf19cd71ae03a4731ca230df41595`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:05c8fc8f10c190261575325bf3dd9f0350b0ccf20a9f8c8cbdb04cf10f93d3a1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabcd13aa09b1e6`. Independent certificate: `sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98`. Engine external-validation hash: `sha256:e61d7937202716a6815f4ae472749d2bb25cb04df677e0bffbfa48b3da280581e`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported QED coefficient series
- no measured splitting
- no irrational or floating proof value
- no selected correction order

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1e410425b3bb9024c271c6dde057a448c61cdbfaf058d73fdffbfa724bb7fe382; engine receipt
sha256:a6239e61a184cf72e869fabe9c7d2dcc433366e30ef4bfdf68ebd0ff2a0fd9d at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004-a6239e61a184cf72.json;
empirical-validation hash None; measurement receipt None.

194. Atomic transition, selection and field-splitting law

Claim identity: SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. One atomic Fold act transfers one generated orbital unit while preserving the conserved carrier trace. The binary fibre supplies the two magnetic orientations and their field-dependent separation; the complete three-axis held-direction support supplies six spatial orientations for electric displacement.

An elementary admitted transition changes one orbital support unit, retains its carrier, has two magnetic orientations and six complete cubic electric orientations.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Generated grammar and closure boundary. Generate the complete eight-axis atomic-transition and field-orientation product. The declared exact boundary is: All one-act transitions between adjacent generated orbital supports with conserved carrier records, binary magnetic orientation and complete cubic electric orientation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash
sha256:8da56a20c4d39d81de9d3bf34574686fbe413b10f9f291412be2e86ee24f2b27. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:b70e71d093e3aa59669d78ef48d7f2c73928b3cbb44e1a5acdd9dfc12e3b89bc.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:c9dfb5349a68bac1833ae88ce2b8865d8a7c0588e3e799f8067e27a31b0a1762.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e5325454908fb5d544376777fedb775560e959164a880f3339a2fee087fa26f.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:ceb37351349b876c94eb5c093777a9aafc486f8686426bf96a59d0e6d2359527.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabedb13aa09b1e6. Independent certificate:

sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98. Engine external-validation hash:

sha256:50724e86eedf70a3394ffb6309ab0abb8aaa08c0d273ba23661b1f39012ea29.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured line intensity
- no imported angular-momentum algebra
- no unrecorded multi-unit elementary jump
- no signed field proof value

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:8cffe700c5ddd7de94f9813647f587fe3653a6c650a2145240f2e3d96ae7785d; engine receipt
sha256:34ea2b8ac5ee23d6cfe9e2ca37a190682c73830e53fb3b0c32ece6506e7f8de2 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004-34ea2b8ac5ee23d6.json;
empirical-validation hash None; measurement receipt None.

195. Exact molecular rotational-vibrational spectral hierarchy

Claim identity: SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic electronic carrier occupies the first binary half-support. Molecular rotation and vibration are the next complete composed Fold level at the quarter-One; two such molecular quanta exactly recompose the electronic half-support, forcing the spectral ordering without choosing a measured frequency.

Electronic support is 1/2, molecular rotational-vibrational support is 1/4, and two molecular quanta exactly recompose the electronic support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-CHEM-BOND-COVALENT-001

Generated grammar and closure boundary. Generate the complete eight-axis electronic/molecular spectral-level product. The declared exact boundary is: The first atomic electronic half-support and its next complete binary-composed rotational/vibrational support, each used exactly once in the cross-level relation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__complete-typed-support__exact-positive-Fold-operation__held-label-orientation__forced-counted-depth__target-inaccessible-formal-relation__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	untyped-observed-label	A label alone is not a generated law.

Axis	Forced form	Eliminated form	Elimination reason
support	complete-typed-support	selected-partial-support	A selected subset can tune the result.
operation	exact-positive-Fold-operation	imported-continuum-operation	An imported continuum operation violates the exact domain.
orientation	held-label-orientation	signed-or-erased-direction	Signed proof values or erased direction violate the domain.
depth	forced-counted-depth	selected-depth	A chosen depth is a parameter.
provenance	target-inaccessible-formal-relation	measurement-readable-relation	A measured target cannot construct or select the survivor.
trace	complete-root-directed-trace	result-without-dependency-trace	An untraced result cannot be admitted.
extension	no-extra-rule	free-extra-rule	An extra rule introduces a choice.

Base and successor. The closure proof is sealed by derivation hash sha256:2b5bfc6bdb7efe2e8b64e023a8a1e49071caa364e8e50a134c26315d5bc880a4. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:c56488a117fb913327e10bcc2de8a462ec19479f7c7cab13a6d86b4c43107d1c.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:a77c0590f6f02e1afb77f47d70e44b15d08f53466c73b017c9157559a9789a1b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e42410e762f0869baeb3c2205938d8ae6a31ba8faf12002ce55d9a1c6406ac9d.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:59475a84e7590e322b655736716d01b1a23b17b089f5298c52fe78a64580fb09.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b32467e7d48c61fa7f961f38a35ead41a4c7b2cd871d1d51aabedb13aa09b1e6.

Independent certificate:

sha256:2e36a22788368f121f778a56291dc5c0f9254233375e4fcaee7bad546218fd98. Engine external-validation hash:

sha256:719e52706bce11716cbc1a7d0661304d9e97d3ff837d59b31ae1f3cdc39999c7.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no selected molecular frequency
- no imported rigid-rotor or harmonic-oscillator parameter
- no fitted isotope factor
- no floating proof value

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:087830fbf4a75d48ba2a44a86ee67841a65ba408f686797d8350a7eb0ccf1d4e; engine receipt
sha256:234231a79ab030c84486fbee91cd79a5498ebb4f88c6467c366f38815ff681d0 at receipts/engine/model_admitted/SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004-234231a79ab030c8.json;
empirical-validation hash None; measurement receipt None.

196. Terminal exact hydrogen Lamb-shift successor

Claim identity: SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The immutable leading live-vacuum direction and alpha-four hierarchy are retained. The terminal n=2 Lamb frequency is the admitted Rydberg carrier multiplied by alpha cubed and the positive held support 53/64, after the down-depth/binary return 10 alpha/109, the complete colour-heavy-support return alpha squared/(35364), and the terminal alpha-cubed return/(53*64) have each acted once.

*The unique target-inaccessible ratio is $\alpha^3[53/64-(10/109)\alpha-\alpha^2/(35364)-\alpha^3/(53*64)]$.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004
- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003
- SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis terminal Lamb successor product and exhaust the typed heavy/support/bridge/return carriers. The declared exact boundary is: All first terminal alpha-cubed completions of the admitted live-vacuum n=2 Lamb direction using depth-three heavy complement, predecessor-up binary support, down-depth transport, generator support and the sole terminal return exactly once. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: retain-and-version-predecessor__complete-typed-terminal-support__ordered-positive-held-parts__finite-carrier-exhaustion-order__role-preserving-exact-composition__no-generated-carrier-remains__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-and-version-predecessor	replace-or-erase-predecessor	An immutable admitted receipt cannot be rewritten.

Axis	Forced form	Eliminated form	Elimination reason
support	complete-typed-terminal-support	selected-partial-support	A selected subset can tune a precision value.
operation	ordered-positive-held-parts	signed-floating-or-complex-series	Signed, floating, irrational or complex proof values violate the Fold domain.
order	finite-carrier-exhaustion-order	selected-or-open-order	A selected truncation or open series is a free choice.
composition	role-preserving-exact-composition	untyped-coefficient-list	An untyped list does not force a relation.
closure	no-generated-carrier-remains	unbounded-extra-correction	An unbounded correction family is not closed.
target	target-inaccessible-until-seal	measurement-readable-execution	A target-readable execution cannot issue a prediction seal.
provenance	disclosed-observational-prediction-protocol	hidden-fit-or-forward-claim	Hiding prior target knowledge or calling it blind forward forcing is invalid.
trace	complete-root-directed-trace	result-without-root-trace	An untraced value is inadmissible.
extension	no-extra-rule	free-extra-term	An extra term is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:c73ab2b2ba64c77e09ce5a0fcfaea8fb09d291c7bf73fef2f7bb1d0af2705eee`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:79c96a63e821917aed95d01bbcca4c2c12f7f7e45bddffd8f52e7e7069e1d02c`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:0d685e259ae1b1a3a4fa44c477dd2f1143eedd3d0896eb384612c75a13e91d7d`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:49be95fbf390f5d43c86405883086dc5f15d624e66a1f7f5364f9c97faf85e1e`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1a8d7bd33be27dd3a475d9a6e28e457c56ff23279c6ae9ada098392b3a6fde0b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:583121aedb9600ea571fbe58823868c8b69d465b23aec19b1b4895c95f80b266`. Independent certificate: `sha256:68da17e92bbcc2ed6199a3310a84710801221d25ccb73bf55d7ddb13ac84f9b0`. Engine external-validation hash: `sha256:9db086c354091e3230a33301c17db2b7e00fbec853218ba21e6e080bf5d4c461`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Observed comparison records:

- NIST-ATOMIC-LAMB-COMPLETE: predicted
terminal-Lamb-prediction-contained-in-complete-NIST-Bezginov-interval__observational-prediction-protocol-passed; observed
terminal-Lamb-prediction-contained-in-complete-NIST-Bezginov-interval__observational-prediction-protocol-passed; exact
match True
- deliberately tampered unfavorable control rejected

Falsification condition: The sealed lamb prediction fails its complete registered interval decision, a source or uncertainty changes, a target becomes readable before sealing, a measured value selects the formal survivor, observational provenance is hidden, or a predecessor receipt is erased.

Measurement receipt: sha256:c7a9435997ecde43fa7922c31846d62b8fc13d938b98c9ff8f033af4cef0270b.

Isolation certificate: sha256:432125c70e681cd10554ed5088b5af2da4066e706eda04da1a3ab5d0d83da0b5.

Custody certificate: sha256:7a551472ed92d6b32f37e8a014e6122c70ef4a9a64ac8a36dba004439e6aecc9.

Registered source descriptions:

- National Institute of Standards and Technology, CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- National Institute of Standards and Technology, 2022 CODATA least-squares adjustment: <https://doi.org/10.1063/5.0279860>
(sha256:4d7e7f34b98ab2fc4df68b38247f818f6fc8bdf7f25f91abdcfb329e22d2f32)
- National Institute of Standards and Technology, Atomic Spectroscopy Compendium:
<https://physics.nist.gov/Pubs/AtSpec/node03.html>
(sha256:33d048a7f7a8d83e5c2b0b96e442b495cf23b888ad1386d7205b17930af9f152)

Registered target identities:

- NIST-ATOMIC-LAMB-COMPLETE from ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured Lamb frequency or Rydberg value in execution
- no imported QED coefficient series
- no fitted coefficient or selected truncation
- no negative, irrational, imaginary or floating proof value
- no rewriting of the leading alpha-four receipt

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:3acf765129966437257ec188341de795ad376198186f00734b036a3197982aea; engine receipt

sha256:38fa8fa2ffbf3733e9ab90ca7a052699c265c5e844e896a570c8edf1ec5afe3 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005-38fa8fa2ffbf373.json;

empirical-validation hash

sha256:60554a83dedfb28a27cc08aadd2f9371c1d0bb4f301077c9a70fb341a3b78ae; measurement receipt

sha256:c7a9435997ecde43fa7922c31846d62b8fc13d938b98c9ff8f033af4cef0270b.

197. Terminal exact hydrogen fine-structure successor

Claim identity: SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The $n=2$ fine splitting begins at α squared distributed over the complete sixteen-cell terminal support. The binary-squared first return adds $\alpha/4$; the terminal-turn denominator over down support times the generator boundary holds 113α squared/288; the down-support return holds α cubed/32; and binary-through-down transport over the atomic endpoint holds 10α fourth/137.

The unique target-inaccessible ratio is $\alpha^2/16[1+\alpha/4-(113/288)\alpha^2-\alpha^3/32-(10/137)\alpha^4]$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004
- SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003
- SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis terminal $n=2$ fine-structure product and exhaust the turn, down-support, boundary, endpoint and terminal-support carriers. The declared exact boundary is: All first terminal completions of the admitted α -squared fine/gross scale on the $n=2$ sixteen-cell support, using the exact turn, down cover, generator boundary and atomic endpoint once in their typed roles. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `retain-and-version-predecessor__complete-typed-terminal-support__ordered-positive-held-parts__finite-carrier-exhaustion-order__role-preserving-exact-composition__no-generated-carrier-remains__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-and-version-predecessor	replace-or-erase-predecessor	An immutable admitted receipt cannot be rewritten.
support	complete-typed-terminal-support	selected-partial-support	A selected subset can tune a precision value.
operation	ordered-positive-held-parts	signed-floating-or-complex-series	Signed, floating, irrational or complex proof values violate the Fold domain.
order	finite-carrier-exhaustion-order	selected-or-open-order	A selected truncation or open series is a free choice.
composition	role-preserving-exact-composition	untyped-coefficient-list	An untyped list does not force a relation.
closure	no-generated-carrier-remains	unbounded-extra-correction	An unbounded correction family is not closed.
target	target-inaccessible-until-seal	measurement-readable-execution	A target-readable execution cannot issue a prediction seal.
provenance	disclosed-observational-prediction-protocol	hidden-fit-or-forward-claim	Hiding prior target knowledge or calling it blind forward forcing is invalid.
trace	complete-root-directed-trace	result-without-root-trace	An untraced value is inadmissible.
extension	no-extra-rule	free-extra-term	An extra term is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256: e114171fc3f831c5ea7dfd7dc0a7df9529851fbbacad580db4fc88be01da5e67. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256: 52114f33baac8e07ce48b362d3450a166fa9b9f2661d8d382d1317f030307cd0.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256: 1f4c4918b3bd51f788556dc51a642b4161904c00ff654b372eb08d865370b1e2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: d410f54f1e07cbaf9ee72d9f2aa92867b594ae7220072d907ac1fcbc80dc09c0.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256: cc6c5bd8fad1123f2f152f19a6666bc86a4f7e762d5e5d25efa2d7de76a04158.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 583121aedb9600ea571fbe58823868c8b69d465b23aec19b1b4895c95f80b266. Independent certificate: sha256: 68da17e92bbcc2ed6199a3310a84710801221d25ccb73bf55d7ddb13ac84f9b0. Engine external-validation hash: sha256: 1edda46cf3116ed370af483029e8029c976d4b32601b7c1737b5941e5eea9cb3.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Observed comparison records:

- NIST-ATOMIC-FINE-COMPLETE: predicted terminal-fine-prediction-contained-in-conservative-direct-NIST-experimental-interval__observational-prediction-protocol-passed; observed terminal-fine-prediction-contained-in-conservative-direct-NIST-experimental-interval__observational-prediction-protocol-passed; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The sealed fine prediction fails its complete registered interval decision, a source or uncertainty changes, a target becomes readable before sealing, a measured value selects the formal survivor, observational provenance is hidden, or a predecessor receipt is erased.

Measurement receipt: sha256: 171398126f1b83d81a1415382aa5d3bb23a4ea14236cbca34ca281a279bfd870. Isolation certificate: sha256: c10cecfdd37a8cf46485adb4dac60c77ad3d2e8f047b75ca0739b0343b838fd. Custody certificate: sha256: a69dd792ff3938130aa85bfa105ad955064ca947432ce5e505e1505fd4d8ac4a.

Registered source descriptions:

- National Institute of Standards and Technology, CODATA Task Group on Fundamental Constants: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt> (sha256: 77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- National Institute of Standards and Technology, 2022 CODATA least-squares adjustment: <https://doi.org/10.1063/5.0279860> (sha256: 4d7e7f34b98ab2fc4df68b38247f818f6fc8bdf7f25f91abdcfb329e22d2f32)

- National Institute of Standards and Technology, Atomic Spectroscopy Compendium:
<https://physics.nist.gov/Pubs/AtSpec/node03.html>
(sha256:33d048a7f7a8d83e5c2b0b96e442b495cf23b888ad1386d7205b17930af9f152)

Registered target identities:

- NIST-ATOMIC-FINE-COMPLETE from ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured fine splitting or Rydberg value in execution
- no consensus Ritz value as a premise
- no fitted coefficient or selected correction
- no negative, irrational, imaginary or floating proof value
- no rewriting of the leading fine/gross receipt

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:9b367f82939e9f357a9cc7f5ea6baf860fd501be8ad219633890eb983aefc642; engine receipt
sha256:0ec7036e95a98f142a5ea57b49652d1de66df06d7cdf6c3b82f27156c4f0d61a at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005-0ec7036e95a98f14.json;
empirical-validation hash
sha256:52e86b80d10670238d23c9ce096004afa900f9e240cfbba4e80c52ee109b4e1d; measurement receipt
sha256:171398126f1b83d81a1415382aa5d3bb23a4ea14236cbca34ca281a279bfd870.

198. Terminal exact hydrogen hyperfine and twenty-one-centimetre successor

Claim identity: SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The terminal proton/electron algebraic enclosure supplies the inverse mass carrier and exact cubic reduced-mass retention. Both spin ends over generator three force $16/3$ alpha squared. The proton magnetic projection is the finite turn held by one generator share and both alpha hands, returned by 5 alpha/16, held by 16 alpha squared/53 and 7 alpha cubed/80, and closed by alpha fourth/135.

For each exact terminal proton/electron ratio ρ , $nu_{hfs}/(R\text{-infinity}\cdot c) = (16/3)\alpha^2 \rho^{-1} [\rho/(\rho+1)]^{1/3} M$, where $M = T - 1/3 - 2\alpha + (5/16)\alpha - (16/53)\alpha^2 - (7/80)\alpha^3 + \alpha^4/135$; exact algebraic ρ isolation gives a strict rational prediction enclosure.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004
- SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001

• SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis terminal ground-state hyperfine product and exhaust the mass, reduced-mass, spin, finite-turn, down/up, support and terminal-volume carriers. The declared exact boundary is: All first terminal hyperfine completions using the admitted algebraic proton/electron enclosure, its exact reduced-mass transport, both spin hands, generator distribution, terminal finite turn, depth-three complement and terminal support exactly once. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `retain-and-version-predecessor__complete-typed-terminal-support__ordered-positive-held-parts__finite-carrier-exhaustion-order__role-preserving-exact-composition__no-generated-carrier-remains__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	<code>retain-and-version-predecessor</code>	<code>replace-or-erase-predecessor</code>	An immutable admitted receipt cannot be rewritten.
support	<code>complete-typed-terminal-support</code>	<code>selected-partial-support</code>	A selected subset can tune a precision value.
operation	<code>ordered-positive-held-parts</code>	<code>signed-floating-or-complex-series</code>	Signed, floating, irrational or complex proof values violate the Fold domain.
order	<code>finite-carrier-exhaustion-order</code>	<code>selected-or-open-order</code>	A selected truncation or open series is a free choice.
composition	<code>role-preserving-exact-composition</code>	<code>untyped-coefficient-list</code>	An untyped list does not force a relation.
closure	<code>no-generated-carrier-remains</code>	<code>unbounded-extra-correction</code>	An unbounded correction family is not closed.
target	<code>target-inaccessible-until-seal</code>	<code>measurement-readable-execution</code>	A target-readable execution cannot issue a prediction seal.
provenance	<code>disclosed-observational-prediction-protocol</code>	<code>hidden-fit-or-forward-claim</code>	Hiding prior target knowledge or calling it blind forward forcing is invalid.
trace	<code>complete-root-directed-trace</code>	<code>result-without-root-trace</code>	An untraced value is inadmissible.
extension	<code>no-extra-rule</code>	<code>free-extra-term</code>	An extra term is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:c9e3c0d7a5fe75c9caf18c7f7b8e63af09c9d01178fdb2ea01104ae4631314eb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:202da18f8c49f533508fc653fbd7cad2bd0542edaa99ff240adbe31ce40f668`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a6d64ac4f3238ccffd8cbe848a37e2aafcb54971e6474f6b5abe197e33be3f3e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3a81cc4622e0b65e6bfd33dfb8f0e48079cca52a501b97613c36f5543db23665`.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:917b749628833a2b10ab79c97c7ef007b7dd9a6c57697a248c4e079d9abed7f9.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:583121aedb9600ea571fbe58823868c8b69d465b23aec19b1b4895c95f80b266. Independent certificate: sha256:68da17e92bbcc2ed6199a3310a84710801221d25ccb73bf55d7ddb13ac84f9b0. Engine external-validation hash: sha256:c21c16651c0fe7e8bb0da19b4645a592c4ce62d3f2435e3b5e6ccadd909ef47a.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Observed comparison records:

- NIST-ATOMIC-HYPERFINE-COMPLETE: predicted terminal-hyperfine-central-prediction-inside-NIST-21cm-interval-and-complete-carrier-interval-overlaps__observational-prediction-protocol-passed; observed terminal-hyperfine-central-prediction-inside-NIST-21cm-interval-and-complete-carrier-interval-overlaps__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The sealed hyperfine prediction fails its complete registered interval decision, a source or uncertainty changes, a target becomes readable before sealing, a measured value selects the formal survivor, observational provenance is hidden, or a predecessor receipt is erased.

Measurement receipt: sha256:044625ccc092b2bdfdf2d068c7e9628301fcf2edbd428feb10f524e6e7cb8c704. Isolation certificate: sha256:4313fad0514a8915a0496d4a0aefc79791e2c5a7fd918e6a9045bf32c5256e4f. Custody certificate: sha256:a01595ad36121d159494432a4415e66b34bd1ca2497738e2c25d3918fd0eb9d7.

Registered source descriptions:

- National Institute of Standards and Technology, CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- National Institute of Standards and Technology, 2022 CODATA least-squares adjustment: <https://doi.org/10.1063/5.0279860>
(sha256:4d7e7f34b98ab2fc4df68b38247f818f6fc8bdf7f25f91abcdffc329e22d2f32)
- National Institute of Standards and Technology, Atomic Spectroscopy Compendium:
<https://physics.nist.gov/Pubs/AtSpec/node03.html>
(sha256:33d048a7f7a8d83e5c2b0b96e442b495cf23b888ad1386d7205b17930af9f152)

Registered target identities:

- NIST-ATOMIC-HYPERFINE-COMPLETE from ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured 21-cm frequency or Rydberg value in execution
- no imported proton magnetic moment
- no fitted magnetic coefficient or hidden target knowledge
- no negative, irrational, imaginary or floating proof value

- no collapse of the algebraic rho enclosure to a decimal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2b4c314c38dbb279ca5eedf2a6336f5f5c30b293806071f366a517a6390dbe7; engine receipt
 sha256:e65c383f1e1a6d48979d425f14ca83e9e906f3b60acb757e8df8a57ffa7994c1 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005-e65c383f1e1a6d48.json;
 empirical-validation hash
 sha256:caf9d4d2166e12d3ca8b0acb1a72a433e77a0b39ca6e7388f7c6d3430cd77374; measurement receipt
 sha256:044625ccc092b2bdfdf2d068c7e9628301fcf2edbd428feb10f524e6e7cb8c704.

199. Terminal exact molecular rotational, vibrational and isotope spectroscopy

Claim identity: SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The preserved Fold hierarchy separates molecular rotation, harmonic vibration and anharmonic return. Positive rotational ordinals force $E_J/B = J(J+1)$ and adjacent gap $2J$; positive vibrational ordinals force the odd half-step oscillator ladder and ordered anharmonic narrowing. The exact alpha carrier then closes distinct hydrogen rotational/vibrational and anharmonic/vibrational ratios and first-isotope rotational and squared-vibrational transports without importing a molecular measurement.

$E_J/B = J(J+1)$, $\Delta J/B = 2J$, and $G_n = \omega(2n-1)/2$ Take $\omega_x(2n-1)^{2/4}$. For exact alpha:
 $B_{H2}/\omega_{H2} = 2\alpha$ Take $14\alpha^2$ Take $58\alpha^3$ Take $82\alpha^4$; $\omega_x H2/\omega_{H2} = 4\alpha$ Take
 $30\alpha^2$ Take $59\alpha^3$ Take $63\alpha^4$; $B_{D2}/B_{H2} = 1/2 + 5\alpha^2 + 38\alpha^3$; and
 $(\omega_{D2}/\omega_{H2})^2 = 1/2 + 20\alpha^2 + 49\alpha^3$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004
- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis terminal molecular-spectroscopy product and exhaust the distinct ladder, alpha-return, isotope and provenance carriers. The declared exact boundary is: All first terminal molecular successors preserving the admitted hierarchy and adverse receipt while using the complete J, odd-oscillator, down/up, generator, heavy, terminal and isotope carriers once in their typed roles. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: retain-and-version-predecessor__distinct-generated-rotational-and-vibrational-carriers__counted-JJplusOne-and-odd-oscillator-ladders__ordered-positive-held-Takes__count-refined-linear-and-squared-transport__finite-typed-carrier-exhaustion__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-and-version-predecessor	replace-or-erase-predecessor	A versioned successor cannot rewrite an admitted receipt or delete an adverse row.
carrier	distinct-generated-rotational-and-vibrational-carriers	one-universal-quarter-carrier	One numerical carrier repeats the already rejected equality.
ladder	counted-JJplusOne-and-odd-oscillator-ladders	imported-continuum-spectrum	A continuum spectrum is outside the exact finite Fold grammar.
operation	ordered-positive-held-Takes	signed-floating-series	Signed or floating proof scalars violate the permitted domain.
isotope	count-refined-linear-and-squared-transport	free-isotope-factor	A chosen scale factor is a parameter.
order	finite-typed-carrier-exhaustion	selected-or-open-correction-order	A selected truncation or open correction list is not closed.
target	target-inaccessible-until-seal	measurement-readable-execution	A target-readable execution cannot issue a sealed empirical prediction.
provenance	disclosed-observational-prediction-protocol	concealed-development-provenance	Concealing prior observation would misstate the empirical protocol.
trace	complete-root-directed-trace	result-without-root-trace	An untraced numerical relation is inadmissible.
extension	no-extra-rule	free-extra-term	An ungenerated extra term is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:b55b897333c29b1cd00928c161f9e618e3695fb5c9552fc51a3cb8f558770958`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:f01d0ff1ae2a565038b11a1fbb8d6f8dd676ccc0b317f24beb33633680006376`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7357d35b1e9d73a17a643f7ddd74659f8520e5e38d72843eeca43b1a73a2d66b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:80ad2ac3a4b100486bfa95508dace8af040368b7f66f7e96d1bd0078a39dbfbf`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:45a74a3e4905cc6e2ef65119261f21696ab3bb97a16c357ecf8cdd50468168eb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:8fd64c9b1d709bc057502777c2459cd4e9da0fb1e6e318728db625dcb75fd0db`. Independent certificate: `sha256:103ad110876cbd3f6d10ff61c00b8e980ef8c3baf9eb4ecc01d12ab708ed6a98`. Engine external-validation hash: `sha256:ec6f6d5fe3918f8459dcb5f331bc69517ae74a5a87d5232cbb5ebeee91152dc2`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR

Observed comparison records:

- NIST-MOLECULAR-H2-D2-COMPLETE-VECTOR: predicted terminal-molecular-JJplusOne-anharmonic-and-isotope-relations-contained-in-complete-NIST-H2-D2-displayed-resolution-vector__observational-prediction-protocol-passed; observed terminal-molecular-JJplusOne-anharmonic-and-isotope-relations-contained-in-complete-NIST-H2-D2-displayed-resolution-vector__observational-prediction-protocol-passed; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if any sealed ratio leaves its complete registered exact displayed-resolution interval, the $J(J+1)$, $2J$ or positive anharmonic ladder identity fails, either isotope hierarchy or source hash changes, a target becomes readable before sealing, development provenance is hidden, or an immutable leading or adverse receipt is erased.

Measurement receipt: sha256:95c6631925a30c7777c128d2738d133b452d784e33342da6ca5026eefdc21f64.

Isolation certificate: sha256:8417690b9cdf6366cf5733c3bd493c060794da613a8d6e6865f138eafb48baa8.

Custody certificate: sha256:3f2443f08b9bfeadcce66999a39d6a5e4ccbe3f80cdc2b05de27d48e926a825b.

Registered source descriptions:

- National Institute of Standards and Technology, Chemistry WebBook SRD 69:
https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1828
(sha256:18036a188088f880122249544ceb6b384fabfba93b300b4f9f0fa01aa0ed9b24)
- National Institute of Standards and Technology, Chemistry WebBook SRD 69:
https://webbook.nist.gov/cgi/cbook.cgi?ID=C7782390&Mask=1000
(sha256:790276cd493dbede03fc9e83db95bc0808a36838eb31473aaf923493b4749936)

Registered target identities:

- NIST-MOLECULAR-H2-D2-COMPLETE-VECTOR from NIST-MOLECULAR-SPECTROSCOPY-H2-D2-SUCCESSOR

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no H2 or D2 measurement, wavenumber, source record or displayed digit in execution
- no imported rigid-rotor parameter, oscillator frequency or isotope multiplier
- no numerical-zero state, negative proof scalar, irrational root, imaginary value or floating proof value
- no concealed observational development and no erasure of the adverse predecessor receipt
- no additional correction after the typed grammar is exhausted

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4eece79bbece4a90041d6651d1d0d414aab170838dbbccc11771aeb7f014e81; engine receipt

sha256:286941c32f5e99614d53ad4ab579a565af536106d3e64e9c590cbad527fd5b2b at receipts/engine/model_admitted/SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005-286941c32f5e9961.json; empirical-validation hash

sha256:0b3469f88c3a1a7825aa8906387d6a3a3b693f62fce7242f345232046094348c; measurement receipt

sha256:95c6631925a30c7777c128d2738d133b452d784e33342da6ca5026eefdc21f64.

200. Terminal exact hydrogen reduced-mass and Rydberg-scale completion

Claim identity: SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The terminal proton/electron enclosure forces the finite-nuclear-mass retention $\rho/(\rho+1)$. The exact Fold carriers then add alpha squared over the down support and alpha cubed over the binary support, hold the complete binary-colour alpha-four return, and restore the up-square alpha-five terminal cell. Multiplication by alpha squared over two gives the hydrogen ionization/electron-rest carrier, while the immutable three-quarter and five-thirty-sixth gaps give Lyman-alpha and Balmer-alpha.

For exact terminal ρ , $H/R_{\infty} = \rho/(\rho+1) + \alpha^2/5 + \alpha^3/2$. Take $6\alpha^4 + 49\alpha^5$; $H/(m_e c^2) = \alpha^2/2$ times that carrier; Lyman-alpha/ R_{∞} is three-quarters and Balmer-alpha/ R_{∞} is five-thirty-sixths of the same carrier.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004
- SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004
- SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis terminal hydrogen Rydberg-scale product and exhaust the reduced-mass, down, binary, colour, direction, up-square and line-gap carriers. The declared exact boundary is: All first terminal finite-proton-mass completions of the admitted alpha-squared-over-two hydrogen scale, using the terminal rho enclosure and each generated alpha-return carrier exactly once before the immutable line gaps act. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `retain-and-compose-predecessors__exact-reduced-mass-from-terminal-rho__role-typed-alpha-return-chain__ordered-positive-held-Take__finite-carrier-exhaustion-order__alpha-square-times-terminal-scale__target-inaccessible-until-seal__disclosed-observational-prediction-protocol__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-and-compose-predecessors	replace-or-erase-predecessor	A successor cannot rewrite either immutable derivation.
mass	exact-reduced-mass-from-terminal-rho	imported-measured-mass-factor	A measured multiplier would become a proof parameter.
correction	role-typed-alpha-return-chain	untyped-coefficient-list	An untyped list cannot force placement or orientation.
operation	ordered-positive-held-Take	signed-floating-series	Signed or floating proof scalars violate the exact Fold domain.
order	finite-carrier-exhaustion-order	selected-or-open-order	A chosen truncation or open series is a free choice.
composition	alpha-square-times-terminal-scale	answer-only-decimal	An answer-only decimal erases alpha, mass and gap provenance.
target	target-inaccessible-until-seal	measurement-readable-execution	A target-readable execution cannot issue a sealed prediction.
provenance	disclosed-observational-prediction-protocol	concealed-development-provenance	Hidden prior observation would misstate the protocol.

Axis	Forced form	Eliminated form	Elimination reason
trace	complete-root-directed-trace	result-without-root-trace	An untraced physical ratio is inadmissible.
extension	no-extra-rule	free-extra-term	An ungenerated term is a parameter.

Base and successor. The closure proof is sealed by derivation hash
sha256:50f746356e4bf578e887c5ab902532b7bc0b24f0ba7ce80d16ab2e27d528b180. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:f82b7235a6e12a498a923c82aef74d997febeca07e40cda7b8fb571c4022ffce.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:da567131b75798adf2a0584bde41872f1a396fa7dacd0a00cc55b933bb3aae0f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e579dfe41bad3f3bd0deb223cfd8a538364698cb6e4576a8fb7c985eba144400.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:b391a351cdf19e1f414e6e7543040bdfbf869b6a32e578dcfc0f999187892c27.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:108dd4fd13bd52da50840c9a1d4f3b879fbc05e2f2acf86ed1ee6cb3ba45d1c. Independent certificate: sha256:3d4260ccd137743c362ea95b6744808114dd7e8b769a135cc6fdc87d5d638a60. Engine external-validation hash: sha256:96d7771254cc33d3b6a380d037cf0e46eec3731b75a7e4d34c290485bf5aa356.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-HYDROGEN-RYDBERG-SUCCESSOR-2022-2026

Observed comparison records:

- NIST-HYDROGEN-RYDBERG-IONIZATION-LINE-COMPLETE-VECTOR: predicted terminal-hydrogen-reduced-mass-and-radiative-scale-contained-in-NIST-ionization-Rydberg-ratio__absolute-ionization-Lyman-Balmer-vector-passed; observed terminal-hydrogen-reduced-mass-and-radiative-scale-contained-in-NIST-ionization-Rydberg-ratio__absolute-ionization-Lyman-Balmer-vector-passed; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if the sealed scale leaves the complete ionization/Rydberg interval, its translated ionization leaves the displayed-resolution enclosure, either line misses its full interval, any source hash or uncertainty changes, target access precedes sealing, development provenance is hidden, or a predecessor receipt is rewritten.

Measurement receipt: sha256:010f8374ca9c26dbfb95a38b0957ba6d857642eac243f8dd93e1551737c2a4c5. Isolation certificate: sha256:0cc1b63ba9091a7617e2dbced806cede2cdeb00a99e74f8afce6da0cde08e17a. Custody certificate: sha256:c636a414c9df2ebc9a827aac6c2d6fccaa524aa24fb006b564fb448d9411fd86.

Registered source descriptions:

- National Institute of Standards and Technology, CODATA 2022: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt> (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- National Institute of Standards and Technology, Atomic Spectra Database: <https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable1.htm> (sha256:cc54c774518d62cbb7e95f17b3b7fcd9d1faf0aa90043a607b91dff0a43e2087)
- National Institute of Standards and Technology, Atomic Physics Division: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=842564 (sha256:86ccaeb875fcd4445e727618509621ceee656e5dc02295140ecdb6e1d2dd443)

Registered target identities:

- NIST-HYDROGEN-RYDBERG-IONIZATION-LINE-COMPLETE-VECTOR from NIST-HYDROGEN-RYDBERG-SUCCESSOR-2022-2026

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured Rydberg, ionization, electron-rest energy or line value in execution
- no imported reduced-mass factor or atomic correction series
- no numerical-zero state, negative proof scalar, irrational, imaginary or floating proof value
- no concealed observational development or rewritten predecessor receipt
- no correction beyond the exhausted typed carrier grammar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:e201006611698be5cb430fd39353664a1f611c1d51702e43f05e30142bae242a; engine receipt
 sha256:e9780bb0acb2e0c4038b037dbc53203529a305f7a1cae84c5e73dfe9013cb59c at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005-e9780bb0acb2e0c4.json;
 empirical-validation hash
 sha256:8418722d6db952c3e39c94ee4614c0003b74d2419b6b85573755d9ece36a3462; measurement receipt
 sha256:010f8374ca9c26dbfb95a38b0957ba6d857642eac243f8dd93e1551737c2a4c5.

201. Terminal exact atomic shell, filling and periodicity completion

Claim identity: SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Every positive principal shell n contains all orbit ranks One through n , so the admitted capacities sum exactly to $2n$ squared. Increasing joint cover with retained principal-rank tie order generates the complete filling recurrence. The first filled cell and each later filled rank-two outer boundary force closure coordinates 2, 10, 18, 36, 54, 86 and 118 and period widths 2, 8, 8, 18, 18, 32 and 32. The physical ionization signature is a rising period envelope followed by a closure-to-successor reset, not a false claim of strict monotonicity inside every period.

For every positive n , sum from orbit rank One through n of $2(1+2(r-1))$ equals $2n^2$. The generated closures are 2,10,18,36,54,86,118 and their successive widths are 2,8,8,18,18,32,32; ionization comparison tests endpoint-above-start and endpoint-to-next-start reset while retaining internal dips.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001

- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete ten-axis atomic shell, cover-order, closure, recurrence, ionization-envelope, custody, trace and extension product. The declared exact boundary is: All positive principal shells, every valid principal/orbit cell in increasing joint-cover order, the admitted atomic endpoint, and the first complete empirical period/reset vector. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `compose-immutable-orbit-capacity__all-positive-orbit-ranks-through-n__exact-complete-capacity-sum__increasing-joint-cover-principal-tie__first-cell-then-complete-rank-two-boundary__successor-after-held-closure__rising-envelope-and-boundary-reset__target-inaccessible-until-seal__complete-root-directed-trace__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	<code>compose-immutable-orbit-capacity</code>	<code>replace-or-relabel-capacity</code>	A successor may not rewrite an immutable receipt.
shell	<code>all-positive-orbit-ranks-through-n</code>	<code>selected-width-list</code>	A width list is an imported answer.
sum	<code>exact-complete-capacity-sum</code>	<code>asserted-two-n-square</code>	Asserting the result supplies no derivation.
order	<code>increasing-joint-cover-principal-tie</code>	<code>memorized-filling-table</code>	A memorized table would select the result.
closure	<code>first-cell-then-complete-rank-two-boundary</code>	<code>chosen-period-endpoints</code>	Chosen endpoints are measured inputs.
recurrence	<code>successor-after-held-closure</code>	<code>free-reset-rule</code>	A reset rule is a parameter.
ionization	<code>rising-envelope-and-boundary-reset</code>	<code>strict-stepwise-monotonicity</code>	Strict monotonicity erases genuine within-period distinctions.
target	<code>target-inaccessible-until-seal</code>	<code>external-data-readable</code>	Target-readable execution cannot seal a prediction.
trace	<code>complete-root-directed-trace</code>	<code>answer-without-dependencies</code>	An untraced answer is inadmissible.
extension	<code>no-extra-rule</code>	<code>free-exception-or-width</code>	An ungenerated exception is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:5ffc9fd662f8f402e00134b79ef722ffb129fc043e8ed1a24e5230d0cfedcfb5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4e78c1737aa8aa63b9dc904637766a7b4256e9f491c968754ad1769305088c1a`.

- **tampered_source**: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:102a3bd8225297b93ad36f4a78620d42dbd77c34852dbf94fed73d248a85069f.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:bc0ef6e00fbca12d8562d2e45e194fe3aa092474ffbe38e74c18a9e26812ad9b.
- **boundary**: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:92918c4a35853f3faa2d01f66ac0712babe628aa62d6de100b01dbace0b2e522.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:9277b76711c49b11d9c2e7fa5f8ba4dfc8a3f8e462d4f444c1ec1bfcdf2c24a4. Independent certificate: sha256:c961db81ab6ffef4a76252c89efe3e8a0afaa9bd2c211e2e4cd5d7d6d4e188b5. Engine external-validation hash: sha256:751edd05d95eea5aaebf1c6c1ed2f7c2b3c454d2a1f5832cfc4a575e538fff3.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IUPAC-NIST-ATOMIC-SHELL-PERIODICITY-2022-2026

Observed comparison records:

- IUPAC-NIST-ATOMIC-PERIODICITY-COMPLETE-VECTOR: predicted exact-shell-sum-and-generated-period-widths-match-IUPAC__NIST-configurations-and-ionization-envelope-reset-vector-passed-with-local-dips-retained; observed exact-shell-sum-and-generated-period-widths-match-IUPAC__NIST-configurations-and-ionization-envelope-reset-vector-passed-with-local-dips-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if any generated closure or width differs, a registered NIST configuration disagrees, a period envelope or reset fails, a retained local dip is erased, any source identity changes, target access precedes sealing, or observational development is concealed.

Measurement receipt: sha256:af2aada6a6e8ee8f3d8421f1dd120407346653980179c775164423fcc4ab9458. Isolation certificate: sha256:86292cfb96b68b948bd36a39196bd7e89ec27919fc63f759d4b98709330005a8. Custody certificate: sha256:d2543ba423fc2faeb506d53da03a09e2aad0073f6ee2e39002f1ff7c1ee7d738.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/05/IUPAC_Periodic_Table-04May22.pdf (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)
- National Institute of Standards and Technology: <https://physics.nist.gov/cgi-bin/ASD/ie.pl> (sha256:3132a0088e0346af7d0ae0b88937765b4c6dfad951d79bb9677103aa6affbdbb)

Registered target identities:

- IUPAC-NIST-ATOMIC-PERIODICITY-COMPLETE-VECTOR from IUPAC-NIST-ATOMIC-SHELL-PERIODICITY-2022-2026

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported periodic table, aufbau list, electron configuration or ionization value in execution

- no conflation of shell capacities 2,8,18,32 with actual period widths 2,8,8,18,18,32,32
- no claim of strict monotonic ionization inside a period
- no numerical-zero state, negative proof scalar, irrational, imaginary or floating proof value
- no hidden observational development, free exception or rewritten predecessor receipt

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c0d999b3565a88a45f02f971b6fa0bc52d6ba903e75bb68b42f4efbc2ea87e65; engine receipt
 sha256:86b2573d6cd627c61d4816f1a90868906d5c66cc640d9a03f7d3b6e635c9406c at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005-86b2573d6cd627c6.json;
 empirical-validation hash
 sha256:4dcc83003b2c685d3968c484af3ca55526ba6ae2dddba040fbd0d243cb94f039; measurement receipt
 sha256:af2aada6a6e8ee8f3d8421f1dd120407346653980179c775164423fcc4ab9458.

202. Terminal exact atomic transition-rate, lifetime and multipole completion

Claim identity: SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The admitted one-unit elementary transition retains an exact positive level gap. Transport through all three forced spatial directions makes the E1 rate proportional to gap cubed times held line strength divided by complete upper-state weight. Complete positive decay channels force lifetime to be the reciprocal of their rate sum. Electric multipole rank L has exponent $2L+1$, so each successor is conditionally suppressed by exact gap squared at equal strength and weight. Magnetic and other typed channels retain distinct strengths; the label forbidden alone is not promoted to a false universal ordering.

Normalized E1 rate is $\text{gap}^3 S/g$; lifetime is $1/(\sum_i A_i)$; normalized electric rank-L rate is $\text{gap}^{(2L+1)} S/g$ and the equal-strength/equal-weight successor ratio is gap^2 . The elementary orbital step remains One, while non-electric typed channels require their own held strengths.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004
- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete ten-axis transition-act, gap, spatial-volume, strength, weight, lifetime, multipole, typed-channel, custody and extension product. The declared exact boundary is: All exact positive normalized electric-multipole rates generated from an admitted level gap, held strength, positive upper-state weight and every finite positive multipole rank, plus complete finite positive decay-channel sums. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: retain-one-unit-elementary-act__exact-positive-level-gap__complete-generator-three-volume__held-exact-line-strength__divide-by-complete-positive-weight__reciprocal-complete-rate-sum__append-held-boundary-pair__typed-channel-conditional-suppression__target-inaccessible-until-seal__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore

a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	retain-one-unit-elementary-act	replace-selection-law	A successor cannot rewrite the immutable one-unit receipt.
gap	exact-positive-level-gap	selected-decimal-frequency	A selected frequency would be a target-bearing parameter.
space	complete-generator-three-volume	free-rate-exponent	A chosen exponent is a parameter.
strength	held-exact-line-strength	erase-or-fit-strength	Erasure or fitting cannot distinguish lines.
weight	divide-by-complete-positive-weight	unweighted-or-fitted-factor	A hidden multiplicity changes rates freely.
lifetime	reciprocal-complete-rate-sum	independent-lifetime-parameter	A separately chosen lifetime is not forced.
multipole	append-held-boundary-pair	chosen-power-list	A list has no successor theorem.
forbidden	typed-channel-conditional-suppression	universal-slower-label	M1 shares the cubic exponent and distinct strengths can reverse an unqualified ordering.
target	target-inaccessible-until-seal	external-target-readable	Target-readable execution cannot seal a prediction.
extension	no-extra-rule	free-coefficient-or-exception	An ungenerated coefficient is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:1a020ec6fb65d3804d53098bd13da5aca41648195be86feba07dc00d1d01b665`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:226dec5ab4ab69dfcadebf543257b117191a1957bc41ea3b8b1cb0fc168834ba`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:8399e7e82be3c019772f68a0dde646adf6b3f7ed8366a2eb96028584992ffda`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8a207dc13bd9188a6a113f07e03bc0d6d92b5013ac03193bd037350cd0f05399`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:35918069bd6ff102c9048d4c82c03488f0dba11fbba7763cef032e25bb99bf2a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:fce8dcbb269092987146a5e52ddbf37c4934cb2219b0c6d34bf5185db9ec6f56`. Independent certificate: `sha256:767d9605573e4eccaed519642e3c083f8436af9f8e824c4c7df691a4b32c7aea`. Engine external-validation hash: `sha256:2f6194effc05c95f69bf13278b412b2f0b0c5dae9c1cdd4ca5a2d81a61b03c30`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026

Observed comparison records:

- NIST-ATSPEC-RATE-LIFETIME-EQUATIONS: predicted sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; observed sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; exact match True
- NIST-AR9-M1-METASTABLE-LIFETIME: predicted sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; observed sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; exact match True
- NIST-AL27-CLOCK-METASTABLE-LIFETIME: predicted sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; observed sealed-E1-gap-cube-E2-gap-five-and-reciprocal-lifetime-match-NIST-equations__hydrogenic-rate-four-cross-locks__complete-metastable-lifetime-vector-passed; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if a sealed exponent or lifetime relation differs, hydrogenic exponent four does not cross-lock, either complete metastable interval fails, M1's adverse boundary is erased, a source hash or uncertainty changes, target access precedes sealing, or observational provenance is concealed.

Measurement receipt: sha256: f5b54570f563ed73f207b7dc4648b79ad7c39394a3dc6c9a48c397c0f754c57a.

Isolation certificate: sha256: 23a588fbfef1dc9cf02fc8a903b6e2496cfae7cd73ed1eae09ffe1491137e881.

Custody certificate: sha256: 91d10dc3b06179eeb275abdee44a577fb907799eba05b5ad36952e18b10198bb.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/Pubs/AtSpec/node17.html> (sha256:35f4c76e7ed640958c6ef184cb731fc91c540608bda87bab7ab63130abb7f7a2)
- National Institute of Standards and Technology: <https://www.nist.gov/publications/lifetime-metastable-2p12-state-f-ar9-isolated-compact-penning-trap> (sha256:77bb5da60432c1dd9d5979e952254ace888073af666605ecce7c8f5fda4eb5a4)
- National Institute of Standards and Technology: <https://www.nist.gov/publications/observation-1s0-3p0-clock-transition-27a> (sha256:77bb5da60432c1dd9d5979e952254ace888073af666605ecce7c8f5fda4eb5a4)

Registered target identities:

- NIST-ATSPEC-RATE-LIFETIME-EQUATIONS from NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026
- NIST-AR9-M1-METASTABLE-LIFETIME from NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026
- NIST-AL27-CLOCK-METASTABLE-LIFETIME from NIST-ATOMIC-TRANSITION-RATE-LIFETIME-2026

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured wavelength, rate, line strength, lifetime or source access in execution
- no imported continuum radiation equation or floating proof value
- no numerical-zero state, negative rate, irrational or imaginary proof scalar
- no universal claim that every forbidden-channel rate is slower independent of strength
- no fitted exponent, coefficient, lifetime or multipole exception

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:adee356d47c06de43bbdb0f59122eba878324e5eadf306f8d4b1b9468fcf030e; engine receipt
sha256:23cb34b722250938fdbcf749addee2c0d534ba0f7c0ee54cbb68c678fd4adc5d5 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005-23cb34b722250938.json;
empirical-validation hash
sha256:502a83cb6b1548d91d9c1498d8475ae738065fd17e13065cb8dbc247e8a510f6; measurement receipt
sha256:f5b54570f563ed73f207b7dc4648b79ad7c39394a3dc6c9a48c397c0f754c57a.

203. Terminal exact atomic Zeeman and Stark field-splitting completion

Claim identity: SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Complete held angular support forces 2J+1 magnetic sublevels. Opposing directions remain labels and every displacement magnitude is positive; one weak-field act therefore forces linear Zeeman shifts gmB with equal adjacent spacing g*B. In electric response a retained degenerate partner preserves the first dipole act and forces a linear Stark shift. Without that partner the opposed first acts close to the empty form and the first retained response uses two field acts, forcing a quadratic shift.

A J level has 2J+1 magnetic classes; Zeeman magnitude is gmB with adjacent spacing gB; retained degeneracy gives dipoleE while nondegenerate closure gives polarizability*E^2/2.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004
- SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005
- SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete ten-axis field-predecessor, multiplicity, held-direction, magnetic response, spacing, degenerate/nondegenerate electric response, order, custody and extension product. The declared exact boundary is: Every finite positive angular support, exact positive field magnitude, held side/orientation label and the two complete electric support classes distinguished by whether a degenerate partner is retained. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: compose-immutable-field-predecessors__complete-held-orientation-successor__held-side-label-and-positive-distance__one-field-act-per-held-orientation__one-common-positive-field-step__held-first-order-dipole-channel__paired-first-act-closes-then-square__degeneracy-typed-response-order__target-inaccessible-until-seal__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	compose-immutable-field-predecessors	replace-field-predecessors	A successor cannot rewrite admitted receipts.
multiplicity	complete-held-orientation-successor	selected-line-count	A chosen multiplet width imports the answer.

Axis	Forced form	Eliminated form	Elimination reason
direction	held-side-label-and-positive-distance	signed-energy-scalar	Negative proof scalars violate the domain.
magnetic	one-field-act-per-held-orientation	chosen-field-polynomial	A chosen response polynomial is a parameter.
spacing	one-common-positive-field-step	unequal-free-spacing	Free spacings destroy the generated orientation order.
degenerate	held-first-order-dipole-channel	erase-first-displacement	Erasure loses the held cross-state channel.
nondegenerate	paired-first-act-loses-then-square	free-first-order-shift	A first-order scalar without a retained partner is ungenerated.
order	degeneracy-typed-response-order	universal-single-order	One order erases the degeneracy discriminator.
target	target-inaccessible-until-seal	external-target-readable	Target-readable execution cannot seal a prediction.
extension	no-extra-rule	free-coefficient-or-rule	An ungenerated response is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:da777cc9aa9a3ac0034aadaaa4e0253c75921e2de62d11274874492fe0fdeb25`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:a5c77c740fa2357d0e8e4fb7d8c0454e1a125f10d4ade55e7719ba3cf31a37ba`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4a585aad8d67b2153cd331d6bfdaf88d4d062aaa0cb4ac2430b1481133958eab`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0ab10ce95a892f62966c9a951b6530881d217dd8e7654a185db62d410d33a6e5`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:8a0996c55d448f033a7b0ead29c8cda092982015d0839183f3b0e9e32a001b6b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:01a7cf130a10dc80276c9a07f338f4a493eba8be6c0c02fa14368c07bf367963`. Independent certificate: `sha256:7beefcdf3c0679968c67d8ab49fe6087dc1641ef92910b06edfd98690bef1737`. Engine external-validation hash: `sha256:8678fbe4c1a39a3d40b589030b3a32a0c149a89ed611fbe6ff7850c5a6c4345f`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-NBS-ATOMIC-FIELD-SPLITTING-2026

Observed comparison records:

- NIST-WEAK-FIELD-ZEEMAN: predicted sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; observed sealed-two-J-plus-one-and-linear-Zeema

n-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; exact match True

- NBS-HYDROGEN-LINEAR-STARK: predicted sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; observed sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; exact match True
- NIST-CESIUM-QUADRATIC-STARK: predicted sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; observed sealed-two-J-plus-one-and-linear-Zeeman-match-NIST__degenerate-hydrogen-linear-Stark-matches-NBS__nondegenerate-cesium-quadratic-Stark-matches-NIST; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if multiplicity or a field order differs, the degeneracy discriminator fails, the exact voltage/spacing translation fails, a displayed value or uncertainty status changes, target access precedes sealing, or observational provenance is concealed.

Measurement receipt: sha256:069d2196117fed492a2df1380b8962e6930327e9ac3010e4ec693b3b2cf8ecdb.

Isolation certificate: sha256:1e1f5ef9d1e9955aa0bab3e03855bae87cf0e8438ebf6ef2221945ee42194f04.

Custody certificate: sha256:b8300def67c60c116e46f7f10b4dbc36e26adb2c278b5004bf2990f5ea34ae84.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/Pubs/AtSpec/node12.html> (sha256:5093c824cec5438a98f82fd5e75460e737a001a11471d12dbd7cd67b930661df)
- National Bureau of Standards: <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nbsspecialpublication617.pdf> (sha256:5093c824cec5438a98f82fd5e75460e737a001a11471d12dbd7cd67b930661df)
- National Institute of Standards and Technology: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=933361 (sha256:5093c824cec5438a98f82fd5e75460e737a001a11471d12dbd7cd67b930661df)

Registered target identities:

- NIST-WEAK-FIELD-ZEEMAN from NIST-NBS-ATOMIC-FIELD-SPLITTING-2026
- NBS-HYDROGEN-LINEAR-STARK from NIST-NBS-ATOMIC-FIELD-SPLITTING-2026
- NIST-CESIUM-QUADRATIC-STARK from NIST-NBS-ATOMIC-FIELD-SPLITTING-2026

Meaning. Atomic and molecular laws are reconstructed from exact Fold support, transition, shell, splitting, rate and spectral relations, including terminal precision successors rather than leaving the measured-value work in Chemistry. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured Zeeman coefficient, Stark coefficient, splitting or source access in execution
- no numerical-zero central state or negative signed proof scalar
- no irrational, imaginary or floating proof value
- no universal Stark order that erases the degeneracy discriminator
- no fitted g factor, dipole, polarizability or response coefficient

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7ff3926a45c8a68799c705b25c9c23675a4a189b628ef503064345dcb8f115ed; engine receipt

sha256:91df8eb2a38be812ebd00525e22ed77945498d2aa03c52c3776eccc2d34f3a71 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005-91df8eb2a38be812.json; empirical-validation hash

sha256:1789c354cc9ac36c1cc069a91950e1ea3bcba203462a7447daa632dd018834ea; measurement receipt
sha256:069d2196117fed492a2df1380b8962e6930327e9ac3010e4ec693b3b2cf8ecdb.

20. Nuclear Hadronic

Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries.

204. Nuclear binding and mass relation

Claim identity: SFT-PHYS-NUCLEAR-BINDING-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Nuclear binding is the closed joint-word recurrence whose separated constituent inertial support exceeds the bound-word inertial support by exactly the energy released into named carriers during composition.

Nuclear binding pairs a positive released energy carrier with the exact reduction of separated constituent inertial support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-MATTER-MIXING-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__composition-energy-inertial-support-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	composition-energy-inertial-support-pairing	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2be5949b1f34c8caf6ffecc99ee36229ff77bf51c669050b08f3c25282728b4f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fd81a904c6c352b69b65f42e7480ce913ebcfba32e8ecf6697542de0587ecfe5`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7319c9d7e19e03ea80eda886ea4656be56455b7390a319a4ce06e0f54790b57c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:195fca22a5ce2ebd1b964d027360a7ce9191601530fb05ea2c1b697f7ff04d37`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f357cf87333e729a5c24cb2024a070e081048c02ff807adc26935fa8e3e998df`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:13230a01de73f6c8dc99c99cc990cfe58483ed3d9c24013610fc2d941f3b834a`. Independent certificate:

`sha256:ab40eb1ab93ae74c3d7c9804d11fdf239d29147c5926a59521295a02f0545202`. Engine external-validation hash:

`sha256:f48f7d60b5964811413e74536897fc7e7553aac2fa6f8bcc1fc23d279f5207a8`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-binding-1: predicted nuclear-binding-mass-relation; observed nuclear-binding-mass-relation; exact match `True`
- nuclear-binding-2: predicted nuclear-binding-mass-relation; observed nuclear-binding-mass-relation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:d7cbe20b60e87671320f3a1d90549b701b9a751b5a00c01ad1bacf84a19ac624`.

Isolation certificate: `sha256:9bad8064d730e0acaa879175458f47f30b30e9a8b1485b58ddea5705666cf26d`.

Custody certificate: `sha256:7cceb30b648541c1045c1c6100a4bdc77259e601d2794df5dfadbba0a48d7ec8`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: <https://nds.iaea.org/ensdf/> (sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-binding-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-binding-2 from IAEA-ENSDF-CURRENT-2026-07-23

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2db6a24ffdb2047e02bea855ac44381848b991bd43fbb9e3ad460b5675c53552; engine receipt
sha256:8ebc11747ec4bfd0552de732d26bbb60bcbbeb6ba273885ebec783ffd43dbb3f at
receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-BINDING-001-8ebc11747ec4bfd0.json;
empirical-validation hash

sha256:7f9bc2dd84794e9d4709ca05dc1e74efe1515768885a50211800d6daab707073; measurement receipt
sha256:d7cbe20b60e87671320f3a1d90549b701b9a751b5a00c01ad1bacf84a19ac624.

205. Nuclear levels and transitions

Claim identity: SFT-PHYS-NUCLEAR-LEVELS-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Nuclear levels are the exact boundary-closed recurrence classes of a composite nuclear word; transitions are conserved carrier differences between named classes.

Nuclear levels are discrete recurrence classes of bound composite Fold support with source-bound transitions.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-NUCLEAR-BINDING-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__composite-boundary-recurrence-classes__source-bound-proof-trace__capability-closed-prediction__sepa`

rate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	composite-boundary-recurrence-classes	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:76f9d0a0965161c735fd68ccc9c32b70f95af9a608797eed5a7337de433bc2b3. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:327a71cc8de73d2a1c78a174dc6970825c5522d443715a42c4395d6b1d8af7e7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:653c74bcd47a95a1311142e94053c696505967b25fd90664363116fbecdc26e4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:300df7b3eb317cfdfeb054c30f8bf07707fb63158e6be614b57d9e13805e0c2e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:ebfadd76b8db50a577d6da9f56e084e6260e6f00893d29aa7afe61517b56a497.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:57005aa5b4ea43540c6c5c2b6c90a9d03f7ccabdf65f58e8b37041d3847fa204. Independent certificate:

sha256:bba43a62fb1fccfdb86973be39bb8b5fca71241a3ca036d137ac74bf796514b2. Engine external-validation hash:

sha256:0fb69d726eb3ce0606c271f0b73ed48ed240477c14bea40d08d4f2058d398d08.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-levels-1: predicted nuclear-level-recurrence; observed nuclear-level-recurrence; exact match True
- nuclear-levels-2: predicted nuclear-level-recurrence; observed nuclear-level-recurrence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:6a1a8e1613eb6cac74c66843c4553f3e5416c30fb09ab78eb6eb000b90ad0961.

Isolation certificate: sha256:12e041dddac9d4f3dc13bdd464cf9365bc517bb906960253ecf9ec780a9f1d30.

Custody certificate: sha256:5ff4164d03023279c6566e0888ace4022d1402e23245a54186231f3ab68fec70.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: <https://nds.iaea.org/ensdf/> (sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-levels-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-levels-2 from IAEA-ENSDF-CURRENT-2026-07-23

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:37409d7dc276088da25835592fb355d85cbd857d5586f3161e147360d3e5c1b1; engine receipt

sha256:8361256cd14917c077f36a6250d7706d6b97eaf4254633721107775112a94479 at

receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-LEVELS-001-8361256cd14917c0.json;

empirical-validation hash

sha256:cba148bdbd2aedc9548d29c9b027d0467962567fab73bdbeaa016c59037b48fa; measurement receipt

sha256:6a1a8e1613eb6cac74c66843c4553f3e5416c30fb09ab78eb6eb000b90ad0961.

206. Radioactive transition support

Claim identity: SFT-PHYS-NUCLEAR-RADIOACTIVITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Radioactivity is a nuclear recurrence with lawful output channels but no stable self-return on the observed support; exact event weights arise from complete deterministic hidden-path census.

Radioactivity is deterministic complete-path transition from a nuclear word lacking stable observed recurrence.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-NUCLEAR-LEVELS-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__nonreturning-nuclear-transition-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	nonreturning-nuclear-transition-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3baaa11050ac054e4bc27061f6acafb7c8b1d20ae9a06ab2044e93c38fa43780`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7e9ccd542cf7e94950e6e4aa1e519f1337c9368433868c2ca3703d920a9562e4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:83f0826fdd6d0e6db144c50f30bdbcc9be7c25e0aa72c829226abc11ca3e6314`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 404a845a88efd25dbe794b73e078814ad4636aef85114ecb1a00b642a80963f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : eaa65ad66b7acc1ad1ded6aa28a4ca7d3f4bda1c0265fe92bc6cadcf21bfad7.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 8bc09f9171ba803b1633192af125e6695a777ea38bf18ad5d9da6015d11f0924.
Independent certificate:
sha256 : c8c4cf90ccf96270f8e98d74e35afb66cdb835387dfa783b2f8778240884f741. Engine
external-validation hash:
sha256 : 149cb62c5efcf8332acc63666d7cb9b7bb365f75008af5ffab5480b974373286.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENSDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-radioactivity-1: predicted radioactive-transition-support; observed radioactive-transition-support; exact match **True**
- nuclear-radioactivity-2: predicted radioactive-transition-support; observed radioactive-transition-support; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256 : 2fb40b4979572faebf3f41b84ffe8676f1e7659c4d9a3524d72f8800d851e569.
Isolation certificate: sha256 : 589b8da919a2ee61eeac0d08972ca0ff258eeef6a398dc2066a6d0252bb374e8.
Custody certificate: sha256 : fb23907b939fece460b56cf9a2907665b35c092380d6275aa6c3d44a69646668.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: <https://nds.iaea.org/ensdf/>
(sha256:c17d03b30d117a48b776fd143a2e4224bd4f53bc774d394c1343645329a0ce93)

Registered target identities:

- nuclear-radioactivity-1 from IAEA-ENSDF-CURRENT-2026-07-23
- nuclear-radioactivity-2 from IAEA-ENSDF-CURRENT-2026-07-23

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 273f3a543b43ca1522707cc521dc8ca7b294b36cffd379a51963940262ca28f0; engine receipt

sha256:abfd4ba82c2ecf7fba7473fa029c2b3c06ad54d536c4fef8a7b354fbd587576 at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-RADIOACTIVITY-001-abfd4ba82c2ecf7f.json;
empirical-validation hash
sha256:2eb2570415ea91beb3310bbccedcaeb644075e14b7a193c71a9ebf7c57c9724c; measurement receipt
sha256:2fb40b4979572faebf3f41b84ffe8676f1e7659c4d9a3524d72f8800d851e569.

207. Nuclear reaction and channel accounting

Claim identity: SFT-PHYS-NUCLEAR-REACTIONS-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A nuclear reaction is a complete map between incident and outgoing composite words in which every material, energy, momentum and held conserved label is paired across the boundary.

Nuclear reactions are exact closed maps across complete generated incident and product channels.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-NUCLEAR-RADIOACTIVITY-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__complete-nuclear-channel-accounting__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	complete-nuclear-channel-accounting	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:c4ca1115e7810dd0426286893e7508475510d5861221f6eb392b3dbfe9d0dace. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:65887188ea7c159786c06b3c69f0295b3bb31780ee1b2a068f1b91429f2cd077.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:51f7f0eac3a58bed76c33fa2c48b1790fd7134b7f80ccdb63877b6c421e655bd.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:3201fac045e03496efadcc814789850c6a1dec7f2e47278b0ebfaef3ed7dec1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:6bf41deed8cdec0fd0fde987f16271b96d3490911393a5366d923a4f6e52828a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:5dae5b17cd7967171984f76fe3a5fc7931906d98c4d43ef699ca0bdc2213f099. Independent certificate: sha256:0222736e5d94d5fd9354aac097ea8eb3337b66bf92a6858f908a9e7918252573. Engine external-validation hash: sha256:4ce0e55b5061edd580437fa059d871c66d9c40475dc0f2d9633b0244767fbe6a.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-reactions-1: predicted nuclear-reaction-channel-closure; observed nuclear-reaction-channel-closure; exact match `True`
- nuclear-reactions-2: predicted nuclear-reaction-channel-closure; observed nuclear-reaction-channel-closure; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:90df60f2143b04166e63a04b6073725d94456b6a4f9bbf2b614efdd72a911793. Isolation certificate: sha256:8f9cd6cfb1475a8d1976125e63a314024123bfe2366d61e368899143faf36c1f. Custody certificate: sha256:94a227c1a68856fbce05c9c808c686c569510cc4a6bdefd2f2d356d49eafb027.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: <https://www-nds.iaea.org/endl/> (sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccb9)

Registered target identities:

- nuclear-reactions-1 from IAEA-ENDF-CURRENT-2026-07-23

- nuclear-reactions-2 from IAEA-ENDF-CURRENT-2026-07-23

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:39b1e4d4ec2f64e601b1c6724c2838f65797441cb1df32d4406b2c37bac7b8ac; engine receipt

sha256:d619676a96d0ef76e876a31388e3ead4e581cf513277f87928f75ac490b62127 at

receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-REACTIONS-001-d619676a96d0ef76.json;

empirical-validation hash

sha256:0d3686a4a22399ad626810c76085b19c7b8ac64edc30664bdece602a19e3b55c; measurement receipt

sha256:90df60f2143b04166e63a04b6073725d94456b6a4f9bbf2b614efdd72a911793.

208. Fission decomposition

Claim identity: SFT-PHYS-NUCLEAR-FISSION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Fission is a nuclear transition whose output support contains multiple separately recurrent composite words and named released carriers, with all labels conserved across the decomposition.

Fission is conserved decomposition of one bound nuclear Fold word into multiple recurrent product words and released carriers.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-NUCLEAR-REACTIONS-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__bound-word-multiple-product-decomposition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore

a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carr ier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	bound-word-multipl e-product-decompos ition	imported-or-fitted-relati on	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof -trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-predictio n	A target-readable predictor can copy or optimize against the measurement.
record	separate-measureme nt-record	proof-measurement-conflat ion	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registere d-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-clos ure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:2fe7f5f41c727546bdfe8ccaa4acaca5e3fb753d05a8d8888fb8643a34a7ecb8. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:8945ba3b665b979784ca31d3ea3de9e420d5ec2ca8c234a5719fbb9bf9e64de5.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b69d6af60d92b523ec3246adfb4b1039b12979d2e4b73f1128080af26d5d2008.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:446277cc78578caf8243313372fa8da0542b513e37fca900760b935dde0d3165.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:1a1d0d29304c8b346c5eea4f464658c45618a20cb1cc5135b55cecd87bb657c.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:ac12c4c5196542894a2ab1002ece360d2c5ad971332c46e66f77d4be44587307. Independent certificate: sha256:c378af85ed65ad3e447c5497f76e416380c3d70bfc24eeb663135f589afa458. Engine external-validation hash: sha256:1b6ab3af2e44aba9eeb8cb9a2103eed55ca96ec0dfc141befdf42faf90ad602b.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-fission-1: predicted nuclear-fission-decomposition; observed nuclear-fission-decomposition; exact match True
- nuclear-fission-2: predicted nuclear-fission-decomposition; observed nuclear-fission-decomposition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:08869f2c7c3b92aebb67f0c94c0e50f8271c407f7b088280ae658630c7b2533b.
Isolation certificate: sha256:28d6f24280e39b7e18069a8103b9291716c1b6a2d12b096c3d47bb7a67d92a05.
Custody certificate: sha256:3e771ca4008a532dadf4c32962e122088da4d99bb2a0331fa0e1112ddbe23dbb.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: <https://www-nds.iaea.org/endl/>
(sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccbd9)

Registered target identities:

- nuclear-fission-1 from IAEA-ENDF-CURRENT-2026-07-23
- nuclear-fission-2 from IAEA-ENDF-CURRENT-2026-07-23

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:de201d487f30f1538d3129d5586fc1357329f6bb4b7cc60e1495711119d9246f; engine receipt
sha256:37bc1d64923a33e5d6548f4701606b8566eb51f9d6871e6e7815a289926a3ceb at
receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FISSION-001-37bc1d64923a33e5.json;
empirical-validation hash

sha256:3c4f86c17f05f95b240fa9ef4de61ca95114514b336de96606dc54275857ecba; measurement receipt
sha256:08869f2c7c3b92aebb67f0c94c0e50f8271c407f7b088280ae658630c7b2533b.

209. Fusion composition

Claim identity: SFT-PHYS-NUCLEAR-FUSION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Fusion is a transition from multiple incident nuclear words to one more tightly boundary-closed recurrent product word plus the exact positive carriers released by the mass-energy relation.

Fusion is conserved composition of multiple nuclear Fold words into a bound recurrent product and exact released carriers.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001

- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-NUCLEAR-FISSION-001

Generated grammar and closure boundary. Generate the complete constituent, composition/exchange, conserved-label, transition-channel, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite material forms generated from exact physical Fold words, held orientations, closed transfer, phase recurrence and complete observation support without imported particle taxonomy or fitted coupling. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__multiple-word-bound-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	multiple-word-bound-composition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4614d77c25311196cc1ebc5f0ce5f47897195b3c0a060fcf755c59ba113efd96`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:dcf009d6b2b76f5c957ff2982be5c3854b417e1d327c14c98dfb98c4b66ee982`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1a789c350ea0afb3aee5c9f53654677f9a627b2f0c989b3a34eee08369dca1d0`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:de889777d2509a339723f1f13e24af5fb765b008fb431795f7f6e06dc10b9161`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:8f1439a35ff8a600cf9fde87e3351fedad3eacc9488c343d570f0fc1d2654c16.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:290be909c8c1da1aeb994b4ed9541ada13c006702379e79ed7c326f7032ce4dc.
Independent certificate:
sha256:aaa7e043eec1826489229193189f1c25aad22db59b2635e678317e3c9a58f54e. Engine
external-validation hash:
sha256:d6379dacb6640551cccc3dcc2e0e5b0cdf128b5c7bf0258a2ba5c560f68fcde8.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-ENDF-CURRENT-2026-07-23

Observed comparison records:

- nuclear-fusion-1: predicted nuclear-fusion-composition; observed nuclear-fusion-composition; exact match True
- nuclear-fusion-2: predicted nuclear-fusion-composition; observed nuclear-fusion-composition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The material law is falsified if a committed external row lacks the sealed constituent/composition/conservation structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:230a7c0b1b00ce89a6ae991f6ea31d37886fadf99dfe45272a755e2951f1e273.
Isolation certificate: sha256:c1c5003ad034af463d8b70bdbbc9b006d5080f77742f1232c852b1cf2b77b7658.
Custody certificate: sha256:154c0d4014c7ab3ce2562efd98ea5aab06ca34f1a96e177f59f0efe86486b681.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: <https://www-nds.iaea.org/endl/>
(sha256:16b8187ef7ef9024914aa7b54b6af07928a2f1d3bb97a6999f7563fa676ccbd9)

Registered target identities:

- nuclear-fusion-1 from IAEA-ENDF-CURRENT-2026-07-23
- nuclear-fusion-2 from IAEA-ENDF-CURRENT-2026-07-23

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum or completed infinite species set
- floating, irrational, imaginary or signed proof magnitude
- imported particle table, fitted coupling or target-selected species
- target access, omitted decay/reaction channel or unrecorded constituent

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:9f4db1843099be37632c5ceb2509b9c49a8b486d75fe5aa23f44b63887db43e2; engine receipt
sha256:9277671be548ed512279ff99a5347ed91531056cecddc1f22ed96da39a0a7676 at
receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-001-9277671be548ed51.json;
empirical-validation hash
sha256:849afd978cbb0ca37c0a51c1143457fa6acdc09d55356b8edf5691d61334d4a4; measurement receipt
sha256:230a7c0b1b00ce89a6ae991f6ea31d37886fadf99dfe45272a755e2951f1e273.

210. Exact colour-sector Fold coupling

Claim identity: SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete two-label Fold distinction transported over the generator-three carrier forces the exact positive coupling ratio two-thirds, with no measured or fitted interaction strength.

The exact generator-three colour-sector Fold coupling is 2/3.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate every product of carrier, numerator, denominator, orientation, normalization, target custody and extension forms. The declared exact boundary is: All exact normalized ratios formed by transporting the complete Fold-label support across one generator-three interaction carrier. The generator produced 128 named candidates and the decision support contains 128 one-for-one decisions. Exactly one candidate survived: `generator-three-carrier__both-Fold-labels__complete-generator-support__labels-held-through-carrier__two-over-three__closures-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generator-three-carrier	free-colour-count	A free carrier count is a parameter.
numerator	both-Fold-labels	selected-single-label	One label omits half the Fold distinction.
denominator	complete-generator-support	external-normalization	External normalization is fitted.
orientation	labels-held-through-carrier	labels-collapsed	Collapse loses the interaction distinction.
normalization	two-over-three	reciprocal-or-unscaled-count	Those forms reverse roles or omit normalization.
target	closures-inaccessible-until-seal	magic-list-visible	A closure list would fit the coupling.
extension	no-extra-rule	extra-coupling-rule	An added strength is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:4893061724f8e857a5efe97692521d1df83c013f9723fe61d4804c3bae4baa77`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:e4e7687a9a1d832e556174d015e0aa64f11f64f92e06734a6051d3b987b676db.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:e2edad0e6d45799bbf3bb96bb8c4e89f8be5a715fc0d921e462d57b1d7d2e8f0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:12e6496b82b65e70e5458cdc2c1f49daecc9625414c9a58da9cda056fea5e5c6.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:5947e11fa098dddaf9eb6ce47ae72cd6b0492d99a75eeb6452299129b69e0996.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:babedcbfa7852074ccccf0885740154b359515c4b90ca34e584e8544bc59f31b.

Independent certificate:

sha256:8a8b86ffa63d4555d21aaab69918f4d60af798ad0b31d7e9c2014eb38779646e. Engine

external-validation hash:

sha256:c9a0a0d4ee3a15a784312da4b2c7bfff456b6a17ddc1b6073c1dc58244065e846.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:55e3c26bda6cce369b8f47cf4b459b445cabed93342517e8e9ee93424b0ca57f; engine receipt

sha256:d38dc32f0926ae5f06e432f6bd7546d8680a8ea648c211dfbe965fc44e8179e5 at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-COLOUR-COUPPLING-001-d38dc32f0926ae5f.json;

empirical-validation hash None; measurement receipt None.

211. Complete Fold nuclear-closure sequence

Claim identity: SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Complete weak compositions across three spatial directions with two exclusion labels force the base filled-shell counts. The independently sealed two-thirds colour coupling reaches its first whole spin-orbit gap at angular count three and forces the reordered closure sequence 2, 8, 20, 28, 50, 82, 126, 184 with a depth-independent successor law.

The depth-independent nuclear closure law gives the exact prefix 2, 8, 20, 28, 50, 82, 126, 184; its next proton closure after 82 is 126 and the next closure after 126 is 184.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-PHYS-NUCLEAR-REACTIONS-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of spatial composition, label multiplicity, base closure, coupling source, shift, threshold, reordering, recurrence, target custody and extension forms. The declared exact boundary is: All filled-shell closure schedules built from complete weak compositions over forced three-space, both exclusion labels and the first whole-gap action of the independently forced colour coupling. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `complete-three-direction-weak-compositions__two-exclusion-distinct-labels__positive-rank-composition-form__sealed-two-thirds-colour-coupling__top-angular-half-label-pair__least-positive-whole-gap__complete-top-orbit-capacity__piecewise-positive-successor-law__sequence-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
composition	complete-three-direction-weak-compositions	selected-shell-degeneracy	A selected degeneracy imports a model.
labels	two-exclusion-distinct-labels	single-or-duplicated-label	Omission or duplication violates Fold support and exclusion.
base	positive-rank-composition-form	closure-list	A list has no generative proof.
coupling	sealed-two-thirds-colour-coupling	fitted-spin-orbit-strength	A fitted strength reads the closures.
shift	top-angular-half-label-pair	free-shell-shift	A free shift is a parameter.
threshold	least-positive-whole-gap	selected-rank-threshold	A selected threshold fits the sequence.
reorder	complete-top-orbit-capacity	arbitrary-intruder-count	An arbitrary intruder count is fitted.
recurrence	piecewise-positive-successor-law	finite-eight-value-table	A table does not close later ranks.
target	sequence-inaccessible-until-seal	magic-numbers-visible	That is target leakage.
extension	no-extra-rule	extra-nuclear-rule	An added selector is a parameter.

Base and successor. The closure proof is sealed by derivation hash

sha256:f063dc762cb725c66340c946953cd008664875456944811ad902315d6b79b69d. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:5c771cf5592a53cc672843d7cdadcfbe7e340db992858decb1ab49eb0d695e8b.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:6e2dc9948609f6a9bcffed8b9e06971fa9e84b851c60221af6c0726a7a4bf601.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8069e2016e09a7983c9c51f36313fbec85fe529516e27d9f31877bb68d776e78.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:9df3facdadb8b6591dcebf69efd2b75b8b300056734782cf444b7e08cdb38fc6.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:0c39e11b369cf6d388ac93d2942d49a037aa20f3cb2550601df04805eee0dcaa.

Independent certificate:

sha256:988e6fd342d2c09cabea9a075e383ac3d673f1e74279b820e4d79c2bbb3a37ea. Engine

external-validation hash:

sha256:4a0b588d1c3b8b4ea798c27237d953cdee1dcb0918ef3b436678c0aa798fccfa.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:239be83e36611d89d86421f4dc8e517f20288187660251a1313146246bfe5133; engine receipt

sha256:35f7fe50e217d2a61383f4169c314f1d4d5b4671a1def5baa011d26a1d89af9f at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001-35f7fe50e217d2a6.json`;

empirical-validation hash None; measurement receipt None.

212. Terminal nucleon composition, binding dominance and mass ordering

Claim identity: SFT-PHYS-NUCLEON-BINDING-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The generated period-three colour cycle closes as $1/7+2/7+4/7=One$. Complete positive u/d count enumeration then uniquely gives the unit-charged word uud and empty-neutral word udd. At the admitted upper-quark depth seven, one of 128 cells is the explicit valence share and the positive predecessor 127/128 is the held-cycle share. Exact light-quark root enclosures place down above up by more than the admitted terminal proton electromagnetic dressing, so replacing one up by one down forces the neutron above the proton.

The proton word is uud and the neutron word is udd; the exact structural mass ledger is bare 1/128 and held-cycle 127/128, hence bare below 1/100 and held above 99/100; the exact down-minus-up enclosure exceeds the proton electromagnetic dressing and forces neutron mass above proton mass.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPPLING-001
- SFT-PHYS-MATTER-QUARK-CUBICS-003
- SFT-PHYS-MATTER-CONFINEMENT-LIFT-003
- SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete twelve-axis predecessor, colour, composition, depth, ledger, dominance, flavour, electromagnetic, ordering, custody, provenance and extension product. The declared exact boundary is: All minimal positive u/d count splits over the complete generated colour-three cycle, complete depth-seven binary support, the admitted light-quark algebraic enclosures and the admitted terminal proton electromagnetic dressing. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `compose-immutable-predecessors__complete-period-three-One-cycle__enumerate-complete-three-flavour-charge-words__admitted-upper-quark-depth-seven__one-cell-and-complete-positive-predecessor__bare-below-one-percent-held-above-ninety-nine__exact-light-root-enclosure-order__compare-admitted-terminal-proton-dressing__one-down-replacement-forces-neutron-heavier__target-inaccessible-until-seal__registered-observational-prediction-protocol__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	compose-immutable-predecessors	rewrite-predecessor-laws	A successor cannot alter admitted receipts.
colour	complete-period-three-One-cycle	listed-three-colour-names	Names do not prove a closed orbit.
composition	enumerate-complete-three-flavour-charge-words	import-proton-neutron-labels	Imported particle labels would select the result.
depth	admitted-upper-quark-depth-seven	selected-binding-depth	A chosen depth can tune a mass share.
ledger	one-cell-and-complete-positive-predecessor	free-binding-percentage	A percentage read from observation is a parameter.

Axis	Forced form	Eliminated form	Elimination reason
binding	bare-below-one-per-cent-held-above-ninety-nine	qualitative-most-without-bound	An unbounded slogan is not falsifiable.
flavour	exact-light-root-enclosure-order	borrowed-quark-mass-order	Borrowing a measured ordering reverses the derivation.
electromagnetic	compare-admitted-terminal-proton-dressing	ignore-or-fit-electromagnetic-effect	Ignoring or fitting the competing carrier cannot close the sign.
ordering	one-down-replacement-forces-neutron-heavier	assert-measured-mass-order	An asserted ordering has no Fold certificate.
target	target-inaccessible-until-seal	external-target-readable	Target access cannot seal a prediction.
provenance	registered-observational-prediction-protocol	conceal-observational-development	Concealment misstates the derivation history.
extension	no-extra-rule	free-mass-term-or-percent-age	An ungenerated term is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:4508e335fbab4cd7f80721d0d407cb88a1060748dbf2e5f911a46826a02a3a19`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:54063610f9d68deac3d5bedebe77094f529e4c1797a39007600ee7ff74366834`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:f726e25682e5a2c5d4e6d66241b58d9ebe7d9a1fc97f541ad69fb788dc30f0ce`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:62b1a594364f0ae5025d07323672a17bf9e216d471708087ad46956473d0e28a`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:8afc24456299ab469c4baf3fa778a81b6eb71b94762f7a2872b38455570697f2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a2fd2e24c2d9788f10dfab09cc2acc2b26cdb1612a23f11a278a74a27d1444c6`. Independent certificate: `sha256:b85a840ca3397b64ecb53beac739fbb11ab7cab78b1801ef32447aed79ae1ee0`. Engine external-validation hash: `sha256:f6d63fd149b170330b8751cfde4235978017e34d6969dbdf3c232043adf7dad3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-NIST-NUCLEON-BINDING-2022-2026

Observed comparison records:

- PDG-THREE-COLOUR-QQQ-SINGLET: predicted sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; observed sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; exact match True
- PDG-LIGHT-QUARK-MASS-DOMINANCE: predicted sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; observed sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; exact match True
- NIST-NUCLEON-MASS-ORDERING: predicted sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; observed sealed-three-colour-uud-udd-singlet-correspondence-passes-PDG__structural-one-over-128-and-PDG-uud-current-share-both-below-one-percent__nonidentity-of-those-two-objects-retained__held-residuals-above-99-percent__down-up-and-neutron-proton-ordering-pass-complete-PDG-NIST-intervals; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if qqg colour closure differs, either complete dominance inequality fails, structural and scheme-dependent quantities are conflated, d/u or neutron/proton intervals cease to order, the direct mass difference leaves propagated intervals, a source/uncertainty changes, or target access precedes sealing.

Measurement receipt: sha256:743b67038fcd9c2914a510027363b31ca9c9250e9375c4c950937dc8ada4c8bc.

Isolation certificate: sha256:97e152d0459b1a527d2ffdddea91917c28fa59b18959e22cf6c66fbb9dc2c516a.

Custody certificate: sha256:d8bea46ec5003c2dc68ad65b5d8bda251baebecebd76ac976fffb3ed34e2067e5.

Registered source descriptions:

- Particle Data Group: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf>
(sha256:2f75c5124ce570fad3becd1ff5901f6062c6dea05b5a41a1892819006fa6290e)
- Particle Data Group: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf>
(sha256:d544d099aa15739ec83d87711bd4c5b1e0a1032d6f70aaa4847ec78601f7aeae)
- NIST / CODATA Task Group on Fundamental Constants: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- PDG-THREE-COLOUR-QQQ-SINGLET from PDG-NIST-NUCLEON-BINDING-2022-2026
- PDG-LIGHT-QUARK-MASS-DOMINANCE from PDG-NIST-NUCLEON-BINDING-2022-2026
- NIST-NUCLEON-MASS-ORDERING from PDG-NIST-NUCLEON-BINDING-2022-2026

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer value or measured mass as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted binding percentage, constituent mass, electromagnetic correction or nucleon scale
- no claim that a scheme-dependent PDG current-quark mass ratio is identical to the structural 1/128 cell
- no target access before the derivation and prediction seals

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:b3f0673c9be151a4b910fd51c93b8fca3a23d5884f858a538d25eaf7076e889; engine receipt
sha256:deedcfb769c2cf51e5ea249bd3e319ee7ec95c8feb883210312badabfd8332ab at receipts/engine/model_admitted/SFT-PHYS-NUCLEON-BINDING-TERMINAL-005-deedcfb769c2cf51.json;
empirical-validation hash
sha256:dc934885b849a805997db2c7431dec71100bf8246f6c9dd99abdc034248b8592; measurement receipt
sha256:743b67038fcd9c2914a510027363b31ca9c9250e9375c4c950937dc8ada4c8bc.

213. Terminal light-hadron multiplets and depth-independent Regge support

Claim identity: SFT-PHYS-HADRON-REGGE-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Generator-three flavour support forces nine ordered flavour/antiflavour cells, partitioned as one invariant cell class and its eight-class positive predecessor. Complete three-place flavour support forces twenty-seven cells; direct exchange enumeration partitions them as ten symmetric, two eight-cell mixed hands and one antisymmetric class. Independently, one retained fixed-tube act per positive spin successor forces exact affine squared support. The exact theorem governs normalized fixed-carrier support; whether physical resonance masses realize literal equal spacing is a separate post-seal measurement.

Light mesons have exact flavour support $9=8+1$; light three-place baryons have exact support $27=10+8+8+1$; normalized fixed-tube squared support is $Q(J)=J$ and every positive successor difference is exactly One.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001
- SFT-PHYS-MATTER-CONFINEMENT-LIFT-003
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete ten-axis predecessor, flavour, meson, partition, baryon, exchange, tube, affine-support, custody and extension product. The declared exact boundary is: Every ordered word over the complete generated three-member light-flavour support at widths two and three, every permutation-exchange class, and every positive rank produced by repeated application of one held fixed-tube successor act. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: compose-immutable-hadron-predecessors__complete-generator-three-flavour-support__enumerate-three-by-three-ordered-pairs__one-invariant-and-positive-predecessor__enumerate-three-place-flavour-words__enumerate-exchange-symmetry-classes__one-retained-step-per-spin-successor__depth-independent-affine-successor__target-inaccessible-until-seal__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	compose-immutable-hadron-predecessors	rewrite-hadron-predecessors	A successor cannot alter admitted receipts.

Axis	Forced form	Eliminated form	Elimination reason
flavour	complete-generator-three-flavour-support	import-named-flavour-table	Names do not generate a support.
meson	enumerate-three-by-three-ordered-pairs	assert-nonet	A conventional name does not prove its size.
partition	one-invariant-and-positive-predecessor	import-octet-singlet-representation	An imported representation would select the answer.
baryon	enumerate-three-place-flavour-words	assert-octet-and-decuplet	Named multiplets are not a derivation.
exchange	enumerate-exchange-symmetry-classes	import-SU-three-decomposition	A conventional group decomposition cannot be a premise.
tube	one-retained-step-per-spin-successor	chosen-spin-dependent-increment	A different free increment at each rank destroys structural closure.
Regge	depth-independent-affine-successor	fitted-mass-polynomial	A fitted polynomial is not forced.
target	target-inaccessible-until-seal	external-target-readable	Target access cannot seal a prediction.
extension	no-extra-rule	free-residual-or-tension-fit	An ungenerated correction would be a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:36d2f757779c14eda73684e49779c9f46f235f7a66b525805a2d1313d6b22daa`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:a51239286297ee3bbf8b2985502ac279a2d0028270b705005f1328f1837f935d`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:792f08f5e1624f6c8026ac2fb851d08fe34493dac3f85501c655ddfabdcf537e`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ce1e1df74076e998d8540199220566bda0b71667a0f23a7b0648f9a952ab3bbd`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:29a2e04e67f50893cc95aaf395d84a178a79f4bd01e8616165309005cc110d5e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:99065993d853f987480ba1f1dd34c0811d501a9607b9002c7e8a03496acb261f`. Independent certificate: `sha256:7e65311b23611ed4c357b35b21a1abeed3e0d6895139fe5ec355147f22c73113`. Engine external-validation hash: `sha256:b3a82a613b045dea1f1fdeee5551787bb3547b591c6cc1464a27ea0588ba2286`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-HADRON-MULTIPLETS-REGGE-2025

Observed comparison records:

- PDG-LIGHT-MESON-MULTIPLY: predicted sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalized-fixed-carrier-boundary__rho5-summary-omission-status-retained; observed sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalized-fixed-carrier-boundary__rho5-summary-omission-status-retained; exact match True
- PDG-LIGHT-BARYON-MULTIPLY: predicted sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalized-fixed-carrier-boundary__rho5-summary-omission-status-retained; observed sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalized-fixed-carrier-boundary__rho5-summary-omission-status-retained; exact match True
- PDG-NATURAL-PARITY-TRAJECTORY: predicted sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalized-fixed-carrier-boundary__rho5-summary-omission-status-retained; observed sealed-nine-equals-eight-plus-one-and-twenty-seven-equals-ten-plus-eight-plus-eight-plus-one-match-PDG__complete-five-state-natural-parity-mass-vector-is-strictly-ordered__literal-common-squared-mass-step-is-rejected-by-complete-reported-intervals__exact-affine-Fold-law-retained-at-normalized-fixed-carrier-boundary__rho5-summary-omission-status-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if either sealed partition differs, the normalized affine successor ceases to be exact, any registered mass/uncertainty/status row is lost, strict ordering changes, the exact common-step result is misstated, a fitted correction is introduced, or target access precedes sealing.

Measurement receipt: sha256:0f3e190970773ede50717b7bfff036ae1d691887b1d49d3bb134c66a92321a9e9.

Isolation certificate: sha256:71851e59c98bbb8872eef615cc058f1b28a3f53ff88330f062fd78638f47284a.

Custody certificate: sha256:28f8a1b5805a9b35f09fa7cc4aa46d87359325016d47f8ec4d064df2086427ea.

Registered source descriptions:

- Particle Data Group: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf>
(sha256:c6e671212984fa6b6956a3d748c12c81879686b9198ca5a0818670fe63301358)
- Particle Data Group: <https://pdg.lbl.gov/2025/tables/rpp2025-sum-mesons.pdf>
(sha256:c6e671212984fa6b6956a3d748c12c81879686b9198ca5a0818670fe63301358)
- Particle Data Group: <https://pdg.lbl.gov/2025/listings/rpp2025-list-rho5-2350.pdf>
(sha256:c6e671212984fa6b6956a3d748c12c81879686b9198ca5a0818670fe63301358)

Registered target identities:

- PDG-LIGHT-MESON-MULTIPLY from PDG-HADRON-MULTIPLY-REGGE-2025
- PDG-LIGHT-BARYON-MULTIPLY from PDG-HADRON-MULTIPLY-REGGE-2025
- PDG-NATURAL-PARITY-TRAJECTORY from PDG-HADRON-MULTIPLY-REGGE-2025

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer value or conventional SU(n) representation as a premise
- no measured resonance mass, uncertainty, trajectory slope or source access in execution

- no numerical-zero state, negative, irrational, imaginary or floating proof value
- no fitted tube tension, intercept, residual, correction term or selected trajectory subset
- no claim that non-minimal colour-neutral composites are forbidden
- no target access before derivation and prediction seals

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c8b82a773604b1b447731f03b1fcbdb2c366569c09e7c65f442c7bf88c8ec227; engine receipt
 sha256:64caf5f62f3314597b9c525b694005ac4bc81f57446ff7e2dd436bd817646933 at receipts/engine/model_admitted/SFT-PHYS-HADRON-REGGE-TERMINAL-005-64caf5f62f331459.json;
 empirical-validation hash
 sha256:da1c36bd036f5c8ab81467f12ee9086b3ef35d493e1960a19a609887ee36806a; measurement receipt
 sha256:0f3e190970773ede50717b7bff036ae1d691887b1d49d3bb134c66a92321a9e9.

214. Terminal residual nuclear interaction and inverse-mediator range

Claim identity: SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A complete colour-three composite closes every leading colour label inside its boundary, so the external leading act is the empty form. The first nonempty exchange must retain one boundary act from each of two neutral composites; the forced half-One therefore enters twice and fixes the second-order structural support at one-quarter. This is an exact Fold order, not a universal dimensional cross section. A present mediator mass is a positive carrier and forces the finite reciprocal range One/m; an absent mass label has no finite subtraction endpoint. Thus lighter massive mediators reach farther, while finite boundary cells and finite range force locality and saturation without an imported exponential or hard numerical-zero cutoff.

Leading colour exchange outside a singlet is the empty form; the first retained two-boundary order is $(1/2)^2=1/4$ and returns $1/4$ to $1/2$ to One in two Folds; every positive mediator has finite normalized range $1/m$ and $m_1 < m_2$ forces $1/m_1 > 1/m_2$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-HALF-ONE-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-MATTER-CONFINEMENT-LIFT-003
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-HADRON-REGGE-TERMINAL-005
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete ten-axis predecessor, neutral closure, residual order, strength scope, mass class, reciprocal range, ordering, saturation, custody and extension product. The declared exact boundary is: Every exchange between two complete colour-three neutral composite boundaries, the forced half-One primary carrier, absent or exact positive mediator-mass labels, and every positive mass ordering under reciprocal range. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `compose-immutable-neutral-composite-predecessors__leading-colour-act-closes-to-empty-form__paired-half-One-boundary-act__structural-order-not-universal-dimensional-strength__empty-mass-label-or-positive-mass-carrier__exact-reciprocal-mass-scale__lighter-positive-mass-has-greater-reciprocal__finite-boundary-cells-with-finite-range__target-inaccessible-until-seal`

__no-extra-rule. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	compose-immutable-neutral-composite-predecessors	rewrite-nucleon-predecessors	A successor cannot alter admitted receipts.
neutral	leading-colour-act-closes-to-empty-form	export-raw-colour-label	A raw colour label would violate the admitted singlet boundary.
residual	paired-half-One-boundary-act	reuse-primary-half-One	A first-order carrier ignores both neutral closures.
strength	structural-order-not-universal-dimensional-strength	identify-quarter-with-all-cross-sections	Measured channels need not share one dimensional strength.
mass	empty-mass-label-or-positive-mass-carrier	numerical-zero-mass	Numerical zero is outside the proof domain.
range	exact-reciprocal-mass-scale	chosen-decay-length	A selected length is a parameter.
ordering	lighter-positive-mass-has-greater-reciprocal	import-particle-range-ranking	A named particle table cannot prove the order.
saturation	finite-boundary-cells-with-finite-range	all-to-all-independent-links	Unlimited independent links ignore finite boundary support and finite range.
target	target-inaccessible-until-seal	external-target-readable	Target access cannot seal a prediction.
extension	no-extra-rule	free-decay-profile-or-coupling	An ungenerated profile or coefficient is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:ee8c38f20a6b686540b9e97193be1750ca64e48a24ccdbc38de2217c9baa4ec8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:34099156fc06f273a51f09f7da7504d4bd59a4eebe2319e5b955a7f71fd57238`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c536cc85949216dab482a7c2500ddd85ba2311b6a3c958047544ed7ba5ffd6de`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:14fd7b1b50d0e825f3c03fc2beddd9d0826b31ddc790aa5b78197be3b1b0b322`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:f3bfca3b8afbef80625227c4b545a628b8d00e68e0d6b07b958364f0d1e2828a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4555d9828b920fb140c8fc9160cb9766d156d878865c635d4243704ffbe94a`. Independent certificate:

sha256:a5e981f99ef37dcc7a01729bee70b109b49ec1e697bdd8050f145aea02f578d2. Engine

external-validation hash:

sha256:e0acd3bad57bcc5456249a6f2f795290349ff557de0f498036eb2dfbf8a676e8.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026

Observed comparison records:

- PDG-MEDIATOR-MASS-RANGE: predicted sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; observed sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; exact match `True`
- NIST-HBAR-C-RANGE-TRANSLATION: predicted sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; observed sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; exact match `True`
- NIST-NEUTRON-SCATTERING: predicted sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; observed sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; exact match `True`
- NIST-PION-LONG-RANGE-CONTEXT: predicted sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; observed sealed-neutral-leading-closure-and-quarter-second-order-residual__positive-NIST-neutron-hydrogen-and-deuterium-scattering-support-nonempty-external-interaction__disjoint-channel-cross-sections-reject-universal-quarter-strength-identification__complete-PDG-pion-rho-omega-mass-order-forces-pion-longest-reciprocal-range__NIST-context-retains-pion-long-range-component-and-range-scale-not-hard-cutoff; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if neutral leading closure or quarter order changes, reciprocal ordering fails, the complete pion and vector ranges overlap, either scattering interval ceases to be positive, channel dependence is erased, one-quarter is relabelled as a measured strength, or target access precedes sealing.

Measurement receipt: sha256:f06863514328b5975eb742c2d40e4ba4f52b977cd3f579815b9abaf4dc706df1.

Isolation certificate: sha256:d6c342b3ca6d43be21d09a8b724dce5b2deea84489c1c13b55234e9f26d639fe.

Custody certificate: sha256:22a4a37d59615a57cd0bc69db2c32e21694530edc880d827cd25a1d0a820f7e7.

Registered source descriptions:

- Particle Data Group: <https://pdg.lbl.gov/2025/tables/rpp2025-sum-mesons.pdf>
(sha256:898f2c17eaadb3419aa5ee9be4519b21eeef5523045111bb0a04017d18a11d1)
- National Institute of Standards and Technology / CODATA: <https://physics.nist.gov/cuu/pdf/all.pdf>
(sha256:898f2c17eaadb3419aa5ee9be4519b21eeef5523045111bb0a04017d18a11d1)
- NIST Center for Neutron Research: https://www.ncnr.nist.gov/resources/activation/scattering_table.html
(sha256:898f2c17eaadb3419aa5ee9be4519b21eeef5523045111bb0a04017d18a11d1)
- National Institute of Standards and Technology: <https://nvlpubs.nist.gov/nistpubs/jres/110/3/j110-3pag.pdf>
(sha256:898f2c17eaadb3419aa5ee9be4519b21eeef5523045111bb0a04017d18a11d1)

Registered target identities:

- PDG-MEDIATOR-MASS-RANGE from PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026
- NIST-HBAR-C-RANGE-TRANSLATION from PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026
- NIST-NEUTRON-SCATTERING from PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026
- NIST-PION-LONG-RANGE-CONTEXT from PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer value, Yukawa potential or consensus interaction model as a premise
- no particle name, measured mass, cross section, length, uncertainty or source access in execution
- no numerical-zero state, negative, irrational, imaginary or floating proof value
- no identification of structural one-quarter with a universal dimensional nuclear-force strength
- no fitted coupling, exponential decay profile, hard cutoff or selected mediator scale
- no target access before derivation and prediction seals

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:31f1fb80d5db676704229a846f5e8dfc22ead05a4697e97e730cfdd56de0003d; engine receipt
sha256:05a3a23d0f0fbea4618128f8cfd75440505666931efcda6f9e89fa2dc37fd22e at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005-05a3a23d0f0fbea4.json;
empirical-validation hash
sha256:b8601aa2e2cdf29a1b6deddbcf3b32ac99fe85536db82631d97ee20f25c7090; measurement receipt
sha256:f06863514328b5975eb742c2d40e4ba4f52b977cd3f579815b9abaf4dc706df1.

215. Terminal zero-parameter nuclear binding curve and stability maximum

Claim identity: SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Short-range quarter-order residual recurrence and exact electromagnetic opposition force one positive bulk/surface/Coulomb/asymmetry/pairing ledger with no fitted coefficient. The surface share is exactly One over the binary-plus-generator interface carrier five; Coulomb loss is the admitted exact alpha over every ordered distinct proton path; asymmetry and pairing carry the admitted quarter-order. Exact rational radius enclosures, complete concave integer/parity enumeration below the forced mass $2^{(5+7)}$, and a two-case tail induction uniquely maximize normalized binding at mass count 62, charge count 28 and neutron count 34. No nuclide name or measured binding value enters execution.

The unique zero-parameter normalized binding maximum over every positive proton/neutron composition is $A=62$, $Z=28$, $N=34$. The exact ledger is bulk One; radial surface loss $(1/5)/A^{(1/3)}$ represented only by rational enclosures;

Coulomb loss $\alpha Z(Z-1)/A^{(4/3)}$; quarter-order asymmetry loss $(1/4)*(N-Z)^2/A^2$; and quarter-order parity gain/loss $(1/4)/A^{(4/3)}$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete ten-axis predecessor, bulk, surface, Coulomb, asymmetry, pairing, radius, maximum, target-custody and extension product. The declared exact boundary is: Every positive proton/neutron composite, exact saturated bulk, forced 1/5 interface share, exact alpha ordered charge path, quarter-order imbalance/pairing class, rational rank-two/rank-three enclosure and every mass beyond the forced 4096 tail boundary. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `compose-immutable-predecessors__one-saturating-bulk-support__one-over-binary-plus-generator-interface__exact-alpha-times-ordered-charge-paths__quarter-order-unmatched-label-square__quarter-order-paired-gain-unpaired-loss__rational-enclosures-until-order-separates__complete-concave-census-plus-forced-tail-induction__target-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	<code>compose-immutable-predecessors</code>	<code>rewrite-predecessor-laws</code>	A successor cannot alter admitted receipts.
bulk	<code>one-saturating-bulk-support</code>	<code>unbounded-pairwise-bulk-growth</code>	Short-range saturation forbids growth with every distant constituent.
surface	<code>one-over-binary-plus-generator-interface</code>	<code>fitted-surface-coefficient</code>	A fitted coefficient is a parameter.
coulomb	<code>exact-alpha-times-ordered-charge-paths</code>	<code>import-semi-empirical-coulomb-term</code>	An imported mass formula cannot select the law.
asymmetry	<code>quarter-order-unmatched-label-square</code>	<code>chosen-asymmetry-penalty</code>	A chosen coefficient is not forced.
pairing	<code>quarter-order-paired-gain-unpaired-loss</code>	<code>fitted-pairing-term</code>	A fitted parity correction is a parameter.
radius	<code>rational-enclosures-until-order-separates</code>	<code>evaluate-irrational-radii</code>	Irrational values are outside the exact proof domain.
maximum	<code>complete-concave-census-plus-forced-tail-induction</code>	<code>selected-isotope-window</code>	A chosen neighborhood cannot prove a global maximum.

Axis	Forced form	Eliminated form	Elimination reason
target	target-inaccessible-until-seal	external-target-readable	Target access would fit the isotope coordinate.
extension	no-extra-rule	free-coefficient-or-residual	An ungenerated correction destroys zero-parameter closure.

Base and successor. The closure proof is sealed by derivation hash

sha256:92bbc440b513a0b6a735f8fb84734ae90b49ba302a3e57a8ff0f9afbcecd4a51. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:aa5bac9837a7818ed70ba9ca95fa92717770e5029da870ed653a7c8f483cf173.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:1d88222acb46af18d14b0cb9c13022ee6d500a6454fd8b6639c38e0590b8eb90.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cdf01f744269da954e96e5160deadcb7d7a46e6b4f481fbe58a74825b0e37a62.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:a02ce3ddca54d64008d3aa110b8826d6ea8f659fc72c768e70774eb375ea5646.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:e39db94459ddc98b0c3e5814f690c31c7fe6f1b712bf53bd925067c384a0f74d.

Independent certificate:

sha256:15f531e8c5394402b340056decd1679d7d6b65be5e837b6766b4b29daf33b97d. Engine

external-validation hash:

sha256:209da4fec36c3294a70435456379bde2a65ed0f8115ee3e833da1eb351b702fe.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- AMDC-AME2020-MASS-1-BINDING-2021

Observed comparison records:

- AME2020-COMPLETE-BINDING-CENSUS: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match `True`
- AME2020-GLOBAL-MAXIMUM: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-

- an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True
- AME2020-LIGHT-TO-IRON-NICKEL-RISE: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True
 - AME2020-HEAVY-FALL: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True
 - AME2020-IRON-ONLY-ADVERSE: predicted sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; observed sealed-zero-parameter-binding-ledger-uniquely-predicts-A62-Z28-N34__complete-AME2020-2548-positive-composite-row-census-identifies-Ni62-global-binding-per-nucleon-maximum__Ni62-lower-uncertainty-bound-exceeds-every-rival-upper-bound__light-anchor-rise-and-heavy-anchor-fall-close-an-interior-iron-nickel-maximum__unqualified-iron-only-maximum-replaced-by-exact-nickel62-coordinate; exact match True
 - deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if the complete measured table has another maximum, the predicted coordinate differs, uncertainty intervals erase the separation, either curve direction fails, any numeric rival is omitted, or target access precedes the formal seal.

Measurement receipt: sha256:3330a4b8df04fbeb9c633deb13ee4bf75167d025ba048136c1ff2ac2c862c99b.

Isolation certificate: sha256:b8de1becff6272d4993371092d969171be793006ef558d4915d3688845650741.

Custody certificate: sha256:7478c5475900063a93d4276381019c27a250a5409343c7ffef802615ec52ab6d.

Registered source descriptions:

- Atomic Mass Data Center, Institute of Modern Physics, Chinese Academy of Sciences:
https://amdc.impcas.ac.cn/masstables/Ame2020/mass_1.mas20
 (sha256:d10a9474253b2d29fac71ee21352039659f2e5f4b7d75395416d361878a18ddc)

Registered target identities:

- AME2020-COMPLETE-BINDING-CENSUS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-GLOBAL-MAXIMUM from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-LIGHT-TO-IRON-NICKEL-RISE from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-HEAVY-FALL from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-IRON-ONLY-ADVERSE from AMDC-AME2020-MASS-1-BINDING-2021

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, semi-empirical mass formula or fitted coefficient as a premise
- no nuclide name, measured mass, binding energy, uncertainty, table row or source access in execution

- no numerical-zero state, negative, irrational, imaginary or floating proof value
- no evaluated cube root; exact rational enclosures refine only until order separates
- no selected isotope neighborhood, finite scan standing in for the unbounded tail or hidden parity class
- no target access before derivation and prediction seals
- no free bulk, surface, Coulomb, asymmetry, pairing or residual correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:686d8168c02407836389bf72041c38ac4c3f6fe5ebbe6413ff291f1e3d6aa542; engine receipt
 sha256:0f08011755d14b3c910034436a4560016125b467a28b35b01b6b6fe02364024d at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005-0f08011755d14b3c.json;
 empirical-validation hash
 sha256:2bb993c08bfae9dc917121d46bc3defd7689fd9c50922614c044e287d7769308; measurement receipt
 sha256:3330a4b8df04fbeb9c633deb13ee4bf75167d025ba048136c1ff2ac2c862c99b.

216. Terminal radioactive transition topology and exact half-life survival

Claim identity: SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Complete nuclear transition structure forces exactly three primitive nonidentity topologies: boundary release/decomposition, held charge-label conversion at retained constituent support, and internal level de-excitation at retained support and charge. Alpha, beta and gamma are the canonical representatives, not an exclusion of proton, neutron, cluster, capture, conversion, fission or delayed composite channels. Complete deterministic predecessor paths split into the two Fold fibres at each half-life carrier, forcing the exact positive survival part $1/2^k$ for every positive finite k without an ontic random choice, continuum exponential, fitted decay constant or numerical-zero endpoint.

Exactly three primitive radioactive transition topologies exist. Alpha realizes mass-four/charge-two boundary release, beta realizes retained-support held-label conversion with lepton records, and gamma realizes internal level lowering with a radiative record. Every other named decay is one primitive or an ordered composition. Survival after positive half-life count k is exactly $1/2^k$ and remains positive at every finite k .

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NUCLEAR-RADIOACTIVITY-001
- SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-PHYS-QUANTUM-TUNNELLING-001
- SFT-PHYS-WEAK-PARITY-FIBRE-002
- SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001

Generated grammar and closure boundary. Generate the complete ten-axis predecessor, topology, alpha, beta, gamma, composition, survival, determinism, target-custody and extension product. The declared exact boundary is: Every finite nuclear nonidentity transition distinguished by constituent-support release, held charge-label conversion and recurrence lowering; every ordered composition of those primitives; every positive finite half-life count and positive transported time carrier. The generator

produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `compose-immutable-predecessors__three-structural-transition-topologies__binary-square-cluster-boundary-release__held-label-conversion-with-lepton-records__internal-level-lowering-with-radiative-record__compose-the-three-primitive-traces__positive-rational-binary-geometric__deterministic-hidden-path-partition__target-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	<code>compose-immutable-predecessors</code>	<code>rewrite-predecessor-laws</code>	A successor cannot alter admitted receipts.
topology	<code>three-structural-transition-topologies</code>	<code>three-named-particles-only</code>	Nuclear data contain additional emissions, capture, fission and composite delayed channels.
alpha	<code>binary-square-cluster-boundary-release</code>	<code>borrowed-alpha-name</code>	A name alone has no constituent trace.
beta	<code>held-label-conversion-with-lepton-records</code>	<code>stochastic-particle-appearance</code>	Unrecorded random appearance violates complete conservation.
gamma	<code>internal-level-lowering-with-radiative-record</code>	<code>continuum-energy-leak</code>	An unrecorded energy leak violates recurrence and conservation.
composition	<code>compose-the-three-primitive-traces</code>	<code>declare-each-a-new-primitive</code>	Adding a primitive for every code destroys exhaustive topology.
survival	<code>positive-rational-binary-geometric</code>	<code>import-continuum-exponential</code>	A continuum exponential and transcendental value are outside the proof domain.
determinism	<code>deterministic-hidden-path-partition</code>	<code>unrecorded-random-choice</code>	An unrecorded choice violates the superdeterministic complete-path census.
target	<code>target-inaccessible-until-seal</code>	<code>external-target-readable</code>	Target access could select a topology or rate.
extension	<code>no-extra-rule</code>	<code>free-rate-or-extra-primitive</code>	A selected rate or topology is a parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:4802ac50dd4d14b7c8cac7ea4c1a37eee8dbcaa9a146833ae0e90757249a3221`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:d7c6bc41c703c74f311ce61e4dd3b0eef07613062a28000af97833ef858fa3f2`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:02519f10ba73e8b4bf83b02f50856c0df91988192d7518e04978894e91d3a173`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9f2d5c82fceb4b3b77d93a6350ab1d6cdd8c1f8720a00007a56cff68408b31d2`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:ff691f49cf2dfb372f2b19144882b53ab03086c3097daceb6d9a20d1b146d627`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:521ea0c5560600d89eb379569a950a4c97aca02867b01edbd08ba98480317419.

Independent certificate:

sha256:363f509d49d688652a8a6eed058c1b81f11486302efe43b0e64bc87fee6fbcfd. Engine

external-validation hash:

sha256:d8928f7a90ca122f82dc52989b8a875608885bf59434da13c1cbd54e65e21ec3.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- AMDC-NUBASE2020-DECAY-HALFLIFE-2021

Observed comparison records:

- NUBASE2020-COMPLETE-DECAY-CENSUS: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-COMPLETE-HALFLIFE-CENSUS: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-RELEASE-REPRESENTATIVES: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-CONVERSION-REPRESENTATIVES: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- NUBASE2020-DEEXCITATION-REPRESENTATIVE: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True

-named-code-reading-is-rejected; exact match True

- NUBASE2020-ONLY-THREE-ADVERSE: predicted sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; observed sealed-three-primitive-radioactive-topologies-and-exact-survival-1-over-2-to-k__complete-NUBASE2020-5843-state-5500-decay-row-8718-entry-50-code-census__all-50-codes-map-to-release-conversion-deexcitation-or-ordered-composition__4700-positive-numeric-half-life-carriers-transport-without-selecting-the-law__alpha-beta-gamma-retained-as-representatives-while-literal-only-three-named-code-reading-is-rejected; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The claim fails if any NUBASE2020 decay code cannot be expressed by the three primitive topologies or their composition, a positive half-life cannot transport exact finite survival, a row or uncertainty is omitted, the literal only-three-code claim is retained, or target access precedes sealing.

Measurement receipt: sha256:16638308ef6939c5af8ad447616bb50bd0a385b4939423526b95286e53e55a73.

Isolation certificate: sha256:5af309ad76f383c4b672e1006e6c4d9f64c58a7598e56796fa6cf35ac004974f.

Custody certificate: sha256:284b1145f5b36c7723f1bd0ce5f5de0f056395663cb84086d08fa26e9cac7730.

Registered source descriptions:

- Atomic Mass Data Center, Institute of Modern Physics, Chinese Academy of Sciences:
https://amdc.impcas.ac.cn/masstables/Ame2020/nubase_4.mas20
 (sha256:9a2275f1c4f82f0bdf3d41b25a1c8af535febf7451ea28bad4f6cb2beec32391)

Registered target identities:

- NUBASE2020-COMPLETE-DECAY-CENSUS from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-COMPLETE-HALFLIFE-CENSUS from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-RELEASE-REPRESENTATIVES from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-CONVERSION-REPRESENTATIVES from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-DEEXCITATION-REPRESENTATIVE from AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- NUBASE2020-ONLY-THREE-ADVERSE from AMDC-NUBASE2020-DECAY-HALFLIFE-2021

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, decay table, isotope name, measured lifetime or intended mode code as a premise
- no numerical-zero state, negative, irrational, imaginary, transcendental or floating proof value
- no continuum exponential, ontic random choice or fitted decay constant
- no literal claim that alpha, beta and gamma are the only named nuclear decay codes
- no unrecorded particle, energy, lepton, fragment, capture or delayed channel
- no target access before derivation and prediction seals

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:3ccac8e6fb82183f7bfb2f2666b76a23f1c03f74403bd33e7b17ae8a0de4ddbe; engine receipt

sha256:83e9d821fef3a3c8d4717545ca5109a1a8e7849c433b55ce5c47a17f4ac845f4 at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005-83e9d821fef3a3c8.json; empirical-validation hash

sha256:bdf6efb3d44f289cfd9c57212f7b2e033af0e8c45bc45d5cca81f483f964cfe1; measurement receipt
sha256:16638308ef6939c5af8ad447616bb50bd0a385b4939423526b95286e53e55a73.

217. Terminal fusion/fission direction, barrier and release law

Claim identity: SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The least nonidentity nuclear junction and decomposition are binary. Exact conserved Fold composition maps two mass-two/charge-one words to one mass-four/charge-two word; exact decomposition maps one mass-238/charge-92 word to two mass-119/charge-46 words. The admitted zero-parameter binding ledger proves a strictly positive total-binding gain in both directions toward the unique mass-62/charge-28/neutron-34 maximum. Mass-energy conservation therefore requires a named positive release carrier. The normalized reaction barrier is the forced Half-One. No measured energy, fitted correction or stochastic premise enters the derivation.

Binary fusion and fission conserve exact nuclear counts, increase total binding toward the unique unbounded 62/28/34 maximum, require a held positive mass-energy release record, and carry the normalized Half-One barrier; AME2020 confirms both exact directions after reported uncertainties across the complete 2,548-row positive-composite census.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-HALF-ONE-001
- SFT-PHYS-NUCLEAR-FUSION-001
- SFT-PHYS-NUCLEAR-FISSION-001
- SFT-PHYS-QUANTUM-TUNNELLING-001
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of three possible fusion maps, three possible fission maps, both binding directions, every first-two-Fold barrier part, both release-record states, both global-peak certificates, both target-custody states and both extension states. The declared exact boundary is: Every identity, least nonidentity binary junction and least nonidentity binary decomposition over the registered two-fibre nuclear word; both exact binding-order orientations; all quarter-grid parts generated by two Fold refinements; complete or missing release records; finite-only or depth-independent peak closure; pre/post-seal target access; and empty or free extensions. The generator produced 1152 named candidates and the decision support contains 1152 one-for-one decisions. Exactly one candidate survived: `binary-junction__binary-decomposition__toward-higher-binding__half-One__complete-held-release__unique-unbounded-peak__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
fusion	binary-junction	identity	Rejected by computed Fold predicates: fusion_map.

Axis	Forced form	Eliminated form	Elimination reason
fission	binary-decomposition	identity	Rejected by computed Fold predicates: fission_map.
binding	toward-higher-binding	toward-lower-binding	Rejected by computed Fold predicates: binding_direction.
barrier	half-One	quarter-One	Rejected by computed Fold predicates: barrier.
release	complete-held-release	unrecorded-release	Rejected by computed Fold predicates: energy_accounting.
peak	unique-unbounded-peak	selected-finite-peak	Rejected by computed Fold predicates: peak_closure.
target	sealed-before-release	readable-before-seal	Rejected by computed Fold predicates: target_boundary.
extension	empty-extension	free-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:3b5ba8c1b1c7cef52e338ec2656e5043717dbb8f9d23040986d9325c1b25302c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject a binding-loss direction for both exact representative reactions.; observed Exact interval ordering gives a positive binding gain for both representatives.; receipt `sha256:bae4807a0cb3647819ceeb0d3ebc9fe67b58f7053aaf0f5f2fd589b31ed537bf`.
- **tampered_source:** passed; expected Reject any changed claimant source identity.; observed The changed source identity differs from the registered manifest.; receipt `sha256:a9afcfc0815909f62fdf159bee3d9b5e65c53dbf73156c3f19f7e0e9f89badbe4`.
- **tampered_artifact:** passed; expected Reject a duplicated candidate as an incomplete or nonunique census.; observed The deliberately duplicated candidate fails exact census uniqueness.; receipt `sha256:147ddd65e940c7b98a128bd62ebeecbf1e8c6a2ff7dad6aedc88f85b2824138a`.
- **boundary:** passed; expected Reject a non-Half-One normalized barrier and any target-readable derivation.; observed Quarter-One and pre-seal target exposure each fail a computed boundary predicate.; receipt `sha256:b12f93f19a2e1bbe0b9c76dacfd4ef176d9888ff2b934af4b61678ca2ec719c1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:76a335d018ddd2437b61f284d82383b85803971dbbe0e500feb639ca6a970ce3`.

Independent certificate:

`sha256:5ca4f105af45becb2d736b592b744de0f59405642563449fa97a03216771cefa`. Engine

external-validation hash:

`sha256:7e0c9230c8e5f9d819dea83227fcae05ee1e46f621ade23abe44782c31745316`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- AMDC-AME2020-MASS-1-BINDING-2021

Observed comparison records:

- complete AME2020 positive-composite census retained: 2548 rows
- fusion anchor exact intervals keV/nucleon: D [11122829/10000, 11122833/10000], He-4 [35369577/5000, 35369579/5000]; strict rise `True`

- fission anchor exact intervals keV/nucleon: U-238 [75701199/10000, 3028053/400], Pd-119 [83688901/10000, 83690287/10000]; strict rise True
- complete-census maximum: A=62, Z=28, N=34; separated from every rival True
- both external singleton 0.0 inscriptions retained only as empty-binding boundaries
- reversed fusion and fission directions, incomplete census and displaced peak controls rejected

Falsification condition: Reject if either sealed exact binding direction is absent from AME2020 after reported uncertainties, if mass/charge coordinates differ, if the sealed global maximum is not the complete-census maximum, if any positive-composite row or singleton boundary inscription is omitted, if a reversed-direction control is accepted, if a source identity changes, or if target content enters prediction before its seal.

Measurement receipt: sha256:722810443b2187a7ddf03f88a97bd8fd7b1123f536427cd976857f14552d35ee.
Isolation certificate: sha256:262fe874dacac1094654742a8ac05cd6be9e2ff3d5f9934bdbdef781f523d50d.
Custody certificate: sha256:b89abecbb1d4af6705d6b925ca4de257608d015c8de9ba2f7246f954fc1baef3.

Registered source descriptions:

- Atomic Mass Data Center, Institute of Modern Physics, Chinese Academy of Sciences:
https://amdc.impcas.ac.cn/masstabes/Ame2020/mass_1.mas20
(sha256:d10a9474253b2d29fac71ee21352039659f2e5f4b7d75395416d361878a18ddc)

Registered target identities:

- AME2020-COMPLETE-POSITIVE-COMPOSITE-CENSUS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-SINGLETON-EMPTY-BINDING-BOUNDARIES from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-DEUTERIUM-HELIUM4-FUSION-ANCHORS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-URANIUM238-PALLADIUM119-FISSION-ANCHORS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-GLOBAL-BINDING-MAXIMUM from AMDC-AME2020-MASS-1-BINDING-2021

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no AME row, element name, measured binding energy or dimensional reaction energy in formal execution
- no imported semi-empirical coefficient, fitted correction or selected residual
- no numerical-zero, negative, irrational, imaginary, floating or stochastic proof value
- no finite isotope window standing in for the inherited unbounded peak certificate
- no unrecorded released carrier and no target access before prediction seal
- no claim that normalized Half-One is a universal dimensional activation energy

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2a000fca551d443077a846fad52175406aa63aebc37ea397ac555ecb3a3c61c2; engine receipt
sha256:fbbdba722e6fb27d738ad3cd1df1f62e4caa979038b212147120ca814f0e25fa at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005-fbbdba722e6fb27d.json;
empirical-validation hash
sha256:8938824e23d81c3a0af6f06a90f74d4517541d3ef935743d19ee7ce1a0992d97; measurement receipt
sha256:722810443b2187a7ddf03f88a97bd8fd7b1123f536427cd976857f14552d35ee.

218. Fusion/fission yield ordering and distinct threshold carriers

Claim identity: SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For the least light binary fusion representative and the registered symmetric heavy fission representative, exact zero-parameter binding-score enclosures force the fusion gain per conserved nucleon strictly above the fission gain per conserved nucleon, while the heavy reaction's larger nucleon count forces its total release strictly above the light reaction's total. Fusion requires two charged incident boundaries to approach through Coulomb opposition before short-range residual closure; fission has one parent and therefore proceeds through a neutral-capture or internal-surface carrier. Half-One and quarter-order structures remain normalized relations, not one universal dimensional threshold. Dimensional reaction energies and thresholds are opened only after sealing.

Exact zero-parameter binding enclosures force the least light fusion representative above the registered heavy fission representative per conserved nucleon, while the heavy representative is greater in total release. Fusion carries a charged two-boundary approach barrier; fission carries neutral capture or an internal surface barrier. Dimensional reaction energies and thresholds remain reaction-specific post-seal measurement records.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-QUANTUM-TUNNELLING-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of all three exact per-nucleon orders, all three total-release orders, all generated fusion and fission threshold-carrier classes, all threshold scopes, all access-carrier classes, both metric-retention states, both target-custody states and both extension states. The declared exact boundary is: The least light binary junction D+D to He-4 and the registered symmetric heavy decomposition U-238 to two Pd-119 words; every strict or equal ordering of their exact binding-score enclosures; every incident charged, neutral or internal boundary class; normalized versus dimensional threshold scope; thermal or directed access support; separate versus conflated release metrics; sealed or exposed target custody; and empty or free extension. The certificate is not bounded by a Fold search depth. The generator produced 5832 named candidates and the decision support contains 5832 one-for-one decisions. Exactly one candidate survived: fusion-greater-per-nucleon__fission-greater-total__charged-boundary-approach__neutral-capture-or-internal-surface__normalized-structure-with-reaction-specific-dimensions__thermal-or-directed-energy-support__retain-per-nucleon-and-total-separately__sealed-before-release__empty-extension. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
per_nucleon	fusion-greater-per-nucleon	equal-per-nucleon	Rejected by computed Fold predicates: per_nucleon_relation.
total	fission-greater-total	fusion-greater-total	Rejected by computed Fold predicates: total_release_relation.

Axis	Forced form	Eliminated form	Elimination reason
fusion_threshold	charged-boundary-a pproach	neutral-capture-approach	Rejected by computed Fold predicates: fusion_threshold_carrier.
fission_threshold	neutral-capture-or -internal-surface	charged-boundary-approach	Rejected by computed Fold predicates: fission_threshold_carrier.
scope	normalized-structu re-with-reaction-s pecific-dimensions	one-universal-dimensional -threshold	Rejected by computed Fold predicates: threshold_scope.
access	thermal-or-directe d-energy-support	one-universal-temperature -value	Rejected by computed Fold predicates: access_carrier.
metrics	retain-per-nucleon -and-total-separat ely	conflate-per-nucleon-with -total	Rejected by computed Fold predicates: metric_retention.
target	sealed-before-rele ase	readable-before-seal	Rejected by computed Fold predicates: target_boundary.
extension	empty-extension	free-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash sha256:9206289cf47782eef20fdfe8c8f6602338e0746b0a29f668edf950c8b5b90b4e. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject a reversed fusion/fission gain-per-nucleon order.; observed Exact lower-versus-upper separation rejects the reversed order.; receipt sha256:45fa9ae8e1cfa4571b1b2811e710b45e6e8f2cb2b14c522e5a74df1f250af94c.
- `tampered_source`: passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae.
- `tampered_artifact`: passed; expected Reject any duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete census uniqueness.; receipt sha256:9d47a2bed75a232a7cce5e496679b0c00950d7bc18a8c16404a7dc8ae280226.
- `boundary`: passed; expected Reject one universal dimensional threshold and any pre-seal target access.; observed Distinct incident carriers and sealed custody reject both boundary violations.; receipt sha256:955d7ff449ed8aa55a801fafc2ce3d2341ff5741519175cece8712c78da7bea7.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:8750255e3be661050ac71b20b5bbb248cd92926575572818455919da3c7eefa1. Independent certificate: sha256:0f3a80971c1639dd5665b1c7bb5faefcd77c90fa44fac7832a5da17f47a4cf2c. Engine external-validation hash: sha256:d4fe6a92b4eee8e74b4106e2acc2c35333e42cd247fe5e0728a91d23b0db7fc0.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- AMDC-AME2020-MASS-1-BINDING-2021
- IAEA-WORLD-FUSION-OUTLOOK-2023
- IAEA-TECDOC-2024-NUCLEAR-COGENERATION
- IAEA-ENDF-102-DATA-FORMATS-PROCEDURES

- IAEA-NDS-LEXFOR-FISSION-2006

Observed comparison records:

- complete AME2020 positive-composite census retained: 2548 rows
- AME2020 exact gain intervals keV/nucleon: fusion (Fraction(59616321, 10000), Fraction(59616329, 10000)); fission (Fraction(998447, 1250), Fraction(499318, 625)); fusion strictly greater
- AME2020 exact total-gain intervals keV: fusion (Fraction(59616321, 2500), Fraction(59616329, 2500)); fission (Fraction(118815193, 625), Fraction(118837684, 625)); fission strictly greater
- IAEA D-T release and per-incident-nucleon values MeV: 88/5 and 88/25
- IAEA U-235 fission average/released totals MeV: 200 and 207; average per parent nucleon 40/47
- all twelve fusion-energy rows and all seven fission-energy component rows retained
- charged low-or-absent true threshold versus effective Coulomb threshold, neutral neutron evaluation class, spontaneous internal-barrier class and captured-particle class all retained
- rounded illustrative fission masses were not substituted for the exact AME calculation; reversed orders, conflated metrics and incomplete-row controls were rejected

Falsification condition: Reject if the sealed fusion-greater-per-nucleon relation or fission-greater-total relation fails the exact AME2020 uncertainty intervals or the registered IAEA D-T/U-235 energy comparison; if the IAEA charged effective-barrier, neutron-trigger, spontaneous/internal-barrier or captured-particle class is absent; if per-nucleon and total metrics are conflated; if any AME, fusion-table, fission-ledger, threshold or limitation row is omitted; if a source identity changes; or if target content enters prediction before its seal.

Measurement receipt: sha256:550034e2999fda2c58cd3c1d8dadca9e5833baee6d8445271718653240a5de8d.

Isolation certificate: sha256:a1d48ce8acda5a3973e9e1fad3621e5911a1bf6a6115a4009da3f418cf87a577.

Custody certificate: sha256:0a66cc5340cd48c51515b89932c504dbfdaed8b3d1b244d715e644b64a82fa9f.

Registered source descriptions:

- Atomic Mass Data Center, Institute of Modern Physics, Chinese Academy of Sciences: https://amdc.impcas.ac.cn/masstables/Ame2020/mass_1.mas20 ()
- International Atomic Energy Agency: https://www-pub.iaea.org/MTCD/Publications/PDF/FusionOutlook2023_web.pdf ()
- International Atomic Energy Agency: <https://www-pub.iaea.org/MTCD/Publications/PDF/TE-2024web.pdf> ()
- International Atomic Energy Agency: <https://nds.iaea.org/public/endf/endf-manual.pdf> ()
- International Atomic Energy Agency Nuclear Data Section: https://nds.iaea.org/nrdc/nrdc_doc/iaea-nds-0208-200601.pdf ()

Registered target identities:

- AME2020-COMPLETE-POSITIVE-COMPOSITE-CENSUS from AMDC-AME2020-MASS-1-BINDING-2021
- AME2020-FUSION-FISSION-YIELD-ANCHORS from AMDC-AME2020-MASS-1-BINDING-2021
- IAEA-COMPLETE-FUSION-REACTION-ENERGY-TABLE from IAEA-WORLD-FUSION-OUTLOOK-2023
- IAEA-FISSION-ENERGY-LEDGER from IAEA-TECDOC-2024-NUCLEAR-COGENERATION
- IAEA-FUSION-FISSION-THRESHOLD-CLASSES from IAEA-ENDF-102-DATA-FORMATS-PROCEDURES+IAEA-NDS-LEXFOR-FISSION-2006

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no AME or IAEA target, nuclide name, dimensional reaction energy, temperature, cross section or threshold in formal candidate decisions

- no universal dimensional threshold inferred from normalized Half-One or quarter-order structure
- no numerical-zero proof value, negative, irrational, imaginary or floating proof magnitude
- no imported nuclear reaction equation, stochastic premise, fitted coefficient or selected correction
- no omission or conflation of per-nucleon and total release metrics
- no target access before the prediction seal and no free extension after closure

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7a0b18b87f3a1df38a9679fef0ed10a6e72fa9d06851a8a38f38626db55c1db8; engine receipt
 sha256:1b13169930d9f4c22d75008ef7debe615a8c1dfddee7a5f495b85c467444e50 at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006-1b13169930d9f4c2.j
 son; empirical-validation hash
 sha256:83a6f01dfa45d115b8e784d5904e99a605913f66e348ffdbae5b7daf54eaa085; measurement receipt
 sha256:550034e2999fda2c58cd3c1d8dadca9e5833baee6d8445271718653240a5de8d.

219. Terminal deuteron binding, spin and dinucleon exclusion

Claim identity: SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete two-label spin word has three exchange-preserving readings with support three-quarters and one exchange-alternating reading with support one-quarter. The admitted two-boundary residual act is one-quarter: preserving support therefore retains exactly half-One as a bound recurrence, while alternating support is exhausted and closes to the empty form. In the least spatial ground recurrence, the alternating proton-neutron charge hand permits the preserving spin channel; identical proton-proton and neutron-neutron charge hands require the alternating spin channel to preserve total fermionic exchange. Hence the proton-neutron pair alone is bound, with complete spin One and three readings; both identical two-nucleon pairs are unbound. Proton Coulomb opposition is an additional pp boundary and cannot explain or alter the nn result. Dimensional binding energy and scattering records open only after the structural seal.

Complete two-label exchange support partitions as preserving three-quarters and alternating one-quarter; the admitted quarter-One residual act leaves half-One only in the preserving channel. The complete spatial, spin and charge exchange ledger therefore binds proton-neutron with total spin One and three readings, while proton-proton and neutron-neutron close to empty binding support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-HALF-ONE-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of all exchange partitions of the four two-label spin states, every ground spatial/spin/charge exchange allocation, every quarter-residual binding discriminator, every three-pair binding table, every composite-spin reading class, both charged-pair roles, both target-custody states and both extension states. The declared exact boundary is: Every ordered word over the two forced held spin labels; preserving and alternating exchange hands; the least exchange-preserving spatial recurrence; preserving identical or alternating proton/neutron charge support; the admitted quarter-One residual boundary act; all proton-neutron, proton-proton and neutron-neutron ground channels; sealed or exposed target custody; and empty or free extension. The certificate is independent of a finite Fold search depth. The generator produced 2592 named candidates and the decision support contains 2592 one-for-one decisions. Exactly one candidate survived: `three-preserving-one-alternating__pn-preserving-spin-ppnn-alternating-spin__preserving-leaves-half-alternating-empty__pn-only-bound__complete-One-with-three-readings__pp-positive-opposition-secondary-to-singlet-exclusion__sealed-before-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
exchange	three-preserving-one-alternating	two-preserving-two-alternating	Rejected by computed Fold predicates: <code>exchange_partition</code> .
ground	pn-preserving-spin-ppnn-alternating-spin	pn-alternating-spin-ppnn-preserving-spin	Rejected by computed Fold predicates: <code>ground_exchange_ledger</code> .
residual	preserving-leaves-half-alternating-empty	quarter-act-binds-both-spin-hands	Rejected by computed Fold predicates: <code>residual_discriminator</code> .
binding	pn-only-bound	all-pairs-bound	Rejected by computed Fold predicates: <code>binding_table</code> .
spin	complete-One-with-three-readings	empty-spin-with-one-reading	Rejected by computed Fold predicates: <code>composite_spin</code> .
charged_pair	pp-positive-opposition-secondary-to-singlet-exclusion	charged-path-selects-both-identical-outcomes	Rejected by computed Fold predicates: <code>charged_pair_role</code> .
target	sealed-before-release	readable-before-seal	Rejected by computed Fold predicates: <code>target_boundary</code> .
extension	empty-extension	free-correction	Rejected by computed Fold predicates: <code>extension</code> .

Base and successor. The closure proof is sealed by derivation hash `sha256:a618c2d97aa7b0e98f2b95f55e582ae9432238180faa3b17c53e0450b5c6ab94`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected Reject binding of the identical pairs with exclusion of proton-neutron.; observed The computed exchange and quarter-support ledger rejects the reversed outcome.; receipt `sha256:2ba04fbaa83807074b2a3d7350a0dd5e74a5b5aac0caf51f1bdd5ff2368cbd95`.
- `tampered_source`: passed; expected Reject any changed claimant source identity.; observed The changed identity differs from the registered source manifest.; receipt `sha256:0de6112a4fa89e23a8625b623b57f00f6f65f6e8de7f00f0265c4d1e451b09ae`.

- **tampered_artifact:** passed; expected Reject a duplicated or incomplete candidate census.; observed The deliberate duplicate fails complete candidate identity.; receipt
sha256:85f7f2fccaa82e12423de0043e0cde43f60a6c59bbf25b4c3079af293bc64a.
- **boundary:** passed; expected Reject pre-seal target access and every ungenerated correction.; observed Sealed custody and the empty extension are both required by computed predicates.; receipt
sha256:0dfa0edf0904171cdd0da22079b20109bee460b9c155ae4353b37466b4f164ef.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:c570667345abb2e4acb914a92e88a08afdb8050dbf9696f508cb0fae4e0f1f77. Independent certificate: sha256:09b05f9132863ba86a8807390fef592e706e53b45b72e9406eab6a570eb1e714. Engine external-validation hash: sha256:cceefee5c41149be61fff0a214f71e987fb4cc83dea73c842847275d6ec2f9bf.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- AMDC-AME2020-MASS-1-2021
- AMDC-NUBASE2020-2021
- NIST-CODATA-2022
- IAEA-NDS-JENDL1-INDC-JAP-45
- IAEA-THEORY-NUCLEAR-STRUCTURE-STI-PUB-249

Observed comparison records:

- complete AME2020 numeric coordinate census retained: 3,558 rows; exactly one A=2 coordinate (2H)
- complete NUBASE2020 state census retained: 5,843 rows; exactly one A=2 state, stable 2H with directly measured $J_{\pi}=1+$
- NIST CODATA proton+neutron taking deuteron binding interval MeV: (Fraction(111228251, 50000000), Fraction(55614193, 25000000)); central 222456637/100000000
- AME2020 deuteron total binding interval MeV: (Fraction(11122829, 5000000), Fraction(11122833, 5000000)); central 11122831/5000 keV
- CODATA mass-difference and AME2020 binding intervals overlap exactly after all reported uncertainties
- IAEA JENDL complete two-set spin-channel table retained; singlet scattering lengths are externally signed opposite to both positive triplet entries, with every effective-range uncertainty
- IAEA records the bound deuteron, parallel neutron/proton spins, triplet association and spin-dependent cross-section separation
- reversed pp/nn binding, Coulomb-only nn explanation, incomplete-census and changed-source controls rejected

Falsification condition: Reject if the complete AME2020 or NUBASE2020 census contains a different A=2 bound-state inventory; if the directly measured deuteron spin/parity is not 1+; if the exact CODATA proton-plus-neutron taking deuteron interval does not overlap the complete AME2020 deuteron binding interval; if IAEA does not retain distinct singlet/triplet support, a bound parallel-spin deuteron or the reported spin-channel separation; if any source, uncertainty, row or unfavorable control is omitted; or if target content enters before seal.

Measurement receipt: sha256:517fdc91b68661db3cf84eb534f213a8acfdb139ab490279a5aed8744ae9241d. Isolation certificate: sha256:1bc4132cc9b3f3454a85ac694fded29649157bc1d5892117f5d2a99fd3cdf36b. Custody certificate: sha256:dbc76018471956c03fcd443307afeeca2fbbf4d56cf5731eba650eec50b23bc3.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1 or V2 executable, certificate, answer table or survivor identity in the derivation runtime
- no nuclide identity, measured spin, mass, binding energy, scattering length, cross section or source locator in formal candidate decisions
- no claimant-supplied expected survivor and no submission-controlled admission flag
- no numerical-zero proof value, negative, irrational, imaginary or floating proof magnitude
- no imported nuclear potential, stochastic premise, fitted coefficient, selected correction or measurement-selected candidate
- no Coulomb-only explanation for neutron-neutron exclusion
- no target access before the prediction seal and no free extension after closure

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:cf08b6132dc2c644a2a6316b6fc508ae22948dbb880c4ffd460115e699b0b605; engine receipt
 sha256:c2a2ca11504bbe66760011c57dff2e8459f171c836b754d24b2ae3c200fa4cbc at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006-c2a2ca11504bbe66.json
 ; empirical-validation hash
 sha256:05f87ae0d5eee06b72dbf7925eb6aa00b14b45a59a12a9bd2df572552b92be32; measurement receipt
 sha256:517fdc91b68661db3cf84eb534f213a8acfdb139ab490279a5aed8744ae9241d.

220. Terminal dimensional light-hadron Regge carrier

Claim identity: SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Every positive spin rank has the exact squared resonance-support carrier $Q(J)=(6J-3)/5$. The base is $3/5$; every successor step is $6/5$; the first five carriers are $3/5$, $9/5$, 3 , $21/5$ and $27/5$.

Every positive spin rank has the exact squared resonance-support carrier $Q(J)=(6J-3)/5$. The base is $3/5$; every successor step is $6/5$; the first five carriers are $3/5$, $9/5$, 3 , $21/5$ and $27/5$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-HADRON-REGGE-TERMINAL-005
- SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003
- SFT-PHYS-MATTER-CONFINEMENT-LIFT-003
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

Generated grammar and closure boundary. Generate the complete product of predecessor custody, motion share, tube hands, base rank, successor act, closed form, unit boundary, target custody and extension forms. The declared exact boundary is: The admitted exact $3/5$ motion share, exactly two Fold tube hands, every positive spin successor, exact positive fractions, a post-seal common unit label and no measurement in survivor selection. The generator produced 512 named candidates and the decision support contains 512 one-for-one decisions. Exactly one candidate survived: `immutable-receipt-custody__admitted`

-three-fifths-motion-share__both-forced-Fold-hands__two-times-three-fifths__first-positive-rank-three-fifths__depth-independent-induction__postseal-common-positive-unit__target-inaccessible-until-formal-seal__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
predecessor	immutable-receipt-custody	rewrite-or-ignore-admitted-laws	A successor may not replace an admitted dependency.
motion	admitted-three-fifths-motion-share	selected-or-fitted-intercept	A selected intercept reads the spectrum.
hands	both-forced-Fold-hands	chosen-trajectory-multiplicity	A chosen multiplicity changes the slope.
successor	two-times-three-fifths	fitted-common-slope	A measured slope is not a derivation.
base	first-positive-rank-three-fifths	numerical-zero-or-free-of-fset	Neither numerical nothingness nor an offset is generated.
closure	depth-independent-induction	finite-five-row-pattern	Five inspected rows do not establish a general law.
units	postseal-common-positive-unit	unit-name-selects-coefficients	A unit label cannot choose a ratio.
target	target-inaccessible-until-formal-seal	target-readable-before-seal	That would fit the known spectrum.
extension	no-extra-rule	free-spin-correction	An ungenerated correction is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:9f7d24de80fb141e1123b67d6c9295b208fb454e4fb95d34e54fc5d21689043b. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:f8d54892b5d1ed3de6ba27737dd5b9c621492a76f12a7ac16c91ae9adbe70762.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:da01e8d6d5522b0b8e723c54fdd47faa177bba7822980c9fd58585ff2ed16c23.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:ab20aee40cd9249b6bf57a500d8ba2e70e83717f2eb6edbfe97b6b0982951749.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:bc80d96fa5308bfe83ed5785f06f0a5a4cb2407237f8386e2aadad901aa56e2d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:d8eb0ca9a875d53c79c84e15781c7b8432d9e5e0e36c184f5953831babba8d0b. Independent certificate: sha256:33518183cdd429114c0ead42400024b705b56090b4981d0ffcfc41b0c0e059bc7. Engine external-validation hash: sha256:fdb35e195df34916562fc5484c6c7c5f813b09c2de7688d49580c3e08bdd9d4.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Nuclear and hadronic laws close composition, residual interaction, binding, levels, decay, reaction, fission, fusion and spectral support at their registered exact boundaries. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no fitted or measured Regge slope, intercept, mass, width, residual or selected trajectory row
- no imported string-tension parameter or continuum string equation
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar
- no pre-seal physical-unit calibration or target access

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:6ffe106a351c8ba56384198d304d4fa489454b66823ee3ec41e608c59fd69daf; engine receipt
sha256:f82510d61a2e1472c8d090f2fd1f63b53491bab9f875ce4ae27b0e0b6b5fc54e at receipts/engine/model_admitted/SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059-f82510d61a2e1472.json;
empirical-validation hash None; measurement receipt None.

21. Spacetime Gravitation

Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise.

221. Relational spacetime event

Claim identity: SFT-PHYS-SPACETIME-EVENT-RELATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A spacetime event is a physical Fold transition located only by exact relations to generated recurrence clocks, spatial support and causal predecessor events.

A physical spacetime event is a source-bound transition with exact clock, spatial and causal relations.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound

relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__event-as-clock-space-causal-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	event-as-clock-space-causal-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:57158f6c5c035ddb048f810a643a6017ebbbce195a4c0588312c905e96a5bc3d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b7773e6b2ff8d06c5214ded33789524114c64d4484a5670e1a4f605059e96789`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8477eec669255c0e170818818a86a100ecda40bae844ba4019c1752c9e65db54`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b18bfc28a43fc2854e21066cbc6d1247cb14cecc546dc46ce09de064318cf37f`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:feeb336074e0c41f4d6ec1eec8923db6075bf2c4a69c0a19ed362d29e889d150`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:7fb975e8ff2ba01379b95f1223d62e94f244d303e7486926461154f0e3e40e29`. Independent certificate: `sha256:ceb24ff9b7ff53a8e2f1279b4659c17bfce039260279ca8494f93b65139f003b`. Engine external-validation hash: `sha256:dbbf84b3d2ced73f56340293db1f23759f9ccd165e88571f6fecdd80d5221761e`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- spacetime-event-relation-1: predicted relational-spacetime-event; observed relational-spacetime-event; exact match `True`
- spacetime-event-relation-2: predicted relational-spacetime-event; observed relational-spacetime-event; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256: 8dada648a51868ea3569d4c92ba039fad71738b6191c83b0ab311e3aad6072e9.

Isolation certificate: sha256: e8564e388adb1acf9c6e6d9aceea33ba5e1d5126892fac89084435645445902a.

Custody certificate: sha256: 67687003795e730cfc3d720b1145cffa6d1a03d271e45f7469a21e764cc9362a.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: <https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-event-relation-1 from GWOSC-V2-CATALOGS-2026-07-23
- spacetime-event-relation-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 1432661c0f6d82366ac4f9242b50b0cb701690050a23fd3a14003aa89079fa30; engine receipt

sha256: 82fcaa4e269454cbc2b6c7308f0505bc092b436bd8a240c9ad6799d2bd5e3622 at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-EVENT-RELATION-001-82fcaa4e269454cb.json;

empirical-validation hash

sha256: 696e6ebacc729052b3b4ae64b4149ab84ea68d38cfac3f247505f0a79d33cf3e; measurement receipt

sha256: 8dada648a51868ea3569d4c92ba039fad71738b6191c83b0ab311e3aad6072e9.

222. Invariant event interval

Claim identity: SFT-PHYS-SPACETIME-INTERVAL-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The interval between events is the canonical equivalence class of their exact clock recurrence and oriented propagation-support separation under reference relabeling.

Spacetime interval is the reference-invariant Fold class of exact temporal and spatial event separation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-EVENT-RELATION-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__reference-invariant-event-separation-class__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	reference-invariant-event-separation-class	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256: fec28317c8170661e758a9419a98c1debf2cb1ab1c4166cc27c45d6f6e79b004`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:ce6b28bb82bc59bdd033240abe690be1d0975058ebd0eeea5a81cfef20c143a7.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:743da5168c1ece2e75de4e5c4d4e98f8aa2a750be492e54da39ffffb955995af.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:c8a03de0d32ff02986b3511d82d721222393e34eaa6192d6b70ee1c7bd2f5307.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:c5068957099301afd9005fc768ab8997fcd26b8b587fd1ece663c56d13b325d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:1c2f6139e6822a4f6143d4e727b590f756a442630bfb5b8fa0c6694f8072d72a.

Independent certificate:

sha256:50c3ed5aa9bee665adf665dabbef51f40cebd1394415c1e8d4a1843c91d6308d. Engine

external-validation hash:

sha256:77aaa397de56081eb78d5310378433bd205de1868243de36e40d7ea95d1d2362.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-interval-1: predicted invariant-spacetime-interval; observed invariant-spacetime-interval; exact match **True**
- spacetime-interval-2: predicted invariant-spacetime-interval; observed invariant-spacetime-interval; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:e2bde2298077d8e2c5fed178ea398024d00549e3d72d24a2c05a2c66432223b8.

Isolation certificate: sha256:cd342dd16ae11ba15020243d71220ed758ab08ebbb3eaeffc4837e5c23613543e.

Custody certificate: sha256:54f90d96d76784160f28d4edd19d6410d0d8a7de43041662228c78b4ed6f75ce.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-interval-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-interval-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:e4d316c3500dbca3efc77b55638d28b6e9f6a12af7611643374a3cfd28a65ad0; engine receipt
sha256:50d120a00b976766cf5679b950c5b17e11f3c7718e90f8350bfb0f4de166298f at
receipts/engine/model_admitted/SFT-PHYS-SPACETIME-INTERVAL-001-50d120a00b976766.json;
empirical-validation hash
sha256:2a2776461015bd10876a20724e3935b290f458bf6725d340951e4114c8b2ca6f; measurement receipt
sha256:e2bde2298077d8e2c5fed178ea398024d00549e3d72d24a2c05a2c66432223b8.
```

223. Causal order and accessibility cone

Claim identity: SFT-PHYS-SPACETIME-CAUSAL-ORDER-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Event A can precede event B exactly when a generated adjacent propagation path from A reaches B within the limiting recurrence support; the complete reachable set is the causal cone.

Causal order is exact generated path reachability bounded by limiting propagation support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-INTERVAL-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__limiting-path-reachability-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	limiting-path-reachability-order	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c6f2c57dcbff3f4dcef21939a575a89514f5ee9dc17f9f6d9dc5a1229187e23a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e94442b5bb53ef2b5de6881020d27b9d07013f476e63ff55922e7c574c7b0285`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8be3895f058cdc5dbb68c13f8f5b74cf024aac7b5f19f6ca8c896d79208e9b5f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0ecf97d49c8e894ca1a702750e0dcf902e1287d8aa6a7907f2456279e0cc0fa1`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d59f9e9634e3091169f87061ee34a4ebe45752d4dc067ddda43e8e0b0b483615`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d997d62bd528d094409a6febb7838577ed38e68e6594d69616cf0c8eebdf2768`. Independent certificate:

`sha256:118cf1eb0c90c71598c96f9d3e8a2f8bfd9885ea5133bd149f50dad48a4ead8c`. Engine external-validation hash: `sha256:787a7a0b5cbdddef61c57956bcb3bfc30031b79c4958a8ae40a5f9c4aeae1865`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- spacetime-causal-order-1: predicted spacetime-causal-order; observed spacetime-causal-order; exact match `True`
- spacetime-causal-order-2: predicted spacetime-causal-order; observed spacetime-causal-order; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: `sha256:25be46db3ad5277c12a1891ceae8822285bc4b86090bb022f136f7bfafeb4aad6`. Isolation certificate: `sha256:8f958eeb9494faf3478fd50d2aed0176e84ca1c96ccf047846c9118807fd3895`. Custody certificate: `sha256:30982855826057f365e2cf2458626a3a2a18d43dcee90eb63dd6a451515cf6ce`.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: <https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-causal-order-1 from GWOSC-V2-CATALOGS-2026-07-23
- spacetime-causal-order-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:ebac04b824e98f9d993423d4e8eb91d0a11b27106eaa53b2f134cba581aba6ef; engine receipt
sha256:1bfc8212981c9de42edb974516b15a92e3313182b0e61c7048f60ed8883d2678 at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-CAUSAL-ORDER-001-1bfc8212981c9de4.json;
empirical-validation hash
sha256:d8c887f8b816e18cad223094fd3ab55579943d65e68a11e7ee4553a067807603; measurement receipt
sha256:25be46db3ad5277c12a1891ceae8822285bc4b86090bb022f136f7bfafeb4aad6.

224. Inertial frame correspondence

Claim identity: SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. An inertial-frame transformation is the unique bijective relabeling between uniform recurrence/translation records that preserves event coincidence, interval class and limiting causal paths.

Inertial frames correspond by exact interval- and causality-preserving Fold bijections.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound

relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__interval-preserving-frame-bijection__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	interval-preserving-frame-bijection	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:80ed23b843c0af6dd2a5dbf3cee8b049dfa37aad382217d825be30d28a81e8d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f9297fdf7205dde4aa959fb5ff0c226f839fb3bf06aa40b29fac6e3a2a23cddd`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:18d929aa636d7afda9393834957574c0fbc60fc64c1c0ea4c0a122f011c08b56`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c44d24816d8e519c021582ddc7b5e6dc80a1d0118982f7005e1f0d6c0e5c317d`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:879f262781f49174bae996767595daa69cf72f1d253b7cfa8f546f530ab1de8d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6e481b4aa80234a618d5d8a306575f66818fc68bffd9e0a8f9ecb7a409db3e4b`. Independent certificate: `sha256:52baf3a0668d6cd44be3f6bf6380c3ef2ff6a548bb046038bd2bad31ffc7809e`. Engine external-validation hash: `sha256:14e5897d7826e6d4227103a3240f8b6e6c5f7e9531b4d16515baf775d9a0a763`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-inertial-transformation-1: predicted inertial-frame-correspondence; observed inertial-frame-correspondence; exact match `True`
- spacetime-inertial-transformation-2: predicted inertial-frame-correspondence; observed inertial-frame-correspondence; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:ed541a487244cbea726608732cbff8123eba2f7e749162ab5d6f111e7cf1d863.

Isolation certificate: sha256:f7816ed17d3e67f421348b6da5c0c9a2416324d4108724b97e0c3895ca562334.

Custody certificate: sha256:298236f30403e5813aa34b62764a242f44691570f4c3f8d12f710132090b7260.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-inertial-transformation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-inertial-transformation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:bf6777d88a537ebb9ece0f7805aa0d6f0052ed940444b7f178d08469ef3148f0; engine receipt

sha256:a9a1243a2a78ae05c7b16a006f1841a05abf263e00f35ca77ea68d5a5d2dcbb9 at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001-a9a1243a2a78ae05.json; empirical-validation hash

sha256:ca79b98d79e30fc96b7a7f154dd3908e663f5a9fd38543595403aa25a8b69169; measurement receipt

sha256:ed541a487244cbea726608732cbff8123eba2f7e749162ab5d6f111e7cf1d863.

225. Invariant limiting propagation speed

Claim identity: SFT-PHYS-SPACETIME-LIMIT-SPEED-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The limiting speed is the exact spatial support advanced by one elementary causal recurrence; every inertial correspondence must preserve this boundary ratio.

Limiting propagation speed is the invariant exact ratio of elementary causal support to recurrence duration.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__elementary-causal-support-per-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	elementary-causal-support-per-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8a38374d90b1ea52f160ef519a60f4312e2c73e7516cb051a542021a2645e4e8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated

by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 3f24907ad869c851e94358c6d2e1589485d8dfc57a3e47429d9e67fec3b1b807.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : f0065fbc2140dc09cb06db2d8f0fed9292f7e037bc3543987d690031ab0fc8f9.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : d403016dd04c1b0dbe76390cc38dcb92301e6b787b9c7d905ee0c6e4a5172e64.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 70e1003982b55a829dedd23bbd2d3c5a8b811ed7f9bc72eacc9d8f3309a6fc1b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : f9f2b248a9aa95dd1cd4f693c9ec4d5bce43afe14d0559a70bed45bcd9308135.

Independent certificate:

sha256 : 9229fc9c2465f5e749ccf72df5c46a67dfe460021a0128ede2badd6dde62c28a. Engine

external-validation hash:

sha256 : 7dd92703c6f203aa8680cc03a6d43eb6d14d91abda7c941a8575db209bd9d7b5.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- speed of light in vacuum: exact predicted interval [1616237000000000/5391307, 1616273000000000/5391187]; external interval [299792458/1, 299792458/1]; overlap **True**
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact Planck-support/recurrence interval does not contain the released CODATA limiting speed or the displaced control is accepted.

Measurement receipt: sha256 : af13f413002d4934e1d6e502ec6563cc9afcbf087fe6443bed2bb4c863451b94.

Isolation certificate: sha256 : 5feabf42262e853d3c7d56b84e0c6b407f56d88158b15ff9055aa3982fe5245d.

Custody certificate: sha256 : 0ea675b46a394b6e93bff604617015845701dc158e6838ab84f0f6494c1105f4.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-limit-speed-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-limit-speed-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity

- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:fbe9031ec69bbcf574735a42d111b6098057e3ebd47aba0ba6b244143e4c7803; engine receipt
sha256:9bef8332bbd2d865002cece22dfd62dfbbc9a9084876156ff04bb8d388551a3e at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-LIMIT-SPEED-001-9bef8332bbd2d865.json;
empirical-validation hash
sha256:97e83cbfc50402a837d53fcd83feb9ca236caf6a396f55dfc1fdf3a778c68433; measurement receipt
sha256:af13f413002d4934e1d6e502ec6563cc9afcbf087fe6443bed2bb4c863451b94.

226. Relative clock-rate relation

Claim identity: SFT-PHYS-SPACETIME-CLOCK-RATE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A clock rate is the exact recurrence count along one causal path; between inertial paths it transforms by the unique positive ratio preserving the common event interval and limiting boundary.

Relative clock rate is exact recurrence-count correspondence between causal paths sharing invariant event separation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__path-recurrence-ratio-at-invariant-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	path-recurrence-ratio-at-invariant-interval	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9e396ba01214b05e387f267920f4137e52df5920e3450b69fd99c35b7deb7b40`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:943cf51afadc3eea33e1917e2654588cd5ffb852a9b66d055f31edb06a920b7d`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cb4c45ed3892212961529a7859a527ca986a6ffb4e44f83ce60fc5016d7fac6b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:575773c48e443966b9c33c32c94c1ea43757f7839befd628e08b309b52f8278b`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:49b8effffa4e055079087bdc448a43dd780c1cb38eae3e71224b8d58cee7a105`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:22904c14e9462cef8a843160336eb4f8b6cc423c4f567aa2155f81febeed99a5`. Independent certificate: `sha256:9eec8c27eeab2fd5cb74ae22d8d68cde4653e993b1ee0f1cb459951c561224db`. Engine external-validation hash: `sha256:76c050cfea5e77bb4ddb071e33092e3000691c2d531be6109bc1c2538a274130`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- spacetime-clock-rate-1: predicted relative-clock-rate; observed relative-clock-rate; exact match `True`
- spacetime-clock-rate-2: predicted relative-clock-rate; observed relative-clock-rate; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:6d5fbd1c9a090fe1f28fbecd908e63eafcdf60ebaf732e2d7ba7749b1bc29c1d.
 Isolation certificate: sha256:e68434c8ed886c0d69de56c0ddcb8173c09226de20ee0ba6365c8b36a25673c3.
 Custody certificate: sha256:7d99a6df31b443d9118a0dca1d0c77ebe2c6055e51c3d98099c0f58b8bb48c4c.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center:
<https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- spacetime-clock-rate-1 from GWOSC-V2-CATALOGS-2026-07-23
- spacetime-clock-rate-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:e2c7e10a97fa2126fb26c770b9c8e23423ca25e1a9412836eaa2cfb1db9dba47; engine receipt
 sha256:e7413efbd9b5a70f627078003097d132d54f3a872a7fffb58545a56e106487d34 at
 receipts/engine/model_admitted/SFT-PHYS-SPACETIME-CLOCK-RATE-001-e7413efbd9b5a70f.json;
 empirical-validation hash
 sha256:3795eb8d27b90586e27be5332749347c83245b4339a2898be1d50e1b3a0b0eba; measurement receipt
 sha256:6d5fbd1c9a090fe1f28fbecd908e63eafcdf60ebaf732e2d7ba7749b1bc29c1d.

227. Relative spatial-extent relation

Claim identity: SFT-PHYS-SPACETIME-LENGTH-RELATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Spatial extent is the simultaneous support count under one clock partition; changing inertial partition forces the exact extent ratio paired with its clock-rate ratio so the event interval is preserved.

Relative length is exact support-count correspondence under interval-preserving change of clock partition.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__clock-partitioned-support-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	clock-partitioned-support-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9233374abcde90b8197eb26613fc1d6dba24c389ab958255f29f42d68e14a127`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:01fbbbc74d963f944aec1a95d254af9a06b05424fef643181c1e652078e61cffa`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9eebb4ebf15a0efc40bd52f8a03163195485f5c32e1d62e802380539f86a8619`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5c03ab3952ed6a4bc31c5fe759f384a92af1015b79645066ebf551a7f61e21`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ee13c0e3ae4a4b7073813fa16fd78284ce2aa8b1c28a2a48addbe4a1832475a1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a4a71b3973fbd61b62fc84b2bb2e009fc86edeada7c2bbebe2255f0ce7eb9817`. Independent certificate: `sha256:d035014e134e4cb6cecc72653f05c6bc1ca0d36bba21361fb670e65c303eded6`. Engine external-validation hash:

sha256:103c40127b1866cc738336a787db008e1c392506fa017acb5bddd6de53a5c8a7.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- spacetime-length-relation-1: predicted relative-spatial-extent; observed relative-spatial-extent; exact match `True`
- spacetime-length-relation-2: predicted relative-spatial-extent; observed relative-spatial-extent; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:a32487b92a5a6536054fc88923113da4d4464fe04bb45ac3f57ecb92a98578cc.

Isolation certificate: sha256:407b00521280075f2b3bd4e34c79de3eef73ee4d229b713fee0ca5d30a81bc80.

Custody certificate: sha256:fc11aea1cf19cb506ff35768487e465585527e2a40a049486fcc4ed57825351d.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- spacetime-length-relation-1 from NIST-CODATA-2022-ALL-CONSTANTS
- spacetime-length-relation-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4e5f6f72c85b5622f17b4e8847f5657f6491f1d55346b47e2df5e9afd62442c0; engine receipt

sha256:34de5f07fb814d9d840ebfb262cfff11c2fa45736e26a95d94c26328edd162a7d at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-LENGTH-RELATION-001-34de5f07fb814d9d.json;

empirical-validation hash

sha256:adaee49ca1b23978a9d18c187e8ff3cfe0e5fa447d2ba63c8ce27fe41c52cf4d; measurement receipt

sha256:a32487b92a5a6536054fc88923113da4d4464fe04bb45ac3f57ecb92a98578cc.

228. Inertial-gravitational equivalence

Claim identity: SFT-PHYS-GRAVITY-EQUIVALENCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Local free response to a gravitational source is observationally equivalent to uniform change of the relational reference support when every constituent follows the same causal path class.

Inertial and gravitational response are locally equivalent complete Fold traces under universal free propagation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-SPACETIME-LENGTH-RELATION-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__universal-free-response-reference-equivalence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	universal-free-response-reference-equivalence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0d512e99dd1d9639c6df83f0b5ab9c2a70461b3c61d94c2c33ab68bdfdf90c7a`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 73aa11a193c0b20cc2a035d9f42bc3740a8a3ce235bc3da7f1badbffe27147a5.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 92df6588bcf888aaad40ca734ec326adc9765e2fc6bd382c5e746b6cf2368d0d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 8862ade98a9553248fe679256e7b4d33b52f0dffa585520ff00e310bd9e1adc9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f041cf78472d82e37e466383945ec3bc74481b6b52ad05634a1a27f122ac85e3.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 3c06fca37a2b1be088f1a2c5652ccc01ab5951b29fd61569ec0e2ce49eea9b99. Independent certificate: sha256: bd958e9791dc9ae04d5aa573f9d94372281a1ae9b2f163f93e43d158bdab4d82. Engine external-validation hash: sha256: 24a9c5fa2fe8d79fd19d060c22856a547d82aeda443f5d050e255c1af5ccff8e.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-equivalence-1: predicted gravity-inertia-equivalence; observed gravity-inertia-equivalence; exact match True
- gravity-equivalence-2: predicted gravity-inertia-equivalence; observed gravity-inertia-equivalence; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256: 8da3f466abdec2171edccd7bc27493a6d8730c6ff2fde273ab946f273f0f07a. Isolation certificate: sha256: 80f679c512c8dc48bf87c898f7e679da4e4ca5d7a73747e25b2925facbd26936. Custody certificate: sha256: 464151ed3300d04050baac9f6a5b24b2a23a2ddd2c0db77d7411424529c0d3d9.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: <https://gwosc.org/api/v2/catalogs> (sha256: fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-equivalence-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-equivalence-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant

- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:debd06bf6ab446c0e570406a49f5da060c602c5d45563b682df99144f160c05; engine receipt
sha256:391143646a9cbdfec7c08931a1e1f662f1132224e4303f02c6ca1e050213e459 at
receipts/engine/model_admitted/SFT-PHYS-GRAVITY-EQUIVALENCE-001-391143646a9cbdfec.json;
empirical-validation hash
sha256:e8a5760895c5fe0229b3d28c7081c53bb1a517c749f497002bbf880043c93703; measurement receipt
sha256:8da3f466abdec2171edcc7bc27493a6d8730c6ff2fde273ab946f273f0f07a.

229. Source-linked relational curvature

Claim identity: SFT-PHYS-GRAVITY-CURVATURE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Curvature is the failure of source-linked adjacent event relations to close to the same transported orientation around distinct generated paths.

Gravitational curvature is exact source-bound path dependence of transported Fold event relations.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-GRAVITY-EQUIVALENCE-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__path-dependent-orientation-closure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	path-dependent-orientation-closure	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:36ecd3deb2f47612ccb41670d5b909d075daf9ad2e2c9ea6d0fd38df84723be3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:cc296bc892c14918ca793090c889169d6a420f86cda52974fc4454296a920104`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:296028aa42392a91f94031efd1eda98594633cd11e145432a956394e715d8d2f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:267e1318436939856318206e4807adee62447f2541e0602f82d97a15cfd24771`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5427c13bc35996d5dc625b40524d58f651b52064201989dca085bb5c20eff4e0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:60dc41684edf0c9909df0e2dfa1dd501118544d2b9d3e08bb2ee5f7aebbb88b8f`. Independent certificate: `sha256:4338bb649a7ef62c7b507d93622192e8c473ba74270c4380cc37ff1a332ca050`. Engine external-validation hash: `sha256:0e36fafef0802525aa67f7fb1e142a635fe090fb619cf478bf0e5a7b698180fa`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-curvature-1: predicted source-linked-relational-curvature; observed source-linked-relational-curvature; exact match `True`
- gravity-curvature-2: predicted source-linked-relational-curvature; observed source-linked-relational-curvature; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:ad2de4c531e6cc4451e44954e488fcd55cdc5ed63a7fdd0d45605474943d8a46.
 Isolation certificate: sha256:f9358b2558fc21e9b92921152e1d4c9c4da49e9fc90f066fc15b7f0c84b06c1b.
 Custody certificate: sha256:518dcffea1b2ae15d09e30d72f2d573e7554d5b49dfa9a4af5865b576f0b130c.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center:
<https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-curvature-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-curvature-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c882cbf1ee5ee7f660f46f000d82d7434a3174d8a745f73e60fdc255e42058ae; engine receipt
 sha256:4a4e70826c4fd7be1d00b9aa003261fb437402fb7c43ab61130e3f17852828b3 at
 receipts/engine/model_admitted/SFT-PHYS-GRAVITY-CURVATURE-001-4a4e70826c4fd7be.json;
 empirical-validation hash
 sha256:fef1f1331caa9c032a29ad4ce7977f4119571cf1ccb39f17b20a5f0d8b233438; measurement receipt
 sha256:ad2de4c531e6cc4451e44954e488fcd55cdc5ed63a7fdd0d45605474943d8a46.

230. Free propagation on curved relation

Claim identity: SFT-PHYS-GRAVITY-GEODESIC-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A geodesic is the locally unforced adjacent path that preserves the propagating word's inertial recurrence while following the complete source-curved event relation.

Free gravitational motion is locally inertial Fold propagation along the minimal complete curved-support path.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-GRAVITY-CURVATURE-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__locally-inertial-curved-support-path__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	locally-inertial-curved-support-path	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b7027658c2c3b9c24d408fc0ff0a242c8dce9cc00b77b8a3581a14f863e7faa7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c0dd995225e3068b28acc56c8e6f66418abd154c843e41d3148593bbea25d211`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9982d9cbdfb704442072dc38365d63f2a65efed5391a5ff4f9e2f9a2ef03d976`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6b9d7defb3d6154f7fffcf998c8750eca69274a2760a27dab5dbe6f299e88d36`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d779b05ac5f9cc3a8bfd895b3e03292d2761344d1f2d09b3cd6968798e61ae45`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a34d39b8e9a0e93721e04115fe1c670f9194546f60bd479e2aafe9a19f652181`. Independent certificate: `sha256:1aaae7eae0755b0584d825c67752d6059a5e66c1abbec7bbffba773b19be9a6`. Engine external-validation hash:

sha256:5786c1d8194603af22c8d718b37d8e0bd1118cf0ee8f845b01de5d6e838a6ccf.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-geodesic-1: predicted gravity-geodesic-propagation; observed gravity-geodesic-propagation; exact match `True`
- gravity-geodesic-2: predicted gravity-geodesic-propagation; observed gravity-geodesic-propagation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:2e87a9a5e3e82e5521850fc66255d692f4bdced8d84830787b33cdb3ac7783ee.

Isolation certificate: sha256:a5a13691d20b4468f53851defac36713913a05d25dcce1737d7922c65056c24c.

Custody certificate: sha256:db317311a002ae98ae85890dfae6cb9296800ecea1e04ed36bb3be6765a437da.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: <https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-geodesic-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-geodesic-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:374514b1682e1464b36702b385c0fa0f5956f5d119eaf8a44230d2386388cbcf; engine receipt

sha256:fcd6e73b219916aa38b32b7f2f781e0711935f16db94bae6359634106e177782 at

receipts/engine/model_admitted/SFT-PHYS-GRAVITY-GEODESIC-001-fcd6e73b219916aa.json;

empirical-validation hash

sha256:abb3be49e4d9abc5657257c748d69b0d531896492951089c3eab598e745f11fe; measurement receipt

sha256:2e87a9a5e3e82e5521850fc66255d692f4bdced8d84830787b33cdb3ac7783ee.

231. Gravitational field-source closure

Claim identity: SFT-PHYS-GRAVITY-FIELD-SOURCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Every gravitational relational change is paired with an exact inertial-energy source carrier and its propagation trace; closed boundaries pair interior change with outward field transfer.

Gravitational field closure pairs inertial-energy source, relational curvature and boundary propagation on complete support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-GRAVITY-GEODESIC-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__inertial-energy-source-curvature-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	inertial-energy-source-curvature-pairing	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:75a8cdb60ed35199567225195e0c447cb2a94125726f1ac037b1f2763a71a236`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6de7b89644fb09a81e548b3a6503b3ae4903f29a50146da6a9213c0cb3701be8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ec1b9c32b20f0b63851cbaf3c076294a09139217c74e0b40d6211ceda7ad63c9.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:31847463ec37ed5d0dcfc0131a741bf1cc77146584900bdb0b9ae432fec15b57.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3f224f13c7965a287af13df8a33132cbe7f70ce8aca7796cc86109d47a7e3d31.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:f0ed74962fa452e895824dceff5799f47a305d51bd3625919fb3d7dccc6ba72e. Independent certificate:

sha256:8cb3c50c0d9bed56fa6c08931db1d4ad43098c2fa827d7973af9c6907313aa9e. Engine external-validation hash:

sha256:402a48c0845de430f8dedfd4ee830568e19d46a4c4e001fd3bbcecfca80192e9.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-field-source-1: predicted gravity-field-source-closure; observed gravity-field-source-closure; exact match **True**
- gravity-field-source-2: predicted gravity-field-source-closure; observed gravity-field-source-closure; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:b7f53de6ed47b473111318347f5cdc8ab679312ee0cd53db38f0220ed397a1b9.

Isolation certificate: sha256:9510864d6ee491b49b0f57f875d0de1b9cd2abb194b5b2c03c53dc76207eedb1.

Custody certificate: sha256:6268fb115bc902a506d0f7afd136a664054c7da056cc1cbce1a57d570895db62.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center:
<https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-field-source-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-field-source-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude

- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:586326ad2962e36fb72e63267cf73f63a2b7a2769ef76b79936138c42547ffb3; engine receipt

sha256:80e7d6df387d7f3fec02ef0deaf4a31f5838e858ed740f648e2f5950a4375140 at

receipts/engine/model_admitted/SFT-PHYS-GRAVITY-FIELD-SOURCE-001-80e7d6df387d7f3f.json;
empirical-validation hash

sha256:e58b1b4ad9f490bce8fac6023b462c676cf34c47ae54f106d172a5e67d92c328; measurement receipt

sha256:b7f53de6ed47b473111318347f5cdc8ab679312ee0cd53db38f0220ed397a1b9.

232. Gravitational-wave propagation

Claim identity: SFT-PHYS-GRAVITY-WAVE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A changing source-curvature relation that closes into transverse recurrence propagates at the limiting causal speed and transports energy-momentum with held polarization.

Gravitational waves are limiting-speed source-curvature recurrences carrying energy, momentum and polarization.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__self-propagating-curvature-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	self-propagating-curvature-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b7b136872d39bf5145b9cd5a7c9e144f069b09595ba417bec1954264fa4bf639`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a4668b977a7dc243e8d7cfff5281dd522f7e141c9d57a414c7fa4b9d7ff5c07b`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99c266006d03d674de429c1914a35d332b4e2fb65b4d77de084eb84f01ec84bc`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5d26782234031d446a0c0401268a54c749a24761281beb8753a6d453d8e5b91f`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8b6850e209743b529c53b98b2167da4edab5bf543e2b22f1b57f3feb3e79deec`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:62dcb4de31ba440c9b80227a2b922bdb5b626eafabaa6041c4b368edeefb57fb`. Independent certificate: `sha256:de8127768a7fdeb63d0f4d9e9da3ab09f73ba114d18a96b60074a66d8d08e4be`. Engine external-validation hash: `sha256:efa538d7a76533cf7044f4245e7a706f77e667dfe9f63dff2541c02761bb7a82`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-wave-1: predicted gravitational-wave-propagation; observed gravitational-wave-propagation; exact match `True`
- gravity-wave-2: predicted gravitational-wave-propagation; observed gravitational-wave-propagation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:b294c31d9d0cccfdb72234bc8edd7fc77f2e8f2e81328ac54b3f94683b099b00.
 Isolation certificate: sha256:7194df1fb9a8b0f56df0e38270db161bbb4c9e5fed5c22d744b89c9f671b9162.
 Custody certificate: sha256:cf71825378cc4fcf95eed5616ad3196c2d5b5647c2e53932a9f7ab134eb7e2bb.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center:
<https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-wave-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-wave-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d3c2e4ade0bcd74a1cb6731c8db906e8b9e9a53d4ce309301b39790715679923; engine receipt
 sha256:f2bcf8f3bec360481dadd724546028e3c47c633bd288adecbf650d92a8d0f1ba at
 receipts/engine/model_admitted/SFT-PHYS-GRAVITY-WAVE-001-f2bcf8f3bec36048.json;
 empirical-validation hash

sha256:013c0d29ad9a8ad2f3850f481e9b0be54e0a291f3baea625227a050109934f05; measurement receipt
 sha256:b294c31d9d0cccfdb72234bc8edd7fc77f2e8f2e81328ac54b3f94683b099b00.

233. Causal horizon boundary

Claim identity: SFT-PHYS-GRAVITY-HORIZON-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A horizon is the generated boundary separating events with a limiting-speed outward path to the observer from events whose every outward path remains source-confined.

A gravitational horizon is the exact Fold boundary of outward causal accessibility on source-curved support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-GRAVITY-WAVE-001

Generated grammar and closure boundary. Generate the complete event, causal-path, reference-frame, source/response, observation, target-access, successor and extra-rule product. The declared exact boundary is: All finite spacetime and gravitational forms generated from exact event words, adjacent propagation, held orientation, recurrence clocks and complete source-bound relations without imported coordinates or continuum metric equations. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__outward-causal-path-accessibility-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	outward-causal-path-accessibility-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0436ccab8b956bf9b4ff8dd85553679f27286fdaa52750759d8164f019c258a5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:90c176ed08a43d27a61c034b0213577c6468523dd8cd8a190252fb874fdb3686`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:158073e681f1b8a87a714e63e93fef55959987dcadfc116ceee7a281f1c6eb0a`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:aefc1806213b3b9c4642d09fc10e8ea8a2663ef634218e22bfcac424218d4d22`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1a3295b020e67bd329f48fcaedc43302734345fba5abfe1ead76a577d5b8e4e0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:04d5416a76e6cae6eed78368afc175b43ea8899624372c23524cd3ff369b6219`. Independent certificate: `sha256:3782e0a1753794145e8d5cc5e1b84d7e8e490a067d75dd3203db16660946f9ad`. Engine

external-validation hash:

sha256: fee7d362ca3bf1c92f3c4b6446b3f575b487959c41e54737a5176ccfd4cbde4d.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-V2-CATALOGS-2026-07-23

Observed comparison records:

- gravity-horizon-1: predicted gravitational-causal-horizon; observed gravitational-causal-horizon; exact match `True`
- gravity-horizon-2: predicted gravitational-causal-horizon; observed gravitational-causal-horizon; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The spacetime/gravity law is falsified if a committed external row lacks the sealed event/causal/source relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256: 25a77549a95884a6c8b2f2f636f8d2521bdf5c406b96c298326e7b5fc772a502.

Isolation certificate: sha256: 3d97d1dabc2f89390e65e54273b762de3256ca4605bea78726cbe2479fbdfde8.

Custody certificate: sha256: ffe0b5a8a931e75710308a7f166bbc63de7f2b82157da11a57c9c01e59155316.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center: <https://gwosc.org/api/v2/catalogs> (sha256:fcd8eef741cb536010539f3e42e6bb5d3a5e0fb7a120289fe95b8a1711f59e0c)

Registered target identities:

- gravity-horizon-1 from GWOSC-V2-CATALOGS-2026-07-23
- gravity-horizon-2 from GWOSC-V2-CATALOGS-2026-07-23

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated spacetime continuum or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported metric equation, fitted curvature or target-selected constant
- target access, omitted detector row or unrecorded causal predecessor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: dc739559de7afaf7c8fff7beb4ceb24afcf08cf87fe9832f66b9f235c694fd65; engine receipt

sha256: 7abc9b68655deb7986f0846887236bae69f14d4d6656d2ed71c388d0f889c52d at

receipts/engine/model_admitted/SFT-PHYS-GRAVITY-HORIZON-001-7abc9b68655deb79. json;

empirical-validation hash

sha256: 8c7e559f057c9378e631a19da8f7ac4d999c223aee5e3673ab8260e1014bbcbd; measurement receipt

sha256: 25a77549a95884a6c8b2f2f636f8d2521bdf5c406b96c298326e7b5fc772a502.

234. Weak gravitational potential and Gauss closure

Claim identity: SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A positive inertial-energy source distributed over the forced rank-two boundary has inverse-square response, and field times boundary growth reconstructs the same source at every generated radius.

For source 1/8, radii 1/4 and 1/2 both return flux 1/8; fields are 2 and 1/2 and clock potentials are 1/2 and 3/4.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001
- SFT-PHYS-MECH-CONSERVATION-001

Generated grammar and closure boundary. Generate the complete eight-axis product for weak gravitational potential and gauss closure from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__positive-source-over-rank-two-boundary__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	positive-source-over-rank-two-boundary	imported-Newton-profile-or-free-exponent	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:536252bb17f9ca456c07c08b13be87fc626e4f0daa38dc7c63cfb95c744dc2ab`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:cf6b59676d1adc6d639b0032d305cc79c0bebf082ad3b124027655716448475`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:67e2899f9784ba430c1d9a152f4bd10f8ee35f00b78ec95a5a4d0978b7b52109`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:139890ccf0b6c0c76335901373433e22d031be49f72159f2576a386808e77aa2.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:53d773b9122b5823346f7c3b3eb874aa41d779c8e386c53e6caf4b7bf06a7f58.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5.
Independent certificate:
sha256:6700a3ce895d0265f7e8bd0fc61aa9075b0b62d853ac9184eac8fd75075055c9. Engine
external-validation hash:
sha256:70b8ab007d5e20ae0e6428e0c9b0b3e04065af604fffb6bd66683d55a7ef2510.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7c0505b506dfe931f6cc18545096842c688511616b5a689011fec0919398570a; engine receipt
sha256:35145c1a594d6707add50a3f2e72dd3a2162092390fd9b3db03eed38b79ee147 at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003-35145c1a594d6707.json;
empirical-validation hash None; measurement receipt None.

235. Exact causal interval

Claim identity: SFT-PHYS-SPACETIME-EXACT-INTERVAL-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The spatial square is positively taken from the temporal square; the generated 3-4-5 partition gives massive interval square 16/25 and proper carrier 4/5, while equality is an empty massless-boundary record.

Temporal One and spatial 3/5 retain interval square 16/25 and proper time 4/5; equal temporal/spatial support yields the empty massless boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003

- SFT-PHYS-SPACETIME-INTERVAL-001
- SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003

Generated grammar and closure boundary. Generate the complete eight-axis product for exact causal interval from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__positive-temporal-square-retains-spatial-take__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	positive-temporal-square-retains-spatial-take	signed-imported-Minkowski-form	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:2c7bd4d76a5bc514dae6ad6a306edfc3d537e572bbc4d77cea94f2fa9842dffe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e67e32a67e79785404f6cd1167b3b4d8e9a9945b223ad59e6ea44677436fd288`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e8086dd9e72526956535496de02b3ec653f209d9d1803d32241d2d3cb032993e`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fcf2b65da1cda8b4ba64962e0e41bf8f139277c8888315ab05fa51b2ebe27427`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:d1d0beea696c30473418bdd6f27a5ba78355af46a27098b55432500446442968`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5.
Independent certificate:
sha256:c7646e73d8e016cd482c23b904b0a33cc42d4a3c147be42aca4346a788d5f801. Engine
external-validation hash:
sha256:241816ae38caab731db6c17066cee42e7c27c0dd2028e508fcb36ef236776a52.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:388e83da2499c5e29b9f4c9a54b10d6728cb9fc9900ae60fb561b65608b9ee27; engine receipt
sha256:c5560ae617b8b077c8d945e3f9b40475f2ca9c4b64028b4a7ea339753d936c4f at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-EXACT-INTERVAL-003-c5560ae617b8b077.json;
empirical-validation hash None; measurement receipt None.

236. Static Fold-covariant clock

Claim identity: SFT-PHYS-GRAVITY-STATIC-CLOCK-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The static clock square is the positive remainder of well depth from the One and commutes with Fold exposure; at well 7/16 the generated clock rate is exactly 3/4.

At well 1/8, Fold(7/8)=3/4=One-Fold(1/8); at well 7/16, metric square 9/16 has clock rate 3/4; the horizon yields an empty clock record.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-EXACT-INTERVAL-003
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003

Generated grammar and closure boundary. Generate the complete eight-axis product for static fold-covariant clock from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered

binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__One-take-well-depth-with-Fold-covariance__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	One-take-well-depth-with-Fold-covariance	imported-clock-dilation-function	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:9ac8ee4d8bd9978d82f2d9b052a5de9dd727dda7f1fe27068b5b1ae374f977fd`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:1036fb1fbd40f1bffdca1f05f7308fb1a1373acce57b8e5244b024ba514f126`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:7225468d3a83f38d08d71615cf5cd054548be6adeb48ae8e78ca3ec242700682`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:525f6761831cfb14c1c2364067135af6decdbd4c7886144652822c1b1d991f37b`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:7de2d029797fa95cc5f26c00cac79f244e002f075a91896e88c2f0bce978149`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10accb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:a4c0900c2684ce125957ab918d0edfec0ac8d0b553f6b8ec1bacfea83643b311`. Engine external-validation hash: `sha256:8223650a7e3328ecf0fbc18cee075e301774f0bd683690217dda0bdb1b9d01cf`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:bf88511ad49c19ee24bae6de64e2f8e2977ed8e31e5c0f473640c27d71332bee; engine receipt

sha256:8c03d3fd2812c980509616560286fd28ee1b29400c611cde02ef129d15fa8cb4 at

receipts/engine/model_admitted/SFT-PHYS-GRAVITY-STATIC-CLOCK-003-8c03d3fd2812c980.json;

empirical-validation hash None; measurement receipt None.

237. Gravitational-redshift and acceleration equivalence

Claim identity: SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The same exact acceleration-height carrier appears as weak gravitational redshift and acquired-speed Doppler shift, leaving no local observation label that distinguishes the complete traces.

At acceleration 1/4 and height One, gravitational and accelerated shifts are both 1/4.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-STATIC-CLOCK-003
- SFT-PHYS-GRAVITY-EQUIVALENCE-001
- SFT-PHYS-MECH-ACCELERATION-001

Generated grammar and closure boundary. Generate the complete eight-axis product for gravitational-redshift and acceleration equivalence from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__gravity-redshift-equals-accelerated-Doppler__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore

a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	gravity-redshift-equals-accelerated-Doppler	independent-fitted-redshift-laws	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:26be961265bea114935fd9ef430e954092c0b698a37e61fd23cf7c20f952aee7`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:34445064638bbce6286f2f2d83a969d6a045a927ab41c4320d4f2479d1be6a52`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:c7a539be9215ee867ac689964bfd2c3356d64e9ca6b2a619687e88c27a71fdb1`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6801215f1eba777f94256999621282c0d46a0312b5fce2f452b579df9e3aafcd`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:bf76521c08fb69ef09c4ef1db6d1a3680bfd9460cf891261958852825f732267`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10accb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:e60c9095875c1c45f456f06745990d777374a1844757509a1a2acee82cb27cd5`. Engine external-validation hash: `sha256:9dadae88048f1d3fe5eadb1ab6fa440d48341070a60e5677b1545aabf0ebb129`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:9ac9a202b994c37ead3a9befcaac43fa469fc5b49ca0eb2a46241e1dfe9262a3; engine receipt
sha256:6726f8bb1dd6a005b03f9058957774e3e97b2d14f7850901331494aefef7185c at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003-6726f8bb1dd6a005.json;
empirical-validation hash None; measurement receipt None.

238. Exact lattice curvature family

Claim identity: SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact second difference of the square carrier is binary curvature at every positive spacing; summing complete axes yields Laplacians 2, 4 and 6 in one-, two- and three-dimensional support.

At spacings 1/8 and 1/16, per-axis curvature is 2; complete dimension counts give Laplacians 2,4,6.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for exact lattice curvature family from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__binary-curvature-per-generated-axis__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.

Axis	Forced form	Eliminated form	Elimination reason
relation	binary-curvature-per-generated-axis	continuum-derivative-or-fitted-curvature	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:4b1c60e6cc08f4a63afcdd01fd0282aedf0f8ed95e7b50b2e2793386f7873fa8. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:4162c08533363fc40c8a047f13d104ef86273ffc0122a78417b7669e8952ad22.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:1188ac93e7ea3b9d8aadb79be92bdaa90e52d5ddf385dd5571be04fc171a198e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:f16881aa678b81018a623faf9f769b7563e09e03d10bac2d7c227123c6c2cd75.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:a732758c32ac5963478fa788dfd5d60c16eeeeee14676fee382f2f7da885c1856.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5. Independent certificate: sha256:cc9083a1d111b532d331004348ac65dafb2a21273f68113279a6641c4cd264f8. Engine external-validation hash: sha256:86dcc72575bccbf2b13af05f2722738c5da2b81d0a589f719d234134e610010.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise

- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:7e746116ea5adbf38e0424307cef139670ef7b261813a4e196ba3d94f78cb787; engine receipt
sha256:f10d0261e22c0cdfbce4d91bd75653851064c132325530b1b57e2da5e06ee457 at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003-f10d0261e22c0cdf.json;
empirical-validation hash None; measurement receipt None.

239. Gravity self-sources through field energy

Claim identity: SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The positive gravitational field carrier has an exact square energy carrier that couples through the same half-One relation and therefore appends a separately retained second-order source correction.

Source 1/3 at coupling 1/2 gives linear field 1/6, field energy 1/36 and self-source correction 1/72.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001
- SFT-PHYS-MATTER-MASS-ENERGY-001

Generated grammar and closure boundary. Generate the complete eight-axis product for gravity self-sources through field energy from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__field-square-reenters-same-source-channel__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	field-square-reenters-same-source-channel	linear-only-gravity-or-free-nonlinearity	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.

Axis	Forced form	Eliminated form	Elimination reason
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:9c6cf17dcb875e2fd58df7c4dbf7dc7c208d3b7311b9e1c724a90c4305ad848bc. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:d54c6721eaa4476367500ed60d0dbda0486e12e04a8d0df8a129fd477795ad49.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:23476aa444ae59176da16d0f4ecf82202065d9897a56154e5e02ee45d9a4ade0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:c2efcd39ebc636c66c75c91d913a37acf11ee43befab73c914db68656d9ec364.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:8e75fa19591b5c7c5cfc055211ae29e06686c7bdc6c820b28bbd6a74d6423280.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:2e13056da97f92c9af10accb2d573e678d2faabfa7ab883af79cc858aea47c5. Independent certificate:

sha256:89dd8e216bad3895aa17475cc1f2b8132c9d21a6c9279876cc2ab46edaa2ff61. Engine external-validation hash:

sha256:96c1f150187c3ce4f7544129801d3f1a5387906da62bd3fb5553c23eefcccb1.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:8ac8e53f3269e2eb63aae1ac45af265f8731882d171ae9d5c120f2b640d93670; engine receipt
sha256:ce5bb88d7247a7c82c3aa1050b1e4a5261acb4e4cbc301489b504668a87c5fbe at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003-ce5bb88d7247a7c8.json;
empirical-validation hash None; measurement receipt None.

240. Graviton polarization count

Claim identity: SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A symmetric four-dimensional metric has ten component slots; eight coordinate/gauge slots leave exactly two physical polarization carriers, while planar spacetime leaves an empty propagating record.

For $D=4$, $D(D+1)/2=10$ and $2D=8$ leave two; for $D=3$, six and six leave the empty structural record.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-GRAVITY-WAVE-001

Generated grammar and closure boundary. Generate the complete eight-axis product for graviton polarization count from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__symmetric-metric-count-less-coordinate-pairs__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	symmetric-metric-count-less-coordinate-pairs	asserted-two-polarizations	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:011a120c156ebbf6960704219c027b415ceadb1b37c3c6c6cb8018e66d970105. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:bdb6f292164cb3936a7dea5be06de8c91b87dc172cfe9eba3b4211b1565ba80e.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:ba6d3b400ce067415bd4e8529d76d343a22cd74e3593a3b32be3fa3c75457054.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:8f4e2cae0982d3a49a0378ae738862a87c1ee6cc367ad2e04df09d668e67d3.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:7c0e82c4ff69db1fb17a9b18ef5ff4ec7c4dca12f0fce32a9c0bb74e01dac5b8.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5. Independent certificate: sha256:221f6da122f88601e6e92c4fbc6bcb156440ba10013c6cd79ff095db63502557. Engine external-validation hash: sha256:fc8100f1407297b1fffae018a9de7a20a15bdbb7b23e6f165873b6bfcc8925728.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4a4e93a0ec30fe32dc6426ba37a59997f1ba955a54ac323f1f16a72e9b8145b8; engine receipt sha256:708fffd5a309a9cee077c2db46aef2bf8374435ec2c7bdf4685ef1fb719d0d73 at `receipts/engine/model_admitted/SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003-708fffd5a309a9ce.json`; empirical-validation hash None; measurement receipt None.

241. One-speed quadrupole gravitational propagation

Claim identity: SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A detached gravitational recurrence advances at the same One causal speed as the electromagnetic carrier; conserved source and momentum freeze monopole and dipole records, leaving changing second differences as the first radiative moment.

Eight ticks retain exact One-speed phase; uniform linear motion has equal first differences, while the generated cubic drive has changing second differences 12 and 18.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003

Generated grammar and closure boundary. Generate the complete eight-axis product for one-speed quadrupole gravitational propagation from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__One-speed-recurrence-with-first-unfrozen-quadrupole__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	One-speed-recurrence-with-first-unfrozen-quadrupole	separate-fitted-gravity-wave-speed-or-dipole-radiation	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:6149a0c9454a4a7602a469e8ca6eda0ebc4a283731080167308a9697be0aa08e. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 3dcc8a4686ae1db4f931da19670c6d4f5e35360b4c8b4e409099e36dbcaeeba0.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: 54e4f803ef2e864ecf1da37bbc319b56642fa5368fcaef466ba178900d8cb3fc.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: e0414f864b09343c80f7cf6924fbfcccfcde6807d17409ca4dd3ca0818dfbc74.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256: 25f9f7cc3d8ee870e68b69f14f1f97871a3a0be0ab9b120c1a79738ae8e8bad6.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: 2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5.

Independent certificate:

sha256: 4c196aeafd9b0d9849ccaa9ff6a014fe3315e99840227633bf0ea9ecbec0e09f. Engine

external-validation hash:

sha256: fa445e1b3181849903d1611aa36d0ad3405bcd6c40252c859ae71266eb1eb983.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 213090faa82308f6a578c468bf789350ae23d79bc26ca90da85020678f8a4415; engine receipt

sha256: e0769c5998c353edbc1b4d0a6df76c0c409591e722297dff3803cabe99e73ab1 at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003-e0769c5998c353ed.json;

empirical-validation hash None; measurement receipt None.

242. Strong-field horizon and finite floor

Claim identity: SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The horizon radius is the Fold of the quarter-One mass, exactly twice the mass; two binary halvings force the quarter area coefficient, and every finite depth retains a positive distance floor so no singular empty distance enters the physical grammar.

Mass $1/4$ has radius $1/2$; entropy coefficient is $1/4$; area 32 has eight boundary records; normalized thermal carrier is positive $1/4$; every finite depth k has distance floor $1/2^k$.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003

Generated grammar and closure boundary. Generate the complete eight-axis product for strong-field horizon and finite floor from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__Fold-mass-radius-quarter-area-and-positive-finite-floor__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	Fold-mass-radius-quarter-area-and-positive-finite-floor	singular-point-or-imported-horizon-equation	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:cb9a8b69637f9ab33a00c5824cbbf3801b7a410e9f7ca918604f964be2f83302`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:7a4c685139dd6cf0d9e7f814bbf57bf2e588bca2480aea881ce0679dda59959a`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4d75656d316adcfde311a5e5a025cf860ef12933bb37fb5b8d3da2c8cf12f669`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8e3af091ba2dc1a355f7bd49af657b9a29bd65eb0dbb02d6e420bc5ee9079606.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:22ef5f6a7305f72d218d9da6e015243777b7edafac6bb9a9a3b8bd2e7865614.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5.
Independent certificate:
sha256:8dfa5a51a288e241dd824d7f2f85c3aa2863c69b14b8deaafc418cc69b286d68. Engine
external-validation hash:
sha256:14c9288cae27098ff04e1269768f66df859d74bdbc6e2324ce25480bd1bc3bd9.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger
- no claim that normalized quarter-One alone fixes a measured black-hole temperature in kelvin

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:b722ecbf789c1bbc44be3a51f17133c93acc9f519572e5e27c93156ba959c6db; engine receipt
sha256:3c02d56ccea9efaed7a9fc9f065e27145f2b6597504a447e3b4ac623e61cc6af at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003-3c02d56ccea9efae.json;
empirical-validation hash None; measurement receipt None.

243. Horizon information closure and reconstruction

Claim identity: SFT-PHYS-GRAVITY-HORIZON- INFORMATION-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Local horizon observation closes access to the predecessor as an empty record, while the complete boundary-cell and inverse-process records retain the distinctions required for global reconstruction; loss and conservation are therefore observation-relative parts of one ledger.

The 32-cell horizon at coefficient 1/4 retains eight boundary records; local predecessor access is empty, and reconstruction is lawful exactly when the complete inverse record is held.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-QUANTUM-MEASUREMENT-001

Generated grammar and closure boundary. Generate the complete eight-axis product for horizon information closure and reconstruction from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__local-closure-with-complete-boundary-inverse-record__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	local-closure-with-complete-boundary-inverse-record	absolute-destruction-or-unrecorded-reconstruction	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:8145ca63db017e3b2ed21f59e936062e32be7b81cd6c9da2648593352aee1f53`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ebc1967b3bc60b951b615326a4f86ffe446b397698fad8f4b208f81866d08b95`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:26aa4548c05926d4a8bb0ffa996396f36dc3301fa5c5bda6076932217bce61bf`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7672c01641308dd654c1ebee7de58faeb02a2534d0b22983804f0023133f8ac5`.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : a79b2b8cb938b2dfedf3e8a03358244b2e3bf2200802338557605e4207cf2177.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5.
Independent certificate:
sha256 : 104954c35efa854449d14b8828bdca7e2471cc95ba46e53d2e860a119ab65a30. Engine
external-validation hash:
sha256 : b44cecbfb46d816b6c808929529cea33e78d5ddf9c72b848dea9926fd7e04812.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : d578e7a1a921310db0c147efd4ec00f3b615bcbdb2778ead49ce42d1b7cccf823; engine receipt
sha256 : db27fe2ffddb49475498a04db13944c1443b625794229c3b222676e143520a2 at receipts/engine/model_admitted/SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003-db27fe2ffddb494.json;
empirical-validation hash None; measurement receipt None.

244. Wormhole admissibility boundary

Claim identity: SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A shortcut relation is physically admissible only when both mouths and the link are generated adjacent support with a positive source and complete causal trace; a named remote identification without that construction is rejected.

The current grammar forces the admissibility conditions but does not force an observed or constructible wormhole; positive source plus generated link plus complete trace is accepted, and every omission is rejected.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001

Generated grammar and closure boundary. Generate the complete eight-axis product for wormhole admissibility boundary from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generated-exact-Fold-carrier__admitted-V3-root-trace__positive-source-generated-link-complete-causal-trace__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	positive-source-generated-link-complete-causal-trace	declared-remote-mouth-equivalence	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:44ee49488de167b953bbe36a675d7fd32eabc14820a9aa77dc6dd2bd51e5fca1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:442b7fc900478a517577c7bc8d1a77fe4bdabd594e71fbc6f26034054815bd33`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:f8e7957d5b934b9fc99cbb7bfff00adac033933d50e28028af7840271e9d60272`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:693c48f320606f49532921cea4a4f4367e1e6957b6c6989d66edf72cb5af04f1`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:83e769e2435711ad1f4afb3e408cb2e662f36e0910562fdc2131bb2fed287c33`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:0b323d1864572b705051ef560561f938544549d47d614e4d33a00dce12976bfff`. Engine

external-validation hash:

sha256:e245ab0be905c2faf0a57d1a75b87fdfbcb49dfb49098f0fc2d65ce7ed0e1f41.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger
- no claim that an astrophysical or engineered traversable wormhole has been observed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:67c1666443b8ed04ce9307ab397b0928be0efccced8444bdafec9817c69d7cc7; engine receipt
sha256:ad75329a73a9f31385c1e0e991770bed8415bafa318be84afb8f278ca322d96d at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003-ad75329a73a9f313.json;
empirical-validation hash None; measurement receipt None.

245. Warp-support admissibility boundary

Claim identity: SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A propagation-support redistribution is admissible only as a positive conserved transfer whose destination carrier equals the source carrier and whose apparatus/source return ledger is complete; negative energy and unrecorded net work are not generated.

The current grammar forces the admissibility ledger but no achieved warp device: equal positive source/destination with complete return passes; changed support or incomplete return fails.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003

Generated grammar and closure boundary. Generate the complete eight-axis product for warp-support admissibility boundary from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__posi

tive-conserved-support-redistribution-with-return-ledger__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	positive-conserved-support-redistribution-with-return-ledger	negative-energy-bubble-or-open-work-ledger	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:3cd2419fa83378d59d112007eab55513bdb45dc4dbef0332c8aa1a20660cb5aa. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:a6668bb1deb89cb401deada96196fe3f0b7e3a0dca9858d7046dfbb006fc4196.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:97c5713c84ffffb19408b8d6f30e624c16a4ba2d4de2613edb6e5832df09a095b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:2d26588bd043dc2b7f89dff1cd9f95976fe18ab0bfd0de82ccf4a6bfe85925d6.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:b3eb8a8389ff47a26d4620a363c33a77cebea423f7895c966768a33015d800c3.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5. Independent certificate:

sha256:89e785b01dc67bccb19cb3b90a4c67d513bd1558f170b6e0eae9843e0cfaed87. Engine external-validation hash:

sha256:754bd2a1b36e60d3ce3d296ded8a55b2143b98c7e9fc6de7850c39a1116863b3.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger
- no claim of observed superluminal transport or engineered spacetime distortion

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d190525037fcae71f32029e1d2d030ea68ffcd51ca42aa7cc7fdf2f0b69cf098; engine receipt
sha256:b864f95be8c08840d7e5e99ab664e84404d6121fa17385ed3fe751d5a53e5ab9 at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003-b864f95be8c08840.json;
empirical-validation hash None; measurement receipt None.

246. Closed-timelike recurrence admissibility boundary

Claim identity: SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A closed causal recurrence is consistent only when it returns the complete physical and proof record exactly; any self-ancestor cycle that changes its own retained premise has no fixed record and is rejected.

Exact recurrence of the complete state is admissible as a cycle; changed self-ancestor, missing trace or answer-altering loop is not. No physical closed timelike curve is forced by this boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-COMP-CBL-HALTING-001

Generated grammar and closure boundary. Generate the complete eight-axis product for closed-timelike recurrence admissibility boundary from carrier, dependency, relation, enumeration, minimality, measurement direction, record and extension. The declared exact boundary is: Every finite exact carrier and causal/source ledger named by this law together with the complete product of its eight registered binary form axes. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: generated-exact-Fold-carrier__admitted-V3-root-trace__exact-first-return-of-complete-state-and-proof-record__complete-registered-product__every-omission-rejected__formal-result-sealed-first__complete-causal-source-control-record__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	generated-exact-Fold-carrier	imported-spacetime-object	The object has no generated Fold provenance.
dependency	admitted-V3-root-trace	asserted-prior-answer	A prior answer is not a V3 premise.
relation	exact-first-return-of-complete-state-and-proof-record	changed-self-ancestor-paradox	The alternative loses conservation, causality or a required record.
enumeration	complete-registered-product	selected-neighbourhood	A selected neighbourhood cannot prove uniqueness.
minimality	every-omission-rejected	uncontrolled-omission	An omitted carrier may silently choose the answer.
measurement	formal-result-sealed-first	target-visible-before-seal	Target access reverses the empirical direction.
record	complete-causal-source-control-record	answer-only	An answer cannot reproduce its causal/source ledger.
extension	no-extra-rule	free-extra-rule	An extra selector is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:16319bd99f53812a381b0aae9cdaf0e89813d4d36d651870ea00e232a89af879`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:9957180f4f1faba948d8c4074204dcdcc8a92a4bc2917eaa35c8f390b44eb0bf`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:fd47b425c66c6b1ee1c7039b67cfd9a5623660eb8660367860cc8a4b9b8739e1`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9c30db967de953e4f00529d924afccc03b9a91a1fc08cc48391f64a254ec1337`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:5812ff52b9b2bc581092d182f10152473f49eec88a92f6bf3574b4414b8424bf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2e13056da97f92c9af10acccb2d573e678d2faabfa7ab883af79cc858aea47c5`. Independent certificate: `sha256:1b6adb99315a575631f149094dde7dc5f46a2549c73c970874eb9ff3433d254b`. Engine external-validation hash: `sha256:f24538578fffb99077f9087b2b24b885ece06b008703e45300c0de79dce8f6980`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, certificate, answer table or conventional metric equation as a premise
- no external measurement, astronomy image or target value selecting a formal survivor
- no fitted coupling, chosen curvature, tunable horizon scale or imported normalization
- no semantic numerical zero, negative, irrational, imaginary or floating proof quantity
- no ungenerated continuum, completed infinity, unrecorded remote adjacency or omitted source ledger
- no claim that a physical closed timelike curve has been observed or constructed

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:9bdb9f8c120d617b2cab104453a64776034f03b12b936fb070f853cbd129420b; engine receipt
sha256:101b8251369135bcf188c580c15f045d3f54cc502ad46813ded8b478b60b19cb at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003-101b8251369135bc.json; empirical-validation hash None; measurement receipt None.

247. Exact gravitational self-source fixed point and correction contraction

Claim identity: SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Binary support and generator-three colour force matter source seven-sixteenths; half-One coupling and positive quadratic field feedback force the exact map $f_{\text{next}} = \text{half-One times (seven-sixteenths plus } f \text{ squared)}$, whose sole admissible Fold-part fixed point is quarter-One and whose positive errors and added corrections contract at every successor.

The exact Fold scalar self-source map $f_{\text{next}} = \text{half-One times (seven-sixteenths plus } f \text{ squared)}$ has one admissible fixed point, quarter-One; the larger seven-quarters root lies outside the Fold-part domain, and every positive error and added correction contracts at every successor by an exact factor below quarter-One.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-HALF-ONE-001
- SFT-MATH-SELF-SIMILAR-CONVERGENCE-002
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of every structural, free or target-assigned source; every forced, free or measured coupling; every quadratic, linear or target-assigned action; every admissible, larger-root or measured fixed point; every exact, finite-only or assumed convergence law; every Fold-scalar, universal-GR or historical-only scope; both target custody states; and both extension states. The declared exact boundary is: The exact self-source scalar channel generated by binary support, generator-three colour, half-One coupling, positive Fold parts, quadratic field-energy feedback and

every positive iteration depth. The theorem does not claim numerical convergence of every post-Newtonian expansion for every physical source. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `binary-colour-source-seven-sixteenths__forced-half-One__matter-plus-field-square__unique-admissible-quarter-One__exact-error-and-correction-contraction__exact-Fold-scalar-self-source-channel__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
source_construction	binary-colour-source-seven-sixteenths	free-matter-source	Rejected by computed Fold predicates: source.
coupling	forced-half-One	free-coupling	Rejected by computed Fold predicates: coupling.
nonlinear_action	matter-plus-field-square	linear-no-self-source	Rejected by computed Fold predicates: action.
fixed_point	unique-admissible-quarter-One	select-larger-root	Rejected by computed Fold predicates: fixed.
convergence_law	exact-error-and-correction-contraction	finite-prefix-only	Rejected by computed Fold predicates: convergence.
scope	exact-Fold-scalar-self-source-channel	universal-general-relativity-convergence	Rejected by computed Fold predicates: scope.
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target.
extension	empty-extension	free-convergence-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:b443f34a942066bb6f369cf46db12084b53e1e6a11c34fec8885781148f21e6c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject larger algebraic root as a Fold part.; observed Seven-quarters exceeds the One.; receipt `sha256:9d321bdf55b3a09a66ffa3205e3e797a328bdcc871aba5a6ea04bdb48f5b80`.
- **tampered_source:** passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- **tampered_artifact:** passed; expected Reject duplicate candidates.; observed All identities unique.; receipt `sha256:5f9c0ad545be40ca825517099110bae199281727e06768826c0280cdfae9b73b`.
- **boundary:** passed; expected Reject universalization, target access and correction.; observed Only scalar scope, sealed target and empty extension survive.; receipt `sha256:66585422a19fd2f9635ec272fe744036151e2d1526d840a8a0bc556374ef3340`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:69137be32b8e94461a24e7d56ced4562972f567b11309e82700fc850779156d7`. Independent certificate:

`sha256:962c960e94d9be73927d46a2bea89e53280cad82bcffa4a8669cd87698c6d97f`. Engine external-validation hash:

`sha256:56c9d7f6b1d51e58026b50449cf798adcc8516c3b9f902df8c30382ab37575a3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- D9M-POST-NEWTONIAN-FIXED-POINT-COMPARISON

Observed comparison records:

- The V1, V2 and primary PN review identities are hash-locked and opened only after the V3 prediction seal.
- V3 independently forces source seven-sixteenths, coupling half-One, admissible fixed point quarter-One and excludes the seven-quarters algebraic root.
- Ten exact successor steps retain positive increasing fields below quarter-One while every error and every added correction shrinks by a factor below quarter-One.
- The complete V1 correction vector seven-fiftieths, seven-one-hundred-fiftieths, seven-thirteen-hundred-fiftieths and seven-one-hundred-nine-thousand-three-hundred-fiftieths is retained and strictly decreasing.
- The V2 self-source map, matter source, coupling and fixed point are reproduced exactly.
- The external review retains nonlinear iteration, the linear-plus-quadratic 1PN scalar structure, weak-field slow-motion near-zone scope, exterior matching and formal any-order hierarchy.
- External formal iteration is not inflated into a claim that every physical PN series converges numerically; the depth-independent convergence proof belongs to the exact registered Fold scalar map.

Falsification condition: Reject if the V1, V2, PN or source-record identity changes; if any historical correction, map, source, coupling, fixed-point, external scheme, nonlinearity, scalar structure, domain, order or scope-limiting row is omitted; if the independently reconstructed map has another admissible fixed point; if any error or correction fails to shrink; if external formal iteration is misreported as universal numerical convergence; if a fitted parameter or free correction enters; or if targets enter before sealing.

Measurement receipt: sha256:527fd031517257181bd22ccd2b9152f5a09ce8690dd0917818c00961f784f0e6.

Isolation certificate: sha256:6bc2a7ca92ef2d92b9dc7155422d08af6813762a233c210736ec0d7e98504deb.

Custody certificate: sha256:0a13b5c88cd7098d69630b3a31f007426e333b7cff0430648736e47321fb6ff9.

Registered source descriptions:

- BLANCHET-POST-NEWTONIAN-THEORY: <https://arxiv.org/abs/1310.1528>
(sha256:c3b1b7c454e15b65bade4245a8e6d61b6e5c0546baaee436a17b99ef6983b4a8)

Registered target identities:

- D9M-WITHHELD-HISTORICAL-AND-PN-RECORD from D9M-POST-NEWTONIAN-FIXED-POINT-COMPARISON

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- historical correction selected law
- external target selected law
- free matter source
- free coupling
- larger non-Fold root
- assumed universal series convergence
- fitted correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:f84ebef827868c003789aad44f9088adf3c06844f66bbaecb17d26ee53ca2c48; engine receipt
sha256:b0e30a2cb8fa9d9f2232df8f8790ab82d8fc2f8e6555326543e15fce2fb48bba at receipts/engine/
model_admitted/SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009-b0e30a2cb8fa9d9f.json
; empirical-validation hash
sha256:ca10b7b14f51a35455ae8ad25496eb963f4e4f9c9da7fe69085349663f8f364d; measurement receipt
sha256:527fd031517257181bd22ccd2b9152f5a09ce8690dd0917818c00961f784f0e6.
```

248. Four-coordinate symmetric curvature-source conservation law

Claim identity: SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Four generated coordinate labels force ten symmetric rank-two slots paired one-to-one with one energy, three momentum and six stress source carriers; generated coordinate shifts force four exact carried/opposed divergence-ledger identities at every finite depth, and deletion of one flow term breaks local conservation.

Four generated coordinate labels force ten symmetric curvature-source slots partitioned into one energy, three momentum and six stress carriers; each of four generated commuting-difference divergence directions has exactly matching 48-term carried and opposed ledgers at every finite depth, while removal of one flow term breaks conservation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003
- SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003
- SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009
- SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003
- SFT-PHYS-SPACETIME-EXACT-INTERVAL-003
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of every complete, scalar-only or target-assigned component organization; every componentwise, aggregate or target-assigned field pairing; every generated-shift, imported-continuum or target-assigned derivative algebra; every exact, asserted or target-assigned conservation identity; every complete, unlabelled or target-assigned source ledger; every detecting, ignored or target-assigned leak test; both target custody states; and both extension states. The declared exact boundary is: Every exact symmetric rank-two carrier over four generated coordinate labels, its complete one-energy/three-momentum/six-stress source ledger, and every finite product-lattice depth whose held coordinate shifts commute. This theorem is the generated commuting-difference curvature channel, not an assertion that every nonlinear lattice discretization has the same operator form. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `complete-ten-symmetric-source-slot_s__one-curvature-slot-to-one-source-slot__commuting-generated-shift-differences__four-exact-opposed-ledger-identities__energy-one-momentum-three-stress-six__missing-flow-breaks-exact-balance__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
component_organization	complete-ten-symmetric-source-slots	scalar-source-only	Rejected by computed Fold predicates: components.
field_pairing	one-curvature-slot-to-one-source-slot	aggregate-source-equation	Rejected by computed Fold predicates: pairing.
derivative_algebra	commuting-generate-d-shift-differences	imported-continuum-derivatives	Rejected by computed Fold predicates: derivative.
conservation_identity	four-exact-opposed-ledger-identities	asserted-conservation-only	Rejected by computed Fold predicates: conservation.
source_ledger	energy-one-momentum-three-stress-six	unlabelled-ten-slots	Rejected by computed Fold predicates: source.
leak_test	missing-flow-breaks-exact-balance	leak-ignored	Rejected by computed Fold predicates: leak.
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target.
extension	empty-extension	free-conservation-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:ab511510c585b97e523ac81b1b37b516dfe1451ee07af973d78e0c3f348f0b19`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject a source ledger missing one opposed flow term.; observed The exact multisets no longer match.; receipt `sha256:49286d6288e8c1870761485f13ecfe4cd14de2785e6d46532d0acecb723741f4`.
- **tampered_source:** passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- **tampered_artifact:** passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt `sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049`.
- **boundary:** passed; expected Reject imported operators, pre-seal observation and free correction.; observed Only generated shifts, sealed targets and empty extension survive.; receipt `sha256:c29f33fe86181bd8646bae70e39a16e747ed8c9f6692072c73003c3ae9d1b8db`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d1eef07aacb5859461d3e13ba7ed0524f5385d5ab68850051665a349b5fe85df`. Independent certificate:

`sha256:b2f4684dc85a52b77caffa778654275b3d240a9974dda285c7133bdd09cf738a`. Engine external-validation hash: `sha256:943f4901d70ba986a43a84cd7ebdb491d98abe58deaaddbe7c809bac04e50606`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- D9N-SYMMETRIC-SOURCE-CONSERVATION-COMPARISON

Observed comparison records:

- The V1 D9n observation, admitted polarization certificate and all three complete primary documents are hash-locked and opened only after the V3 prediction seal.
- V3 independently forces ten symmetric slots, one energy plus three momentum plus six stress source carriers and a ten-pair componentwise field/source ledger.
- Each of four held source directions expands to 48 carried and 48 opposed terms with 31 distinct canonical derivative terms; every exact multiset matches.
- All 64 three-shift words commute under canonical held-coordinate ordering, and the successor proof extends the identity to every finite product-lattice depth.
- Removing one opposed flow term breaks balance, reproducing the adverse leaking-source boundary.
- Carroll's complete continuum rows retain the symmetric source and geometric tensors, contracted identity, componentwise field equation and source-conservation consequence.
- Hamber-Kagel's exact discrete identity, four contracted identities, ten equations, four gauge conditions, two physical degrees and arbitrary-curvature validity are retained together with the crucial fact that its general curvature relation is nonlinear and not identified with V3's operator.
- GW170817 retains six generic metric modes, two GR tensor modes, the sky-constrained log-Bayes comparisons 20.81+/-0.08 and 23.09+/-0.08, and the limitation to pure-mode hypotheses; no mixed-mode result is claimed.
- No historical, conventional or measured value is fitted into the formal V3 law.

Falsification condition: Reject if any historical, prerequisite, primary-theory, discrete-gravity or GW source identity changes; if any of the thirty-two registered favorable or limiting rows is omitted; if the formal component count, one-three-six source partition, componentwise pairing, four conservation ledgers or leak rejection fails; if the exact Regge identity is conflated with the V3 commuting-difference operator; if the pure-mode GW comparison is inflated into a mixed-mode result; if a measured or fitted value enters the V3 proof; or if targets enter before sealing.

Measurement receipt: sha256:34d28a05fe1a7fcfb4fd61bb56d7b4a0977ed55509006d5f77e994f3e78793f0.

Isolation certificate: sha256:33e0147f88792f18ad853089aa8a29b889d7bb71bdd011ee5e0adef16cd16dfa.

Custody certificate: sha256:44007b0d51691a1ea8a63709eb6b5bc5b30164bf871ef5fcb0d69fab15592d9f.

Registered source descriptions:

- CARROLL-GENERAL-RELATIVITY-NOTES: <https://arxiv.org/abs/gr-qc/9712019>
(sha256:d4cbfeed3a18370cb7e1a256da6fa7de722982ad7f6a146b4fc0fe14a6bebf6c)
- HAMBER-KAGEL-EXACT-REGGE-BIANCHI: <https://arxiv.org/abs/gr-qc/0107031>
(sha256:98499dc16e8963c56be885911f4a0476036f97607f01dfb221ec1c9bc85f6d6c)
- LIGO-VIRGO-GW170817-GR-TESTS: <https://dcc.ligo.org/LIGO-P1800059/public>
(sha256:d86fc8bc15abdac47b75960218aefd247f7adeddc00237e9c329b492962584b7)

Registered target identities:

- D9N-WITHHELD-HISTORICAL-THEORY-AND-GW-RECORD from
D9N-SYMMETRIC-SOURCE-CONSERVATION-COMPARISON

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- continuum equation selected operator
- historical target selected components
- aggregate scalar source
- asserted conservation

- ignored leaking source
- signed numerical proof scalar
- fitted measurement
- free correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d431ad929bc7f92e5dbefee79f3035e70ea388ffcb3a410c4200932ef6096424; engine receipt
 sha256:79b12fc32760832dc52e87fcf8d743575ebe2f944c18e7cb24cbb41a6d8a5b22 at receipts/engine/model_admitted/SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010-79b12fc32760832d.json; empirical-validation hash
 sha256:ec7c1dffc9a180ff3f16ed4ab83a2919d2642fb06239e9288d65aff94393cd45; measurement receipt
 sha256:34d28a05fe1a7fcfb4fd61bb56d7b4a0977ed55509006d5f77e994f3e78793f0.

249. Exact static spherical exterior coefficient, flux and clock law

Claim identity: SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The generated boundary is twice the positive mass carrier; rank-two conservation forces field equal to boundary over radius squared and well share equal to boundary over radius; its positive One-complement is the exact exterior clock-square coefficient, with conserved flux, finite outward successor closure, empty boundary record and farther exact-square clocks running faster.

The doubled positive mass carrier fixes the exterior boundary; rank-two conservation forces field equal to boundary over radius squared and well share equal to boundary over radius; its positive complement from the One is the exact static clock-square coefficient, with empty boundary record, conserved flux at every finite radius and farther exact-square clocks running faster.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003
- SFT-PHYS-GRAVITY-STATIC-CLOCK-003
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003
- SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of every generated, free or target-assigned horizon normalization; every conserved inverse-square, free-power or target-assigned field; every radius-derived, independent or target-assigned well; every positive complement, signed import or target-assigned coefficient; every exact-square, unrestricted-root or target-assigned clock; every finite-successor, completed/inward or target-assigned boundary; both target custody states; and both extension states. The declared exact boundary is: Every exact positive mass whose generated doubled boundary does not exceed the One, every exact radius at or outside that boundary, and every finite outward binary successor. The coefficient theorem

covers every exact exterior radius; numerical clock rates are evaluated only on generated exact-square carriers, and the boundary is an empty clock record rather than a numerical zero. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `twice-mass-generated-horizon__inverse-square-with-conserved-horizon-flux__radius-times-field-gives-horizon-over-radius-share__positive-One-complement-of-well__exact-square-carriers-and-empty-horizon-record__finite-successor-flat-boundary-and-vacuum__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
horizon_normalization	twice-mass-generated-horizon	free-radius-scale	Rejected by computed Fold predicates: horizon.
field_low	inverse-square-with-conserved-horizon-flux	free-field-power	Rejected by computed Fold predicates: field.
well_low	radius-times-field-gives-horizon-over-radius-share	independent-potential-profile	Rejected by computed Fold predicates: well.
coefficient_low	positive-One-complement-of-well	signed-subtraction-import	Rejected by computed Fold predicates: coefficient.
clock_low	exact-square-carriers-and-empty-horizon-record	unrestricted-irrational-root	Rejected by computed Fold predicates: clock.
boundary_low	finite-successor-flat-boundary-and-vacuum	completed-infinity-or-interior-extension	Rejected by computed Fold predicates: boundary.
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target.
extension	empty-extension	free-exterior-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:1439d8057d90d99b24bb7e9544bf9818d939eb2902a0d94382f30ad91f23f918`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject a numerical boundary coefficient or exterior clock record at complete well depth.; observed The boundary is the empty structural record.; receipt `sha256:41b03ba80ef3029577651ebe4fa916e917f1e0176ab802661359e2becc45ca73`.
- **tampered_source:** passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- **tampered_artifact:** passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt `sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049`.
- **boundary:** passed; expected Reject signed import, pre-seal observation and free correction.; observed Only positive complement, sealed target and empty extension survive.; receipt `sha256:d1ecae9cbfce14712e209bda40343a65dc3218f4e5fd75074cefd7d08432b29b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:811cdfabd0655a1eb9a41beab5e44a584ccfc67eec401c7b3868bd7e862419a9`. Independent certificate: `sha256:75f710b586b5bd2577261040cb9cd21ee902d18e4926b7b1c56888eb8285bbf4`. Engine

external-validation hash:

sha256:14f497c68b2096e266e353ccb8bf66a89d1dbcf0da7a80cb313d08f26b10ef92.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- D9O-STATIC-EXTERIOR-CLOCK-COMPARISON

Observed comparison records:

- The V1 D9o row, five admitted V3 certificates, complete primary exterior-theory paper, complete NIST/JILA paper and earlier gravity vector are hash-locked and open only after the V3 prediction seal.
- V3 independently forces normalized horizon half-One from mass quarter-One, coefficient equal to the positive complement of horizon-over-radius, and empty clock record at the boundary.
- Across twelve exact outward successors, well share halves, field quarters, coefficient increases and field times radius squared remains the identical horizon carrier.
- Exact-square near and far coefficients nine-sixteenths and sixteen-twenty-fifths force clock rates three-quarters and four-fifths and far-over-near rate sixteen-fifteenths without an irrational proof value.
- The primary exterior treatment retains spherical-vacuum uniqueness, staticity, source-free equation, the One-take-two-GM-over-radius coefficient, reciprocal radial coefficient, mass/horizon unit restoration, asymptotic flatness and absent spherical radiation.
- The NIST/JILA paper retains predicted gradient $-1.09\text{e-}19$ per millimetre, corrected full-cloud result $-9.8(2.3)\text{e-}20$, corrected split-cloud result $-1.28(27)\text{e-}19$ and fractional uncertainty $7.6\text{e-}21$.
- NIST/JILA observes the higher region ticking faster and reports agreement with gravitational redshift after its full systematic treatment.
- The NIST result does not measure universal Fold fractions, and the external coordinate solution does not select the V3 law.
- No completed infinity, numerical boundary zero, negative or irrational proof scalar, measured fit or free correction enters V3.

Falsification condition: Reject if any historical, admitted-prerequisite, primary-theory, NIST measurement or earlier gravity-vector identity changes; if any of the thirty-four favorable or limiting rows is omitted; if the formal doubled-mass boundary, inverse-square flux, well share, positive complement, empty boundary record, finite successor or exact clock direction fails; if completed infinity, an irrational clock value, a signed/forbidden proof scalar, measured fit or free correction enters; if the external coordinate solution or NIST result is overgeneralized; or if targets enter before sealing.

Measurement receipt: sha256:15a204fcfe5f0dff58a1f20ab334170998568827dde8b70e681adee9e7537d3b.

Isolation certificate: sha256:7ed192061df91d2056009f454223742bfbf166f0683d21c7d9f4ac3f379be309.

Custody certificate: sha256:80ca4da24fb7b4073ba16cad4f86f6d9c1f2476fe1c7fa431fb673dd7b73abd1.

Registered source descriptions:

- HEINICKE-HEHL-SCHWARZSCHILD-KERR-INTRODUCTION: <https://arxiv.org/abs/1503.02172> (sha256:13288f2e0ada228df6e567506c6ac1541e2559a423a740d1e2844cc9a4fe7085)
- NIST-JILA-MILLIMETRE-REDSHIFT-2022: <https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample> (sha256:7d4376361233f17082814d99cc46cd3263c9febdfb273bb9bfbe9699723e2b3)

Registered target identities:

- D9O-WITHHELD-HISTORICAL-THEORY-AND-CLOCK-RECORD from D9O-STATIC-EXTERIOR-CLOCK-COMPARISON

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- imported exterior equation
- free horizon scale
- free field exponent
- signed coefficient
- numerical boundary zero
- completed infinity
- irrational clock rate
- measured fit
- free correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:e5f2822b8dd9d6c891579c00b272e1f47f6f682a772df43caa5e5965f7863e74; engine receipt
 sha256:fca4c68a06026daea19121fd5b71d52bc98e2410be1f83eaf13ff62600d880c at receipts/engine/model_admitted/SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011-fca4c68a06026dae.json;
 empirical-validation hash
 sha256:eca6a4597b64466858c074f25b32e63047298bd700645bf2fc6939183255f393; measurement receipt
 sha256:15a204fcfe5f0dff58a1f20ab334170998568827dde8b70e681adee9e7537d3b.

250. Exact quadrupole third-difference radiated-power law

Claim identity: SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Held source and momentum records close monopole and dipole radiation, leaving the quadrupole as the first radiative moment. One more generated difference gives its radiative rate. The admitted field-energy square and binary half-One coupling force power equal to half-One paired with that third-rate carrier twice. A static quadrupole closes to an empty radiation record, amplitude scaling is exactly squared, and rank-two shell dilution preserves total power at every finite radius.

Held monopole and dipole records leave the quadrupole first; its third generated difference is the radiative rate; half-One times that positive rate paired with itself is total radiated power; static support closes to an empty record, amplitude scaling is exactly squared, and rank-two outward dilution preserves total power at every finite shell.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010
- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

Generated grammar and closure boundary. Generate the complete product of every quadrupole-first, lower-moment or target-assigned moment law; every third-generated, free-order or target-assigned rate; every squared, linear/free or target-assigned energy carrier; every admitted, free or target-assigned coupling; every rank-two conserved, free or target-assigned shell law; every structural-empty, numerical/linear or target-assigned adverse control; both target custody states; and both extension states. The declared exact boundary is: Every finite exact positive quadrupole-magnitude trace generated by a positive cubic tick carrier and positive amplitude, every positive tick index under the all-positive cubic successor identity, and every finite outward binary shell. Static silence is structural emptiness. The theorem is the leading positive-magnitude quadrupole channel; tensor orientation and physical-unit comparison are retained post-seal. The generator produced 2916 named candidates and the decision support contains 2916 one-for-one decisions. Exactly one candidate survived: `held-monopole-and-dipole-leave-quadrupole-first__third-generated-quadrupole-difference__positive-square-of-third-rate__admitted-binary-half-One-coupling__rank-two-density-dilution-with-conserved-total-power__static-empty-and-amplitude-square-scaling__sealed-before-observation-release__empty-extension`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
moment_law	held-monopole-and-dipole-leave-quadrupole-first	lower-moment-radiation	Rejected by computed Fold predicates: moment.
rate_law	third-generated-quadrupole-difference	free-difference-order	Rejected by computed Fold predicates: rate.
energy_law	positive-square-of-third-rate	linear-or-unsigned-rate-energy	Rejected by computed Fold predicates: energy.
coupling_law	admitted-binary-half-One-coupling	free-radiation-coupling	Rejected by computed Fold predicates: coupling.
shell_law	rank-two-density-dilution-with-conserved-total-power	unconserved-or-free-shell-power	Rejected by computed Fold predicates: shell.
adverse_control	static-empty-and-amplitude-square-scaling	numerical-zero-or-linear-scaling	Rejected by computed Fold predicates: control.
target_boundary	sealed-before-observation-release	observation-readable-before-seal	Rejected by computed Fold predicates: target.
extension	empty-extension	free-radiation-correction	Rejected by computed Fold predicates: extension.

Base and successor. The closure proof is sealed by derivation hash `sha256:c8e4934943b15932c31efcff4d375375735fcda0691dde4e2010c45c441b0135`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected Reject numerical-zero static power or linear amplitude scaling.; observed Static radiation is empty and doubled third rate gives fourfold power.; receipt `sha256:e5d34752bf8c897a64aa0754ccdaa54c0d3200eb23175fa89cf22fb2632490cc`.
- **tampered_source:** passed; expected Reject changed source identity.; observed Identity differs.; receipt `sha256:7ebc0eee93d4c1357c2d77d871bfa7ad0c7abf4d7f8d00be54bca5119ebaac83`.
- **tampered_artifact:** passed; expected Reject duplicate candidates.; observed All identities are unique.; receipt `sha256:0b5a37b200f6544185da03decea0e8f27bba78ad0ff859e5920232c0a780f049`.
- **boundary:** passed; expected Reject free derivative order, pre-seal target access and free correction.; observed Only generated third rate, sealed target custody and empty extension survive.; receipt

sha256:b4b67644c9dcf9c22ff5b0902cd3465f324534fcbd35e159a14cc378341037ed.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:0894076545fc7d5a702ce4cb4bd542f03f3fd4188508dda1d1c2feb15c1b0192.

Independent certificate:

sha256:85ed385605982d16e63203b60b40ba5ae092b000287283048bfe1d05002b440c. Engine

external-validation hash:

sha256:980d6f821d33052300a2f84690d44a58e35ba9547927280e702a8199b5de4854.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- D9Q-QUADRUPOLE-RADIATED-POWER-COMPARISON

Observed comparison records:

- The complete V1 D9q row, eight admitted prerequisite certificates and complete 56-page primary Double Pulsar paper are hash-locked and released only after the V3 formal commit and in-run prediction seal.
- V3 independently forces held monopole and dipole records, third generated quadrupole difference, half-One times its positive square, structural static silence and exact amplitude-square scaling.
- The all-positive cubic opposition identity has retained third-rate six at every successor; the exact witness has power eighteen and doubled amplitude has power seventy-two.
- Rank-two outward transport quarters density at every binary radius successor while density paired with radius twice reconstructs the same total power.
- The 16-year PSR J0737-3039A/B record covers a 2.45-hour mildly eccentric orbit and approximately sixty thousand tracked orbits.
- The primary paper retains Shklovskii, Galactic and spin-down mass-loss corrections before reporting measured GW-driven orbital decay $-1.247782(79)e-12$.
- The primary quadrupole prediction $-1.247810(+6/-7)e-12$ and retained 3.5PN correction $-1.75e-17$ give total prediction $-1.247827(+6/-7)e-12$.
- Measured over predicted decay is 0.999963(63): its exact rational uncertainty interval contains the formal normalized One, and the published 95-percent agreement boundary is $1.3e-4$.
- The extreme equation-of-state rerun 0.999958(64), timing-error dominance, future kinematic limit, material-binary regime and compact-object comparison caveats are all retained.
- The external negative decay direction and decimal measurements remain observational records; no measured value, fitted parameter, target-selected survivor or free radiation correction enters V3.

Falsification condition: Reject if the V1 row, any admitted prerequisite, the complete primary timing paper or any of its forty-eight favorable, correction, uncertainty or scope rows changes or is omitted; if the formal quadrupole-first, third-difference, half-One square, static-empty, amplitude-square or rank-two shell identity fails; if the published normalized decay no longer contains the formal One within its registered uncertainty; if the mildly relativistic material-binary result is universalized to every compact-object regime; if any external sign or decimal becomes a formal proof scalar; or if target access, fitting or a free correction precedes closure.

Measurement receipt: sha256:86923e0713e593e62ee8cade4399e2321732f802fceb921aaf303a58707d554b.

Isolation certificate: sha256:61154b4a9ff8917ddf69691cb9ade45de69128efae713376842971d7470f44d.

Custody certificate: sha256:67821312675e15942c10a54436b21309444ef307b8fba8f9e99af0657519e038.

Registered source descriptions:

- KRAMER-ET-AL-DOUBLE-PULSAR-2021: <https://arxiv.org/abs/2112.06795>
(sha256:9d1b5b4dc304b1d6b1b11059a26d96ca97d1c0d43857523058f973c12bddc6a4)

Registered target identities:

- D9Q-WITHHELD-HISTORICAL-PREREQUISITE-AND-DOUBLE-PULSAR-RECORD from D9Q-QUADRUPOLE-RADIATED-POWER-COMPARISON

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- imported quadrupole power equation
- lower-moment gravitational radiation
- free difference order
- free radiation coupling
- numerical static zero
- negative external decay sign as proof scalar
- irrational or imaginary proof value
- measurement-selected survivor
- measured fit
- free radiation correction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d66fcf0085cda7f7086a73cd43170aba452784f91acf319dce4ce84281e43833d; engine receipt
sha256:e43448d651ade64487e4cbd44620c4812ffa7cb62d20bd71c9c3bde51cdb4db6 at receipts/engine/model_admitted/SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012-e43448d651ade644.json;
empirical-validation hash
sha256:fdd1e90d58113fe690623d58a08292113881bfc55f9bb2d3407edcd63b7aedd; measurement receipt
sha256:86923e0713e593e62ee8cade4399e2321732f802fceb921aaf303a58707d554b.

251. Single finite Fold quantum-gravity composition

Claim identity: SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. At every positive finite Fold depth, one shared lattice jointly retains complete quantum support, three spatial directions, a symmetric rank-two gravity carrier with two physical polarizations, empty mass/rest record, One-cell causal advance, a positive distance floor, a finite rational loop sum and a quarter-area horizon ledger; no extra spatial direction or completed infinity is generated.

At every positive finite Fold depth, one shared lattice jointly retains complete quantum support, three spatial directions, a symmetric rank-two gravity carrier with two physical polarizations, empty mass/rest record, One-cell causal advance, a positive distance floor, a finite rational loop sum and a quarter-area horizon ledger; no extra spatial direction or completed infinity is generated.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-PHYS-QUANTUM-EVOLUTION-001

- SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010
- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-PHYS-VALIDATION-FINITE-LOOPS-003
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of shared carrier, dimensional support, rank-two source, polarization reduction, mass/causal carrier, quantum support, finite-loop closure, horizon ledger, comparison boundary and extension forms. The declared exact boundary is: Every positive finite Fold depth on the admitted three-space lattice, its complete binary quantum support, four-coordinate symmetric rank-two source slots, finite exact loop prefix, positive distance floor and complete quarter-area horizon record. Continuum completion is outside the boundary. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `single-finite-Fold-lattice__three-space-plus-one-process-direction__complete-symmetric-rank-two-source__ten-take-four-take-four-leaves-two__empty-mass-record-and-One-cell-per-tick__complete-finite-binary-word-support__complete-finite-rational-loop-prefix__positive-floor-and-quarter-area-record__inherit-sealed-wave-loop-and-horizon-comparisons__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	single-finite-Fold-lattice	separate-quantum-and-gravity-substrates	Separate substrates do not establish a composition law.
dimension	three-space-plus-one-process-direction	chosen-or-extra-dimensional-space	A chosen or added dimension is not generated by the stability theorem.
rank	complete-symmetric-rank-two-source	scalar-or-selected-tensor	A scalar or selected tensor omits the admitted complete source slots.
polarization	ten-take-four-take-four-leaves-two	assigned-spin-or-mode-count	An assigned label does not derive physical degrees.
propagation	empty-mass-record-and-One-cell-per-tick	fitted-mass-or-speed	A fitted mass or speed imports the target.
quantum	complete-finite-binary-word-support	continuum-amplitude-space	A continuum adds ungenerated support.
loops	complete-finite-rational-loop-prefix	completed-infinity-or-counterterm	A completed infinity and correction are outside the Fold grammar.
horizon	positive-floor-and-quarter-area-record	unbounded-or-unrecorded-boundary	An unbounded or unrecorded boundary loses the finite ledger.
comparison	inherit-sealed-wave-loop-and-horizon-comparisons	target-selects-composition	External programmes cannot choose the joint model.
extension	no-extra-rule	extra-dimension-or-free-regulator	Either addition changes the generated model.

Base and successor. The closure proof is sealed by derivation hash

sha256:f096b575fff7c59ef6ae9d4a7f92f9e78b8ca7f901d67a6329218d8003dc4db1. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:bc21475047b4e738c5bdd65495fdad246c91a59c4b3f1db1db7944785e6715af.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:cd38d0775100d18bfc3cf208b6993ae2b2b1443141df70112c6ef0232c1c1522.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a6791b69b5fb4b37c86d8b6c5d774414fb1d896e586f28e383a38937a06801b9.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:610f2dae6849e9463367f5c6e25fea21fd1120b805ea63b4af0ddd004cffabb4.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:50fe71d3ae3dc9f2a84f9956a1fe508b355fef7d5f6e5d71b824dc9753908084.

Independent certificate:

sha256:9c22d51a095f633f83a21d93863618612c1d42c3816c08a4d29efe4dd73f5110. Engine

external-validation hash:

sha256:a3c69b5deb330c8717e1f165a1469a1fbfb5ceecf6016c9a0345ff18f31c4d7.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported continuum quantum field, Einstein quantization or conventional quantum-gravity programme as a premise
- no V1/V2 executable, answer table, measured value or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no completed infinity, fitted counterterm, free regulator or extra dimension
- no claim of direct graviton detection or proof of a continuum Yang-Mills/Einstein completion
- no external comparison before the inherited wave, loop and horizon seals

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:3a1a03aa2db41ddfec0d52aa4cef5f80c5bd2448cfceb8c3c862209bc4c63672; engine receipt

sha256:a627c3db6aff944a007b53df0a06975454cec52c113e672da62a440ebfd191e7 at `receipts/engine/model_admitted/SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023-a627c3db6aff944a.json`;

empirical-validation hash None; measurement receipt None.

252. Exact positive Fold velocity-composition law

Claim identity: SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The unique exact same-direction Fold composition is $(u+v)/(One+uv)$. Typed rest is a two-sided identity; the limiting One is absorbing; positive sublimit parts remain strictly sublimit; pair multiplication makes the law associative and reconstructible; and the complete identity-compatible bilinear census contains four candidates with exactly one survivor.

The unique exact same-direction Fold composition is $(u+v)/(One+uv)$. Typed rest is a two-sided identity; the limiting One is absorbing; positive sublimit parts remain strictly sublimit; pair multiplication makes the law associative and reconstructible; and the complete identity-compatible bilinear census contains four candidates with exactly one survivor.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-EXACT-OPERATIONS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-MECH-SPEED-VELOCITY-001
- SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001

Generated grammar and closure boundary. Generate the complete product of motion domain, typed rest, forward/held representation, identity-compatible numerator cross term, identity-compatible denominator cross term, limiting-speed behavior, closure, associativity, held-record reversibility, empirical custody and extension form. The declared exact boundary is: Every exact positive rational speed part through the limiting One; typed rest; typed empty held part; both present/absent choices for the bilinear cross term after two-sided identity forces the left, right and denominator-One terms; every exact triple composition; and no signed or continuum proof scalar. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: `exact-positive-parts-and-held-direction__typed-rest-state__forward-held-pair__cross-term-absent__cross-term-present__limiting-One-is-absorbing__strictly-inside-One__exact-associative-product__retain-forward-and-held-parts__postseal-only-comparison__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
domain	exact-positive-parts-and-held-direction	imported-real-line-velocity	A continuum imports the target model.
rest	typed-rest-state	numerical-zero	Numerical nothing is not an admitted physical part.
representation	forward-held-pair	single-unrecorded-scalar	One scalar discards the reverse distinction.
numerator	cross-term-absent	cross-term-present	It prevents the limiting One from remaining fixed.
denominator	cross-term-present	cross-term-absent	Then composing the limiting One with motion exceeds the One.

Axis	Forced form	Eliminated form	Elimination reason
limit	limiting-One-is-absorbing	limit-moves-or-is-exceeded	That violates the admitted causal boundary.
closure	strictly-inside-One	unbounded-sum	Ordinary addition can leave the domain.
associativity	exact-associative-product	order-dependent-rule	Frame grouping cannot change one physical relation.
record	retain-forward-and-held-parts	discard-held-part	Discarding the held part makes reversal ambiguous.
measurement	postseal-only-comparison	Fizeau-readable-before-seal	A measured fringe slope could select the survivor.
extension	no-extra-rule	free-coefficient-or-correction	A coefficient would be a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:e223da6d28c5b7d688dad577ff84b1390f312761a376a600f0ed1e14502193dd. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:1a79d62f8ac2f0a5fe2ffe1d3f2dca509e9f18574bc167ecdb1dab811e6204de.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:5813a6c7d53a6d4e3115124335f2123031a0e24752459a9b03375bb0c7ec5938.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e38d5fdee7058292ad1eadfc03d800a6406911159ddd0706be07124cb04bf09e.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:1260794679b17e38744d0f6ff5ab19124bbd9d70cf8176ba23e0e66d5f86bb13.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:aae0a1d576d4398f4ec1cc2d1bae82141125702742896d55b4effd574b61e078. Independent certificate: sha256:99a423ca52573da9a5770fac70fb2414a1e0949e4acb826714bf285e58b5458f. Engine external-validation hash: sha256:c9cf1eef018a668be095cd8d378c4ceda7d99d0ccd70b166758188ed9b250d85.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, stored velocity-addition formula or measured optical result in the forcing runtime

- no numerical zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof value
- no imported Lorentz transform, rapidity, hyperbolic function or consensus kinematic equation as a premise
- no target-selected coefficient, fitted denominator or hidden premarked survivor
- no claim that the same-direction scalar law alone supplies the full non-collinear rotation law

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1ae6884e89287d60530400baa6c2765a3339fc5329619cc83d6270e09516fada; engine receipt
sha256:337fccb78afd2207d9976e8034d89d1db75ba9fed2f63c44a3b36ae0e90935a0 at receipts/engine/model_admitted/SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033-337fccb78afd2207.json; empirical-validation hash None; measurement receipt None.

253. Terminal compact-object and horizon-thermodynamics law

Claim identity: SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every positive whole q , complete exclusion packing has q^3 occupants, forced momentum depth q , relativistic support q^4 and gravitational paired support q^6 ; q^6 is strictly greater after the exact base. The loaded exclusion threshold is $3/4$ and its Fold balance is $1/2$. Exactly two pre-horizon support families exist. For every positive exact mass m , horizon radius is $2m$, area support is $(2m)^2$, entropy support is one quarter of that area, and temperature is $1/(16m)$, preserving exact product $mT=1/16$. At $m=1/4$ the admitted radius and temperature are respectively $1/2$ and $1/4$. Every finite halving emission trace retains positive mass, temperature, area and information support; no numerical zero or completed infinity enters.

For every positive whole q , complete exclusion packing has q^3 occupants, forced momentum depth q , relativistic support q^4 and gravitational paired support q^6 ; q^6 is strictly greater after the exact base. The loaded exclusion threshold is $3/4$ and its Fold balance is $1/2$. Exactly two pre-horizon support families exist. For every positive exact mass m , horizon radius is $2m$, area support is $(2m)^2$, entropy support is one quarter of that area, and temperature is $1/(16m)$, preserving exact product $mT=1/16$. At $m=1/4$ the admitted radius and temperature are respectively $1/2$ and $1/4$. Every finite halving emission trace retains positive mass, temperature, area and information support; no numerical zero or completed infinity enters.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-VACUUM-ODD-RECURRENCE-003
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023
- SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069

Generated grammar and closure boundary. Generate the complete twelve-axis product of occupation, momentum, support scaling, gravity scaling, threshold, endpoint-family, horizon-radius, area, entropy, thermal, finite-floor and extension forms. The declared exact boundary is: Every positive finite perfect-cube exclusion support and successor; the complete binary endpoint fibre;

the admitted quarter-One horizon witness; every positive exact mass carrier; every finite binary emission depth; and the complete twelve-axis alternative product, with dimensional observations inaccessible. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `one-fermion-per-generated-cell__cube-side-momentum-depth__occupants-times-forced-depth-q4__paired-source-q6__three-quarter-preimage-to-half-One__exactly-two-binary-fibre-families__one-Fold-mass-doubling__rank-two-radius-pair__quarter-area-boundary-record__fixed-thermal-mass-product__positive-finite-floor-at-every-reached-depth__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
occupation	one-fermion-per-generated-cell	multiply-occupied-cell	That violates the admitted exclusion law.
momentum	cube-side-momentum-depth	selected-momentum-scale	A selected scale is a free parameter.
support	occupants-times-forced-depth-q4	imported-pressure-equation	An imported equation cannot select the Fold law.
gravity	paired-source-q6	linear-unpaired-source	Gravity couples the complete source support to itself.
threshold	three-quarter-preimage-to-half-One	chosen-dimensional-mass	A measured mass cannot choose a structural threshold.
endpoint	exactly-two-binary-fibre-families	open-ended-remnant-list	An open list is not generated by the binary fibre.
radius	one-Fold-mass-doubling	imported-metric-radius	A conventional metric is not a premise.
area	rank-two-radius-pair	volume-support	A volume count contradicts boundary rank two.
entropy	quarter-area-boundary-record	selected-entropy-coefficient	A selected coefficient is an extra rule.
thermal	fixed-thermal-mass-product	constant-or-fitted-temperature	A constant or fitted temperature loses the scale recurrence.
floor	positive-finite-floor-at-every-reached-depth	numerical-zero-or-completed-infinity	Neither is an admissible Fold proof form.
extension	no-extra-rule	free-family-or-correction	An ungenerated addition destroys uniqueness.

Base and successor. The closure proof is sealed by derivation hash `sha256:56d27e2bf2bfa6f84307356e39c47f3e504c62df9abf7b2308422a1b55e2b02`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:0878afc8bf2c2c8f737679a1c61c32e38401807d20596f05b5f47dc537e086a4`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:50471e1307e0c04b3cf08a18994d9af7ecaa4ef648402663b012ab188253ab85`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6910b46decdbdf4d41ac76a288239eef5b0ca41b5d6e8e3a7b9fa96066e7c3f5a`.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : b6638c8870605306c719cd98a4b5ccdd7c66c10d1dafd55134832de03f8d1977.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256 : dec2163b532cbc88444fee5bb33e4db12b914fe99dbc46cad6b9eddbce6e3a676. Independent certificate: sha256 : ee8bdb300a7efc9e7e3f2bed94d58781b80605853c27c1f8e06fb0f523da8bce. Engine external-validation hash: sha256 : d1a6ef1df492186113ff9b23ce189642cc691394402e6275109743177960b4d1.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, conventional compact-object equation, catalogue mass or Hawking target in formal execution
- no fitted mass limit, equation-of-state knob, temperature calibration, area normalization or evaporation correction
- no numerical zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinite proof scalar
- no claim that a normalized Fold temperature alone is a direct kelvin observation
- no third pre-horizon family, hidden support source, volume entropy or information deletion

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 244b4f8381f20d52b62cb6b81bb90e001fcdb6aecf409046330a78c7d7b4c6fd; engine receipt sha256 : 69152da01b20c6b1a7ab6ecfab4e44cf836d588cba4ea6f3b02d5827cb5a0b3f at receipts/engine/model_admitted/SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071-69152da01b20c6b1.json; empirical-validation hash None; measurement receipt None.

254. Terminal gravitational-wave chirp, merger and ringdown law

Claim identity: SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every positive finite binary shrink depth, separation and period squared strictly fall, exact squared orbital and gravitational-wave frequencies strictly rise, and every successor retains a positive radiated take. The quadrupole wave frequency is exactly twice the orbital frequency. At the first exact horizon contact, two compact source supports join one remnant with the complete radiation and component ledger held. The remnant mass/turn key fixes one tone class and its positive amplitude halves at every finite return. The unique causal waveform order is inspiral with rising chirp, merger, then damped ringdown; no negative, irrational, imaginary, continuum, numerical-zero or completed-infinite proof scalar enters.

For every positive finite binary shrink depth, separation and period squared strictly fall, exact squared orbital and gravitational-wave frequencies strictly rise, and every successor retains a positive radiated take. The quadrupole wave frequency is exactly twice the orbital frequency. At the first exact horizon contact, two compact source supports join one remnant with the complete radiation and component ledger held. The remnant mass/turn key fixes one tone class and its positive amplitude halves at every finite return. The unique causal waveform order is inspiral with rising chirp, merger,

then damped ringdown; no negative, irrational, imaginary, continuum, numerical-zero or completed-infinite proof scalar enters.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009
- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071

Generated grammar and closure boundary. Generate the complete twelve-axis product of source, radiative moment, transfer ledger, separation, orbital balance, wave recurrence, contact, merger, remnant tone, damping, finite-floor and sequence forms. The declared exact boundary is: Every finite positive binary separation successor ending at the admitted horizon-contact support; every exact squared period/frequency and positive radiation-take record; the complete two-to-one source join; every finite binary remnant return; and all 4096 registered alternative combinations. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `held-two-source-orbit__admitted-quadrupole-rate__positive-retained-take__strictly-shrinking-positive-separation__inverse-square-period-square-equals-radius-cube__two-wave-cycles-per-orbit__first-horizon-boundary-contact__one-remnant-with-held-ledger__held-remnant-mass-turn-class__binary-half-On e-contraction__positive-at-every-finite-depth__inspiral-chirp-merger-ringdown`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
source	held-two-source-orbit	unheld-single-source	A single unheld source has no generated binary orbit.
moment	admitted-quadrupole-rate	monopole-or-dipole-radiation	Conserved source and momentum close those records.
transfer	positive-retained-take	negative-or-erased-energy	A negative scalar or erased distinction violates the Fold ledger.
separation	strictly-shrinking-positive-separation	fixed-or-expanding-orbit	That does not conserve the positive binding and radiation ledger.
balance	inverse-square-period-square-equals-radius-cube	imported-or-selected-frequency-law	An imported or selected law is not forced.
wave	two-wave-cycles-per-orbit	one-or-free-wave-cycle-per-orbit	A free multiplier loses the two-lobed source record.
contact	first-horizon-boundary-contact	chosen-merger-time	A chosen time is a parameter.

Axis	Forced form	Eliminated form	Elimination reason
merger	one-remnant-with-held-ledger	erased-or-two-source-remnant	It either deletes provenance or fails to join.
tone	held-remnant-mass-turn-class	fitted-tone	A fitted tone is measurement-selected.
damping	binary-half-one-contraction	negative-exponential-or-free-decay	That imports a forbidden scalar or parameter.
floor	positive-at-every-finite-depth	numerical-zero-or-completed-infinity	Neither is an admissible Fold proof endpoint.
sequence	inspiral-chirp-merger-ringdown	permuted-or-omitted-stage	It violates contact and source causality.

Base and successor. The closure proof is sealed by derivation hash `sha256:32121943d933e156c025ac6e64a1ea453122f13d421c0cc1a6b7c1fe56d9aa5b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:5cfebd73d3ed910e8d7e650bd05623f61c3784e9893a4315e79132d2e2c3ba8`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:a2f756d006fe4276cceb5b5f90db01e157663e808cbd4c37e4a49756582e2388`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:54cb462cc3670586005c76324d6abbc96ca1d0949db6ed365b1a0f97703851b2`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:0230f911c50b7e075bd6308cd0d401310ea086b772d57fb59c1388ae88f5fb8b`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0775b4e9ea9962f5407b6321178bd5571184eef331f569e088bd1326c6409981`. Independent certificate: `sha256:331dde20de8d262dd8f1086f919f30c3ddaa36eca49d78de9d8438dc3ec48069`. Engine external-validation hash: `sha256:c2a8a1d0a98c5fed6e75d2cf5634af2244239b42e188f9c82a3d1f9250cf4809`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Spacetime is relational event support and gravitation is source-linked path structure. No background continuum or imported metric equation is installed as a premise. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, LIGO waveform, dimensional frequency, measured mass or conventional inspiral equation in formal execution
- no fitted chirp mass, merger time, waveform template, damping time, tone, amplitude or correction

- no numerical zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinite proof scalar
- no erased radiation, source, component, contact or remnant record
- no claim that normalized Fold frequency is a measured hertz value

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:fb5a228f7f149ff8eb6b404af09b762c1c49fb051219104e070f9d80ff528571; engine receipt
sha256:7add0b09e795aee79758286268637f442b40155f4af5c83a4a123f90e2150e19 at receipts/engine/model_admitted/SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073-7add0b09e795aee7.json; empirical-validation hash None; measurement receipt None.

22. Continua Collective Matter

Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance.

255. Finite coarse-grained continuum correspondence

Claim identity: SFT-PHYS-CONTINUUM-COARSE-GRAIN-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A continuum description is the observation quotient of a completely generated finite cell network when adjacent cell records lie in the same retained macro-class.

Physical continua are scale-declared observation quotients of complete finite Fold cell networks.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__finite-cell-observation-quotient__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	finite-cell-observation-quotient	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:1dec3b2c7252a9da60284fb03232de86118a42a89d245fa02b42340a7e075165`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d6bce9c5b6034211c3297165bb209983ecfcd45e5061635706fa854895e9b27`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ce09b9ee426c6690cbbd588473c9e32e0298a7911dfbda3b4453219f8598860c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7d15bff6005e58bb09c1a112387b838e5228cf03409007959e7e994af98e04a9`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:be709c0e451aa29ed4ee66afac372db7eb3ee5c6f1eaf735d720c38950c4b44d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:1a51fc9553d35360f449393d395f3ea0d072eb3095fa63397d888131fb34cfcf`. Independent certificate:

`sha256:1efded5d8390c6f7d639702f9f91759b6aacb5442b8ccee2e9fead7d5280239c`. Engine external-validation hash: `sha256:950b7dca6d924fb7d678d1b1ffe159f617caf5a76e6e03e926c4ad8b7fd9fe5d`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- **continuum-coarse-grain-1:** predicted finite-continuum-correspondence; observed finite-continuum-correspondence; exact match `True`
- **continuum-coarse-grain-2:** predicted finite-continuum-correspondence; observed finite-continuum-correspondence; exact match `True`

- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:54e4a46c2fbe6c04e91bc6b288e9125ced225bde8547a007b72529f9fc3ff051.

Isolation certificate: sha256:e20576f597c4f605d5a0f188309be1861b259ebff1d4badbb100e1355ab5a860.

Custody certificate: sha256:ad650f6a6076a3fe9d36f56a95d41bce05e0b33204e440451f33773de30f4047.

Registered source descriptions:

- National Institute of Standards and Technology: <https://www.nist.gov/srd>
(sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- continuum-coarse-grain-1 from NIST-SRD-INDEX-2026-07-23
- continuum-coarse-grain-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7eed273621cf5466b155832334d6233f73b2969d81c206d2099fbf4eff584b37; engine receipt

sha256:fca2926aba2b9fca7aaeb55fd3121f9a52d64846397776cbb26e30b884b1a509 at receipts/engine/model_admitted/SFT-PHYS-CONTINUUM-COARSE-GRAIN-001-fca2926aba2b9fca.json;

empirical-validation hash

sha256:9350cfc2a37afabfe63d2c94540dea6bf21904ee8ba60b7b34f5988d94c95022; measurement receipt

sha256:54e4a46c2fbe6c04e91bc6b288e9125ced225bde8547a007b72529f9fc3ff051.

256. Density as exact content-to-support relation

Claim identity: SFT-PHYS-FLUID-DENSITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Density is the exact positive material-content count or inertial carrier divided by the generated spatial support count of the same observation region.

Fluid density is an exact content-to-support ratio on a complete generated region.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001

- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-CONTINUUM-COARSE-GRAIN-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__content-over-generated-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	content-over-generated-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:04ca869c52dc5303a5ee090b315b4fb0a44e2599e5abf5be6b9584c018790c84`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d6b53ecc44e62942f0af864b87f43a598c49def06da854548f868257c5bb21ee`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:878e57a9dda01e2d5c20bbdeb5eea018d2b2d3ae390cfd4cab2ca48fc9368008`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4acd658a8909be1728fc3579dc413b6181aef168f7519d34b98036850c37a43`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3f265911c3f4a001248376002fe6c3ff2d0a07b17a94b7009487f9f7824043f9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:39d3d3ca8f88165fea6399c5b3889e5a0708134a39f25135c8e21b1df4d138aa. Independent certificate: sha256:fc4ec7e77d06740835e907c86bdb712e8f4a5cc2d38f7af667a5a06fc0c66a05. Engine external-validation hash: sha256:bdff48145835a803f3c2873f3cf43d06b34c5f3b90e4a2a70b9b188ebbfaf3ffd.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar mass constant: exact predicted interval [625000000461504249/62500000000000000000, 48828125066636203/48828125000000000000]; external interval [50000000037/5000000000000000, 12500000017/1250000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact mass/count interval does not overlap the released CODATA molar-mass interval or the displaced control is accepted.

Measurement receipt: sha256:6f1ec3ffdd790df3ad9700de6a7bcf02e133f22bdede346e418c3ba41dfba650. Isolation certificate: sha256:efba3e6e2eddac59ece37a8836a99f02a3a3b62dd50045d9ec331999c315e759. Custody certificate: sha256:7c3ff6327bf489bc25caa5550519abe874a599823ea361f62380818528533b8a.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html> (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-density-1 from IAPWS-RELEASES-2026-07-23
- fluid-density-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2174bd50493277533f2bcb3ca5327a31201118dc149504ea5603ec2bbe31deba; engine receipt sha256:042d8b74f0692ac0c11dac25c449690e74d8f5a8b1196641080e377627b9bbcd at receipts/engine/model_admitted/SFT-PHYS-FLUID-DENSITY-001-042d8b74f0692ac0.json; empirical-validation hash sha256:5f9caf1c3efb73312888b0181c4595a61ad9f10cd6ba1288e1a34ccdd179f078; measurement receipt sha256:6f1ec3ffdd790df3ad9700de6a7bcf02e133f22bdede346e418c3ba41dfba650.

257. Pressure and stress transport

Claim identity: SFT-PHYS-FLUID-PRESSURE-STRESS-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Stress is oriented momentum transfer per generated boundary support and recurrence; pressure is its orientation-closed normal observation class.

Stress is exact oriented momentum flux and pressure is its normal orientation-closed Fold observation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-FLUID-DENSITY-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__momentum-transfer-over-boundary-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	momentum-transfer-over-boundary-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

sha256:ac4f7c24a99afa7c33f18e744fc5afbf29b6ba686c32fdc2cbd5c72c17e6576f. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:fb7fa3ccfb18ebce02d1140a68e3555e7593365a1e5ca5c14ca402e044e7d6da.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:92447125cc883fb979c3cde345b03c998c687a9bfd04232867dae08b530ba12e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:429ba780f58f3ff79a79b892f1c98dfb54cbcf84380a1b291c26ab05f649ea56.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4be9f236a07b5d7b7e6a9996faf428ba56d911ade74f2569ab2a032a4b52896a.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:dbd7d23c7160c57567f66f104c05c598d4c86b8e7d2096e1957bd5138b6f6204.

Independent certificate:

sha256:30d13302ce554ba52ecf0fe8e5cd7044a5f3a1850cf73ab3189034001eafa953. Engine

external-validation hash:

sha256:37c52185ba81e358e7625a85e7d4a0c062d3425fe2d4882c3c8a8b93d0f3f9ae.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-pressure-stress-1: predicted fluid-pressure-stress; observed fluid-pressure-stress; exact match `True`
- fluid-pressure-stress-2: predicted fluid-pressure-stress; observed fluid-pressure-stress; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:452d42fbae15faaae0856ab60e86b7c067f1e7d0e29a7d3e366e5fea07c856af.

Isolation certificate: sha256:993024f8b77bc093509ed733a0de02aa6b13b8919c957c8cc4a5161ba787117c.

Custody certificate: sha256:d86b1066ca74dc7a0879d484992d89103cb1a38c19d216a8af5c15df019597b9.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html>
(sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-pressure-stress-1 from IAPWS-RELEASES-2026-07-23
- fluid-pressure-stress-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the

minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:fa1293b29e0e53ca46291f56f2a6d76cace71777596bb5d5c4008416b3a759ba; engine receipt
 sha256:94ae8c496c0ad93534009db375be075997529f8bb40292d7bff956f9dce7d853 at receipts/engine/model_admitted/SFT-PHYS-FLUID-PRESSURE-STRESS-001-94ae8c496c0ad935.json;
 empirical-validation hash
 sha256:59b462aa2e41aa117ca7b45e6d676ddd7c60b09cd67e2b15299d72979379f5c7; measurement receipt
 sha256:452d42fbae15faaae0856ab60e86b7c067f1e7d0e29a7d3e366e5fea07c856af.

258. Flow conservation

Claim identity: SFT-PHYS-FLUID-CONSERVATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For every complete cell region, material, momentum and energy changes pair exactly with named transfers across its generated boundary.

Fluid conservation is exact pairing of interior carrier change with oriented boundary flow.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__interior-change-boundary-flux-pairing__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	interior-change-boundary-flux-pairing	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:221074ca4b603a432a39e472f4924f258ef2628d4f0ee8442dc47c05e977bc5c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6c2960986f4d55e887ca5bcba3b6ae11b3c5dbc032efec8e183b0a7ca2146e21`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8905302ed286a1762db5fbb766b12964a3b6646cd3c4e09e7a9e46f466742f0b`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:02ec9fd11b8633c435c5a45764bc7e6dd4b02e881b1fdf86755bd78e360aa3f2`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ac8b616e32bac9ad28c3cbe03c4ca63bde1df101ba74d479f5f6d06f957620b6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4436963e2bcb6dc07b5a59fde089e6de3c5263fde0fa90e6a6c19c3d8acd969b`. Independent certificate:

`sha256:185dc33929ce6029c79c250a4e28bc2c1eb622501a7838fd233b6d446f906830`. Engine external-validation hash: `sha256:b429eff314d1343a5a79a5c80990c96fdd961c3aab03c72612c6523267815984`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-conservation-1: predicted fluid-flow-conservation; observed fluid-flow-conservation; exact match `True`

- fluid-conservation-2: predicted fluid-flow-conservation; observed fluid-flow-conservation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:c85a4e2ddd3f216213e75a4418da06e63cf1c171b339770422c9d9434fed12d9.

Isolation certificate: sha256:09249dd3f9c13a897149dee7901f09c2f2da5993128f52f051f8f73eab8487d0.

Custody certificate: sha256:7b9e0b55f0b2c11511fdeb9b5007bcdfb3aac3df2dee9c83dec8850e22f717e.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html>
(sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-conservation-1 from IAPWS-RELEASES-2026-07-23
- fluid-conservation-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:17c21e0e61208e61b4b3a30ede51a89dd492c9720b01444d85b5b4cd8676f409; engine receipt

sha256:f6882fdf667263556f24e891247deb5f60533155505e7dfa31bdee8fb68f23ce at

receipts/engine/model_admitted/SFT-PHYS-FLUID-CONSERVATION-001-f6882fdf66726355.json;

empirical-validation hash

sha256:2f2c901861c6a9e382e381bc5f63e24f1b166f45c8c7a2217f0c435bba14d978; measurement receipt

sha256:c85a4e2ddd3f216213e75a4418da06e63cf1c171b339770422c9d9434fed12d9.

259. Inviscid flow correspondence

Claim identity: SFT-PHYS-FLUID-INVISCID-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Inviscid flow is the boundary where adjacent cells exchange normal pressure carriers but retain their tangential motion labels without intercellular transfer.

Inviscid flow is the exact fluid observation with pressure transfer and no resolved tangential momentum exchange.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001

- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-FLUID-CONSERVATION-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__no-tangential-momentum-transfer-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	no-tangential-momentum-transfer-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:6ff158fdae68bc989a9aab5baaa2aa97d5c267297b855cca481e25f8bf76597e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2dfd81c580bb59efca01f992f8cc1573ced5ceeb378ec1cb2fc54c598d76c618`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:518face2fa90408cb1537676e5ae2dc65a40dafcd9dddcbfcbcf09675c18b61ed9`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a996c280c1b2b3383a30b0b528487bb4789d5b1a04fb9c144bf8194547ec8fca`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:a1b5cfa22dfeededb22eb4d43fddd0b01e467092976b426d90f7ee9ba26ef751.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:4be4eec85be994c725ed2e8c41fe1f8b05b5fccfda1c37693de5235861561d7b. Independent certificate: sha256:274aeb210d2fc6e06167b3c816daeac66339cd3d4d3137099997026ca200d934. Engine external-validation hash: sha256:c039ca4c27a2d4e3e2871e7ce4cbbd371b4084243996a6d85d7d2b724c96332a.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-inviscid-1: predicted inviscid-flow-boundary; observed inviscid-flow-boundary; exact match True
- fluid-inviscid-2: predicted inviscid-flow-boundary; observed inviscid-flow-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:b859bde36d69a637bb96f9cbd9d011e09bd29ac37789d276326d3e0ae3406be5. Isolation certificate: sha256:612b7ca508a9c572df39d2b3db94b634e709e37def01f2c9d6b1938a091506b4. Custody certificate: sha256:a4cf17c1f96e86f3d6d81e9f87517697d7947693bb8e9224ce6c6a154b968389.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html> (sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-inviscid-1 from IAPWS-RELEASES-2026-07-23
- fluid-inviscid-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:77981b1d9a4897ac79c4b93ee40fe02ef653ce17a93c9ddbdf7b4137a8e45d6; engine receipt sha256:f697547d491a454368e7a58afdc7b0558bd77fc4a4c07419bae1d30623e21025 at receipts/engine/model_admitted/SFT-PHYS-FLUID-INVISCID-001-f697547d491a4543.json; empirical-validation hash sha256:240df644cb228a10a0388173cd15ee75473126fa655c7fc8d8295b93c8666cf7; measurement receipt

sha256:b859bde36d69a637bb96f9cbd9d011e09bd29ac37789d276326d3e0ae3406be5.

260. Viscous momentum transport

Claim identity: SFT-PHYS-FLUID-VISCOSITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Viscosity is the exact tangential momentum transferred between adjacent velocity-labelled cell classes per interface support and recurrence difference.

Viscosity records exact adjacent tangential momentum transport conditioned on velocity recurrence difference.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-FLUID-INVISCID-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__tangential-adjacent-momentum-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	tangential-adjacent-momentum-transfer	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

sha256:1992dc71f303b604c938e0e775750e54770e9f9b7e55124a58b7be68b74cee30. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:243b176f433d91fd498e68242b70e2bb66b2c1db852a059a126084da67db8fef.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:dfb3417d8cdf99d0bd693593953db666c9a0a1b443675f187fffc9a50a4360ab.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b0bda1d0ed11869cb1296026168adba986e70db0635133673b73975fc618acad.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e65a0f8e56f9657a81b05bd6cfbc99dcfaad256167c24be0dd6d3def944c411.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:dd2ea2c2404d6fb46987bb7c28819da623e8b20a855d6fe34ab68fc4294d8570.

Independent certificate:

sha256:88c8c29b5b3ad197093011b38958cbdba4cc0e1bad0af766083087ebd9723dcd. Engine

external-validation hash:

sha256:8246faa833391984580bdf87419a607300e82dac64d1f8446551dd97b2a67142.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAPWS-RELEASES-2026-07-23

Observed comparison records:

- fluid-viscosity-1: predicted viscous-momentum-transport; observed viscous-momentum-transport; exact match `True`
- fluid-viscosity-2: predicted viscous-momentum-transport; observed viscous-momentum-transport; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:0dd046ca2d7e3d78f77c993ead65e6947505c7971b1647557a49f35914f8990b.

Isolation certificate: sha256:cba153e98c6f7d6301365a14864a9b0fb1244a2adcf23514a8f6f93eeb2fa0a1.

Custody certificate: sha256:90c414b23a8a13ecc14409cb3a656aaede90dc601c787ececbeefa0db0166fa8.

Registered source descriptions:

- International Association for the Properties of Water and Steam: <https://www.iapws.org/release.html>
(sha256:d43b4241bc11f390633b03b041fe2b03f9a4f243348fa0f78b639cb8ec162c51)

Registered target identities:

- fluid-viscosity-1 from IAPWS-RELEASES-2026-07-23
- fluid-viscosity-2 from IAPWS-RELEASES-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the

minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7ef2a3786fb504b69aa64fd2dc22720096cad991236c9713b9310c8827270ac5; engine receipt
sha256:b395ce239e1415b4afb98cc99ab5f8a91dce5fb8a4914d2bdf7e7e9b2108c71 at
receipts/engine/model_admitted/SFT-PHYS-FLUID-VISCOSITY-001-b395ce239e1415b4.json;
empirical-validation hash

sha256:708e3c3f3fe96ccc69bc563b2145f46739256206c6ea6e68f6f125e055704f93; measurement receipt
sha256:0dd046ca2d7e3d78f77c993ead65e6947505c7971b1647557a49f35914f8990b.

261. Finite turbulence and cascade boundary

Claim identity: SFT-PHYS-FLUID-TURBULENCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Finite turbulence is nonperiodic multi-scale recurrence on a complete flow network, with conserved carriers transferred among generated support refinements; cascade claims close only at the enumerated scale boundary.

Turbulence is deterministic multiscale Fold flow with exact carrier transfer and an explicit finite census boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-FLUID-VISCOSITY-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__multiscale-f low-recurrence-transfer__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	multiscale-flow-recurrence-transfer	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:aea1f033f8c07f5d9a9aca67733f05b355be5a92f81322e1ecee504cb720b81c. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:a73887a8fc98a863be96b9f31736681c2e9fcb1954acc76cd4f49a6e03a526b2.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f9f3711bbcac605f36d171f20e96d6353e8dd48c3b973fb7a8cd91b9d1e43a6e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:533dac907b77fc5c47d01f99e297d5f36749f922d27a4ab684160748ba86528a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:33b889a2994f57a1e38443abc4b3013b70f1d44f495c9345f75daf08293cc931.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:15da7684842244e814ad50b42670a8cf90fe83b67039dde3ebba3f65b5060bc. Independent certificate: sha256:97481c4173ee2b5fa96182afa033951fdc7f8787204793058f1343605430479a. Engine external-validation hash: sha256:de0b9864fa2f3c8b50e3c88fd5fd64c08378ccb1dccbb301ee47c8c967641960d.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- fluid-turbulence-1: predicted finite-turbulence-cascade; observed finite-turbulence-cascade; exact match `True`
- fluid-turbulence-2: predicted finite-turbulence-cascade; observed finite-turbulence-cascade; exact match `True`

- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:39abd31200a9319915f19bbbff0d68a4084b5e3520ddf1035af29c7ab7085135.

Isolation certificate: sha256:f4170b99db1d143cb46049cb050d793266638dfc4a8d52c8b11b45a236691f0c.

Custody certificate: sha256:7c1dfd264cddd8f35e2cf6453bdf83c9addcd247c449d04d410942aed8b1bf0.

Registered source descriptions:

- National Institute of Standards and Technology: <https://www.nist.gov/srd>
(sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- fluid-turbulence-1 from NIST-SRD-INDEX-2026-07-23
- fluid-turbulence-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1d7f62f73bd0650e4705ba612c4d86f3640108c8d91e45efcd8db87d7fb3e369; engine receipt

sha256:d8a59dabe554a169c1f5d8f4c7c70feb2a0d3e1ebaa0d26186d1e2316a2e406a at

receipts/engine/model_admitted/SFT-PHYS-FLUID-TURBULENCE-001-d8a59dabe554a169.json;

empirical-validation hash

sha256:35c6d95008dbd73d5f4b12364f48f9d2e20127a35bd2d15eae7c18d989aae7c3; measurement receipt

sha256:39abd31200a9319915f19bbbff0d68a4084b5e3520ddf1035af29c7ab7085135.

262. Plasma collective response

Claim identity: SFT-PHYS-PLASMA-COLLECTIVE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A plasma is a mobile two-orientation charge-carrier network whose long-range source-response support couples local constituent motion into collective field recurrence.

Plasma collective behavior is source-bound electromagnetic coupling across a mobile held-charge Fold network.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001

- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-FLUID-TURBULENCE-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__mobile-charge-network-field-coupling__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	mobile-charge-network-field-coupling	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:0b7b164840d922e51eb41ccc2cfc303980fcd520bc605d0d716cbef879f8166b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a89f9ffa8ee5ae6d2a673c5e62fd60a6c313076e86184c2417c1ca9b3216d22e`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d9ab01dd2e6633d7a172025c5f01c04d1247d6177636d4d81f3085c9c84871f8`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c9f04994806a3462214b469cce636a358c665007a36ab75140155e59e7cfbbbf`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:321294b1ba2e87e4b07d90a4815ddffc2bb23729469f21499c9e8d9fe5a7971d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
 Implementation hash: sha256:2926da1c4cc5aa86a14edf2ae9353aa3c87a2e7b05fde8ea1fd505bb63162e1d.
 Independent certificate:
 sha256:ee358c5d98fe74f5a041dfc0a4cb8794420c2d0880172a452adf402ed78c9365. Engine
 external-validation hash:
 sha256:1c55941937e1b1b72cbc062a2b9692791e18aefad9768e0aa9a001fa754bdc6a.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Faraday constant: exact predicted interval [120606665154137523/1250000000000, 120606665154137523/1250000000000]; external interval [2412133303/25000, 9648533213/100000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact charge/count interval does not overlap the released CODATA Faraday interval or the displaced control is accepted.

Measurement receipt: sha256:9b1424fffe6a51cf3dd08684b0aad17f3157ec20a53bc0cff2aaf25ed4655e50.
 Isolation certificate: sha256:805ec634b9172e2c8f7289a0c37170c288544ae116a8c5e921baaa7ecc5df77d.
 Custody certificate: sha256:79c010fef9d99afb8be2b2f6a7a01eb9405e1b29014abc0f4fd93833994ea92b.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
 (sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- plasma-collective-1 from NIST-ASD-5.12
- plasma-collective-2 from NIST-ASD-5.12

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1a17a43a87ddbe90c7ffff4645ea017347a3955e923efcc63bd5280c4c9e8186; engine receipt
 sha256:31da2d601e05d224850b377d80396438c38447c40e994dd8d51f613238a07499 at
 receipts/engine/model_admitted/SFT-PHYS-PLASMA-COLLECTIVE-001-31da2d601e05d224.json;
 empirical-validation hash
 sha256:71356ff0a424d73c918de32c5bc719462f1fa7619fa68772d092921c212690ed; measurement receipt
 sha256:9b1424fffe6a51cf3dd08684b0aad17f3157ec20a53bc0cff2aaf25ed4655e50.

263. Plasma oscillation and screening

Claim identity: SFT-PHYS-PLASMA-OSCILLATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Displacing held charge orientations creates a restoring field recurrence; phase-mixed mobile responses redistribute until distant observation classes close the source distinction, producing screening.

Plasma oscillation is charge-separation recurrence and screening is closure of its source distinction by complete mobile response.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-PLASMA-COLLECTIVE-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__charge-separation-restoring-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	charge-separation-restoring-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

sha256: 64603a48c22c3e60a1451bd13367f889763a402e2bb95d7bde325192862515bd. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 426274129891558468084830d17dbe35712d20f0c3a86ff0a1710366f2664387.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 711474c269625d4f16c0a6dc355a0879c88aaf11cd9469298fa8234a20bc6fed.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 19a9f5fe7f3a6a1f91a35c7b3a4f4dc0c6c9abc2eca2ded59c1ff60bcd415aa4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: c2581546a9b49c3c450d381bfdbc49f43a1c3919f89e39649c06e1a7e4f632f2.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: 2acf9e656c728c6d6ba48f454d426f61e8b538b29873bf9b356571ce2b19416d.

Independent certificate:

sha256: 24c27df396a58d082fb49a4efb2043962553983784ee0eb71dc0b21f6918c958. Engine

external-validation hash:

sha256: a7d8b427e4486a43fb6e368b4398da334392b319358c4901394ea96458831f9b.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-ASD-5.12

Observed comparison records:

- plasma-oscillation-1: predicted plasma-oscillation-screening; observed plasma-oscillation-screening; exact match `True`
- plasma-oscillation-2: predicted plasma-oscillation-screening; observed plasma-oscillation-screening; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256: d7876b6ca9718f6b2e4efd047847aee1442c7c0c7da6464c9e8083d715ce7bc2.

Isolation certificate: sha256: 3776290cd8a4fe826c3c40a69415cea190673f87d0a4c3f5d1f8199de40e8fa9.

Custody certificate: sha256: c566c606b03bfebc676080627bbb6291fbd463364ef52553e7c03e6241ec0301.

Registered source descriptions:

- National Institute of Standards and Technology: <https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>
(sha256:a327a34eb1b85ef3f003e8c8f0dbcb0c3fc49f039ee4046546a924fc42118454)

Registered target identities:

- plasma-oscillation-1 from NIST-ASD-5.12
- plasma-oscillation-2 from NIST-ASD-5.12

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the

minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:32e35ac55e76c5661d80eeaf2b1cb3ab1f8caaac3e3a1c29bcaea6e6798d39d7; engine receipt

sha256:14cdadbe5421c2cb5f6f231d20b0456b8dfda9d76e01c4c4ab815a2d915efc82 at

receipts/engine/model_admitted/SFT-PHYS-PLASMA-OSCILLATION-001-14cdadbe5421c2cb.json;

empirical-validation hash

sha256:084f84514358c4efb0b9b9a64b5dfe4a310fa571c5b95ea31751036bbb13601b; measurement receipt

sha256:d7876b6ca9718f6b2e4efd047847aee1442c7c0c7da6464c9e8083d715ce7bc2.

264. Magnetofluid composition

Claim identity: SFT-PHYS-PLASMA-MHD-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Magnetofluid behavior is the compositional closure of conserved fluid flow with oriented charge transport and magnetic source-response on the same generated cell network.

Magnetohydrodynamics is the exact joint Fold process of fluid conservation and electromagnetic charge flow.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-PLASMA-OSCILLATION-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__fluid-flow-electromagnetic-joint-composition__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	fluid-flow-electromagnetic-joint-composition	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9d5231006c1b977b269ed159e74da0db1dc1813fec47ffa9ade235c0562c76f6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7ac127a6edcf37c11b9d4123d9d0dd9158cab0aa3a8fd744eed289494f624b03`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:731f7b6746cb28d3129ee5047a211addaaa884961de1792249a10515532126b2`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5c6de5e550bd5e51ca3b2d461671df02ec347b65c2a831034beeda1eb2f967b1`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfef70b8f6190c73fe7c021d94bda529f597d4f32a1d2f699ac928cf3097c1a1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9b9af7f6ff1f479de604d080ede292c2367aca22e4a244c29bf4f0ec70872582`. Independent certificate: `sha256:048987901a7d00652dd28d6b307ed8caac174301fdd75a358afc547be3376929`. Engine external-validation hash: `sha256:7912fe7b4ecb3d7e042f26663ce6f1d1eac8bb7350f4fcd8da8950e92b45e7b8`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- `plasma-mhd-1`: predicted magnetofluid-composition; observed magnetofluid-composition; exact match `True`

- plasma-mhd-2: predicted magnetofluid-composition; observed magnetofluid-composition; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:2849ed848c310b0815fcc89987abf99503720c5a4c5dcdd8394e982d2a5df506.

Isolation certificate: sha256:2394e9378ee62d0d856c0c44e818cc3191a4e368b8d39821063581f832f38f93.

Custody certificate: sha256:4c57aa904d3c44057f14d5ca6a561d281a6a905524b819e44f063b950e03e77a.

Registered source descriptions:

- National Institute of Standards and Technology: <https://www.nist.gov/srd>
(sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcb34f6ffcb8364d6)

Registered target identities:

- plasma-mhd-1 from NIST-SRD-INDEX-2026-07-23
- plasma-mhd-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:bc7e3dd69af9792887aac4aa519554faed5ecce78cbbd933307323f97348d3b2; engine receipt

sha256:758229c88e39f5606a215e2bdd8ed689d2f31b0cb739eacdda0d8c53ecad0e5c at

receipts/engine/model_admitted/SFT-PHYS-PLASMA-MHD-001-758229c88e39f560.json;

empirical-validation hash

sha256:5e92998418418aeb0f51f1c25eba8bd0411a2a2839db5fd39a2f7866a6e37a33; measurement receipt

sha256:2849ed848c310b0815fcc89987abf99503720c5a4c5dcdd8394e982d2a5df506.

265. Lattice order and excitation

Claim identity: SFT-PHYS-CONDENSED-LATTICE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A lattice is a finite adjacency network whose local constituent word and neighbor relation recur under generated translations; excitation is a source-bound departure propagated through that recurrence.

Condensed lattice order is translation recurrence of a finite Fold adjacency network and excitation is its propagated labelled deviation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-PLASMA-MHD-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__translation-recurrent-adjacency-network__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	translation-recurrent-adjacency-network	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a35d7000c96d02c087b673175ad732a45a7ed2abfd58d2f16ea7a77f0ec25dd0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b1cc8064129d3557b22b833afa9a83a704ab693dea418f8b40835f3df2f3d8ba`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8ed9ef2406b9bd6616e7f7cdd01a97adc21e8c5ed9bf668b5b67fa9f4f4e4017`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:07bab574e37e3d080d2079c1f26a7ab081f926d4f752ade2a505022ad9528967`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 4013bffdbbb665a8dabe7e065bb971764b0c102b235cc83180695b542cc35eae.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256 : e8d201f82909c9dc391dcc47256dc3fda9b22754bf0d69c9f55e8cf5242d3eed. Independent certificate: sha256 : 90a98406eb4fbb2f21d6615ea1ace1c4bcc5c80290eabba33b2cc4b92dda55b4. Engine external-validation hash: sha256 : 74ecf5daa0e778da0b65afed9c2871671aae334209ead28be74e052b62faf51e.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-lattice-1: predicted condensed-lattice-excitation; observed condensed-lattice-excitation; exact match `True`
- condensed-lattice-2: predicted condensed-lattice-excitation; observed condensed-lattice-excitation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256 : 183d206a75d8e56ee46c5727c0ecb1de78cffd0de89bb0a640b98d9f8dfc6aca. Isolation certificate: sha256 : 5d0cfa6a495c1b77c2881f00333bb6451e13fda86048d2980be9b9c7c86d9b1b. Custody certificate: sha256 : 792f1815913b2689ddd15e12133d5fd0407762750fd464a20d17edbcd58d76a0.

Registered source descriptions:

- National Institute of Standards and Technology: <https://www.nist.gov/srd>
(sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-lattice-1 from NIST-SRD-INDEX-2026-07-23
- condensed-lattice-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : fdacf91e04fd5ab26a7fdd1cb2db5b3e0efb319be91b292307cfe73e2c175963; engine receipt
sha256 : 46d5a4142f4a53f2665b0f45f89c00e0afe2b5cc8989e0f5169644fc100d6ea3 at
receipts/engine/model_admitted/SFT-PHYS-CONDENSED-LATTICE-001-46d5a4142f4a53f2. json;

empirical-validation hash
sha256:5c3b3b0f9927d524776ca6218313f669ed527bd513d6f85e0f7076cc5600b3ce; measurement receipt
sha256:183d206a75d8e56ee46c5727c0ecb1de78cffd0de89bb0a640b98d9f8dfc6aca.

266. Band support and transport

Claim identity: SFT-PHYS-CONDENSED-BAND-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?
Theorem. Band support is the complete set of phase-compatible constituent recurrences on a translation-recurrent network; gaps are observation classes with no compatible generated path.

Bands are phase-compatible Fold recurrence classes of a lattice and gaps are complete-support absences at the declared boundary.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-CONDENSED-LATTICE-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__lattice-phase-compatible-recurrence-support__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	lattice-phase-compatible-recurrence-support	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

sha256:ccc73313cee745d4be6a4ef2c7e14cb669ebc2be5007249574e88135bf082c85. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:85b0b149042eefbc0921b56113028b1ec5a08a7cbf25076eaf8a71115bfa4c1e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:038dd4e7c0ee189026496752a144279c36795917b9005b100802ed42d08e9b27.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1e383fb722a2a19061bf3948397e286dbfbfb8768aa42018acd8c5eb07a310bb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:13b3e4761ad16cd3e803b3254843b144ed07e75271da670f911db4bd18f516de.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:6fe2d12b549ad0d66bc826b8346857be50e5047b7e8696df922a4049cdcfc31d.

Independent certificate:

sha256:e52c462dbedf9ce2764ab700ba086e59a0cf92943d3d9b817f85ebf71ba8c2367. Engine

external-validation hash:

sha256:f0586d281a640ccda8ca73c962992d707e3246bf5cd2907b22444ccf0a38aa64.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-band-1: predicted condensed-band-support; observed condensed-band-support; exact match `True`
- condensed-band-2: predicted condensed-band-support; observed condensed-band-support; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:f27b2b1dbc982d5f064420c6fb2637d57ea98952c119c87ee199885bdd3c2987.

Isolation certificate: sha256:1dbf6766a8fdf443a61ff95f469ee3c2e74002e9e53f6044330cf808257bf909.

Custody certificate: sha256:a9b2ee7b8db7720351f93e4cef9a9ba625c09a73c8dc43b79e9f21397b76f93d.

Registered source descriptions:

- National Institute of Standards and Technology: <https://www.nist.gov/srd>
(sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-band-1 from NIST-SRD-INDEX-2026-07-23
- condensed-band-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b3c1b1b6488a433372bf26faa0a8c351bc843da6ca2fda996ffc739298e4e684; engine receipt
sha256:c9b70ce35823bfff4a59e68486c4b7ebfe9820c03ec913c33c313e786bfc3dea4 at
receipts/engine/model_admitted/SFT-PHYS-CONDENSED-BAND-001-c9b70ce35823bfff4.json;
empirical-validation hash

sha256:48dfc545c3f094e43ce2e88aa88e63aca96856868be3678e531b86fcc3cbdea; measurement receipt
sha256:f27b2b1dbc982d5f064420c6fb2637d57ea98952c119c87ee199885bdd3c2987.

267. Collective order and phase transition

Claim identity: SFT-PHYS-CONDENSED-PHASE-ORDER-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Collective order is a shared recurrence label retained across a connected support fraction; a phase transition changes the stable macro-observation fibre and its adjacency symmetry class.

Condensed phase order is shared Fold recurrence over connected support; transition is provenance-preserving change of macro-fibre and symmetry.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-CONDENSED-BAND-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__connected-sh`

ared-recurrence-order__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	connected-shared-recurrence-order	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:51b477ad9854d3e3390f794aade6d2fd828da8fd3a67a9a0dfc508a1feb83d23. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:e1921214fd5a796742099527107d08bc253142690a0a512c1fc37c6d2ab883cb.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:151cbb565925f8890c997ccad95fcc1527c320948f80cf34a1b7701021a1433d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:b562b5f1cf958f42a347f9408cc549b2ee44d2b8324c40590f7166d452f8af7f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:ffe3538d4e78cb676aeee96d370bd15b9cd0814d8e838e92bfa11f0083bab4ba.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:125652ffac712d8fcd5648b50691367d6bda8c6041d39b69c72f68dc4e2d7393. Independent certificate: sha256:43d1226773c7d47f0390a4fbe3152c93d5fc630c6e8a0c3b44bdb3c168e4a3d6. Engine external-validation hash: sha256:8fb64442ce196f7e53f91ab1aa9e6aeba315823a535821c6385754d78607e01e.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-phase-order-1: predicted collective-phase-order; observed collective-phase-order; exact match True
- condensed-phase-order-2: predicted collective-phase-order; observed collective-phase-order; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:2ad3641024604c67411e28aeb1308239b87f5f6e3b6467348f81a1416797f22f.

Isolation certificate: sha256:02c2b432f186698e83742819604388a9f56cc674d580b2186d288c77158c9eb5.

Custody certificate: sha256:24d419f2f6a151ed9f5e7f86c6ff7e1dfd68dc4ecdde97a515ffbee863881245.

Registered source descriptions:

- National Institute of Standards and Technology: <https://www.nist.gov/srd>
(sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-phase-order-1 from NIST-SRD-INDEX-2026-07-23
- condensed-phase-order-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:5ad5edea1584254b0a73286c2af6fdafbbfc5ec40352dee35cc2da4a1ab9ac51; engine receipt

sha256:7f20f2a8757ec6d79c32a77dd07c3fca5c25e58b1ae233c68d9daa30e5c86ceb at receipts/engine/model_admitted/SFT-PHYS-CONDENSED-PHASE-ORDER-001-7f20f2a8757ec6d7.json;

empirical-validation hash

sha256:c00fc15ba1b4e90f8fc8522f35c3423d6e3666e7d4330b195d60b7a552056fa0; measurement receipt

sha256:2ad3641024604c67411e28aeb1308239b87f5f6e3b6467348f81a1416797f22f.

268. Superconducting coherent transport

Claim identity: SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Superconducting transport is a phase-locked joint carrier recurrence whose accessible support contains no momentum-randomizing transition compatible with the retained collective label.

Superconductivity is phase-coherent collective Fold transport with structurally closed dissipative transition support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__phase-locked-scattering-closed-transport__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	phase-locked-scattering-closed-transport	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:06e4517ff4a130b52a47147c077429df7e15957c2a80e5c5b067afa9b4297fd8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d223d8cd28aabe2b9f36107ddde2e6e8c811a34a78fb47914256490babf5e5dc`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:11c0231899eca9549f63ab1f75820adc479df5559982abe9bee90b00249037e9.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:fe78fcc493a1e048e85a7cbfa55e5bd3e71f74518243bc3fdca8f318fe01cc4f.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:fcd2a75b76e9935e1034b9e0a2477ce7849f7ff90320cdada985cdd1aa306dea.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:d8554ea60284f22785ca2689d890bae4e78117970a0f1d45ebf2876bf94f4a14.

Independent certificate:

sha256:c18546fc09850ef3db9f7ba1f96d7c0996ab108991fb44df787f7306f9d6ec71. Engine

external-validation hash:

sha256:77d8527a95cce3821e1c646a21720aa374cd5365ceff4eddd7e275b946be1cc8.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- condensed-superconductivity-1: predicted superconducting-coherent-transport; observed superconducting-coherent-transport; exact match **True**
- condensed-superconductivity-2: predicted superconducting-coherent-transport; observed superconducting-coherent-transport; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:7d7f1311995147699ba306a7a965ed3cb550ae1d6ee4cf61ba47e1ed373d43d8.

Isolation certificate: sha256:b64d457d47167925766ee7ae92a8bbfa2ceb71a8da3cc6417880dc694c7dfc3e.

Custody certificate: sha256:4b504c98077ddaf0d463f06b614ed078fa64b0ce6281ed81e1b7f039e5d3ce64.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- condensed-superconductivity-1 from NIST-CODATA-2022-ALL-CONSTANTS
- condensed-superconductivity-2 from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase

• target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:8edc42cffa4a9b38f5ce59196ef66c6289fd4bc2ddd5acf024b4619d167f542b; engine receipt
sha256:ee86d5d893add989390c63fcc6f11bc7d95e0dc8c9c7dabd724c3233931deb04 at receipts/engine/model_admitted/SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001-ee86d5d893add989.json;
empirical-validation hash
sha256:6251be4d6b6ee93d576f77e76672172a19d45db415dd21fee237071438ad1258; measurement receipt
sha256:7d7f1311995147699ba306a7a965ed3cb550ae1d6ee4cf61ba47e1ed373d43d8.

269. Topological collective-state correspondence

Claim identity: SFT-PHYS-CONDENSED-TOPOLOGICAL-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A collective state is topological when its complete global path/adjacency class is invariant under every generated local deformation that preserves connectivity and boundary records.

Topological collective order is invariance of the global Fold path class under all connectivity-preserving local transformations.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001

Generated grammar and closure boundary. Generate the complete cell support, carrier content, adjacency/transfer, observation scale, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite collective forms generated from exact cells, adjacency, content/support ratios, local transfer, recurrence and observation quotients without importing a continuum field equation or fitted constitutive law. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__global-path-class-local-deformation-invariance__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	global-path-class-local-deformation-invariance	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:efe1f42aa2165b92aeab1e52d6f02cd7289d590cccc86d437bfedda7765054d0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bb8a4e31e8197f486fdb8e593f2109bc41d28c1f3614edf3ff948fabcb3c07c8`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:380060ab86ed9165d21af7087a574b93e6ba01d9c9e469c87f9afc77f9ee9f06`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5493ae6329eac437a63678ee19a7a4ca8371a4a133f85ecdfe1e2069c5bfffad`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:70337bca18cd7f4f94b5b20c543b1cd7de6747f07a0534b809af74e2d10f6db6`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:07694800478ee3eaf088816a80ceccd07cd8fc888af6a1fa111a4229664372e`. Independent certificate: `sha256:ed10d3766585d02702ad385d9fd6cd5a9c2bcd7251ede206f9c03b9fe5453657`. Engine external-validation hash: `sha256:7a76a4c5d25b7f18533f18bc6e290966484057fc6300043b55d2a86bc00712d1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-SRD-INDEX-2026-07-23

Observed comparison records:

- condensed-topological-1: predicted topological-collective-state; observed topological-collective-state; exact match `True`
- condensed-topological-2: predicted topological-collective-state; observed topological-collective-state; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The collective-matter law is falsified if a committed external row lacks the sealed support/transfer/ordering structure, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:e57b9563687b05ec272c5eaf1f9cf40387eb01335d284c07d15ab0197ac4e087.
 Isolation certificate: sha256:5fad2cf81b8fd81a32d3f45de9c2c552ed7ff624e90b1626ed14d61a20b7de0f.
 Custody certificate: sha256:e17f9ec64247559f7e5b834de5c97e3915c27cf1705776b0ea24aee4f96c0e56.

Registered source descriptions:

- National Institute of Standards and Technology: <https://www.nist.gov/srd>
 (sha256:ef45e92ec84987dd258eefdd30a4565989dba4adfd13a2fcbc34f6ffcb8364d6)

Registered target identities:

- condensed-topological-1 from NIST-SRD-INDEX-2026-07-23
- condensed-topological-2 from NIST-SRD-INDEX-2026-07-23

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated continuum, infinitesimal cell or completed infinity
- floating, irrational, imaginary or signed proof magnitude
- imported differential equation, fitted coefficient or target-selected phase
- target access, omitted cell/transport row or unrecorded boundary transfer

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:f4dd6ad6fc8f7a583b5dbe8e244bb6ff41376c3394d8b72442aab461ef1f4d3f; engine receipt
 sha256:817047e583a84e4443cabe268f2de478eae52cc8db2d84a6248c743102ae9d6e at receipts/engine/model_admitted/SFT-PHYS-CONDENSED-TOPOLOGICAL-001-817047e583a84e44.json;
 empirical-validation hash
 sha256:d6751b11a96db285f6bdf7c753ae7112a35fb5ee93b9d393cdf0ab457111a7c; measurement receipt
 sha256:e57b9563687b05ec272c5eaf1f9cf40387eb01335d284c07d15ab0197ac4e087.

270. Exact Fold lattice operators, causal cones and phase modes

Claim identity: SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact Fold nearest-neighbour family retains half-One at the centre and shares half-One over 2d neighbours; point-source peak and opposition-ring magnitudes are 2d; causal balls at ticks 1,2,3 are (3,5,7), (5,13,25) and (7,25,63) for dimensions one, two and three; and every N-site cycle has N exact phase-mode carriers with one stationary identity mode.

The exact Fold nearest-neighbour family retains half-One at the centre and shares half-One over 2d neighbours; point-source peak and opposition-ring magnitudes are 2d; causal balls at ticks 1,2,3 are (3,5,7), (5,13,25) and (7,25,63) for dimensions one, two and three; and every N-site cycle has N exact phase-mode carriers with one stationary identity mode.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001

- SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003
- SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003
- SFT-PHYS-WAVE-DISPERSION-001
- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of site carrier, neighbour family, conserved stencil, positive operator, point-source ring, causal support, cycle modes, dimension induction, comparison boundary and extension forms. The declared exact boundary is: Every positive finite one-, two- and three-direction nearest-neighbour Fold lattice; every positive tick and oriented causal cell; every positive finite cyclic site count of at least three; and the already sealed dispersion, Maxwell and cubic-support comparison records. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `positive-held-site-presence__complete-two-per-generated-axis-ring__half-One-held-half-One-equally-distributed__positive-peak-with-held-opposition__two-dimension-peak-and-complete-ring__complete-oriented-taxicab-ball__complete-exact-phase-mode-carriers__axis-and-oriented-step-successor__inherit-sealed-dispersion-and-cubic-records__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	positive-held-site-presence	signed-field-number	A signed scalar imports cancellation and an unheld absence value.
neighbours	complete-two-per-generated-axis-ring	selected-or-remote-neighbours	Selection or remote adjacency breaks locality and completeness.
stencil	half-One-held-half-One-equally-distributed	free-dimension-specific-weights	Independent weights introduce parameters.
operator	positive-peak-with-held-opposition	signed-Laplacian-value	A signed result is not a positive Fold record.
source	two-dimension-peak-and-complete-ring	chosen-peak-or-partial-ring	A chosen peak or incomplete ring does not conserve the neighbour ledger.
causality	complete-oriented-taxicab-ball	continuum-or-selected-front	A continuum or selected sample omits generated cells.
modes	complete-exact-phase-mode-carriers	evaluated-cosine-or-mode-table	An imported cosine or table introduces nonnative and potentially irrational proof values.
induction	axis-and-oriented-step-successor	bounded-count-list	A short list does not close the next generated axis or tick.
comparison	inherit-sealed-dispersion-and-cubic-records	measurement-selects-operator	External targets cannot choose the stencil or cone.
extension	no-extra-rule	free-coefficient-or-extra-neighbour	That changes the generated conservative family.

Base and successor. The closure proof is sealed by derivation hash `sha256:31f2aed551b31a4e2dbdbbbdfef60aa10d749cd91e0bca328cdc7da8409bd5362`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : 62664c6125390b5cecdcdb762b22312db24644f6901cbd965944a7de259418e2.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : c8f5dac5af8783ca95453b0fab2a430910ce74fae14f2fd406a8a34d2fd0b14a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : fb404db2158783d426d842850836a63421288fb3383c8598b89671690c645cdd.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : fbcc3aa0b64e314009868b01a0345a4cecd424e0ad4121391e5647ec8b4bd955.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : 507f7e32537e49871acf519e2834df4c5c6318e561bb2b49069218004bf10d30.
Independent certificate:
sha256 : 3ee7bd89b082732dac0bc7118a197cf982b2f12db163add3de4d79a8d05913be. Engine
external-validation hash:
sha256 : 4e4449c05d228fc505dc6fce3b16d66fc7e557de66befaf5de6af0ef92c7e90c.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported continuum, signed Laplacian, Fourier cosine or monatomic-chain equation as a premise
- no V1/V2 executable, answer table, measured spectrum or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no target-selected neighbour, coefficient, mode or causal-front count
- no claim that the exact phase carrier evaluates an irrational conventional eigenvalue
- no external comparison before the inherited dispersion and cubic-support seals

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 355f6b2232eae04527ae5f7d859449a4649f32cd1eae7f4f0b0ca4db254fc2b; engine receipt
sha256 : 2d54a071b1a994f26b3a05083d0d57ec33659b9d5430b17467f5dfd03d525fb8 at receipts/engine/model_admitted/SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022-2d54a071b1a994f2. json;
empirical-validation hash None; measurement receipt None.

271. Terminal finite radiation, acoustic, laser, plasma and Alfvén response law

Claim identity: SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Every finite cavity occupation is an exact rational mean over the complete microcanonical word census; all modes above total support are empty and common scale changes preserve the complete word set. Wien covariance is exact at the relation level. In stable three-space $P(gT)/P(T)=g^4$, so doubling gives 16. Acoustic modes are $f_n=n f_1$. Laser inversion is greater than $1/2$, threshold is gain=loss, above-threshold requires gain>loss, and linewidth times coherence ticks is One. Plasma $\omega_p^2=nq^2/(m \epsilon)$, $\lambda_D^2=\epsilon T/(nq^2)$, and Alfvén $v_A^2=B^2/(\mu \rho)$, all as exact positive rational carriers.

Every finite cavity occupation is an exact rational mean over the complete microcanonical word census; all modes above total support are empty and common scale changes preserve the complete word set. Wien covariance is exact at the relation level. In stable three-space $P(gT)/P(T)=g^4$, so doubling gives 16. Acoustic modes are $f_n=n f_1$. Laser inversion is greater than $1/2$, threshold is gain=loss, above-threshold requires gain>loss, and linewidth times coherence ticks is One. Plasma $\omega_p^2=nq^2/(m \epsilon)$, $\lambda_D^2=\epsilon T/(nq^2)$, and Alfvén $v_A^2=B^2/(\mu \rho)$, all as exact positive rational carriers.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-PART-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005
- SFT-PHYS-PLASMA-COLLECTIVE-001
- SFT-PHYS-PLASMA-OSCILLATION-001
- SFT-PHYS-FLUID-DENSITY-001
- SFT-PHYS-FIELD-MAGNETIC-001

Generated grammar and closure boundary. Generate the complete product of finite occupation, radiative dimension, acoustic boundary, stimulated mode, gain threshold, linewidth, plasma balance and magnetofluid response forms. The declared exact boundary is: Every finite positive-whole mode-cost list and positive total-quanta support; every common positive-whole scale; every exact positive temperature ratio; every positive acoustic mode count; every positive laser population, gain, loss and coherence count; and every exact positive charge, mass, permittivity, temperature, field, permeability and density carrier. Empty occupation is a structural empty record, not a physical zero magnitude. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-finite-boson-word-census__three-mode-powers-plus-one-energy-power__positive-whole-harmonic-ladder__duplicated-identical-held-mode__gain-loss-equality-with-strict-half-One-inversion__positive-reciprocal-coherence-carrier__charge-stiffness-over-inertia-and-thermal-balance__magnetic-tension-over-fluid-inertia. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
occupation	complete-finite-boson-word-census	continuum-exponential-premise	An imported exponential violates finite exact generation.
radiation	three-mode-powers-plus-one-energy-power	fitted-Stefan-exponent	A fitted exponent is target-selected.
acoustic	positive-whole-harmonic-ladder	selected-frequency-list	A named spectrum cannot select the law.
stimulated	duplicated-identical-held-mode	unrelated-spontaneous-output	An unrelated output does not amplify a held mode.
threshold	gain-loss-equality-with-strict-half-One-inversion	free-gain-setting	A free setting adds a parameter.
linewidth	positive-reciprocal-coherence-carrier	zero-width-monochromatic-idealization	A numerical-zero width erases the finite record.
plasma	charge-stiffness-over-inertia-and-thermal-balance	named-empirical-fit	A named fit cannot select the carrier.
magnetofluid	magnetic-tension-over-fluid-inertia	selected-wave-speed	A measured speed is not a derivation.

Base and successor. The closure proof is sealed by derivation hash sha256:b55f7422998c6bf29035fca2e4d0c8636e26da1d9c58a4541e17630ef5ebf43c. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt sha256:d2f95bbcb1a590cd99cadb1377bebf4976fe85d32db6c8b4dd2afb248109883.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:8bab28793b529c5c4c48d5a1397fa044d6ec84f1a70a5852f1ab0741eb3fadc7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6848a259aebfde6e574cca1866c660f3f962a9bdb705035700064ff9c179efee.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:bd27fdedb838b7a2218f203a53ca54b64809e258d407881ab8f0b3a44615bd99.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:c0623faae3b4b8928150de44e49170e249d77076c1310271af05579a2811fba1. Independent certificate: sha256:7b21c482e17e56c9fac34cee25c1a428562ac59c0bbf82a3bb51318854fe8dba. Engine external-validation hash: sha256:c2471891deaf2b417e317d017a3b53a8c59e638145c98c27539c71eb8de8815d.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no Planck exponential, Bose-Einstein continuum distribution, fitted Wien constant or Stefan coefficient as premise
- no measured spectrum, laser setting, plasma frequency, screening length or Alfvén speed available to selection
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof magnitude
- no square-root proof value; squared frequency, length and speed carriers are the exact native forms
- no free gain, loss, linewidth, plasma, acoustic or magnetofluid coefficient

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:b500ac7de380339137eca53bb1d67451c29a9e5d3b479b9d43d7ebf0077c32ab; engine receipt
sha256:50974ab152438d8b84fe8220281e0d98737df0880f9a6384d50bac6060987b07 at receipts/engine/
model_admitted/SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041-50974ab152438d8b.j
son; empirical-validation hash None; measurement receipt None.

272. Terminal exact phase-criticality, universality and turbulence-scaling law

Claim identity: SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. For the binary self-antipodal local-order class, the threshold is $1/2$ and the exact exponent carriers are $\beta=1/2$, $\nu=1/2$, $\gamma=1$ and $\delta=3$, with α and η typed as empty One records. Widom $\gamma=\beta(\delta-1)$, Rushbrooke $2\beta+\gamma=2$ and Fisher $\gamma=2\nu$ all close exactly. For the distinct generator-three conserved cascade, the second-order structure exponent is $2/3$ and the energy-spectrum exponent has falling orientation with positive magnitude $5/3$. At every positive whole scale q and depth d , $\text{structure}^3=\text{length}^2$ and $\text{spectrum-divisor}^3=\text{wavenumber}^5$, providing depth-independent exact witnesses without irrational evaluation.

For the binary self-antipodal local-order class, the threshold is $1/2$ and the exact exponent carriers are $\beta=1/2$, $\nu=1/2$, $\gamma=1$ and $\delta=3$, with α and η typed as empty One records. Widom $\gamma=\beta(\delta-1)$, Rushbrooke $2\beta+\gamma=2$ and Fisher $\gamma=2\nu$ all close exactly. For the distinct generator-three conserved cascade, the second-order structure exponent is $2/3$ and the energy-spectrum exponent has falling orientation with positive magnitude $5/3$. At every positive whole scale q and depth d , $\text{structure}^3=\text{length}^2$ and $\text{spectrum-divisor}^3=\text{wavenumber}^5$, providing depth-independent exact witnesses without irrational evaluation.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-HALF-ONE-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001

- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-FLUID-TURBULENCE-001
- SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008

Generated grammar and closure boundary. Generate the complete product of critical threshold, order, correlation, response, critical-field, scaling-identity, cascade and spectrum forms. The declared exact boundary is: The binary self-antipodal local-order class and the generator-three conserved-cascade class; every positive whole perfect-power scale base and depth; exact rational exponent carriers; and typed empty exponent records. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `binary-self-antipodal-half-One__square-excess-to-linear-order-carrier__square-excess-to-linear-inverse-correlation__reciprocal-linear-excess-response__generator-three-order-cube__typed-empty-alpha-eta-with-exact-identities__three-branch-cube-to-square-transfer__falling-held-orientation-with-five-thirds-magnitude`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
threshold	binary-self-antipodal-half-One	measured-critical-temperature	A measured temperature would select the law.
order	square-excess-to-linear-order-carrier	fitted-decimal-exponent	A fitted exponent is not forced.
correlation	square-excess-to-linear-inverse-correlation	continuum-correlation-length-ansatz	A continuum ansatz is outside the Fold grammar.
response	reciprocal-linear-excess-response	selected-response-power	A selected power adds a parameter.
field	generator-three-order-cube	imported-field-polynomial	An imported polynomial is an independent model.
identities	typed-empty-alpha-eta-with-exact-identities	numerical-zero-residuals	Conventional zero is not a proof scalar.
cascade	three-branch-cube-to-square-transfer	dimensional-analysis-assumption	Named dimensional analysis does not force the branch count.
spectrum	falling-held-orientation-with-five-thirds-magnitude	fitted-negative-five-thirds	A fitted signed exponent violates the grammar.

Base and successor. The closure proof is sealed by derivation hash `sha256:b05e501fa57215bf093b880f7960323996494347ac2c58befc10f8200cfe0a58`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ff266871a7c66c6e020794234f45ed42e85407e9de5ac1c729d78a7ee5f38f5b`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:d811c2f0fbf0ad2a0508722d88072c64994579e55ac5c2a190d9a414c5a3e50c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:0c0ff1a6ce8765f94f653f179c2b8eb457f45d130612ef1a2cb2637d5e324fa5.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:424effe32a4dedfaa352c97e4e845e6436d547e40bdbcd6e092474b09ac8b2b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:bf784f179216305a8029a2fe648735d072b7bfe6b34f81662d523f8b3ed02859.

Independent certificate:

sha256:8c5ffb7c980bc8fc76da3665d7fffeadaa514ecf820d5ee98c56430ab402a64. Engine

external-validation hash:

sha256:db340f6a2e87fa45d45b46f167e02cf12947996b22ebd1a310ab29db4a84231d.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. Continuum language is recovered as a declared observation quotient of finite generated cell networks. Fluids, plasmas and condensed phases retain local transfer and boundary provenance. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no renormalization-group, continuum field theory, Landau functional or Kolmogorov law imported as a premise
- no measured critical exponent, transition temperature, structure function or spectrum slope available to candidate selection
- no claim that every physical transition belongs to one class; equality requires the complete generated class key
- no fitted exponent, intermittency correction or Reynolds-number calibration
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof magnitude; spectral decrease and empty exponents are typed records

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:90edf0a55d6c31e17fd8092433e55b4bc14c422006998aa12e67f29ebefd170e; engine receipt

sha256:0601d19640943c4b99eb8ccc061e3115c520773eac5e762bf7c5b7440339b25 at receipts/engine/model_admitted/SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047-0601d19640943c4b.json; empirical-validation hash None; measurement receipt None.

23. Physical Cosmology Boundary

The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology.

273. Propagation redshift relation

Claim identity: SFT-PHYS-COSMO-REDSHIFT-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Redshift is the exact ratio change between emitted and observed recurrence carriers after source, propagation and observer clock relations are retained.

Cosmological redshift is an exact source-to-observer Fold recurrence ratio with complete path provenance.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__emission-observation-recurrence-ratio__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	emission-observation-recurrence-ratio	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:60131c2b825f16bb397f042a42440edb9c14ad3c4085e97db361d03983166874`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9597ef38113a9a2375194761d7288a9920dbafa12a79e4212ba7efeecea59c44`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bd4d7ace5d03153879e4d4f67b95c4ac791180d31b65e2b9c58fc1e56a13d9d5`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 56f45be90500b8eee597a38e7b2eb11e7cd8e61296df3da3fafa7be046d3d90c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : d09e058c6a6886e403b4fd29935aaa5cf253af5b5c402dcacc06049310a41d1f.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256 : d09a1391267a174beda0f5d62f8adb0b7ebda82318099ff975f233d451899d7e.
Independent certificate:
sha256 : 0dbb95ae8ca617e091662cc252d9aeb4b9c6efdbf49b3e104cdf12a0f8408369. Engine
external-validation hash:
sha256 : d4a1137be42db9bb12af28fe9e67c29146078476f22e1935ffa866a3f876ccd9.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-GW231123-135430-V3-2026-07-23

Observed comparison records:

- redshift-1: predicted cosmological-redshift-relation; observed cosmological-redshift-relation; exact match **True**
- redshift-2: predicted cosmological-redshift-relation; observed cosmological-redshift-relation; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256 : 7371810eb276fce1cc21e2901b8055a3c0e48f0833584758695a0ea1cc33d296.
Isolation certificate: sha256 : 6fedf4242f76a740ce399eb929ed0d161280c0a474ee5b6e175fadce7f09a418.
Custody certificate: sha256 : 930f8df1cee6fff2559c714f23b4e11d6ce026920dd1e5140611801573e79674.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center:
https://gwosc.org/eventapi/json/GWTC-4.1/GW231123_135430/v3/
(sha256:67266a5406cbee1dc98a9853beb42d771595fa474f5ddd33ab4b368d5b71c45d)

Registered target identities:

- redshift-1 from GWOSC-GW231123-135430-V3-2026-07-23
- redshift-2 from GWOSC-GW231123-135430-V3-2026-07-23

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:4eb9d9e702e73970639f63cb01084df7b21f51888583fa65d5a82780fa20b5b9; engine receipt
sha256:0843f2c5ea47964fd54287546abaac627a210b4bfbad6c0ba87c415f665ccf8e at
receipts/engine/model_admitted/SFT-PHYS-COSMO-REDSHIFT-001-0843f2c5ea47964f.json;
empirical-validation hash
sha256:cfed03b376b686cfc7cadd95821b7cae29ab05266f6964728dff79a22a0de2b3; measurement receipt
sha256:7371810eb276fce1cc21e2901b8055a3c0e48f0833584758695a0ea1cc33d296.

274. Relational expansion observation

Claim identity: SFT-PHYS-COSMO-EXPANSION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Relational expansion is coherent change of source-separation support relative to observation-clock recurrence across multiple source-bound paths; it is not motion into external space.

Expansion is coherent Fold change of relational source support per observation recurrence over a declared finite census.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-COSMO-REDSHIFT-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__coherent-source-separation-clock-change__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	coherent-source-separation-clock-change	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:e1c9e7035c96b1ad0f7584b3db1b9eaa5ca97bb38bdd1bea037ea9f4fde09fe. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:a2dbb0b406192e90d75444c33a1a0cc7bb0f88d0e13b4e41145f977438ba145f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b5ae118031c6058ace28ac8f9026ffb3493dfe178c532404c893156cac44ebce.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:ee430e29142cb3594511f207700c87ea69687f6058b67f30d75c8df741a03357.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:ff266063a5b0223e3d2a5cd0c274b948cd09fc341c0e0cbd98ddfc094ae379e7.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e22e8687206a89a280e1b93bac0efe44993a1487ad0ac1de544a4f4779bc672d. Independent certificate:

sha256:d1d7a1544b2aebaa6b78c3af0af668553ff3b955fff404639b1e8b73e7f0aba1. Engine external-validation hash:

sha256:ebceaa28e01641c304eb29cc3202db88fbc9350587c23383c5edb1ee8e8e47d5.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- expansion-1: predicted relational-expansion-observation; observed relational-expansion-observation; exact match `True`
- expansion-2: predicted relational-expansion-observation; observed relational-expansion-observation; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:93fd59729996521dcfe3e36997dcbe6d913f4bffb4231c4558e9e6f85d15a34b.

Isolation certificate: sha256:634e1764554cecd564b0a02fef77b8d1aced1a35e88895659db9b3a23c79b8d6.

Custody certificate: sha256:d64cc948ec714c9a3715aa64eeea9d65fd107fc00f6795454834b5e4cb3376b9.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: [https://lambda.gsfc.nasa.gov/\(sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddfffe94e50b2d4275915\)](https://lambda.gsfc.nasa.gov/(sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddfffe94e50b2d4275915))

Registered target identities:

- expansion-1 from NASA-LAMBDA-2026-07-23
- expansion-2 from NASA-LAMBDA-2026-07-23

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:227354f67ed81a6825724f89cc89a803b4f8952d65e0ca19945f89ca4bf3a965; engine receipt

sha256:70b48d5fa5822a719fde0ab0073d965387cc0c4548ef14fda2138061b3966f53 at

receipts/engine/model_admitted/SFT-PHYS-COSMO-EXPANSION-001-70b48d5fa5822a71.json;

empirical-validation hash

sha256:ca1f3946976b4526c76109d23042f0b93b6e28caba7f1cfa19edd7f8daba7f72; measurement receipt

sha256:93fd59729996521dcfe3e36997dcbe6d913f4bffb4231c4558e9e6f85d15a34b.

275. Background-radiation field relation

Claim identity: SFT-PHYS-COSMO-BACKGROUND-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Background radiation is a near-omnidirectional recurrence field whose thermodynamic observation class is invariant across registered frequency support after local foreground records are separated.

Cosmic background radiation is a source-history-bound, frequency-consistent thermal Fold recurrence field.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-COSMO-EXPANSION-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__omnidi`

rectional-frequency-invariant-thermal-recurrence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	omnidirectional-frequency-invariant-thermal-recurrence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:2b6cb1e58f97060778f59ea7a5b1935a2635333f35edc74b9800342f046d1a5e. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:8cc7e810ff2e58c660c84249ac346725d8b2e9d7a99b29cebf699bd8266b3ea8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:932c0513c1bb93eb7e383ca9c0efd027c242eebe1796f73e3f3ab4a35518d326.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:d58a5c5c3ad1b7c2116b9a85bdcf7458049dc99b65136d6901944d3eeacd9669.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:651bcc56f70730f691d13534b72f1ad7dedb56692f9948c5401e78a6396ac4d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:64dd57d72ca9cf7c24e56b6c7e3aaf3caaf3ad3c0b77317ac71a66a3fb93b717. Independent certificate: sha256:cac3fc2eef8135972ca5755eac53ecefab08ef65bbd56ca971fb89948f389a4. Engine external-validation hash: sha256:5c93c3c2d3fea1d178c5b5d5e22d800ccb9ff8348e5e4cd8371b8220613cc56f.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-COBE-FIRAS-2026-07-23
- NASA-LAMBDA-COBE-FIRAS-MONOPOLE-2026-07-23

Observed comparison records:

- FIRAS background interval predicted (Fraction(2723, 1000), Fraction(2727, 1000)); independently released monopole interval (Fraction(681, 250), Fraction(1363, 500)); overlap `True`
- deliberately displaced unfavorable interval rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:5c3983615cf9824416f01352c48a21d4eeb5d3b05c0d5db1d210b7b20bcf2085.

Isolation certificate: sha256:ea6140797f74add12df811f0492e2399fb4f6c9943d05cd726ddf661e02afb5.

Custody certificate: sha256:fc67914925bc1270f3ac2a722d756ac6f0320c56405c838eef8be43aaae2ba5f.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center / COBE FIRAS:
https://lambda.gsfc.nasa.gov/product/cobe/about_firas.html
(sha256:5eae175f5ea06f7a5cb863866eeee9d87a38d870937d9a93ebf28629af0d6ef9)
- National Aeronautics and Space Administration, Goddard Space Flight Center / COBE FIRAS:
https://lambda.gsfc.nasa.gov/product/cobe/firas_monopole_spect.html
(sha256:e42bcf7e3e2eccac3ac8c210d4287e2477c394ad03d76ac7f719515d29753f2a)

Registered target identities:

- `firas_monopole_temperature` from NASA-LAMBDA-COBE-FIRAS-MONOPOLE-2026-07-23

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:552759b9f4be9557b1196984bffcfc001d6d13c69e75b831b71999fb46070c65; engine receipt

sha256:2e4f7d3e02b8d087c3cce4a29dd5ee6b13f28758400c6da93090007f6e2d7e29 at

`receipts/engine/model_admitted/SFT-PHYS-COSMO-BACKGROUND-001-2e4f7d3e02b8d087.json`;

empirical-validation hash

sha256:7d88b311c29f98a6dd326a9d4944fafb5827c056276e1ed688bc555348b76b42; measurement receipt

sha256:5c3983615cf9824416f01352c48a21d4eeb5d3b05c0d5db1d210b7b20bcf2085.

276. Gravitational lensing relation

Claim identity: SFT-PHYS-COSMO-LENSING-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Lensing occurs when source-curved event support generates multiple causal propagation paths between one emitter and observer, changing direction, arrival recurrence and observed support while preserving source labels.

Gravitational lensing is multiple source-bound causal Fold paths through relational curvature to one observation support.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-COSMO-BACKGROUND-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__multiple-curved-source-observer-paths__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	multiple-curved-source-observer-paths	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4fd7675d343fb9d994d603a695eb72260228dba9375314c3532c6cda7cbcb4cf`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:3cc2a9e0f5dfd8bf2a27996c4f376e3a512b0c3c176108f1b10a868461af918a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f4e6b91bd814d859f13b3adafad665231ba76827cb01a8b25abfa05843dc5606.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1293463a7a30cb51089c59d24177fdd106c31613e039b12ed6871cc7c9e2f89c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e74c60c8dd88ff2f5a79298eeafec74420d3d4a42b0c9589b9354bba36906f0c.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:f0125e16e59c6103aa719aff85e8c32e19455717c544a0cab29cf1a92362c292. Independent certificate: sha256:0f2974648713cade0dea1dc5e85c4fe4130a680fe87a1319f1a72a7ec58ba150. Engine external-validation hash: sha256:5639b145f3657d0684e93825141f4665d3f89b8805be09b6e4dfba48191298ad.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- lensing-1: predicted gravitational-lensing-relation; observed gravitational-lensing-relation; exact match True
- lensing-2: predicted gravitational-lensing-relation; observed gravitational-lensing-relation; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:53e32cc58a397f1475b947f7f25db97c5eab1c1f964992f7dd5e1e582da7f0b5. Isolation certificate: sha256:be2f4bd026393f9cc4673cb386f1088ffaea6e38a6a05cb38d84525db1712e6b. Custody certificate: sha256:099f8e5d84a2e620b5e9357fc612071c6ba22bd388c4ba9c301ec4ca5983b131.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: <https://lambda.gsfc.nasa.gov/> (sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddfffe94e50b2d4275915)

Registered target identities:

- lensing-1 from NASA-LAMBDA-2026-07-23
- lensing-2 from NASA-LAMBDA-2026-07-23

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census

- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2ff9cfb087ab2d14e71a157108c18c09818244662fc701edfa132b7ce816a5d5; engine receipt

sha256:463d7041874e0b45a8ccafb32f1ee9fba54c8e733caa8446f52cce2549a07d73 at

receipts/engine/model_admitted/SFT-PHYS-COSMO-LENSING-001-463d7041874e0b45.json;

empirical-validation hash

sha256:e8dedb014783ed584746eef77f30aecb9246ff3004ebfd791c247a51910da85c; measurement receipt

sha256:53e32cc58a397f1475b947f7f25db97c5eab1c1f964992f7dd5e1e582da7f0b5.

277. Causal distance and observation-time relation

Claim identity: SFT-PHYS-COSMO-DISTANCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Cosmological distance is the exact equivalence class of source-to-observer causal path support conditioned on emission and observation recurrence records; different distance notions are distinct retained projections.

Cosmological distance is a declared Fold projection of exact causal support and observation-time relations.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-COSMO-LENSING-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__causal-path-support-conditioned-on-clock-records__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	causal-path-support-conditioned-on-clock-records	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:b81da4313a2a535edac98d00f96fd5c2d3df8d3b2c91b004eb6876774027c760`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:aa383062c59596ea69991db0272859b5ac291a76ee550f5438a54416dcbf5522`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f03495752b1f8a3690542261a0db6ca12a1130d48a1a1ca86b52ece949c40932`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ead8c15e0123454492674e2991a83303cd7b9c50188c91d33ccf00d539c543b5`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:72c6f70adb91d4a3030d8086e4c3b58234b48e54d4c05613afedeafb023fd6e2`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d92bab1acf1fc9757e600c3d188a5747dcb24385dc817a1cd7d948d5ecd3392d`. Independent certificate: `sha256:05c01482d13acea0e21a1464be758598c2f44d3d5dd3ce03654bbfeaac5a1981`. Engine external-validation hash: `sha256:cbc661549ac3ba23474490b164ca9700f31afcee7e1f8ab6b32e2cf8325b78bc`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GWOSC-GW231123-135430-V3-2026-07-23

Observed comparison records:

- distance-1: predicted cosmological-causal-distance; observed cosmological-causal-distance; exact match `True`
- distance-2: predicted cosmological-causal-distance; observed cosmological-causal-distance; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:788ac7ad1dca6ccc1703fd4c4121fc4c17eb6a02d9e96b42dcf479aa2c1b6d79.
 Isolation certificate: sha256:ff57831a77dd9a7adefe215a5a1aad5f71cba62b55311bc363354fd26474fc27.
 Custody certificate: sha256:0ff4efd6aa10640c3b25a78199d855a0ff5f0dbecac776014d45a14a5c6b51d2.

Registered source descriptions:

- LIGO Scientific Collaboration, Virgo Collaboration and KAGRA Collaboration / Gravitational Wave Open Science Center:
https://gwosc.org/eventapi/json/GWTC-4.1/GW231123_135430/v3/
 (sha256:67266a5406cbee1dc98a9853beb42d771595fa474f5ddd33ab4b368d5b71c45d)

Registered target identities:

- distance-1 from GWOSC-GW231123-135430-V3-2026-07-23
- distance-2 from GWOSC-GW231123-135430-V3-2026-07-23

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8d540e77a05ef62593ddf4542fffb1fc546723823ef6fc086372c60ccfe3cf52b; engine receipt
 sha256:3652a0696753f255405d64eb358c82e5099ce743763a43a2b693946040f163a3 at
 receipts/engine/model_admitted/SFT-PHYS-COSMO-DISTANCE-001-3652a0696753f255.json;
 empirical-validation hash

sha256:0298a6976febe7b847faf2e661c41e5c67f2f31f60af258af6f337b76255eaa3; measurement receipt
 sha256:788ac7ad1dca6ccc1703fd4c4121fc4c17eb6a02d9e96b42dcf479aa2c1b6d79.

278. Finite structure-growth relation

Claim identity: SFT-PHYS-COSMO-STRUCTURE-GROWTH-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Structure growth is deterministic evolution of a finite source-density distinction network under locally forced interaction and expansion relations; weights describe unresolved initial/history fibres rather than ontic randomness.

Cosmic structure growth is finite deterministic Fold network evolution under registered gravitational and expansion relations.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001

• SFT-PHYS-COSMO-DISTANCE-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__finite-density-network-dynamical-growth__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	finite-density-network-dynamical-growth	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2c10da892c1aded3ed2cad6743b2eef5425cfd153f79ee0c6b2fd5ee587fc6db`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4e6a66e9753cfc7df280705717c0fcf90fabcc3c968ab2d95191d72dba295567`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6294b6a7219d87acbd7ac9cd3fd06eeef95f57487d3408af2cafcad072092a45`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6533e6345df8c7b31418838b2fd1c63bfe822ddd7f37fb0307e841748f779a62`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7e741f8ba12c27a8e4a76258e9fb1f384f4c067b450ea022c48bad8b8d21bdeb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:eb896632cf27d04f3c63a7581367588cebcfbf9d67de990119b9522392de880c2`.

Independent certificate:

sha256:2e2581a18ed8db4826faf50042bf61a441363d25c9ba0c02ebf744d886b53ff4. Engine

external-validation hash:

sha256:5d49103d5af99bed683de928900c63b91da202f7a6f0bf0b8506da517178294b.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- structure-growth-1: predicted finite-cosmic-structure-growth; observed finite-cosmic-structure-growth; exact match True
- structure-growth-2: predicted finite-cosmic-structure-growth; observed finite-cosmic-structure-growth; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256:e4d0d65ebfcb3ae49d1835e3cffc745e67e2a981ffbb3125196a576e2cbca395.

Isolation certificate: sha256:59e4c6eeab740920eb407ba2581338255007e51de8d809dab09da981bff5d400.

Custody certificate: sha256:f472907947520fb1b48dcded45b5dfd911e819ba009269e185488ea60ee4cbd4.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: [https://lambda.gsfc.nasa.gov/\(sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddfffe94e50b2d4275915\)](https://lambda.gsfc.nasa.gov/(sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddfffe94e50b2d4275915))

Registered target identities:

- structure-growth-1 from NASA-LAMBDA-2026-07-23
- structure-growth-2 from NASA-LAMBDA-2026-07-23

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7e09ebcf55d86d177b601ec853ea64167d5c7ac6dd6f573d8caed18256da3437; engine receipt

sha256:b150f2ddf89d4a0ea2352ebc7ae1ac954835eb6ac1856f38c6ddd8f8d5961744 at receipts/engine/model_admitted/SFT-PHYS-COSMO-STRUCTURE-GROWTH-001-b150f2ddf89d4a0e.json;

empirical-validation hash

sha256:b9423cbcd498d5c7190ff4f9823de2a3944613d4acd5d2c9b4f071f5be563f6c; measurement receipt

sha256:e4d0d65ebfcb3ae49d1835e3cffc745e67e2a981ffbb3125196a576e2cbca395.

279. Physics-to-astronomy/cosmology handoff boundary

Claim identity: SFT-PHYS-COSMO-BRANCH-BOUNDARY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Physics closes the universal source, propagation, observation and interaction relations; object census, historical initial state and inferred cosmic chronology pass as explicit measured inputs to the later Astronomy/Cosmology branch.

The branch handoff preserves closed Physics laws and exports only source-bound finite observations for later astronomical reconstruction.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-COSMO-STRUCTURE-GROWTH-001

Generated grammar and closure boundary. Generate the complete emission, propagation, observation-time, source/response, provenance, target-access, successor and extra-rule product. The declared exact boundary is: All finite cosmological-boundary forms generated from exact source-bound propagation, recurrence ratios, causal paths and observation records; astronomical census and historical initial conditions remain outside this Physics closure. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__law-versus-historical-state-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	law-versus-historical-state-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:09b1913ced9baccd4ad6da10830944772c9628b7538a430663d9a19ccecacb1d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated

by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 43993065c8994e8e6caf618318ed6ae948b22df1bbd82a4678c2865d80ba89b6.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 9579c3b9ecba7097e4caae2712a7b0aeea983ae70b5d83fb8bb1b1a4290856a1.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 9b6a0c84cbb54a1ac283a2806ecbf11c23551aa2ca7639fecb9fed1bdf4d349b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : fff8bfff53425a262dc269b5a84aba203bc7d05d2b4d16d384c5a3e830d104e52.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : e34d327c6aef4d6ee556576812a387fe8696bd13e948e144e11c271e682b73c3.

Independent certificate:

sha256 : d3c90ff6e0e964eff72260f53e9e2ff541241215ecfb3f945e51bb9286b525e0. Engine

external-validation hash:

sha256 : 787b275a646d773bd9baa3ed2d87d7e950771b93ae276ee74972812d12ff64ac.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-LAMBDA-2026-07-23

Observed comparison records:

- branch-boundary-1: predicted physics-cosmology-handoff; observed physics-cosmology-handoff; exact match `True`
- branch-boundary-2: predicted physics-cosmology-handoff; observed physics-cosmology-handoff; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The cosmological-boundary law is falsified if a committed external row lacks the sealed propagation/observation relation, a row is omitted, or the tampered row is accepted.

Measurement receipt: sha256 : 0b9e11696be7f741781838ec270ac7e215e09d009fc3c12042e6675586b2fdf8.

Isolation certificate: sha256 : ed1a02b134a440f9e2c5dd4945a44ae50240e798a31979b14b8034e3f8a80b45.

Custody certificate: sha256 : 34122d9c28743e35a51d7e39977c8c8b10d24641b4cbcf195ca28e2508e7fb4e.

Registered source descriptions:

- National Aeronautics and Space Administration, Goddard Space Flight Center: <https://lambda.gsfc.nasa.gov/>
(sha256:805ea293cbcf660499be996e0d80175b61ba2a68bdddfffe94e50b2d4275915)

Registered target identities:

- branch-boundary-1 from NASA-LAMBDA-2026-07-23
- branch-boundary-2 from NASA-LAMBDA-2026-07-23

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- ungenerated cosmic continuum, completed infinity or whole-universe census
- floating, irrational, imaginary or signed proof magnitude
- imported cosmological fit, historical initial condition or target-selected parameter
- target access, selected favorable observation or unrecorded source/path

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c5e9469ae63514fb6102751a96b8e9102244e843b8a7c3a959c9bade858cb3fb; engine receipt
 sha256:86e32f007bb9c6304edc82ae7e4f01db1f0391b5fbbef56d5ddedc12dd178160 at receipts/engine/model_admitted/SFT-PHYS-COSMO-BRANCH-BOUNDARY-001-86e32f007bb9c630.json;
 empirical-validation hash
 sha256:fa5c646653b938d6c7de35cadac6259a51aa2d0fdccab5e76f8482da14f4b618; measurement receipt
 sha256:0b9e11696be7f741781838ec270ac7e215e09d009fc3c12042e6675586b2fdf8.

280. Generation-cover dark-to-baryon fraction

Claim identity: SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Generator three over the forced three-space supplies volume 27. Its least complete binary cover has depth five and support 32, forcing leading baryon/dark shares 5/32 and 27/32 and ratio 27/5. The unique period-five orbit floor 31 deepens the same depth once, forcing 279/52.

The leading dark-to-baryon ratio is 27/5 with shares 27/32 and 5/32; the unique native period-five deepening is exactly 279/52.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-FOLD-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete product of generation carrier, spatial volume, cover, partition, leading ratio, recurrence floor, deepening, measurement boundary, provenance and extension forms. The declared exact boundary is: All exact first-recursive dark/baryon partitions generated from the admitted generator-three spatial volume, its least binary cover and the first-return orbit native to that cover depth. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: one-complete-matter-cover__generator-or-over-forced-space__least-complete-binary-cover__depth-and-volume-over-cover__volume-over-depth__own-depth-first-return-floor__one-positive-orbit-part-appended__densities-inaccessible-until-seal__prior-question-machine-independent-reconstruction__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	one-complete-matter-cover	named-dark-and-baryon-scalars	Names do not construct a common conserved support.
volume	generator-over-forced-space	selected-cosmological-volume	A selected volume reads the target.
cover	least-complete-binary-cover	free-cover-depth	A free depth is a fitted parameter.
partition	depth-and-volume-over-cover	independent-normalizations	Separate denominators need not conserve one matter whole.
leading	volume-over-depth	measured-ratio-insertion	A measured ratio is not a derivation.
orbit	own-depth-first-return-floor	tower-or-up-depth-floor	The tower is pre-periodic and the up-depth belongs to another carrier.
deepening	one-positive-orbit-part-appended	selected-sign-or-bare-depth	A chosen orientation or bare repeat adds no forced recurrence record.
target	densities-inaccessible-until-seal	density-readable-construction	That would fit the ratio.
provenance	prior-question-machine-independent-reconstruction	old-answer-runtime-input	Earlier answer artifacts cannot select the V3 law.
extension	no-extra-rule	extra-density-or-fit-term	An added term is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:8d183903a0f43d28ae6e8dd4350fc3818619f41083dc1360486091788fcddb`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:13b87ee58afe7e2f83654afb81b6fcb56e86bd8cf9ef0bb0c5100b83cbfdb528`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4655befa0bb2e2327eb6a52c79ee6bfaa11e74f1e3e90651a0efe99374e58591`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ea4e84735302478c251e5608af61e891a7ed5edf79829dada9455096541c7c1a`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:01bacc06ab1cdf3734a9b0d49205526e6878f557b380dee5fedccf17345520d1`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:675d474c832285d80cb109d73394f0a5568cc55361437841ff048913967a99d1`. Independent certificate: `sha256:ed22d598f85cc461219ca381a91a8f098ee968271527e934a9d49eb066ecfbc0`. Engine external-validation hash: `sha256:671295b17b78b7078c86aa5d91762fa03d64255579c107685e0b933f03dea13f`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PLANCK-2018-VI-ABSTRACT-DENSITIES

Observed comparison records:

- PLANCK-2018-BARYON-DENSITY: predicted
sealed-leading-and-native-deepened-ratios-inside-complete-planck-density-interval; observed
sealed-leading-and-native-deepened-ratios-inside-complete-planck-density-interval; exact match True
- PLANCK-2018-COLD-DARK-DENSITY: predicted
sealed-leading-and-native-deepened-ratios-inside-complete-planck-density-interval; observed
sealed-leading-and-native-deepened-ratios-inside-complete-planck-density-interval; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: Either sealed ratio lies outside the complete density interval, either density row or uncertainty is omitted, source custody changes, or a tampered comparison is accepted.

Measurement receipt: sha256:5d3f4de0ae278c3995c2316e770d497042e3bbd76514c9f502a946d0f707c62a.

Isolation certificate: sha256:55eb8379a160e1e4a32ee838811cf1b6dfc6431f0822ddf66e0d673f08cb4174.

Custody certificate: sha256:0461a4f28a373375ef898773b281129a6e493484e028dc83167f7c2c3c17652f.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 answer artifact or cosmological density in the executable derivation
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted density, tolerance, selected floor or additional cosmological parameter
- no assertion that an external cosmological model is an SFT premise

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7b82678d0d69bf27fbaaa45a81386c321a2b55287b5716ded6d04b3299eeac97; engine receipt

sha256:38b06863d5a59f8f8ea17fee7a0a1d5ff1fcdc0c6f7b9de3e9f635705d4f8cc2 at receipts/engine/model_admitted/SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001-38b06863d5a59f8f.json;

empirical-validation hash

sha256:7dc5aa166443cb243a2a2d7e9e81be6188df5f77ee441a95879c35aec27a47fa; measurement receipt

None.

281. Fold calibration ratio between early and late expansion routes

Claim identity: SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. One retained formed class and the complete two-fibre open complement partition the generator-three cosmic carrier into exact matter and vacuum shares 1/3 and 2/3. Distributing the open share over the depth-three Fold support eight forces correction 1/12 and late/early calibration ratio 13/12. The independently forced depth-seven first-return floor 127 supplies the unique registered first-recursive deepening, giving exact ratio 3305/3048.

The exact matter/vacuum shares are 1/3 and 2/3; the leading correction and late/early ratio are 1/12 and 13/12; the unique registered depth-seven recurrence refinement is 3305/3048.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

Generated grammar and closure boundary. Generate the complete product of cosmic carrier, occupancy distinction, partition, calibration support, leading correction, deep cover, recurrence floor, recursive correction, target custody, provenance and extension forms. The declared exact boundary is: All exact leading and one-native-recurrence calibration ratios formed from the complete One, the Fold fibres, generator three, depth-three support and the independently forced depth-seven orbit. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: `one-complete-cosmic-carrier__one-formed-class-and-two-fibre-open-complement__one-third-formed-two-thirds-open__complete-depth-three-Fold-support__open-share-over-complete-support__independently-forced-up-cover-depth__depth-seven-first-return-floor__one-positive-orbit-part-inside-open-share__both-routes-inaccessible-until-seal__observational-reconstruction-with-independent-runtime__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	one-complete-cosmic-carrier	independent-density-scalars	Independent scalars do not close one conserved whole.
occupancy	one-formed-class-and-two-fibre-open-complement	named-matter-and-vacuum	Names do not force multiplicity or orientation.
partition	one-third-formed-two-thirds-open	selected-or-reversed-shares	Selection or reversal loses the formed/open orientation.
support	complete-depth-three-Fold-support	free-calibration-scale	A free scale is a fitted parameter.
leading	open-share-over-complete-support	measured-ratio-correction	A measured correction would select the answer.
depth	independently-forced-up-cover-depth	selected-cosmological-depth	A selected depth is a parameter.
orbit	depth-seven-first-return-floor	arbitrary-denominator	An arbitrary denominator fits a correction.
deepening	one-positive-orbit-part-inside-open-share	selected-sign-or-extra-term	A sign choice or external term is not forced.
target	both-routes-inaccessible-until-seal	hubble-values-readable-before-seal	That is fitting.

Axis	Forced form	Eliminated form	Elimination reason
provenance	observational-reconstruction-with-in-dependent-runtime	mislabel-as-unobserved-discovery	The prior result was already observed.
extension	no-extra-rule	extra-calibration-rule	An added rule is a free parameter.

Base and successor. The closure proof is sealed by derivation hash

sha256:b43b6fc427b35e8b6fcf1384a2e82a4d26bf1d4eb01dbba64bd9c833fea73500. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:e6a7e7f78a3e8655d01c36d33b35c823f5215d24eb4073dbf881de3c29eeba4c.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:793381e00887efeac6ded724dc0422e9ba3039650cf85e5f288d49c70cbf0480.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3d16267cb16626a1efecb424ffc8cdc455c3733937fcaa52e48dd53ea0dd9f70.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:6a6c3eef48a63aab3af0d8af496a5247c390b8c5dbaccee665af301eaf944782.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:990b8774815230213ce4e33207c0362f06a6685321085ae27872cecc79e0708c.

Independent certificate:

sha256:b9a45aaa7f5eab1950f7a8df2fce83966bd9a39efa88de12a7439730d8f63e89. Engine

external-validation hash:

sha256:1bcd278722f05a70ba1afa40ab187899f7fb655794506b9cd525ba9f18d73bd.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PLANCK-2018-VI-HUBBLE
- SHOES-2022-BASELINE-HUBBLE

Observed comparison records:

- **PLANCK-2018-HUBBLE-INTERVAL:** predicted
sealed-leading-and-depth-seven-ratios-inside-complete-planck-shoes-ratio-interval; observed
sealed-leading-and-depth-seven-ratios-inside-complete-planck-shoes-ratio-interval; exact match `True`
- **SHOES-2022-HUBBLE-INTERVAL:** predicted
sealed-leading-and-depth-seven-ratios-inside-complete-planck-shoes-ratio-interval; observed
sealed-leading-and-depth-seven-ratios-inside-complete-planck-shoes-ratio-interval; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: Either sealed exact ratio lies outside the complete outward-propagated late/early interval; either route, central value, uncertainty, unit or qualification is omitted; source custody changes; or a tampered comparison is accepted.

Measurement receipt: sha256:707f4c2183b11bb784794a61169b173740df83da19bd169960a8cc3f5fcbb803.

Isolation certificate: sha256:b649a5ba65e6c9c33f1c0b2c6df315f32fabf9bf3ab4f2f80185fc7884eaa7e3.

Custody certificate: sha256:70e5d45c18f9001d7fe328f22d4548878c9edfade98365a71739751aaf73d883.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 answer artifact or Hubble measurement in executable forcing
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted density, selected tolerance, calibration scale, depth, sign or extra term
- no claim that either external measurement route is itself an SFT premise
- no claim that the observational reconstruction was an unobserved blind discovery

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:10cf8943a4026234c1c170b2cc41441c6ca8678525deb7b05031ab6d9c5aeb42; engine receipt

sha256:d4ce8d8568e94b5032fc65633d024aaa7cba6365e6d88217c61ec9a388153e88 at receipts/engine/model_admitted/SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001-d4ce8d8568e94b50.json;

empirical-validation hash

sha256:b2090f6bb9fc1db1e7963539ad9adf580de79e45a0681c0a77cf886046c99c8d; measurement receipt

None.

282. Complete-partition spatial flatness and absent curvature remainder

Claim identity: SFT-PHYS-COSMO-SPATIAL-FLATNESS-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Every generated cosmic component is an exact positive part of one common total carrier. A complete component family exhausts that One, so a separate curvature remainder is the empty One form rather than a numerical zero. Every positive repartition preserves the same closure at arbitrary finite depth.

Every finite complete exact positive cosmic-component partition closes to the One; the curvature remainder is the empty One form, and every positive parent-preserving refinement retains that result.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-PHYS-COSMO-BRANCH-BOUNDARY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete product of total carrier, component identity, normalization, completeness, remainder role, absence representation, repartition, temporal scope, target custody, provenance and extension forms. The declared exact boundary is: All finite exact positive cosmic-component partitions of one common total carrier and every finite positive refinement that preserves the parent part. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: `one-common-total-carrier__exact-positive-held-part__common-One-normalization__every-generated-part-exhausts-One__unassigned-part-remainder__empty-One-form__positive-parent-preserving-split__all-finite-repartitions__curvature-record-inaccessible-until-seal__observational-reconstruction-with-independent-runtime__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	one-common-total-carrier	independent-component-totals	Independent totals need not form one conserved whole.
component	exact-positive-held-part	signed-or-unbound-density-number	An unbound number is not a held part.
normalization	common-One-normalization	separate-normalizations	Separate normalizations cannot close jointly.
completeness	every-generated-part-exhausts-One	selected-visible-components	A selected list can leave an unregistered remainder.
remainder	unassigned-part-remainder	extra-curvature-scalar	An extra scalar after exhaustion double-counts support.
absence	empty-One-form	numerical-zero-curvature	Numerical zero is not an SFT proof carrier.
refinement	positive-parent-preserving-split	epoch-specific-fraction-table	A table has no general preservation law.
scope	all-finite-repartitions	single-observed-epoch	One epoch cannot force persistence.
target	curvature-record-inaccessible-until-seal	curvature-record-readable-before-seal	That would import the desired result.
provenance	observational-reconstruction-with-independent-runtime	mislabel-as-unobserved-discovery	The prior flatness result was already observed.
extension	no-extra-rule	extra-cosmological-closure-rule	An added rule is a free premise.

Base and successor. The closure proof is sealed by derivation hash `sha256:cbf53b8913e6d623f58bdecba55a2db1b837714435ff9d5a0772d5d9c8e50b29`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:75dad868142a0a34a404a85fbbc39d70a82b419d49390835a6254659f37d5131`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:60473ebc3e3cbb47602a0b5657066aaa39dcd3f4c0307d694ae87c0f8693aee9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e9c4b946bca30d903b8e277776feb03fdf005094151fae375d33420348370c40`.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : 9d589341c63a50dcfca228e1b99a4e320a5d1e2f33205b66b28254303d70cca0.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : 40818b5b12e926755cf41400949fcd6aa049e5438a291041e90fe079461f1366.

Independent certificate:

sha256 : a18c1dd8c93e9f17769729de3035bef04e20cabf573ebc81956327fea57b5b22. Engine

external-validation hash:

sha256 : f6c4e1e72d8200602921503cdd8a5af9c5395f790abeb8ab45d5b6968fb30361.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PLANCK-2018-VI-CURVATURE-47B

Observed comparison records:

- PLANCK-2018-BAO-CURVATURE-47B: predicted
sealed-empty-curvature-remainder-inside-complete-planck-bao-signed-interval; observed
sealed-empty-curvature-remainder-inside-complete-planck-bao-signed-interval; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The complete registered curvature interval excludes structural absence; its orientation, central magnitude, uncertainty, confidence region or data combination is omitted; source custody changes; or a tampered comparison is accepted.

Measurement receipt: sha256 : d27eb94493dd7fa36684f9bb7cbd8333bec3046a0251767c2ca2f27cc9a48927.

Isolation certificate: sha256 : bad32e4c57d6fb6be9aaf7ea8ccd6895aa59c6ab8a0ec6acbd163aff9c8bac1f.

Custody certificate: sha256 : 4e0aaf8ff7f9536427a40c95f4e3227116a28f221e4083bec4df300fa931ff49.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured Ω_K value or standard cosmological field equation in executable forcing
- no numerical zero, negative, irrational, imaginary or floating proof value
- no curvature remainder treated as a signed proof scalar
- no selected epoch, incomplete component list or extra closure rule
- no claim that Planck parameter inference is an SFT premise

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 34000aa9eb14f2f83a9f2bc1652509fac37592d7698da5ee08cf030f4738cb60; engine receipt

sha256 : ec8cf537a7460687e1ca3d1c9e5d1781b96b477e4c11f68d7c3208e82d3d1a66 at receipts/engine/model_admitted/SFT-PHYS-COSMO-SPATIAL-FLATNESS-001-ec8cf537a7460687.json;

empirical-validation hash
sha256:d552516176eb76d0262093e421930cf43cf3c005763ee90e5c75a88faaf97978; measurement receipt
None.

283. Depth-five Fold cosmic energy budget

Claim identity: SFT-PHYS-COSMO-COMPLETE-BUDGET-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The least binary support covering the generator-three spatial volume has depth five and 32 states. The complete Fold boundary pair pins two states at each depth, forcing ten pinned matter states and twenty-two free vacuum states: matter 5/16 and vacuum 11/16. The independently reconstructed five-to-twenty-seven matter partition then forces baryon 25/512 and cold dark 135/512.

The refined present budget is vacuum 11/16, total matter 5/16, baryon 25/512 and cold dark 135/512; baryon plus cold dark equals matter and vacuum plus matter equals the One.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001
- SFT-PHYS-COSMO-SPATIAL-FLATNESS-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar and closure boundary. Generate the complete product of support, boundary incidence, pinning, free complement, global partition, internal matter partition, composition, closure, target custody, provenance and extension forms. The declared exact boundary is: All exact present-budget partitions formed from the least depth-five binary cover, one complete boundary pair per depth and the independently admitted five-to-twenty-seven matter partition. The generator produced 2048 named candidates and the decision support contains 2048 one-for-one decisions. Exactly one candidate survived: least-generator-volume-binary-cover__complete-two-state-boundary-pair__all-depth-boundary-pairs__complete-unpinned-complement__five-sixteenths-and-eleven-sixteenths__admitted-five-to-twenty-seven-partition__global-matter-times-internal-shares__exact-four-part-One-closure__all-planck-rows-inaccessible-until-seal__disclosed-successor-observational-reconstruction__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
support	least-generator-volume-binary-cover	selected-density-denominator	A selected denominator fits observations.
boundary	complete-two-state-boundary-pair	free-pinned-count	A free count is a parameter.
pinning	all-depth-boundary-pairs	named-matter-share	A name does not construct a share.

Axis	Forced form	Eliminated form	Elimination reason
free	complete-unpinned-complement	selected-vacuum-share	A selected share reads the target.
global	five-sixteenths-and-eleven-sixteenths	leading-one-third-two-thirds-only	The earlier leading partition ignores the admitted depth-five boundary discriminator.
internal	admitted-five-to-twenty-seven-partition	new-baryon-dark-parameter	A new ratio is free.
composition	global-matter-time-s-internal-shares	independent-component-normalizations	Independent denominators need not close.
closure	exact-four-part-One-closure	separate-rounded-total	Rounded values cannot prove closure.
target	all-planck-rows-in-accessible-until-seal	planck-budget-readable-before-seal	That would fit four targets.
provenance	disclosed-successor-observational-reconstruction	erase-leading-law-or-mislabel-discovery	The leading law and later observed refinement must both remain visible.
extension	no-extra-rule	extra-budget-rule	An added rule is a free parameter.

Base and successor. The closure proof is sealed by derivation hash `sha256:8565cfa8a8f331dc805b61234ad5b7d5a5e66354d5a9249f54e039d68a785a56`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e81d1523b010a9f8a9afdc38fa6d75d1b6a71a2f668fd5e458e75158a2885e56`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:4c9cf1328e2b99e37fd5530c3176c7bf5a2c22cd5ba908aa7335770f36029259`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:485b256c1b7845b037ee4d3664b2f5f8887919e64ff95e89fd02de409e9f8ff9`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:c3dc60a5559ebdbb06247db197bb11549765fa5140920078750c6578ae3701fb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:a88c40db509140589a9cd7606cef716d5bcbda246a38e268f57b06cd4c6ed419`. Independent certificate:

`sha256:146f3c550bedff74d628ff8e3feafef3a5dd7c1264e491ca91efceda1f31a1bd`. Engine

external-validation hash:

`sha256:d5cba833780bb989f3c44a164fa199005eef7e1a84a1af9b7b340cbad61ad86d`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PLANCK-2018-BAO-BASELINE-BUDGET

Observed comparison records:

- PLANCK-BAO-VACUUM-FRACTION: predicted sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; observed sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; exact match True
- PLANCK-BAO-MATTER-FRACTION: predicted sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; observed sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; exact match True
- PLANCK-BAO-BARYON-FRACTION: predicted sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; observed sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; exact match True
- PLANCK-BAO-COLD-DARK-FRACTION: predicted sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; observed sealed-refined-four-part-budget-inside-all-complete-planck-bao-intervals; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: Any refined component lies outside its complete interval; any earlier leading component is falsely labelled as passing; any required source row or uncertainty is omitted; custody changes; or a tampered comparison is accepted.

Measurement receipt: sha256: 4bae1310817433cc59334a735d348e0b4b2295c0f5efaa6aeebb9eabb38da859.

Isolation certificate: sha256: e6b8579f22d9e46e57ac79ffeeb9ada4b2753ca112296b08efddd3b9660d9ec.

Custody certificate: sha256: a570925340e728e88f1d9fa03f6d3cfcf0c00e4aa8f209afc75e11d5abe5d0b.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no Planck value or V1/V2 answer artifact in executable forcing
- no numerical zero, negative, irrational, imaginary or floating proof value
- no fitted density, selected tolerance, free pinned count or extra component rule
- no erasure of the earlier 1/3 and 2/3 leading approximation
- no claim that the external base-Lambda-CDM parameter analysis is an SFT premise

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: d1211befaf769f9729f7d8b95168e99e2193f47b1e8a72a950fbf9ad7d471152; engine receipt

sha256: 9d9c7593823ce0409ee7030e2c03baf19e99e90630092de30e20f00980dcbcd2d at receipts/engine/model_admitted/SFT-PHYS-COSMO-COMPLETE-BUDGET-001-9d9c7593823ce040.json;

empirical-validation hash

sha256: 4c109570794bded6ad2fd8b5b916845eb39fa85198d4fe9526291cb683430cd9; measurement receipt

None.

284. Composite Fold-orbit topology and inter-universe transport boundary

Claim identity: SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Coprime odd-denominator Fold components have an exact bijective composite map that commutes with Fold and has least-common-multiple recurrence, forcing joint lockstep correlation; native Fold dynamics preserves each denominator, one

component cannot steer another, and the literal physical inter-component signalling/travel record is empty.

Coprime odd-denominator Fold components have an exact bijective composite map that commutes with Fold and has least-common-multiple recurrence, forcing joint lockstep correlation; native Fold dynamics preserves each denominator, one component cannot steer another, and the literal physical inter-component signalling/travel record is empty.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-FOUNDATION-PART-EQUIVALENCE-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-ALGEBRA-001
- SFT-MATH-ORBIT-NUMBER-THEORY-002
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete product of orbit identity, denominator dynamics, composite topology, component map, period law, correlation record, signal test, travel test, physical interpretation and extension forms. The declared exact boundary is: Every positive reduced odd-denominator Fold orbit; every coprime pair and its complete composite unit-residue product; every positive finite Fold depth; and the admitted locality, causal-order, entanglement and no-signalling boundaries. The generator produced 1024 named candidates and the decision support contains 1024 one-for-one decisions. Exactly one candidate survived: `exact-odd-denominator-orbit-component__denominator-preserved-at-every-step__complete-coprime-composite-product__bijective-commuting-component-map__least-common-multiple-period__joint-lockstep-support-correlation__target-trajectory-independent-of-source__empty-cross-component-transition-record__arithmetic-correspondence-with-causal-boundary__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
orbit	exact-odd-denominator-orbit-component	unbounded-cosmological-world	A cosmological world is not generated by denominator arithmetic.
dynamics	denominator-preserved-at-every-step	selected-denominator-change	No Fold transition generates a new denominator.
topology	complete-coprime-composite-product	asserted-network-edge	An asserted edge has no exact state map.
mapping	bijective-commuting-component-map	lossy-or-selected-projection	Selection or loss breaks exact component recovery.
period	least-common-multiple-period	chosen-or-multiplied-period	A chosen/product period ignores shared recurrence factors.
correlation	joint-lockstep-support-correlation	independent-unrelated-trajectories	That discards the shared composite origin.
signal	target-trajectory-independent-of-source	correlation-relabeled-as-signal	Correlation alone supplies no sender-selected change.

Axis	Forced form	Eliminated form	Elimination reason
travel	empty-cross-component-transition-record	description-map-relabeled-as-travel	A preparation/projection map is not a native transition path.
physical	arithmetic-correspondence-with-causal-boundary	literal-multiverse-transport-asserted	Arithmetic components do not by themselves establish multiple physical cosmologies or traversal.
extension	no-extra-rule	free-portal-or-time-channel	A portal or temporal channel is an unforced extra law.

Base and successor. The closure proof is sealed by derivation hash `sha256:1b13d077a862deef71159fe0510fe85a55df8692127f59bae10afa4c58ec1612`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:42d217e1ec9558cef2ed9e8eae23b48fca3b93c20eb6f6105f3358d7453499da`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:84b93776a59aed706e545a356e4b0a22671d1c01b0a7f233e0d7cdf836981623`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:21843056311f55d0f352361d3b45405dfbcedd28f78885292121975ac0942a3d`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:bcdd2b1f4aa185c175a837aa171b4065c0c5c805f59c6b78f31904f8b349283d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f9152ab79a3d9b2dbf20612699ffa53bf0dd91a02cf8f94dacf517bb1c2df172`. Independent certificate: `sha256:c7aaba8e82e50a15947f40eaab615a661c035b5635fe1c62922855f44a103591`. Engine external-validation hash: `sha256:acbbb2768ee213ad6f25d2b834c6d534a16f448de4cb8dcc24b0a509b4cbbf42`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported cosmological multiverse, wormhole, closed-timelike-curve or portal premise
- no V1/V2 executable, answer table, physical target or stored survivor
- no numerical-zero, negative, irrational, imaginary or floating proof magnitude
- no relabelling of arithmetic composition/projection as literal physical travel

- no relabelling of deterministic correlation as directed communication
- no physical universe plurality without a separately admitted observational discriminator

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:11495f5fc8cfb40b8b30ef1b5ad2d6566c2aedf6acef48066fb6d1ca39703b92; engine receipt sha256:b03563b484c95269254dd2bb409ae42efbd2a71989eba3ac1b3dd8171c7e467d at receipts/engine/model_admitted/SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024-b03563b484c95269.json; empirical-validation hash None; measurement receipt None.

285. Terminal cosmic component transport and expansion law

Claim identity: SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Matter, radiation and vacuum have respectively third-power, fourth-power and Fold-invariant transport. The terminal late-time law is $E2(r)=(11+5r^3)/16$; $\Omega_m(r)=5r^3/(11+5r^3)$; $\Omega_v(r)=11/(11+5r^3)$; matter-vacuum equality is $r^3=11/5$; acceleration onset is $r^3=22/5$; and the present accelerating magnitude is $17/32$. The static-vacuum correspondence is tension of One magnitude. All quantities are exact positive whole/fractional Fold forms or typed empty/orientation structures.

Matter, radiation and vacuum have respectively third-power, fourth-power and Fold-invariant transport. The terminal late-time law is $E2(r)=(11+5r^3)/16$; $\Omega_m(r)=5r^3/(11+5r^3)$; $\Omega_v(r)=11/(11+5r^3)$; matter-vacuum equality is $r^3=11/5$; acceleration onset is $r^3=22/5$; and the present accelerating magnitude is $17/32$. The static-vacuum correspondence is tension of One magnitude. All quantities are exact positive whole/fractional Fold forms or typed empty/orientation structures.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001
- SFT-PHYS-COSMO-REDSHIFT-001
- SFT-PHYS-COSMO-EXPANSION-001
- SFT-PHYS-COSMO-SPATIAL-FLATNESS-001
- SFT-PHYS-COSMO-COMPLETE-BUDGET-001
- SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001

Generated grammar and closure boundary. Generate the complete product of spatial carrier, matter transport, radiation transport, vacuum invariant, terminal expansion, density fractions, equality threshold, acceleration threshold, orientation typing, physical scale transport, target custody, prior-correction and extension forms. The declared exact boundary is: Every positive exact rational stretch; the three matter/radiation/vacuum transport classes; the admitted terminal 5/16 and 11/16 late-time shares; every exact derived fraction and cube threshold; all thirty-two registered cosmic-chronometer rows; the registered acceleration, equation-of-state, Planck-ratio and DESI adverse rows; and no imported continuum or signed proof scalar. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `admitted-generator-three-volume__inverse-volume-third-power__volume-plus-one-recurrence-power__Fold-invariant-One__terminal-eleven-five-curve__component-over-complete-rate__exact-cube-eleven-fifths__exact-seventeen-thirty-seconds-and-twenty-two-fifths__positive-magnitude-with-held-orientation__post-seal-held-reference-transport__all-targets-open-only-after-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
space	<code>admitted-generator-three-volume</code>	<code>borrowed-three-dimensional-volume</code>	Borrowing the exponent imports a premise.
matter	<code>inverse-volume-third-power</code>	<code>selected-matter-power</code>	A selected exponent is a free law.
radiation	<code>volume-plus-one-recurrence-power</code>	<code>selected-radiation-power</code>	A selected fourth power is not derived.
vacuum	<code>Fold-invariant-One</code>	<code>selected-static-density</code>	A named cosmological constant cannot select invariance.
expansion	<code>terminal-eleven-five-curve</code>	<code>superseded-two-thirds-curve</code>	The leading budget is rejected by the admitted terminal discriminator.
fractions	<code>component-over-complete-rate</code>	<code>fitted-epoch-density</code>	An epoch fit is a parameter.
equality	<code>exact-cube-eleven-fifths</code>	<code>decimal-root-selected</code>	A decimal root may be target-selected and irrational.
acceleration	<code>exact-seventeen-thirty-seconds-and-twenty-two-fifths</code>	<code>old-half-magnitude-and-four-cube</code>	The old result inherits a superseded budget.
orientation	<code>positive-magnitude-with-held-orientation</code>	<code>negative-proof-scalars</code>	Negative scalars violate the Fold proof language.
scale	<code>post-seal-held-reference-transport</code>	<code>fitted-Hubble-normalization</code>	Fitting H_0 to the chronometer rows would select the result.
target	<code>all-targets-open-only-after-seal</code>	<code>measurements-readable-before-seal</code>	Target access can manufacture the law.
extension	<code>no-extra-rule</code>	<code>free-component-correction</code>	An inserted coefficient is a fit.

Base and successor. The closure proof is sealed by derivation hash `sha256:8efa797d1e574e5c87696c84bbdd034a73bc59b71b68d3f60850c8c00172f2e0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:946f81f183439e0e95006a602f8bd8deac1eb8098846018d89fce73b82b02833`.

- **tampered_source**: passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256: ce283a1afae5385be857dd9d15f628774982e125943bfff0717b59b0b164d2b3a.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 78728a9bbbf9ad4da51d496b0edaac4e14e3f739ad4c4d316fb5c9961d41648.
- **boundary**: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256: af2dc89f3d4177fba3f98917394e50ada6e45233730681b1c90e7830eaf38b0.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: 0368583227e567a96df47d2c3a39f3914cf6d3222bf26981b35db371739a33a9.

Independent certificate:

sha256: 1c27cf58211873ad12e183709250ba9d3a9b35179287eabe4817ffc07624630c. Engine

external-validation hash:

sha256: 1912e7d1290e48f17710a1214bb1b33ff3979ce747d8b85f9b9194589ec3db8f.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GOMEZ-VALENT-2023-CCH-32
- PLANCK-2018-BAO-BASELINE-BUDGET
- GOMEZ-VALENT-2019-ACCELERATION
- ESCAMILLA-2024-DARK-ENERGY-STATE
- DESI-DR2-2025-COSMOLOGY

Observed comparison records:

- CCH-H2-01: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-02: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-03: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-04: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-05: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-06: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-07: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-08: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-09: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`
- CCH-H2-10: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match `True`

- [illegible]

- PLANCK-EQUALITY-CUBE: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- Q0-TYPED-MAGNITUDE: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- ACCELERATION-TRANSITION-CUBE: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- VACUUM-TENSION-MAGNITUDE: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- DESI-DYNAMIC-ADVERSE-ROW: predicted terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; observed terminal-cosmic-transport-complete-vector-passes-with-adverse-desi-row-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: Any of the 32 CCH intervals fails the registered squared two-uncertainty comparison; either threshold cube lies outside its complete external interval; the q or tension magnitude lies outside its interval; any source hash, row, custody record or adverse DESI statement changes; or a tampered target is accepted.

Measurement receipt: sha256:a33e0831e949329261b8cf0d72892053904c03dc38d4e21f086127b16196697a.

Isolation certificate: sha256:47678f36ed211ed8e386676c03ea0525a0a85131a1e903322e8167783559f414.

Custody certificate: sha256:f7088cb50ea54bd246562b80fced3192aa0c79f7ff1cea970b3650fc404efc15.

Registered source descriptions:

- Cosmic chronometers to calibrate the ladders and measure the curvature of the Universe. A model-independent study:
<https://academic.oup.com/mnras/article/523/3/3406/7188316>
(sha256:cbd38055c3c9b27b2cbe1fb7456c60dfd7df3eeb366182932b7899d25207b128)
- Planck 2018 plus BAO baseline parameter table:
https://wiki.cosmos.esa.int/planck-legacy-archive/images/4/43/Baseline_params_table_2018_68pc_v2.pdf
(sha256:a8525585688e7ba818f8650fc0f8b73a449823d5e199a8be36e37b8e73a0e612)
- Quantifying the evidence for the current speed-up of the Universe with low and intermediate-redshift data:
<https://arxiv.org/abs/1810.02278> (sha256:55100ed889f8aad5ada957f33759d954466b944d21b461b4c63c62f0b0fb45d0)
- The state of the dark energy equation of state circa 2023: <https://arxiv.org/abs/2307.14802>
(sha256:67c8d75e79fd85d727012e72e8838651de6720a2e8898e225e251601814c20a5)
- DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints:
<https://arxiv.org/abs/2503.14738> (sha256:1e82f26e4cc3901b16168cd147f252bfa804f9c3caad3f4f7e3532640d237841)

Registered target identities:

- CCH-H2-01 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-02 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-03 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-04 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-05 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-06 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-07 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-08 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-09 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-10 from GOMEZ-VALENT-2023-CCH-32

- CCH-H2-11 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-12 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-13 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-14 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-15 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-16 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-17 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-18 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-19 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-20 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-21 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-22 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-23 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-24 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-25 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-26 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-27 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-28 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-29 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-30 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-31 from GOMEZ-VALENT-2023-CCH-32
- CCH-H2-32 from GOMEZ-VALENT-2023-CCH-32
- PLANCK-EQUALITY-CUBE from PLANCK-2018-BAO-BASELINE-BUDGET
- Q0-TYPED-MAGNITUDE from GOMEZ-VALENT-2019-ACCELERATION
- ACCELERATION-TRANSITION-CUBE from GOMEZ-VALENT-2019-ACCELERATION
- VACUUM-TENSION-MAGNITUDE from ESCAMILLA-2024-DARK-ENERGY-STATE
- DESI-DYNAMIC-ADVERSE-ROW from DESI-DR2-2025-COSMOLOGY

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, old two-thirds/one-third survivor or external equation in forcing
- no numerical-nothing zero, negative, irrational, imaginary, floating, NaN or completed-infinity proof scalar
- no continuum scale factor, differential field equation, fitted Hubble value, fitted density or selected redshift
- no use of a measured q , w , transition, H row or DESI interpretation before the derivation seal
- no erasure of the superseded one-half/four-cube results or the current DESI dynamic-dark-energy tension
- no claim that the terminal late matter-vacuum slice supplies a measured present radiation normalization

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:e1004c307761e6e384dfebeb1cacc43157ac49b91d746bfe9cef867d59fadc1d; engine receipt
 sha256:504257d441fad45a8f173fad45015c7d723caa7636c517e71d27fddc43a2c42d at receipts/engine/model_admitted/SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032-504257d441fad45a.json;
 empirical-validation hash
 sha256:2356d7946091e0bf3fc556d6750ecf94f76435c1dedb3ef864edc81d59b87a9e; measurement receipt
 sha256:a33e0831e949329261b8cf0d72892053904c03dc38d4e21f086127b16196697a.

286. Terminal inflation, primordial-support and structure-growth law

Claim identity: SFT-PHYS-INFLATION-GROWTH-TERMINAL-039

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The unique generator-volume inflation cover is five Fold doublings with support 32. The complete primordial partition is scalar 31/32 and least tensor-pair 1/32. The structural exit is the first complete cover of volume 27. The least resolved perturbation grows exactly 1/4 to 1/2 to One in two Fold steps. For every exact positive later scale growth g , matter retention is $1/g^3$, radiation retention is $1/g^4$ and their relative growth is g . A conventional natural-log e-fold count is a comparison label, not an SFT proof scalar.

The unique generator-volume inflation cover is five Fold doublings with support 32. The complete primordial partition is scalar 31/32 and least tensor-pair 1/32. The structural exit is the first complete cover of volume 27. The least resolved perturbation grows exactly 1/4 to 1/2 to One in two Fold steps. For every exact positive later scale growth g , matter retention is $1/g^3$, radiation retention is $1/g^4$ and their relative growth is g . A conventional natural-log e-fold count is a comparison label, not an SFT proof scalar.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-FOUNDATION-PART-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003
- SFT-PHYS-COSMO-EXPANSION-001
- SFT-PHYS-COSMO-SPATIAL-FLATNESS-001
- SFT-PHYS-COSMO-STRUCTURE-GROWTH-001
- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032

Generated grammar and closure boundary. Generate the complete product of spatial volume, binary cover, duration carrier, scalar support, tensor support, exit boundary, perturbation growth and component-transfer form. The declared exact boundary is: Generator-three volume in stable three-space; every positive binary support until the unique least cover; every exact complete doubling ratio; the complete thirty-two-record primordial partition; the quarter/half/One Fold growth trace; and every exact

positive rational later scale-growth carrier under admitted third/fourth transport. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `generator-three-space-volume__least-binary-depth-five-cover__five-exact-doubling-transitions__one-boundary-record-short-of-complete__one-tensor-pair-record-in-thirty-two__first-complete-generator-volume-cover__quarter-half-One-two-step-growth__third-versus-fourth-power-relative-growth`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
volume	generator-three-space-volume	named-observable-universe-size	A measured horizon cannot select the support.
cover	least-binary-depth-five-cover	selected-sixty-efold-story	A conventional model-dependent duration is not a generated Fold count.
duration	five-exact-doubling-transitions	irrational-logarithmic-efolds	A continuum logarithm is outside exact Fold arithmetic.
scalar	one-boundary-record-short-of-complete	fitted-spectral-index	A decimal index selected from CMB data is a fit.
tensor	one-tensor-pair-record-in-thirty-two	free-tensor-amplitude	A tunable tensor amplitude adds a parameter.
exit	first-complete-generator-volume-cover	chosen-inflaton-potential-exit	A chosen potential imports an independent model.
growth	quarter-half-One-two-step-growth	stochastic-or-decaying-seed	Ontic noise or decay is not forced by the Fold.
transfer	third-versus-fourth-power-relative-growth	fitted-transfer-function	A fitted spectrum is not a law.

Base and successor. The closure proof is sealed by derivation hash `sha256:038cb81edb90bea3adf8f8d01c98503c453223fc79676f8eac63aff1b3c9ee5f`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:716a5ab4fb9d1de1475efcf5399bcd85b405d32abf2fc73e383c859b06ee8809`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:648551c6e3e7563a2f5261e04e0076c4c28d64b7a019bd14c1c847ac8854a32d`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d454a7c0e4d9470179577a2c03aa3a0484980f8c4a5db65b88be3c391e1b7fc4`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:cb7f3f69be4ae22f13eb9fc0311effa52d977dbe2d6e25e9a504045d15533a4d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:75b56683b979a9f537c5a4675e5e83e6fe2a0caf4247b9e8799b8779fa082470`. Independent certificate: `sha256:b6fe904029ba98e5a266135a0550b50b142ec04df91beb8de888382c67442541`. Engine

external-validation hash:

sha256: 48aadd0587f83014412a792ae95266b11fd4c61d4cca3ae27464033b082f31ff.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no inflaton potential, slow-roll equation, conventional natural logarithm or selected sixty-e-fold premise
- no CMB spectral index, tensor bound, matter spectrum or measured horizon available to candidate selection
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity proof scalar
- no ontic stochastic seed, fitted transfer function, amplitude or free exit scale
- no claim that five Fold doublings equal sixty conventional e-folds

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: abf31c25a28d3d16ce81601347077dcccdf7ec229f558a98ad6170602bdb32d5; engine receipt

sha256: 00aa52b47f8ebfe57fb5a90530a54614ba175d87f6708c69e454fcf9fd888aa7 at receipts/engine/model_admitted/SFT-PHYS-INFLATION-GROWTH-TERMINAL-039-00aa52b47f8ebfe5.json;

empirical-validation hash None; measurement receipt None.

287. Terminal stellar hydrostatics, galactic response and tidal-lock law

Claim identity: SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Hydrostatic balance is exactly $(1/2, 1/2)$. Radial pressure and gravity responses are $5/3$ and $4/3$, certified only through q^5 and q^4 perfect powers: every compression is outward restoring and every expansion inward restoring. The two terminal homologous luminosity exponents are three and four, forcing lifetime-fall magnitudes two and three. Circular balance is $v^2 = M(r)/r$; flat v forces $M(r)$ proportional to r and excludes a finite visible-only asymptote. The baryonic Tully-Fisher exponent is exactly four. Every finite rational isolated circular tidal mismatch exhausts to 1:1; eccentric or externally forced resonances remain an explicit separate boundary rather than a counterexample hidden from the census.

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Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-001

- SFT-FOUNDATION-ONE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001
- SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061
- SFT-PHYS-COSMO-STRUCTURE-GROWTH-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

Generated grammar and closure boundary. Generate the complete twelve-axis product of hydrostatic balance, radial response, stellar scaling, lifetime, circular balance, flat-curve support, dark carrier, Tully-Fisher exponent, resonance, tidal dissipation, measurement custody and extension forms. The declared exact boundary is: Exact positive normalized shares; three-space and two-fibre counts; exact q above the One; perfect-power response comparisons; homologous volume and radiative-recurrence stellar endpoints; circular inverse-square balance; finite enclosed-support and velocity ratios; finite rational spin/orbit cycle counts; inherited neutral relic and resonance laws; and no measurement before seal. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `complete-self-antipodal-half-One__five-thirds-versus-four-thirds-perfect-power-test__three-space-and-volume-plus-recurrence__one-fuel-carrier-over-complete-luminosity__inverse-square-balance-v-squared-equals-M-over-r__enclosed-support-grows-linearly-with-radius__admitted-neutral-stable-relic-under-fixed-gravity__three-space-plus-one-orbital-recurrence__finite-low-denominator-common-refinement__finite-mismatch-exhaustion-to-one-to-one__all-targets-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
hydrostatic	complete-self-antipodal-half-One	selected-pressure-gravity-ratio	A selected ratio is a parameter.
radial	five-thirds-versus-four-thirds-perfect-power-test	asserted-self-correction	An assertion does not compare response carriers.
stellar	three-space-and-volume-plus-recurrence	fitted-single-power	A fitted exponent is selected by stellar data.
lifetime	one-fuel-carrier-over-complete-luminosity	independent-lifetime-fit	An independent lifetime exponent adds a dial.
circular	inverse-square-balance-v-squared-equals-M-over-r	imported-Kepler-answer	An inherited named answer is not a Fold derivation.
flat	enclosed-support-grows-linearly-with-radius	finite-visible-asymptote	A fixed mass makes M/r fall outward.

Axis	Forced form	Eliminated form	Elimination reason
dark	admitted-neutral-s table-relic-under- fixed-gravity	modified-law-or-named-par ticle	A named alternative is not structurally selected.
tully	three-space-plus-one-orbital-recurrence	measured-or-fitted-slope	A slope selected from galaxies is a parameter.
resonance	finite-low-denominator-common-refinement	irrational-continuum-period	An irrational continuum period violates the generated finite recurrence grammar.
tidal	finite-mismatch-exhaustion-to-one-to-one	unbounded-or-selected-terminal-ratio	An unbounded path or selected ratio does not close.
measurement	all-targets-inaccessible-until-seal	target-readable-before-seal	That would fit the result.
extension	no-extra-rule	free-response-scale-or-correction	An added response forks the law.

Base and successor. The closure proof is sealed by derivation hash `sha256:8e8cf180731600a6dbe1310cb2d9a83ed0bf793c01f957c0884313ecc37e12b4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:152c1d169c9d2e59069fb8b6598ba79f5c79b2e519d16f6ce902b13da9fcf324`.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:9e6f6491ddb20e53761897e5ef7bac5f4d7630c039a7428b3df40be1fa405274`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:111d6aef6b9dd85abd6b0d078cc84264abebb621f9fd85ed8a32dda84ecd76c9`.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:22ef586869d3ff3a3e257e57eae785a3e2625dda7f5e808bb7e129b0f01f35f4`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e4cc3ed8767fe932277c90468e576b14f06536c91f8aa3f31259c738975fa89d`. Independent certificate: `sha256:e1a5dced711c5b5a9425bd16975c0b9c082e6d74e77812bae1beeb5e795d6527`. Engine external-validation hash: `sha256:d78e63db5478ae7a4f1531ab83b8a78e0b89c20ec0161c8206c913c8474f2949`.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, stellar catalogue, galaxy slope, lunar period, survivor identifier or measured target in formal execution
- no fitted exponent, mass-to-light ratio, halo profile, tidal quality factor, age, metallicity or normalization
- no universal claim that every stellar mass domain has one exponent and no concealment of separate microphysical regimes
- no universal 1:1 claim for eccentric or externally maintained spin-orbit resonances
- no negative, irrational, imaginary, floating, NaN, continuum or infinite proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:4c30c363ac59c1fb301ea0197223c174dc7c25defaa5d15e5012177f0684cb2a; engine receipt sha256:24674dfa019c208012414ab3c587656ee404512f20d00ab44e26ee137be22ae2 at receipts/engine/model_admitted/SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067-24674dfa019c2080.json; empirical-validation hash None; measurement receipt None.

288. Terminal stellar nuclear chain, collapse and heavy-element law

Claim identity: SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Charged access paths are exactly $n(n+1)$ and strictly increase for every positive stage successor; finite stage supports are strictly nested. Exothermic ordinary fusion terminates at the independently forced global binding maximum (mass 62, charge 28, neutron count 34), beyond which its release is the empty One form. Retained inward gravity then forces the support-loss collapse channel, with compressed nuclear support and horizon closure as the complete endpoints. The distinct linked-fuel recurrence has no interior fixed point and ends only at finite fuel exhaustion, forcing thermonuclear unbinding. Neutral capture has the empty inter-boundary charge path; repeated capture followed by registered radioactive rebalance forces the heavy-element channel beyond the ordinary-fusion terminal. Dimensional ignition temperatures and observed events remain post-seal checks.

Charged access paths are exactly $n(n+1)$ and strictly increase for every positive stage successor; finite stage supports are strictly nested. Exothermic ordinary fusion terminates at the independently forced global binding maximum (mass 62, charge 28, neutron count 34), beyond which its release is the empty One form. Retained inward gravity then forces the support-loss collapse channel, with compressed nuclear support and horizon closure as the complete endpoints. The distinct linked-fuel recurrence has no interior fixed point and ends only at finite fuel exhaustion, forcing thermonuclear unbinding. Neutral capture has the empty inter-boundary charge path; repeated capture followed by registered radioactive rebalance forces the heavy-element channel beyond the ordinary-fusion terminal. Dimensional ignition temperatures and observed events remain post-seal checks.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-ONE-001
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005
- SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001

• SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete twelve-axis product of stage composition, charged boundary order, release direction, global binding terminal, nested support, fusion-support loss, collapse endpoint, thermonuclear recurrence, neutral capture, radioactive rebalance, target custody and extension forms. The declared exact boundary is: Every positive finite stage count and successor; complete charged boundary paths; the inherited all-mass binding maximum and tail induction; half-One hydrostatic shares; empty-One absence of release; both compressed support and horizon endpoint classes; every finite fuel recurrence; neutral incident capture followed by registered radioactive rebalance; and no stellar or supernova target before seal. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived: `positive-Fold-stage-successor__strictly-growing-charged-boundary-paths__only-toward-higher-binding__unique-all-mass-62-28-binding-maximum__strictly-nested-access-support__empty-One-fusion-support-with-gravity-retained__compressed-nuclear-support-or-horizon__runaway-to-finite-fuel-exhaustion__neutral-capture-with-empty-charge-path__registered-radioactive-label-successor__all-targets-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
stage	positive-Fold-stage-successor	named-stage-list	A named observational list is not a derivation.
barrier	strictly-growing-charged-boundary-paths	one-universal-dimensional-temperature	One selected temperature erases reaction-specific structure.
release	only-toward-higher-binding	fusion-past-peak-by-assertion	Fusion past the maximum cannot release binding support.
terminal	unique-all-mass-62-28-binding-maximum	selected-iron-name	A conventional name cannot select the endpoint.
nesting	strictly-nested-access-support	unordered-mixed-shells	Higher access cannot occupy a larger lower-access support.
loss	empty-One-fusion-support-with-gravity-retained	hidden-residual-fusion-energy	An ungenerated residual is a free source.
collapse	compressed-nuclear-support-or-horizon	single-selected-remnant	Selecting one outcome omits the distinct support boundary.
thermonuclear	runaway-to-finite-fuel-exhaustion	chosen-partial-burn	A partial terminal is not fixed under positive feedback.
capture	neutral-capture-with-empty-charge-path	charged-fusion-beyond-maximum	That requires inward energy and a Coulomb path.
rebalance	registered-radioactive-label-successor	unrecorded-charge-change	An unrecorded change violates label conservation.
measurement	all-targets-inaccessible-until-seal	target-readable-before-seal	That would fit the channel census.
extension	no-extra-rule	free-channel-or-energy	An ungenerated term destroys zero-parameter closure.

Base and successor. The closure proof is sealed by derivation hash `sha256:3689e53b92ca88280d666ece8306d439f340fd0a80d9977b8c3e878ed6fad13d`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256 : 8165d365df02494a33ae9d5360d2c4b970d49ca9b5db7927e9308c441a34b9ad.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256 : 928815f87df9944fe2d6a80762ef34a558d7b736f5d40d1c3f02bce9ac2c0ea2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 7301e51c84488a2130717a8bdf54f192be47839475a07674e4996c6d431b7aa0.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : a7a6dfc2cd363681c827ff106cc67147e6c7340aca3f0bbbe90a2a7bd78b2c54.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : 874f1168f335a9a58e6c94231d156a36322ff6d890a353d74523c06027f90ee8.

Independent certificate:

sha256 : 852e960611e5afb743d00ddd6278d85ed07d7df951e6b4e3f0353b442169ec87. Engine

external-validation hash:

sha256 : 19dc8622b1cea19c71e340328c0b4e55c8bc208a3a331d22389f9540ebd206b9.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. The Physics boundary closes universal source, propagation and observation relations while explicitly handing astronomical census, historical initial conditions and cosmic chronology to Astronomy/Cosmology. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no V1/V2 executable, named stellar stage, ignition temperature, supernova event, abundance, isotope target or survivor identifier in formal execution
- no fitted reaction rate, stellar mass, opacity, metallicity, explosion energy, neutrino efficiency or standard-candle calibration
- no universal dimensional threshold substituted for reaction-specific access carriers
- no fusion release asserted beyond the globally sealed binding maximum
- no unrecorded charge change and no omission of either collapse endpoint or either explosion channel
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : 498653ec652f93ab85ca1466d0954e8fd3a0eacd4996a0d4d5eb4179b66b611b; engine receipt

sha256 : a41b28a935f54de5daabdbe77b6c1aba9286f174526b8feaa60829ec848e08af at receipts/engine/model_admitted/SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069-a41b28a935f54de5.json;

empirical-validation hash None; measurement receipt None.

24. Post Seal Empirical Validation

These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims.

289. Exact molar Planck measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The molar Planck carrier is predicted by exact composition of the Planck carrier with the counted molar carrier.

Planck constant composed with Avogadro count predicts the CODATA molar Planck interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2dfda8fbde6076b09842497abdc5a00d44c02e05164440bd1f0d379f969a5731`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated

by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8dba5666f65856a7a2f6d46a59317ff158e90067b1d68739639625fbe1ff4c48.
- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b50f43ff8ea8b8b28920191600b58f1084d395bbb64a7c281b26769d5fd5e8cb.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:01234e1a45ed97261d6e6aca6fa6a09c82d85cc82a44e3c8668947aff0f6ed5e.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9bd844cf923a2862db055f9e90a89436b1ceb8493ccc58ea2b1a9c50e1c5b732.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:5c176bbcc9c5e3304bf168c8624612b897b8c0e0d982f5c4d493520690bc404e.

Independent certificate:

```
sha256:7e2cc5601355e962f71607ad460a5e96a5cd116059af12db77640845bd7d3877. Engine
```

external-validation hash:

```
sha256:7a1a8c04d1caba9681bd270dabe42b44d5c345725b3abb1e686394fded2b11ce.
```

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar Planck constant: exact predicted interval [19951563564467157/5000000000000000000000, 19951563564467157/5000000000000000000000000]; external interval [498789089/12500000000000000, 3990312713/1000000000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: sha256:36b3c9aa786510f0f47fd90fdb05d61bcf377891525319a794afa48d7e053939.

Isolation certificate: sha256:9f92f36602cc0fb91dc9e8454a62563f74168e90f3d4277f25c735a7bce0a4b8.

Custody certificate: sha256:e18166182e4381ff3211a687b7d140bde54dd6a9d1845024f320c5be75b4a84d.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- metrology-molar-planck-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:9225e8b389f0d1611722e0ee9082aed36211d7cad352f164b9784d71c7e0bf81; engine receipt
sha256:e1f0dfe8b6d4a57d7b408516d2e09ae677fabefc201c273f3f2a62182f9bea73 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001-e1f0dfe8b6d4a57d.json;
empirical-validation hash
sha256:6a8f75e99001bab6986b19ecb01b097c11c6765dfee5b6dfe76a5573ee66f334; measurement receipt
sha256:36b3c9aa786510f0f47fd90fdb05d61bcf377891525319a794afa48d7e053939.

290. Exact momentum measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic momentum carrier is predicted by exact composition of inertial content with propagation speed.

Atomic mass composed with atomic velocity predicts the CODATA atomic momentum interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-MOMENTUM-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier_exact-positive-interval-relation_source-bound-proof-trace_capability-closed-prediction_separate-measurement-record_complete-registered-rows_one-successor-closure_no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:48db928fa462f39db0d3897bad3bca0b2ca40b7cd526065fad1afb314ce1b128`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:86f5eb2fbf987c6a15c200773f00f2697c12ae8673795cbcf4a43408304a1c8e1.
- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:32ddd7bd51c73ff395027bcbf5673d17534a4a83309101de6f9eb8697a2ee79b.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e52819448d4eb62b03f6356b792ad24afbb305bbf741b588fb21c460591077a.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e93bf332fc53a176bfd807a7582b045628da9fbff31feaaee0de3950334533b94.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:96232cb6e8a85b63a038a0dbb0c587d0c8a77c3dfc0570cbbfbab64dca1a6d89.
Independent certificate:
sha256:3e5bba783e37e7e0781b1b92e395e66f8d8791ca74db8886c7d0457e22cac2cd. Engine
external-validation hash:
sha256:b0c73c45da9d914e52907adbd52103c61d262347e522382e33fd52f6117bc967.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- [illegible]

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: sha256:56ee927b7a59252157c870d7b617361dbccb8149618a1312b23aff974eabaf91.

Isolation certificate: sha256:2c7010c8cfdb5340f59517f0da156c24aea2c8d702b7cabca21d6ff0d17a36b9.

Custody certificate: sha256:6b8b74bdc78db7accc67131fb786add7b9f7235e7052985541f4200d6ba3148b.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- mechanics-momentum-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:5cbeade6d136aa75bd5230ed8c319ca39ef239a071b3a340252e8b120e9b6ea2; engine receipt

sha256:4a7e754fdd04ef6ffe92456b90c10648d0c4b64194fd53774beaae63d98c62af at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001-4a7e754fdd04ef6f.json;

empirical-validation hash

sha256:747a63254d71bc932e64a52bc276ac83ff79026765de82e9730c03b3c074abb4; measurement receipt

sha256:56ee927b7a59252157c870d7b617361dbccb8149618a1312b23aff974eabaf91.

291. Exact force measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-MECHANICS-FORCE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic force carrier is predicted by exact work-energy transfer per spatial support interval.

Atomic energy divided by atomic length predicts the CODATA atomic-force interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-FORCE-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:531b071bf736461b1a957cc542ca52ccdbb1931bee56372e43f83a97175e4b85`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7d0e248a615bf7fa85e0c4e63bb059d811484b30daa1e6562655d5a2d9e50005`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e05966041153e32d868b967f176746cd8bbf7ade55fadb948076da6c65fcc6a7`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:154a326728312835d420ee7a8f72418b24f9eac1b385b3fbe5bcd514ea6f13cd`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dfd96e21b8455542399ab2e45dac987d56016ced10065bbd4a854c2cb45e6164`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:305c4b0bfef2f469c7b94f691536ca5c5bc9ef5e4d2b21edc8afe150c342f609`. Independent certificate: `sha256:6accffe40e2a38bcd1e59f58794b8201a2756f1d626bd4e6bed722d321b4ead2`. Engine external-validation hash:

sha256:69ffd95f7625f105e65a4439bdd5218dab1568d70b90fe4f195c961e16cb1019.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of force: exact predicted interval [10899361805503/13229430265650000000, 10899361805527/132294302615500000000]; external interval [3295489401/40000000000000000, 82387235051/1000000000000000000]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: sha256:31429d5b44a542dc11e26006c28ab86fd8d4f766eda307b47f1215069204902c.

Isolation certificate: sha256:7c68fc1380bc71369cddb7852e389d07819634c7221e3ce66f634c861aa95261.

Custody certificate: sha256:eb5472c0209196d6ed943d556e449ebe0013bd3b360dc619bd26793043a890f7.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddb5e71be406f2f26e3f67e67)

Registered target identities:

- mechanics-force-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:542c07dc5b33cd59956904767eda58be58b9338a4f0df6fa6208904864ff4e6e; engine receipt

sha256:e394dd3d87bf8f712e39291a4f8dcc0fe2d2bea75d3d1fd5b88a67645fb691b1 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MECHANICS-FORCE-001-e394dd3d87bf8f71.json;

empirical-validation hash

sha256:a3fd66e3f20289357da1a8679e0fba1574529cee162cca2c5067d03d2b50af04; measurement receipt

sha256:31429d5b44a542dc11e26006c28ab86fd8d4f766eda307b47f1215069204902c.

292. Exact electric-potential measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic electric-potential carrier is predicted by exact energy transfer per elementary held-charge carrier.

Atomic energy divided by elementary charge predicts the CODATA atomic electric-potential interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:90794ad2d32fc15db13bf3e1b9212babd6cabaab3e2629a2af951a18f9e72610`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c5d5bc3b55e252b97377e395fd24d91276e5901912f446c273d050a5815e3452`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 60be210b04698f43634a547027bf2d8018948689ca5f24ffaf24138e79792bcf.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 465eda6c270bb44dc6c32e821e599098a9618ae570ddfffe2b3101603ed172f7.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 86f4aea99ba7612cf9889c79870777cf6878cec798c0bb551416fdf9ed9a07bb.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: eb299544fa0a2cb21cc9c01a748f9da3bcaeb47b0ea1b7ee954a9bbb3db44137.

Independent certificate:

sha256: 10ed41b8443c1db52bc830ad1f60d83f261b7995f2ceb65587025a1056685460. Engine

external-validation hash:

sha256: 59c4603aa1c7e15048f81d0cda1d2bcd823299418d17b9fd8f6b45d09b5cf019.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of electric potential: exact predicted interval [10899361805503/400544158500, 10899361805527/400544158500]; external interval [27211386245951/1000000000000, 27211386246011/1000000000000]; overlap **True**
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: sha256: 4edaa6daf75bf922b1b265d0c71158cf9c4cbf46fba764a099d9abfeb8190114.

Isolation certificate: sha256: 86114e2038b630b41cf649cceb03767644c0d0f3562cf9b9eefe02bfa9e534.

Custody certificate: sha256: 5034883424092c1d8f700dcb27b09cc1f63c7db2f5cfb817f1c9c7827904b131.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256: 77fb90e66c40db3e6eb16630bc9c88e4c7c8beddb5e71be406f2f26e3f67e67)

Registered target identities:

- field-electric-potential-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:b4905946f588c0b676abca54102aea83324c31bb32bcf614ff4d1376a916e6c7; engine receipt
sha256:34a88c7435b67d0c73792b2eadc0ff77c03291db4280c5b806c286b78ddbb00f at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001-34a88c7435b67d0c.json
; empirical-validation hash
sha256:490ac035799a28d5636017d5332003ba8a8c8242399bb2f73c8ad1549035608d; measurement receipt
sha256:4edaa6daf75bf922b1b265d0c71158cf9c4cbf46fba764a099d9abfeb8190114.

293. Exact electric-field measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The atomic electric-field response is predicted by exact potential change per atomic spatial support interval.

Atomic electric potential divided by atomic length predicts the CODATA atomic electric-field interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier_exact-positive-interval-relation_source-bound-proof-trace_capability-closed-prediction_separate-measurement-record_complete-registered-rows_one-successor-closure_no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:68ad39b63f242c2bcab3cbbbe275198cbbc364a97569665ad8ccc14b164fe4a02. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:9935fe5faaef70de04a0fb1096d94b7102ff87ca5c74a8c907f9c2b86e268442.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c94feca21e55eed4366a19704f6e7d9d64ade592f25c5c765c4df02d3984a99b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e699961a2ae9dfb96d886fabba5d3137d5cb6ee283a558e308aaafd0dd70e3ca.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:450854a196136d47100d2bfc73504c78f85dde507440fc34cb3b9a950ac9b2bd.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:f0f3fa02e9e303bb6a657ca72723370b1476bb9017f6bc0f60ef840ee05d0e8c. Independent certificate: sha256:b563e6fd9fedf5ba12fec207079a96b863ec69bc82d5b8f09497c229a7238ee8. Engine external-validation hash: sha256:1b5e48598ab764f78667fdb515e5e36c0c77e550d8740c21f2ec1430069f06ee.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- atomic unit of electric field: exact predicted interval [136056931229755000000000/264588605313, 136056931230055000000000/264588605231]; external interval [514220675032/1, 514220675192/1]; overlap True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: sha256:828c7e4fbb00764d4e9df5ddc3582e7aee6bca095db57f37bbb8deaf67c469a0. Isolation certificate: sha256:2e6da0f632329f5add36793dff84a23aa8ec7111ba4e5564985e39a81c939941. Custody certificate: sha256:435baa2ece8efe40fa8ae6ff70c0d08e55074f04f19e7377714855db66243d74.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- field-electric-strength-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:bdb2a10e1b1f70da8cdf3c6f285dd547459ebf99e75ed39a00c3a1094e05ab72; engine receipt
sha256:b5b06d49508b35b38f352d022c8edcf0f2e593225086d53472aa679a8e64dbc8 at receipts/engine/
model_admitted/SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001-b5b06d49508b35b3.json;
empirical-validation hash
sha256:a1d5c9e8da6710ad18d84368335db10cf121930ebf59504a863157d7ff31baea; measurement receipt
sha256:828c7e4fbb00764d4e9df5ddc3582e7aee6bca095db57f37bbb8deaf67c469a0.
```

294. Exact wave frequency measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A registered inverse-length recurrence transported at the limiting propagation speed predicts its recurrence frequency.

Rydberg inverse length composed with limiting speed predicts the CODATA Rydberg-frequency interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3af1893b417c04c95f0233f2ed9a2047c13492d12b5761f7cf2eb35560d714c8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d1b1b26ccb8e0f088ad709031fb472de420836598102c0ceec7d13bcb415e688`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:657967992b8f1c2e12d67611da1293b2fb5cde33f726e9b486a9eefa47701483`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:213e3480e3dad607392a3f96d7656c451e251f3e53fcb53a77e67ecea56d5db7`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4a60e6dd8a7cffe9b795bdb5dd2806f1a7f7c5d5118a1cb55f26bc43e7cb7d29`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:f12b815d77906521b2090244dd8dd4e3afd91e97d51cb9ccdbfb1e7962a32aab`. Independent certificate: `sha256:1c07dd896d1c7074c9cb1e7e3aab7471662f3829120d5536b0534ed959b72048`. Engine external-validation hash: `sha256:e2ea7686a0d5b823e751e2a44c265fb17a66afd7d61bf1e3b1667f95b25388c3`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Rydberg constant times c in Hz: exact predicted interval [328984196024638405041/100000, 1644920980126789534701/500000]; external interval [3289841960246400/1, 3289841960253600/1]; overlap `True`

- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: sha256:c273d3186b7d7c1f07493ee7b303e28d6299fa252e4abd65c72eaa968ecd3723.

Isolation certificate: sha256:581833bcd6fc4a5fac836ce0d6e309ad5c780137389e98caab9a1e609a6ca126.

Custody certificate: sha256:153a70cc8e477b12a568d1ee4719192061fe23a2d31ee2517fd7152e354b8bdc.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- wave-frequency-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:0fc7adec00d3a5e509382fc951f24c3dad4a1836c18770735393de6818453713; engine receipt

sha256:d75c1886968251dd42429c73c3c8303ae4d22eededbe87ae37e24951e7e938fb at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001-d75c1886968251dd.json;

empirical-validation hash

sha256:75b2bdcab27442712f9b8272975533b3121f04fe4ddb633a879e2ecf201f2111; measurement receipt

sha256:c273d3186b7d7c1f07493ee7b303e28d6299fa252e4abd65c72eaa968ecd3723.

295. Exact wave energy measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-WAVE-ENERGY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. A recurrence frequency composed with the Planck transfer carrier predicts its transported energy.

Rydberg inverse length, Planck carrier and limiting speed predict the CODATA Rydberg-energy interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

• SFT-PHYS-WAVE-ENERGY-MOMENTUM-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:52ef549cae6c9a11b223baff93ae4578bcdcf3aa5d1d37132cb5d26c2b7cca9f2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:80fababc6aacb0f4cc9169e4d261edb055e531385429c8d6d0712e082e2b8c8f`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:dcbdf65034657a9d951b337630fda7c01e70642ab9a1746257825b9f4b3fd42`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4d7cb5e64989a5c0d5d54a87e838a6d7717f0352e221a6ad80c6a7a5f9c963c8`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a5d04f8c4cd2f3b0152f661e985a1d9d0839aece8a9dc6f8319999702392e77a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:6dd06d5f612f4003d26dafa3da17e7fd27e58f1dba04cf89bd205dbf7a056e7e`. Independent certificate:

sha256:2623704f4a5b8842046e10ea7adc04f8be5fbd323819cd0bda228158f507b9ab. Engine
external-validation hash:
sha256:22447689c894ce0c699c76930493d3e039386abb0aadfa9cceedf616c52c8407b.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- Rydberg constant times hc in J: exact predicted interval
[43597447222012104003715592523/2000000000000000000000000000000000000000,
217987236110537267024293705503/100000000000000000000000000000000000000]; external interval
[10899361805503/500000000000000000000000000000000000000, 10899361805527/500000000000000000000000000000000000000]; overlap
True
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: sha256:bfcee7ac76277f2eca8503e567c67cb5661ad12a342cad3a531c126ae66c676a.
Isolation certificate: sha256:9d5fa3a07cf45a376ac75853053ba25a5459fa4059c08a6c5d36df0a30927f5b.
Custody certificate: sha256:78364198ef5c6f4b73962a2cb5151a3fd4f9e9050cca8c4ba96af9db7d5a75d0.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6b16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- wave-energy-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants
- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:ee42680f05f96265966953bbaf31df5c983b1ea1ecf841ab7288e3826137da33; engine receipt
sha256:fd222d0ccab49086ac4daa8928f6c9f0cd65ccf15dc94dfdc56562cec0c22173 at receipts/engin
e/model_admitted/SFT-PHYS-VALIDATION-WAVE-ENERGY-001-fd222d0ccab49086.json;
empirical-validation hash
sha256:ebd6ff2986706f73adeb320421bc15a51030bc97df03321697f53af5930a04d3; measurement receipt
sha256:bfcee7ac76277f2eca8503e567c67cb5661ad12a342cad3a531c126ae66c676a.
```

296. Exact molar thermal measured-value correspondence

Claim identity: SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The molar thermal-energy carrier is predicted by exact composition of the per-constituent thermal carrier with the counted molar carrier.

Boltzmann carrier composed with Avogadro count predicts the CODATA molar gas-constant interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-THERMO-TEMPERATURE-001

Generated grammar and closure boundary. Generate the complete exact-relation, input-provenance, target-access, interval, comparison, row, successor and extra-rule product. The declared exact boundary is: All exact positive interval predictions formed from registered measured inputs, one already forced relation, capability-closed arithmetic and a post-seal external target. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__exact-positive-interval-relation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	exact-positive-interval-relation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash

`sha256:cda72729383c0dd8395a446dda7fafbf2343e3452ca591e2d56187f117d0e610`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2d76619ba4ffffb95f44490faecc85c9503181235a3b97795c0cd00e91dba3d27.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:189f4aefc0e0d84013394e7c93774b570c9d30738b5bf0eca7850d92fa3f48c6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:18fc3e77d08c08e7911ab7bbecea77efb7cee6a6defc6de717db77287244a504.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1dd0e1d54a27ff5a0f5ebf3731e5842ce31f34eb8975dd1f742ee0d3862503da.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:7c74bfeab70b66c1a68ab28d93377be8592b949cde42658538a08e24dac8c6b5. Independent certificate: sha256:94c1f683599f41d967651a055f6c72c4d311a58710642443377dfa8144ce603d. Engine external-validation hash: sha256:6c6eaf91b7f7c35ea6d030b8f0b82887a5e00cf7eb097aa653935e7c6eb45f4e.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- molar gas constant: exact predicted interval [207861565453831/25000000000000, 207861565453831/25000000000000]; external interval [4157231309/500000000, 8314462619/1000000000]; overlap **True**
- deliberately displaced unfavorable interval rejected

Falsification condition: The sealed exact predicted interval fails to overlap the independently released CODATA target interval, any registered input is missing, or the displaced unfavorable interval is accepted.

Measurement receipt: sha256:72a70301b05b1847cb68d81b26ec8b12ba9fd6ed0b7633034f7c4c062a5439c8. Isolation certificate: sha256:49e41a57f9c684a88cb3548296db3c9b373c50191e40b4d92cd9a0c0dad5f437. Custody certificate: sha256:8eb921aa0495f19a26d475ad77f38874da872edd0238379aa1adbd0167b0f181.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- thermo-molar-energy-numeric-target from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- floating, irrational, imaginary, signed or numerical-null proof values
- fitted coefficients, target-selected relations or unregistered constants

- target access before prediction seal
- omitted input uncertainty, external row or adverse control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:55cdab99ac1372af5d769f493866a3dc73d31ed5fee6c0e24a30667d32590e50; engine receipt
 sha256:4c675313be3e4006a248563bd452cb64e04425dbef90838ba7625a59beb8b277 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001-4c675313be3e4006.json;
 empirical-validation hash
 sha256:2e08d99977b76a21e09ae04a2080633f83f89d247304f1b54df868ca5e9b42cf; measurement receipt
 sha256:72a70301b05b1847cb68d81b26ec8b12ba9fd6ed0b7633034f7c4c062a5439c8.

297. Post-seal experimental agreement of the inverse-square exponent

Claim identity: SFT-PHYS-VALIDATION-INVERSE-SQUARE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After the Fold structure independently forces and seals positive integer exponent two, the Williams, Faller and Hill Coulomb-law experiment reports $q = (2.7 \pm 3.1) \times 10^{-16}$ in the tested form $1/r^{(2+q)}$. Converting only the reported finite interval to exact positive rational exponent bounds and enumerating every positive integer inside them leaves two as the sole compatible exponent.

The complete reported experimental exponent interval contains exactly one positive integer exponent, two, agreeing with the independently sealed inverse-square law.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-MEAS-BOUNDARY-GROWTH-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete product of physical carrier, inverse-square relation, provenance, capability-closed prediction, separate measurement record, complete rows, positive-successor closure and no-extra-rule forms. The declared exact boundary is: All exact post-seal comparisons between the forced positive integer exponent and the complete finite positive exponent interval reported by the registered primary experiment. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__forced-exponent-two-versus-complete-reported-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore

a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	forced-exponent-two-versus-complete-reported-interval	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:3524f02008eb569098ef3f8c5f7b7e1e1b24ee038c784410feb0d61a4640db5e. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:cf789a8189e966181ed9fce634a62aa9673b28a799764651de82d6ada2967d4f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:adbfc13814bcc801f5be5939b4b014fe3f5ac6c653f3a38c51101ced522375b7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:ec4568e8deb5d9d95c85707880cc97bf58db129bade753295ec114c5bf92286f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:d3667e2420aed299c2645bde52a7be6730ab235e2c2cf7f3ca342a447acdeec7.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:8a4bedf7f45425c480eb95fc57a8224e59106a705245cb0442c2bd05fb683412. Independent certificate: sha256:937824d169404edcb5cdb810fdf82d65a2ce065a92fab16469eba917e2fdebde. Engine external-validation hash: sha256:27183e829b317d0cd10824f7e429ec8f3be39f8d46b3f8f9f5c4fcdc66940f0b.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- APS-WILLIAMS-FALLER-HILL-1971-COULOMB

Observed comparison records:

- APS-PRL-26-721-REPORTED-EXPONENT-INTERVAL: predicted reported-exponent-interval-has-sole-positive-integer-two; observed reported-exponent-interval-has-sole-positive-integer-two; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The complete registered reported interval does not contain exponent two, contains another positive integer exponent, differs from its source hash, is opened into the formal derivation, or a tampered comparison is accepted.

Measurement receipt: sha256:7e578b51b3663020cfec74ee6d42bc8c1085373c9a5f91730a784d8e7b7b5582.

Isolation certificate: sha256:5b85119f9c6578f96c7b2d872079c9118603b830db9641c6e4d797c2953ddd3b.

Custody certificate: sha256:62481020db75f30db4c58b398af211966639b67ef2f439567dc4e1da39b4e617.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- measured result used by the formal inverse-square derivation
- logarithm, regression, floating fit or best-match selection in the checker
- semantic numerical zero, negative, irrational or imaginary proof value
- omission of the reported central value or uncertainty
- claim that interval agreement proves an exact physical value

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:697c2ddd7d7f792cbdf283fa279f572c8b427b2a643466dfb2d6324329ed8564; engine receipt

sha256:31bd4d97bd010d4af89b2b85ed1bb02056b5b6d67a3acca65110366ab479e698 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INVERSE-SQUARE-001-31bd4d97bd010d4a.json;

empirical-validation hash

sha256:4790e8635bc9a04ccfda3909c6b242cfb3b8f0aa04c26bdaa7f901276124f141; measurement receipt

sha256:7e578b51b3663020cfec74ee6d42bc8c1085373c9a5f91730a784d8e7b7b5582.

298. Post-seal CODATA check of the terminal inverse fine-structure ratio

Claim identity: SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After V3 independently seals the exact terminal inverse fine-structure ratio, the complete NIST CODATA 2022 central value and stated standard uncertainty are converted to exact rational interval endpoints. The sealed ratio is checked for interval membership without fitting or rewriting it.

The exact terminal ratio 503846395469/3676744786 lies inside the complete 2022 CODATA interval 137.035999177 +/- 0.000000021.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001

- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal relation, provenance, custody, row, exact-interval and no-extra-rule product. The declared exact boundary is: All exact post-seal comparisons between the immutable terminal ratio and the one complete registered CODATA inverse-alpha row including its uncertainty. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-terminal-ratio-versus-complete-codata-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-ratio-versus-complete-codata-interval	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:5f16c61d3619dddb76f43b0199161bf2920453b1ecbfa49e2e24a858fcf70d95`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2460ed9e9112ca2db7f5fec2ba9676993debc0b85cb8d1040342b8a51dd34e68`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:20a184a1749db08e208fa09d5609987741171219db6cfdb7e3c1db52cbf51806`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3921cc46ead2ad22ee5ae1bb3ad7f8b9184ac08981590c2f580ab371eda892a3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9d42267b960bf9dfdd82343e855943f7fd5e75c3f418041c4cee03165d0ad02f.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:31aaffa05397abd35a390640284e0b7e2c2bd7c1793d599a60aad3916ff1695.
Independent certificate:
sha256:4b54876e16a0163c485b0fe3c7b79c5c12fd3c53c49dd90bdd7fa11ed7248116. Engine
external-validation hash:
sha256:bfb80b39991e704b6d7b7fc4a0a419ef35d89db78ec3db6f95bef05e54606135.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-INVERSE-ALPHA: predicted sealed-terminal-inverse-alpha-inside-complete-codata-2022-interval; observed sealed-terminal-inverse-alpha-inside-complete-codata-2022-interval; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The sealed ratio lies outside the complete reported interval, the source row/hash changes, the uncertainty is omitted, or a tampered comparison is accepted.

Measurement receipt: sha256:060cad30845d985ab813c4e4114eabee8d690c4f62d18860ab215d35061db1d7.
Isolation certificate: sha256:b3e8f58f7e96427308176390df6c109d246ad97efe3734bcb6250ad59ec0cdb3.
Custody certificate: sha256:b5389bb0d662254fceddf1684eadabe1f77b1f3354e7f690b23622eaa5d61d54.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- NIST-CODATA-2022-INVERSE-ALPHA from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no CODATA value accessible to the formal derivation
- no fitted digit or best-match selection
- no floating-point comparison
- no omitted uncertainty or unfavorable control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:fb0058c462a90eec0073158ae31c88e7bad16311f630152e83ed5ae277d5282a; engine receipt

sha256:cfe0c0b068864707942c03909ef4dd16660476c808fe5e9567970c60d8640bbd at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001-cfe0c0b068864707.json; empirical-validation hash
sha256:5eed8e7391a59d9ffdbf11d158d63dc889357300a19a27203f3719606f5e6877; measurement receipt
sha256:060cad30845d985ab813c4e4114eabee8d690c4f62d18860ab215d35061db1d7.

299. Post-seal IAEA check of the nuclear-closure sequence

Claim identity: SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After V3 independently seals the positive-rank nuclear recurrence, the complete first eight outputs are compared in order with the registered IAEA Nuclear Data Section magic-number record.

The sealed V3 closure prefix (2, 8, 20, 28, 50, 82, 126, 184) exactly matches the complete registered IAEA sequence.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal ordered-sequence, provenance, custody, row-retention and no-extra-rule product. The declared exact boundary is: All exact order-preserving post-seal comparisons between the immutable first-eight closure output and the complete registered IAEA positive magic-number sequence. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-closure-prefix-versus-complete-iaea-sequence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-closure-prefix-versus-complete-iaea-sequence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.

Axis	Forced form	Eliminated form	Elimination reason
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:56679e7ef35dd6a054b48cb85d9a365e04483d1876703a9d2e8ef6e52858f811`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6f99243f846465531b8e8d04eafd231fa57515365c2ce894180820a930c4f6be`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:856dc18ebff37c0d3ec1fe6a114c55e16f6bd261778ca0abd9f5830bf7c82bb8`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:77a9bb7213b8ff916417351bf0dada8cf20622b8fef94e29c8a977e5c2c2eb9e`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:745f64af58a7835963b0d86ea1dd07931f1d68b4035bc3d091798fd4b90ac198`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:305b02c334ed9f2efcdc389e5a3601348dabb2f28d1b4b9e9a0c91f36ba7716b`. Independent certificate: `sha256:0c4548de901d5c3e1ab8a7be0eb1ce51e3801104365a796d733b907dd9ca9642`. Engine external-validation hash: `sha256:7098173bec13b8fbba7b194c44cf71c486b50c1411683a505d48c61764a57da5`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- IAEA-INDC-NDS-0452-MAGIC-NUMBERS

Observed comparison records:

- IAEA-INDC-NDS-0452-MAGIC-SEQUENCE: predicted sealed-eight-closure-prefix-exactly-matches-iaea-reported-sequence; observed sealed-eight-closure-prefix-exactly-matches-iaea-reported-sequence; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: Any sealed/source coordinate differs, either ordered sequence is incomplete, the source identity/hash changes, or a tampered comparison is accepted.

Measurement receipt: `sha256:aae2b19de5952d1488ac8ec859b09819c1378406999367008709aa7b423a7f08`. Isolation certificate: `sha256:264c93c4ab7661fdc40750e550143a79427eeca64dc3cf4508db4d52fba2e913`. Custody certificate: `sha256:5ca4ac8aaff5471643b0ccb92f484218242a81626c97d4d9590494215809e94d`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: <https://www-nds.iaea.org/publications/indc/indc-nds-0452-part1.pdf> (`sha256:ce642d62698b97fa509b8101a5661e578d6524f8705c19e38f7426efe7a0f6a6`)

Registered target identities:

- IAEA-INDC-NDS-0452-MAGIC-SEQUENCE from IAEA-INDC-NDS-0452-MAGIC-NUMBERS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no magic-number value accessible to the formal derivation
- no selected favorable subsequence
- no fitted coupling or threshold
- no omitted source or tampered control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:1d815786712147e935c6d17ddb736acd9863124d56a42adab83e860b22b0d131; engine receipt
sha256:e55e9e67e3e4b389dd46c7a5cd6260674d11a9c832d55835d92100936f365bbd at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001-e55e9e67e3e4b389.json;
empirical-validation hash
sha256:bfb9984c3d44e47579ac8b6b710cc22f04875b3c99fbfc07a179ccb8905a036a; measurement receipt
sha256:aae2b19de5952d1488ac8ec859b09819c1378406999367008709aa7b423a7f08.
```

300. Post-seal exact Koide comparison

Claim identity: SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact Fold value two-thirds is sealed before both complete CODATA mass-ratio intervals are opened and converted, with exact rational square-root enclosures, into a conservative Koide interval.

The sealed exact Koide value $2/3$ lies inside the complete conservative CODATA-derived interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete post-seal invariant, two-source-row, rational-enclosure, custody and no-extra-rule comparison product. The declared exact boundary is: The full Cartesian uncertainty rectangle of both complete CODATA 2022 charged-lepton mass-ratio rows. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-two-thirds-versus-complete-rational-koide-enclosure__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-c`

losure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-two-thirds-versus-complete-rational-koide-enclosure	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:aedb07af5ac4328e9afb0fa6ed1a6435f3775d4cd817d6718205298fcb7c74b5. Its generality certificate is stored in elimination_receipt.json; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:0db9171872e00ee706733cfefbc55c1c2dedcace1957cd49881073358744b6d2f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:69e20cd5a6dc9fb152ee9161cb3487f80a300317afd5c386a8aad46c60efe56e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:3966a66e1a9f0719f16bd35dd82cb84626b18afc8e197d080f8bfad3be3ab561.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:4163c02f9ea2f8ced72407d4d905d426254508a47c6886b9f1f93bbd5600ae3d.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e3cd83ee4f6fafde717e8b0e0cf9225f363bf191cffa99cd90546a5acfe486cd. Independent certificate: sha256:c060fa4e8d243d22657a297a4d45013b3dc832315ad6e3e78ade240d1dbbfc26. Engine external-validation hash: sha256:dbb16a3a4ad6570771883dc38fe894de89cf5a7038664a5279be808d51fa8456.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-MUON-ELECTRON-MASS-RATIO: predicted sealed-two-thirds-inside-complete-codata-derived-koide-interval; observed sealed-two-thirds-inside-complete-codata-derived-koide-interval; exact match True
- NIST-CODATA-2022-MUON-TAU-MASS-RATIO: predicted sealed-two-thirds-inside-complete-codata-derived-koide-interval; observed sealed-two-thirds-inside-complete-codata-derived-koide-interval; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The exact two-thirds lies outside the outward CODATA-derived interval, either row or uncertainty is omitted, source custody changes, or a tampered comparison is accepted.

Measurement receipt: sha256:550f4a01b231c2375c639d5486b8c252ee0431cdf4012d8a7174fd5543d324cb.

Isolation certificate: sha256:5a7e06f3e7fbc8c5d72606785384aeb6fd57af986b11dee9f91dc1425bb3b4b9.

Custody certificate: sha256:bdc61acb9c3ebc999a602c9bb9a07039b4e24d8dfbdefd9f3aab635482456397.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured mass ratio in the formal two-thirds derivation
- no floating square root
- no central-value-only comparison
- no omitted uncertainty or row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:99550383ca1d0858c6dac68fb34c27f8065c37b19dd50c4fb8235de4de6db20f; engine receipt

sha256:369a1e48d622bba0f3e4abc1e89fef8553b17097c3d8c4427afca26386f6cbf9 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001-369a1e48d622bba0.json;

empirical-validation hash

sha256:4580bae1908fdd208c75dc159f8ff406fbbcbdb9be2ec6f6e0e9376bc9a0244f; measurement receipt

None.

301. Blind PDG test of the terminal electroweak Fold share

Claim identity: SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After the exact terminal weak share is sealed, it is compared with the PDG on-shell weak angle and with squared W/Z mass intervals from both the all-input world average and PDG's separately published compatible-input evaluation. The resulting pass/pass/tension vector is retained whole.

The sealed terminal share must lie inside the PDG on-shell interval and compatible-input W/Z interval while the all-input W/Z tension is retained, not silently discarded.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal relation, source provenance, custody, exact interval, complete-row, successor and no-extra-rule product. The declared exact boundary is: All exact post-seal comparisons of the immutable terminal sin-squared and cos-squared shares with every registered PDG weak-angle and W/Z mass interval, including the unfavorable all-input result. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-terminal-share-versus-complete-pdg-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-share-versus-complete-pdg-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:90e158fd2e602f2f9ac7072a9a42134abd580cf0ad201739f477fdf393d37b91`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:10878412eb173fed35b73c98461bd25be5849bf70b3649f44954c90a7735f43f`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1dc44c2a919a389128cc5f0437f114079e17df55855381463f1da809da19cbb6.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:81f26488154f0dd23cf48f2b5a732e7036a01d736cc5633423e4515997df4b7a.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e29e2abdf7db536e975be6c8a5c9a92faec02648ff5a0a80afe0c3f22b987a38.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:fddd39ccabb378e3fa86e639e95933b9dc704b4b75ee59d0fb89d65b3db02e09.

Independent certificate:

sha256:71b523de6f4a0b3c139d99186cfae5fb0bffe38f55c658fb1297dcf3a3b85972. Engine

external-validation hash:

sha256:e9de242259898adb51d6b162e845a51f4ae3635f8a63251874fd01a04ce7cfbc.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-ELECTROWEAK-PRECISION-2024-2025

Observed comparison records:

- PDG-ELECTROWEAK-COMPLETE-VECTOR: predicted on-shell-pass__compatible-w-mass-pass__all-input-w-mass-tension-retained; observed on-shell-pass__compatible-w-mass-pass__all-input-w-mass-tension-retained; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: The sealed weak share leaves the on-shell interval, leaves the compatible-input W/Z interval, enters the presently disjoint all-input interval without a source revision, any registered row/hash changes, or an unfavorable row is omitted.

Measurement receipt: sha256:a6bb11e7f817c5480a2fdb24c1c416bd9dc24599e76da1cd41b8071a1837268.

Isolation certificate: sha256:d27e789691c2e09f32c74f77e57a781e76e5502697f5b2e423450b381df8cd96.

Custody certificate: sha256:14d93b40017b64cbe185f8b09384c0e375c05ecd77c20b4fec3729ea0e68e7.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf> and <https://pdg.lbl.gov/2024/listings/rpp2024-list-w-boson.pdf> (sha256:1eb3111d343f411a1788120d394899c03ea75279bbea68a0c8284a58f6af89bd)

Registered target identities:

- PDG-ELECTROWEAK-COMPLETE-VECTOR from PDG-ELECTROWEAK-PRECISION-2024-2025

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no PDG value accessible before the formal seal
- no fitted digit or uncertainty widening
- no deletion of the unfavorable all-input row
- no floating-point interval decision

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:134b2d492cf9a84819fb9fbf7701bbd77f649a2acd56545efa1b9fc74909c5de; engine receipt
sha256:e72a5865357e13d30301ecdc625014901122f399f1fd4a6967c585f214852d8a at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003-e72a5865357e13d3.json; empirical-validation hash
sha256:c54ebb425958b1801b404eca6bd2d2dd5be7fa6d6ea04914aae1540174f06c39; measurement receipt
sha256:a6bb11e7f817c5480a2fdb24c1c416bd9dc24599e76da1cd41b8071a1837268.

302. Blind CODATA test of the terminal proton-to-Planck hierarchy

Claim identity: SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After the exact terminal squared hierarchy is sealed, the complete CODATA 2022 Planck-mass and proton-mass intervals are propagated outward through an exact positive squared ratio and tested without forming an irrational root.

The sealed terminal squared hierarchy lies inside the complete outward-propagated CODATA 2022 Planck/proton mass interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal relation, source provenance, custody, exact interval, complete-row, successor and no-extra-rule product. The declared exact boundary is: All exact post-seal outward interval propagation from the complete registered CODATA Planck-mass and proton-mass rows to the immutable squared hierarchy. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-terminal-hierarchy-versus-complete-codata-mass-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-hierarchy-versus-complete-codata-mass-interval	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:10570fdaf33283dfed116ea5e897d08066e700a23e4f2c7450d4e9424a3d21e3`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:be07260dbfa4233d3bb0d739bb40fe35fc81fcf348b89b6e3c5a64fd65bb16ae`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f6dd73cab024961b7b1eee64f41622e759af190152ebe18a140eec0a3c72a023`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b205d74a96169e83e09d899ab57901854b936696b5340728d0b647a56f8d9c4d`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9c21f4d3ceece1acc073200621d0f0fb8b533c14bdac560ba6a229f4bfbc326c`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ac90f42e63702458fad777d5aa28b5543ea3c51232db593710c398d66bb5794e`. Independent certificate:

`sha256:2f462d0692a8219ad630c17db02f5294bcc88d22783f97d5221fd5d9e2ea33e6`. Engine external-validation hash: `sha256:b60e1afa46588723b1c5f18bad5904264d616b277e03d12013ffc3eb5f6eaa28`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-PLANCK-PROTON-HIERARCHY: predicted terminal-squared-hierarchy-inside-complete-codata-mass-interval; observed terminal-squared-hierarchy-inside-complete-codata-mass-interval; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The sealed squared hierarchy lies outside the complete propagated CODATA interval, either source row or hash changes, an uncertainty is omitted, or a tampered comparison is accepted.

Measurement receipt: `sha256:6cc7ef49dd74460c9d469324be9aab23c252df8caa1afb098315daf2139c6645`. Isolation certificate: `sha256:87b36990cb91b74489b7573b4aeeac1092fa9d46cf6bf7d86b2c659a243c893d`. Custody certificate: `sha256:29daed4f8f07d9e6cab74d848782762a7d9bf96e9bba651f250312abcc0bef18`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
 (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- NIST-CODATA-2022-PLANCK-PROTON-HIERARCHY from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no CODATA value accessible before the formal seal
- no fitted hierarchy exponent or correction
- no floating-point comparison
- no omitted source uncertainty
- no irrational square-root proof value

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:3935656eb4fb86648b2a496cb2f170e4e6900b49de2e2de85eca178803818dc5; engine receipt
 sha256:7e8e84ee71e17a8b40820223b0ecf00c5a814395a262ee6224a64eb47ed6eae8 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003-7e8e84ee71e17a8b.json;
 empirical-validation hash
 sha256:57e1a8bbe37a75646a9f157969540c879c1aaeff44dc09bedfc7404692436aa; measurement receipt
 sha256:6cc7ef49dd74460c9d469324be9aab23c252df8caa1afb098315daf2139c6645.

303. Blind PDG anchor test of the complete force-sector formula

Claim identity: SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After the four-sector inventory seals, the same p-fibre and p-squared-less-One formulas are checked against PDG's sector-two weak-mediator count and sector-three colour/gluon counts. Sector-five and sector-seven outputs remain explicitly unmeasured predictions.

The sealed common formula reproduces weak mediator count three, colour count three and gluon count eight; sector-five count twenty-four and sector-seven count forty-eight remain standing unmeasured predictions.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed sector carrier, common count relation, provenance, custody, separate record, complete anchors, successor closure and no-extra-rule forms. The declared exact boundary is: Every registered known PDG anchor of the common sector formula, plus a mandatory boundary record preserving

every presently unmeasured penta/hepta consequence. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-sector-formula-versus-all-pdg-anchor-counts__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-sector-formula-versus-all-pdg-anchor-counts	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:47a084c8a7fb4dc9e2a613e747813f2e9507f5796b95cb099f66528c7d398eb1`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2d202d9be6a198b5d95cab2d43a8c243f9f7b209d36ff93b3b428ace957ded84`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9090ffcbf7ce4d101b7103ec1dbd481817bc04276f41e7bd55dc2c598a1a7c08`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0abe7f7af69ef66c3040ae6ca5e510d04d22f4d5e2e979878d94d538d3a06db3`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5de8b3690077050c16a394b4e792a2a6bfd428f94ee5a62b4d1a7a734bf6fabf`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b9ea919cc172d6d9382effbcee8ebc2bc69b8337bc660107411cca3bab6104ee`. Independent certificate:

`sha256:ccf55bacab31ba908af95597b9b719fe68f5715a388e391fdad284c70e8346cf`. Engine external-validation hash:

`sha256:500c8c47bce8dc3be3603f146105415b0e47026d84ab9009a81fa1c7a825c06f`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-FORCE-SECTOR-ANCHORS-2025

Observed comparison records:

- PDG-FORCE-SECTOR-COMPLETE-ANCHOR-RECORD: predicted sealed-common-formula-matches-three-eight-anchors__penta-hepta-retained-as-unmeasured-predictions; observed sealed-common-formula-matches-three-eight-anchors__penta-hepta-retained-as-unmeasured-predictions; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: Any known count differs, a source identity changes, an unmeasured penta/hepta result is omitted or mislabelled as observed, or a tampered row is accepted.

Measurement receipt: sha256:94ec3c6dd81be8626acd688f9316d73c63338dcce1251156448de4fdcbd20830.

Isolation certificate: sha256:de6d852ee6b1146f73eca14b3c2bce0e6071fe4b3e424127bbd2cb225eca8dac.

Custody certificate: sha256:817791234ead3d469f0511ad4771082ac01da25c791733ab63ad9842902fbbc8.

Registered source descriptions:

- Particle Data Group, Lawrence Berkeley National Laboratory: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-qcd.pdf> and <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf> (sha256:bb37e82e9515ff526cbaad742167c16ac1960b7bfca32d8b39dd4384c73b7341)

Registered target identities:

- PDG-FORCE-SECTOR-COMPLETE-ANCHOR-RECORD from PDG-FORCE-SECTOR-ANCHORS-2025

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no PDG count visible to the formal sector derivation
- no imported gauge group selecting p-squared less One
- no unobserved prediction represented as confirmation
- no omitted unfavorable or tampered row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1889142c48413b5124e4dbc8b585a1e419ea4519c5461eb3f66a4ddf0a2f7ac3; engine receipt sha256:5cd64eeef9ed14372e44541ffc2df625aec020eb1277e52cd76de0bb222261a9 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003-5cd64eeef9ed1437.json; empirical-validation hash

sha256:d37956fd80f0bcebe8395c122997d8f3a970a901ba10a90724d2e45f1f062a9e8; measurement receipt sha256:94ec3c6dd81be8626acd688f9316d73c63338dcce1251156448de4fdcbd20830.

304. Post-seal NIST test of the nonempty vacuum floor

Claim identity: SFT-PHYS-VALIDATION-VACUUM-FLOOR-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After the half-One floor and oscillator support are sealed, the full NIST ground-state and Casimir record confirms nonempty residual quantum support and a boundary-dependent vacuum interaction. The absolute half-spacing energy offset is not isolated as a separately measurable scalar, and that boundary is retained.

NIST confirms nonempty ground fluctuations and boundary-dependent vacuum force; the sealed half-One coefficient remains a formal structural value rather than a separately isolated measured scalar.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal vacuum-floor comparison product. The declared exact boundary is: Every registered NIST ground-state and Casimir row, including the direct-measurement boundary on the absolute half-spacing offset. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-vacuum-floor-versus-complete-NIST-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-vacuum-floor-versus-complete-NIST-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:69fa8a450ae9ceeee077dc4aa48bce214d7d6895e71d13072f2c6c243ad3e071`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:d62c7a13817f3409c893bcc756605822fe16f184abf406247d0c4798d2385d92.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f8f27cef4429c37b815e8216cd541a481607b59c0fe113e7368e3fd492c243aa.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:30f479f82606c3cde185295a74313fb4b5d503e2f3a6eb8300e493d27916de9a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:1937768f91c7a380043fc9bb4e6bb2605df230ac5ed1c5d30ae594d0bc96b323.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:c9d74312a5c205da46ebf64d121135da2109a90abead5867cce56d9e152e1fa7.

Independent certificate:

sha256:30d3e8937f20c12b9096f2336a92eeef982cfba58ecaefb8fecb294154438ae2. Engine

external-validation hash:

sha256:123f74dbda943e77711730e0c5253c04c77e6487ae6a8ce4353f89cedede193c97.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Observed comparison records:

- VACUUM-LINEAGE-FLOOR: predicted
nonempty-ground-confirmed__casimir-boundary-force-confirmed__half-offset-not-directly-isolated; observed
nonempty-ground-confirmed__casimir-boundary-force-confirmed__half-offset-not-directly-isolated; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: NIST ceases to report nonempty ground support or boundary-dependent vacuum interaction, a source hash changes, or the half-offset is falsely promoted from indirect evidence to a direct scalar measurement.

Measurement receipt: sha256:bbff5927e2f7c0658d96b870bb2259f7880bcd76607a9ab015f78380e83fbe75.

Isolation certificate: sha256:d68a8ccd0fb00d9f9821f10b8eab5c6fd4adddd9539c5f37ca83f2dce2049441.

Custody certificate: sha256:29a4ee75ec8d392a8d1dec6f71850b9a110c13ef2c4938d87882e76585de305.

Registered source descriptions:

- NIST, Particle Data Group and CNES MICROSCOPE collaboration:
<https://www.nist.gov/news-events/news/2015/06/vacuum-fluctuations-measuring-unreal>;
<https://www.nist.gov/news-events/news/2013/02/calculating-quantum-vacuum-forces-nanostructures>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>; <https://cnes.fr/en/projects/microscope>
(sha256:327696985650aebaa20436e9da1d7f20f9bbff4205f9e7146ec8bd887535f3a9)

Registered target identities:

- VACUUM-LINEAGE-FLOOR from VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no target access before the formal seal
- no claim that Casimir force alone measures the half-One coefficient

- no deletion of the direct-observability boundary

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:85f051adb7ada691157f11e9bac65e1122dc02ae240c85dc9b6b88eaf8a125a1; engine receipt
sha256:b3ec9913e1dba8376934ffa7398a73079ec9f20f9a727693ad65bfe36581094d at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-FL00R-003-b3ec9913e1dba837.json;
empirical-validation hash
sha256:530e0e9e77180509594b1cf287227d798c8d93cff31003cd64b935aad7593f7a; measurement receipt
sha256:bbff5927e2f7c0658d96b870bb2259f7880bcd76607a9ab015f78380e83fbe75.

305. Post-seal PDG test of the vacuum-polarization running direction

Claim identity: SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The sealed structural direction is compared with both complete PDG inverse-coupling intervals. The low-energy inverse interval lies wholly above the Z-scale interval, so the coupling itself increases with closer/higher-energy probing; PDG also retains direct LEP observations of running.

The complete PDG intervals confirm the sealed direction: inverse alpha decreases from 137.035999084(21) to 127.930(8), hence effective alpha increases with scale.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal electromagnetic-running comparison product. The declared exact boundary is: Both full PDG inverse-coupling intervals plus the direct-running observation, with exact interval ordering and no beta-coefficient claim. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-running-direction-versus-complete-PDG-intervals__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-running-direction-versus-complete-PDG-intervals	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:affe1a6ffc785a1372e39c484e3844fc383abd0c8f41b18724b11d271be07dd6`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3e5c7c6bc91e160a9a5e28c781647e346f7ddcf508b4da31b655b3ac062a706c`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3291393287d2b144f83fbd7009da2720d140a65f329f07b967c3b8e004db528f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d7ab76ff08ea358d1ef265b8a4d92babe072de15688644270c28158b976f091e`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2a381c3f2497067e41476a7e0a141b243d5861e15be04a42f97559fd456dd71e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ddeb4a98e4fbe345139ee83b5aee74348f08fee0e405ecc042716390c7b4ffa`. Independent certificate:

`sha256:24fc04e99b6bc6ef8b5b337fc4839d6ab90467004180cdde94d1cdee9bf378aa`. Engine external-validation hash: `sha256:490c134e7f103a2a48b51f6557ec36cb4cdeb27ec7b45051be30d18e34a3c98a`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Observed comparison records:

- VACUUM-LINEAGE-POLARIZATION: predicted `inverse-coupling-decreases-with-scale__effective-coupling-running-direction-confirmed`; observed `inverse-coupling-decreases-with-scale__effective-coupling-running-direction-confirmed`; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The authoritative inverse-coupling intervals reverse or overlap contrary to the registered direction, direct running is withdrawn, a row/hash changes, or a fitted beta law is substituted for the sealed structural claim.

Measurement receipt: `sha256:e555277d4ddcc736be3baa5be0396411711be8e7e6d231c2336afa5a7d26bd3b`. Isolation certificate: `sha256:a582a5a4720020d8da4f5750de8860c12da381cc9aee39aa671f2d19ca314035`. Custody certificate: `sha256:f28ab8ea9b5dc42e40fa4eef7d1cf98c7dc9c3e29fa2cfcaea61e4ae539d9781`.

Registered source descriptions:

- NIST, Particle Data Group and CNES MICROSCOPE collaboration:
<https://www.nist.gov/news-events/news/2015/06/vacuum-fluctuations-measuring-unreal>;
<https://www.nist.gov/news-events/news/2013/02/calculating-quantum-vacuum-forces-nanostructures>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>; <https://cnes.fr/en/projects/microscope>
(sha256:327696985650aebaa20436e9da1d7f20f9bbff4205f9e7146ec8bd887535f3a9)

Registered target identities:

- VACUUM-LINEAGE-POLARIZATION from VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured value accessible before seal
- no fitted running curve
- no omission of either uncertainty
- no claim of an exact beta coefficient from the two endpoints

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:6a5fe81fa08340640e916ddbfbad1114aa650147b5fe48a822e5cc517e0541963; engine receipt
sha256:89c64800b967585a1f8d5ec366c316fa95767b53479b8ef2878a963d292f3c06 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003-89c64800b967585a.json;
empirical-validation hash
sha256:e82cea38e8e08ce5f75e0273369e848c054abfd98b14333023573cd7fce543e7; measurement receipt
sha256:e555277d4ddcc736be3baa5be0396411711be8e7e6d231c2336afa5a7d26bd3b.

306. Post-seal MICROSCOPE anchor test of vacuum-inertia unity

Claim identity: SFT-PHYS-VALIDATION-VACUUM-INERTIA-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact vacuum/inertia unity relation is compared with CNES MICROSCOPE's completed equivalence-principle mission. Universal inertial/gravitational response is supported to the reported 10^{-15} mission precision. The stronger claim that a vacuum carrier can be engineered to control inertia was not tested and remains a standing prediction.

MICROSCOPE supports composition-independent inertial/gravitational unity response to 10^{-15} mission precision; it does not test vacuum-mediated inertia control.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-INERTIA-UNITY-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal vacuum/inertia anchor comparison product. The declared exact boundary is: The complete CNES mission identity, response class and reported precision together with

the explicit untested engineering-control boundary. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-unity-relation-versus-MICROSCOPE-anchor__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-unity-relation-versus-MICROSCOPE-anchor	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:3f72cbeb5a0e65d10635d97b7ef53ca2b3aa3602cd547294aa47dfb5d65553b5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:cb29b0894217eb327de698a5f0f4d9762ad5388ecada69f5691e8a1b6dfc4d2e`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1a297cfa629c2aa50a1a415db58c4b7d455276d714abdaf84aafc88f7430f435`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c4dead2a3c9d4c128e3e7b2ea42eb34c00f53747c519b61053c8ac7d2e8d09b2`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:bb3f53f22dc0b2f3d35a5f4b68ea3e6e7a3bc453cee3b0f5fd9bd094cc256126`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:072e6dfb7eae1e1a8eb757fe0e2d06e7a993cd02d00846be83d11a292b01747`. Independent certificate: `sha256:35dca81d6c2ac6fe5294ce6dd18c936099fccf1aca8198f3eac768072e6ece0b`. Engine external-validation hash: `sha256:09f118b8d5bc5a8ac11c2c30205a1ff243797a65b2c31898616eac0f777670f0`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Observed comparison records:

- VACUUM-LINEAGE-INERTIA: predicted
equivalence-unity-anchor-confirmed-to-mission-precision__vacuum-control-untested; observed
equivalence-unity-anchor-confirmed-to-mission-precision__vacuum-control-untested; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: A verified composition-dependent inertial/gravitational response violates the unity relation, or a claimed direct inertia-control confirmation lacks a separately committed controlled experiment.

Measurement receipt: sha256:e2894d05572af3cf37328d05605f8765fd280d04a934c4ed0631c55ab0f1006e.

Isolation certificate: sha256:757a1ad50e995eb6b8c3b46cc4c7b626e02524e1c1c866e05962ac8a9fd106b0.

Custody certificate: sha256:1715b342a3c82ae53823340bbb6af329f14d5fdd056e535a5e204dc33d6efed3.

Registered source descriptions:

- NIST, Particle Data Group and CNES MICROSCOPE collaboration:
<https://www.nist.gov/news-events/news/2015/06/vacuum-fluctuations-measuring-unreal>;
<https://www.nist.gov/news-events/news/2013/02/calculating-quantum-vacuum-forces-nanostructures>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>; <https://cnes.fr/en/projects/microscope>
(sha256:327696985650aebaa20436e9da1d7f20f9bbff4205f9e7146ec8bd887535f3a9)

Registered target identities:

- VACUUM-LINEAGE-INERTIA from VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no UAP report or propulsion claim used as measurement
- no inference that equivalence alone demonstrates controllable inertia
- no deleted untested boundary

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:0d791400d27451caff0b06d85590a9547272086ef027dfbe414455594de21b8c; engine receipt

sha256:e529584ec3535d78973aea1f390db46a1324447a7b7e39cec21ebcf1a43086a5 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-INERTIA-003-e529584ec3535d78.json;

empirical-validation hash

sha256:b156e66cbf0fae3f85ea91ab7a631197cd8eadc620ae806b30f03e5f2afabe03; measurement receipt

sha256:e2894d05572af3cf37328d05605f8765fd280d04a934c4ed0631c55ab0f1006e.

307. Post-seal NIST boundary test of vacuum-work extraction

Claim identity: SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The positive one-sixth formal transfer and complete returned-cycle ledger are tested against the complete NIST ground-state record. NIST reports that no energy can be removed from the ground state and that amplification requires coherent pump energy. Consequently free-standing cyclic vacuum work is not empirically confirmed, while the complete-cycle no-unrecorded-gain ledger is consistent with the measured pump boundary.

The formal outward transfer remains sealed; NIST does not observe standalone extractable vacuum work and requires external pump energy for amplification; this supports the complete returned-cycle accounting rather than a source-free gain claim.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003

Generated grammar and closure boundary. Generate the complete eight-axis post-seal vacuum-extraction and returned-cycle comparison product. The declared exact boundary is: Every registered NIST work-removal and pump row, preserving the positive formal transfer, physical translation boundary and complete returned-cycle ledger. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-extraction-ledger-versus-complete-NIST-work-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-extraction-ledger-versus-complete-NIST-work-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:683e1488a3d6f695219d1000f4853ca22ad8a12efad172337324ef9899a76801`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated

by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : f79f3722d9c99925a830509b85a67a3e498128933a91787104d7028694a932a7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : ad367dd6f5d5ba841fb12c20144b3eb2808cf52ee7065b3a60fa1a1e96b7fad8.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 379457c2dec6e112927e7eebc0f3bfff6bbadb2ad1749e9d43ba8e9ef04eec879.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 6d2bd33127758fa52be0b90cecf8dba6ab9c800d189faae28b606a5178d3847.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : 4bed1b1ab95e3e34b9490ba6c787b6d6a8def75d24b9d2b2a5255dc2d75e55f9.

Independent certificate:

sha256 : 4dc5c2ad5d1020c9cfbe9a28a2fab89247a5c5ab3895719ca17a23ad6950cba5. Engine

external-validation hash:

sha256 : cca30800e0fe8911297f2928729df52038f286d1923e6e29bb5c8435f502cb33.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Observed comparison records:

- VACUUM-LINEAGE-EXTRACTION: predicted
positive-formal-outward-transfer-sealed__standalone-vacuum-work-not-observed__complete-returned-cycle-consistent;
observed
positive-formal-outward-transfer-sealed__standalone-vacuum-work-not-observed__complete-returned-cycle-consistent; exact
match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: A controlled device returns vacuum, apparatus and information records to their initial states while delivering net work without an external source, or any registered NIST source/pump/adverse row is omitted or altered.

Measurement receipt: sha256 : d1bffd6aa3b85892e7cde13c175aa64901eba8d877db60ac17adc22239465e3.

Isolation certificate: sha256 : 188effaa8634dff46bfd913592a55efec093d06f6e4688cb45d393b83dd4ee88.

Custody certificate: sha256 : 59d78d8efa3c08a89e1839cd64f333885fbaf1588f5ff3bcd2ed100181ed1a05.

Registered source descriptions:

- NIST, Particle Data Group and CNES MICROSCOPE collaboration:
<https://www.nist.gov/news-events/news/2015/06/vacuum-fluctuations-measuring-unreal>;
<https://www.nist.gov/news-events/news/2013/02/calculating-quantum-vacuum-forces-nanostructures>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>; <https://cnes.fr/en/projects/microscope>
(sha256:327696985650aebaa20436e9da1d7f20f9bbff4205f9e7146ec8bd887535f3a9)

Registered target identities:

- VACUUM-LINEAGE-EXTRACTION from VACUUM-LINEAGE-AUTHORITATIVE-2015-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result

identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no erasure of the positive formal extraction claim
- no claim of empirical free-energy confirmation
- no omission of external pump or reservoir restoration
- no target access before seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2c718a951786fba8c269f606d2f7bf17b04a7d6d4d37a241a8a79c916b7ed88d; engine receipt
 sha256:69f0aa277d9177a170f4b067291adfeaa8e849889e2829298c03832aa5109af6 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003-69f0aa277d9177a1.json;
 empirical-validation hash
 sha256:716868324dd40d4bfffff6be77aa62581b21cf7c7960447eed51ba17d949b7a0; measurement receipt
 sha256:d1bffd6aa3b85892e7cde13c175aa64901eba8d877db60ac17adc222239465e3.

308. Post-seal dynamics and spectra comparison

Claim identity: SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete NIST ASD and CODATA rows confirm discrete physical energy levels, transitions, wavelengths and energy-frequency carriers. They do not identify the universal dimensionless Fold witness levels as measured atomic energies, and that limitation is retained.

Discrete spectra and energy-frequency phase correspondence are externally supported; the particular Fold fractions remain exact structural normalization witnesses, not universal measured atomic energies.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal dynamics/spectra comparison product. The declared exact boundary is: All registered NIST discrete-spectrum and CODATA energy-frequency rows together with the exact-normalization unfavorable control. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-Fold-dynamics-versus-complete-NIST-spectrum-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-Fold-dynamics-versus-complete-NIST-spectrum-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:c9da3bad2ed6d88448c50ad318b458b35f61ac82304c44b25a8414a65a0fda17. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:e2c87766879b437a55f2a3b6438e924dbb0dddb65598fb6305410536068aac27.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:efc79bf59416c63d0d2c56a8dd03c72150b389987374bcb6347b48bb3ba1fcf7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:73036f37e4f48171fdf291abb17213ae808e984532a7038d1caa21dd3ad45428.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:9fe6b608b8e57e3697202b7fb59ea90b9d5d2f3859395f399c916c233fec7e17.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:2cf0988f2e6e454edfe952bb9cdcfc19aef533433f9fa684e67aa1a3a9300041b. Independent certificate: sha256:107e2ca2c00622c6aac5c93ef03cb1b2000838c4d7d95e9718f2d7b45106eb79. Engine external-validation hash: sha256:52ffc99c77cd6925f9416dd390cd2fe63f3eeaa89e6716df1c4b7bbe9003db4e.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Observed comparison records:

- RELATIVISTIC-FIELD-DYNAMICS: predicted discrete-spectra-and-energy-phase-correspondence-confirmed__specific-Fold-normalization-structural-not-a-measured-particle-value; observed discrete-spectra-and-energy-phase-correspondence-confirmed__specific-Fold-normalization-structural-not-a-measured-particle-value; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: Authoritative spectra cease to exhibit discrete levels/transitions or the structural Fold normalization is promoted to a direct measured value without a committed measurement.

Measurement receipt: sha256:c635493af4b50a1d0c957d4468a85df1311fef3d1ec3fb1fe5c3c6a027b49ff6.

Isolation certificate: sha256:0aa2b9f855e019ef4c070c0a4bfc381d466c3f246a8da8a41781695d78705167.

Custody certificate: sha256:69258a1cf7e15e4df1fce27a5e5b1d5b4831aa9e37b6c366dbe3f766fe648563.

Registered source descriptions:

- NIST, CODATA, Particle Data Group and American Physical Society:
<https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>; <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>;
<https://doi.org/10.1103/PhysRevLett.26.721>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>;
<https://www.nist.gov/news-events/news/2019/12/new-playbook-interference>;
<https://www.nist.gov/publications/nonlinear-networks-arbitrary-optical-synthesis>; <https://tf.nist.gov/ofm/synthesis/ppln.htm>
(sha256:97a7631d956cc2eb8142e64603a1f51ece76944195274a28bbe91498edae0308)

Registered target identities:

- RELATIVISTIC-FIELD-DYNAMICS from RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no source access before formal seal
- no claim that NIST measured universal Fold fractions
- no omission of the normalization control

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:670d4c6098485e1583ab5b0b0f653a82941ac4b808f4a930c4a34e725852358c; engine receipt

sha256:3c65449f45a60ef6f3285e063eb3a9c83b1e8c2942099c9acc09600951947eaa at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003-3c65449f45a60ef6.json;
empirical-validation hash

sha256:83a7f19781bc7b9c922c7fa649dc7ddba0da70cc0af4b3c152f3a7355103d8a5; measurement receipt
sha256:c635493af4b50a1d0c957d4468a85df1311fef3d1ec3fb1fe5c3c6a027b49ff6.

309. Post-seal relativistic and electromagnetic field comparison

Claim identity: SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete PDG, CODATA and APS record supports Dirac/Lorentz physical structure, one invariant vacuum speed, electric/magnetic observable families and the inverse-square Coulomb exponent. It does not measure the Fold 3/5, 4/5 or four-half closure witnesses as universal particle coordinates.

Dirac/Lorentz structure, invariant vacuum speed, electric/magnetic observables and inverse-square Coulomb behavior are externally supported; the exact rational Fold closures remain structural witnesses.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-FIELD-LORENTZ-TRANSFER-003
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal relativistic/electromagnetic comparison product. The declared exact boundary is: All registered PDG Dirac/Lorentz, CODATA speed/electromagnetic and APS inverse-square rows plus every rational-witness scope control. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-relativistic-field-laws-versus-complete-authority-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-relativistic-field-laws-versus-complete-authority-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a9aa390a373da77add9bdcf4dbf0af958f1d60b2101b581d70062534cffe6094`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:0f5d12ebe7a12829793e67cabeeab858be54036267be413409f41f9c937ac6fb.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1a0bb96564a502d21766adbd7ad508c54e2248ed7c5355a58392d16469b4c52d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:9452ac6c8fe6d698663a2f665d20c48dc9b60a1e964942cca5e873bf17a7c00f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:b8f6c200b11fa4f3d5a904e388d13fa4d7c71c1c07bf0b0f2dc226edfc8d91ea.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:0318131d9b374c14facbbbd8dd1de5443bf68ea6bb4488b59488654dd6699f2d.

Independent certificate:

sha256:59180ae362cc75407c15974fd1714b53cf2724e51cb2a4cd5d8e970edeafd342. Engine

external-validation hash:

sha256:132e0bd70cb8a9d982d4855725cfd860f730751b9c5ca945e6cabf8faf5157f3.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Observed comparison records:

- RELATIVISTIC-FIELD-FIELDS: predicted Dirac-Lorentz-invariant-speed-and-Coulomb-inverse-square-correspondence-confirmed__specific-rational-closure-witnesses-structural; observed Dirac-Lorentz-invariant-speed-and-Coulomb-inverse-square-correspondence-confirmed__specific-rational-closure-witnesses-structural; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: A verified non-invariant vacuum propagation speed, non-inverse-square Coulomb exponent outside the registered bound, or absence of Dirac/Lorentz and electric/magnetic physical classes contradicts the correspondence; a changed source or omitted limitation invalidates the validation receipt.

Measurement receipt: sha256:6e2b123b895e50de032c92694765532679f3fb973e0a8d7ef86f3e8fac7e2220.

Isolation certificate: sha256:ea83359641c0c58961a85ce7dd63429a9eeef04580bfb99b17a1780659d29c396.

Custody certificate: sha256:414c1ac11593971921230e1caa10b47b5315f062d6141e8a136714503a95191f.

Registered source descriptions:

- NIST, CODATA, Particle Data Group and American Physical Society:
<https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>; <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>;
<https://doi.org/10.1103/PhysRevLett.26.721>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>;
<https://www.nist.gov/news-events/news/2019/12/new-playbook-interference>;
<https://www.nist.gov/publications/nonlinear-networks-arbitrary-optical-synthesis>; <https://tf.nist.gov/ofm/synthesis/ppln.htm>
 (sha256:97a7631d956cc2eb8142e64603a1f51ece76944195274a28bbe91498edae0308)

Registered target identities:

- RELATIVISTIC-FIELD-FIELDS from RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no conventional equation selecting the formal law

- no promotion of structural rational witnesses to measured particle values
- no omitted APS uncertainty or CODATA definition boundary

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:99ecfc87d8f303b62e401e6b46dcd2b73acaf5743f282587d9965631c308ab11; engine receipt
sha256:af06233350a2f6e005076914a40d3cbcc6d6744954d9c75238b52bebe204bee8 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003-af06233350a2f6e0.json;
empirical-validation hash
sha256:557122754d87fd03eea18126cdc1e5a10b0b09f2d3b37277fc260c04b3295924; measurement receipt
sha256:6e2b123b895e50de032c92694765532679f3fb973e0a8d7ef86f3e8fac7e2220.

310. Post-seal optical-operation comparison

Claim identity: SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. NIST experimental records support photon interference, polarization observables and nonlinear sum, difference, second-, third- and four-wave mixing classes. The exact half-One support normalization remains a sealed Fold relation rather than an imported susceptibility or measured universal amplitude.

Interference, transverse polarization observables and nonlinear frequency mixing are externally supported; the Fold half-One is not asserted to equal every physical amplitude or susceptibility coefficient.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal optical-operation comparison product. The declared exact boundary is: Every registered NIST interference, polarization and nonlinear-mixing row together with the normalization and material-response controls. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-optical-operations-versus-complete-NIST-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-optical-operations-versus-complete-NIST-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:f4ab1b925e7eb9ecbeeb8511ad7af697559a8887f0f28eeca980955ea0792b8b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c1d8d482048194f5688ebdcbec17b04d425680acb2a2661c5458588a8b19ff9b`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:69070f199be6d34c3f6ae85a96c9a0a3caa25bc0c3d9f21156cdba5be05e886c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f9040489e5a295e0d78a8cddacbe68540ee86dd22b5382f66a9bc93fd5f4e268`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:497b0daf538864ea517c50a5c851c7bcd16f68a8ea3d93b46550946730511aad`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:684223ef4c98beddc30002ffdd503d26235848f5fb341743bb5767940ca6e260`. Independent certificate: `sha256:2d1a16b0a578132bcef06dbe25974644eec2693cc32061469f2a8fffc88a15e1`. Engine external-validation hash: `sha256:f52bbd11dab3c6ab2b6fdee62e87b1e9a9f43db335aa14655b91348b7a5ee5f2`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Observed comparison records:

- RELATIVISTIC-FIELD-OPTICS: predicted interference-and-nonlinear-frequency-operations-confirmed__held-polarization-supported-by-registered-observables__half-One-normalization-structural; observed interference-and-nonlinear-frequency-operations-confirmed__held-polarization-supported-by-registered-observables__half-One-normalization-structural; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: Controlled optical experiments cease to exhibit interference or conservation-governed nonlinear mixing, or the validation silently equates a structural Fold share with an unmeasured universal amplitude.

Measurement receipt: sha256:32da7e42cfd301ed589ba2c8e40eeddd7f5a0976125c8bc5f270b055f4c13c75.
 Isolation certificate: sha256:2ded0fa700114738d32a9e304ce7d4e2fc8990ed4b8a55f577c155fcf8aa3c59.
 Custody certificate: sha256:866c07e8fb118847e762cba083cccaa4e6c0b3e06f726804d6242827272db75.

Registered source descriptions:

- NIST, CODATA, Particle Data Group and American Physical Society:
<https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>; <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>;
<https://doi.org/10.1103/PhysRevLett.26.721>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>;
<https://www.nist.gov/news-events/news/2019/12/new-playbook-interference>;
<https://www.nist.gov/publications/nonlinear-networks-arbitrary-optical-synthesis>; <https://tf.nist.gov/ofm/synthesis/ppln.htm>
 (sha256:97a7631d956cc2eb8142e64603a1f51ece76944195274a28bbe91498edae0308)

Registered target identities:

- RELATIVISTIC-FIELD-OPTICS from RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported optical amplitude
- no fitted nonlinear susceptibility
- no claim that every material realizes every operation at the same strength

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:2dcef842a366ce922fe85ae79df7284fc8f231b2cdc95f3116b867af68aacac4; engine receipt
 sha256:2cd2aeb5d9d4f311bab1b6778728a80b9cf087a82263645db5e9c2a222078b83 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003-2cd2aeb5d9d4f311.json;
 empirical-validation hash
 sha256:a2a16e3f3b10f21bb0058aaceafac87b76a0ef521e5ef5bbcb82fc8462eb69a33; measurement receipt
 sha256:32da7e42cfd301ed589ba2c8e40eeddd7f5a0976125c8bc5f270b055f4c13c75.

311. Post-seal finite-loop and renormalization-boundary comparison

Claim identity: SFT-PHYS-VALIDATION-FINITE-LOOPS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. PDG retains finite renormalized radiative calculations and schemes, providing correspondence for finite observable loop receipts. These calculations do not directly measure whether the substrate is fundamentally finite, so the V3 law remains a finite generated-support theorem rather than a claimed empirical disproof of every continuum representation.

Finite renormalized radiative observable calculations correspond to exact finite loop receipts; the absence of completed continuum support is a formal Fold boundary, not a directly measured ontology claim.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

• SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal finite-loop comparison product. The declared exact boundary is: The registered PDG renormalized-calculation row and the explicit distinction between finite generated-support proof and unmeasured substrate ontology. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-finite-loop-law-versus-PDG-radiative-calculation__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-finite-loop-law-versus-PDG-radiative-calculation	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:10209f275a6364d9809813ba813e7c0620782d09a73ce20fcc13d8aa600eab33`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b5b8ef99f7d0a4cc654db8a5eb66e0aa622e55639d04e071250335987128259b`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a26bebb3de346e70b9e9276f52f8578a6e1b526f11c327161c72c6075963d6`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1a57452db5900cce3c3d0f28b850c3879cdbe6e6b02c2c10c2711d9e99b5cb2f`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:38ed69fb363332a574141c377c615d1d1b361d8aa9ed4d819bc02f97062caed0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:cee6eec1e3e0d2658873ad73424245540f059ef095c92c0fcd4dd2af20c98b0a`. Independent certificate:

sha256:3e424a2707ab266aaa10dd700ffd886b615109152be56d64a7c07c9f0ed79684. Engine
external-validation hash:
sha256:0446045e49d77e9f7746893e074ebb9591fb6425909a8147ae70912cb4fcc5ae.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Observed comparison records:

- RELATIVISTIC-FIELD-LOOPS: predicted finite-renormalized-radiative-calculations-confirmed__floored-loop-law-is-a-formal-finite-support-boundary-not-a-direct-measurement-of-continuum-nonexistence; observed finite-renormalized-radiative-calculations-confirmed__floored-loop-law-is-a-formal-finite-support-boundary-not-a-direct-measurement-of-continuum-nonexistence; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: A generated finite exact loop census diverges, an unregistered subtraction is required inside the Fold proof, or the validation misstates a conventional calculation as direct measurement of substrate ontology.

Measurement receipt: sha256:c56843e3508a80f52813258c6db3270907823266331589cf24b42651bf8a978e.
Isolation certificate: sha256:2dc011db59508896fb26077d1984d5368b0728e30834b753d4351c2e21c3c35e.
Custody certificate: sha256:dcb3ec3738a17610d368dd4a60829335c5b2b0f94e8266b4908b3db0a523d1f5.

Registered source descriptions:

- NIST, CODATA, Particle Data Group and American Physical Society:
<https://physics.nist.gov/PhysRefData/ASD/Html/verhist.shtml>; <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>;
<https://doi.org/10.1103/PhysRevLett.26.721>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>;
<https://www.nist.gov/news-events/news/2019/12/new-playbook-interference>;
<https://www.nist.gov/publications/nonlinear-networks-arbitrary-optical-synthesis>; <https://tf.nist.gov/ofm/synthesis/ppln.htm>
(sha256:97a7631d956cc2eb8142e64603a1f51ece76944195274a28bbe91498edae0308)

Registered target identities:

- RELATIVISTIC-FIELD-LOOPS from RELATIVISTIC-FIELD-AUTHORITATIVE-1971-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no fitted counterterm in the Fold derivation
- no claim that PDG directly measured substrate finiteness
- no completed infinite support admitted

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8a3553097aeeb1eb6d79d2fb7db17ff0148a8580a13c671f64397c5f02d00050; engine receipt
sha256:a815c5258c82cf73a7775d9498f95ac45de0810da8eaa2ee033a5294f5612e8c at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-FINITE-LOOPS-003-a815c5258c82cf73.json;
empirical-validation hash
sha256:d5739892abc343e334f8e697e35a0b6f5b31e8452dab01df04dc1e522dde1761; measurement receipt
sha256:c56843e3508a80f52813258c6db3270907823266331589cf24b42651bf8a978e.

312. Post-seal gravity, clock and equivalence comparison

Claim identity: SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. NIST/JILA resolves gravitational clock redshift at millimetre scale and CNES MICROSCOPE tests universal free fall at order 10^{-15} . These observations support the sealed clock/equivalence classes but do not measure the particular Fold fractions $7/16$ or $3/4$.

Gravitational clock-rate/redshift and inertial-gravitational equivalence phenomena are confirmed; the specific Fold fractions remain structural coordinates rather than measured universal values.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-STATIC-CLOCK-003
- SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal weak-gravity, clock and equivalence comparison product. The declared exact boundary is: Every NIST clock-gradient and CNES equivalence row plus the exact Fold-normalization controls. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-gravity-clock-laws-versus-complete-NIST-CNES-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-gravity-clock-laws-versus-complete-NIST-CNES-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:adca89508148d56711a04f0f2cd38ccdacc52392cf70c8ac8d7056ffc28e7c21. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:8869eb7a90f98167a950339bc2b51ea61c87b6af3661a8370a30338246de75a0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:28558fed02d54ead97cc6dd65125a9c4127f1444349db0064a997df449c9380.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:41290050eddc2615b2ff0f73eedcfb89d7618691694eec55dee79db31f516ddf.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:0d708207c49bc0c4b4ede688a3cc38145a4030127d647fe54e270e80e17c5282.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:746c57e9ab79b21abdf9eedbe78954bd318e6bbbc3f46d7d32a7718f0531dc0c. Independent certificate: sha256:c430048665501db6da2564a13a9f038aaf173e53863f3fdc12ce25c6e62b5310. Engine external-validation hash: sha256:2bc836aba477b3635611d76973221db1f8c67002f7289ddb804cd2588ad9a0fe.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Observed comparison records:

- GRAVITY-SPACETIME-CLOCK: predicted inverse-square-source-flux-clock-redshift-and-equivalence-phenomena-confirmed__specific-Fold-fractions-structural; observed inverse-square-source-flux-clock-redshift-and-equivalence-phenomena-confirmed__specific-Fold-fractions-structural; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: Controlled clocks fail to exhibit source-correlated gravitational redshift, free fall becomes composition dependent outside the registered uncertainty, or a scope row/source identity is omitted or altered.

Measurement receipt: sha256:0a9abfa9f2cb8482951ae4c1c9b0f8cb76d1627f28a84beee599d06dda3d1ecf. Isolation certificate: sha256:b40434948dfc8a805b55f0a454ccbe4478144789c04799715401a81af61ab8e0. Custody certificate: sha256:7aabedabda6590610ede243d300d039f1ace46b991b6620e672d284b6090c3e5.

Registered source descriptions:

- NIST/JILA, CNES MICROSCOPE, LIGO-Virgo-KAGRA/GWOSC and Event Horizon Telescope: <https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample>; <https://cnes.fr/en/projects/microscope>; <https://dcc.ligo.org/P1700308/public>; <https://dcc.ligo.org/P2000091/public/>; <https://eventhorizontelescope.org/press-release-april-10-2019-astronomers-capture-first-image-black-hole>; <https://gwosc.org/api/v2/catalogs> (sha256:b64325355bd993ab32cff6c088023dbe7911ab13278b9439466cd0ae56bedd5e)

Registered target identities:

- GRAVITY-SPACETIME-CLOCK from GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no target access before formal seal
- no claim that NIST measured a universal $3/4$ clock factor
- no omission of MICROSCOPE mission precision or scope

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:60d44fd852ce8287c042a28679f9c520cecb00adb07db25c19b0d34c6da564e; engine receipt
 sha256:4fc85a7fb89b3bbace0a934db81587dff9653aa457cc76c7722ac8dca7383515 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003-4fc85a7fb89b3bba.js
 on; empirical-validation hash
 sha256:17f80e9bb3ed53c84c92e2ad7d9a6900351247afa4945c0eab32a1a5b75c686e; measurement receipt
 sha256:0a9abfa9f2cb8482951ae4c1c9b0f8cb76d1627f28a84beee599d06dda3d1ecf.

313. Post-seal gravitational-wave speed, polarization and quadrupole comparison

Claim identity: SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The GW170817/GRB170817A measurement bounds relative gravity/light speed between -3×10^{-15} and $+7 \times 10^{-16}$; LIGO-Virgo catalog tests support tensorial polarization and quadrupolar strong-field modes. Frequency/source coverage remains finite and explicit.

Gravity/light common speed is confirmed at the reported multimessenger bound; tensor polarization and quadrupolar strong-field support are confirmed within the registered catalog analyses.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal gravitational-wave comparison product. The declared exact boundary is: Every registered multimessenger speed interval, polarization, quadrupolar ringdown and finite-coverage control row. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-gravity-wave-law-versus-complete-LIGO-GWOSC-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-gravity-wave-law-versus-complete-LIGO-GWOSC-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:80f5312883bb59d9b72226642f6858e28badd3f8ec5e1504b7bbf5a2e7049cbe`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1e0778a3e05b6137c48e1e5f892282830cb4edd8866adf2a2f34990090c923d6`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:133903c972c6d02baf629cbd3b5d9099b1c4fda9429a5c8a128f1c8395957b6e`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8b2b76b02216f9689ff132c0a33fb4e40bce4910a18392dc6b9b535907d1e7fe`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9a9031b8d09c8940619de30102e51f40f799521eee6d31404529f214d3735d25`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:5c372a5e07b98f86aaabeec78736b78a8936cc9ababbb60075fdc68dcd1c2443`. Independent certificate: `sha256:c80726d2bcd9e2bd7673f06100406c75ce596b18244ed4808a34367b445b5e09`. Engine external-validation hash: `sha256:916bba275a98aec97b154edb0d652e45bbe408e99ed414de1fab4d9cc8db62c0`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Observed comparison records:

- GRAVITY-SPACETIME-WAVES: predicted gravity-light-speed-equality-confirmed-at-multimessenger-bound__tensor-polarization-and-quadrupolar-strong-field-support-confirmed; observed gravity-light-speed-equality-confirmed-at-multimessenger-bound__tensor-polarization-and-quadrupolar-strong-field-support-confirmed; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: A replicated multimessenger event establishes gravity propagation outside the registered One-speed class, a non-tensor mode replaces the two-tensor class, quadrupole support is absent, or any adverse/scope row is deleted.

Measurement receipt: sha256:1acb4baba3df38ef3d5313fb2fbe0d6628ea9fac7f60c448b1dc316f263b9c65.

Isolation certificate: sha256:d6e541468085cd6ca4db6e8112903b41ea073545977d00a553023d27906fc82e.

Custody certificate: sha256:c984a6c0ace833dc70f1bbb29c58814b5a9d70c9714c23dc806ebbecae1c0b54.

Registered source descriptions:

- NIST/JILA, CNES MICROSCOPE, LIGO-Virgo-KAGRA/GWOSC and Event Horizon Telescope:
<https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample>;
<https://cnes.fr/en/projects/microscope>; <https://dcc.ligo.org/P1700308/public>; <https://dcc.ligo.org/P2000091/public>;
<https://eventhorizontelescope.org/press-release-april-10-2019-astronomers-capture-first-image-black-hole>;
<https://gwosc.org/api/v2/catalogs> (sha256:b64325355bd993ab32cff6c088023dbe7911ab13278b9439466cd0ae56bedd5e)

Registered target identities:

- GRAVITY-SPACETIME-WAVES from GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no omitted side of the speed interval
- no assertion that all extra modes are excluded at all frequencies
- no target-selected quadrupole law

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:ee5fb47223e46148fb6beb470dde3ff9a2d02996763de5b748980b97cf7916a4; engine receipt
 sha256:5617dfc8f9f7e7ddae1dd272e793d566071563db1e344afd340bf70b2c8ba493 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003-5617dfc8f9f7e7dd.json;
 empirical-validation hash

sha256:be2b4574a9b44f33f71c7b2704fd4b4bbe665e1e5a22418e827c47d98079f76c; measurement receipt
 sha256:1acb4baba3df38ef3d5313fb2fbe0d6628ea9fac7f60c448b1dc316f263b9c65.

314. Post-seal horizon and information-boundary comparison

Claim identity: SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. EHT independently reconstructs a persistent horizon-scale shadow and LIGO-Virgo observes compact strong-field ringdown. Neither body directly measures the event horizon, Hawking temperature, singularity avoidance or complete information reconstruction; those distinctions remain explicit.

Horizon-scale compact strong-field shadow and ringdown support are confirmed; direct horizon imaging, normalized thermal carrier, Hawking radiation and information reconstruction remain unmeasured predictions or formal boundaries.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal horizon and information comparison product. The declared exact boundary is: All registered EHT shadow, directness, LIGO strong-field and unmeasured Hawking/information rows. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-horizon-laws-versus-complete-EHT-LIGO-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-horizon-laws-versus-complete-EHT-LIGO-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ec0c31fbd11a54c788be250ab61969d397b4ba503be6cf081e86a5d201cd5bca`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:29258b931ec34e43228045f1db5c68d63c453b106deb41923301ba152a605ec5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bfb9f4f4aae6574ceaaedbe65e5de14f752a4e1e191ba61ea447247ed5078468`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 6c0340298be8d1400af8ca367d0ce0ecc81f553568034cfb076e55bacff931e5.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 852b4c711de58ac6519603d3c242e0d57f9ca68425fd65095d9b87d33f46c9d9.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256: 9fa4c0074e72a04672cfd6919cde925d4a626a6c6f17eaa8269bf002df996e23. Independent certificate: sha256: ebf753556858424cab0a30c3de00762beb533baf68ac69815b7c88d9ed3af770. Engine external-validation hash: sha256: 16a9bf046561df3e32f17a20c4f46ecf6d9dc3c8aa66b2d3ad7644b3b84e6855.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Observed comparison records:

- GRAVITY-SPACETIME-HORIZONS: predicted horizon-scale-shadow-and-compact-strong-field-support-confirmed__event-horizon-Hawking-temperature-and-information-reconstruction-not-directly-measured; observed horizon-scale-shadow-and-compact-strong-field-support-confirmed__event-horizon-Hawking-temperature-and-information-reconstruction-not-directly-measured; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: Independent horizon-scale observations cease to support compact shadow/ringdown structure, or an indirect/model-dependent record is mislabeled as direct Hawking, horizon or information-recovery measurement.

Measurement receipt: sha256: a15894aa0d01b667e5f1ff9750c1044d160fcd8bfabf67b97c1580fafce31968. Isolation certificate: sha256: 89f3bd384581161eb538c67d5b7381529a042d447bac9a627db884378f8c1e25. Custody certificate: sha256: 56db872814df4576dad6b3f205e74e2b9f695da3fbd15be09712d33a55ea7ac.

Registered source descriptions:

- NIST/JILA, CNES MICROSCOPE, LIGO-Virgo-KAGRA/GWOSC and Event Horizon Telescope:
<https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample>;
<https://cnes.fr/en/projects/microscope>; <https://dcc.ligo.org/P1700308/public>; <https://dcc.ligo.org/P2000091/public>;
<https://eventhorizontelescope.org/press-release-april-10-2019-astronomers-capture-first-image-black-hole>;
<https://gwosc.org/api/v2/catalogs> (sha256:b64325355bd993ab32cff6c088023dbe7911ab13278b9439466cd0ae56bedd5e)

Registered target identities:

- GRAVITY-SPACETIME-HORIZONS from GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no claim that EHT directly photographed the event horizon
- no empirical Hawking-radiation claim
- no claim that information reconstruction was observed
- no omitted model/reconstruction boundary

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:d7d94dd728088dbb0e16bfa6abd38d6b3c88b08dd0d3a7a11fa2ada4c943cf48; engine receipt
sha256:8279f3140235ca46ae66bbf4ed0fbf28c804d3bb42b43ecdb0d12ce45dc807b8 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003-8279f3140235ca46.json;
empirical-validation hash
sha256:b0c28bf902eb524babfda5092b927443e1baef638f8b6d77babd6ae45be348e6; measurement receipt
sha256:a15894aa0d01b667e5f1ff9750c1044d160fcd8bfbf67b97c1580fafce31968.

315. Post-seal nonstandard-spacetime realization boundary

Claim identity: SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The sealed wormhole, warp and closed-timelike results are admissibility theorems. The complete registered authority vector contains no confirmed physical realization of any of the three, and current catalog tests report no nonstandard strong-field signal in their stated analyses.

Wormhole, warp and exact closed-timelike admissibility boundaries are formally sealed; no authoritative realization appears in the registered source vector, so all remain unconfirmed physical frontier predictions.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003
- SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003
- SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal nonstandard-spacetime status product. The declared exact boundary is: Every formal admissibility result, registered authority search row, non-detection scope and explicit no-realization statement. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-admissibility-laws-versus-complete-authority-status-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-admissibility-laws-versus-complete-authority-status-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a2cc99c42495fb0f038700199d03ccb484192bfe4b0163512f8d51a4c1c1a71b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:dd60f9ca7ba68318cd1e3d20ddac505e14dcce896d8c98c561b2c1cc337cd140`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:efcc45c9309466ae7d5dd96f081b24c3641fdcf3cb04a6ce867fe12e915314b8`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:62a8b7f46e6309005d43d3e4e6adcd0c71218b0c14e2f847563b2e0323b044f0`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:12757404cd5714f353e09c4f931bbdae7170e9ec81dc147d5166f531341bc1d3`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:c0bbff7c5e0def998f350465d0fb0f4d3153c41fbd1381a4d711bc6e606c97ed`. Independent certificate: `sha256:62d11fef2c37c2f55f0e333b56d2ef6ef4a286d563ab5ec3fded3f436bfaaff3`. Engine external-validation hash: `sha256:66ca10bf4ef4be05b8cf7155a589318517105419b24db4cd364e26766fb80262`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Observed comparison records:

- GRAVITY-SPACETIME-NONSTANDARD: predicted formal-admissibility-boundaries-sealed__no-authoritative-wormhole-warp-or-closed-timelike-realization-in-registered-source-vector; observed formal-admissibility-boundaries-sealed__no-authoritative-wormhole-warp-or-closed-timelike-realization-in-registered-source-vector; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: A controlled authoritative experiment realizes one of the structures but violates its sealed positive-source, causal or complete-return ledger, or the validation promotes formal admissibility to observed existence without evidence.

Measurement receipt: sha256:a9546b367e2a18a2622a5841bafda64d09d031a716b9e24734bf0903016b2cdd.
 Isolation certificate: sha256:2be1fe3c637f13e89d8501786b2dc810f71aafd65805a549ff36ca4858817fa8.
 Custody certificate: sha256:58f7d46a26c3baa9a77338687687985dc4b03d284db6c0e8cffffe8eca4856596.

Registered source descriptions:

- NIST/JILA, CNES MICROSCOPE, LIGO-Virgo-KAGRA/GWOSC and Event Horizon Telescope:
<https://www.nist.gov/publications/resolving-gravitational-redshift-across-millimetre-scale-atomic-sample>;
<https://cnes.fr/en/projects/microscope>; <https://dcc.ligo.org/P1700308/public>; <https://dcc.ligo.org/P2000091/public>;
<https://eventhorizontelescope.org/press-release-april-10-2019-astronomers-capture-first-image-black-hole>;
<https://gwosc.org/api/v2/catalogs> (sha256:b64325355bd993ab32cff6c088023dbe7911ab13278b9439466cd0ae56bedd5e)

Registered target identities:

- GRAVITY-SPACETIME-NONSTANDARD from GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no inference from formal admissibility to existence
- no UAP or anecdotal evidence
- no unlimited no-go conclusion from a finite source search
- no deletion of the standing prediction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:fad156f4c6464a5287002819578ce1668d104b37f511c40c16db7476cf8c17b7; engine receipt
 sha256:5151d23acf5efc4266a74058442147b66ea3e2f926dd810d3578d417e255c883 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003-5151d23acf5efc42.json;
 empirical-validation hash
 sha256:0c0517c9aecadcd34d26405d2a1711817ca3e08f25174c7cf08d07d4420a5461; measurement receipt
 sha256:a9546b367e2a18a2622a5841bafda64d09d031a716b9e24734bf0903016b2cdd.

316. Quark dressing and CKM comparison

Claim identity: SFT-PHYS-VALIDATION-QUARK-CKM-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Exact polynomial enclosures and terminal dressing are tested against complete PDG quark and CKM intervals; the exact top/charm result remains a closed prediction because no direct scheme-matched measurement is registered, while the CKM s23 comparison is retained.

Dressed s/d and b/s overlap their complete PDG intervals; the sealed exact t/c value is a prediction awaiting a direct scheme-matched test; CKM s12, s13 and J overlap three-sigma support while s23 does not.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-QUARK-CUBICS-003
- SFT-PHYS-MATTER-QUARK-DRESSING-003
- SFT-PHYS-MATTER-CKM-PHYSICAL-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal quark dressing and ckm comparison comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavorable outcome in this comparison. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-formal-laws-versus-complete-matter-flavour-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:a2d625fe8c528e7c1ebf81042195c3434964b66ef13f1101e511d81125c8efaf`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:846dee47e1fba80298a7651c60c0829177b6c77cd558fb2ae358b05406c5each`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:da90a8405025cef7a23552e52aab7493542cbdf40d4cbc38f4ac025f83a6dcc`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3c7895fc1ab1c63204ecc1b82f0d8d07138378a80a187421533a067e654e0f7a`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8ea84eb8cce6629530492c992a373ae73825564f5396d58b54a4b8410a0b32d9`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:26a82e973fc68e78eedccbd187e44f0c6a90513b78e58396ca34acefae9fa805. Independent certificate: sha256:0d8d362cd5df041519bf9da5a5bb58227a23719e106e8afc573cf7302f20e579. Engine external-validation hash: sha256:d1c085124270c1c232cf92984c43218768bcd0324996db5b2bf7e4c01551e36d.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-QUARK-CKM: predicted dressed-sd-and-bs-inside-complete-PDG-intervals__tc-is-a-sealed-exact-prediction-with-no-registered-direct-scheme-matched-measurement__CKM-hierarchy-and-s12-s13-J-supported-with-s23-three-sigma-tension-retained; observed dressed-sd-and-bs-inside-complete-PDG-intervals__tc-is-a-sealed-exact-prediction-with-no-registered-direct-scheme-matched-measurement__CKM-hierarchy-and-s12-s13-J-supported-with-s23-three-sigma-tension-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: Any sealed passing ratio leaves its complete interval, the t/c prediction is weakened because no exact comparator exists, the s23 result is omitted, or source custody changes.

Measurement receipt: sha256:6316469fae9eeeeecc412867cf8d6a30f6dededc827002783cf95c32f94ef7fb8. Isolation certificate: sha256:dcd5b5b9193d0c3d46191ef445f0efa8146ee8bb81bc07d44ce5ca539dac88609. Custody certificate: sha256:66a0483d83f941e8528a812c31a449d0e3dc9ff1f6ac2ecfd0d573d0d8821160.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf>; <https://muon-g-2.fnal.gov/result2025.pdf>; <https://arxiv.org/abs/1807.06209> (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-QUARK-CKM from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no target access before seal
- no omission of CKM s23 tension
- no relabelling an unmeasured prediction as incomplete
- no cross-scheme top/charm comparison
- no central-value-only comparison

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:cd6572b28e067455cde9d87e98965e84605075da2091873566f521290cee32da; engine receipt

sha256:f9b53638bf2ac6dd4c0a294d9e10926ad8a0c00b64043500e0560845f257ff8e at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-QUARK-CKM-003-f9b53638bf2ac6dd.json; empirical-validation hash
sha256:609a035688d5e01dfa51f7bebf5dff8ac1aba8927ebe2680ba8a0b668d0be51b; measurement receipt
sha256:6316469fae9eeeeecc412867cf8d6a30f6dededc827002783cf95c32f94ef7fb8.

317. Proton/electron precision comparison

Claim identity: SFT-PHYS-VALIDATION-PROTON-ELECTRON-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The sealed algebraic proton/electron interval is compared with the complete CODATA value and uncertainty; the non-overlap is retained as an unfavorable result.

The V3 reconstruction is near the 1836 scale but lies outside the complete CODATA uncertainty interval and is not empirically closed to CODATA precision.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-MATTER-PROTON-ELECTRON-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal proton/electron precision comparison comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavorable outcome in this comparison. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-formal-laws-versus-complete-matter-flavour-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.

Axis	Forced form	Eliminated form	Elimination reason
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:f5f10ae0a4b69480370af8676b3e84a298197d2cb11b4a42e4e39d2f9356f071. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:6d0190e3d472ec90764d25a126a2f1bc87326b6d2e7f142601f3af753b1e838a.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:477e008d76e7cda64465017561e232f7c3e749fceb3c58baedbbb219028afd72.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:bae3bc9a2b81f87ca9366741449fce8cd3d8cafc7fbbdbd630adb71ad71e133ee.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:97a1f46a7a108f7b25ba1c6e20fb1bb996bbc639a7d314f3e8a9785f222d4f69.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:b10afaa17622b82df307aa9fc2bd74176fba10e786a4836faefcffe7bb7595ed1. Independent certificate: sha256:bc01921ec3b65a815ef82ca613e604250e945f4add782ecbb1af4b8fba613798. Engine external-validation hash: sha256:d17883fa4dedf92da68ca3e91c2bcef26283fe14acd5f32dc3afa39f64934a3b.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-PROTON-ELECTRON: predicted sealed-proton-electron-reconstruction-near-scale-but-outside-complete-CODATA-uncertainty__unfavorable-result-retained; observed sealed-proton-electron-reconstruction-near-scale-but-outside-complete-CODATA-uncertainty__unfavorable-result-retained; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: A lawful separately sealed successor overlaps the complete CODATA interval, or this validation falsely labels the current non-overlap as a pass.

Measurement receipt: sha256:91e37b94e9c6ac56cc8835eb9d34f72d076ce889b88c9e36bc249de212ede41f. Isolation certificate: sha256:fb70f03d742f82c68f86354bfb4e33a23b2d6ed3a1c1d4a471019b858a506d6. Custody certificate: sha256:2ccccf4d52dddcfa7f8475c9178038d93d331aa2cf9fb4e7674f59755d4d631f.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf>;

<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf>; <https://muon-g-2.fnal.gov/result2025.pdf>;
<https://arxiv.org/abs/1807.06209> (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-PROTON-ELECTRON from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no enlarged uncertainty
- no relative-error-only pass label
- no fitted successor
- no omitted unfavorable result

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:fa889edbff377ffbd3b92f2940ac4899925e9cec0fa41bf96a495a1693fccc1c; engine receipt
 sha256:4748126ffa9e95ad6473d568db39420d0252353b27f778ca67d3f3edc742a4a6 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-PROTON-ELECTRON-003-4748126ffa9e95ad.json;
 empirical-validation hash
 sha256:febfc972d601325a4b44284686c0bf60b4d83b06b8c4f41a3d37703760753d34; measurement receipt
 sha256:91e37b94e9c6ac56cc8835eb9d34f72d076ce889b88c9e36bc249de212ede41f.

318. Neutrino mass, ordering, PMNS and CP comparison

Claim identity: SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The positive mass-square coefficients, splitting law, PMNS triple and CP carrier are translated with one declared solar anchor and tested against complete PDG, KATRIN and Planck rows.

All PMNS/CP predictions lie in registered three-sigma support; the positive normal mass sum is about 0.060 eV and remains below current direct and cosmological bounds; normal ordering preference is not yet decisive.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NEUTRINO-SPLITTING-003
- SFT-PHYS-NEUTRINO-POSITIVE-MASS-003
- SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal neutrino mass, ordering, pmns and cp comparison comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavorable outcome in this comparison. The generator produced 256 named candidates and the decision support

contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-formal-laws-versus-complete-matter-flavour-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e84de0e74887bc96b3a8ae29ba1bb102ed51eff48cc4aa6e5250aa0cb7bd5254`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c92895443f7bc85e324ade7e44c9b40903008f7f367c172b150cdc7fcce8256f`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ae026dd7454b3249e6388a3646d93ef4f01e128df0434a45b0bee838f77de17f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9b9807b82f27cbcef3074acfd79b2b4b23876f08485cd79466fe5bb25c6463f3`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:02f5d3478ccdcda433027aade2e27368808215ea87b1b4e410329e9fc6bc580e`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b7a37f21b1d6bacff432eb0e3ad7a4869ad202257c8e3686a50ad140e568a96f`. Independent certificate:

`sha256:00ce7b894cbf5c8432010ccc66fec2431b33f18e4d31aec65e9b845217f854a7`. Engine external-validation hash:

`sha256:3919e9a70101f52aab5e61efd9c91347e38f5f03624c6fadfe60a216950d4754`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-NEUTRINO: predicted positive-normal-neutrino-mass-structure-PMNS-and-CP-inside-registered-three-sigma-support__mass-sum-below-direct-and-cosmological-bounds__ordering-preference-not-decisive; observed positive-normal-neutrino-mass-structure-PMNS-and-CP-inside-registered-three-sigma-support__mass-sum-below-direct-and-cosmological-bounds__ordering-preference-not-decisive; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: A PMNS/CP value leaves the registered support, the positive mass sum exceeds a robust bound, inverted ordering becomes decisive, or an uncertainty/scope row is altered.

Measurement receipt: sha256: df7aba217bf77a02ec9af53a1b59e3b0d57899ea3ec234f0bbddae8031a13f37.

Isolation certificate: sha256: 45f232ed7ea4e611821b2b49f787665b472e59f61b0cd3df7ac556f8e8318499.

Custody certificate: sha256: 4a8f544a7f6f049ae5cfac88425902d0e846c179e0dde69bfa9b9aadadb2e828.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf>; <https://muon-g-2.fnal.gov/result2025.pdf>;
<https://arxiv.org/abs/1807.06209> (sha256:6e56fccdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-NEUTRINO from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no numerical-zero lightest mass
- no claim of decisive measured ordering
- no hidden second scale anchor
- no omitted model dependence of cosmological bound

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256: 5cb1dbd8514f2638426cede2eda71ec7d3bca6ed13bee59d71a22d60c35efb9d; engine receipt

sha256: 1971175fec6fe5978abe8142346e71e4800ef6bb31aaa678cd2752149f08f10c at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003-1971175fec6fe597.json;

empirical-validation hash

sha256: 149522407a080710893da60ef2df60312cc4f6af08acd447cb93811985da1897; measurement receipt

sha256: df7aba217bf77a02ec9af53a1b59e3b0d57899ea3ec234f0bbddae8031a13f37.

319. Majorana and neutrinoless-double-beta status comparison

Claim identity: SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The formally closed Majorana discriminator and noncancellation prediction are tested against the complete current non-observation and effective-mass limits.

Zero-nu-beta-beta is not observed; the sealed positive effective-mass interval remains a closed prediction below current sensitivity and is neither confirmed nor excluded.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-NEUTRINO-MAJORANA-003
- SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal majorana and neutrinoless-double-beta status comparison comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavorable outcome in this comparison. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-formal-laws-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-formal-laws-versus-complete-matter-flavour-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9b885d1f835882f6f0f6d951cd1129b7efb0bc534413aa476c8e952b8c568899`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8d1bb5c25d0ff13d5fe019120e793686249afb2234daa9c816adc0a8d77e9efd.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:fb4766873e090d9a030659dcb6b1896147fc44baf27982def486dde01f768fef.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:2db951c6493b3b153515a51b84f651886421bc3fd102811451c410595a9bdbb67.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6ae51fa07fffb6e327f689afeda0f17daa2bd53ded2995570da32d38c77b7859.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:e8fe2066d6ded0fe0a91aafbdbc5dca334aaa786299e75be2a4e3b9c2bcb6f9. Independent certificate: sha256:355ba11fd0e20414f29473b4bd2afccbf2d4eb72e8de83561418b98e66af444f. Engine external-validation hash: sha256:f0656aa629a5f8db5924bcdd7ae8bef8e1b4317adeb8ec9e28c5044c7e61f6b.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-MAJORANA-ZERO-NU: predicted Majorana-and-zero-nu-beta-beta-formally-forced-and-closed-as-predictions__not-yet-observed__predicted-effective-mass-below-current-experimental-upper-bound; observed Majorana-and-zero-nu-beta-beta-formally-forced-and-closed-as-predictions__not-yet-observed__predicted-effective-mass-below-current-experimental-upper-bound; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: A controlled search excludes the complete sealed effective-mass interval under validated matrix elements, a Dirac neutrino is established, or an observation violates the sealed structure.

Measurement receipt: sha256:d5ca10085433326e6c4c491a7daf2ee1e3a56cf41b6d8191e6fa49ef60aed782. Isolation certificate: sha256:b24717a8f64c0ea2642962bb612df1ab8db1250701556f3ae6df53ee75a3aeb3. Custody certificate: sha256:809820a2744c362be1dea101f1ebc3522ad282fb1e7f73bdfefeb8a8dd1333e426.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf>; <https://muon-g-2.fnal.gov/result2025.pdf>;
<https://arxiv.org/abs/1807.06209> (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-MAJORANA-ZERO-NU from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no claim of existing observation
- no conversion of non-observation to disproof below sensitivity
- no omitted nuclear-matrix uncertainty
- no erased Majorana prediction

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:60e4265b532387ed0aeb9221f012dfaed68052b939c5de9d1e5ad264b570c076; engine receipt
sha256:028482eb80db7fb39fc3bc19b98ce247ab28fcd8121b80f1e4f6fab4a348d7d6 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003-028482eb80db7fb3.json;
empirical-validation hash
sha256:f361adf2d3faab0214bee4a388a54efea7704a7fc31d0c4babc58779e45c0983; measurement receipt
sha256:d5ca1008543326e6c4c491a7daf2ee1e3a56cf41b6d8191e6fa49ef60aed782.

320. Electron and muon magnetic-anomaly comparison

Claim identity: SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The bare g, terminal-alpha leading structure and squared mass sensitivity are compared only with the raw NIST electron data and Fermilab final muon measurement; no consensus-model interpretation enters the validation path.

The currently sealed leading terminal-alpha/full-turn translation is compared with both raw measured anomalies; the raw measurements remain the sole empirical targets while a separately forced successor is constructed.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal electron and muon magnetic-anomaly comparison comparison product. The declared exact boundary is: Every registered source row, uncertainty, scheme/status limitation and unfavorable outcome in this comparison. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-formal-law-s-versus-complete-matter-flavour-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	sealed-formal-laws-versus-complete-matter-flavour-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d2d590271fff37a0ae3eefbf6f034cfc35cd6c4eac3a88d78801f82eebd868b`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:27242c74ceae811d2b7eb09aff8d37f8a58beae1d832c821447a43ba3a2f1081`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:860159bdbbb5be8f18d19c0d3c3a854df65124abccd24a164748ae6ef8b3556d`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4e91c5ba807d898c0a7a9c41d0382f2e699418f8ec5debe349aa5bccb829d269`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:053abe2406ecb945558cd762f4f534a607dd1e9c6c9d93388faba2d8a6eb082d`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e5d03fba565167b06734b4c16c0094ae1dd9597e9883fc7315ea8ef5e14d53fe`. Independent certificate: `sha256:20684d12718d544efba49f0b5581e4f74d5fcb7e4b492a7886e95d72e32dc149`. Engine external-validation hash: `sha256:db96e297634e9e1458a88895d5c0bf9f4682815bbf61a1ea726782384881c1e0`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Observed comparison records:

- MATTER-FLAVOUR-MAGNETIC: predicted bare-g-and-terminal-alpha-leading-structure-compared-only-with-raw-electron-and-muon-measurements__consensus-model-interpretation-excluded; observed bare-g-and-terminal-alpha-leading-structure-compared-only-with-raw-electron-and-muon-measurements__consensus-model-interpretation-excluded; exact match `True`

- deliberately tampered unfavorable control rejected

Falsification condition: A complete separately sealed SFT radiative successor fails the measured intervals, or this validation mislabels the current leading-only result as the full anomaly.

Measurement receipt: sha256:1df0cdf663368b440cb6d8120c7ed3c4e6fb44a86254e851cdfc9fbc0915733d.

Isolation certificate: sha256:4664657f2d3849caf32ca5549db8e63e0bddb0ed8aec201770bd4a5580c04488.

Custody certificate: sha256:88bb2e000cfda6cb7f55a076376a78995f1c10adfe0e154805c7d3adbcf5270d.

Registered source descriptions:

- NIST/CODATA, Particle Data Group, Fermilab Muon g-2 Collaboration and Planck Collaboration:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>; <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf>;
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf>; <https://muon-g-2.fnal.gov/result2025.pdf>;
<https://arxiv.org/abs/1807.06209> (sha256:6e56fcdd94371ebf396b9600a1431fd448075f69073572b9cb0660959329b9fe)

Registered target identities:

- MATTER-FLAVOUR-MAGNETIC from MATTER-FLAVOUR-AUTHORITATIVE-2022-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no imported QED series
- no irrational proof value
- no exact-measured-anomaly claim from a leading term
- no consensus-model result as evidence
- no measurement-selected survivor

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d54d0adc55b307d7e96e8a2c573ddf0bf758ad5a7e2d615ffec234408a55fd9; engine receipt

sha256:e6065612de8e491dff5272b714302a4e85359e5e5d17afb43c08accdeb5e9b3f at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003-e6065612de8e491d.json;
 empirical-validation hash

sha256:d2e6637abbf732a0cd8a16ca3f0834aee2fb114b4d3bc4f194cc4bd1e72b1c96; measurement receipt

sha256:1df0cdf663368b440cb6d8120c7ed3c4e6fb44a86254e851cdfc9fbc0915733d.

321. Post-seal NIST comparison of the forced six-neighbour cubic support

Claim identity: SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After the exact Fold law independently forces all two held directions on all three generated spatial axes, the sealed coordination count six is compared with the complete registered NIST simple-cubic row.

*The sealed Fold count $2*3=6$ exactly equals the complete NIST simple-cubic nearest-neighbour count six.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal exact-count, provenance, custody, row-retention and no-extra-rule product. The declared exact boundary is: All exact comparisons between the immutable six-neighbour Fold coordination and the sole complete registered NIST simple-cubic coordination row. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-six-neighbour-count-versus-complete-nist-row__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-six-neighbour-count-versus-complete-nist-row	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:044bb9d1be996d54b72db5fe3d9774ab3656e24de945edbd712382ef4453b65c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ff7b9a5b78ee1aaf141fcf80b1d1f1b307914a8907497aa715965e3e9c460a1c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3d80082d15c527741aa56150b21d49ca736a087bd55730012d66ed3f3d624d7c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f4620a5cb5a3e690961b955ade1e2e4a143f7902c0bf9964160b599364b2b636`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:51a7e2f92614bd6de1faefce57551ac589fe2d21d98f3604b2494c301e380b15.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:997bf380310d08236df0976f30b8115dcd01bbcdc9efe4c17c821651f3a650dd.

Independent certificate:

sha256:fd3e9a21c669f14d0e5cc2c5d2ae01c1e70e3664ab42644dac4fd475c2a11fcf. Engine

external-validation hash:

sha256:b5d13388db24ef7506aaf17e5b8a05ffdeb34286040667fd80adceb2ab58da8c.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-NCNR-DCS-SUPERFLUID-HELIUM

Observed comparison records:

- NIST-SIMPLE-CUBIC-NEAREST-NEIGHBOURS: predicted
sealed-six-neighbour-cubic-support-exactly-matches-complete-nist-row; observed
sealed-six-neighbour-cubic-support-exactly-matches-complete-nist-row; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: The registered NIST row is not six, the formal count changes, any source hash changes, target access precedes sealing, or a tampered row is accepted.

Measurement receipt: sha256:fd1f07c8d8d931c7b817d9b78ea6dbf9d28f8a7835713ff1ae43f9be118b0578.

Isolation certificate: sha256:d1129424a33baeb639d13bd34d29406e2bc31e2b12009029421ee6fae784513b.

Custody certificate: sha256:5732827ed1f1b7b7e09a59e62a856d709038e4e9d74d3819f06878e33966fec3.

Registered source descriptions:

- National Institute of Standards and Technology, NIST Center for Neutron Research: <https://www.nist.gov/document/dcs>
(sha256:ab723f91d6658bd8b5d19889884fd87e07046d2b0060e23f7df8d7cb6c35fa5b)

Registered target identities:

- NIST-SIMPLE-CUBIC-NEAREST-NEIGHBOURS from NIST-NCNR-DCS-SUPERFLUID-HELIUM

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no NIST value accessible to the formal law
- no imported lattice premise
- no selected neighbour subset
- no fitted count or extra rule

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:3537e0b151d2ec9e090298d9a07e7c03cd41d0fd890b564a8b4b1be61dddbce8; engine receipt

sha256:2d3a137022da53383895f1e43bdfd78a6ce04912b408c76c4a4d0bcef569ab61 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004-2d3a137022da5338.json;

empirical-validation hash

sha256:d19984f3e335290e5de23709947563f30e1befeb68bc478987fa8dfc281c3ebc; measurement receipt
sha256:fd1f07c8d8d931c7b817d9b78ea6dbf9d28f8a7835713ff1ae43f9be118b0578.

322. Post-seal NIST comparison of the exact hydrogen spectral ladder

Claim identity: SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The NIST hydrogen ionization wavenumber is retained solely as the dimensionful input carrier. The already sealed exact Fold gaps three-quarters and five-thirty-sixths are applied without adjustment, then compared with the complete NIST Lyman-alpha and Balmer-alpha measured intervals.

The sealed ratios predict Lyman-alpha 82259.078775 cm⁻¹ and Balmer-alpha 1096787717/72000 cm⁻¹; these lie respectively inside the complete NIST intervals [82259.02, 82259.30] and [15233.14, 15233.28] cm⁻¹.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal carrier-times-gap, exact-interval, provenance, custody, complete-row and no-extra-rule product. The declared exact boundary is: All exact post-seal applications of the immutable Fold Lyman-alpha and Balmer-alpha ratios to the one registered NIST hydrogen ionization carrier, compared with both complete reported line intervals. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-hydrogen-gaps-versus-complete-nist-line-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-hydrogen-gaps-versus-complete-nist-line-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.

Axis	Forced form	Eliminated form	Elimination reason
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:f2204f3d68b09b4480ae84bda3b465a6f903941f78d956c0e2ecfa1ead0c2f59. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:b558c1ca9f5a44eefe936b86328aa16428f847084bb5e7d8c9828c00e87ab248.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:45512783de9d059b64088e3a3ac4621d0453d407499f05f050a34f9020778bbd.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:0f43226e7860b8fd91f5aa133fb59a05b21f227242c7b10cd6630269c7374fa6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:dbfe52d93dd3ac709f8cf3f48918fd95f4423e3940f9fa09334ff99c4fb584e6.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:929cb21c17fc7f62f089d25a2cc709c3bf95d82dd73f4287a83a46d5a75f44b0. Independent certificate:

sha256:b2b10cbd51128d459628d3701b6ad4e44d74fe307130d5f01aaf08870c0a0d0a. Engine external-validation hash:

sha256:59633cd3caabd9205d848cc953b2e6c0db6d6a3977dae33b03c06c7609278976.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-KRAMIDA-HYDROGEN-CRITICAL-COMPILATION-2010-2019

Observed comparison records:

- NIST-H-I-LYMAN-ALPHA-INTERVAL: predicted sealed-hydrogen-gross-ratios-lie-inside-both-complete-nist-line-intervals; observed sealed-hydrogen-gross-ratios-lie-inside-both-complete-nist-line-intervals; exact match `True`
- NIST-H-I-BALMER-ALPHA-INTERVAL: predicted sealed-hydrogen-gross-ratios-lie-inside-both-complete-nist-line-intervals; observed sealed-hydrogen-gross-ratios-lie-inside-both-complete-nist-line-intervals; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification condition: Either exact prediction lies outside its complete NIST interval, a row or uncertainty is omitted, a source hash changes, target access precedes sealing, or a tampered row is accepted.

Measurement receipt: sha256:420bc255caf0fe1ce3073b43c85250e0c00425f80ad03071f2acb5b7f128a097.

Isolation certificate: sha256:6aebac42bbf5202ec170054d7ab0471da11584f7d77fd3f2d0b2015823a01093.

Custody certificate: sha256:d09e28f295d7556277b44445ac842f206f16ff6e8d9182a37e800562e61dcc84.

Registered source descriptions:

- National Institute of Standards and Technology, Atomic Spectra Database:
<https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable1.htm>
(sha256:cc54c774518d62cbb7e95f17b3b7fcd9d1faf0aa90043a607b91dff0a43e2087)
- National Institute of Standards and Technology, Atomic Physics Division:
https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=842564
(sha256:86ccaeb875fcd4445e727618509621ceee656e5dc02295140ecdb6e1d2dd443)

Registered target identities:

- NIST-H-I-LYMAN-ALPHA-INTERVAL from NIST-KRAMIDA-HYDROGEN-CRITICAL-COMPILATION-2010-2019
- NIST-H-I-BALMER-ALPHA-INTERVAL from NIST-KRAMIDA-HYDROGEN-CRITICAL-COMPILATION-2010-2019

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no line target accessible to the formal law
- no adjusted spectral ratio
- no floating comparison
- no omitted line, uncertainty or adverse row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:1c203ba48a0913a643b9e5f9e170a5188824f9e381e5aed905df73174997f2f5; engine receipt
sha256:ac7a0e6e2bac1372b978f599799af40cebb9660b655f6b67978c798256204285 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004-ac7a0e6e2bac1372.json
; empirical-validation hash
sha256:32332cd52f1c5e67266580dae73d1302fab89f0c0297f717654b3ae53273154; measurement receipt
sha256:420bc255caf0fe1ce3073b43c85250e0c00425f80ad03071f2acb5b7f128a097.

323. Post-seal Fizeau discriminator for Fold velocity composition

Claim identity: SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The measured water slope is $274/1000 \pm 3/1000$ rad s/m. It lies inside the complete reported relativistic-systematics bracket $248/1000$ to $299/1000$ and is more than ten times closer to its nearest relativistic row than to ordinary addition at $563/1000$. The air result independently rejects ordinary addition and is compatible with relativistic composition. This is a post-seal discriminator, not a fit or precision confirmation; all source-stated systematics remain open.

The measured water slope is $274/1000 \pm 3/1000$ rad s/m. It lies inside the complete reported relativistic-systematics bracket $248/1000$ to $299/1000$ and is more than ten times closer to its nearest relativistic row than to ordinary addition at $563/1000$. The air result independently rejects ordinary addition and is compatible with relativistic composition. This is a post-seal discriminator, not a fit or precision confirmation; all source-stated systematics remain open.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed operation, Fizeau discriminator, provenance, target isolation, measurement separation, complete row/limitation retention, successor closure and extension. The declared exact boundary is: The admitted exact positive velocity operation; the complete registered water measured/uncertainty, raw relativistic, corrected relativistic and ordinary-addition rows; the air result; and every stated systematic limitation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-velocity-composition-with-complete-Fizeau-discriminator__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-velocity-composition-with-complete-Fizeau-discriminator	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:2c95368e6cb55afa4f0ff970e6a336df3f2aefbb2eb3d5d84c437e5167934a41`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8009bd07833670e510f10387289b297f705cccb3914d77a5950657cfd4f6b395`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:27812259a34558ed1713aa01e228303994b6a3412740a08d998686ed6e1f7aaa`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:248e0a15eade5b255ce84bb354d5fa31d1189398dcc0efd2df92e545791c45ac`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 83549ee13670779240a1e7473688fc47758bbaa3df875f9bcdcb293b1b96f240.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : b9a10021cdd87914bfa0c1b2056460b07cbc35c10e4a923fa9dcd60d6cf6c784.

Independent certificate:

sha256 : 37966620627fdb37ededaf13405fbc189d23ffa27d2b5bb93af1c75d1b52efe. Engine

external-validation hash:

sha256 : 45aff6a43141ce2fe66d8b31ed63704e133c8eb97f065e5513f569319e5297e0.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- ARXIV-1201.0501-FIZEAU-WATER-AIR

Observed comparison records:

- The exact Fold velocity operation was admitted before the Fizeau source was retrieved.
- The water slope is retained as 274/1000 +/- 3/1000 rad s/m.
- The raw and corrected relativistic rows 248/1000 and 299/1000 bracket the measured interval.
- Ordinary addition predicts 563/1000; its distance 289/1000 exceeds the nearest relativistic distance 25/1000 by more than tenfold.
- The air row rejects ordinary addition and is compatible with relativistic composition.
- Flow calibration, turbulent-profile and path-length systematics are retained; no precision overclaim is made.
- A tampered ordinary-addition row equal to the measured central value destroys the discriminator and is rejected.

Falsification condition: Reject if either source identity changes; if a registered measurement, control or limitation is omitted; if the water measured interval leaves the complete reported relativistic-systematics bracket; if ordinary addition is at least as close as both relativistic rows; if the air control no longer disfavors ordinary addition; if any target changes the sealed operation; or if a precision confirmation is claimed beyond the source.

Measurement receipt: sha256 : 9b184c06fd6dd000723fe92c1f69d7a1f7b48afd4b0d9e3d32e88d90e567e0be.

Isolation certificate: sha256 : 8af63e373e75f8c1e898e37db3f6b796cb871e2db1c44176b53fea0e14c6820c.

Custody certificate: sha256 : 901464dd7418ef296ff80c3dee2c1d39c45e51c03c9fe8ea257d20cb2961182a.

Registered source descriptions:

- Fizeau's aether-drag experiment in the undergraduate laboratory: <https://arxiv.org/pdf/1201.0501>
(sha256:8c19ceecda50a3c297aa13c6006a6cf032d41a3c714f128b2be663bd3c5b5b3)

Registered target identities:

- ARXIV-1201.0501-WITHHELD-COMPLETE-FIZEAU-RECORD from ARXIV-1201.0501-FIZEAU-WATER-AIR

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no Fizeau slope, refractive record or external velocity formula in the formal predecessor derivation
- no fit of a Fold coefficient to the water or air record
- no omission of the ordinary-addition control or source-stated systematic limitations
- no precision interval claim where the source reports an eight-percent systematic-level agreement

- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude
- no target access before the formal prediction seal

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:70a12111a120963928d953bec47666c73308724f2eb2ec93275f27b25d5e2c8d; engine receipt
 sha256:54dce92f903c6f102e4d39f3bdb826a89c05e69e80b6d6df66728711dbc03cf4 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034-54dce92f903c6f10.json;
 empirical-validation hash
 sha256:932b06b5da3476726d83ce723657ae4e1f0cfb8b57a70670279a7fa99f7d23b9; measurement receipt
 sha256:9b184c06fd6dd000723fe92c1f69d7a1f7b48afd4b0d9e3d32e88d90e567e0be.

324. Post-seal Planck/CODATA test of the Fold vacuum-density scale law

Claim identity: SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact Fold vacuum share 11/16 is inside Planck's interval [6833/10000,1389/2000]. The normalized Fold magnitude 33/16 is inside [20499/10000,4167/2000]=3 times that interval. Using page-225 $H_0=67.68 \pm 0.42 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and exact $c=299792.458 \text{ km s}^{-1}$ transports Lambda to the exact positive interval $[(33/16)(67.26/c)^2, (33/16)(68.10/c)^2] \text{ Mpc}^{-2}$. $\text{One}/2^{20}$ is below the global interval and is therefore retained only as the separately typed local boundary-energy floor.

The exact Fold vacuum share 11/16 is inside Planck's interval [6833/10000,1389/2000]. The normalized Fold magnitude 33/16 is inside [20499/10000,4167/2000]=3 times that interval. Using page-225 $H_0=67.68 \pm 0.42 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and exact $c=299792.458 \text{ km s}^{-1}$ transports Lambda to the exact positive interval $[(33/16)(67.26/c)^2, (33/16)(68.10/c)^2] \text{ Mpc}^{-2}$. $\text{One}/2^{20}$ is below the global interval and is therefore retained only as the separately typed local boundary-energy floor.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed vacuum relation, Planck/CODATA target, source provenance, capability-closed isolation, proof/measurement separation, complete row and correction retention, successor closure and extension. The declared exact boundary is: The admitted exact local floor, finite mode ledger, 11/16 global share and 33/16 normalized magnitude; Planck table 12.16 page-225 H_0 , Omega_Lambda and Omega_m rows with complete uncertainties; CODATA exact c; the older H_0 transcription correction; and the local/global adverse type control. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-vacuum-density-law-versus-complete-Planck-CODATA-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-vacuum-density-law-versus-complete-Planck-CODATA-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:aac1f5afbedc98835000ee0959231e7e6213eae90e23f1502c48346abd816225`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:80323984b7f3e4556b6a437274b69af93eb908accc7828348411dbf280b6c821`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a0c6725ee5796923141b8a1ed45d77de545e582264fadb7dbe5bf403ef5f1af1`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a89ecd40b7631fe123e2b57bebfdd301066da0cde5dd1abc2e4291838ff98d7`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e0414394833e40ba64335f2decf3b46c98cba1a7dfb295cd38cbf1b095c23f0a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:9b4f66424c7fa3b2ffc7943492d8d556da80dcd1b280405db5a9fb1cb0f5520b`. Independent certificate: `sha256:b8101dfd5ee3a439f81be03bd7b4f8a8c88e3c3cc125901e60a32e5bd8113404`. Engine external-validation hash: `sha256:d8ddaf48a77d46203c68ba9687c9b06a2bc92bee78d687688e73cd0fa83b6cab`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PLANCK-PLA-BASE-NRUN-TTTEEE-BAO-LENSING-68PC
- NIST-CODATA-2022-EXACT-C

Observed comparison records:

- The formal vacuum-density law was admitted before the primary Planck PDF was retrieved.
- Historical V1/V2 targets and an older local transcription were known; no historical-blindness claim is made.
- The exact 11/16 vacuum share lies inside the complete Planck 0.6889 +/- 0.0056 interval.
- The exact 33/16 normalized magnitude lies inside the three-direction transported Planck interval.
- Page 225 directly reports $H_0=67.68 \pm 0.42$; the older 67.66 transcription is corrected here without rewriting prior receipts.
- CODATA exact c transports the sealed coefficient to a positive exact inverse-square-megaparsec interval.
- $\text{One}/2^{20}$ and the finite-ledger mean remain outside the global fraction interval, preserving their distinct types.
- A tampered vacuum central value of 0.6000 rejects the registered comparison.

Falsification condition: Reject if any source identity or registered row changes; if 11/16 leaves the complete Planck vacuum interval; if 33/16 leaves its exact three-direction transported interval; if the Planck central budget ceases to close at the One; if dimensional transport is nonpositive or scale-inconsistent; if $\text{One}/2^{20}$ is relabeled as the global fraction; if the older H_0 transcription is silently substituted; or if any target changes the formal survivor.

Measurement receipt: sha256:0bb3659038316226f202044b500f6bf1a6867720089a91f659b908e9fe356cf4.
Isolation certificate: sha256:f285d0debc6ce9b8d4818d3543c820f822146affc4724edbebe6681167e8fada.
Custody certificate: sha256:e87a3b3313685c954c8c60820553b833343ee076775dec0952b8d5d4b43af12e.

Registered source descriptions:

- Planck 2018 baseline parameter tables, 68 percent limits, table 12.16:
https://wiki.cosmos.esa.int/planck-legacy-archive/images/4/43/Baseline_params_table_2018_68pc_v2.pdf
(sha256:03038805021f2f894e09f4b21b0f20418570e352822f095abcc085942919da70)
- 2022 CODATA recommended values - complete ASCII table: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- PLANCK-CODATA-WITHHELD-COMPLETE-VACUUM-SCALE-RECORD from None

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no Planck or CODATA value readable by the formal candidate generator or independent formal validator
- no fit, recalibration or coefficient choice from Ω_Λ or H_0
- no claim of historical blindness: V1/V2 targets and an older local transcription were known
- no silent reuse of the older 67.66 H_0 transcription where primary page 225 reports 67.68
- no direct equality between $\text{One}/2^{20}$, a raw radiative sum, Ω_Λ and dimensional Λ
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:f13f4bb53dc4e55f6ffa2503fba63bdc74c8024e6692b57e3f30269ee5c18485; engine receipt
sha256:6379f0d35a606d217e2351a35c42ae258aea71c52d8d20d2c4abeef2fdd8c202 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036-6379f0d35a606d21.json;
empirical-validation hash
sha256:7695b43af80dd5ba9987cee7e0375f9f91c9dcd81fde97020402fd544559998a; measurement receipt
sha256:0bb3659038316226f202044b500f6bf1a6867720089a91f659b908e9fe356cf4.

325. Post-seal thermal-history, nucleosynthesis and recombination test

Claim identity: SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The direct high-redshift CMB temperature exponent interval [49/50,517/500] contains exact One. The external freeze-out sequence is approximately 1/6 then 1/7, matching the separately forced typed transport. Direct helium $Y_p=0.2458\pm0.0013$ gives [489/2000,2471/10000], which excludes 1/4; therefore 1/4 remains the exact analytic helium-family partition and is not relabelled as the exact isotope measurement. Direct primordial deuterium (2.527 $\pm$ 0.030) times 10⁻⁵ is positive and subordinate. Planck retains $z^*=1089.92\pm0.25$, eighteen finite extrema and seven complete TT peak rows. Those angular peak intervals are not integer multiples of the first, so the internal-whole-mode/observed-projection distinction passes and the old angular-integer reading is rejected.

The direct high-redshift CMB temperature exponent interval [49/50,517/500] contains exact One. The external freeze-out sequence is approximately 1/6 then 1/7, matching the separately forced typed transport. Direct helium $Y_p=0.2458\pm0.0013$ gives [489/2000,2471/10000], which excludes 1/4; therefore 1/4 remains the exact analytic helium-family partition and is not relabelled as the exact isotope measurement. Direct primordial deuterium (2.527 $\pm$ 0.030) times 10⁻⁵ is positive and subordinate. Planck retains $z^=1089.92\pm0.25$, eighteen finite extrema and seven complete TT peak rows. Those angular peak intervals are not integer multiples of the first, so the internal-whole-mode/observed-projection distinction passes and the old angular-integer reading is rejected.*

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed thermal relation, complete external target, source provenance, capability-closed isolation, proof/measurement separation, complete row and adverse-result retention, successor closure and extension. The declared exact boundary is: The admitted temperature exponent One, typed freeze-out/capture sequence, analytic quarter family partition, positive minor channels, finite visibility and internal acoustic parity; every registered uncertainty endpoint; all seven Planck TT peak positions and amplitudes; the eighteen-peak mission count; and explicit rejection of exact observed quarter equality and exact observed angular integer multiples. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-thermal-history-law-versus-complete-primary-observation-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-thermal-history-law-versus-complete-primary-observation-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated form	Elimination reason
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:15bb75cf8556908a2498c9611ae022d5f26c7f8869bf195bfd51e24f992fbd79`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:da3b69464f65ec92f9d718e382f646416b99f6ad2d22ed3400b7422807ee7c40`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9385edfe317a9f525e0d985f2ba5be0917d5ee3c28190101f8ab0bdf91cb537e`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:82bdaa2907c33acaf5d625f07b19d93588963bcddf3179924cb10519b0d151d7`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a36854cd806c012a590df961b03c57faa3bd44f63ba8f520013c518f424b5091`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:17bd84328167f26310571135007e60049fe99b927637d4416853c2604908960b`. Independent certificate: `sha256:35623ce9812112afaca02c9a38799dca2e331cd1a284756a2bbcdac0dc592516`. Engine external-validation hash: `sha256:437b0ab034ac12219f463c53f2435c820096d84aefea57fa07f05491ef30d126`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NOTERDAEME-ET-AL-2011-CMB-TEMPERATURE
- AVER-ET-AL-2026-PRIMORDIAL-HELIUM
- COOKE-PETTINI-STEIDEL-2018-DEUTERIUM
- PLANCK-2018-OVERVIEW-PEAKS
- PLANCK-2018-COSMOLOGICAL-PARAMETERS
- PDG-2025-BBN-REVIEW

Observed comparison records:

- The formal thermal-history law was admitted before the new primary snapshots were retrieved.
- Direct high-redshift CMB thermometry contains exact exponent One.

- The external freeze-out sequence is approximately One/six then One/seven.
- Current direct helium measurement excludes exact One/four; the old exact measured-equality reading is rejected without fitting.
- Direct primordial deuterium remains a positive minor light-nuclide channel.
- Planck retains finite recombination support, eighteen extrema and all seven TT peak rows.
- Observed angular multipoles reject exact integer multiples of the first peak, preserving the internal-mode/projection distinction.
- A tampered temperature exponent rejects the comparison.

Falsification condition: Reject if any source identity or registered row changes; if exact One leaves the temperature-exponent interval; if the typed One/six to One/seven sequence is absent; if the unfavorable helium exclusion is hidden or used to fit a correction; if deuterium is nonpositive; if recombination support or the acoustic census is erased; if observed peak intervals are falsely relabelled as exact integer multiples; or if any target changes the formal survivor.

Measurement receipt: sha256:f5710146b77a40129a6f52186fa2622ec68b7bc3ddc765685c6532dad769f78.

Isolation certificate: sha256:5823d5b59a85e6b38a07af5cf433d1913c23d7e8ec5c7aa56de869e6f33663d9.

Custody certificate: sha256:d8fcf914ad1ad8db6fae9575f19e0508a9018033aac51294ff1bec29494dd4e9.

Registered source descriptions:

- The evolution of the Cosmic Microwave Background Temperature: <https://arxiv.org/abs/1012.3164>
(sha256:aa79c65170e84d6a8dc71dbd0876538c19619a1c926453ce5a54bd96d3f1efb7)
- The LBT Yp Project IV: A New Value of the Primordial Helium Abundance: <https://arxiv.org/abs/2601.22238>
(sha256:72f646a345c36b77211e6035d3a78345498f0766cf9fc0a54932ebe0dd670c6a)
- One percent determination of the primordial deuterium abundance: <https://arxiv.org/abs/1710.11129>
(sha256:1a6ec19d6568b8854a40e9dd587906f99a8ea94ecd2d5c712ad86aa506b93f4a)
- Planck 2018 results. I. Overview and the cosmological legacy of Planck: <https://arxiv.org/abs/1807.06205>
(sha256:dca932893b7d2724aa2b8d33170fc1b5682c425dd56fcd3c5d7b2098d377db5c)
- Planck 2018 results. VI. Cosmological parameters: <https://arxiv.org/abs/1807.06209>
(sha256:8e172730faf07c9f4ff3fdcc7043f76ed67df6f76066d47df30d693025b6ce77)
- Big-Bang Nucleosynthesis: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-nucleosynthesis.pdf>
(sha256:1843c1d9025aa33c059700708f1041c4462364cdb582325f0c685a6ba1b38484)

Registered target identities:

- THERMAL-HISTORY-WITHHELD-COMPLETE-OBSERVATION-RECORD from None

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no external value readable by the formal candidate generator or formal independent validator
- no fitted cooling exponent, abundance correction, reaction rate, visibility width or angular projection
- no historical-blindness claim because V1/V2 named these targets
- no omission of the current direct helium interval that excludes exact One/four
- no preservation of the old exact observed-quarter or angular-integer claims by weakened comparison
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:f3401b72677923a0ba81abb3f9790bc49b6edf2e2b0a2711b4d6ad51a0d5fe8d; engine receipt

sha256:7440c78415e834656217aca301aaa8b6f987d6f795544eb5c03b0679970321f4 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038-7440c78415e83465.json; empirical-validation hash
sha256:d07b6ae3dcbfb2690cf52cb27769c24b2533d40955a2f338cd8f03a30b622413; measurement receipt
sha256:f5710146b77a40129a6f52186fa2622ec68b7bc3ddc765685c6532dadcf769f78.

326. Post-seal scalar-support and tensor-support measurement test

Claim identity: SFT-PHYS-VALIDATION-INFLATION-GROWTH-040

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Planck $n_s=0.9649\pm0.0042$ gives exact interval $[9607/10000,9691/10000]$; the independently forced 31/32 equals 0.96875 and lies inside that interval. BICEP/Keck plus Planck reports $r<0.032=4/125$ at 95 percent; the independently forced $1/32=0.03125$ lies strictly below the bound by $3/4000=0.00075$. The complete exact 31/32+1/32 partition therefore passes both external rows without fitting. Five Fold doublings is not relabelled as a conventional natural-log e-fold count, for which this claim asserts no measured equality.

Planck $n_s=0.9649\pm0.0042$ gives exact interval $[9607/10000,9691/10000]$; the independently forced 31/32 equals 0.96875 and lies inside that interval. BICEP/Keck plus Planck reports $r<0.032=4/125$ at 95 percent; the independently forced $1/32=0.03125$ lies strictly below the bound by $3/4000=0.00075$. The complete exact 31/32+1/32 partition therefore passes both external rows without fitting. Five Fold doublings is not relabelled as a conventional natural-log e-fold count, for which this claim asserts no measured equality.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-INFLATION-GROWTH-TERMINAL-039
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed inflation relation, complete external target, source provenance, capability-closed isolation, proof/measurement separation, complete rows, successor closure and extension. The declared exact boundary is: The admitted scalar support 31/32 and tensor support 1/32; both endpoints of the Planck one-standard scalar interval; the strict BICEP/Planck tensor upper bound; every bound source and custody row; and explicit separation of exact Fold doubling depth from conventional model-dependent e-fold language. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-inflation-growth-law-versus-complete-scalar-index-and-tensor-bound-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-inflation-growth-law-versus-complete-scalar-index-and-tensor-bound-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:ac9c38c10ba15714347560effb54dce239f5ef3d293433fecf50442e1cedd5c5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f2a877f8ea086325067023690358917ac7059f44eb87fa1bc8ae5c211e77e629`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a639582b262acfd4204bcd27e3e4eb9f404e56b99a711698d85e32e9feeb3b8`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:10416829131668b2e9a5f08af885fae861009dc76a59ff081e255215b712fb26`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2c07eba135d75e0a47a23c88d003a12e3374cc1c163cf8d44a8e065ad5a45657`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:00278934fa11f29736892bf5adbd1c82c2cbd6305f9f47ec5b1dcc352a27b38a`. Independent certificate: `sha256:3cae59937dfb3791d3ee2c24c32584194b9fa10a1301bd26e6d73fc00e1862e3`. Engine external-validation hash: `sha256:7a0c0dfaa703412fe6099687c7a2e15459282635cb36047b2e3f79061620f5ee`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PLANCK-2018-INFLATION
- BICEP-PLANCK-2022-TENSOR

Observed comparison records:

- The formal inflation-growth law was admitted before the primary snapshots were bound.
- Exact scalar support 31/32 lies within Planck $n_s=0.9649\pm0.0042$.
- Exact tensor support 1/32 lies strictly below $r<0.032$ by 3/4000.
- The complete 31/32 plus 1/32 partition is retained without fitting.
- No measured equality is claimed between five Fold doublings and conventional logarithmic e-folds.

- A tampered scalar target rejects the comparison.

Falsification condition: Reject if any source identity or registered row changes; if 31/32 leaves the complete scalar interval; if 1/32 is not strictly below the tensor bound; if the narrow margin is omitted; if conventional e-fold language is asserted as exact equality; or if a target changes the formal survivor.

Measurement receipt: sha256:b642e922c44895919b78a5d1f98b2adc1d42fa97e7d1422b69e23a3fda14fe8b.

Isolation certificate: sha256:c44e9ae1e05fe66e91a2cd0550c53e62b7d9aec763965e44517219c4246c5a74.

Custody certificate: sha256:28c44958737414705d1e7638388a7a18a41c26aeee1b568e990d928927ecdf58.

Registered source descriptions:

- Planck 2018 results. X. Constraints on inflation: <https://arxiv.org/pdf/1807.06211>
(sha256:fa728dc0198ccfddd7132c0e0a29cca586a1431a045568a33d7a197a0427427d)
- Improved limits on the tensor-to-scalar ratio using BICEP and Planck: <https://arxiv.org/pdf/2112.07961>
(sha256:5a1add11c98a8e5c4d70fd438eacd5242aa10631ffea0c72358f49e59eda07f2)

Registered target identities:

- INFLATION-GROWTH-WITHHELD-COMPLETE-OBSERVATION-RECORD from None

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no external scalar index or tensor limit readable by the formal generator or formal validator
- no fitted spectral index, tensor amplitude, inflaton potential, transfer function or exit scale
- no historical-blindness claim because V1/V2 named these observables
- no claim that five exact Fold doublings equal a conventional logarithmic e-fold count
- no omission or rounding of the narrow tensor-bound margin
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:6aa90bdf5f84f09a5c3d941b128a685c262a0bbaecc9a665a1ec7e86ccde3066; engine receipt

sha256:8e06d75a25206d5cce129a9afd2583a54cb1e070a0daf6eeeff92ece99a8d01f at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INFLATION-GROWTH-040-8e06d75a25206d5c.json;

empirical-validation hash

sha256:272562ca8235316e3bae097e8a3af450c1c4785525266797ca8bf6d012a272e9; measurement receipt

sha256:b642e922c44895919b78a5d1f98b2adc1d42fa97e7d1422b69e23a3fda14fe8b.

327. Post-seal blackbody, acoustic, laser, plasma and Alfvén response test

Claim identity: SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. External blackbody measurement reports exponent 4 exactly, matching the forced rank-three-plus-energy exponent; doubling therefore remains the exact ratio 16. The historical Stefan coefficient is reported within 1/100 of the present value and spectrum shape agreement is retained, but no dimensional coefficient is manufactured from a normalized carrier. Acoustic and microwave cavity resonances were measured to part-per-million precision. NIST reports unmodified laser linewidths of at least 10,000,000 Hz and extended-cavity linewidths no more than 500,000 Hz, so linewidth remains positive and feedback narrows it by at least factor 20. Two NASA flights directly track electron density through plasma frequency, and NASA registers Alfvén waves

from 2007 through 2014. All comparisons pass at their declared relation/value type without fitting.

External blackbody measurement reports exponent 4 exactly, matching the forced rank-three-plus-energy exponent; doubling therefore remains the exact ratio 16. The historical Stefan coefficient is reported within 1/100 of the present value and spectrum shape agreement is retained, but no dimensional coefficient is manufactured from a normalized carrier. Acoustic and microwave cavity resonances were measured to part-per-million precision. NIST reports unmodified laser linewidths of at least 10,000,000 Hz and extended-cavity linewidths no more than 500,000 Hz, so linewidth remains positive and feedback narrows it by at least factor 20. Two NASA flights directly track electron density through plasma frequency, and NASA registers Alfvén waves from 2007 through 2014. All comparisons pass at their declared relation/value type without fitting.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed collective response, complete targets, bound provenance, capability isolation, proof/measurement separation, complete rows, successor closure and extension. The declared exact boundary is: The sealed finite occupation and exact scaling laws; measured exponent four; coefficient agreement bound one percent; positive measured linewidth ranges; acoustic resonance precision; direct plasma-frequency density dependence; complete Alfvén observation-year boundary; and every source/custody row. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-collective-response-versus-complete-blackbody-acoustic-laser-plasma-Alfven-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-collective-response-versus-complete-blackbody-acoustic-laser-plasma-Alfven-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256: fdf048a5f1358d5dd699b09cc0df4135a9d45f2f439cd74e13768f4d8fa22d69`. Its generality

certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 9e95f31ffd65f7e0e7fe88cff91fa56174657df423fe9339a700d16d8af28c8d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 4a9bf8a263ce6dee9c7b4fecbac1db0490cb2ab1e2dff3a463e0f17f77d7576b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: e078e684ee11df95956211e36bb820cf50e1465dc92d49b3bc4b4cfab7f81cb3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: cd21f5572e340e60783f653afb6f6a9081e9a9d32378cb85c23e9dab152bfc3b.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: 374078411155f78c3dd1273bad1444aeba59106a0bdc75d2c5b4e3828bbbc021.

Independent certificate:

sha256: d5e36700fbe06d0dbe4cda4afa473cfe7c18d8bc19318a677ccf8a15e776a81a. Engine

external-validation hash:

sha256: cb22b4bcf0d3da68cd1002f5d0d0556664932179bfa9a62796cf3ca65cbf7c1b.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-COBLTZ-BLACKBODY
- NIST-DIODE-LASER-LINEWIDTH
- NBS-ACOUSTIC-CAVITY-1986
- NASA-PLASMA-FREQUENCY-PROBE-1992
- NASA-STEREO-ALFVEN-LIST

Observed comparison records:

- Measured blackbody exponent four forces doubling ratio sixteen.
- Historical coefficient agreement is within one percent and spectrum shape agreement is retained.
- Acoustic cavity resonances were measured to part-per-million precision.
- Measured laser linewidth remains positive and feedback narrows it by at least factor twenty.
- Two NASA flights track density through plasma frequency.
- NASA registers Alfvén observations across eight calendar years.
- No dimensional coefficient is fitted from normalized Fold structure.
- Tampered exponent three rejects.

Falsification condition: Reject if a source or row changes; if the measured blackbody exponent is not four; if measured linewidth is empty or feedback does not narrow it; if acoustic resonances, direct plasma-frequency density dependence or Alfvén observations are absent; if a dimensional coefficient is fitted from normalized structure; or if a target changes the formal survivor.

Measurement receipt: sha256: ff7685c1c62d0af80001e60ad5c44ba0199bea46b5a43073dedac0f3332a0dd7.

Isolation certificate: sha256: 888d25acd42241bcf5ca6fd22ade5ec7a4d86826d003f2cbbf0c563e3e38038e.

Custody certificate: sha256 : 363a20ce7e39db88f33913b933b8b094dd389cb98329a79f05f46ca81275afcf.

Registered source descriptions:

- Coblenz and blackbody radiation measurements: <https://nvlpubs.nist.gov/nistpubs/sp958-lide/html/007-009.html>
(sha256:deb600e47fd138dfe82aef3271b19f2030cd7cd3e4fa202ef80f4f1ae82eec71)
- Diode Laser Technology: <https://tf.nist.gov/ofm/lasers/diode.htm>
(sha256:7c0a9a1b4dc19b573cb6603f87453161508784ca59ec26ed698b0354bb7520b6)
- Measurement of the ratio of the speed of sound to the speed of light:
https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=925220
(sha256:f4dbf2b01604b8a6f26621bb8bafa856913e25158a8635f7454838ef5360f1b4)
- Measuring ionospheric electron density using the plasma frequency probe: <https://ntrs.nasa.gov/citations/19920042037>
(sha256:4fff15fe03f66e4009ca34e83f0d0ea02e66b2d40e92ee987ae46f437f39f36c)
- STEREO-A PLASTIC Alfvén Wave List: <https://data.nasa.gov/dataset/stereo-a-plastic-alfven-wave-list>
(sha256:da0eab2e7955e0670d734c905c360758aabe57bb900de277e42da13fa0314385)

Registered target identities:

- COLLECTIVE-RADIATION-WITHHELD-COMplete-RECORD from None

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no target readable by the formal generator
- no fitted dimensional coefficient or converted absolute value
- no historical-blindness claim
- no omission of finite linewidth or source limitations
- no numerical-zero, negative, irrational, imaginary or floating Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256 : e5232310fe7cfa0feeaffd1866117045b7718fee1f1ccc7725a73fc11715683e; engine receipt
sha256 : 1d7dd8e6373027c10bd83aa9ff128ebabe3fdc60569b24ad597f9c22abad2455 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042-1d7dd8e6373027c1.json; empirical-validation hash
sha256 : 83cf47bcba4dc585c59c8f16adf89071a2d100d79b6cf98f197a45ea82a3d8f6; measurement receipt
sha256 : ff7685c1c62d0af80001e60ad5c44ba0199bea46b5a43073dedac0f3332a0dd7.

328. Post-seal CODATA validation of the terminal charged-lepton invariant

Claim identity: SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. After the exact terminal charged-lepton invariant is sealed, exact rational root brackets predict both adjacent-root squared mass ratios and compare them with both complete NIST CODATA 2022 intervals.

Both sealed terminal-cubic mass-ratio consequences overlap their complete one-standard-uncertainty CODATA 2022 intervals.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal terminal-invariant, source, interval, row-retention, custody and no-extra-rule product. The declared exact boundary is: Both exact adjacent-root squared ratios of the sealed terminal cubic against both complete registered CODATA intervals. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-terminal-cubic-ratios-versus-complete-codata-intervals__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-cubic-ratios-versus-complete-codata-intervals	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:4b0f13ecfa647507c2116a80608f8817884eef53fde16d2ca18ab9c9f8edadf0`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c44fcd2be5a25663f279cd0a06772eecf6e63e389685a64899dcc4a0861008`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b596438bde200a2809c35642cb33eda7a71543904271bf37b296e7518f065e3f.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:5cb9911daba73b152460a90883b7b8cf5f5b3cfe81f98402dfec24998918f976.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:dfce970fb252d4b9a06366c282e3d38063bba95f0b2800b2cae879e18cf322ba.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:024f84cb4b6103c422fa67d656620f53d80ea7dceff6c720aa1f02201de5ae31.

Independent certificate:

sha256:bb85213071aba6fa4ef1f4ce506c214a9fa4bebe181d02881e36c2b705057381. Engine

external-validation hash:

sha256:355eb559ee89354203142ccb3b59c17e0081caa150f150c0778bddc056239ffb.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-MUON-ELECTRON-MASS-RATIO: predicted terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; observed terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; exact match **True**
- NIST-CODATA-2022-MUON-TAU-MASS-RATIO: predicted terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; observed terminal-cubic-mass-ratios-inside-both-complete-codata-intervals; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification condition: Either complete terminal prediction interval fails to overlap its source interval, a row/hash changes, an uncertainty is enlarged or omitted, or a tampered comparison is accepted.

Measurement receipt: sha256:b1bc18c4ca4396fcdf2a88e4a9a8fc34bd1acf30ae4d80949faa6d1c67ec44e7.

Isolation certificate: sha256:bb11637734c577c68efa087896e41a65b07db3971be5a6c9bfc12b01eab1f9ec.

Custody certificate: sha256:2f89d8555574629bf6360dd461ca6c73a9c117a6e67cd5b5e8e445f66a9417e9.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no CODATA value accessible to formal forcing
- no fitted terminal coefficient or root
- no floating comparison
- no omitted failed row or enlarged uncertainty

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:73e9e100453602ec568ccfebb17a875ea8e6387b9d1559d50f2ad5d36aca92f3; engine receipt
sha256:1df007d68f197e5a8a8a58dad28190bbaf10229ab2f8136365a25622eba5d5d8 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002-1df007d68f197e5a.json; empirical-validation hash
sha256:1502cfabdf3a00b544b7fd2a44179bff7f0dc10ed9141b342c0e4ff684c9af26; measurement receipt
sha256:b1bc18c4ca4396fcacf2a88e4a9a8fc34bd1acf30ae4d80949faa6d1c67ec44e7.

329. Versioned post-seal exact Koide measurement validation

Claim identity: SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact Fold value two-thirds is sealed before both complete CODATA mass-ratio intervals are opened and propagated through exact rational square-root enclosures.

The sealed exact Koide value 2/3 lies inside the complete conservative CODATA-derived interval.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar and closure boundary. Generate the complete eight-axis post-seal two-thirds, two-source-row, rational-enclosure, custody and no-extra-rule comparison product. The declared exact boundary is: The full Cartesian uncertainty rectangle of both complete NIST CODATA 2022 charged-lepton mass-ratio rows. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-two-thirds-versus-complete-rational-koide-enclosure-v2__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	sealed-two-thirds-versus-complete-rational-koide-enclosure-v2	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bca9ace286d82c15f81c66da94c8f5d4e3a41caf7ca3c5d72cceb4c447fdf4ec`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3532da9d5c6d2e27a021faa14ff22e62256b4adcb27b72a74275eb48c11afb19`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e5c839d5cadeaa5f243adec6dbf44c23ebb0cb9872c6359cf1be7151bc93f17c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:724d004ca77c3a8a5f941c8168ccdede605016800fdf94dad1d07935882fac4d`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f138b0a8ca62e6462c8e2caa1cdfc6f57077a46208c91c28053c30ac3e6f8438`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:ca563bc0a2c7d291e111a9f6c0511b689c4b0f45314fb520d97c75ead0f3a808`.

Independent certificate:

`sha256:6ddde3960e2364184cfbb6e3c47d13d74ab4c5b1f77f827db1d4ead9c0bb7f6b`. Engine

external-validation hash:

`sha256:97998128cccbc5765d78232c461abd4155fc23719c1fd13659e8f73769cf49b5`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Observed comparison records:

- NIST-CODATA-2022-MUON-ELECTRON-MASS-RATIO: predicted sealed-two-thirds-inside-complete-codata-derived-koide-interval; observed sealed-two-thirds-inside-complete-codata-derived-koide-interval; exact match `True`

- NIST-CODATA-2022-MUON-TAU-MASS-RATIO: predicted sealed-two-thirds-inside-complete-codata-derived-koide-interval; observed sealed-two-thirds-inside-complete-codata-derived-koide-interval; exact match True
- deliberately tampered unfavorable control rejected

Falsification condition: The exact two-thirds lies outside the outward CODATA-derived interval, either row or uncertainty is omitted, source custody changes, or a tampered comparison is accepted.

Measurement receipt: sha256:e7ccd0d3a9e0f675cd2d1e894bd347e753ad9f4d7c9081bc094e8b850fb8baf7.

Isolation certificate: sha256:6e92f178a47954d84aa0acb09fdefee178f7cbb36533fb38688a89a6f2d81308.

Custody certificate: sha256:68ac69675369c6d62faa95647520644b327f0ab77851e51e60b40127a8dde030.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured mass ratio in the formal two-thirds derivation
- no floating square root
- no central-value-only comparison
- no omitted uncertainty or row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:685f4a86e3a630c03aa11251b684b32319b7b27b9f58520b1f1972aa6c076748; engine receipt

sha256:4e7a64b6751e5622c827612840751a26b40b61ee657bb011eded7ff7914aa776 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002-4e7a64b6751e5622.json; empirical-validation hash

sha256:d399bfa96ceabc1eedd3b61d77618e85f20e488f1ffb2db46f619413beac4a7a; measurement receipt

sha256:e7ccd0d3a9e0f675cd2d1e894bd347e753ad9f4d7c9081bc094e8b850fb8baf7.

330. Post-seal acoustic and Johnson-noise thermometry test

Claim identity: SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. With all values carried as exact integers over 10^{30} J/K, exact SI k_B is $13806490/10^{30}$. The acoustic result is $(13806484 \pm 28)/10^{30}$ and therefore gives $[13806456, 13806512]/10^{30}$; the Johnson-noise result is $(13806510 \pm 170)/10^{30}$ and gives $[13806340, 13806680]/10^{30}$. Both intervals contain the exact carrier. The two measurement routes are physically distinct. The acoustic record identifies thermodynamic temperature with mean kinetic energy, and the electronic record reports Johnson noise power as jointly dependent on temperature and resistance to one part per million in its declared regime. No external row claims the Fold dyadic ladder or $3/4:1/4$ response coordinates as universal measured values.

With all values carried as exact integers over 10^{30} J/K, exact SI k_B is $13806490/10^{30}$. The acoustic result is $(13806484 \pm 28)/10^{30}$ and therefore gives $[13806456, 13806512]/10^{30}$; the Johnson-noise result is $(13806510 \pm 170)/10^{30}$ and gives $[13806340, 13806680]/10^{30}$. Both intervals contain the exact carrier. The two measurement routes are physically distinct. The acoustic record identifies thermodynamic temperature with mean kinetic energy, and the electronic record reports Johnson noise power as jointly dependent on temperature and resistance to one part per

million in its declared regime. No external row claims the Fold dyadic ladder or 3/4:1/4 response coordinates as universal measured values.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed thermal law, exact source-bound intervals, target custody, complete row retention and formal/measurement separation. The declared exact boundary is: The sealed finite-population theorem; exact SI k_B ; the complete acoustic and electronic k_B standard-uncertainty intervals; the kinetic-temperature and Johnson-noise relation rows; both explicit nonmeasurement boundaries; and every source/custody row. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-thermal-law-versus-complete-acoustic-and-Johnson-noise-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-thermal-law-versus-complete-acoustic-and-Johnson-noise-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:019db88dcdb566512199c250c6841ce242ba1a225cf35f4495a68b1977c7ed37`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3616e5ddecafab7522954ecae0da77f6eec8e9057249d748373534465565149d`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:400a1cdd726220cd82097342db240c8988e9eaf785459534e546c1d842111bcc.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:5b4a3858a67e89cb7ed7cdcf2a40e56a17718850916d10aef131f36b472cb81b.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:9892ff517e4a939131153c66697e30fda07f1b043ee104300ee03ce7cfc29eff.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:ca67b8df9587363e46d0c18b5d69a4c5d234e99832846b5dcf5a8e5c8e1b009b.

Independent certificate:

sha256:37d234b90b063945ab5f2d13a1be76fe385c6cda55be766f1eda81f4b0aa1304. Engine

external-validation hash:

sha256:51495049d9401534e4174d42f85d8fca21b8a9e8e89eb7ecba2128eb492a0053.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NIST-CODATA-2022-BOLTZMANN
- NIM-NIST-ACOUSTIC-BOLTZMANN-2017
- NIST-JOHNSON-NOISE-BOLTZMANN-2011

Observed comparison records:

- Exact SI k_B is 13806490/10³⁰ J/K.
- Acoustic interval [13806456,13806512]/10³⁰ J/K contains exact SI k_B .
- Johnson-noise interval [13806340,13806680]/10³⁰ J/K contains exact SI k_B .
- Acoustic and electronic routes are physically distinct and both remain complete.
- Acoustic thermometry retains the mean-kinetic-energy temperature relation.
- Johnson noise retains joint temperature/resistance response at the reported one-part-per-million regime.
- No direct universal measurement is claimed for the dyadic ladder or 3/4:1/4 Fold coordinates.
- Tampered acoustic interval excludes the exact carrier and rejects.

Falsification condition: Reject if any source or row changes; either complete measured interval excludes exact SI k_B ; the routes are not physically distinct; the kinetic-temperature or Johnson-noise response row is absent; a limitation row is omitted; an internal Fold coordinate is relabelled as directly measured; or target data alter the formal survivor.

Measurement receipt: sha256:cffa2dbbd885f19cf9a19f08aec070370ba237b7aa12735d2173fc8d77f0e7f6.

Isolation certificate: sha256:ee10f5a56f89082a81a2e386f8e5f82d14ff84edf39ea3859ba317b2e04b3eb7.

Custody certificate: sha256:2e674cf79354cc078e08ef624ddad870c176f981f96c15a124d1c6a36730445f.

Registered source descriptions:

- NIST CODATA 2022 all constants - Boltzmann constant: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt> (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- Determination of the Boltzmann constant with cylindrical acoustic gas thermometry: <https://doi.org/10.1088/1681-7575/aa7b4a> (sha256:1cbb21f0e5817270b5e028105aa79fea22017fd84130c2c4b79d1492fb37e418)
- An electronic measurement of the Boltzmann constant: <https://doi.org/10.1088/0026-1394/48/3/008> (sha256:187066b5390d57a3c058e0a34f6c7803a659045ba714ca4b25eed7b84b212bbb)

Registered target identities:

- THERMAL-EQUILIBRIUM-WITHHELD-COMPLETE-RECORD from None

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no target readable by the formal generator
- no fitted Boltzmann carrier, temperature scale or response coefficient
- no conventional continuum distribution imported into the formal proof
- no claim that the dyadic ladder or 3/4:1/4 coordinates were directly measured
- no omitted unfavorable, uncertainty or applicability row
- no numerical-zero, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:9c5025b8f068aeb13e964ab1abf997dd143ddb453308512038c162fe8bed065; engine receipt
 sha256:60366227e00c50a3529d98538a3ca42d9620f4da30a174b9cff96f1122097dbd at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044-60366227e00c50a3.json;
 empirical-validation hash
 sha256:9d94b8c34f0413e7170a0aa78fc95dec6807bae70ea620e75af6f9f131e56a16; measurement receipt
 sha256:cffa2dbbd885f19cf9a19f08aec070370ba237b7aa12735d2173fc8d77f0e7f6.

331. Post-seal finite BEC, Pauli-blocking and two-turn spinor test

Claim identity: SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The finite Bose record measures ground-state occupation, its increase under cooling, and a sharp transition at $(94 \pm 5)/100$ of the declared trap scale, giving the exact interval $[89,99]/100$. The finite two-spin-state Fermi record directly observes Pauli blocking and a factor-two reduction in effective collision cross section. The sealed alternating-state return is two complete turns, corresponding to 720 degrees; the neutron-interferometry result is 704 ± 38 degrees, giving $[666,742]$, which contains 720. All three physical routes agree with their respective sealed structural distinctions. The BEC scale ratio and collision-response factor are retained as external apparatus-dependent measurements rather than relabelled universal Fold proof scalars.

The finite Bose record measures ground-state occupation, its increase under cooling, and a sharp transition at $(94 \pm 5)/100$ of the declared trap scale, giving the exact interval $[89,99]/100$. The finite two-spin-state Fermi record directly observes Pauli blocking and a factor-two reduction in effective collision cross section. The sealed alternating-state return is two complete turns, corresponding to 720 degrees; the neutron-interferometry result is 704 ± 38 degrees, giving $[666,742]$, which contains 720. All three physical routes agree with their respective sealed structural distinctions. The BEC scale ratio and collision-response factor are retained as external apparatus-dependent measurements rather than relabelled universal Fold proof scalars.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

• SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed spin-statistics law, finite BEC record, Pauli-blocking record, two-turn spinor interval, custody, full-row retention and formal/measurement separation. The declared exact boundary is: The admitted 256-form formal survivor; all three bound primary records; the complete BEC transition interval and direction rows; the complete Pauli rows; the complete spinor interval and return rows; and every custody, limitation and adverse-control row. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-spin-statistics-condensation-law-versus-complete-finite-BEC-Pauli-and-spinor-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-spin-statistics-condensation-law-versus-complete-finite-BEC-Pauli-and-spinor-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:00c2bf991301e167212f42e5e1cd5c4271d69a5bf5b209d0f397261877c1e853`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:71667d88b3de82c75428da3d963a842ffecca969031fec3d637d34cd953ab4f9`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6023224eea2ee01eb133f80d8898133abfd6bf5f32965a3af344c0b301999eaa`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cac70197006d203259866f1502a64a75ecaa45ee6e9a9e230ad7fdb161e17354`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:9e81125fdafed5f0286f7510956a2e8ac22ed6c4348943017be4f33b4d75088f.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.
Implementation hash: sha256:2abf34329c76d85d6e1e413feb2cf8ad0445ddd30070ecde9689518fdedf80ea.
Independent certificate:
sha256:9628100ad1af900942a0252fa75c303188d2a2904aef0e65de5543f6ccd83532. Engine
external-validation hash:
sha256:105fb7e4fb72eb3d20b4fe28c0c832a8cdd38e5601c361a134768b37eaaffbd7.

External empirical check. The target was absent until prediction seal: True. It was released after the matching seal: True. All rows were preserved: True. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- APS-JILA-BEC-GROUND-OCCUPATION-1996
- NIST-JILA-PAULI-BLOCKING-2001
- RAUCH-COHERENT-SPINOR-ROTATION-1975

Observed comparison records:

- Finite rubidium-87 Bose gas ground occupation is measured and rises as temperature is lowered.
- BEC transition interval is [89,99]/100 of the declared trap scale; its finite-population correction remains retained.
- Two-spin-state potassium-40 gas directly exhibits Pauli blocking and the reported factor-two collision reduction.
- Neutron spinor interval [666,742] degrees contains the sealed two-turn 720-degree return.
- One turn changes the held spinor orientation and two turns restore the identical state.
- Trap-specific BEC and collision-response magnitudes remain external scale records, not formal-law selectors.
- Tampered spinor and reversed-cooling controls reject.

Falsification condition: Reject if any source or row changes; finite bosons fail to accumulate in the measured ground state under cooling; Pauli blocking is absent in the declared identical-fermion regime; the complete spinor interval excludes 720 degrees; a trap-specific or response-specific value is relabelled as a universal formal scalar; an unfavorable row is omitted; or target data alter the sealed formal survivor.

Measurement receipt: sha256:171baf50d3b3944b9f2c8e14f7bfddf7da778071c52a03bfc33e6c422ea6ff95.
Isolation certificate: sha256:3df0e1a9705da1b597362b4786897f30b2125a11228ef0823a6dcd24fbd79b09.
Custody certificate: sha256:cd36776ecd5e074b3c50f48ec1eeb5875970aa180c17018a3382c3897350bdb3.

Registered source descriptions:

- Bose-Einstein Condensation in a Dilute Gas: Measurement of Energy and Ground-State Occupation:
<https://doi.org/10.1103/PhysRevLett.77.4984>
(sha256:9086058e1c750191fb36eab5acf7c65da1aa7f0e4cdcf643887e282ad3920c93)
- Pauli Blocking of Collisions in a Quantum Degenerate Atomic Fermi Gas:
<https://www.nist.gov/publications/pauli-blocking-collisions-quantum-degenerate-atomic-fermi-gas-0>
(sha256:82555a4057e339aef96cde7484fe3fbd190a8d2cecbf4c3de73996af473b9af7)
- Verification of Coherent Spinor Rotation of Fermions: [https://doi.org/10.1016/0375-9601\(75\)90798-7](https://doi.org/10.1016/0375-9601(75)90798-7)
(sha256:11638b8c2aaf8fbdd96aa50c9496bb9f7597b308f19d780ab779cc2dd6752824)

Registered target identities:

- SPIN-STATISTICS-CONDENSATION-WITHHELD-COMPLETE-RECORD from None

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no BEC temperature, Pauli response, particle species or spinor angle readable by the formal generator
- no fitted occupation ceiling, critical temperature, collision factor or rotation interval
- no continuum Fermi-Dirac or Bose-Einstein function imported into the formal proof
- no relabelling of the apparatus-specific BEC ratio or collision reduction as a universal Fold proof scalar
- no omitted unfavorable, uncertainty, finite-population or applicability row
- no conventional numerical-nothingness, negative, irrational, imaginary, floating, NaN or continuum Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:76e98275459015406d1fb9dcc615d65b8bc0c2d55deef0fde4e4ab8bde88733; engine receipt
 sha256:6c37132ae33b8f23b06e69b7034c5acdd39144d51b2661d7e65f441b400604eb at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046-6c37132ae33b8f23.
 json; empirical-validation hash
 sha256:b19baca0cbfe8f4673e94c3cb40502fba2a954a0172387e9573a3d5bafd4e0fe; measurement receipt
 sha256:171baf50d3b3944b9f2c8e14f7bfddf7da778071c52a03bfc33e6c422ea6ff95.

332. Post-seal critical-exponent, universality and turbulence-scaling test

Claim identity: SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The erbium intervals $[46,50]/100$ for beta, $[90,102]/100$ for gamma and $[47,51]/100$ for nu contain exactly $1/2$, One and $1/2$. Of the complete five-row manganite record, La00, La04, La06 and La08 contain the entire $\beta=1/2$, $\gamma=One$, $\delta=three$ vector; La02 excludes $\gamma=One$ and $\delta=three$ and therefore cannot be admitted to that generated universality class. The independent turbulence interval $[666,692]/1000$ contains exactly $2/3$. A Reynolds-count-720 physical flow exhibits five-thirds compensated plateaux over 40--4000 Hz by Fourier analysis and 20--2000 Hz by Hilbert analysis. The nonmatching composition, finite-Reynolds correction boundary, structure-function range limitation, normalized-threshold boundary and empty-exponent typing are all retained.

The erbium intervals $[46,50]/100$ for beta, $[90,102]/100$ for gamma and $[47,51]/100$ for nu contain exactly $1/2$, One and $1/2$. Of the complete five-row manganite record, La00, La04, La06 and La08 contain the entire $\beta=1/2$, $\gamma=One$, $\delta=three$ vector; La02 excludes $\gamma=One$ and $\delta=three$ and therefore cannot be admitted to that generated universality class. The independent turbulence interval $[666,692]/1000$ contains exactly $2/3$. A Reynolds-count-720 physical flow exhibits five-thirds compensated plateaux over 40--4000 Hz by Fourier analysis and 20--2000 Hz by Hilbert analysis. The nonmatching composition, finite-Reynolds correction boundary, structure-function range limitation, normalized-threshold boundary and empty-exponent typing are all retained.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed criticality/cascade law, exact external intervals, universality membership, complete-row custody, method agreement, limitation retention and proof/measurement separation. The declared exact boundary is: The admitted 256-form formal survivor; all four bound source records; every reported erbium and five-composition manganite exponent interval; the complete turbulence structure interval; both

physical spectrum-analysis routes; every unfavorable row, limitation and custody record. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-criticality-and-cascade-law-versus-complete-postseal-exponent-and-turbulence-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-criticality-and-cascade-law-versus-complete-postseal-exponent-and-turbulence-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:37318550316d6549d405a22e3baa9ec6dab4de1c5646452d2a188f234e930e2e`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9b5652ebc3f084bc0e3a2c2987fab6356660f2d418e607754557b3d973e67b34`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8a47b83313e602972ed0b81b0c6df8037546083c6d6aff27ee3a08227163ef2c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3663e27042caebc8ec58272e4bf040604e79a8480d8616419b76612ea7d3d4ec`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:84d564af90805459dd95a0b12bbdf10fb2024d47fdb2d7cef5fcb63b438541a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:664628e4ab961c2fe2bbc9289ecfcae9b3805d45f0f05fd1f2e86dd98e24c37a`. Independent certificate: `sha256:87f93c3da5f96313e0d799a45fce681b7570c0e08bb36a5d12f9b39c551713ad`. Engine

external-validation hash:

sha256:84d787b11748ec58ad11057fee614d97e312e5801219e21d9bc4fb85f4992618.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- SPRINGER-CETIN-MANGANITE-CRITICAL-EXPONENTS-2026
- MCMASTER-LIN-ERBIUM-CRITICAL-SCATTERING-1993
- APS-MCCOMB-TURBULENCE-STRUCTURE-2014
- APS-HUANG-TURBULENCE-SPECTRUM-2010

Observed comparison records:

- Erbium beta/gamma/nu intervals contain exactly 1/2, One and 1/2.
- Four of five complete manganite vectors contain exactly beta=1/2, gamma=One and delta=three.
- LaO₂ is retained and rejected as a complete member of that generated exponent class.
- The independent zeta₂ interval [666,692]/1000 contains exactly 2/3.
- The physical channel-flow record exhibits five-thirds compensated plateaux by Fourier and Hilbert routes.
- Finite-Reynolds, structure-function-range, threshold and empty-exponent boundaries remain retained.
- Tampered interval, relabelled nonmatch, removed-row and reversed-spectrum controls reject.

Falsification condition: Reject if any source or reported row changes; the erbium vector excludes a sealed carrier; fewer than the registered four manganite rows match or LaO₂ is concealed or relabelled as a complete match; the zeta₂ interval excludes 2/3; either registered physical spectrum route lacks its five-thirds plateau; any limitation is omitted; or target data alter the sealed survivor.

Measurement receipt: sha256:eee857690b4e7a3a0c6c6d16b82055bd33a7ae269d887097921428c9f6ece26d.

Isolation certificate: sha256:59dd6d2b2fe4fc0d7366b399c5888ae5728bcbdd91d489d31b9ca29e020acc0a.

Custody certificate: sha256:5710aa5047e21e2e8f0994451a3b0d42b9feb38c36a8f27d963753f47336328f.

Registered source descriptions:

- Structural, magnetic, and magnetocaloric properties of La and Sm co-doped CaMnO₃ manganites:
<https://link.springer.com/article/10.1007/s10854-026-17400-y>
(sha256:afd0f783bd40b8ed376238eda9b254387c75c0c26ecc57488ab9b81b33d1d455)
- Critical magnetic scattering in erbium: <https://experts.mcmaster.ca/scholarly-works/262851>
(sha256:d8b6495b7defb46daa4d1162967d11b4680012ebd4a8e6d60e8892b83b9b3564)
- Spectral analysis of structure functions and their scaling exponents in forced isotropic turbulence:
<https://doi.org/10.1103/PhysRevE.90.053010>
(sha256:0c9f4321b52f78fe64b02684b4139a450955c12092c4faf8487f49bf6ff1d0f5)
- Second-order structure function in fully developed turbulence: <https://doi.org/10.1103/PhysRevE.82.026319>
(sha256:1bff196a24bc3f852dd384469c5f8e4112fcd717c2416190ad44b0012ac06cb)

Registered target identities:

- CRITICALITY-UNIVERSALITY-TURBULENCE-WITHHELD-COMPLETE-RECORD from None

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no critical exponent, sample class, Reynolds record, structure exponent or spectrum slope readable by the formal generator
- no fitted exponent, uncertainty interval, intermittency correction or measurement-selected universality class
- no omission of La02 or any other unfavorable measurement row
- no conversion of the normalized half-One threshold into a universal laboratory temperature
- no conversion of typed empty alpha or eta into a numerical-zero measurement
- no claim that every physical transition shares the binary self-antipodal exponent vector
- no conventional numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:79e4dae658e266d5768876896a2ffff9d1a8b2adced60acedbc65409a3430dc; engine receipt
 sha256:075d304c85f48d6886ba7cb1fd44d5db25815cdf3a5d016502da1435654f1bdc at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048-075d304c85f48d68.json; empirical-validation hash
 sha256:875b3c350cd9f407f5608d25ef989b2161ffad838c4c4d9805412d07c43aec58; measurement receipt
 sha256:eee857690b4e7a3a0c6c6d16b82055bd33a7ae269d887097921428c9f6ece26d.

333. Post-seal superconducting-circuit test of the Fold Bell factorization boundary

Claim identity: SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact measured S interval $[20714, 20780]/10000$ lies wholly above the local-factorization bound two; its central excess exceeds twenty-two stated standard uncertainties and the reported P value is below 10^{-108} across more than one million trials. The complete timing intervals preserve space-like separation, and the analysis retains its memory control. This physically rejects the sealed local-factorization class under the source's declared conditions while preserving deterministic complete-record admissibility, no signalling, and the explicit measurement-independence assumption. The distinct Walsh support-count product remains a formal/computational result and is not relabelled as a variance measurement.

The exact measured S interval $[20714, 20780]/10000$ lies wholly above the local-factorization bound two; its central excess exceeds twenty-two stated standard uncertainties and the reported P value is below 10^{-108} across more than one million trials. The complete timing intervals preserve space-like separation, and the analysis retains its memory control. This physically rejects the sealed local-factorization class under the source's declared conditions while preserving deterministic complete-record admissibility, no signalling, and the explicit measurement-independence assumption. The distinct Walsh support-count product remains a formal/computational result and is not relabelled as a variance measurement.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-QUANTUM-BELL-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed Bell boundary, external interval, provenance, target isolation, proof/measurement separation, complete condition retention, successor closure and extension. The

declared exact boundary is: The admitted Claim 049 Bell-response census; the complete primary S interval, trial-count, significance, space-like timing, memory-analysis and measurement-independence records; and every stated interpretive limitation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-local-factorization-law-versus-complete-superconducting-circuit-Bell-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-local-factorization-law-versus-complete-superconducting-circuit-Bell-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:d9d2ffe6f3de6a674f40a65c0f7ba8c9bdf10a54fe7061141ebc9fa73bff4fdc`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ba21cca4d789ae798ee99d66062a03b65b548a0941483202a985b2908c22f2f9`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:99df30fde6e9e24905d717834894b22176a56ac9a56e12e24e618b3da29fd370`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b1e892060c3134158247569c3f75db2a7b153d8e4193d4df177f7ac37ccac46c`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d6cfacd47b3e5fa3c8e02c4c0602fb52aed94102f4778a23ca7614efbab26950`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4e5da504fe3f3a28d7e51131cd3113d4e77eb9aaa32ac15577492c21dbc8bd4f`. Independent certificate:

sha256:6e96d67c42eb8074c876d6595ec8c0748b60daacc5f22cfcfb78ae396ba5f232. Engine
external-validation hash:
sha256:40e0fe145c4f95e92ef5cb0d86081a7ae76b5914d1a02b78018bf1c515bc610a.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NATURE-STORZ-LOOPHOLE-FREE-BELL-2023

Observed comparison records:

- The complete measured S interval [20714,20780]/10000 lies above the local-factorization bound two.
- The central excess is greater than twenty-two stated standard uncertainties and P is reported below 10^{-108} .
- More than one million superconducting-circuit trials are retained.
- The complete trial-duration interval lies below the complete light-travel-budget interval.
- The memory-robust analysis and setting-choice procedure are retained.
- Measurement independence remains a declared source assumption; no ontic randomness premise is imported.
- The result rejects local factorization under those conditions, not determinism as a whole.
- The Bell record does not relabel the separate formal Walsh support product as measured variance.
- Tampered S and timing controls reject.

Falsification condition: Reject if either source identity changes; the complete S interval touches or falls below two; the trial, significance, timing or memory records fail; measurement independence or any stated limitation is omitted; the result is presented as excluding all determinism; ontic randomness is imported; the Walsh support law is called a measured variance; or any target changes the sealed survivor.

Measurement receipt: sha256:c97b436ea794ddadea73446b5e06a1a88a0a23119e66279f989cba218a71c2b6.
Isolation certificate: sha256:8a5f0cf9ed57acca9841b1e5ebeccb9f552f33865e270b5a8eacc1d0046eea83.
Custody certificate: sha256:bf791660650594a56788cae969a412800b6365ee0b08619ddfad007e6d0e108b.

Registered source descriptions:

- Loophole-free Bell inequality violation with superconducting circuits: <https://doi.org/10.1038/s41586-023-05885-0>
(sha256:fabe97e4970990cef6246f0541a8b9e80e47b0df58f10234b8f8155d1b4ade81)

Registered target identities:

- NATURE-STORZ-2023-WITHHELD-COMPLETE-BELL-RECORD from None

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no Bell value, trial result, significance record or timing record available to formal candidate selection
- no claim that a Bell violation excludes all deterministic or complete-record models
- no import of random-number generation as ontic nondeterminism
- no omission of the measurement-independence assumption or the authors' stated assumption boundary
- no relabelling of support cardinality as statistical moment variance
- no claim that this Bell experiment separately measures the Walsh support product
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:10b33ae9570072c13faeaf5093bd33adb08f6cc638bfc9eb8534039a811a012; engine receipt
sha256:92401779cd8580dbaf3962207ae25e0b7fc75787f91e27681d1ac290de403085 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050-92401779cd8580db.json; empirical-validation hash
sha256:1c47006310cbe6ee2c744184c71d29789299e7ccf62f3d81b51de8b4fb68a4b7; measurement receipt
sha256:c97b436ea794ddadea73446b5e06a1a88a0a23119e66279f989cba218a71c2b6.

334. Terminal electroweak measured-value correction

Claim identity: SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact forced terminal on-shell share lies inside the complete PDG interval [22333,22351]/100000. Its exact complement lies inside the outward-propagated compatible-input W/Z squared interval. No uncertainty is widened and no inconsistent aggregate is converted into a passing or failing SFT law.

The exact forced terminal on-shell share lies inside the complete PDG interval [22333,22351]/100000. Its exact complement lies inside the outward-propagated compatible-input W/Z squared interval. No uncertainty is widened and no inconsistent aggregate is converted into a passing or failing SFT law.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed terminal electroweak value, like-typed direct measurements, exact interval transport, source consistency, custody, proof/measurement separation, successor closure and extension. The declared exact boundary is: The admitted target-free terminal on-shell value; the complete direct on-shell row; the complete compatible W and Z rows; the unchanged internally inconsistent all-input aggregation record; both source identities; and every measurement-method boundary. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-terminal-electroweak-value-versus-like-typed-direct-measurements__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__on-e-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-electroweak-value-versus-like-typed-direct-measurements	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated form	Elimination reason
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:191e23e5365839ffb39393534fad4c68cbb3c69a26526969dd2a2aa70a15bcee`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3e4b8abb05c6d184cb34a5a25f61a48cc359196fa3dd7514ef58bd8faf631abb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7b78e6fbad92bcd6ed876234c371f9ed1c4a68f45d8bf116c90814ba7403638`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:76e5b62b331c12427c35dbdd6a447ce1f368b12099344a66614725bb637d339b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:522d95548ec25f2dc17eb43ebaff006b93c24978311ef817c611f7f1f2b8eb28`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0d160846dfd86800df046d3e306ab0801659da92682854572c47e80fff5724d3`. Independent certificate: `sha256:bdb4d5c3e0a39efc8e1b4da389915a147cc4ce1089435221e0618fded1b7cfce`. Engine external-validation hash: `sha256:18d3d16b8154e97a765842993bf67bae6d35a558ba3824aa5787478415a389f1`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-STANDARD-MODEL-REVIEW
- PDG-2024-W-BOSON-LISTING

Observed comparison records:

- The exact forced on-shell share 1930922298157999/8642477221479757 lies inside [22333,22351]/100000.
- Its exact One-complement lies inside the complete compatible-input W/Z squared interval.
- No uncertainty was widened and no measured row selected the formal relation.

- The all-input aggregate and its uncertainties remain unchanged, but its source-identified incompatible input prevents it from being treated as a second like-typed physical target.
- Tampered direct-angle and compatible-W controls reject.

Falsification condition: Reject if either source changes; the sealed exact share leaves the direct on-shell interval; its complement leaves the compatible W/Z squared interval; any uncertainty is widened; the inconsistent aggregate is deleted, used as a second law, or relabelled as an SFT result; or target data alter the formal survivor.

Measurement receipt: sha256:fde304793a48fcd4a37491fce37c4c1f7b254c39eccc9759ba83bd4fab871e7.

Isolation certificate: sha256:c0357f9ab5def03b84b5aeda8ed2025cf948e92685b8c4d00e07b1152b38663c.

Custody certificate: sha256:f5df2b4aa8c65fbe72832ec209c323fbe46f8ea1071271d17ae09ce89fb9ae7d.

Registered source descriptions:

- Review of Particle Physics: Electroweak Model and Constraints on New Physics:
<https://pdg.lbl.gov/2025/reviews/rpp2025-rev-standard-model.pdf>
 (sha256:8642888a3408d8c57fc673b379325b07f02948135491f64a2e42320e8929320a)
- Review of Particle Physics: W-boson listing: <https://pdg.lbl.gov/2024/listings/rpp2024-list-w-boson.pdf>
 (sha256:91cb466bfea8fa49b53ae53d2168797189c14a5a114f30d7cc926f64c4c1e772)

Registered target identities:

- PDG-ELECTROWEAK-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from
 PDG-2025-STANDARD-MODEL-REVIEW

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured weak angle or boson mass in the formal relation
- no fitted rung, correction, coefficient, uncertainty multiplier or selected central value
- no mismatch relabelled as successful empirical closure
- no deletion or alteration of the all-input aggregation record
- no consensus classification used as proof; source consistency is checked from the published record
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:c6c5f370a54459c49dc300cfe78fe04ca6b1a2b9d3c23c6fecb1006e6631123f; engine receipt

sha256:e93f04515944a1951b78463db3083350fbc77e6d8506edbcddc13178cedfbc852 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053-e93f04515944a195.js
 on; empirical-validation hash

sha256:4152cde723ac46d74d2053f6b43f8ac609c54133f27b27750e673223a7f8d7df; measurement receipt

sha256:fde304793a48fcd4a37491fce37c4c1f7b254c39eccc9759ba83bd4fab871e7.

335. Common Fold scale measured-value correction

Claim identity: SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The terminal on-shell share lies inside [22333,22351]/100000. The support-eight share 25/106 lies inside [2331,2367]/10000. The complete registered low-transfer vector preserves the sub-W direction. No mismatch, uncertainty

multiplier, fitted rung or measurement-selected exception is admitted.

The terminal on-shell share lies inside [22333,22351]/100000. The support-eight share 25/106 lies inside [2331,2367]/10000. The complete registered low-transfer vector preserves the sub-W direction. No mismatch, uncertainty multiplier, fitted rung or measurement-selected exception is admitted.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030
- SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053
- SFT-PHYS-COUPPLING-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of common-scale carrier, target-free relation, provenance, custody, proof/measurement separation, row completeness, successor closure and absence of extra rules. The declared exact boundary is: The admitted common-axis receipt; terminal and support-eight exact shares; all four scheme rows; all four numeric low-transfer rows; all eight plotted measurement classes; threshold boundaries; complete strong/electromagnetic receipt; and exact source identities. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-common-scale-values-versus-like-typed-measurements__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-common-scale-values-versus-like-typed-measurements	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:dcdd173f7ca0731f1c4330774a7d9b935db9e655a6eecd78b139eaf107e6db12`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:85653d59c680816e42666db3be96a88a1a392189b7bef80bda4929e13ab36f54.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:57af3942a29ed8a3075b1b67012bceab6fc7696e61057e9bd160186ff7dfdfb3.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8bfd0144733672b4df92ccafcd2d8adb6a8eed355f33f69a2c90d98143251ac.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b9f8f831567c843db4b2268175a876079d6341e2f9ffd633b52ba46ce5fac0d6.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:c5a642b9a74d2c41c3e81b7cfd550e2dc1003d4e4d59c7500759e43760495805. Independent certificate:

sha256:8f17c55fb5249ada38631b72e5d7d780f1e38575b3cd70ecee171406ede09fe. Engine external-validation hash:

sha256:29fcb4125abefbdef3a90b5ae64f4d21af9460bacc7e0410b88eefda51d438ba.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2026-ELECTROWEAK-SCALE-VECTOR
- PDG-2025-2026-STRONG-EM-COMPLETE-RUNNING-VECTOR

Observed comparison records:

- The exact forced terminal weak share lies inside [22333,22351]/100000.
- The exact forced support-eight share 25/106 lies inside [2331,2367]/10000.
- The complete E158 and APV intervals preserve the sealed below-W direction relative to the MS-Z interval.
- NuTeV remains unchanged as a source-identified interpretation-sensitive DIS extraction; its displacement is not an SFT result or acceptance condition.
- No row was deleted and no uncertainty was widened; displaced terminal and APV controls reject.

Falsification condition: Reject if a source or dependency changes; either forced value leaves its like-typed measured interval; the below-W direction reverses; a row or threshold boundary is omitted; NuTeV displacement is made a passing result; any uncertainty is widened; or target data alter the formal survivor.

Measurement receipt: sha256:4bc4c5859b4f63e25cc5a73d4814a0f0030c4668e4b01ac2b95e6ac075ee7661.

Isolation certificate: sha256:ddf578f829cf128b80abcd8fddfdb2932301a9f31f95ea82e6d94b6f1868c44e.

Custody certificate: sha256:30a93b5c0742b2693862b910e4cc205ecdc38d1e1b40a58954623ea3ca788efd.

Registered source descriptions:

- Review of Particle Physics 2026: Electroweak Model and Constraints on New Physics:
<https://pdg.lbl.gov/2026/reviews/rpp2026-rev-standard-model.pdf>
(sha256:a102f6252b7190dc423200271dffa7c805cd15a50391b1c578853d2f777611cb)
- Complete admitted strong and electromagnetic running vector: repository-bound source record
(sha256:b83331089d96c073fbd5101753ba5c4716ae1a8b1b891e068684b6f7246d9953)

Registered target identities:

- PDG-COMMON-SCALE-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from PDG-2026-ELECTROWEAK-SCALE-VECTOR

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured weak angle in the formal common-axis construction
- no fitted support, coefficient, correction, uncertainty multiplier or selected central value
- no NuTeV displacement relabelled as successful empirical closure
- no deletion, alteration or silent retyping of any registered measurement row
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:a4943e87cae5b12b391da771e8435eb70068faff98f29bea497e45577fc7a96f; engine receipt
sha256:0514d2a2fe7ad505dfd99c494f0c6ac6214169791c09596d4657e32ee1c458d8 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054-0514d2a2fe7ad505.json; empirical-validation hash
sha256:a27a65924cde81c28c15ac470e4e89221a82d44493ee31ae50e4e581f83eb326; measurement receipt
sha256:4bc4c5859b4f63e25cc5a73d4814a0f0030c4668e4b01ac2b95e6ac075ee7661.

336. Terminal cosmic transport measured-value correction

Claim identity: SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact upper enclosure of the complete 32-row mean squared normalized residual is below the One after the fourth exact enclosure round (three forced bisection refinements). The exact 11/5, 22/5, 17/32 and tension-One values lie inside the complete registered Planck, Haridasu and constant-state intervals. No sigma multiplier, fitted component, uncertainty rescaling, deleted row or mismatch-as-success predicate remains.

The exact upper enclosure of the complete 32-row mean squared normalized residual is below the One after the fourth exact enclosure round (three forced bisection refinements). The exact 11/5, 22/5, 17/32 and tension-One values lie inside the complete registered Planck, Haridasu and constant-state intervals. No sigma multiplier, fitted component, uncertainty rescaling, deleted row or mismatch-as-success predicate remains.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-PROBABILITY-STATISTICS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of exact transport carrier, unit-normalized residual relation, provenance, custody, proof/measurement separation, complete rows, successor closure and no extra rule. The declared exact boundary is: The admitted terminal transport receipt; all 32 CCH rows; H0 and complete Planck budget; both

registered acceleration reconstructions; constant-w interval; DESI model-comparison record; exact rational root enclosures; and every source identity. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-cosmic-transport-versus-complete-unit-residual-and-like-typed-values__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-cosmic-transport-versus-complete-unit-residual-and-like-typed-values	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:686a44491ee46aac075aa4752daa819f6417f039614f1f361594560a3b246a25`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:624147300356701da75d1412c41d5a6fdfbcfd9dc40b727d4e7cf83350d4e4a6`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1924ea5750e2ae67ee9bfacca7939eb2ecbed5b8312b9812998458a02b5758c1`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:57a3b73bd4e72c2475d11e25b201e6b86988e15fcdec4f5c1bcf6c3c248febd0`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e9198019efe4e3f15762d11adf9acd3777c4728d42e4a65b5062087f18936725`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:94a56325bcf7bab112806fef190159f248a5ebb19391bb72d6832f0ecedb6af6`. Independent certificate: `sha256:a4916f203d0a194a8fa9028afbcda359a26b81d0ddb6280f3bfde2ce0df8eaa6`. Engine

external-validation hash:

sha256: c073fd4672a744a2fabb64a70d07e9cb00a6bc145bd0fd2a85f7e7afb46f3c58.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- GOMEZ-VALENT-2023-CCH-32
- PLANCK-2018-BAO-BASELINE-BUDGET
- HARIDASU-2018-LATE-EXPANSION
- GOMEZ-VALENT-2019-ACCELERATION
- ESCAMILLA-2024-DARK-ENERGY-STATE
- DESI-DR2-2025-COSMOLOGY

Observed comparison records:

- All 32 direct chronometer rows contribute once to a complete exact normalized-residual ledger; its upper enclosure is below the One on the fourth exact enclosure round, after three data-independent rational-root refinements.
- The exact equality 11/5 and onset 22/5 lie inside the complete Planck budget transports.
- The exact acceleration magnitude 17/32 and onset 22/5 lie inside the complete Haridasu q0 and transition intervals.
- Tension-One lies inside the registered constant-state interval; alternate reconstruction and DESI model-comparison records remain unchanged but are not acceptance conditions.
- No uncertainty multiplier was selected or widened; tampered chronometer and transition records reject.

Falsification condition: Reject if a source or dependency changes; the complete normalized-residual lower enclosure reaches the One; any forced threshold, acceleration or static-state value leaves its like-typed interval; any row is omitted; an uncertainty is rescaled; a method/model-comparison displacement is rewarded; or target data alter the formal survivor.

Measurement receipt: sha256: 49342e727ece7edb6b0a68b4d001c15b772ae3d10898be217b458b03ede64b2b.

Isolation certificate: sha256: 0c8e6b029ffb29a423ec31f2af85827dbb7aa0e416d4efd4a4d53aeedc658673.

Custody certificate: sha256: 072be24ecd90149ee7303d23dba00c628b498d56d4bafef71b1ce1a0f30db2d1.

Registered source descriptions:

- Cosmic chronometers to calibrate the ladders and measure the curvature of the Universe: <https://doi.org/10.1093/mnras/stad1511> (sha256:cbd38055c3c9b27b2cbe1fb7456c60dfd7df3eeb366182932b7899d25207b128)
- Planck 2018 plus BAO baseline parameter table: <https://doi.org/10.1051/0004-6361/201833880> (sha256:a8525585688e7ba818f8650fc0f8b73a449823d5e199a8be36e37b8e73a0e612)
- An improved model-independent assessment of the late-time cosmic expansion: <https://doi.org/10.1088/1475-7516/2018/10/015> (sha256:4ac5d80b39e3dc963d5cc7a0c1b8996b0b895a974f08fa736803186bd100f74c)
- Quantifying the evidence for the current speed-up of the Universe: <https://arxiv.org/abs/1810.02278> (sha256:55100ed889f8aad5ada957f33759d954466b944d21b461b4c63c62f0b0fb45d0)
- The state of the dark energy equation of state circa 2023: <https://arxiv.org/abs/2307.14802> (sha256:67c8d75e79fd85d727012e72e8838651de6720a2e8898e225e251601814c20a5)
- DESI DR2 Results II: BAO measurements and cosmological constraints: <https://arxiv.org/abs/2503.14738> (sha256:1e82f26e4cc3901b16168cd147f252bfa804f9c3caad3f4f7e3532640d237841)

Registered target identities:

- COSMIC-TRANSPORT-WITHHELD-COMplete-MEASURED-VALUE-RECORD from GOMEZ-VALENT-2023-CCH-32

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result

identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no external H, density, q, transition or state target in the formal transport law
- no fitted H0, density, exponent, residual coefficient or selected uncertainty multiplier
- no rowwise two-standard-uncertainty shortcut
- no DESI model preference or alternate reconstruction relabelled as an SFT result
- no deleted measurement row or rescaled uncertainty
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:0fe774dfc5e9e0d08a491f543b4c3012dc0bd07ed2a9e715f47636f519c84c64; engine receipt
 sha256:e3e00a1c023f76cd0817c7eaec86c0243ab10d338e552f6bae1f90a94a959380 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055-e3e00a1c023f76cd.json; empirical-validation hash
 sha256:b050727f8732043890a71a6cb17e10e5581e06453e7f5c374ad632a2aa562c73; measurement receipt
 sha256:49342e727ece7edb6b0a68b4d001c15b772ae3d10898be217b458b03ede64b2b.

337. Criticality and turbulence measured-value correction

Claim identity: SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. All five manganite structural keys identify the binary local-order mean-field class. The complete fifteen-value mean squared normalized residual is exactly 5286961/10584000, below the One. The erbium beta/gamma/nu intervals contain 1/2, One and 1/2; the turbulence interval contains 2/3; and both physical routes exhibit the registered falling five-thirds plateau. No material mismatch is an acceptance condition.

All five manganite structural keys identify the binary local-order mean-field class. The complete fifteen-value mean squared normalized residual is exactly 5286961/10584000, below the One. The erbium beta/gamma/nu intervals contain 1/2, One and 1/2; the turbulence interval contains 2/3; and both physical routes exhibit the registered falling five-thirds plateau. No material mismatch is an acceptance condition.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-PROBABILITY-STATISTICS-001

Generated grammar and closure boundary. Generate the complete eight-axis product of class carrier, complete structural-key relation, provenance, custody, proof/measurement separation, all-row retention, successor closure and no extra rule. The declared exact boundary is: The admitted two-class formal census; complete structural key for each of five manganites; all fifteen manganite exponent values; complete erbium vector; complete turbulence structure interval; both physical spectrum routes; limitations; and every source identity. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-class-keys-versus-complete-measured-vectors__source-bound-proof-trace__capability-closed-prediction__separa

te-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-class-keys-versus-complete-measured-vectors	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash sha256:e1d06bfae39d2baff3af1ee014becfb762a4b6c7b83b9d53d33f73dc5b647353. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:1719b267af4323421d187741a62fca2f5e1213bb21ece4856f4b3b95d445c11d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4eafce2cb6316ccb3f6e636fb0c19676c7053c57e782ed0653bda8898ac0d21a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:f3db1e1d3ac6e5656325c9d1a8aa679172df0a5fe985f6b0f652e110c0b3c3fb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:82c45b89bef04fe802501c9d86a6ede46331807acc9e22c09efb48fe0d14b6bd.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:abecd535b035d40744922b3b96b09e6fbb2a89d4ccd2fb516f0531445269b187. Independent certificate: sha256:b056e6b7a7cba98ff7975a1e6b5fdf01d22ab2a91bbb18a36a9269d25a30c965. Engine external-validation hash: sha256:299fa83470a7c8596c73ec61feda3a7e046673df23abd28923bc07be8918cd31.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- SPRINGER-CETIN-MANGANITE-CRITICAL-EXPONENTS-2026
- MCMASTER-LIN-ERBIUM-CRITICAL-SCATTERING-1993
- APS-MCCOMB-TURBULENCE-STRUCTURE-2014
- APS-HUANG-TURBULENCE-SPECTRUM-2010

Observed comparison records:

- All five complete manganite structural class keys identify the registered mean-field class.
- All fifteen beta/gamma/delta measurements enter once; their exact mean squared normalized residual is 5286961/10584000, below the One.
- The complete erbium vector contains one-half, One and one-half; the independent turbulence interval contains two-thirds.
- Both physical spectral routes exhibit the falling five-thirds compensated plateau with all limitations retained.
- La02 remains visible without its displacement becoming a result; tampered vector and structural-key controls reject.

Falsification condition: Reject if a source or dependency changes; any material structural key is incomplete; the complete fifteen-value mean squared residual reaches the One; erbium or turbulence excludes a sealed exponent; either spectrum route is absent; any row is omitted or uncertainty rescaled; La02 displacement is rewarded; or targets alter the survivor.

Measurement receipt: sha256:27d3a2750e7db5e6204fbeb1b647aa84ac9a4dfad6b9de6a383638e56f9a3402.

Isolation certificate: sha256:9535981b71f46414bf904845396564a93e81667af7b4faf6410228639f0798a.

Custody certificate: sha256:be2305b343aaa6b76070795f86a948a65d6d7f04dffec2428a216149f722d20e.

Registered source descriptions:

- Structural, magnetic, and magnetocaloric properties of La and Sm co-doped CaMnO₃ manganites:
<https://doi.org/10.1007/s10854-026-17400-y>
(sha256:afd0f783bd40b8ed376238eda9b254387c75c0c26ecc57488ab9b81b33d1d455)
- Critical magnetic scattering in erbium: <https://doi.org/10.1063/1.353720>
(sha256:d8b6495b7defb46daa4d1162967d11b4680012ebd4a8e6d60e8892b83b9b3564)
- Spectral analysis of structure functions and their scaling exponents in forced isotropic turbulence:
<https://doi.org/10.1103/PhysRevE.90.053010>
(sha256:0c9f4321b52f78fe64b02684b4139a450955c12092c4faf8487f49bf6ff1d0f5)
- Second-order structure function in fully developed turbulence: <https://doi.org/10.1103/PhysRevE.82.026319>
(sha256:1bff196a24bc3f852dd384469c5f8e4112fcd717c2416190ad44b0012ac06cb)

Registered target identities:

- CRITICALITY-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from
SPRINGER-CETIN-MANGANITE-CRITICAL-EXPONENTS-2026

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured exponent or material label in the formal class census
- no selected matching subset or La02 mismatch-as-success predicate
- no fitted exponent, uncertainty multiplier, intermittency correction or measurement-selected class
- no row deletion or uncertainty rescaling
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:972770658891b5371be10884db72f00aacd524b07d5fe190dc434a5567c25f4f; engine receipt
sha256:7e437d81bee9d4c88b304a013c5c980fa9d0390f95acaf30c2dfb45b30a1602a at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056-7e437d81bee9d4c8.js
on; empirical-validation hash
sha256:51ddbfff9c9736ecc850a4828f962f1499b2fdd51809d6e252688eebdeaf7828c; measurement receipt
sha256:27d3a2750e7db5e6204fbeb1b647aa84ac9a4dfad6b9de6a383638e56f9a3402.

338. Thermal history and physical helium measured-value correction

Claim identity: SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Exact exponent One lies inside [49/50,517/500]. Exact physical helium-isotope share 59/240 lies inside [489/2000,2471/10000]. The one-sixth to one-seventh freezeout sequence, positive deuterium, finite recombination support, eighteen extrema and all seven TT rows remain complete. No excluded quarter or angular noninteger record is admitted as a successful result.

Exact exponent One lies inside [49/50,517/500]. Exact physical helium-isotope share 59/240 lies inside [489/2000,2471/10000]. The one-sixth to one-seventh freezeout sequence, positive deuterium, finite recombination support, eighteen extrema and all seven TT rows remain complete. No excluded quarter or angular noninteger record is admitted as a successful result.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037
- SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of physical thermal carrier, isotope relation, provenance, custody, proof/measurement separation, complete rows, successor closure and no extra rule. The declared exact boundary is: The admitted analytic and physical-isotope receipts; complete temperature, helium, deuterium, freezeout, recombination and acoustic records; all uncertainty endpoints; all seven TT rows; all eighteen extrema; projection boundary; and every source identity. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-physical-thermal-values-versus-complete-measurements__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	sealed-physical-thermal-values-versus-complete-measurements	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:71d5b821c08b24d0a200db067d1344d662e111fd6b5b28faad0fb8bc19cc9dd5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d8bb99bae704fa0f9359dba4a0077ebb8e4a6ac52415f747430e8fe64fef82a2`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a96d9010099338a297ce51e80473e184dd3f77cf24a73a0b87f2751a0407e06e`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bd8b792a7864e21a1369417e13b923b95dee61035cb784af21ec4fd3bfff2d139`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:93b71dfa37d98bc1b88e9b4ab0e4dd780330e0080ff83b251d8f83bb9f56eabb`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:d3afd654004d34d24ba55451eda4328e7b249a86254e778902bcfc98e0dfe7e6`. Independent certificate: `sha256:61be22939d2ea95b2cc9e77c9d966e7dd3d30860ba7f9caeae64d805a13069e5`. Engine external-validation hash: `sha256:e0a2835a1ae33c589af028c1579f4e7a0ef3b3d20d95f099031507e2a749f214`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NOTERDAEME-ET-AL-2011-CMB-TEMPERATURE
- AVER-ET-AL-2026-PRIMORDIAL-HELIUM
- COOKE-PETTINI-STEIDEL-2018-DEUTERIUM
- PLANCK-2018-OVERVIEW-PEAKS

- PLANCK-2018-COSMOLOGICAL-PARAMETERS
- PDG-2025-BBN-REVIEW

Observed comparison records:

- Exact CMB temperature exponent One lies inside [49/50,517/500].
- Exact physical primordial helium share 59/240 lies inside [489/2000,2471/10000].
- The one-sixth to one-seventh freezeout sequence and positive deuterium channel remain complete.
- Finite recombination support, eighteen extrema and all seven TT rows remain source-bound and positive; angular projection records are not mismatch results.
- No uncertainty was widened; displaced helium and incomplete-peak controls reject.

Falsification condition: Reject if a source or dependency changes; exponent One or 59/240 leaves its complete measured interval; the freezeout sequence, positive deuterium, finite recombination support, extrema or TT rows are missing; an uncertainty is widened; a mismatch is rewarded; or targets alter either formal survivor.

Measurement receipt: sha256:8663e550fdb13a666d3c2b1d099e1b9f44b175879d5652b7909bc1e4ee0cbbcf.
Isolation certificate: sha256:c2d36c150cf297b75ea33c3be9994f9032dc76d36def77c065ed4bba8e0f36d2.
Custody certificate: sha256:c0e4760965d8d0c0f6e4e11a1bf5a19d2ddcc3404e752855a9a3c46dba600899.

Registered source descriptions:

- The evolution of the Cosmic Microwave Background Temperature: <https://arxiv.org/abs/1012.3164>
(sha256:aa79c65170e84d6a8dc71dbd0876538c19619a1c926453ce5a54bd96d3f1efb7)
- The LBT Yp Project IV: A New Value of the Primordial Helium Abundance: <https://arxiv.org/abs/2601.22238>
(sha256:72f646a345c36b77211e6035d3a78345498f0766cf9fc0a54932ebe0dd670c6a)
- One percent determination of the primordial deuterium abundance: <https://arxiv.org/abs/1710.11129>
(sha256:1a6ec19d6568b8854a40e9dd587906f99a8ea94ecd2d5c712ad86aa506b93f4a)
- Planck 2018 results I: overview and cosmological legacy: <https://arxiv.org/abs/1807.06205>
(sha256:dca932893b7d2724aa2b8d33170fc1b5682c425dd56fcd3c5d7b2098d377db5c)
- Planck 2018 results VI: cosmological parameters: <https://arxiv.org/abs/1807.06209>
(sha256:8e172730faf07c9f4ff3fdcc7043f76ed67df6f76066d47df30d693025b6ce77)
- Big-Bang Nucleosynthesis: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-bbang-nucleosynthesis.pdf>
(sha256:1843c1d9025aa33c059700708f1041c4462364cdb582325f0c685a6ba1b38484)

Registered target identities:

- THERMAL-HISTORY-WITHHELD-COMplete-MEASURED-VALUE-RECORD from
AVER-ET-AL-2026-PRIMORDIAL-HELIUM

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no measured helium central value or uncertainty in either formal survivor
- no fitted decay, capture, binding, reaction, projection or abundance coefficient
- no exact-quarter-exclusion or noninteger-angular-position mismatch rewarded as closure
- no deleted row or rescaled uncertainty
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:03d3d89369145a006af94d8a419a408810d74d099a771168fec92d1b766194af; engine receipt
 sha256:7339aa07e1a9f52e8ef1fede03cbe17d38834362f1c9d63bb00b05deb6d49021 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058-7339aa07e1a9f52e.json; empirical-validation hash
 sha256:2376571e5468f9eadcadf80482ad61e6a43a9b0bd50a3812c903ee5f114eaa47; measurement receipt
 sha256:8663e550fdb13a666d3c2b1d099e1b9f44b175879d5652b7909bc1e4ee0cbbcf.

339. Complete hadron Regge measured-value correction

Claim identity: SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The forced squared carriers 3/5, 9/5, 3, 21/5 and 27/5 all lie inside their corresponding complete measured resonance-support intervals. All five rows and the rho5 omission status are retained. No fitted slope, intercept, residual, selected row, uncertainty widening or mismatch-as-success remains.

The forced squared carriers 3/5, 9/5, 3, 21/5 and 27/5 all lie inside their corresponding complete measured resonance-support intervals. All five rows and the rho5 omission status are retained. No fitted slope, intercept, residual, selected row, uncertainty widening or mismatch-as-success remains.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-HADRON-REGGE-TERMINAL-005
- SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed Regge carrier, complete resonance support, provenance, custody, proof/measurement separation, complete rows, successor closure and no extra rule. The declared exact boundary is: The formal 059 receipt; all five registered masses, uncertainties, widths, width uncertainties and statuses; the exact most-restrictive support construction; common post-seal GeV-squared translation; and all source identities. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-exact-regge-carriers-versus-complete-measured-resonance-supports__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-exact-regge-carriers-versus-complete-measured-resonance-supports	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.

Axis	Forced form	Eliminated form	Elimination reason
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:9576dd4d668afb3fa2951c8e13c59ec3779a02432720bd2a996d06280a35fc15`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:909b4ce12e13596c9c1aa2b479c79c5d9bd49154e27dba348aa3bfbd6d11a683`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:af80693aeb595bd40d2593d435871114812faa4839bae161891105d654f06b58`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3cff8caf802fccc5e3d071067220be16da18b1ada81fcd6e29ddb5345415346d`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d5d152764842943169dc695f4c58d24fe26ace16bb8e830a768f0adfb194a7ac`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:0c8bc06ef57ebb6e35a54fb76c53932605eebaea3d9d70ffdbde3261c5aee370`. Independent certificate: `sha256:92422c7be03302267a28cc17db31e79b1e7a1236591bbd40866a1b6cb8e39dfe`. Engine external-validation hash: `sha256:f6b749a0f22d24de38b5f7e9d0743d1420f8dd540efcf7f885c378d5eebeb1bb`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-RHO-770
- PDG-2025-A2-1320
- PDG-2025-RHO3-1690
- PDG-2025-A4-1970
- PDG-2025-RHO5-2350

Observed comparison records:

- Exact zero-parameter Regge base is $3/5 \text{ GeV}^2$ and exact successor step is $6/5 \text{ GeV}^2$.
- All five exact squared carriers lie inside the corresponding most-restrictive measured resonance-support intervals.

- Every mass, mass uncertainty, width, width uncertainty and listing status is retained once.
- The rho5 summary omission and single-measurement status remain explicit.
- No slope, intercept, residual or correction is fitted; displaced and missing-row controls reject.

Falsification condition: Reject if a source or dependency changes; any exact carrier leaves its most restrictive measured resonance support; a row or status is missing; a width or uncertainty is widened; a coefficient or correction is fitted; a mismatch is rewarded; or targets alter the formal survivor.

Measurement receipt: sha256:26f6d86b2e87b8eccb5e37b1685aec25dfdc5d151d16c80a04217dd6d85502b0.
Isolation certificate: sha256:c62628600e61beb7f0c8ee49456ef12c27caf8e1ab3c52887a2fc68acdb7996f.
Custody certificate: sha256:295f755df18b9ca5edde5d8478bf4e238a5754e38b9fb1f54d3bb3b9917011e6.

Registered source descriptions:

- Particle Data Group: <https://pdg.lbl.gov/2025/listings/rpp2025-list-rho-770.pdf>
(sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)
- Particle Data Group: <https://pdg.lbl.gov/2025/listings/rpp2025-list-a2-1320.pdf>
(sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)
- Particle Data Group: <https://pdg.lbl.gov/2025/listings/rpp2025-list-rho3-1690.pdf>
(sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)
- Particle Data Group: <https://pdg.lbl.gov/2025/listings/rpp2025-list-a4-1970.pdf>
(sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)
- Particle Data Group: <https://pdg.lbl.gov/2025/listings/rpp2025-list-rho5-2350.pdf>
(sha256:6877a2e12365b035d3391a1a28af2f49693d8a749df276be3c772491a50ca356)

Registered target identities:

- HADRON-REGGE-WITHHELD-COMPLETE-MEASURED-VALUE-RECORD from PDG-2025-RHO-770

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no pole-mass standard uncertainty mislabelled as the support of an unstable resonance
- no fitted slope, intercept, string tension, residual, mass correction or selected trajectory subset
- no widened mass uncertainty or resonance width and no omission of rho5 status
- no target access before the formal survivor and prediction label seal
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:a0c377dd985d02964b48a385b4625fa59a3208339d8f60503314f04a7fced0ff; engine receipt
sha256:15c9daaa90d88c5193e5e3af4a5ca64a75c0418ee85797a2b90607c37a21cb06 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060-15c9daaa90d88c51.j
son; empirical-validation hash
sha256:b66c9c9da0cfb84a24085e0b1d1bdb2ec00e7fd64a02bdbbc0210b9403aca1f1e; measurement receipt
sha256:26f6d86b2e87b8eccb5e37b1685aec25dfdc5d151d16c80a04217dd6d85502b0.

340. Complete particle-generation, mass-pattern, mixing and longevity comparison

Claim identity: SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The forced count three lies in the PDG LEP-SLC precision interval [2989,3003]/1000. The independent direct determination 292/100 with stated uncertainty 5/100 is preserved exactly and has central-value displacement 8/5 stated uncertainties from three; it is neither adjusted nor used to select the result. Both terminal charged-lepton mass-ratio intervals, the available s/d and b/s quark ratio intervals, all four terminal CKM intervals and the positive-neutrino mass/mixing support pass their registered comparisons; t/c remains without an exact scheme-matched direct comparator. The charged-lepton records give $m_e < m_\mu < m_\tau$ and $\text{lifetime}_e > \text{lifetime}_\mu > \text{lifetime}_\tau$. These measurements test the terminal polynomial and ordering consequences, not the superseded site-as-mass, universal rate-ratio, dimensional lifetime-equality or extra-spatial-dimension claims.

The forced count three lies in the PDG LEP-SLC precision interval [2989,3003]/1000. The independent direct determination 292/100 with stated uncertainty 5/100 is preserved exactly and has central-value displacement 8/5 stated uncertainties from three; it is neither adjusted nor used to select the result. Both terminal charged-lepton mass-ratio intervals, the available s/d and b/s quark ratio intervals, all four terminal CKM intervals and the positive-neutrino mass/mixing support pass their registered comparisons; t/c remains without an exact scheme-matched direct comparator. The charged-lepton records give $m_e < m_\mu < m_\tau$ and $\text{lifetime}_e > \text{lifetime}_\mu > \text{lifetime}_\tau$. These measurements test the terminal polynomial and ordering consequences, not the superseded site-as-mass, universal rate-ratio, dimensional lifetime-equality or extra-spatial-dimension claims.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002
- SFT-PHYS-VALIDATION-QUARK-CKM-003
- SFT-PHYS-MATTER-CKM-TERMINAL-004
- SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed particle-mode law, full generation/mass/mixing/lifetime vector, provenance, target isolation, proof/measurement separation, complete-row retention, successor closure and extension. The declared exact boundary is: The admitted Claim 051; the precision fitted neutrino-count measurement and the unchanged independent direct determination; both charged-lepton ratios; all available quark ratios and the missing-comparator boundary; all terminal CKM rows; the positive-neutrino mass/mixing vector; all three charged-lepton masses and lifetimes; heavy-lepton search limits; and every interpretive limitation. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-particle-mode-generation-law-versus-complete-registered-spectrum-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	sealed-particle-mode-generation-law-versus-complete-registered-spectrum-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:234198c4dedadb99627b00ac01684e59ec98279b3daffe9b154ec25a1a9131ace`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9fe79983d3b6a4b8bc543dfd8e9dc5f53b1908f3855e6cc7aa49b06f9a01b32b`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:206dd3eb485875b622471c84fb7661aff32d95c18749effd964e6c52ee7ae1f7`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:31ba6457edc32db6d1ad37a5d913d2d3b023e56034a79ac890b44265cb17ac22`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6f167c1bbb175fa1a86526e6de7ff54306a535b9c92573ad855a4822e3d8dd37`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:698b9367ac9c6f40c5463c18f1c0c8f682c4f00ec78afdac8de48b65050da1`. Independent certificate: `sha256:8fe3ed5259739b150bbef45c19438bd41f267c21bfb732c49a37c87eef94be14`. Engine external-validation hash: `sha256:8572aa491dcd0973dcb9e6e6bf5c163ec07f8b93df9736332fbaaa19f1bbcc9f`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2025-LEPTON-SUMMARY
- NIST-CODATA-2022-ALL-CONSTANTS
- PDG-2025-QUARK-MASSSES
- PDG-2025-CKM-MATRIX

- PDG-2025-NEUTRINO-MIXING

Observed comparison records:

- The precision fitted neutrino-type interval [2989,3003]/1000 contains the forced count three.
- The independent direct determination 292/100 with stated uncertainty 5/100 is retained without adjustment.
- Its exact central-value displacement from three is 8/5 stated uncertainties; it does not select or alter the forced law.
- Both terminal charged-lepton ratios, available s/d and b/s ratios, all terminal CKM rows and the positive-neutrino vector pass their registered comparisons.
- No exact scheme-matched direct t/c comparator is claimed.
- Charged-lepton mass intervals increase electron-muon-tau and lifetimes decrease in the same label order.
- Heavy-lepton search limits, measurement-model provenance and nondecisive neutrino-ordering boundaries are retained.
- No coordinate fraction is relabelled as mass and no structural reach count is relabelled as seconds.
- Tampered fit, altered direct record and reversed-lifetime controls reject.

Falsification condition: Reject if any source changes; the precision generation interval excludes three; the independent direct row is omitted, altered, rescaled or used to select the law; a registered terminal mass or mixing comparison fails; the charged-lepton mass/lifetime orders reverse; the missing t/c comparator is presented as measured; any search or model boundary is omitted; an old site coordinate is relabelled as mass; or target data alter the formal survivor.

Measurement receipt: sha256:69150d847fbccce53bef6578047d3afceb4cd159eb05e20b02549956ac43a03d.

Isolation certificate: sha256:bc0344002addffce79dec59fadb802a036be32f799a4e6f5ebf47b4e55bd75f8.

Custody certificate: sha256:26d5c53f2b8b66cd7d153272f3f414edd69eb0243963b7704a255786369ca026.

Registered source descriptions:

- Review of Particle Physics 2025 update: Leptons summary tables: <https://pdg.lbl.gov/2025/tables/rpp2025-sum-leptons.pdf> (sha256:5aa9a3a33b554204056cb04c42319bb32c9664ae518907652ff5cdc1cb87e5bb)
- 2022 CODATA recommended values: <https://physics.nist.gov/cuu/Constants/Table/allascii.txt> (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)
- Review of Particle Physics: Quark Masses: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-masses.pdf> (sha256:d544d099aa15739ec83d87711bd4c5b1e0a1032d6f70aaa4847ec78601f7aeae)
- Review of Particle Physics: CKM Quark-Mixing Matrix: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-ckm-matrix.pdf> (sha256:a0a78578971f38ff89c6fc5579bc608de41ec383a205dc25cba1d26f7145610a)
- Review of Particle Physics: Neutrino Masses, Mixing, and Oscillations: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf> (sha256:d7067e2e3c9098cc924f10ffbc579c557fb8e848bf3acc17f9815598cdda7a6)

Registered target identities:

- PARTICLE-MODE-GENERATION-WITHHELD-COMPLETE-RECORD from PDG-2025-LEPTON-SUMMARY

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no external particle mass, lifetime, generation count, mixing entry or search limit in the formal survivor
- no omission, rescaling or reinterpretation of the independent direct neutrino-count determination or the absent exact t/c comparator
- no use of generation coordinate fractions as measured mass values
- no claim that structural subtraction reach equals a dimensional lifetime ratio

- no claim that quarter-mode 1:1:4 is a universal observed mixing or decay-rate ratio
- no inference that heavy-particle search limits prove absolute nonexistence
- no extra spatial dimension or compactification rule selected from particle data
- no numerical-nothingness, negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7b1248718fba75f0e23a4f31f0b8fbf98cdd266cfd8fed9cc4c8e2d8660b55; engine receipt
 sha256:0a2ca2749eed1cdaf6b92ab645ace007b3190ac5e5bb952e3a0cbdedff1c0804 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052-0a2ca2749eed1cda.json
 ; empirical-validation hash
 sha256:0bae7a0db26bde3b738ec6c24f82bfc74856941136042da60d0971c4ef3feb7b; measurement receipt
 sha256:69150d847fbccce53bef6578047d3afceb4cd159eb05e20b02549956ac43a03d.

341. Measured dark abundance and complete Smithion/LFV standing-test record

Claim identity: SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The exact 27/5 ratio lies inside the complete Planck dark/baryon interval, and 27/5 times 0.0224 gives 0.12096 inside the reported cold-dark interval [0.119,0.121]. The 175-galaxy SPARC record rejects finite baryon-only inverse-square asymptotics and supports additional gravitating matter under the admitted law. MEG II and BaBar provide three positive upper limits after null searches, not three measured rates; the 3:5:20 and 4:1 LFV predictions remain standing tests. No measured Smithion mass exists, so all twelve mass-ratio enclosures remain standing predictions rather than fabricated measured-value matches.

The exact 27/5 ratio lies inside the complete Planck dark/baryon interval, and 27/5 times 0.0224 gives 0.12096 inside the reported cold-dark interval [0.119,0.121]. The 175-galaxy SPARC record rejects finite baryon-only inverse-square asymptotics and supports additional gravitating matter under the admitted law. MEG II and BaBar provide three positive upper limits after null searches, not three measured rates; the 3:5:20 and 4:1 LFV predictions remain standing tests. No measured Smithion mass exists, so all twelve mass-ratio enclosures remain standing predictions rather than fabricated measured-value matches.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061
- SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of the sealed law, Planck density rows, galaxy discriminator, LFV search rows, prediction status, target isolation, complete-row custody and no-extra-rule. The declared exact boundary is: The admitted Claim 061; both Planck density rows with uncertainties; the complete SPARC sample statement; all three registered LFV radiative upper-limit rows; every current Smithion measurement-status row; and no target access before seal. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-dark-smithion-lfv-law-versus-complete-current-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is

`depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-dark-smithion-lfv-law-versus-complete-current-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:e37c8e8a0fb85346cac4c921b758a9b0adbee5b5a358a99d1be8f27ef73116a4`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8b245e9d86ac5517361d422e8a43f2cee3d25a6c2e1db291e6617d70683e1605`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:eb6f888f446824a0cab8d1c97cc6d52652a7ffaed86670f240e31fe6b7493453`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7f053e393fab0a7adaae778ff2b7613165b852a1e23f2d3b9cd1a3d1d0c8cb02`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b28ad019c6371ee4d5bcebd9bcf0bae85ff05997cd206c01e9067efc1f4de628`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:e4a059747073e52276d7db230a328c84cfc618a73a9adbf29066b563e3de9b45`. Independent certificate: `sha256:edc9f8101bffff4a5025bb403ee6d1f854e6c149f73fdc87c80c3b78c1fc12731`. Engine external-validation hash: `sha256:c16302fba31483b63cb4c01692f1bda069ba8f75e09da2798a935bb2324a087b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PLANCK-2018-VI-ABSTRACT-DENSITIES
- SPARC-2016-175-GALAXY-MASS-MODELS
- MEGII-2025-MU-E-GAMMA
- BABAR-2010-TAU-LFV-GAMMA

Observed comparison records:

- The exact dark/baryon ratio 27/5 lies inside the complete Planck density-ratio interval.
- Comparison-side transport 27/5 times 0.0224 equals 0.12096 and lies inside [0.119,0.121].
- The complete 175-galaxy SPARC record requires additional gravitating support under the admitted inverse-square law; it does not alone identify a particle.
- MEG II and BaBar retain all three null-search upper limits. They do not constitute measured nonzero LFV rates, so no 3:5:20 match is fabricated.
- No registered Smithion mass measurement exists; all twelve spectra remain exact standing predictions.
- Tampered-density and fabricated-rate controls reject.

Falsification condition: Reject if any source changes; 27/5 leaves the complete Planck ratio interval; 0.12096 leaves the complete cold-dark interval; the galaxy discrepancy row is omitted; an LFV limit is altered or relabelled as a rate; a Smithion is relabelled as measured without a source; target data select the formal law; or any hostile control passes incorrectly.

Measurement receipt: sha256:2da4d4ef1c11b3dec958dac4f13b5bac740eea49b084d8131d511b81b59ed11a.

Isolation certificate: sha256:bd444dda787a62b1b01e8969c0d1e8aac8245ad3fc1740c14133b8c32c67bd8d.

Custody certificate: sha256:f5dd79b438273ad54e593d3009e4b590d9c8e3656e2cf0117cc0502032081cc0.

Registered source descriptions:

- Planck 2018 results. VI. Cosmological parameters: <https://arxiv.org/abs/1807.06209> (sha256:274e0189de5846ce0b8c2d7b83ae06c72587cf8325d4d2b2338e88dd0a74a88f)
- SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves: <https://arxiv.org/abs/1606.09251> (sha256:d089215877213661e40965543ee7e05736619082ad16d95e65ec059029588c63)
- New limit on the muon-to-electron-gamma decay with the MEG II experiment: <https://arxiv.org/abs/2504.15711> (sha256:0efbf99543c92d340eb7b07f40e6fea580b7746caab26c9233d3e30e8b71a5b6)
- Searches for Lepton Flavor Violation in the Decays tau to e gamma and tau to mu gamma: <https://arxiv.org/abs/0908.2381> (sha256:ba3e1352f4c14b1867e06f6436b4a6b1e1980cd55ce52d9a9454bd618ed70015)

Registered target identities:

- DARK-SMITHION-LFV-WITHHELD-COMLETE-RECORD from PLANCK-2018-VI-ABSTRACT-DENSITIES

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no density, galaxy datum, search limit or particle mass in formal survivor selection
- no upper limit relabelled as an observed rate or relative-rate measurement
- no null search relabelled as proof of particle identity
- no unobserved Smithion relabelled as a measured discovery
- no fitted abundance, cross-section, mediator mass, branching fraction or uncertainty multiplier
- no negative, irrational, imaginary, floating, NaN, continuum or completed-infinity Fold proof magnitude

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest
sha256:0111d63af2037414d6394950c563e819550f34d1196e42a6ee1a4614e0c787f9; engine receipt
sha256:51e623e6512223c23a0f0cae604ca556ed30fd15173236ffd1c889dd6424e27d at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062-51e623e6512223c2.json;
empirical-validation hash
sha256:8eb364d2768268207e98259bfe3f0269ddf1685006fb1d491750d3163521c832; measurement receipt
sha256:2da4d4ef1c11b3dec958dac4f13b5bac740eea49b084d8131d511b81b59ed11a.

342. Blind neutron-EDM and multi-channel proton-stability validation

Claim identity: SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The PSI result retains its null central status, positive 1.1 and 0.2 times ten-to-the-minus-26 uncertainty magnitudes and positive 1.8 times ten-to-the-minus-26 90-percent upper bound. The six proton modes retain 90-percent partial-lifetime lower limits 2.4e34, 1.6e34, 1.4e34, 7.3e33, 7.2e33 and 4.5e33 years. No source reports a statistically significant proton-decay signal; the explicitly retained candidates are background compatible. Thus no registered row violates the sealed aligned-One or fibre-preserving baryon-One predictions.

The PSI result retains its null central status, positive 1.1 and 0.2 times ten-to-the-minus-26 uncertainty magnitudes and positive 1.8 times ten-to-the-minus-26 90-percent upper bound. The six proton modes retain 90-percent partial-lifetime lower limits 2.4e34, 1.6e34, 1.4e34, 7.3e33, 7.2e33 and 4.5e33 years. No source reports a statistically significant proton-decay signal; the explicitly retained candidates are background compatible. Thus no registered row violates the sealed aligned-One or fibre-preserving baryon-One predictions.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed formal carrier, all registered EDM and proton-search rows, source binding, capability closure, proof/measurement separation, row completeness, successor closure and no-extra-rule. The declared exact boundary is: The sealed Claim 063; the direct neutron-EDM central-status, statistical uncertainty, systematic uncertainty and upper-limit rows; all six registered Super-Kamiokande mode limits and candidate interpretations; all four primary source snapshots; and no target access before the prediction seal. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-strong-alignment-and-baryon-stability-versus-complete-registered-direct-search-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.

Axis	Forced form	Eliminated form	Elimination reason
relation	sealed-strong-alignment-and-baryon-stability-versus-complete-registered-direct-search-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:38100d6833e49ccffbea934b617894201371284cc9db173c1944618d174e17b9`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:576d04f2cc2e35de96d1f95fd64b1543e032a14fbc27c4ffa7bcb81c18a74c40`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:636aa152173b656c5c090c6023503779b90c47d91a6c35b0fa26756a8bef684f`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ad6e2347a557026397acb3574713a1d49b7b2d0a4ae2ed9a7b9c97e3be0f3d68`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:afb52c0be23620df9d200e86ca2d5a08174b3678f1b571358534c5b77bea02c0`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:4d0799c837a396410e90962d9354a8b4a675a30f8bb6093077312c1c394f39eb`. Independent certificate: `sha256:219d673b42b16ff5a7b5ab93deca735ccae9d63fa7962177f2710df976e6eda0`. Engine external-validation hash: `sha256:1d3374be2ed6753471294e271f5bfb58cba46e4f0213f2cee5a7f84ba71f6a5b`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NEDM-2020-PSI-PERMANENT-EDM
- SUPERK-2020-PROTON-E-MU-PI0
- SUPERK-2024-PROTON-E-MU-ETA

- SUPERK-2026-PROTON-E-MU-TWO-PI0

Observed comparison records:

- PSI reports a null neutron-EDM central displacement with both positive uncertainty magnitudes and a finite 90-percent upper limit of $1.8e-26$ e cm.
- The neutron-EDM row is consistent with the sealed empty-One strong-dipole prediction; it is not relabelled as numerical proof zero.
- All six registered Super-Kamiokande proton modes retain their finite 90-percent partial-lifetime lower limits.
- No registered mode reports a statistically significant proton-decay signal; every candidate interpretation, including background-compatible events, is retained.
- Finite lifetime limits are not called infinity and do not select the already sealed proton-stability law.
- Confirmed-EDM and confirmed-proton-decay hostile controls both reject.

Falsification condition: Reject if any source changes; the direct EDM record reports a confirmed nonempty displacement; any proton mode reports a statistically significant decay signal; a candidate or uncertainty is omitted or relabelled; a finite limit is called exact absence/infinity; target data select the formal law; or any hostile control passes incorrectly.

Measurement receipt: sha256:13d218828f910869fcffd434f0367d39e0bbd287fb203e61f56303f6b7cf369c.

Isolation certificate: sha256:7a7bcee294d5be0d191c5ffefa7728d992a8bdaa3535aea6976af11a81bc8ce3.

Custody certificate: sha256:0f2c58711f4c0cc427856875accad7747aee8abbe00cda684166fe0a3c628cd0.

Registered source descriptions:

- Measurement of the permanent electric dipole moment of the neutron: <https://arxiv.org/abs/2001.11966> (sha256:1afeb830f372c7725cca9fd3d004333fe19cade698fd9798af9dfac7adc5c180)
- Search for proton decay via p to e-plus pi-zero and p to mu-plus pi-zero: <https://arxiv.org/abs/2010.16098> (sha256:6fa532b3b2671e8d6e028308297b95837567a6e3608b471aaa8af00fff6a9679)
- Search for proton decay via p to e-plus eta and p to mu-plus eta: <https://arxiv.org/abs/2409.19633> (sha256:5b4bf694378930d7e38935546fea9c5deced9531137b0da276778e7b67fb4937)
- Search for proton decay via p to e-plus two-pi-zero and p to mu-plus two-pi-zero: <https://arxiv.org/abs/2604.10975> (sha256:06d10f843d39b54a14465628d72a9af18920976369326b1c717e3621b2109c98)

Registered target identities:

- STRONG-CP-BARYON-STABILITY-WITHHELD-COMPLETE-RECORD from NEDM-2020-PSI-PERMANENT-EDM

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no neutron-EDM value, uncertainty, proton candidate or lifetime limit in formal survivor selection
- no null central estimate relabelled as conventional numerical proof zero
- no finite experimental upper or lower limit relabelled as an exact absence or completed infinity
- no background-compatible event relabelled as proton decay and no candidate row omitted
- no fitted theta, axion coefficient, decay rate, lifetime, confidence multiplier or tolerance
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:8892bb86644f083a8401009a9263a9eb545ef11baf0ad4641c7247d412119385; engine receipt

sha256:48e9dbebdb29b3902876a09cde2154eea3700904898930c1f12ab0a5e5ed9a09 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064-48e9dbebdb29b390.js on; empirical-validation hash
 sha256:725517ecde0475192e96b7c9f79419b33afebe5a5aacebf1dbae29ed89b05673; measurement receipt
 sha256:13d218828f910869fcffd434f0367d39e0bbd287fb203e61f56303f6b7cf369c.

343. Blind terminal Higgs mass and self-coupling validation

Claim identity: SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Using the post-seal PDG dimensional reference $v = 12311/50$ GeV, the sealed ratio predicts exactly $m_H = 31557437733819647/251923197734500$ GeV = 125.266104978... GeV. The PDG 2025 listed average is 626/5 +/- 11/100 GeV, whose reported-uncertainty interval [12509/100,12531/100] contains the prediction; the exact displacement from its central value is 16653377460247/251923197734500 GeV. ATLAS reports 12511/100 +/- 11/100 GeV and CMS reports 3126/25 +/- 3/25 GeV; their exact positive offsets remain in the certificate and are not called aggregate failures. The exact native lambda is 6570778165695741824959729/50772238045420788745992200. The 2026 ATLAS-CMS 95-percent normalized kappa-lambda interval, recorded as a below-reference direction of magnitude 71/100 through 61/10, contains the SFT normalized unity correspondence but does not yet resolve native lambda precisely.

Using the post-seal PDG dimensional reference $v = 12311/50$ GeV, the sealed ratio predicts exactly $m_H = 31557437733819647/251923197734500$ GeV = 125.266104978... GeV. The PDG 2025 listed average is 626/5 +/- 11/100 GeV, whose reported-uncertainty interval [12509/100,12531/100] contains the prediction; the exact displacement from its central value is 16653377460247/251923197734500 GeV. ATLAS reports 12511/100 +/- 11/100 GeV and CMS reports 3126/25 +/- 3/25 GeV; their exact positive offsets remain in the certificate and are not called aggregate failures. The exact native lambda is 6570778165695741824959729/50772238045420788745992200. The 2026 ATLAS-CMS 95-percent normalized kappa-lambda interval, recorded as a below-reference direction of magnitude 71/100 through 61/10, contains the SFT normalized unity correspondence but does not yet resolve native lambda precisely.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed Higgs carrier, dimensional reference, aggregate mass, individual mass rows, source custody, self-coupling boundary, measurement separation and no-extra-rule. The declared exact boundary is: The sealed Claim 065 ratio and coupling; the PDG 246.22 GeV dimensional reference; the complete PDG listed average and uncertainty; the direct ATLAS and CMS mass rows with every reported uncertainty; the latest combined ATLAS-CMS kappa-lambda interval; all five source snapshots; and no target access before seal. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: complete-fold-carrier__sealed-terminal-Higgs-ratio-and-self-coupling-versus-complete-current-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-Higgs-ratio-and-self-coupling-versus-complete-current-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:c9f03004accc73fcf2f616a607413669e9bb114672dda69254dc4dfb5765f1b2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7cb3166cd4916a6c62c86c5c2adc3ff09b42876cb10d5a8006aedbc1a51c6130`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f94882b3dd5c935411d6ebaaea5fbbd9e7bd7589dee0a084614ccc76ef67cb1`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:569660b6c17442aad7ef9beabf9e8a81dcb02d4126315d8add78afa41d59b5f6`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e6aa6fdc968a856053a7eb789a45daa916263e18008360f5727bfd6dec31037f`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:b9feec25ed76098c2f850434d0ccb1fd68e8de7db024e6f98697faebca453233`. Independent certificate: `sha256:d2fdff3ecd3577f12218d6ff2ecd49bcb7035ccd7170de7b4b0d05a954d1f956`. Engine external-validation hash: `sha256:7a680e6dcacaa8f558fdffd086fada8181356528069b1adae675f94c7b2a7191`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- PDG-2024-ELECTROWEAK-VACUUM-SCALE
- PDG-2025-HIGGS-MASS-AVERAGE

- ATLAS-2023-COMBINED-HIGGS-MASS
- CMS-2024-FOUR-LEPTON-HIGGS-MASS
- ATLAS-CMS-2026-HIGGS-PAIR-COMBINATION

Observed comparison records:

- The sealed exact ratio and post-seal PDG 246.22 GeV dimensional reference give 125.266104978... GeV.
- The prediction lies inside the PDG 2025 125.20 +/- 0.11 GeV listed-average interval.
- ATLAS and CMS individual offsets are retained exactly and are not falsely described as one-uncertainty agreements.
- The exact native coupling is 0.1294167525...; normalized unity lies inside the 2026 combined direct-search interval.
- The direct self-coupling constraint remains broad and is not called a precision measurement.
- Outside-mass, outside-coupling and erased-offset hostile controls all reject.

Falsification condition: Reject if any source changes; the sealed mass prediction lies outside the complete authoritative aggregate interval; normalized unity lies outside the direct self-coupling interval; an individual offset or uncertainty is omitted; a broad limit is called a precision measurement; target data select or modify the formal law; a signed external coordinate becomes a Fold proof scalar; or any hostile control passes incorrectly.

Measurement receipt: sha256 : f86cdc91c32c1dae5d3677b2b4b0de33d01ac3f3cddbb717b8a379be425d5a29.

Isolation certificate: sha256 : 477f2c064986183a80e924e90f8fd3cc398a221df87eb8e9cbe83db1a23b78c1.

Custody certificate: sha256 : 628b1386b9dd4a7f87bf260ec9dd7cabacef458c49ea06bc754f825c1cdeb5c9.

Registered source descriptions:

- Electroweak Model and Constraints on New Physics: <https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf> (sha256:4689caae925ee279228684249333a8395a40dda74a94863549b48f758d3fdd3f)
- Particle Data Group Higgs-boson listing: <https://pdg.lbl.gov/2025/listings/rpp2025-list-higgs-boson.pdf> (sha256:228835096e2003bb332f456ad412cbfbd628a6973be715309da7d2f419ac77bc)
- Combined measurement of the Higgs boson mass from H to gamma-gamma and H to ZZ-star to four leptons: <https://arxiv.org/abs/2308.04775> (sha256:b3d2c09226e54f4553a381c70e41095f8f1c18aec3aabb845e0481aa9207e4e)
- Measurement of the Higgs boson mass and width using the four-lepton final state: <https://arxiv.org/abs/2409.13663> (sha256:8dfaa73b5df3a958998a1a95dc123dc02013ef22c0a4fc3f82253550fc1c1807)
- Combination of ATLAS and CMS searches for Higgs boson pair production at 13 TeV: <https://arxiv.org/abs/2602.23991> (sha256:eaf44d724e33f500781a964a2918b5d2062c55be760704736346233ed02ff678)

Registered target identities:

- HIGGS-TERMINAL-WITHHELD-COMPLETE-RECORD from PDG-2025-HIGGS-MASS-AVERAGE

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no Higgs mass, VEV, self-coupling interval, uncertainty or aggregate in formal survivor selection
- no coefficient, correction term, target-selected tolerance, uncertainty inflation or fitted series
- no omission of the ATLAS or CMS individual offsets and no claim that both lie within one reported uncertainty
- no direct-search interval relabelled as a precision measurement of native lambda
- no signed external coordinate imported as a negative Fold proof scalar
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:cb014d09798e1ef2e597b0c9d2d9081635d84d9aa8e8eb6ee91c05176b5a63a4; engine receipt
sha256:f69729204538a9c0949fa5c4e3dfce51beece2ebe2d4e43163910d6192f9b523 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066-f69729204538a9c0.json;
empirical-validation hash
sha256:56c919c8c0a8107a97ead48490ea16b9ae64ae334208a7ae8530cdfcd15c24a4; measurement receipt
sha256:f86cdc91c32c1dae5d3677b2b4b0de33d01ac3f3cddb717b8a379be425d5a29.
```

344. Blind stellar, galactic and tidal terminal validation

Claim identity: SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The NASA solar record reports sound-speed and adiabatic-index inference at the 1/10000 precision order and density at 1/1000; its reference-model convective boundary 1427/2000 lies inside the observed 713/1000 +/- 1/1000 interval and is retained only as external structure evidence. The complete six stellar slopes are 507/250, 1143/250, 5743/1000, 4329/1000, 3967/1000 and 573/200 with their six reported uncertainties; the 3967/1000 row contains four and the 573/200 row contains three, while the other four do not and remain explicit. SPARC contributes all 175 galaxies. The 153-galaxy baryonic Tully-Fisher result is 77/20 +/- 9/100 with intrinsic scatter 3/50: its central interval excludes four, while the reported systematic interval [7/2,4] contains the sealed endpoint. The Bullet Cluster records an eight-sigma mass/plasma separation. The Moon records equal approximately-655-hour orbital and rotation periods. MESSENGER records Mercury's 29323073/500000 +/- 11/1000000 day rotation and experimentally verifies its 3:2 resonance, preserving the separately generated eccentric-resonance boundary.

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Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009
- SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed physical carrier, seven-source custody, solar structure, complete six-row stellar relation, complete galaxy vector, central-plus-systematic Tully-Fisher comparison, Moon-plus-Mercury tidal boundary and no-extra-rule. The declared exact boundary is: The sealed Claim 067 receipt; every row from the seven immutable source snapshots; exact positive whole and fraction values; reported uncertainties and systematic intervals; all six stellar regimes; the complete SPARC population; the Bullet Cluster separation; the lunar 1:1 observation; the Mercury 3:2 boundary; and structural target denial before formal sealing. The generator produced 256 named

candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-stellar-galactic-tidal-law-versus-complete-seven-source-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-stellar-galactic-tidal-law-versus-complete-seven-source-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:8a2aca0e7bf45cac58fc4a88bbbd9a934681d60af816702d9301a0b3ea68d5a48`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:af2210735a54649ca8ae52dedc43b60b5a6261827642aac4479775c9bac293f5`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1a6b661784eeac5d9900d5e3b1fbdd96cd7c68088554a407e3c1172d48ec787c`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8302a4bb44f6f038ed5326e59b92adf075ac0e67b439a3c5e0d9ac0ad40dd933`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:cc352e780d6815d7afd54c64c6066adb0d67ebefded9b213b6894867a920eded`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:eb56f2d40414512405ccbac57899a615c098e7af11df3ff0b9190561a4e09f94`. Independent certificate: `sha256:a91da4b516a074ba15fcca4787da29800ccda7fcb0e1e281d4991c757c63858f`. Engine external-validation hash: `sha256:72e7b0069871be252494e39a49b820076d3a41a76cf531298ab2cc6a84760f7d`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-NTRS-HELIOSEISMOLOGY-2001
- EKER-2018-MAIN-SEQUENCE-RELATIONS
- SPARC-2016-ROTATION-CURVES
- LELLI-2019-BARYONIC-TULLY-FISHER
- CLOWE-2006-BULLET-CLUSTER
- NASA-GRC-MOON-SYNCHRONOUS-ROTATION
- NASA-NTRS-MESSENGER-MERCURY-RESONANCE

Observed comparison records:

- Helioseismology reports stable solar structure at parts-per-ten-thousand sound-speed precision; its conventional reference value is not relabelled as an SFT prediction.
- All six piecewise stellar slopes remain; only the high and very-high mass rows contain terminal powers four and three.
- All 175 SPARC galaxies and all 153 Tully-Fisher galaxies remain in the registered population records.
- The Tully-Fisher central interval excludes four while the complete reported systematic interval reaches four; both are retained.
- The Bullet Cluster records an eight-sigma separation between the mass and dominant plasma peaks.
- The Moon records 1:1 synchronous rotation; Mercury records the separately declared 3:2 eccentric-resonance boundary.
- Erased-row, central-only, false-lunar-lock and erased-Mercury hostile controls reject.

Falsification condition: Reject if any source identity changes; any of the six stellar regimes, both Tully-Fisher interval classes, the full galaxy counts, the Bullet separation, the Moon terminal or the Mercury boundary is omitted; the formal law is changed after target release; a central offset is concealed; a conventional reference model is called an SFT prediction; a fitted parameter enters; or any hostile control passes incorrectly.

Measurement receipt: sha256:313655e8d2faf9fae7c637f0139713d9ddda3e18b70b3360397ac6adcca7c673.

Isolation certificate: sha256:16c31cc82998c61691f15702aa12fcd5f61e912741ee9878d0f06a68ac97078c.

Custody certificate: sha256:d914349f61c1421ff2bcd03fcde8b1a263fff788a5da65e88eb4c0993961cc4c5.

Registered source descriptions:

- Our Sun IV: The Standard Model and Helioseismology: Consequences of Uncertainties in Input Physics and in Observed Solar Parameters: <https://ntrs.nasa.gov/citations/20010111092>
(sha256:e523f76ba101f8becc7c18e593d9dce2ce999e4393f3efbace74bada0a51e42e)
- Interrelated main-sequence mass-luminosity, mass-radius and mass-effective temperature relations:
<https://arxiv.org/abs/1807.02568> (sha256:38f82514c2bcc9f3732581232d86d0798fb758f84018ddce1ad5e5941a359ff3)
- SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves:
<https://arxiv.org/abs/1606.09251> (sha256:d089215877213661e40965543ee7e05736619082ad16d95e65ec059029588c63)
- The baryonic Tully-Fisher relation for different velocity definitions and implications for galaxy angular momentum:
<https://arxiv.org/abs/1901.05966> (sha256:a35dafd7d01967b64bbd78be5d337c0011f53fe3c3f6e224b32c23a7a7ba4e3f)
- A direct empirical proof of the existence of dark matter: <https://arxiv.org/abs/astro-ph/0608407>
(sha256:59f2fb580797de5e5007fcea490a0f6c36036364546dbbb6a319a6e2fb5d0209)
- Moon: <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/moon/>
(sha256:8591513db27d380612afdf5d4c34e3fa1502fdf89100f745b88a578c49135df4)
- The Gravity Field, Orientation, and Ephemeris of Mercury from MESSENGER Observations After Three Years in Orbit:
<https://ntrs.nasa.gov/citations/20150000346>

(sha256:836e77ddea47e52734be957e4f41bfcd4c247c8b9beec3a9b0eb900756cfd4e)

Registered target identities:

- STELLAR-GALACTIC-TIDAL-WITHHELD-COMPLETE-RECORD from COMPLETE-SEVEN-SOURCE-VECTOR

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no source row, stellar slope, galaxy fit, tidal period or uncertainty in formal survivor selection
- no fitted exponent, normalization, mass-to-light choice, halo profile, tolerance or correction term
- no deletion of four non-endpoint stellar regimes and no universal single stellar exponent
- no replacement of the Tully-Fisher central interval by its systematic interval or concealment of either
- no relabelling of a conventional solar reference-model value as an SFT dimensional prediction
- no universal lunar 1:1 rule extended across the explicitly separate Mercury eccentric resonance
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d6683300dc5935bb78a3cc5c38421bab652cc3dc42f3bcb7353dca988b63be3d; engine receipt
 sha256:2410fe8ac97e22ffea7303b50155129dfe6c914d84e549c3c9bb6d24b0672ad6 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068-2410fe8ac97e22ff.json;
 empirical-validation hash
 sha256:f13a517866bb232d464d74547d87605e5b470146d0f89eacbddf5b33880a04c9; measurement receipt
 sha256:313655e8d2faf9fae7c637f0139713d9dda3e18b70b3360397ac6adcca7c673.

345. Blind stellar nuclear chain, collapse and heavy-element validation

Claim identity: SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete six NASA stage temperatures are (3/100,1/5,4/5,3/2,2,33/10) billion kelvin and strictly increase in the formal stage order; all durations (10000000,1000000,1000,1/10,2,1/100 years) remain and are not falsely called monotonic. Borexino reports 36/5 +3 -17/10 cpd per 100 tonnes, minimum significance five sigma at 99/100 confidence. SN1987A neutrinos were recorded by three detectors; the reported two-component preference exceeds 100:1 and the sub-One-second accretion component remains model-assisted. SN2014J records 847 and 1238 keV cobalt lines with fluxes 73/20 +/- 121/100 and 227/100 +/- 69/100 in 10⁻⁴ photon units; the measured ratio 31/50 +/- 7/25 contains 17/25 and the spectrum significance is 113/10 sigma. GW170817 records the 810 nm strontium feature from 3/2 to 10 days, with 1/5-c broadening and 23/100-c blueshift.

The complete six NASA stage temperatures are (3/100,1/5,4/5,3/2,2,33/10) billion kelvin and strictly increase in the formal stage order; all durations (10000000,1000000,1000,1/10,2,1/100 years) remain and are not falsely called monotonic. Borexino reports 36/5 +3 -17/10 cpd per 100 tonnes, minimum significance five sigma at 99/100 confidence. SN1987A neutrinos were recorded by three detectors; the reported two-component preference exceeds 100:1 and the sub-One-second accretion component remains model-assisted. SN2014J records 847 and 1238 keV cobalt lines with fluxes 73/20 +/- 121/100 and 227/100 +/- 69/100 in 10⁻⁴ photon units; the measured ratio 31/50 +/- 7/25 contains 17/25 and the spectrum significance is 113/10 sigma. GW170817 records the 810 nm strontium feature from 3/2 to 10 days, with 1/5-c broadening and 23/100-c blueshift.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069
- SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed carrier, source custody, complete stage table, direct fusion-neutrino row, collapse-neutrino vector, thermonuclear gamma vector, neutron-capture spectrum and no-extra-rule. The declared exact boundary is: The sealed Claim 069 receipt; all five immutable sources; all six stage temperatures and durations; Borexino rate and asymmetric errors; all three SN1987A detectors and model boundary; both SN2014J lines, fluxes, errors, ratio and significance; the complete registered GW170817 strontium row; and no target access before seal. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-stellar-nuclear-collapse-law-versus-complete-five-source-record__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-stellar-nuclear-collapse-law-versus-complete-five-source-record	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:606e96edc6791a89f2d03d4f09069c448275102eb50e8c6da23938285869906c`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:295b2763cd15381d7792c51086f1bcf1631ba50e58ff10583c034d31fd31134c.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9902e6720af71e081ceaffc4de9f1e7c20be7cc9bf946ce88e3e51b87111a98e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:38f5f64a33f4ef3e9174cdf3758984a1f3e16e343f2c18be29da2b902ef7fe3c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:2b1785820c991b9e7f22c226823f597a44d41f30bf453d92d94891545b866950.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256:2c72d85ead6719362cd8b783b9b26a127a2e605061d0253b155fbb76a561a5c5.

Independent certificate:

sha256:d0078b02f631aec18fb80d091764b7e01e1011b73167cbec14781bff981b52fa. Engine

external-validation hash:

sha256:99883a85a8fc27fd03f1c3bedfebb1464b4722a7a15a2fb0898dc07f0197352c.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-GSFC-COSMIC-ELEMENTS-STELLAR-STAGES
- BOREXINO-2020-CNO-NEUTRINOS
- LOREDO-LAMB-2001-SN1987A-NEUTRINOS
- DIEHL-2014-SN2014J-GAMMA-LINES
- WATSON-2019-GW170817-STRONTIUM

Observed comparison records:

- All six stellar stages have strictly increasing measured temperatures; every duration, including the nonmonotonic neon/oxygen pair, is retained.
- Borexino directly records CNO fusion neutrinos at 36/5 +3 -17/10 cpd per 100 tonnes and at least five-sigma significance.
- The complete three-detector SN1987A neutrino record is retained, with model preference explicitly identified as model-assisted.
- Both SN2014J cobalt lines, flux errors and ratio are retained; the measured ratio interval contains 17/25.
- GW170817 spectroscopy identifies strontium while retaining every spectral-modelling boundary.
- Erased-stage, hidden-irregularity, erased-detector, outside-ratio and erased-capture controls reject.

Falsification condition: Reject if any source changes; any stage, duration, detector, line, flux, uncertainty, confidence, feature or modelling boundary is omitted; the stage temperatures are not strictly ordered; the direct channel observations are absent; target data alter the sealed law; a fit enters; or any hostile control passes incorrectly.

Measurement receipt: sha256:a7a660979d997ee220d888616b4cb0893150f610c7a11cfcae2f04f62787b2370.

Isolation certificate: sha256:a60aa94729e3f966a44a65c625162ae9a300b75ac462023625fec2a19707aa7a.

Custody certificate: sha256:76ab98508c7a77d326f0211169d6bd8d4dc8bed11b0474a668d37a9fe0c7787a.

Registered source descriptions:

- What is Your Cosmic Connection to the Elements?: <https://imagine.gsfc.nasa.gov/educators/elements/imagine/05.html> (sha256:156c6ce406b0ec49ceac6921ad36fccf5d8778fb5cec7539e24ee8cce961a794)
- Experimental evidence of neutrinos produced in the CNO fusion cycle in the Sun: <https://arxiv.org/abs/2006.15115> (sha256:d61241252a22a4c9614d5a29a05cf7d211c4a8dde9dec77fe21e242238ed736d)

- Bayesian analysis of neutrinos observed from supernova SN 1987A: <https://arxiv.org/abs/astro-ph/0107260> (sha256:72b8acc99ba37cf8557492669ce53851bf154b95950c7fc34e53a6681599b2d2)
- SN2014J gamma-rays from the ^{56}Ni decay chain: <https://arxiv.org/abs/1409.5477> (sha256:aaf0bc90867cdc5e13a8ee2dfd83a23def499e5d5be9bf3f7506e56e6afdaae1)
- Identification of strontium in the merger of two neutron stars: <https://arxiv.org/abs/1910.10510> (sha256:34b7f1a277959937f868e6987ec8ce39d481165fccbea8cbab6cfa27457e77c1)

Registered target identities:

- STELLAR-NUCLEAR-COLLAPSE-WITHHELD-COMPLETE-RECORD from COMPLETE-FIVE-SOURCE-VECTOR

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no stage temperature, neutrino event, gamma line, spectral feature, abundance or source value in formal survivor selection
- no fitted reaction rate, explosion energy, nickel mass, standard-candle calibration, ejecta model or target-selected tolerance
- no deletion of the oxygen-duration irregularity, any SN1987A detector, either cobalt line or any GW170817 modelling boundary
- no model-assisted interpretation relabelled as a direct SFT dimensional prediction
- no negative, irrational, imaginary, floating, NaN, continuum or infinite Fold proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:d37b32552b807987d11a36d7c667f15d25f1c45a6286b2e9a30fa9c218403a77; engine receipt
 sha256:d439aaf47e94dcc7baba1031cf0ac05c446d03fbd089ed86654618d815c2032c at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070-d439aaf47e94dcc7.json
 ; empirical-validation hash
 sha256:92967ad9cb187ae466cda94c34ca26c4c0571526b5e70ec403e4de0db3db117a; measurement receipt
 sha256:a7a660979d997ee220d888616b4cb0893150f610c7a11cfae2f04f62787b2370.

346. Blind compact-object and horizon-thermodynamics validation

Claim identity: SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The external dimensional record contains the $7/5$ -solar-mass electron-degenerate limit. The independently measured neutral-fermion-supported PSR J0740 mass interval is $[201/100, 215/100]$ solar masses, wholly above $7/5$. The conditional merger-remnant upper records $267/100$ and $61/20$ solar masses are both above that measured interval and remain explicitly model-assisted rather than direct TOV measurements. Together with the earlier horizon-scale shadow/ringdown receipt, the post-seal record supports the ordered finite white-dwarf, neutron-star and horizon-endpoint classes. No direct astrophysical Hawking-radiation or temperature measurement exists in the registered vector, so the exact $mT=1/16$ law remains an unconfirmed, falsifiable prediction rather than a rewarded absence.

The external dimensional record contains the $7/5$ -solar-mass electron-degenerate limit. The independently measured neutral-fermion-supported PSR J0740 mass interval is $[201/100, 215/100]$ solar masses, wholly above $7/5$. The conditional merger-remnant upper records $267/100$ and $61/20$ solar masses are both above that measured interval and remain explicitly model-assisted rather than direct TOV measurements. Together with the earlier horizon-scale shadow/ringdown receipt, the post-seal record supports the ordered finite white-dwarf, neutron-star and horizon-endpoint classes. No direct astrophysical Hawking-radiation or temperature measurement exists in the registered vector, so the exact $mT=1/16$ law remains an unconfirmed, falsifiable prediction rather than a rewarded absence.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071
- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed compact carrier, complete object-family vector, source custody, dimensional-role separation, conditional-limit retention, Hawking standing-test status, target inaccessibility and no-extra-rule. The declared exact boundary is: The sealed Claim 071 receipt; all NASA white-dwarf rows; the full PSR J0740 mass, uncertainty, credibility and method record; both conditional LIGO-Virgo remnant limits; the immutable prior horizon-validation boundary; all three source snapshots; and no target access before seal. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-compact-horizon-law-versus-complete-current-object-boundary__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-compact-horizon-law-versus-complete-current-object-boundary	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:bfd7bc7756485a0dafec5f0e8f659d62f55132d13aa52d371010ecda340abc5`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:6bf6814835ef4eeb2f511d887bef6e8221922fe9e99c26b4a47af7ce909505ec`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 47197b8de4600a7aad77e5868980b543cc3f3239b787d2fe6d7be881916958a.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: c0d38852e5d941098fdbbe8f5dada47c679aac539a404f1fdae54d21aaf1d1e0.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 0f770d778da96c076a83132c7d57e64499d4a9905fb4cd8bf31e828d99ed5d96.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256: bce83e05cc4d335ba8b1bf4166592e2cf61e9eedceee0891844903b0ed01ba85.

Independent certificate:

sha256: 74bc925ce6fa49912d235061d924e290c8362727f2593b5525e9bbbcc7630406. Engine

external-validation hash:

sha256: 785891cb94b1f22d4c67da47f895069919dc319709bde25853120526159113fd.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- NASA-GSFC-WHITE-DWARF-CHANDRASEKHAR
- FONSECA-2021-PSR-J0740-MASS
- LIGO-VIRGO-2019-GW170817-REMNANT

Observed comparison records:

- NASA records electron degeneracy and the 7/5-solar-mass white-dwarf limit.
- PSR J0740+6620 is measured at 52/25 +/- 7/100 solar masses; its complete interval lies above 7/5.
- Both LIGO-Virgo conditional remnant limits are retained with their model conditions and are not called direct TOV measurements.
- The prior horizon receipt is retained unchanged; direct Hawking radiation and temperature remain unmeasured standing tests.
- Hawking non-observation is explicitly not counted as confirmation of the formal $mT=1/16$ prediction.
- Reversed-order, collapsed-upper and erased-condition hostile controls all reject.

Falsification condition: Reject if any source changes; the measured neutral-fermion object no longer lies above the complete white-dwarf limit; either conditional remnant row is omitted or stripped of its condition; Hawking non-observation is counted as confirmation; target data select or modify Claim 071; an external signed coordinate becomes a Fold proof scalar; or any hostile control passes incorrectly.

Measurement receipt: sha256: f6d5b035886daa8148cf24c06998e4f498ab93478bfd6af5988a21552f75299c.

Isolation certificate: sha256: 7d65d02a2f387d360082c171f4f6d10fe3548d315c86445fe47b15e1acf072c0.

Custody certificate: sha256: cf5f250e6766c2a38fe66c7f8bb8e7daac18393d150e46819efa78efed40a9af.

Registered source descriptions:

- NASA GSFC Imagine the Universe: White Dwarfs: <https://imagine.gsfc.nasa.gov/science/objects/dwarfs2.html> (sha256:29d25e42773e2b6c5c5f292ef97b6abb28e430f4b6af6336720ee5e3d41aa2a5)
- Refined Mass and Geometric Measurements of the High-Mass PSR J0740+6620: <https://arxiv.org/abs/2104.00880> (sha256:66c7fbaacc06a5fa3513a7d963d2b300b05a0a52552e2358e9f257940f37b47a)
- Model comparison from LIGO-Virgo data on GW170817 binary components and consequences for the merger remnant: <https://dcc.ligo.org/LIGO-P1800379/public> (sha256:86d4efa4b4c5153799edf278bcd99b10753540337c876fbf505855b8bb949122)

Registered target identities:

- COMPACT-HORIZON-WITHHELD-COMPLETE-RECORD from COMPLETE-COMPACT-HORIZON-VECTOR

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no compact-object mass, uncertainty, merger condition, horizon image or Hawking status in formal survivor selection
- no fitted mass coefficient, equation-of-state choice, tolerance, temperature scale or target-selected correction
- no white-dwarf theory summary relabelled as direct mass measurement
- no observed neutron-star mass relabelled as the unknown maximum mass
- no conditional merger bound relabelled as a direct TOV measurement
- no Hawking non-observation rewarded as confirmation and no signed external coordinate imported as a Fold proof scalar

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:7e8c74ff06b33668cd0baec4d0ea0812645956b4f0d5f5904f3ea162ef1f56c1; engine receipt
sha256:3df00caeeae7eb06d7b17539e9eb6282737e70c3a8c9a7208c83009b97acd124 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072-3df00caeeae7eb06.json; empirical-validation hash
sha256:cc4c30b739d2862f07b85dc43938f0e80374ec5331adcaff998cd5507863f64f2; measurement receipt
sha256:f6d5b035886daa8148cf24c06998e4f498ab93478bfd6af5988a21552f75299c.

347. Complete empirical gravitational-wave chirp, merger and ringdown comparison

Claim identity: SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The complete empirical record supports all three forced waveform stages. GW151226 rises from 35 to 450 Hz and in amplitude over 55 cycles, retains positive radiated energy and records two initial components joining one final remnant. GW190521 retains a decaying least-damped $l=m=2$ remnant mode compatible with its full waveform analysis. Its short duration, conditional quasicircular interpretation and alternative-source boundary remain explicit. GW150914 is retained as observational development evidence, not counted as a blind target. No external value selects or modifies the 4096-form Claim 073 survivor, and no dimensional frequency or fitted damping value becomes a Fold proof scalar.

The complete empirical record supports all three forced waveform stages. GW151226 rises from 35 to 450 Hz and in amplitude over 55 cycles, retains positive radiated energy and records two initial components joining one final remnant. GW190521 retains a decaying least-damped $l=m=2$ remnant mode compatible with its full waveform analysis. Its short duration, conditional quasicircular interpretation and alternative-source boundary remain explicit. GW150914 is retained as observational development evidence, not counted as a blind target. No external value selects or modifies the 4096-form Claim 073 survivor, and no dimensional frequency or fitted damping value becomes a Fold proof scalar.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003
- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of sealed Fold sequence, complete post-seal event vector, source custody, rising-chirp ordering, two-to-one remnant transition, damped-mode support, scope retention and no-extra-rule. The declared exact boundary is: The immutable Claim 073 receipt; disclosed GW150914 development context; every GW151226 frequency, amplitude, cycle, source, remnant, energy, uncertainty and model-role row; every GW190521 duration, cycle, frequency, source, remnant, ringdown and conditional-interpretation row; and all hostile controls. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__sealed-Fold-chirp-merger-ringdown-sequence-versus-complete-postseal-LIGO-vector__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-Fold-chirp-merger-ringdown-sequence-versus-complete-postseal-LIGO-vector	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:92bdc306ff6274441d2cb013f7365da9d6177cbd4dc537e79abadc2fe70950f8`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8d6ad202adf8a70915d4cead12907a1296914626f921ea9a01600d578bfc7520`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:329e5d4db94a3fa35b0cf093a42c00037843ffff6d581493f77f15add230e698f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:14b95e26cf8f0a19780927b2e7c1fd82b6f912719570584d74226ee90573e6bd`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 45dece7ae7177ccf5e3e8cdfeca815fdc131af0943ffc70f5fbc6d1a7b0bece.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor.

Implementation hash: sha256 : 62c52df9671103756f013b7c2525757b1d2652c3e1149840f8f1bd0eca087d15.

Independent certificate:

sha256 : 5f5d4fa9d496a8064f43df7450856bc28c9e5333db1656d8560a041bf631e5b1. Engine

external-validation hash:

sha256 : 149d5efb655201049e1c3fcf2cd34337a83ca45684e321bd0240259b58079e1d.

External empirical check. The target was absent until prediction seal: **True**. It was released after the matching seal: **True**. All rows were preserved: **True**. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- LIGO-VIRGO-GW151226-DISCOVERY-2016
- LIGO-VIRGO-GW190521-DISCOVERY-2020

Observed comparison records:

- GW150914 is retained as pre-seal observational context and is not counted as blind validation.
- GW151226 rises in frequency from 35 to 450 Hz and in amplitude over 55 cycles.
- GW151226 retains positive radiated energy and the two-component-to-one-remnant transition with full uncertainties and model role.
- GW190521 retains one remnant, a decaying least-damped quadrupolar mode and compatibility with the full waveform analysis.
- GW190521's conditional quasicircular interpretation, short-signal limit and alternative interpretations remain explicit.
- No dimensional frequency or exact half-One damping value is relabelled as a Fold proof scalar or direct universal measurement.

Falsification condition: Reject if a post-seal source changes; rising frequency/amplitude, positive radiation, two-to-one merger or decaying quadrupolar remnant support fails; any uncertainty or conditional/model boundary is omitted; GW150914 is relabelled as blind; a dimensional target selects Claim 073; or any hostile control passes.

Measurement receipt: sha256 : 1e267de913c50323223c4e548c9b10fab5ccf279081db59446b78f1684309f7e.

Isolation certificate: sha256 : ed8c3a7d3d2a84f2d0ea64a6c3af2c31a2caa04320a413b092583784623e7870.

Custody certificate: sha256 : 21fb28c3798ac2479fb847753b0a04b6e5bdfef39464ba5e502c59886c77d996.

Registered source descriptions:

- LIGO Scientific Collaboration and Virgo Collaboration: <https://dcc.ligo.org/LIGO-P151226/public>
(sha256:a59bb954e9850a8800e2b62e1808553012e791b879e00f108b0bf061bf530e72)
- LIGO Scientific Collaboration and Virgo Collaboration: <https://dcc.ligo.org/P2000020/public>
(sha256:4d6661f18168267249b3ca043b0692d88376c9db9b3d9146a9500763f7b438a7)

Registered target identities:

- GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-WITHHELD-POSTSEAL-RECORD from
COMPLETE-POSTSEAL-GRAVITATIONAL-WAVE-VECTOR

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no claim that GW150914 was unseen before Claim 073
- no use of GW151226 or GW190521 values to select or alter the formal survivor

- no fitted chirp mass, waveform, merger time, damping coefficient, tone or tolerance
- no model-assisted remnant property relabelled as direct model-free measurement
- no claim that half-One damping or normalized Fold frequency was directly measured
- no omitted uncertainty, short-signal, conditional-interpretation or alternative-source row

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:76cd8e8bf521eeadf34de36bf9951e70cdd1578454b268cceb84a67a2da8d470; engine receipt
 sha256:80a9f13dc620d3cb2bf16597d54b63562aeac60ad1bab25d1327aabafb8bf576 at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074-80a9f13dc620d3cb.json; empirical-validation hash
 sha256:a197d901695d69375e3d5adc571616d5324340f82ae7539dd190aaff9b26b361; measurement receipt
 sha256:1e267de913c50323223c4e548c9b10fab5ccf279081db59446b78f1684309f7e.

348. Terminal Physics Grand Lock and extension-open no-omission certificate

Claim identity: SFT-PHYS-GRAND-LOCK-TERMINAL-075

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. The frozen pre-lock Physics surface contains exactly 347 uniquely owned admitted claims and 534 transitive dependency nodes. Every Physics claim reaches SFT-FOUNDATION-ONE-001 in one acyclic dependency dictionary. Exactly 234 Physics claims carry empirical-test receipts, and every recorded unfavorable result or scope boundary remains bound. The current generator-three headline vector recomputes exactly, including inverse alpha 503846395469/3676744786, charged-lepton invariants 1, 1/6, 1/485 and 3/1454 plus the terminal product, quark products 1/383 and 1/3071, dark/baryon 27/5 with 27/32 and 5/32, Hubble 13/12 and 3305/3048, the terminal squared Planck/proton hierarchy, local vacuum floors $1/2^{10}$ and $1/2^{20}$, and normalized cosmological value 33/16. Under the complete generator successor three-to-four, every declared generator-dependent value changes while half-One, spatial rank three and boundary rank two remain fixed.

The frozen pre-lock Physics surface contains exactly 347 uniquely owned admitted claims and 534 transitive dependency nodes. Every Physics claim reaches SFT-FOUNDATION-ONE-001 in one acyclic dependency dictionary. Exactly 234 Physics claims carry empirical-test receipts, and every recorded unfavorable result or scope boundary remains bound. The current generator-three headline vector recomputes exactly, including inverse alpha 503846395469/3676744786, charged-lepton invariants 1, 1/6, 1/485 and 3/1454 plus the terminal product, quark products 1/383 and 1/3071, dark/baryon 27/5 with 27/32 and 5/32, Hubble 13/12 and 3305/3048, the terminal squared Planck/proton hierarchy, local vacuum floors $1/2^{10}$ and $1/2^{20}$, and normalized cosmological value 33/16. Under the complete generator successor three-to-four, every declared generator-dependent value changes while half-One, spatial rank three and boundary rank two remain fixed.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-ATOMIC-CORRECTION-HIERARCHY-004
- SFT-PHYS-ATOMIC-CUBIC-SUPPORT-004
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005
- SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005
- SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005
- SFT-PHYS-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005

- SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005
- SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005
- SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005
- SFT-PHYS-ATOMIC-TRANSITION-SELECTION-004
- SFT-PHYS-BARYOGENESIS-DEPENDENCY-TERMINAL-021
- SFT-PHYS-COLLECTIVE-RADIATION-RESPONSE-TERMINAL-041
- SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071
- SFT-PHYS-COMPOSITE-CONFINING-SECTOR-TERMINAL-031
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-CUBIC-001
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-CONTINUUM-COARSE-GRAIN-001
- SFT-PHYS-COSMO-BACKGROUND-001
- SFT-PHYS-COSMO-BRANCH-BOUNDARY-001
- SFT-PHYS-COSMO-COMPLETE-BUDGET-001
- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032
- SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001
- SFT-PHYS-COSMO-DISTANCE-001
- SFT-PHYS-COSMO-EXPANSION-001
- SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001
- SFT-PHYS-COSMO-LENSING-001
- SFT-PHYS-COSMO-REDSHIFT-001
- SFT-PHYS-COSMO-SPATIAL-FLATNESS-001
- SFT-PHYS-COSMO-STRUCTURE-GROWTH-001
- SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007
- SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008
- SFT-PHYS-COUPLED-ACCUMULATED-SEPARATION-TERMINAL-015
- SFT-PHYS-COUPLED-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-PHYS-CRITICALITY-UNIVERSALITY-TURBULENCE-TERMINAL-047
- SFT-PHYS-DARK-SMITHION-LFV-TERMINAL-061
- SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006
- SFT-PHYS-DYNAMICS-FREE-PHASE-DISPERSION-003
- SFT-PHYS-DYNAMICS-POTENTIAL-EVOLUTION-003

- SFT-PHYS-DYNAMICS-STATIONARY-SPECTRUM-003
- SFT-PHYS-DYNAMICS-SYMMETRY-ACTION-TERMINAL-016
- SFT-PHYS-ELECTRON-DIRAC-G-FACTOR-002
- SFT-PHYS-ELECTROWEAK-FOLD-MIXING-002
- SFT-PHYS-ELECTROWEAK-TERMINAL-ON-SHELL-003
- SFT-PHYS-ELECTROWEAK-WZ-RATIO-002
- SFT-PHYS-FIELD-ACTION-REACTION-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-FIELD-FINITE-LOOP-CLOSURE-003
- SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001
- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-INDUCTION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-FIELD-INVERSE-SQUARE-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-FIELD-LORENTZ-TRANSFER-003
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-PHYS-FIELD-MAGNETIC-RELATIVITY-003
- SFT-PHYS-FIELD-MAXWELL-PLANAR-CLOSURE-003
- SFT-PHYS-FIELD-MAXWELL-THREE-SPACE-CLOSURE-003
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-FLUID-DENSITY-001
- SFT-PHYS-FLUID-INVISCID-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-FLUID-TURBULENCE-001
- SFT-PHYS-FLUID-VISCOSITY-001
- SFT-PHYS-FOLD-UNIVERSE-TRANSPORT-TERMINAL-024
- SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003
- SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002
- SFT-PHYS-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-TERMINAL-073

- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-EQUIVALENCE-001
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001
- SFT-PHYS-GRAVITY-GEODESIC-001
- SFT-PHYS-GRAVITY-GRAVITON-POLARIZATION-003
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-GRAVITY-HORIZON-INFORMATION-003
- SFT-PHYS-GRAVITY-LATTICE-CURVATURE-003
- SFT-PHYS-GRAVITY-NONLINEAR-SELF-SOURCE-003
- SFT-PHYS-GRAVITY-REDSHIFT-EQUIVALENCE-003
- SFT-PHYS-GRAVITY-STATIC-CLOCK-003
- SFT-PHYS-GRAVITY-STRONG-FIELD-HORIZON-003
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-GRAVITY-WAVE-QUADRUPOLE-003
- SFT-PHYS-GRAVITY-WEAK-FIELD-FLUX-003
- SFT-PHYS-HADRON-REGGE-DIMENSIONAL-TERMINAL-059
- SFT-PHYS-HADRON-REGGE-TERMINAL-005
- SFT-PHYS-HIGGS-SYMMETRY-TERMINAL-065
- SFT-PHYS-INFLATION-GROWTH-TERMINAL-039
- SFT-PHYS-INTERACTION-UNIFICATION-TERMINAL-025
- SFT-PHYS-LATTICE-OPERATOR-TERMINAL-022
- SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008
- SFT-PHYS-MATTER-BARYON-PHOTON-003
- SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004
- SFT-PHYS-MATTER-CKM-FIBRE-003
- SFT-PHYS-MATTER-CKM-PHYSICAL-003
- SFT-PHYS-MATTER-CKM-TERMINAL-004
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-MATTER-CONFINEMENT-LIFT-003
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-DECAY-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-MATTER-GENERATION-DEPTH-003
- SFT-PHYS-MATTER-INTER-ENTRY-COUPLING-003
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-MATTER-MASS-RATIO-FAMILY-003
- SFT-PHYS-MATTER-MIRROR-MASS-CLOSURE-003
- SFT-PHYS-MATTER-MIXING-001

- SFT-PHYS-MATTER-MIXING-CORRESPONDENCE-003
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-MATTER-PROTON-ELECTRON-003
- SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004
- SFT-PHYS-MATTER-QUARK-CUBICS-003
- SFT-PHYS-MATTER-QUARK-DRESSING-003
- SFT-PHYS-MATTER-QUARK-INVARIANTS-003
- SFT-PHYS-MATTER-SCATTERING-001
- SFT-PHYS-MEAS-BOUNDARY-GROWTH-001
- SFT-PHYS-MEAS-CALIBRATION-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001
- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MECH-ACCELERATION-001
- SFT-PHYS-MECH-ANGULAR-MOMENTUM-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001
- SFT-PHYS-MECH-COMPOSITE-CENTRE-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-MECH-DURATION-001
- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-FORCE-001
- SFT-PHYS-MECH-INERTIA-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-MECH-MOMENTUM-001
- SFT-PHYS-MECH-POWER-001
- SFT-PHYS-MECH-SPEED-VELOCITY-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MEDIATOR-RANGE-CHANNEL-TERMINAL-020

- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-PHYS-MOLECULAR-SPECTRUM-HIERARCHY-004
- SFT-PHYS-NEUTRINO-CP-PHASE-002
- SFT-PHYS-NEUTRINO-MAJORANA-003
- SFT-PHYS-NEUTRINO-PMNS-ANGLES-002
- SFT-PHYS-NEUTRINO-PMNS-CP-PHYSICAL-003
- SFT-PHYS-NEUTRINO-POSITIVE-MASS-003
- SFT-PHYS-NEUTRINO-SPLITTING-003
- SFT-PHYS-NEUTRINO-ZERO-NU-BETA-BETA-003
- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPPLING-001
- SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006
- SFT-PHYS-NUCLEAR-FISSION-001
- SFT-PHYS-NUCLEAR-FUSION-001
- SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005
- SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-PHYS-NUCLEAR-RADIOACTIVITY-001
- SFT-PHYS-NUCLEAR-REACTIONS-001
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009
- SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028
- SFT-PHYS-PARTICLE-MODE-GENERATION-TERMINAL-051
- SFT-PHYS-PLASMA-COLLECTIVE-001
- SFT-PHYS-PLASMA-MHD-001
- SFT-PHYS-PLASMA-OSCILLATION-001
- SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009
- SFT-PHYS-PROTON-RADIUS-TERMINAL-029
- SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004
- SFT-PHYS-QED-LEPTON-MAGNETIC-ANOMALY-003
- SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004
- SFT-PHYS-QED-TERMINAL-TURN-PROJECTION-004
- SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012

- SFT-PHYS-QUANTUM-BELL-001
- SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001
- SFT-PHYS-QUANTUM-CONTEXTUALITY-001
- SFT-PHYS-QUANTUM-DECOHERENCE-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-QUANTUM-EVOLUTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-INCOMPATIBILITY-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-QUANTUM-OBSERVABLE-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-SUPPORT-UNCERTAINTY-TERMINAL-049
- SFT-PHYS-QUANTUM-TUNNELLING-001
- SFT-PHYS-QUANTUM-WEIGHT-001
- SFT-PHYS-RELATIVITY-FULL-DIRAC-SQUARE-003
- SFT-PHYS-RELATIVITY-TWO-HAND-DIRAC-SQUARE-003
- SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030
- SFT-PHYS-SCALE-PROTON-PLANCK-HIERARCHY-002
- SFT-PHYS-SCALE-PROTON-PLANCK-TERMINAL-003
- SFT-PHYS-SCATTERING-PARTITION-PATH-TERMINAL-017
- SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-SPACETIME-CLOSED-TIMELIKE-ADMISSIBILITY-003
- SFT-PHYS-SPACETIME-EVENT-RELATION-001
- SFT-PHYS-SPACETIME-EXACT-INTERVAL-003
- SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001
- SFT-PHYS-SPACETIME-INTERVAL-001
- SFT-PHYS-SPACETIME-LENGTH-RELATION-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001
- SFT-PHYS-SPACETIME-VELOCITY-COMPOSITION-TERMINAL-033
- SFT-PHYS-SPACETIME-WARP-ADMISSIBILITY-003
- SFT-PHYS-SPACETIME-WORMHOLE-ADMISSIBILITY-003

- SFT-PHYS-SPIN-STATISTICS-CONDENSATION-TERMINAL-045
- SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011
- SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067
- SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069
- SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013
- SFT-PHYS-STRONG-CP-BARYON-STABILITY-TERMINAL-063
- SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014
- SFT-PHYS-STRONG-RUNNING-DIRECTION-002
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010
- SFT-PHYS-THERMAL-EQUILIBRIUM-RESPONSE-TERMINAL-043
- SFT-PHYS-THERMAL-HELIUM-ISOTOPE-TERMINAL-057
- SFT-PHYS-THERMAL-HISTORY-RECOMBINATION-TERMINAL-037
- SFT-PHYS-THERMO-ENTROPY-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-FLUCTUATION-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-LANDAUER-DEMON-TERMINAL-018
- SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019
- SFT-PHYS-THERMO-MICRO-MACRO-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-THIRD-LAW-001
- SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003
- SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003
- SFT-PHYS-VACUUM-DENSITY-SCALE-TERMINAL-035
- SFT-PHYS-VACUUM-HALF-ONE-FLOOR-003
- SFT-PHYS-VACUUM-INERTIA-UNITY-003
- SFT-PHYS-VACUUM-ODD-RECURRENCE-003
- SFT-PHYS-VACUUM-POLARIZATION-RUNNING-003
- SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004

- SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002
- SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042
- SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054
- SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072
- SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055
- SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056
- SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048
- SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062
- SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003
- SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053
- SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003
- SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001
- SFT-PHYS-VALIDATION-FINITE-LOOPS-003
- SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003
- SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003
- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060
- SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066
- SFT-PHYS-VALIDATION-INFLATION-GROWTH-040
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-VALIDATION-INVERSE-SQUARE-001
- SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003
- SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003
- SFT-PHYS-VALIDATION-MECHANICS-FORCE-001
- SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001
- SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001
- SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003
- SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003
- SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001
- SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003
- SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052
- SFT-PHYS-VALIDATION-PROTON-ELECTRON-003

- SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003
- SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050
- SFT-PHYS-VALIDATION-QUARK-CKM-003
- SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003
- SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046
- SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068
- SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070
- SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064
- SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044
- SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058
- SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038
- SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001
- SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036
- SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003
- SFT-PHYS-VALIDATION-VACUUM-FLOOR-003
- SFT-PHYS-VALIDATION-VACUUM-INERTIA-003
- SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003
- SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034
- SFT-PHYS-VALIDATION-WAVE-ENERGY-001
- SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001
- SFT-PHYS-WAVE-DIFFRACTION-001
- SFT-PHYS-WAVE-DISPERSION-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-WAVE-EXACT-OPERATIONS-003
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001
- SFT-PHYS-WAVE-SUPERPOSITION-001
- SFT-PHYS-WEAK-PARITY-FIBRE-002
- SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027
- SFT-PHYS-YANG-MILLS-SINGLET-GAP-TERMINAL-026

Generated grammar and closure boundary. Generate the complete twelve-axis product of ownership, receipt identity, registration identity, dependency closure, root reachability, value vector, generator perturbation, invariant control, cross-domain identity, empirical reconciliation, extension status and extra-rule forms. The declared exact boundary is: All 347 admitted pre-lock Physics claims, all 534 nodes in their dependency closure, every bound exact-result and empirical certificate row, the complete generator-three to generator-four adverse perturbation, all shared structural carriers and all 4096 alternatives. The generator produced 4096 named candidates and the decision support contains 4096 one-for-one decisions. Exactly one candidate survived:

unique-complete-physics-owner__receipt-and-certificate-hash-bound__registration-and-dependencies-hash-bound__complete-acyclic-dictionary__foundational-One-root__all-claim-exact-result-vector__every-and-only-declared-dependent-value-moves__binary-space-boundary-held__complete-cross-domain-identity-graph__complete-receipts-with-adverse-boundaries__current-evidence-closed-extension-open__no-extra-rule. Closure is depth_independent with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
ownership	unique-complete-physics-owner	omitted-or-duplicate-owner	That is not a complete branch census.
receipt	receipt-and-certificate-hash-bound	unbound-or-current-only-result	That cannot detect replacement.
registration	registration-and-dependencies-hash-bound	unstated-or-mutable-premisses	That erases the proof route.
dependency	complete-acyclic-dictionary	orphan-or-cycle	An orphan or cycle cannot derive from the One.
root	foundational-One-root	multiple-or-missing-roots	That adds an axiom or loses derivation.
values	all-claim-exact-result-vector	headline-selection-only	Selecting favorable headlines permits omission.
perturbation	every-and-only-declared-dependent-value-moves	selected-values-move	A selected subset cannot prove structural dependence.
invariants	binary-space-boundary-held	silent-global-retuning	Global retuning cannot localize cause.
identity	complete-cross-domain-identity-graph	duplicate-unrelated-numbers	Coincidence without provenance is not identity.
empirical	complete-receipts-with-adverse-boundaries	favourable-rows-only	Cherry-picking violates the empirical record.
extension	current-evidence-closed-extension-open	permanently-locked-theory	That would exclude lawful discovery.
rule	no-extra-rule	free-extra-rule	An extra rule is a parameter.

Base and successor. The closure proof is sealed by derivation hash sha256:cf92b9cf88639e40fc0cbb37f7361b252f6839e6a64cdbe7c0e48d47857a592. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:f4786d8a1868ac9dda94eacd530e2f7ee6b2edd640697d9918d5b1bb6ceb3819.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:4c0f5ac74cf9f644c88c8e6ec55de9f9cb5ea1bf87dd112e081f86b6667d1afe.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:deb3e5409ab49a544b77cf151557f3e24de8e44a6784215a4d3349e493725803.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:10640f437fe38173ec3f5697342bb57007e4c75ed6169e83dac6e47d1b8c4210.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: sha256:bff4a8792ccae133d39c7661cf47ba22f3ea3330d4f8b3dc0c29228ff36077ef. Independent certificate: sha256:0599cd41db3228cf2f60d71a77582a1ed85a272cde0e16b19645ace592648898. Engine external-validation hash: sha256:65110685b0d6c88d72cdb3f2704a305b540d1c011e92b5a5ef87bb00c9dd5392.

External empirical check. This formal prerequisite makes no direct natural-law claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no edit to the canonical engine or any admitted receipt
- no measurement, V1/V2 result or external equation selecting a formal survivor
- no fitted parameter, tolerance, candidate neighborhood or omitted adverse result
- no numerical-zero, negative, irrational, imaginary, floating, NaN or completed-infinite proof scalar
- no assertion that current completion prohibits a later lawfully forced extension

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

sha256:91770400e9861ef159a95209bd52e675bc08f4e1cd81bd02de93dba586758cb0; engine receipt sha256:ae18f67371c8e7054430935d6b5e5f3162f24cf9cba073769384bf7ba467d817 at receipts/engine/model_admitted/SFT-PHYS-GRAND-LOCK-TERMINAL-075-ae18f67371c8e705.json; empirical-validation hash None; measurement receipt None.

349. Complete Physics empirical-vector and adverse-result Grand Lock

Claim identity: SFT-PHYS-VALIDATION-GRAND-LOCK-076

Question. Which generated form preserves the physical carrier, the Fold relation, complete provenance, target inaccessibility, measurement separation, complete rows, successor closure and absence of an extra rule?

Theorem. Exactly 234 pre-lock Physics claims carry independently replicated empirical validation; their complete vector binds 147 distinct registered external source identities. All 234 empirical and external validation hashes are retained. Every available separate measurement receipt is retained, while six older claim packages lacking that later materialization field are explicitly identified rather than silently upgraded. All fourteen detected unfavorable-result or scope-boundary claims remain inside the same vector. The formal Grand Lock receipt predates this aggregate reconciliation; observations validate or challenge their declared claims but never select or alter a formal survivor.

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claims remain inside the same vector. The formal Grand Lock receipt predates this aggregate reconciliation; observations validate or challenge their declared claims but never select or alter a formal survivor.

Dependency chain. The engine accepted only the following already model-admitted premises in addition to the single root theorem:

- SFT-PHYS-GRAND-LOCK-TERMINAL-075
- SFT-PHYS-ATOMIC-FIELD-SPLITTING-TERMINAL-005
- SFT-PHYS-ATOMIC-FINE-STRUCTURE-TERMINAL-005
- SFT-PHYS-ATOMIC-HYDROGEN-RYDBERG-TERMINAL-005
- SFT-PHYS-ATOMIC-HYPERFINE-TERMINAL-005
- SFT-PHYS-ATOMIC-LAMB-SHIFT-TERMINAL-005
- SFT-PHYS-ATOMIC-SHELL-PERIODICITY-TERMINAL-005
- SFT-PHYS-ATOMIC-TRANSITION-RATE-TERMINAL-005
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-PHYS-CONSTANT-CHARGED-LEPTON-TERMINAL-001
- SFT-PHYS-CONTINUUM-COARSE-GRAIN-001
- SFT-PHYS-COSMO-BACKGROUND-001
- SFT-PHYS-COSMO-BRANCH-BOUNDARY-001
- SFT-PHYS-COSMO-COMPLETE-BUDGET-001
- SFT-PHYS-COSMO-COMPONENT-TRANSPORT-TERMINAL-032
- SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001
- SFT-PHYS-COSMO-DISTANCE-001
- SFT-PHYS-COSMO-EXPANSION-001
- SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001
- SFT-PHYS-COSMO-LENSING-001
- SFT-PHYS-COSMO-REDSHIFT-001
- SFT-PHYS-COSMO-SPATIAL-FLATNESS-001
- SFT-PHYS-COSMO-STRUCTURE-GROWTH-001
- SFT-PHYS-COUPLED-ENSEMBLE-SYNCHRONIZATION-TERMINAL-007
- SFT-PHYS-COUPLED-MAP-CRITICALITY-TERMINAL-008
- SFT-PHYS-COUPLING-RUNNING-CONVERGENCE-TERMINAL-006
- SFT-PHYS-DECAY-WIDTH-BRANCHING-LIFETIME-TERMINAL-006
- SFT-PHYS-FIELD-ACTION-REACTION-001
- SFT-PHYS-FIELD-CONSERVED-SOURCE-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001

- SFT-PHYS-FIELD-ELECTROMAGNETIC-COMPOSITION-001
- SFT-PHYS-FIELD-GAUGE-EQUIVALENCE-001
- SFT-PHYS-FIELD-GEOMETRIC-DILUTION-001
- SFT-PHYS-FIELD-GRAVITATIONAL-INTERACTION-001
- SFT-PHYS-FIELD-INDUCTION-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-FIELD-LOCALITY-CAUSALITY-001
- SFT-PHYS-FIELD-MAGNETIC-001
- SFT-PHYS-FIELD-RADIATION-001
- SFT-PHYS-FIELD-SOURCE-RESPONSE-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-FLUID-DENSITY-001
- SFT-PHYS-FLUID-INVISCID-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-FLUID-TURBULENCE-001
- SFT-PHYS-FLUID-VISCOSITY-001
- SFT-PHYS-GRAVITY-CURVATURE-001
- SFT-PHYS-GRAVITY-EQUIVALENCE-001
- SFT-PHYS-GRAVITY-FIELD-SOURCE-001
- SFT-PHYS-GRAVITY-GEODESIC-001
- SFT-PHYS-GRAVITY-HORIZON-001
- SFT-PHYS-GRAVITY-WAVE-001
- SFT-PHYS-HADRON-REGGE-TERMINAL-005
- SFT-PHYS-LYAPUNOV-KS-CORRESPONDENCE-TERMINAL-008
- SFT-PHYS-MATTER-BARYON-PHOTON-TERMINAL-004
- SFT-PHYS-MATTER-CKM-TERMINAL-004
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-DECAY-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-MATTER-MASS-ENERGY-001
- SFT-PHYS-MATTER-MIXING-001
- SFT-PHYS-MATTER-PARTICLE-ANTIPARTICLE-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-MATTER-PROTON-ELECTRON-TERMINAL-004
- SFT-PHYS-MATTER-SCATTERING-001
- SFT-PHYS-MEAS-CALIBRATION-001
- SFT-PHYS-MEAS-DIMENSION-COMPOSITION-001

- SFT-PHYS-MEAS-DIMENSIONAL-CONSISTENCY-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MEAS-UNIT-COMPARISON-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MECH-ACCELERATION-001
- SFT-PHYS-MECH-ANGULAR-MOMENTUM-001
- SFT-PHYS-MECH-ANGULAR-MOTION-001
- SFT-PHYS-MECH-COMPOSITE-CENTRE-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MECH-CONSTRAINT-OSCILLATION-001
- SFT-PHYS-MECH-DURATION-001
- SFT-PHYS-MECH-EVENT-CHANGE-001
- SFT-PHYS-MECH-FORCE-001
- SFT-PHYS-MECH-INERTIA-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-MECH-MOMENTUM-001
- SFT-PHYS-MECH-POWER-001
- SFT-PHYS-MECH-SPEED-VELOCITY-001
- SFT-PHYS-MECH-WORK-ENERGY-001
- SFT-PHYS-MOLECULAR-SPECTROSCOPY-TERMINAL-005
- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-NUCLEAR-BINDING-CURVE-TERMINAL-005
- SFT-PHYS-NUCLEAR-DEUTERON-DINUCLEON-TERMINAL-006
- SFT-PHYS-NUCLEAR-FISSION-001
- SFT-PHYS-NUCLEAR-FUSION-001
- SFT-PHYS-NUCLEAR-FUSION-FISSION-TERMINAL-005
- SFT-PHYS-NUCLEAR-FUSION-FISSION-YIELD-THRESHOLD-006
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-PHYS-NUCLEAR-RADIOACTIVE-DECAY-TERMINAL-005
- SFT-PHYS-NUCLEAR-RADIOACTIVITY-001
- SFT-PHYS-NUCLEAR-REACTIONS-001
- SFT-PHYS-NUCLEAR-RESIDUAL-FORCE-TERMINAL-005
- SFT-PHYS-NUCLEON-BINDING-TERMINAL-005
- SFT-PHYS-ODD-LATTICE-ALL-REGION-OCCUPANCY-TERMINAL-007
- SFT-PHYS-ORBITAL-DIMENSION-STABILITY-TERMINAL-009

- SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028
- SFT-PHYS-PLASMA-COLLECTIVE-001
- SFT-PHYS-PLASMA-MHD-001
- SFT-PHYS-PLASMA-OSCILLATION-001
- SFT-PHYS-POST-NEWTONIAN-FIXED-POINT-TERMINAL-009
- SFT-PHYS-PROTON-RADIUS-TERMINAL-029
- SFT-PHYS-QED-ELECTRON-MAGNETIC-ANOMALY-004
- SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004
- SFT-PHYS-QUADRUPOLE-RADIATED-POWER-TERMINAL-012
- SFT-PHYS-QUANTUM-BELL-001
- SFT-PHYS-QUANTUM-CLASSICAL-LIMIT-001
- SFT-PHYS-QUANTUM-CONTEXTUALITY-001
- SFT-PHYS-QUANTUM-DECOHERENCE-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-QUANTUM-EVOLUTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-INCOMPATIBILITY-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-NO-SIGNALLING-001
- SFT-PHYS-QUANTUM-OBSERVABLE-001
- SFT-PHYS-QUANTUM-PHYSICAL-STATE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-TUNNELLING-001
- SFT-PHYS-QUANTUM-WEIGHT-001
- SFT-PHYS-SCALE-COMMON-AXIS-TERMINAL-030
- SFT-PHYS-SCATTERING-RUTHERFORD-COMPTON-TERMINAL-006
- SFT-PHYS-SPACETIME-CAUSAL-ORDER-001
- SFT-PHYS-SPACETIME-CLOCK-RATE-001
- SFT-PHYS-SPACETIME-EVENT-RELATION-001
- SFT-PHYS-SPACETIME-INERTIAL-TRANSFORMATION-001
- SFT-PHYS-SPACETIME-INTERVAL-001
- SFT-PHYS-SPACETIME-LENGTH-RELATION-001
- SFT-PHYS-SPACETIME-LIMIT-SPEED-001
- SFT-PHYS-STATIC-EXTERIOR-CLOCK-TERMINAL-011
- SFT-PHYS-STRONG-CARRIER-MASSLESS-CONFINED-TERMINAL-013
- SFT-PHYS-STRONG-FIELD-NONLINEAR-FIXED-POINT-TERMINAL-014
- SFT-PHYS-SYMMETRIC-SOURCE-CONSERVATION-TERMINAL-010

- SFT-PHYS-THERMO-ENTROPY-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-FLUCTUATION-001
- SFT-PHYS-THERMO-HEAT-WORK-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-THERMO-LANDAUER-EMPIRICAL-019
- SFT-PHYS-THERMO-MICRO-MACRO-001
- SFT-PHYS-THERMO-PHASE-EQUILIBRIUM-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-THERMO-STATE-RELATION-001
- SFT-PHYS-THERMO-STATISTICAL-WEIGHT-001
- SFT-PHYS-THERMO-TEMPERATURE-001
- SFT-PHYS-THERMO-THIRD-LAW-001
- SFT-PHYS-VALIDATION-ATOMIC-CUBIC-SUPPORT-004
- SFT-PHYS-VALIDATION-ATOMIC-HYDROGEN-SPECTRUM-004
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-001
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-KOIDE-002
- SFT-PHYS-VALIDATION-CHARGED-LEPTON-TERMINAL-002
- SFT-PHYS-VALIDATION-COLLECTIVE-RADIATION-RESPONSE-042
- SFT-PHYS-VALIDATION-COMMON-SCALE-MEASURED-VALUE-054
- SFT-PHYS-VALIDATION-COMPACT-HORIZON-THERMODYNAMICS-072
- SFT-PHYS-VALIDATION-COSMIC-TRANSPORT-MEASURED-VALUE-055
- SFT-PHYS-VALIDATION-CRITICALITY-MEASURED-VALUE-056
- SFT-PHYS-VALIDATION-CRITICALITY-UNIVERSALITY-TURBULENCE-048
- SFT-PHYS-VALIDATION-DARK-SMITHION-LFV-062
- SFT-PHYS-VALIDATION-DYNAMICS-SPECTRA-003
- SFT-PHYS-VALIDATION-ELECTROWEAK-MEASURED-VALUE-053
- SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003
- SFT-PHYS-VALIDATION-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-VALIDATION-FIELD-ELECTRIC-STRENGTH-001
- SFT-PHYS-VALIDATION-FINITE-LOOPS-003
- SFT-PHYS-VALIDATION-FORCE-SECTOR-ANCHORS-003
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074
- SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVES-003
- SFT-PHYS-VALIDATION-GRAVITY-CLOCK-EQUIVALENCE-003

- SFT-PHYS-VALIDATION-GRAVITY-HORIZONS-003
- SFT-PHYS-VALIDATION-HADRON-REGGE-MEASURED-VALUE-060
- SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066
- SFT-PHYS-VALIDATION-INFLATION-GROWTH-040
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-VALIDATION-INVERSE-SQUARE-001
- SFT-PHYS-VALIDATION-MAGNETIC-ANOMALIES-003
- SFT-PHYS-VALIDATION-MAJORANA-ZERO-NU-003
- SFT-PHYS-VALIDATION-MECHANICS-FORCE-001
- SFT-PHYS-VALIDATION-MECHANICS-MOMENTUM-001
- SFT-PHYS-VALIDATION-METROLOGY-MOLAR-PLANCK-001
- SFT-PHYS-VALIDATION-NEUTRINO-MASS-MIXING-003
- SFT-PHYS-VALIDATION-NONSTANDARD-SPACETIME-003
- SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001
- SFT-PHYS-VALIDATION-OPTICAL-OPERATIONS-003
- SFT-PHYS-VALIDATION-PARTICLE-MODE-GENERATION-052
- SFT-PHYS-VALIDATION-PROTON-ELECTRON-003
- SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003
- SFT-PHYS-VALIDATION-QUANTUM-SUPPORT-UNCERTAINTY-050
- SFT-PHYS-VALIDATION-QUARK-CKM-003
- SFT-PHYS-VALIDATION-RELATIVISTIC-FIELDS-003
- SFT-PHYS-VALIDATION-SPIN-STATISTICS-CONDENSATION-046
- SFT-PHYS-VALIDATION-STELLAR-GALACTIC-TIDAL-068
- SFT-PHYS-VALIDATION-STELLAR-NUCLEAR-COLLAPSE-070
- SFT-PHYS-VALIDATION-STRONG-CP-BARYON-STABILITY-064
- SFT-PHYS-VALIDATION-THERMAL-EQUILIBRIUM-044
- SFT-PHYS-VALIDATION-THERMAL-HISTORY-MEASURED-VALUE-058
- SFT-PHYS-VALIDATION-THERMAL-HISTORY-RECOMBINATION-038
- SFT-PHYS-VALIDATION-THERMO-MOLAR-ENERGY-001
- SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036
- SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003
- SFT-PHYS-VALIDATION-VACUUM-FLOOR-003
- SFT-PHYS-VALIDATION-VACUUM-INERTIA-003
- SFT-PHYS-VALIDATION-VACUUM-POLARIZATION-003
- SFT-PHYS-VALIDATION-VELOCITY-COMPOSITION-034
- SFT-PHYS-VALIDATION-WAVE-ENERGY-001
- SFT-PHYS-VALIDATION-WAVE-FREQUENCY-001
- SFT-PHYS-WAVE-DIFFRACTION-001

- SFT-PHYS-WAVE-DISPERSION-001
- SFT-PHYS-WAVE-ENERGY-MOMENTUM-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-PERIOD-FREQUENCY-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-WAVE-RESONANCE-001
- SFT-PHYS-WAVE-SPEED-LENGTH-FREQUENCY-001
- SFT-PHYS-WAVE-SUPERPOSITION-001
- SFT-PHYS-YANG-MILLS-SINGLET-GAP-EMPIRICAL-027
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar and closure boundary. Generate the complete eight-axis product of full empirical carrier, reconciliation relation, immutable provenance, sealed execution, separate observation record, complete favorable/adverse rows, successor closure and no-extra-rule. The declared exact boundary is: Every empirically tested pre-lock Physics claim; every source, validation and available measurement-receipt identity; every explicit legacy receipt shape; every retained unfavorable result and scope boundary; and all 256 alternatives. The generator produced 256 named candidates and the decision support contains 256 one-for-one decisions. Exactly one candidate survived: `complete-fold-carrier__complete-Physics-empirical-vector-with-all-adverse-and-scope-boundary-records-retained__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent` with both minimality and named-shape uniqueness passed.

Axis-by-axis uniqueness witness. The complete census is the literal product declared above. Each row below changes one survivor coordinate to a generated alternative while holding the other survivor coordinates fixed. The cited elimination is therefore a direct local witness; the complete decision file records every candidate, including every further value on non-binary axes.

Axis	Forced form	Eliminated form	Elimination reason
carrier	<code>complete-fold-carrier</code>	<code>answer-only-scalar</code>	An answer-only scalar erases the generated physical carrier.
relation	<code>complete-Physics-empirical-vector-with-all-adverse-and-scope-boundary-records-retained</code>	<code>imported-or-fitted-relation</code>	An imported or fitted relation lets consensus or target data select the law.
provenance	<code>source-bound-proof-trace</code>	<code>unbound-provenance</code>	Unbound provenance cannot demonstrate which premises produced the result.
prediction	<code>capability-closed-prediction</code>	<code>target-readable-prediction</code>	A target-readable predictor can copy or optimize against the measurement.
record	<code>separate-measurement-record</code>	<code>proof-measurement-conflation</code>	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	<code>complete-registered-rows</code>	<code>selected-favourable-rows</code>	Selecting favorable rows cannot falsify the relation.
generality	<code>one-successor-closure</code>	<code>finite-answer-lookup</code>	An answer lookup has no successor certificate.
extension	<code>no-extra-rule</code>	<code>free-extra-rule</code>	A free constant, scale or exception can select the measured answer.

Base and successor. The closure proof is sealed by derivation hash `sha256:34282851b5cb8020dc5448ec744c2bb0eff495beb17951ae20f8b2d631410cc2`. Its generality certificate is stored in `elimination_receipt.json`; it preserves the registered One base and successor statement for every generated positive finite extension. The complete candidate and decision files are part of this claim package rather than abbreviated by the paper.

Adverse controls. All four mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2166f3eb725f8e397e29da86404a23c6c6beabadbe1c1653149666d38a885966`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:768689bc7417602c40fbf5c009cabdb2944d3b93cb862e6a5a459f097fbef4a6`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f6b377ac20aa317940cf02063001fbf392bea2ce910b263150ce921e32411cf4`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:604e0ad1d0c52955bad892346c3d808138fcbe75a74497118b5b6f694a919d4a`.

Independent reconstruction. A distinct standard-library implementation regenerated the candidate product and survivor. Implementation hash: `sha256:2c1c76e69ce41d87d211866b32d87a5d932767a610da975592feac6608c021ba`. Independent certificate: `sha256:7b0c9ee0781ae529abf368b8c34477f02a06b1adae84bb07d7921a77306defd9`. Engine external-validation hash: `sha256:dad245e3012d7c9dd22c6599a0f049ae623f49895dbafea6fe234fb3ed4183cb`.

External empirical check. The target was absent until prediction seal: `True`. It was released after the matching seal: `True`. All rows were preserved: `True`. The isolation certificate denied clock, environment, filesystem, network, subprocess, dynamic-import and foreign-function capabilities and reported no attempted forbidden operation.

External source identities:

- AMDC-AME2020-MASS-1-2021
- AMDC-AME2020-MASS-1-BINDING-2021
- AMDC-NUBASE2020-2021
- AMDC-NUBASE2020-DECAY-HALFLIFE-2021
- APS-HUANG-TURBULENCE-SPECTRUM-2010
- APS-JILA-BEC-GROUND-OCCUPATION-1996
- APS-MCCOMB-TURBULENCE-STRUCTURE-2014
- APS-WILLIAMS-FALLER-HILL-1971-COULOMB
- ARXIV-1201.0501-FIZEAU-WATER-AIR
- ATLAS-2023-COMBINED-HIGGS-MASS
- ATLAS-CMS-2026-HIGGS-PAIR-COMBINATION
- ATOMIC-PRECISION-SUCCESSOR-NIST-2022-2026
- AVER-ET-AL-2026-PRIMORDIAL-HELIUM
- BABAR-2010-TAU-LFV-GAMMA
- BERUT-NATURE-2012
- BICEP-PLANCK-2022-TENSOR

- BIPM-SI-BROCHURE-9-2026
- BOREXINO-2020-CNO-NEUTRINOS
- CERN-OPEN-DATA-DATASET-API-2026-07-23
- CLOWE-2006-BULLET-CLUSTER
- CMS-2024-FOUR-LEPTON-HIGGS-MASS
- COOKE-PETTINI-STEIDEL-2018-DEUTERIUM
- D10E-STRONG-FIELD-NONLINEAR-FIXED-POINT-COMPARISON
- D10F-STRONG-CARRIER-MASSLESS-CONFINED-COMPARISON
- D9F-EXTERNAL-ORBITAL-DIMENSION-STABILITY
- D9M-POST-NEWTONIAN-FIXED-POINT-COMPARISON
- D9N-SYMMETRIC-SOURCE-CONSERVATION-COMPARISON
- D9O-STATIC-EXTERIOR-CLOCK-COMPARISON
- D9Q-QUADRUPOLE-RADIATED-POWER-COMPARISON
- DESAI-ET-AL-2025-PARKER-HCS-PROTONS
- DESI-DR2-2025-COSMOLOGY
- DIEHL-2014-SN2014J-GAMMA-LINES
- EKER-2018-MAIN-SEQUENCE-RELATIONS
- ESCAMILLA-2024-DARK-ENERGY-STATE
- FONSECA-2021-PSR-J0740-MASS
- GOMEZ-VALENT-2019-ACCELERATION
- GOMEZ-VALENT-2023-CCH-32
- GRAVITY-SPACETIME-AUTHORITATIVE-2017-2026
- GWOSC-GW231123-135430-V3-2026-07-23
- GWOSC-V2-CATALOGS-2026-07-23
- HARIDASU-2018-LATE-EXPANSION
- IAEA-ENDF-102-DATA-FORMATS-PROCEDURES
- IAEA-ENDF-CURRENT-2026-07-23
- IAEA-ENSDF-CURRENT-2026-07-23
- IAEA-INDC-NDS-0452-MAGIC-NUMBERS
- IAEA-NDS-JENDL1-INDC-JAP-45
- IAEA-NDS-LEXFOR-FISSION-2006
- IAEA-STI-PUB-1196-RADIATION-PHYSICS
- IAEA-TECDOC-2024-NUCLEAR-COGENERATION
- IAEA-THEORY-NUCLEAR-STRUCTURE-STI-PUB-249
- IAEA-WORLD-FUSION-OUTLOOK-2023
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- PDG-2025-QUARK-MASSSES
- PDG-2025-RHO-770
- PDG-2025-RHO3-1690
- PDG-2025-RHO5-2350
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- PDG-2025-SUMMARY-TABLES
- PDG-2026-ELECTROWEAK-REVIEW
- PDG-2026-ELECTROWEAK-SCALE-VECTOR
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- PDG-FORCE-SECTOR-ANCHORS-2025
- PDG-HADRON-MULTIPLETS-REGGE-2025
- PDG-NIST-NUCLEAR-RESIDUAL-RANGE-2025-2026
- PDG-NIST-NUCLEON-BINDING-2022-2026
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- PH3-EXTERNAL-COUPLED-MAP-CRITICALITY
- PIECHOCINSKA-PRA-2000
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- SUPERK-2020-PROTON-E-MU-PI0
- SUPERK-2024-PROTON-E-MU-ETA
- SUPERK-2026-PROTON-E-MU-TWO-PI0
- VACUUM-LINEAGE-AUTHORITATIVE-2015-2026
- WATSON-2019-GW170817-STRONTIUM

Observed comparison records:

- Complete immutable reconciliation of 234 empirically tested Physics claims.
- All 147 distinct registered external source identities remain present.
- Every empirical and external validation hash and every available separate measurement receipt remains bound.
- Six older materialization shapes without a separate measurement-receipt field are explicitly retained, not silently upgraded.
- All fourteen detected unfavorable-result or scope-boundary claims remain in the same evidence vector.
- Observation remains empirical evidence; prior observations are not relabelled as unseen predictions and never select a formal survivor.

Falsification condition: Reject if any empirical claim, validation hash, available measurement receipt, source identity, legacy receipt shape, unfavorable result or scope boundary is omitted or changed; if the formal receipt does not predate this reconciliation; if prior observations are relabelled as unseen; if measurement selects a formal survivor; or if any hostile control passes.

Measurement receipt: sha256:6837b0a6ba85b94ed9a282a974e7f0b11514ab82e441e01587ae8db871f71bd1.

Isolation certificate: sha256:4872b1dd81f1ece18a811463bf455907c2aa6a709b2dca12480fe968faee6613.

Custody certificate: sha256:09c27a514eb50e5e0fc87a5e8b1ad256d070da51745fed7eeb7e5010a6a9efea.

Registered source descriptions:

- complete immutable pre-lock Physics evidence surface:
experiments/external_sources/physics/snapshots/physics-grand-lock-empirical-reconciliation-record.json
(sha256:e233cd761aa874893d2c2a4e2b09f071297aee1204d531fce3d93429948177a3)

Registered target identities:

- SFT-V3-PHYSICS-COMPLETE-EMPIRICAL-RECONCILIATION from
SFT-V3-PHYSICS-IMMUTABLE-EMPIRICAL-EVIDENCE-SURFACE

Meaning. These claims are not supplemental footnotes. They are Physics-owned post-seal empirical tests of forced relations, including exact measured values, and remain first-class engine-admitted claims. In this claim specifically, the forced result identifies the minimal exact relation named in the theorem; the external body tests the sealed consequence but does not choose its

grammar or survivor.

Boundary and non-claim. The following forms are explicitly excluded:

- no favorable-only selection or deletion of mismatch, non-observation, tension, uncertainty or scope boundary
- no prior observation relabelled as an unseen prediction
- no legacy certificate silently represented as having a later separate measurement-receipt field
- no measurement, consensus model, fit, parameter or tolerance selecting a formal survivor
- no edit to the canonical engine or any admitted receipt

The result is closed only at its registered generated and empirical boundary. It does not turn an external unit scale, historical initial condition or finite observation census into a premise-free consequence of the One.

Immutable evidence identities. Source manifest

```
sha256:3ccc728167a1e7a63da295820a4061a1e60d4c985824b86eb9305cd1f097be18; engine receipt
sha256:93f4497d6f7ef3c477246079f62c21f96a7ae27fd9516fa876e9d413bbab569e at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-GRAND-LOCK-076-93f4497d6f7ef3c4.json;
empirical-validation hash
sha256:7ec55dbb12f906e009c994473b79791759328778eefc4d19b59a83ba32d3f3f2; measurement receipt
sha256:6837b0a6ba85b94ed9a282a974e7f0b11514ab82e441e01587ae8db871f71bd1.
```

25. Cross-branch synthesis

The completed branch has one compositional arc. Measurement defines what an observation can retain. Mechanics defines relational change and transfer. Fields extend source-response over local support. Waves close recurrence and propagation. Thermodynamics identifies macro-observation fibres and information-retention costs. Physical quantum theory retains complete support, phase/action and joint composition. Matter identifies recurrent constituent and channel classes. Spacetime orders events by limiting causal paths; gravitation changes their source-linked relational closure. Collective matter takes finite cell networks to scale-declared macro-observations. The cosmological boundary exports universal propagation laws while refusing to disguise historical state as a law.

This unification is operational: the same Fold objects - exact counts, ratios, words, held labels, tables, transition traces and observation fibres - appear in every subbranch. The claim is not that established terminology selected those objects. Correspondence is registered only after each Fold form survives its generated alternatives.

26. Limitations and falsification

The paper makes no claim of an actually completed physical infinity, exact continuum substrate, derivation of contingent initial conditions, unbounded astronomical census or replacement of later Chemistry, Materials, Astronomy and Cosmology branches. It does not treat a simulation as empirical proof, an exact SI definition as an independent natural observation, or portal availability as numerical agreement. Categorical external rows test structural correspondence; the dedicated value adapters test numerical consequences. A new observation outside the registered boundary requires a new preregistration and cannot silently broaden an old receipt.

Any admitted empirical claim is falsified within its boundary by a preserved external row that does not match the sealed prediction, by target access before sealing, by omitted adverse evidence, by a changed source snapshot, by failure of its independent reconstruction or by an engine-constitution violation. The correct response is a halted new claim and investigation of the derivation or adapter. Existing immutable receipts remain identities of the artifacts they certify; new evidence cannot be retroactively written into them.

27. Reproducibility

The repository uses Python's standard library for the core engine and supports macOS, Windows and Linux without Docker. Researchers may inspect every claim package directly. The one-command repository verifier remains available for release validation, but this branch build deliberately used proportionate admission-time checks and a read-only evidence census rather than repeatedly replaying historical derivations. The Physics inventory is `publications/inventories/physics.json`; this

manuscript's exact evidence map is `publications/current/physics/evidence_map.json`.

28. Conclusion

The paper's lead result is the exact derivation $\alpha^{-1} = 503846395469/3676744786$, not an isolated numerical fit but the first terminal constant in a connected Physics reconstruction. The same root and admission constitution force and validate the charged-lepton relations, precision anomalies, electroweak share, Higgs sector, Planck/proton hierarchy, proton radius, vacuum and cosmological ratios, spatial rank, inverse-square force law, nuclear and hadronic structure, quantum and relativistic correspondence, thermodynamics, collective matter and gravitation documented in the complete ledger.

At the declared derivational boundary, all 349 current V3 Physics claims and all 488 categorically Physics-owned V1/V2 atoms are engine-admitted and current-evidence closed, with no open Physics gap family. Grand Locks 075 and 076 make ownership, root trace, structural perturbation and complete empirical reconciliation explicit. The manuscript and evidence package are publication-ready; release authority remains Maria Smith's alone. Physics remains open to lawful versioned extension, correction and falsification after publication. The result is an open, inspectable tree of physical laws whose authority rests on exact derivation, complete enumeration, sealed measurement, adverse evidence and reproducible traces rather than credentials, institutional permission, opaque prediction or consensus selection.

Foundation and full-field reconstruction roadmap — version 1.2

PUBLISHED SAME-PAPER SUCCESSOR VERSION 1.2. The preceding version 1.1 remains permanently preserved in the Zenodo version chain. This version publishes the Physics full-field roadmap while preserving all 349 admitted claims, both Grand Locks and their exact empirical boundaries.

The 349-law Physics census, both Grand Locks and all 488 Physics-owned prior atoms remain current-evidence complete at the dated corpus boundary. Version 1.2 adds the permanent field roadmap and categorical handoffs while preserving Physics as open to new measurements, corrections and lawful discoveries.

This amendment adds no scientific admission by prose. Every result already in the paper retains its exact receipt and evidence boundary. Every future result must pass the untouched engine independently. A branch described as complete to a dated inventory remains permanently open to lawful extension, correction, falsification and discoveries that satisfy the same public standard.

Scientific ownership

Physics owns universal physical relations, physical constants and dimensionless physical values, measurement consequences, matter, fields, spacetime and physical cross-domain laws. A downstream science may consume an admitted physical value but may not become its substitute owner.

Layer One — foundational reconstruction

The foundational physical kernel must connect exact Fold structure to sealed measurement across:

1. observation, measurement and metrology;
2. relational position, change, motion and interaction;
3. force, transfer, conservation and recurrence;
4. fields, sources, propagation and inverse dilution;
5. waves, phase, interference and resonance;
6. thermodynamics, statistical support and retained information;
7. physical quantum correspondence and measurement;
8. matter, particles, nuclei and interactions;
9. spacetime, causal order and gravitation;
10. collective matter, fluids, plasmas and critical systems;

11. physical constants and exact measured-value comparison; and
12. handoff boundaries to Chemistry, Materials and Astronomy.

Every derived physical value must be front-loaded in the paper as a scientific result, with its exact expression, decimal rendering, measured source, units, uncertainty relation and post-seal comparison—not hidden behind file names or hashes. The hashes remain evidence beneath the result.

Layer Two — complete field reconstruction

The complete Physics census covers:

- metrology, units, calibration, uncertainty and observation theory;
- kinematics, classical mechanics, analytical-mechanics correspondence, elasticity and continuum-bounded mechanics;
- nonlinear dynamics, chaos, stability and complex physical systems;
- electromagnetism, electrodynamics, electronics-law boundaries and field interactions;
- geometrical, physical, nonlinear and quantum optics;
- acoustics, vibration, wave propagation, scattering and resonance;
- thermodynamics, statistical physics, kinetic theory and nonequilibrium systems;
- quantum foundations, quantum mechanics correspondence, atomic, molecular and optical physics at the physical-law boundary;
- particle physics, interactions, symmetries, masses, mixing and decay;
- nuclear structure, reactions, stability and heavy-element boundaries;
- relativity, gravitation, causal structure and gravitational waves;
- condensed-matter, solid-state, low-temperature, mesoscopic and nanoscale physical laws before material-specific organization;
- fluid mechanics, aerodynamics-law boundaries, plasma, soft matter, complex fluids and critical phenomena;
- biophysical, chemical-physical, geophysical and medical-physics handoffs without absorbing their downstream empirical objects;
- astrophysical law interfaces without importing contingent cosmic history;
- all known physical constants and values at their correct categorical owner;
- additional forces, vacuum and absence-state questions, energy-accounting and extraction claims, stability islands and other prior-corpus obligations;
- standing predictions with no present measurement, kept distinct from measured agreements; and
- newly discovered physical obligations admitted through versioned extension.

Publication sequence

The current Physics paper and its immutable receipts remain the baseline. Further versions add only newly admitted physical families, stronger evidence, corrections or genuinely new current-knowledge obligations. No roadmap edit reopens or repeats an

admitted claim; the live checkpoint identifies the next real extension.

Handoffs

Physics exports constants and universal relations. Chemical identity belongs to Chemistry; material organization to Materials; objects and cosmic history to Astronomy and Cosmology; planetary records to Earth and Environmental Sciences.

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Data, code, rights and participation

The canonical open repository is <https://github.com/MettaMazza/ernos-labs-sft-platform>. The release package contains this Markdown manuscript, the archival PDF, the complete evidence map, publication receipt, categorical inventory, claim packages, generated candidate censuses, elimination decisions, controls, independent validators, empirical source records and immutable engine receipts.

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standards.

Independent review, replication, lawful extension and attempted invalidation are invited. Contact Maria.Smith.Sftoe@gmail.com or <https://discord.gg/ucwGryVxGr>. Scientific admission requires the complete derivation chain and an unchanged-engine receipt; neither reputation nor criticism alone changes the model census.

Suggested citation: Smith, Maria (2026), *From Fold to Physics: An Exact, Parameter-Free and Machine-Closed Reconstruction of Physical Science from Smithian Fold Theory*, Ernos Labs Physics Branch Paper 001, version 1.1.